# FUNDAMENTALS OF PHYSICS

## Part Two

# Electromagnetism

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# Electromagnetism

**Stefan Wurm**

A·T·I·C·E

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# Contents

**Preface** i

**1 Introduction** 1

**2 Mathematical Toolkit** 9
- 2.1 Scalars, Vectors and Tensors . . . 9
- 2.2 Scalar, Vector, and Tensor Fields . . . 16
- 2.3 Vector Fields and Integral Theorems . . . 26
  - 2.3.1 Vector Field of a Conservative Force . . . 31
  - 2.3.2 Sources and Vortices of Vector Fields I . . . 32
- 2.4 The Delta Function . . . 33
  - 2.4.1 Sources and Vortices of Vector Fields II . . . 37

**3 Fundamentals of Electrostatics** 43
- 3.1 Fundamental Equations of Electrostatics . . . 44
  - 3.1.1 The Electrostatic Field . . . 44
  - 3.1.2 The Electrostatic Potential . . . 49
  - 3.1.3 Electrostatic Field Energy . . . 53
- 3.2 Fields of Bound Charge Distributions . . . 57
  - 3.2.1 The Electric Dipole . . . 59
  - 3.2.2 The Multipole Expansion of the Potential . . . 67

**4 Electrostatic Fields** 77
- 4.1 Uniformly Charged Solid or Hollow Sphere . . . 78
- 4.2 The Classical Electron Radius . . . 84
- 4.3 Charge Carrying Wire and Coaxial Cable . . . 85
- 4.4 Infinitely Extended, Planar Surface Charge . . . 89
- 4.5 Uniformly Charged Ring . . . 93
- 4.6 Uniformly Charged Disk . . . 101

**5 Conductors in Electrostatic Fields** **111**
5.1 Direct Solution of the Poisson Equation . . . 111
5.1.1 The Radially Symmetric Charge Distribution . . . 112
5.2 The Ideal Conductor in an Electrostatic Field . . . 115
5.2.1 Uniqueness of Solutions to the Poisson Equation . . . 118
5.2.2 Separation of Induction Charges . . . 119
5.2.3 The Faraday Cage . . . 121
5.3 The Principle of Image Charges . . . 122
5.3.1 Point Charge Near a Grounded Metal Plate . . . 122
5.3.2 Point Charge Near a Grounded Metal Sphere . . . 124
5.3.3 Metal Sphere in a Homogeneous Electrostatic Field . . . 128
5.4 Green's Functions . . . 134
5.4.1 Point Charge in Front of a Grounded Metal Plate . . . 137
5.4.2 Point Charge in Front of a Grounded Metal Sphere . . . 140

**6 Electrostatic Fields in Matter** **147**
6.1 Dielectric Polarization . . . 148
6.1.1 Dielectric Displacement . . . 152
6.1.2 Interfaces of and within Dielectrics . . . 159
6.2 Insulators in an Electrostatic Field . . . 163
6.2.1 The Field of a Point Charge in Dielectrics . . . 163
6.2.2 Insulating Sphere in a Homogeneous Electrostatic Field . . . 169
6.2.3 Polarized Dielectric Sphere . . . 172

**7 Dielectric Matter** **177**
7.1 Field Energy in Dielectric Materials . . . 177
7.1.1 Field Energy of the Insulator Sphere . . . 179
7.1.2 Force on Dielectric Materials in an Electrostatic Field . . . 182
7.2 Properties of Dielectric Matter . . . 185
7.2.1 Polarizable Gases . . . 186
7.2.2 Orientational Polarization . . . 189
7.2.3 Dielectric Behavior of Liquids . . . 194
7.2.4 Dielectric Behavior of Solids . . . 199

**8 Direct Electric Current** **207**
8.1 Charge Conservation and Electrical Circuits . . . 208
8.1.1 The Continuity Equation . . . 210

8.1.2 Ohm's Law and Electrical Power . . . 213
8.1.3 Simple Electrical Circuits . . . 216
8.2 Charge Transport in Electrical Conductors . . . 223
8.2.1 The Drude Model of Conductivity in Metals . . . 223
8.2.2 Electron-Hole Conduction in Semiconductors . . . 229
8.2.3 Ionic Conduction in Electrolytes . . . 232

**9 Fundamentals of Magnetostatics 237**
9.1 The Lorentz Force . . . 237
9.1.1 Force on a Current-Carrying Wire . . . 239
9.2 Ampère's Law . . . 240
9.2.1 Magnetic Field Inside a Current-Carrying Conductor . . . 244
9.2.2 Magnetic Field of a Current-Carrying Coil . . . 246
9.3 The Biot-Savart Law . . . 247
9.3.1 Straight, Infinitely Long, Current-Carrying Wire . . . 249
9.3.2 Force Between Two Current-Carrying Wires . . . 251
9.3.3 Magnetic Field of a Current Loop . . . 253
9.4 Magnetic Fields and Special Relativity . . . 256
9.4.1 Moving Charges in a Conducting Wire . . . 257
9.4.2 The Moving Plate Capacitor . . . 259
9.4.3 Point Charge in Motion . . . 263

**10 Magnetic Moment and Electromagnetic Induction 269**
10.1 Vector Potential and Magnetic Moment . . . 269
10.1.1 The Magnetic Dipole . . . 275
10.1.2 Force on a Magnetic Dipole . . . 281
10.2 Faraday's Law of Induction . . . 287
10.2.1 The Hall Effect . . . 288
10.2.2 A Relativistic Perspective . . . 290
10.2.3 Lenz's Law . . . 295
10.2.4 Self-induction . . . 297

**11 Alternating Electric Current 303**
11.1 The Generation of Alternating Current . . . 303
11.2 Complex Resistances – Impedances . . . 309
11.3 Alternating Current Power . . . 313
11.4 Simple Circuits . . . 320

11.4.1 Four-Terminal Networks . . . 320
11.4.2 The Electrical Resonant Circuit . . . 328
11.5 Effective Resistance and Ampère's Law . . . 334
11.5.1 The Frequency Dependence of Effective Resistance . . . 334
11.5.2 The Extension of Ampère's Law . . . 338

**12 Magnetism and Matter** **341**
12.1 Magnetization . . . 341
12.1.1 Diamagnetism . . . 347
12.1.2 Paramagnetism . . . 350
12.1.3 Ferromagnetism . . . 352
12.2 Magnetic Fields in Matter . . . 358
12.2.1 Interfaces of and within Magnetic Materials . . . 358
12.2.2 The Homogeneously Magnetized Sphere . . . 361
12.2.3 Magnetic Sphere in a Homogeneous Magnetic Field . . . 367

**13 Electromagnetic Waves I** **371**
13.1 Maxwell's Equations in a Vacuum . . . 371
13.1.1 Monochromatic Plane Waves . . . 373
13.1.2 Polarized Plane Waves . . . 376
13.1.3 Spherical Waves . . . 380
13.1.4 TE and TM Waves . . . 384
13.2 Field Energy and Field Momentum . . . 388
13.2.1 Electromagnetic Energy Density . . . 388
13.2.2 Electromagnetic Field Momentum . . . 393

**14 Electromagnetic Waves II** **397**
14.1 Electromagnetic Waves in Matter . . . 397
14.1.1 Reflection and Refraction of Light . . . 399
14.1.2 Electromagnetic Waves in Conductors . . . 404
14.2 Generation of Electromagnetic Waves . . . 411
14.2.1 The Oscillating Dipole . . . 412
14.2.2 Electromagnetic Radiation . . . 423

**Appendix A** **427**

**Appendix B** **443**

# Preface

Most colleges and universities around the world offer students a four-semester course in fundamental physics. These courses are usually obligatory for natural science or engineering students as well as for aspiring physicists and in many countries also for those studying to become high school physics teachers. For all those students, these two-year courses stretching over four semesters will form the basis of their physics education after which their path take different directions. The budging physicists among them will move on to take additional courses in various branches of theoretical physics or specific fields of physics, deepening their knowledge gained in the first two years in experimental and applied physics. For all others, such two year introductory courses in fundamental physics will remain a foundational element in their respective fields of specialization, be it in natural sciences other than physics, engineering or teaching. One can find such two-year college or university courses in fundamental physics labeled with various titles, but whatever their labels are, their objectives are usually quite similar. Physics is a fundamental science and as such it underlies many other sciences. Hence, it is essential for many students to master the foundations of physics over a broad range. Typically, the first of these four semester physics courses will introduce students to the fundamentals of motion physics, forces, and energy, essentially the breadth of Newtonian and relativistic mechanics, variously also including chapters on the physics of gases, fluids, vibrations and waves. The second semester course then often follows on with the physics of electrical phenomena, the fundamentals of electrostatic and electrodynamics. The third and fourth semester courses will usually focus either on the physics of heat, that is thermodynamics and aspects of statistical mechanics, and an introduction to atomic physics or on the physics of optics and optical phenomena. For most lecturers, optics is the natural third semester follow-up to the electrodynamics of the second semester, which then leaves thermodynamics, atomic and molecular physics for the fourth semester. Roughly, the present four-semester course of physics lectures, of which this volume comprises the second semester course, follows this scheme. There are of course many excellent physics textbooks out there, readily available to students. The present four semester physics lecture series does not seek to compete with any of those even though it inadvertently may do just that. What it however does seek is to provide an additional learning component which students can access easily in a hopeful highly readable and entertaining form.

Few students will enjoy slogging around heavy textbooks or working through physics compendiums which essentially still only replicate the same textbook reading experience. However, an e-book physics lecture course such as the present one may be something students will be happy to take along wherever their activities may take them while they still have to prepare for their exams.

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Albany, New York
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# 1. Introduction

Just a few centuries ago, very few people could have said anything meaningful about the concepts of electricity and magnetism. Not with respect to what these referred to and certainly not as to what their physical nature was. Today, however, for most of us electric current flows as a matter of course from the outlets in our homes. We use this electricity, the availability of which we take for granted, to power countless devices. Only when the electrical power supply fails, which, thankfully, happens rarely, do we realize the vital role it plays in our lives. In Greek mythology, Prometheus stole fire from the gods to help humanity, thereby becoming the architect of human civilization. One could quite rightly view the modern-day harnessing of electric and magnetic forces for the benefit of mankind as a second Promethean moment, one that transformed human civilization just as profoundly as, or perhaps even more fundamentally so, than the mastery of fire did. It was also the ancient Greeks who gave gave their names to what we today call electricity and magnetism. The ancient Greeks already knew that amber could become statically charged when rubbed with a wool cloth. A fact which manifested itself in the ability of such prepared amber to attract substances such as ash or dust. The ancient Greek word for amber was electron (ήλεκτρον), and thus the process of rubbing amber became associated with what was called an electrical phenomenon. The name stuck, and so we speak today of things like electric charges, electrons, electric currents, electric and magnetic forces, electrostatics and electrodynamics, and electromagnetism. In some places, this charging of amber or other suitable objects through friction is referred to as frictional electricity. However, this phenomenon of static charging has absolutely nothing to do with the mechanical friction discussed in chapter 7 of volume 1. The ancient Greeks also provided the name for magnetism. In antiquity, magnis lithos (μάγνης λίθος) referred to a stone from the Greek region of Magnesia in Thessaly that possessed a special property: a fragment of this rock would align itself in a north-south direction if placed, for instance, on a leaf floating on the surface of still water. But even though the ancient Greeks knew about those things they did not invent the magnetic compass. That in

fact happened several hundred years before the golden age of Classical Greece in ancient China, of which the Greeks however had no knowledge.

Electric and magnetic forces had been known for a long time, but not until the early 18th century was it understood that there existed two kinds of electric charges, positive and negative ones. This terminology of positive and negative charges is simply a matter of historical convention; the labels could just as easily have been reversed, with what we now call positive charge defined as negative and vice versa. The crucial point is that if a body carries equal amounts of positive and negative electric charge, it is uncharged, i.e., it is electrically neutral. A neutral state, meaning a net electrical charge of zero, implies either the complete absence of electric charge on the body or the presence of equal positive and negative charges that cancel each other out. To this day, we do not know exactly what an electric really charge is. However, we do know what characterizes electric charges:

1. There are two types of electric charge, which we refer to as positive and negative charges.
2. Two like charges, that is two negative or two positive charges, repel each other, while two unlike charges, that is a negative and a positive charge, attract each other.
3. There exists a smallest electrical charge, known as the so-called elementary charge. This elementary charge is the charge carried by an electron (negative) or a proton (positive). All other charges are integer multiples of this positive or negative elementary charge.
4. In a closed system, total electrical charge is always conserved, i.e., it remains constant. Charges can only be shifted but not created or destroyed.
5. Any change in the total charge contained within a given volume as a function of time is associated with the flow of charges during that time, i.e., the flow of an electric current out of or into that volume.

The statement regarding the creation and annihilation of charges in point 4 does not consider quantum fluctuations, but these play no role in classical electromagnetism anyway. We have already encountered Coulomb's law of electrostatic force in volume 1. Discovered by Charles-Augustin de Coulomb (1736–1806) and named after him, it describes the interaction between two charges. It shares the same structure as Newton's law of universal gravitation with the difference that, unlike this always attractive force

between two masses, the Coulomb interaction between two electric charges can be either attractive or repulsive:

$$\boldsymbol{F} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{|\boldsymbol{r}|^3}\boldsymbol{r} = q_1\boldsymbol{E}(\boldsymbol{r}) \tag{1.1}$$

In this equation $q_1$ and $q_2$ are the charges, $r$ is the distance between them, and $\boldsymbol{E}(\boldsymbol{r})$ is the electric field strength. The electrostatic force between two positive or two negative charges is always repulsive, whereas the force between a positive and a negative charge is always attractive. Like the Newtonian force between two masses, the electrostatic force between two charges decreases with the square of the distance (i.e., as $r^{-2}$); however, the Coulomb force is orders of magnitude stronger than the gravitational force. If one calculates the force between two hydrogen nuclei, that is two protons, sitting at a distance of 1 Å, the electrostatic repulsion between them is greater than their mutual gravitational attraction by a factor of $10^{36}$. However, we generally do not notice this in our daily lives because the objects surrounding us are typically electrically neutral, which means that they carry equal amounts of negative and positive charge. An adult human consists of approximately $10^{27}$ atoms. Let us assume that our bodies held slightly more negative than positive charges and that we had to carry this excess around as a surface charge. An excess charge of, say, one-millionth of those atoms, would amount to a surface charge of some 160 C per person, resulting in a repulsive force of about $2.6\cdot10^4$ N between two people standing one meter apart. For comparison: Lightning discharges transfer charges to the ground ranging from a few tenths of a coulomb to several hundred coulombs.

It is the force of gravity that binds masses such as or bodies to the much more massive Earth, but it is the forces between electric charges that give the world around us its stability. Gravitational attraction is far too weak to hold atoms, molecules, or solid bodies together. We are only able to build structures like the places we call home because these forces between electric charges exist and because the macroscopic components we use to build our homes, despite being composed of countless positive and negative electric charges, are electrically neutral. The fundamental relationships describing the interaction between stationary electric charges or moving electric charges can be summarized in just a few equations: Maxwell's equations. It is no coincidence that these were already mentioned during the discussion of special relativity in volume 1. As we shall see, and as Einstein already demonstrated, magnetism is a purely relativistic effect. How can this be? After all, natural magnets do exist, such as the lodestone mentioned earlier from the Greek region of Magnesia in Thessaly. The simple reason for this is that all magnetism stems from moving electric charges; there is no such thing as magnetic

charges. However, not all interactions between electric charges that we are aware of can be explained without recourse to the physics of the very small, i.e., to quantum mechanics. Despite the electron's negative charge and the proton's positive charge within the hydrogen atom, and the strong attractive force associated with them, the hydrogen atom remains stable.

Why doesn't the electron simply crash into the proton, as would inevitably happen with freely moving, macroscopic pairs of opposite charges? The reason lies in Heisenberg's uncertainty principle, which describes a purely quantum mechanical phenomenon. This principle states that it is impossible to know both the position and the momentum of a particle with arbitrary precision at the same time. In any case, the exact location of the electron within the hydrogen atom is unknown. From a quantum mechanical perspective, one can only specify the probability of finding the electron at a particular location within the atom. It is this probability distribution surrounding the proton that is often schematically depicted in illustrations as a cloud around the hydrogen proton. Suppose one could somehow compress this electron cloud so that it becomes increasingly localized around the proton. This, however, would amount to the equivalent of being able to measure the electron's position and momentum simultaneously with arbitrary precision. But according to the Heisenberg uncertainty principle this is impossible and thereby this principle prevents the electron from crashing into the proton.

We could have added another observation to the list of characteristics describing the interaction of electric charges above: an electric charge is always bound to particles possessing mass. There are no elementary particles that are electrically charged yet massless. However, since classical electrodynamics does not deal with massless particles, this point is usually not included among the properties of charges. Regarding point 3, we still owe an explanation as to why the discussion focuses on free carriers of charge. To this end, let us first explicitly state the principle of charge quantization as it applies in classical electrodynamics: electric charge is quantized and occurs only as an integer multiple of the electric charge $-e$ of the electron or $+e$ of the proton:

$$e = 1.602\,176\,634 \cdot 10^{-19}\,\mathrm{C} \tag{1.2}$$

For a long time, the discrete integer quantization rule of electrical charge held true beyond just the realm of classical electromagnetism. However, since Murray Gell-Mann (1929 - 2019) and George Zweig (born 1937) independently proposed the quark model in 1964 to explain the internal structure of protons and neutrons, this rule applies only with the qualification that it refers to freely occurring charge carriers. The model's original

three quarks eventually grew to six, all of which were experimentally confirmed over the following decades, the last one in 1995. Three of these quarks carry a charge of $-1/3\,e$, while the other three carry a charge of $+2/3\,e$. As with other elementary particles, corresponding antiparticles exist for these six quarks, namely six antiquarks, possessing charges of the opposite sign. Quarks and antiquarks do not exist as free particles but only in a bound state: in odd numbers, referred to as baryons, such as protons or neutrons, or in even numbers, referred to as mesons, such as pions. This explains the restriction in point 3 regarding charge quantization, which applies to free carriers of electric charges which can only occur as positive or negative integer multiples of the elementary charge. Hence, point 3 does not apply to quarks and antiquarks bound within hadrons (baryons and mesons).

William Gilbert (1544 – 1603), an English physician and natural philosopher with a particular interest in magnetic phenomena, gave electricity its name. He was the first to distinguish between magnetic forces and forces caused by static electricity, which he called *vis electrica* in Latin. In 1733, the French naturalist Charles Dufay (1698 – 1739) discovered that there existed two distinct kinds of electricity. He named one vitreous electricity (*électricité vitreuse* in French) and the other resinous electricity (*électricité résineuse* in French). At the time, both forms were still envisioned as types of electric fluids rather than as charged particles. It was Benjamin Franklin (1706 – 1790) who introduced the concept of electric charge and first distinguished between positive electricity (corresponding to Dufay's vitreous electricity) and negative electricity (corresponding to Dufay's resinous electricity). We therefore owe the definition of positive and negative electric charges to Benjamin Franklin. Franklin is also credited with discovering the conservation of electric charge and, arguably of greater practical significance at the time, with inventing the lightning rod. It was not until more than a century after Franklin's discovery that electric charges are associated with discrete particles that the electron received its name. However, the magnitude of this charge and the mass of the electron remained unknown. This changed towards the end of the 19th century when, in 1897, Joseph John Thomson (1856–1940) succeeded in determining the ratio of electric charge to mass for the particles in cathode rays — now referred to as electron beams. To achieve this, he subjected his cathode-ray tubes to electric and magnetic fields. From the deflection of the cathode rays, he was able to determine the sign of their charge and the charge-to-mass ratio of the corpuscles, the term he used for the particles within the cathode ray. For these corpuscles, this ratio was a thousand times greater than the charge-to-mass ratio of the hydrogen ion $H^+$, i.e., the proton, which was already known from electrolysis. Consequently, the cathode-ray corpuscles discovered by Thomson had

to be either extremely light or carry a very large charge. Thomson inclined toward the view that the much higher charge-to-mass ratio of his cathode-ray corpuscles was more likely to be attributable to their mass being far lower than that of the hydrogen ion rather than his corpuscles being heavy and carrying a very large electric charge. A few years after Thomson's discovery of the cathode-ray corpuscle, several different electron sources were already known. In addition to the electrons observed in gas discharges (cathode-ray tubes), there were thermionic electrons discovered in 1883 by Thomas Alva Edison (1847 - 1931), the electrons produced in the photoelectric effect discovered in 1887 by Heinrich Hertz (1857 – 1894), or the beta radiation emitted during radioactive decay associated with highly energetic electrons discovered in the years 1896 to 1899 by Ernest Rutherford (1871 - 1937) and Antoine Henri Becquerel (1852 - 1908). Although Thomson continued to insist on the term corpuscles for his unknown particles, the similarities among these charged particles from diverse sources gradually led to the adoption of the concept of the charged and particle-like electron. Given the known charge-to-mass ratio of the electron, it then became necessary to determine either the electron's charge experimentally to find its mass, or to determine its mass experimentally to find its charge.

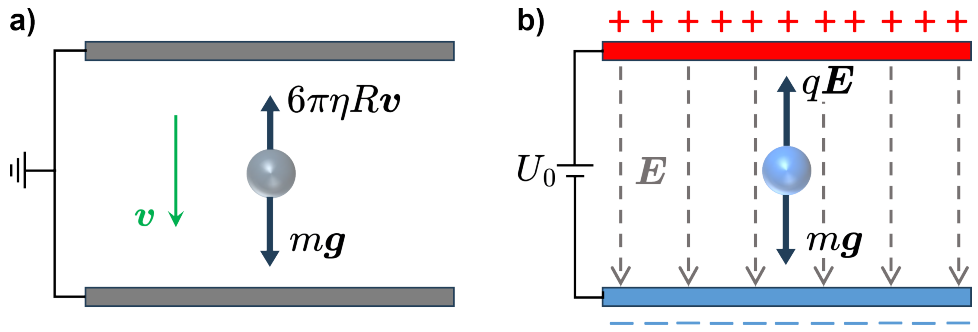



**Fig. 1.1:** The Millikan experiment for determining the elementary charge: **a)** Determination of the oil droplet's mass with the capacitor field switched off. **b)** With the capacitor field switched on, using a variable direct-current voltage source $U_0$, the oil droplet hovers when the electric force counterbalances the gravitational force.

In 1908, Robert Andrews Millikan (1868–1953) determined both values using his oil-drop experiment. Fig. 1.1 shows the experimental setup for Millikan's experiment to determine the charge and mass of the electron. Small oil droplets are sprayed into the space between the electrodes of a parallel-plate capacitor; when the capacitor is uncharged, two forces act upon them: the gravitational force $F_G = m \cdot g$ and the Stokes drag force $F_R^\eta = 6\pi\eta r v$ (already familiar to us from volume 1, section 7.1.3). Just as the settling velocity of an air molecule was determined in volume 1, section 9.4.1, the settling velocity of an oil droplet of radius $R$ in air is given by:

$$m \cdot g = \frac{4\pi}{3} R^3 \cdot \rho \cdot g = 6\pi\eta \cdot R \cdot v \tag{1.3}$$

The viscosity of air ($\eta$) and the density of the oil ($\rho$) were known. Thus, Millikan was able to determine the radius $R$ and consequently the mass $m$ of the oil droplets from their measured falling velocity (fig. 1.1a). To charge the oil droplets, Millikan used the ionizing radiation from an X-ray tube. Charging can occur either through direct ionization or through the ionization of air molecules. In the first case, the oil droplets become positively charged, whereas in the second case, they become negatively charged as electrons originating from ionized air molecules charge the droplets. With the capacitor plates polarized as shown in fig. 1.1b, the gravitational force and the electric force act in the same direction on positively charged oil droplets produced within the electric field of the parallel-plate capacitor. The situation differs however for negatively charged oil droplets, where through appropriate adjustments to the direct-current voltage $U_0$ applied between the capacitor plates the electric force can counteract the gravitational force acting on the droplets, thereby keeping them suspended. If the polarity of the capacitor in fig. 1.1b is reversed, a stable suspended position can be achieved for positively charged oil droplets, but not for negatively charged ones. In both cases, the condition for the suspension of an oil droplet with charge $q$ is given by eq. (1.3):

$$F_{el} = q \cdot E = m \cdot g \quad \Rightarrow \quad q = \frac{m \cdot q}{E} \tag{1.4}$$

In this equation $m$ is the mass of the oil droplet and $q$ the electric charge it carries. All charges of oil droplets - both positive and negative - determined in this way must be integer multiples of the elementary charge $e$ given in eq. (1.2). Naturally, with the technical means available in the early 20th century, Millikan's experiment could not determine the value of $e$ with the precision expressed in eq. (1.2). The measurement setup illustrated in fig. 1.1 suggests a simple experiment, but that is not the case. For instance, Millikan had to thermally insulate the entire apparatus in a water bath to reduce convection effects. He could only observe the movement of very small oil droplets with diameters of around $0.5\,\mu$m or less through a microscope. Millikan published the initial result of his measurements, $e = 1.63 \cdot 10^{-19}$, in 1910; following further series of measurements, his final 1913 result was $e = 1.5924 \cdot 10^{-19}$. The value given in eq. (1.2) is based on modern precision measurements that utilize quantum electrical effects. For example, single-electron transistors make it possible to observe the quantum-mechanical tunneling of individual electrons and to use this phenomenon for precise electrical charge measurements.

Today, several other methods are available for experimentally determining the elementary electric charge with high precision. An indirect method of determining $e$ is worth mentioning here, however, because it highlights the fundamental significance of the elementary charge. In quantum electrodynamics, the strength of the electromagnetic interaction — specifically, how strongly the exchange particle of the electromagnetic interaction (the photon) couples to an electrically charged elementary particle such as the electron — is described by the fine-structure constant $\alpha$. This constant is given by:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \tag{1.5}$$

The fine-structure constant $\alpha$ is determined solely by fundamental constants: the elementary charge $e$, Planck's constant $\hbar$, the speed of light $c$, and the electric field constant $\epsilon_0$. Thus, by measuring $\alpha$ and knowing these fundamental constants, one can determine the elementary charge:

$$e = \pm\sqrt{4\pi\epsilon_0\hbar c\alpha} \tag{1.6}$$

As fundamental as the elementary electric charge is to the electromagnetic interaction, we know very little about it. What is its origin? We can answer this question as to the nature of the electric charge no more than we can answer such a question with respect to the nature of mass. Yet, we can understand the mutual interactions of electric charges just the same as we can understand the mutual interactions of masses. An introduction to electrostatics, magnetostatics, and electrodynamics can begin with the historical development of these fields and the evolution of our understanding of electric and magnetic phenomena, ultimately leading to Maxwell's equations. Alternatively, it can take Maxwell's equations as a starting point and progressively deepen the understanding of these fundamental equations through the examples of a variety of electric and magnetic phenomena. The first approach has the advantage of not immediately requiring the full mathematical toolkit associated with Maxwell's equations. The second approach, however, has the merit of making the physics inherent in Maxwell's equations increasingly tangible to students as the course progresses. A middle path has been chosen here: Maxwell's equations are introduced early on, though not in their entirety. Nevertheless, the relevant mathematical tools, to which the following chapter is dedicated, are employed from the outset wherever possible.

# 2. Mathematical Toolkit

Before we can begin to explore Maxwell's equations we first need to equip ourselves with the required mathematical tools. This chapter will give a brief introduction to selected elements of vector calculus and the theory of vector fields which we then will be able to call on in the following chapters as needed. The first volume of this physics course, which dealt with Newtonian and relativistic mechanics, already presupposed some knowledge of vector calculus and also provided a brief introduction to vector fields. However, for a thorough study of electro- and magnetostatics and, above all, electrodynamics, and especially for the application of the insights gained therein, a more advanced understanding of vector calculus and vector fields is required. The latter is often introduced to students in a second semester college course, which typically includes an initial introduction to electrostatics, magnetostatics, and electrodynamics for physics students. Instead of placing such an introduction, including relevant excerpts on vector fields and vector analysis, in an appendix for reference, as is often done, it is presented here before the introduction to Maxwell's equations. The message is: one should acquire this mathematical toolkit before delving deeper into the physics of electrodynamics. However, since this is an introduction to the fundamentals of electrodynamics, we will limit ourselves here to just the essential mathematical tools required for that purpose.

## 2.1 Scalars, Vectors and Tensors

When considering physical systems, one must always begin with the measurable quantities necessary to describe such systems. These measurable quantities appear in various forms:

- In the form of a number, such as the temperature $T$ of something.
- In the form of three numbers to describe the position or velocity of an object or the local orientation of the electric or magnetic fields.

- In the form of an $n$-tuple of numbers, such as in velocity fields $\boldsymbol{v}(\boldsymbol{x})$ whose changes $\partial v_i/\partial x_j$ represent a $3 \times 3$ matrix.

These three forms of measurable quantities are referred to as

1: Scalars: Quantities that are given by a single numerical value.

2: Vectors: Quantities that have a magnitude and a direction, whose components are represented by an index.

3: Tensors: An extension of the scalar and vector concepts, but one that lacks comparable geometric clarity. Tensors of $n$-th rank are described by $n$ indices. For our purposes, only second-rank tensors are relevant here, as we already encountered them in volume 1 of this physics course, including the tensor of inertia, the stress and strain tensors, and the velocity gradient tensor.

As the special theory of relativity tells us, physical laws must hold true in different inertial frames of reference. Therefore scalars, vectors, and tensors must possess well-defined transformation behavior, which we will now examine more closely. The two fundamental operations that determine the transformation behavior of these quantities are translations and rotations.

Obviously, a scalar quantity does not change when translated or rotated. So we do not need to look at that one any further. However, when translating or rotating a vector quantity, we do need to examine it more closely. If one translates a vector, its starting and ending points shift in the same way; the magnitude, that is, the length of the vector, does not change. The rotation of a vector is a linear transformation

$$\boldsymbol{x} \xmapsto{\boldsymbol{R}} \boldsymbol{x}'$$

defined by its effect on vectors $\boldsymbol{x}$. Here, $\boldsymbol{R}$ is a so-called rotation matrix, as we already encountered it, for example, in volume 1, section 6.2.1 in the context of the Euler angles:

$$\boldsymbol{x} \longmapsto \boldsymbol{x}' = \boldsymbol{R}\boldsymbol{x}$$

As we saw there, because of $\det \boldsymbol{R} \neq 0$ an inverse matrix $\boldsymbol{R}^{-1}$ exists, and because the column vectors of these rotation matrices are pairwise orthogonal, $\boldsymbol{R}^{-1} \equiv \boldsymbol{R}^T$ holds. This has the consequence that the length of a vector, i.e., its magnitude, does not change under a rotation:

$$\boldsymbol{x}'\cdot\boldsymbol{x}' = (\boldsymbol{R}\boldsymbol{x})(\boldsymbol{R}\boldsymbol{x}) = \boldsymbol{x}^T\boldsymbol{R}^T\boldsymbol{R}\boldsymbol{x} = \boldsymbol{x}^T\boldsymbol{R}^{-1}\boldsymbol{R}\boldsymbol{x} = \boldsymbol{x}\cdot\boldsymbol{x}$$

What does the situation look like in the case of freely movable vectors, i.e., vectors whose starting point is not the origin of our coordinate system? For this, we consider the connecting vector $\Delta\boldsymbol{x} = \boldsymbol{q} - \boldsymbol{p}$ between the points $P$ and $Q$ as shown in fig. 2.2. The rotation of this vector is given by:

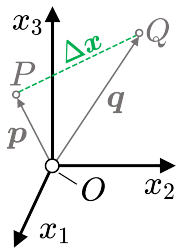



**Fig. 2.1**

$$\boldsymbol{R}\,\Delta\boldsymbol{x} \;=\; \boldsymbol{R}\,\boldsymbol{q} - \boldsymbol{R}\,\boldsymbol{p} \;=\; \boldsymbol{q}' - \boldsymbol{p}' \;=\; \Delta\boldsymbol{x}'$$

And for the length of the rotated vector $\Delta\boldsymbol{x}'$ the following applies:

$$\Delta\boldsymbol{x}'\cdot\Delta\boldsymbol{x}' \;=\; (\boldsymbol{R}\,\Delta\boldsymbol{x})(\boldsymbol{R}\,\Delta\boldsymbol{x}) \;=\; \Delta\boldsymbol{x}^T\boldsymbol{R}^T\boldsymbol{R}\,\Delta\boldsymbol{x} \;=\; \Delta\boldsymbol{x}^T\boldsymbol{R}^{-1}\boldsymbol{R}\,\Delta\boldsymbol{x} \;=\; \Delta\boldsymbol{x}\cdot\Delta\boldsymbol{x}$$

Or written out in components:

$$\sum_i \Delta x_i'^2 \;=\; \sum_i \Delta x_i^2 \qquad \text{(Length invariance)}$$

We have demonstrated herewith that rotating a vector only changes its direction but not its length. This applies to all kinds of vectors, regardless of whether they originate at the origin of a coordinate system or not. Instead of translating or rotating a vector within its coordinate system, we can also view it from another coordinate system translated or rotated with respect to the vector's own coordinate system. The result is the same. As a prominent example of length invariance, we only need to look at the 4-dimensional space-time $x_0 = ct$, $x_1 = x$, $x_2 = y$ and $x_3 = z$ of special relativity. The invariance of the space-time interval, which was discussed in volume 1, sections 12.2.4 and 12.3.3, follows directly from this length invariance discussed here:

$$\sum_{i=3}^{3} \Delta x_i^2 - c^2\Delta t^2 \;\equiv\; \sum_{i=3}^{3} \Delta x_i'^2 - c^2\Delta t'^2 \;\equiv\; \Delta s^2$$

Before we summarize, a brief note on the notation of vector products or vector rotations. Vector products are written as

$$\boldsymbol{a}\cdot\boldsymbol{b} \;=\; \sum_{i=1}^{3} a_i\cdot b_i \;\equiv\; a_i b_i$$

and vector rotations as

$$\mathbf{x}' \;=\; \boldsymbol{R}\,\boldsymbol{x} \qquad \Longleftrightarrow \qquad x_i' \;=\; \sum_{j=1}^{3} R_{ij}x_j \;\equiv\; R_{ij}x_j$$

where the $R_{ij}$ are the components of the rotation matrix $\boldsymbol{R}$. The expression following the "≡" symbol indicates the summation convention, which states that when the summation symbol is suppressed, summation is always performed over the two identical indices.

In summary, scalars, vectors, and tensors are defined using the transformation properties just described as follows:

***Scalars***
*Scalars are quantities $a$ that remain invariant under rotations (and translations), i.e., $a' = a$.*

***Vectors***
*3-dimensional vectors are quantities consisting of three components $\boldsymbol{a} = (a_1, a_2, a_3)$, which transform under rotations of the coordinate system in the same way as the coordinate difference $\Delta x_i$ of two points in space:*

$$\Delta x_i' = \sum_{j=1}^{3} R_{ij} \Delta x_j$$

***Tensors***
*Second-rank tensors are quantities $T_{ij}$ that transform under rotations like products $T_{ij} = \Delta x_i \Delta x_j$. The transformation of these products is given by:*

$$T_{ij}' = \sum_{k,l} R_{ik} R_{jl} T_{kl}$$

A scalar quantity that is familiar to almost everyone is the temperature $T$. If we measure the temperature $T$ with a thermometer suspended in a fixed location, someone else measuring it in a coordinate system that is shifted, rotated, or both relative to ours will always obtain the same temperature $T$.

As an example of a vector, one needs only to look at the velocity $\boldsymbol{v}$. In fact, almost everyone intuitively recognizes that velocity is a vector, regardless of whether it is explicitly called a vector or not. It transforms as follows:

$$v_i' = \frac{\mathrm{d}x_i'}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(R_{ij} x_j) = R_{ij} \frac{\mathrm{d}x_j}{\mathrm{d}t} = R_{ij} v_j$$

The scalar product (dot product) of two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ is a scalar, and their vector or cross product is a vector. We look at the dot product first:

$$\boldsymbol{u}'\boldsymbol{v}' = (\boldsymbol{R}\,\boldsymbol{u})(\boldsymbol{R}\,\boldsymbol{v}) = \boldsymbol{u}^T \boldsymbol{R}^T \boldsymbol{R}\,\boldsymbol{v} = \boldsymbol{u}^T \mathbf{1}_3\,\boldsymbol{v} = \boldsymbol{u}^T \boldsymbol{v} = \boldsymbol{u}\,\boldsymbol{v}$$

If $\alpha$ and $\alpha'$ denote the angles between $\boldsymbol{u}$ and $\boldsymbol{v}$ and $\boldsymbol{u}'$ respectively, then the following applies to the magnitude of the scalar product:

$$|\boldsymbol{u}'\boldsymbol{v}'| \;=\; |\boldsymbol{u}'||\boldsymbol{v}'|\cos\alpha' \;=\; |\boldsymbol{u}||\boldsymbol{v}|\cos\alpha \;=\; |\boldsymbol{u}\boldsymbol{v}|$$

The length of vectors, i.e., their magnitude, does not change under rotation. Therefore, $|\boldsymbol{u}'| = |\boldsymbol{u}|$ and $|\boldsymbol{v}'| = |\boldsymbol{v}|$, and thus $\alpha' = \alpha$. Not only the lengths of the vectors, but also the included angle remains unchanged under rotation. For the cross product $\boldsymbol{w}' \;=\; \boldsymbol{u}' \times \boldsymbol{v}'$, the following holds:

$$\boldsymbol{w}' \;=\; \boldsymbol{R}\,\boldsymbol{w} \;=\; \boldsymbol{R}(\boldsymbol{u} \times \boldsymbol{v})$$

and with $\boldsymbol{u}' = \boldsymbol{R}\,\boldsymbol{u}$ and $\boldsymbol{v}' = \boldsymbol{R}\,\boldsymbol{v}$ thus applies:

$$\boldsymbol{R}\,\boldsymbol{u} \times \boldsymbol{R}\,\boldsymbol{v} \;=\; \boldsymbol{R}(\boldsymbol{u} \times \boldsymbol{v})$$

Considering the magnitude of the vector product

$$|\boldsymbol{u}' \times \boldsymbol{u}'| = |\boldsymbol{u}'||\boldsymbol{v}'|\sin\alpha' \;=\; |\boldsymbol{u}||\boldsymbol{v}|\sin\alpha \;=\; |\boldsymbol{u} \times \boldsymbol{u}|$$

one quickly sees that angle conservation must also hold for the vector product, since $|\boldsymbol{u}'| = |\boldsymbol{u}|$ and $|\boldsymbol{v}'| = |\boldsymbol{v}|$, i.e., $\alpha' = \alpha$. Two frequently occurring vector operations are the scalar triple product:

$$(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) = \boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) \;=\; \boldsymbol{c} \cdot (\boldsymbol{a} \times \boldsymbol{b}) \;=\; \boldsymbol{b} \cdot (\boldsymbol{c} \times \boldsymbol{a}) \tag{2.1}$$

and the double vector product (the so-called Grassmann identity):

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) \;=\; \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b}) \tag{2.2}$$

or respectively:

$$(\boldsymbol{a} \times \boldsymbol{b}) \times \boldsymbol{c} \;=\; \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{a}(\boldsymbol{b} \cdot \boldsymbol{c}) \tag{2.3}$$

Because of the sequence of letters on the right-hand side of the first version of the Grassmann identity, this equation is also known as the *BAC-CAB* formula. Another useful formula is that for the scalar product of two vector products. To obtain this, we first rewrite the scalar product of two vector products using the cyclic invariance of the scalar triple product:

$$(\boldsymbol{a} \times \boldsymbol{b})(\boldsymbol{c} \times \boldsymbol{d}) \;=\; [(\boldsymbol{a} \times \boldsymbol{b}) \times \boldsymbol{c}] \cdot \boldsymbol{d}$$

We now apply the second version of the Grassmann identity to the expression inside the square brackets on the right-hand side of this equation and obtain:

$$[(\boldsymbol{a} \times \boldsymbol{b}) \times \boldsymbol{c}] \cdot \boldsymbol{d} \;=\; (\boldsymbol{b}\boldsymbol{d})(\boldsymbol{a}\boldsymbol{c}) - (\boldsymbol{a}\boldsymbol{d})(\boldsymbol{b}\boldsymbol{c})$$

Therefore, the so-called Lagrange identity applies to the scalar product of two vector products:

$$(\boldsymbol{a} \times \boldsymbol{b})(\boldsymbol{c} \times \boldsymbol{d}) \;=\; (\boldsymbol{bd})(\boldsymbol{ac}) - (\boldsymbol{ad})(\boldsymbol{bc}) \tag{2.4}$$

Another useful vector identity is:

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) + \boldsymbol{b} \times (\boldsymbol{c} \times \boldsymbol{a}) + \boldsymbol{c} \times (\boldsymbol{a} \times \boldsymbol{b}) \;=\; 0 \tag{2.5}$$

Tensors are a generalization of the concepts of scalars, vectors, and matrices. In the simplest case, tensors are two-dimensional square matrices. Such tensors are called second-rank tensors, whereas vectors are sometimes referred to as first-rank tensors and scalars as zero-rank tensors. A second-rank tensor is represented by two indices, for example as $T_{ij}$ or $T_{kl}$. An $n$-th-rank tensor is represented by $n$ indices, i.e., $T_{i_1 i_2 i_3 \cdots i_n}$. While the elements $T_{ijk}$ of a third-rank tensor can still be associated with the components of the three spatial axes, $n$-th-rank tensors with $n > 3$ are abstract mathematical objects. However, the same mathematical rules apply to them as to second- or third-rank tensors; one simply has to account for more indices. Thus, the results of the scalar multiplication of a tensor and of tensor addition or tensor subtraction

$$\lambda \cdot T_{i_1 i_2 i_3 \cdots i_n} \;=\; (\lambda T)_{i_1 i_2 i_3 \cdots i_n}$$

and

$$\mu \cdot S_{i_1 i_2 i_3 \cdots i_n} \pm \lambda \cdot T_{i_1 i_2 i_3 \cdots i_n} \;=\; (\mu S \pm \lambda T)_{i_1 i_2 i_3 \cdots i_n}$$

are themselves tensors. The scalars $\mu$ and $\lambda$ can be real or complex, as can the tensor elements of $S$ and $T$ themselves. For $n = 0$, the second equation represents the rule for scalar addition; for $n = 1$, the rule for vector addition; and for $n = 2$, the rule for matrix addition. In this context, a single rule applies to tensors of any rank: only tensors of the same rank can be added.

The direct product of tensors of rank $n$ and $m$ is a tensor of rank $n + m$, i.e., a tensor of higher rank (except in the case where $n = m = 0$):

$$S_{i_1 i_2 i_3 \cdots i_n} \otimes T_{i_1 i_2 i_3 \cdots i_m} \;=\; U_{i_1 i_2 i_3 \cdots i_{n+m}} \qquad (*)$$

The symbol "$\otimes$" specifically indicates a tensor operation and not a mapping between vector spaces of the kind represented by the multiplication symbol "$\cdot$" or the vector product symbol "$\times$". In the case where $n = m = 0$, i.e., if we are dealing with two scalars on the left-hand side of $(*)$, the right-hand side of the equation is, of course, also a scalar. The tensor product of two zeroth-rank tensors is a zeroth-rank tensor. The situation differs for first-rank tensors. In this case, the operation is referred to as the so-called dyadic product, which is given by:

$$\boldsymbol{u} \otimes \boldsymbol{v} \;=\; \boldsymbol{u} \cdot \boldsymbol{v}^T \;=\; \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} \cdot \begin{pmatrix} v_1 \cdots v_n \end{pmatrix} \;=\; \begin{pmatrix} u_1 v_1 & \cdots & u_1 v_n \\ \vdots & \ddots & \vdots \\ u_n v_1 & \cdots & u_n v_n \end{pmatrix} \tag{2.6}$$

The tensor product of two first-rank tensors is thus a second-rank tensor, i.e., a matrix. In the case of $n = m = 2$, the tensor product of two matrices, the elements of $U_{i_1 i_2 i_3 \cdots i_{n+m}}$ are then:

$$u_{iklm} \;=\; s_{ik} \otimes t_{lm} \;=\; s_{ik} t_{lm} \tag{2.7}$$

The $u_{iklm}$ are thus the components of a fourth-rank tensor. Summing these $u_{iklm}$ over two indices results in a contraction of the tensor:

$$u_{iilm} \;=\; \sum_i u_{iilm} \;=\; a_{lm} \tag{2.8}$$

Summation over two indices transforms a fourth-rank tensor into a second-rank tensor. It is also said that summation over two indices reduces the rank of the fourth-rank tensor or contracts it into a second-rank tensor. This contraction rule applies generally, not just to fourth-rank tensors. For second-rank tensors, i.e., matrices, we are familiar with this rank reduction under the term trace formation:

$$\mathrm{Tr}(t_{ik}) \;=\; t_{ii} \;=\; \sum_i t_{ii} \tag{2.9}$$

The trace of a matrix (Tr is the trace operator) is a scalar. Therefore, taking the trace contracts a second-rank tensor to a zeroth-rank tensor. A special tensor is the Kronecker symbol

$$\delta_{ij} \;=\; \begin{cases} 1 & \text{if} \quad i = j \\ 0 & \text{if} \quad i \neq j \end{cases} \tag{2.10}$$

which transforms into itself via

$$\delta'_{ij} \;=\; R_{ik} R_{jl} \delta_{kl} \;=\; R_{ik} R_{jk} \;=\; R_{ik} R^T_{kj} \;=\; (R R^T)_{ij} \;=\; \delta_{ij}$$

and hence represents an invariant tensor. This property of the Kronecker symbol allows, for example, the scalar product of two vectors $\boldsymbol{u} \cdot \boldsymbol{v}$ to be written as the product of two second-rank tensors:

$$\boldsymbol{u} \cdot \boldsymbol{v} \;=\; u_i v_j \delta_{ij} \tag{2.11}$$

Here, $u_i v_j = w_{ij}$ is the first second-rank tensor, and $\delta_{ij}$ is the second second-rank tensor. Once again, as already shown above, we see that the product of two second-rank tensors is a zero-rank tensor, i.e., a scalar.

This concludes our brief overview of the transformation properties of scalars, vectors, and tensors. The focus of this discussion was to gain a fundamental understanding of the key transformation properties of vectors (first-rank tensors) and matrices (second-rank tensors). Naturally, there is much more to learn on this subject than what was presented here and such extended knowledge is indispensable in theoretical electrodynamics. However, for an introductory course in electrodynamics, what we have learned in this section about scalars, vectors, and tensors suffices to allow us to turn our attention now to the corresponding fields of such quantities.

## 2.2 Scalar, Vector, and Tensor Fields

A function

$$a(\boldsymbol{x}) = a(x_1, x_2, x_3)$$

that assigns a real number, i.e., a scalar, to every point in space describes a scalar field. Perhaps the most familiar example of such a scalar field is temperature, where a specific temperature $T$ is assigned to each point in space. Hiking maps with contour lines are representations of a scalar field that depicts elevation above sea level. Another example is air pressure. When we view temperature maps or so-called isobar maps, i.e., maps showing lines of equal air pressure, during the evening weather forecast, we are likewise looking at two-dimensional scalar fields. Simply put, we are dealing with scalar fields whenever there is a mapping from an $n$-dimensional space to a one-dimensional space of numbers.

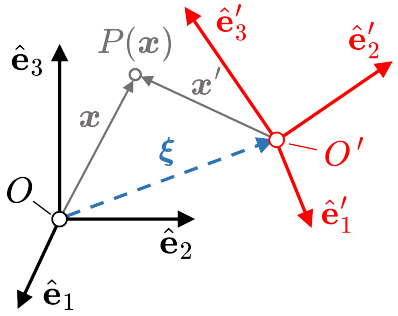



**Fig. 2.2**

How do such scalar fields behave under translation and rotation of the coordinate system? To answer this question, we consider the coordinates of the point $P(\boldsymbol{x})$ in fig. 2.2 in the original coordinate system $O$ and in the translated and rotated coordinate system $O'$. The vectors $\boldsymbol{x}$ and $\boldsymbol{x}'$ describe the same point in both coordinate systems and they are related by:

$$\boldsymbol{x}' = \boldsymbol{x} - \boldsymbol{\xi}$$

Expressed in terms of the components of the unit vectors along the respective coordinate axes in both systems, this relationship reads:

$$\sum_{j=1}^{3} x'_j \hat{\boldsymbol{e}}'_j = \sum_{j=1}^{3} x_j \hat{\boldsymbol{e}}_j - \sum_{j=1}^{3} \xi_j \hat{\boldsymbol{e}}_j \tag{2.12}$$

Let $\boldsymbol{R}$ be the orthogonal transformation that maps the unit vectors $\hat{\boldsymbol{e}}_j$ at $O$ to the unit vectors $\hat{\boldsymbol{e}}_i'$ at $O'$, that is:

$$\hat{\boldsymbol{e}}_i' \;=\; \sum_{j=1}^{3} R_{ij}\hat{\boldsymbol{e}}_j$$

Using this relationship between the unit vectors, if we now multiply equation eq. (2.12) from the right by the unit vector $\hat{\boldsymbol{e}}_i$, we obtain

$$\sum_{j=1}^{3} x_j' \underbrace{\hat{\boldsymbol{e}}_j'\hat{\boldsymbol{e}}_i}_{R_{ji}} \;=\; \sum_{j=1}^{3}(x_j-\xi_j)\underbrace{\hat{\boldsymbol{e}}_j\hat{\boldsymbol{e}}_i}_{\delta_{ji}}$$

giving us the result:

$$x_i-\xi_i \;=\; \sum_{j=1}^{3} R_{ji}x_j' \tag{2.13}$$

As the next step, we now multiply eq. (2.12) from the right by the unit vector $\hat{\boldsymbol{e}}_k'$ and thereby obtain

$$\sum_{j=1}^{3} x_j' \underbrace{\hat{\boldsymbol{e}}_j'\hat{\boldsymbol{e}}_k'}_{\delta_{jk}} \;=\; \sum_{j=1}^{3}(x_j-\xi_j)\underbrace{\hat{\boldsymbol{e}}_j\hat{\boldsymbol{e}}_k'}_{R_{kj}}$$

giving us the result:

$$x_k' \;=\; \sum_{j=1}^{3} R_{kj}(x_j-\xi_j) \tag{2.14}$$

In vector form, eq. (2.13) and eq. (2.14) read:

$$\boldsymbol{x}-\boldsymbol{\xi} \;=\; \boldsymbol{R}^{-1}\boldsymbol{x}' \qquad \text{and} \qquad \boldsymbol{x}' \;=\; \boldsymbol{R}(\boldsymbol{x}-\boldsymbol{\xi}) \tag{2.15}$$

The scalar field $a$ in $O$ and the scalar field $a'$ in $O'$ must have the same value at point $P$ in fig. 2.2; in view of what has just been shown, it therefore holds that:

$$a(\boldsymbol{x}) \;=\; a'(\boldsymbol{x}') \;=\; a'\big(\boldsymbol{R}(\boldsymbol{x}-\boldsymbol{\xi})\big) \tag{2.16}$$

We can now directly apply what we learned about the transformation of the scalar field to vector fields. To do this, we consider the vector field

$$\boldsymbol{a}(\boldsymbol{x}) \;=\; \boldsymbol{a}(x_1,x_2,x_3)$$

and rewrite this expression as

$$\boldsymbol{a}(\boldsymbol{x}) \;=\; \sum_{i=1}^{3} a_i(\boldsymbol{x})\hat{\boldsymbol{e}}_i \;=\; \sum_{i=1}^{3} a_i'(\boldsymbol{x}')\hat{\boldsymbol{e}}_i' \;=\; \sum_{i,j=1}^{3} a_i'(\boldsymbol{x}')R_{ij}\hat{\boldsymbol{e}}_j \;=\; \sum_{j=1}^{3} a_j(\boldsymbol{x})\hat{\boldsymbol{e}}_j$$

It follows directly from this that

$$a_j(\boldsymbol{x}) \,=\, \sum_{i=1}^{3} R_{ij} a'_i(\boldsymbol{x}') \qquad \text{and} \qquad a'_k(\boldsymbol{x}') \,=\, \sum_{i=1}^{3} R_{ki} a_i(\boldsymbol{x}) \tag{2.17}$$

or, respectively, in vector form

$$\boldsymbol{a}(\boldsymbol{x}) \,=\, \boldsymbol{R}^{-1}\boldsymbol{a}'(\boldsymbol{x}') \qquad \text{and} \qquad \boldsymbol{a}'(\boldsymbol{x}') \,=\, \boldsymbol{R}\,\boldsymbol{a}(\boldsymbol{x}) \tag{2.18}$$

The above results for the transformation of scalar and vector fields can be readily extended to second-rank tensor fields. As can be easily shown with a little calculation, the relationship between transformed and original second-rank tensor fields is given by the expressions:

$$T_{j_1 j_2}(\boldsymbol{x}) \,=\, \sum_{i_1 i_2} R_{i_1 j_1} R_{i_2 j_2} T'_{i_1 i_2}(\boldsymbol{x}') \,=\, R_{i_1 j_1} R_{i_2 j_2} T'_{i_1 i_2}(\boldsymbol{x}') \tag{2.19}$$

and

$$T'_{k_1 k_2}(\boldsymbol{x}') \,=\, \sum_{i_1 i_2} R_{k_1 i_1} R_{k_2 i_2} T_{i_1 i_2}(\boldsymbol{x}) \,=\, R_{k_1 i_1} R_{k_2 i_2} T_{i_1 i_2}(\boldsymbol{x}) \tag{2.20}$$

Having seen how scalar, vector, and second-rank tensor fields transform, the question arises as to how changes in these fields themselves transform. To this end, we first consider a scalar field and its change. Let $a(\boldsymbol{x})$ be a scalar field that is differentiable at the points $P(\boldsymbol{x})$ and $Q(\boldsymbol{x} + \mathrm{d}\boldsymbol{x})$. The infinitesimal change $\mathrm{d}a$ of the scalar field between $P$ and $Q$ is then

$$\mathrm{d}a \,=\, a(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) - a(\boldsymbol{x}) \,=\, a(\boldsymbol{x}) + \sum_i \frac{\partial a}{\partial x_i}\mathrm{d}x_i + \ldots - a(\boldsymbol{x})$$

If we neglect higher-rank terms in the Taylor expansion of $a$ then

$$\mathrm{d}a \,=\, \sum_i \frac{\partial a}{\partial x_i}\mathrm{d}x_i$$

The quantity $\mathrm{d}a$ is a scalar, and the $\mathrm{d}x_i$ are the components of the vector $\mathrm{d}\boldsymbol{x}$. Therefore, the $\partial a/\partial x_i$ must also be components of a vector. This vector is the so-called gradient of the scalar field $a$ at the point $\boldsymbol{x}$, and for $i = 3$, i.e., in three-dimensional Cartesian space, this gradient vector is given by:

$$\left(\frac{\partial a}{\partial x_1}, \frac{\partial a}{\partial x_2}, \frac{\partial a}{\partial x_3}\right) \,=\, \mathrm{grad}\, a \,=\, \nabla a$$

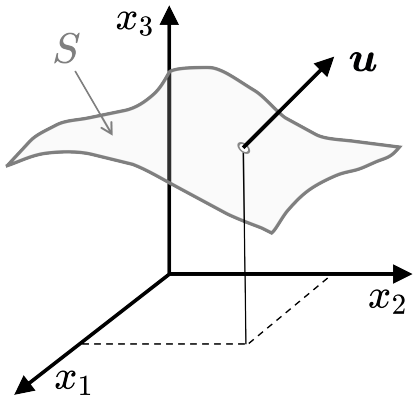



**Fig. 2.3**

The symbol $\nabla$ denotes the so-called nabla operator, a row vector defined as (here for the case $n = 3$):

$$\nabla = \sum_{i=1}^{n} \hat{\boldsymbol{e}}_i \frac{\partial}{\partial x_i} \tag{2.21}$$

To understand what $\nabla a$ means, let us now consider a scalar field $a(\boldsymbol{x})$ that is constant, meaning:

$$a(x_1, x_2, x_3) = c$$

Solving this equation for $x_3$ yields:

$$x_3 = S(x_1, x_2, c) = x_3(x_1, x_2, c)$$

Fig. 2.3 shows such a surface $S$ as a function of the variable parameters $x_1$ and $x_2$ and the constant parameter $c$. We now make the assertion that the vector

$$\boldsymbol{u}(\boldsymbol{x}) = \frac{\nabla a}{|\nabla a|}$$

in fig. 2.3 is the unit vector normal to the surface $S$. To prove this, we consider the constant scalar field $a$ at the points $a(\boldsymbol{x})$ and $a(\boldsymbol{x} + \Delta\boldsymbol{x})$:

$$a(\boldsymbol{x}) = c$$

$$a(\boldsymbol{x} + \Delta\boldsymbol{x}) = c = a(\boldsymbol{x}) + \nabla a \Delta\boldsymbol{x} + \dots$$

Here, we have again neglected the higher-rank terms of the Taylor expansion. A comparison of the two equations shows that the following must hold:

$$\nabla a \Delta\boldsymbol{x} = 0 \qquad \Rightarrow \qquad \nabla a \perp \Delta\boldsymbol{x}$$

$\nabla a$ is thus perpendicular to the vector $\Delta\boldsymbol{x}$ lying in the surface $a(\boldsymbol{x}) = c$. This proves the above assertion that $\boldsymbol{u}$ is the unit normal vector to the surface $S$ for which $a(x_1, x_2, x_3) = c$. Let us now consider a $\Delta\boldsymbol{x}$ that does not lie in the surface $a(\boldsymbol{x}) = c$, i.e., for which applies:

$$a(\boldsymbol{x} + \Delta\boldsymbol{x}) = c + \Delta c = a(\boldsymbol{x}) + \nabla a \Delta\boldsymbol{x} + \dots$$

However, the condition $a(\boldsymbol{x}) = c$ still holds, so obviously $\nabla a \Delta\boldsymbol{x} = \Delta c$. Let us assume that this $\Delta\boldsymbol{x}$, which points away from the surface $a(\boldsymbol{x}) = c$ as sketched in fig. 2.4, is parallel to $\nabla a$, i.e., $\Delta\boldsymbol{x} \parallel \nabla a$. Then

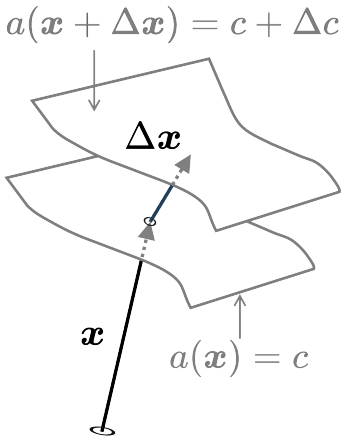



**Fig. 2.4**

$$\nabla a \Delta \boldsymbol{x} \;=\; |\nabla a||\Delta \boldsymbol{x}| \;=\; \Delta c$$

However, this is only possible if $\Delta c > 0$. Thus, the gradient of $a$ points in the direction of the surface $c + \Delta c$ when $\Delta c > 0$. The significance of the gradient of a scalar field lies in the fact that it allows one to determine the field's rate of change in any arbitrary direction at any point within the field. Let $\boldsymbol{\xi}$ be an arbitrary constant vector and $a(\boldsymbol{x})$ a scalar field; then the derivative of the field in the direction of the vector $\boldsymbol{\xi}$ is defined by:

$$\frac{\mathrm{d}a}{\mathrm{d}\boldsymbol{\xi}} \;=\; \frac{\partial a}{\partial x_1}\xi_1 + \frac{\partial a}{\partial x_2}\xi_2 + \frac{\partial a}{\partial x_3}\xi_3 \tag{2.22}$$

Expressed in terms of the gradient of the scalar field $a(\boldsymbol{x})$, this becomes:

$$\frac{\mathrm{d}a}{\mathrm{d}\boldsymbol{\xi}} \;=\; \nabla a \cdot \boldsymbol{\xi} \;=\; \boldsymbol{\xi} \cdot \nabla a \tag{2.23}$$

The gradient is a vector and differential operator that acts on the argument to its right. This argument need not be merely a scalar field $a(\boldsymbol{x})$, as just considered; it can, of course, also be a vector field $\boldsymbol{a}(\boldsymbol{x})$. To understand what the gradient of a vector or vector field is, let us consider an infinitesimal change in the vector field, that is:

$$\mathrm{d}\boldsymbol{a} \;=\; \boldsymbol{a}(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}) - \boldsymbol{a}(\boldsymbol{x})$$

or, respectively:

$$\sum_i \hat{\boldsymbol{e}}_i \mathrm{d}a_i \;=\; \sum_i \hat{\boldsymbol{e}}_i \left( a_i(\boldsymbol{x}) + \sum_j \frac{\partial a_i(\boldsymbol{x})}{\partial x_j} \mathrm{d}x_j \right) - \sum_i \hat{\boldsymbol{e}}_i a_i(\boldsymbol{x})$$

It follows immediately from this equation:

$$\mathrm{d}a_i \;=\; \sum_j \frac{\partial a_i(\boldsymbol{x})}{\partial x_j} \mathrm{d}x_j \;=\; \sum_j \left( \mathrm{d}x_j \frac{\partial}{\partial x_j} \right) a_i(\boldsymbol{x}) \;=\; (\mathrm{d}\boldsymbol{x} \cdot \nabla) a_i$$

The expression on the far right of this equation looks exactly like the right-hand side of eq. (2.23). It is thus a type of directional derivative of the component $a_i(\boldsymbol{x})$ of the vector field $\boldsymbol{a}(\boldsymbol{x})$. Viewed in isolation, the $a_i(\boldsymbol{x})$ are indeed scalar fields; consequently, the expression obtained here can be interpreted as the directional derivative of the scalar field $a_i(\boldsymbol{x})$ in the direction of $\mathrm{d}\boldsymbol{x}$. Accordingly, the directional derivative of a vector field $\boldsymbol{a}(\boldsymbol{x})$ in the direction of the constant vector $\boldsymbol{\xi}$ is defined as:

$$\frac{\mathrm{d}\boldsymbol{a}}{\mathrm{d}\boldsymbol{\xi}} \;=\; \frac{\partial \boldsymbol{a}}{\partial x_1}\xi_1 + \frac{\partial \boldsymbol{a}}{\partial x_2}\xi_2 + \frac{\partial \boldsymbol{a}}{\partial x_3}\xi_3 \tag{2.24}$$

Using the vector gradient

$$\nabla \boldsymbol{a} = \begin{bmatrix} \frac{\partial a_1}{\partial x_1} & \frac{\partial a_1}{\partial x_2} & \frac{\partial a_1}{\partial x_3} \\ \frac{\partial a_2}{\partial x_1} & \frac{\partial a_2}{\partial x_2} & \frac{\partial a_2}{\partial x_3} \\ \frac{\partial a_3}{\partial x_1} & \frac{\partial a_3}{\partial x_2} & \frac{\partial a_3}{\partial x_3} \end{bmatrix} \tag{2.25}$$

eq. (2.24) can be written as:

$$\frac{\mathrm{d}\boldsymbol{a}}{\mathrm{d}\boldsymbol{\xi}} = \nabla \boldsymbol{a} \cdot \boldsymbol{\xi} = \boldsymbol{\xi}^T \cdot [\nabla \boldsymbol{a}]^T \tag{2.26}$$

This equation for the directional derivative of a vector field $\boldsymbol{a}(\boldsymbol{x})$ has the same form as the equation for the directional derivative of a scalar field in eq. (2.23), with the difference that the gradient and the vector on the right-hand side of eq. (2.26) cannot be swapped as easily as in eq. (2.23), since the gradient of a vector field is a matrix. Using eq. (2.9), the trace of the vector gradient of $\boldsymbol{a}$ in eq. (2.25) is

$$\mathrm{Tr}([\nabla \boldsymbol{a}]_{ik}) = \nabla \boldsymbol{a}_{ii} = \frac{\partial a_1}{\partial x_1} + \frac{\partial a_2}{\partial x_2} + \frac{\partial a_3}{\partial x_3} = \nabla \cdot \boldsymbol{a} \tag{2.27}$$

since, with the nabla operator from eq. (2.21), it holds that

$$\nabla \cdot \boldsymbol{a} = \hat{\boldsymbol{e}}_1 \frac{\partial \boldsymbol{a}}{\partial x_1} + \hat{\boldsymbol{e}}_2 \frac{\partial \boldsymbol{a}}{\partial x_2} + \hat{\boldsymbol{e}}_3 \frac{\partial \boldsymbol{a}}{\partial x_3} = \frac{\partial a_1}{\partial x_1} + \frac{\partial a_2}{\partial x_2} + \frac{\partial a_3}{\partial x_3} \tag{2.28}$$

The operation $\nabla \cdot \boldsymbol{a}$, also frequently written as $\mathrm{div}\, \boldsymbol{a}$, is the so-called divergence of the vector field $\boldsymbol{a}$. We previously addressed the divergence of vector fields in the context of fluid motion in volume 1 of this physics course. There, we saw that the divergence of a vector field provides information about the source density of that field. As illustrated in fig. 2.5a, the divergence around a region in a vector field is positive if that region acts as a source of field lines. The divergence is negative around a region where field lines disappear, that is, where there is a sink for field lines, as shown in fig. 2.5b. And if a region contains neither sources nor sinks of field lines, as shown in fig. 2.5c, then the divergence vanishes in that area of the vector field. The divergence of a vector field is a scalar field and it serves, as illustrated in a simplified manner in fig. 2.5a, as a measure of the source density of the vector field within a specific region. To consider the three-dimensional case, we revisit the concept of the flux of a vector field, a topic we previously encountered in volume 1 in the context of fluid motion where we dealt with the flow velocity field $\boldsymbol{v}$. There, we saw that the flux of a vector field, here that of the electric field $\boldsymbol{E}$, through a surface $\mathrm{d}A$ is given by $\boldsymbol{E} \cdot \hat{\boldsymbol{n}}$, where $\hat{\boldsymbol{n}}$ is the unit vector in the direction of the surface normal of $\mathrm{d}A$, as sketched in the upper part of fig. 2.6.

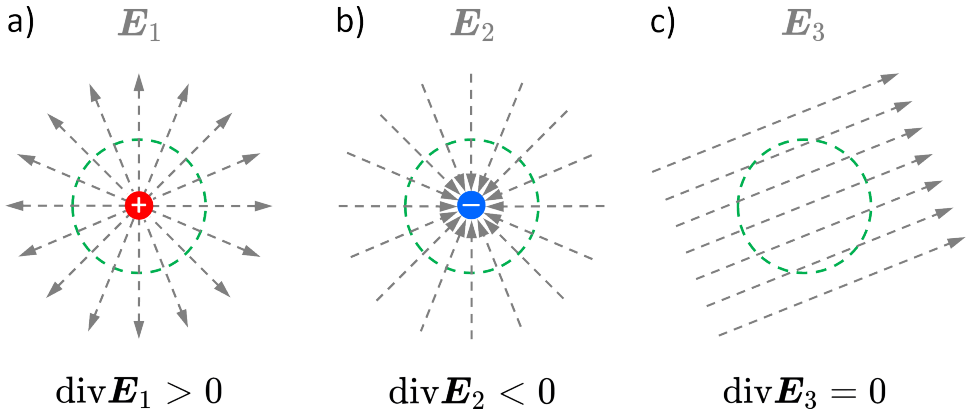



**Fig. 2.5:** The divergence of three vector fields $\boldsymbol{E}_1, \boldsymbol{E}_2, \boldsymbol{E}_3$ within the regions enclosed by the dashed green circles. (a) If the region contains a source of field lines, the divergence of the vector field is positive; (b) if it contains a sink of field lines, the divergence is negative; and (c) if it contains neither sources nor sinks, the divergence of the vector field vanishes in this region.

If one wants to determine the field flux of a given field out of a volume or into a volume, one must calculate the net flux of the field through the bounding surface of that volume. Let us examine how this is done using a simple example. To this end, as shown in the lower part of fig. 2.6, we consider an infinitesimal rectangular volume element, a cuboid, with side lengths $\mathrm{d}x_1$, $\mathrm{d}x_2$, and $\mathrm{d}x_3$, centered at the point $P(x_1, x_2, x_3)$. If, for instance, we are interested in the flux of $\boldsymbol{E}$ in the positive $x_1$-direction, then the relevant entry face of the cuboid volume is the area $\mathrm{d}x_2\mathrm{d}x_3$ located at

$$x_1 - \frac{\mathrm{d}x_1}{2} \quad \text{with surface normal vector} \quad -\hat{\boldsymbol{e}}_1$$

and the exit face of the cuboid is located at

$$x_1 + \frac{\mathrm{d}x_1}{2} \quad \text{with surface normal vector} \quad +\hat{\boldsymbol{e}}_1$$

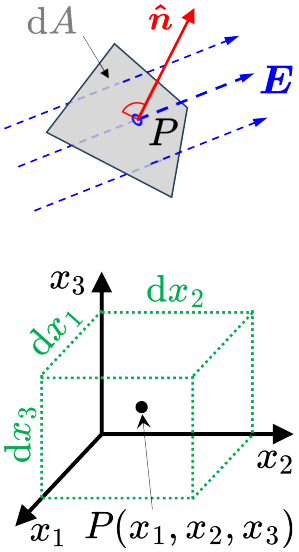



**Fig. 2.6**

The difference between the incoming and outgoing flux of the vector field $\boldsymbol{E}$ in the positive $x_1$-direction, i.e., the net flux of $\boldsymbol{E}$ in this direction, is thus:

$$\left[\boldsymbol{E}\left(x_1 + \frac{\mathrm{d}x_1}{2}, x_2, x_3\right)\hat{\boldsymbol{e}}_1 - \boldsymbol{E}\left(x_1 - \frac{\mathrm{d}x_1}{2}, x_2, x_3\right)\hat{\boldsymbol{e}}_1\right]\mathrm{d}x_2\mathrm{d}x_3$$

$$= \left[E_1\left(x_1 + \frac{\mathrm{d}x_1}{2}, x_2, x_3\right) - E_1\left(x_1 - \frac{\mathrm{d}x_1}{2}, x_2, x_3\right)\right]\mathrm{d}x_2\mathrm{d}x_3$$

Using a first-rank Taylor expansion, this becomes:

$$\left[\left(E_1(x_1,x_2,x_3)+\frac{\partial E_1(x_1,x_2,x_3)}{\partial x_1}\frac{\mathrm{d}x_1}{2}-\ldots\right)-\right.$$
$$\left.\left(E_1(x_1,x_2,x_3)-\frac{\partial E_1(x_1,x_2,x_3)}{\partial x_1}\frac{\mathrm{d}x_1}{2}+\ldots\right)\right]\mathrm{d}x_2\mathrm{d}x_3 \;=\; \frac{\partial E_1}{\partial x_1}\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3$$

The same considerations apply to the flow of $\boldsymbol{E}$ through the cuboid in the $x_2$ and $x_3$ directions. The net inflow and outflow of $\boldsymbol{E}$ into and out of the volume $\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3$ is therefore:

$$\left(\frac{\partial E_1}{\partial x_1}+\frac{\partial E_2}{\partial x_2}+\frac{\partial E_3}{\partial x_3}\right)\cdot\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3 \;=\; \mathrm{div}\,\boldsymbol{E}\cdot\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3$$

If the divergence of $\boldsymbol{E}$ vanishes, i.e., $\mathrm{div}\,\boldsymbol{E}=0$, then there are neither sources nor sinks of the vector field $\boldsymbol{E}$ within the volume $\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3$. In the case of the electric field $\boldsymbol{E}$, this naturally implies that there are no charges whatsoever within this volume.

We have now seen what happens when a scalar or a vector appears to the right of the $\nabla$ operator. In the first case, the result is the gradient of a scalar field, which gives us a vector quantity, and in the second case, we obtain the divergence of a vector field, involving the scalar product of two vectors, the result of which is naturally a scalar. However, it is also possible to form a vector product with the $\nabla$ operator, resulting in a new vector:

$$\nabla\times\boldsymbol{a} \;=\; \mathrm{rot}\,\boldsymbol{a} \;=\; \begin{pmatrix}\partial_1\\ \partial_2\\ \partial_3\end{pmatrix}\times\begin{pmatrix}a_1\\ a_2\\ a_3\end{pmatrix} \;=\; \begin{pmatrix}\partial_2 a_3-\partial_3 a_2\\ \partial_3 a_1-\partial_1 a_3\\ \partial_1 a_2-\partial_2 a_1\end{pmatrix} \tag{2.29}$$

The three $\nabla$ applications

$$\begin{aligned}
\nabla\cdot a \;&=\; \mathrm{grad}\,a \quad &\rightarrow\quad &\text{Result: Vector}\\
\nabla\cdot\boldsymbol{a} \;&=\; \mathrm{div}\,\boldsymbol{a} \quad &\rightarrow\quad &\text{Result: Scalar}\\
\nabla\times\boldsymbol{a} \;&=\; \mathrm{rot}\,\boldsymbol{a} \quad &\rightarrow\quad &\text{Result: Vector}
\end{aligned}$$

can also take combinations of scalar and vector quantities as arguments. If $V=V(\boldsymbol{x})$ is a scalar and $\boldsymbol{a}=\boldsymbol{a}(\boldsymbol{x})$ and $\boldsymbol{b}=\boldsymbol{b}(\boldsymbol{x})$ are vectors, then:

$$\nabla\cdot(V\boldsymbol{a}) \;=\; V(\nabla\cdot\boldsymbol{a})+(\nabla\cdot V)\boldsymbol{a} \tag{2.30a}$$

$$\nabla\times(V\boldsymbol{a}) \;=\; V(\nabla\times\boldsymbol{a})+(\nabla\cdot V)\times\boldsymbol{a} \tag{2.30b}$$

$$\nabla\cdot(\boldsymbol{a}\times\boldsymbol{b}) \;=\; \boldsymbol{b}\cdot(\nabla\times\boldsymbol{a}) - \boldsymbol{a}\cdot(\nabla\times\boldsymbol{b}) \tag{2.30c}$$

$$\nabla\times(\boldsymbol{a}\times\boldsymbol{b}) \;=\; (\boldsymbol{b}\cdot\nabla)\cdot\boldsymbol{a} - (\boldsymbol{a}\cdot\nabla)\cdot\boldsymbol{b} + \boldsymbol{a}\cdot(\nabla\cdot\boldsymbol{b}) - \boldsymbol{b}\cdot(\nabla\cdot\boldsymbol{a}) \tag{2.30d}$$

$$\nabla\cdot(\boldsymbol{a}\cdot\boldsymbol{b}) \;=\; (\boldsymbol{b}\cdot\nabla)\cdot\boldsymbol{a} + (\boldsymbol{a}\cdot\nabla)\cdot\boldsymbol{b} + \boldsymbol{a}\times(\nabla\times\boldsymbol{b}) + \boldsymbol{b}\times(\nabla\times\boldsymbol{a}) \tag{2.30e}$$

Naturally, the $\nabla$ operator can also be applied twice in succession. However, it must be noted that the same commutation rules do not apply to it as to ordinary vectors, since it always acts on the argument to its right but never on what stands to its left. The simplest case of a double application of the $\nabla$ operator is the so-called Laplace operator $\Delta$. For the scalar quantity $V = V(\boldsymbol{x})$ and the vector quantity $\boldsymbol{a} = \boldsymbol{a}(\boldsymbol{x})$, the following applies to the scalar field:

$$\Delta V \;=\; \nabla\cdot(\nabla\cdot V) \;=\; (\nabla\cdot\nabla)\cdot V \;=\; \left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} + \frac{\partial^2}{\partial x_3^2}\right)\cdot V \tag{2.31}$$

and to the vector field:

$$\Delta\boldsymbol{a} \;=\; \hat{\boldsymbol{e}}_1\cdot\Delta a_1 + \hat{\boldsymbol{e}}_2\cdot\Delta a_2 + \hat{\boldsymbol{e}}_3\cdot\Delta a_3 \tag{2.32}$$

As can be easily verified component-wise using eq. (2.29), applying the curl twice to a vector field yields:

$$\nabla\times(\nabla\times\boldsymbol{a}) \;=\; \nabla\cdot(\nabla\cdot\boldsymbol{a}) - \Delta\boldsymbol{a} \tag{2.33}$$

Since $\Delta = (\nabla\cdot\nabla)$, this equation looks very much like the Grassmann identity from eq. (2.2), with the difference that instead of $\boldsymbol{c}\cdot(\boldsymbol{a}\cdot\boldsymbol{b})$, we have $(\boldsymbol{a}\cdot\boldsymbol{b})\cdot\boldsymbol{c}$. That is because the operator $\nabla\cdot\nabla$ acts only on whatever stands to its right. Next, we consider what happens with mixed grad-, div- and rot-operations. Let us first look at the curl of a gradient. If we set $\boldsymbol{a} = \operatorname{grad} V$ in eq. (2.29), we quickly find that:

$$\nabla\times(\nabla\cdot V) \;=\; \begin{pmatrix} \partial_2\partial_3 V - \partial_3\partial_2 V \\ \partial_3\partial_1 V - \partial_1\partial_3 V \\ \partial_1\partial_2 V - \partial_2\partial_1 V \end{pmatrix} \;=\; 0 \tag{2.34}$$

Just as the cross product of a vector with itself is zero, the curl of a gradient, which is the cross product of $\nabla$ with itself, is also always zero. It follows directly from eq. (2.34) that for any vector field $\boldsymbol{a}$ that can be written as the gradient of a scalar field (i.e., $\boldsymbol{a} = \nabla V$), the relation $\nabla\times\boldsymbol{a} = 0$ always holds. As can be easily verified component-wise, by taking the divergence of eq. (2.29), it is also always true that:

$$\nabla\cdot(\nabla\times\boldsymbol{a}) \;=\; 0 \tag{2.35}$$

**Tab. 2.1:** Important identities in vector algebra and vector analysis.

**Some vector identities**

Scalar triple product: $(\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}) = \boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b}) = \boldsymbol{b}\cdot(\boldsymbol{c}\times\boldsymbol{a})$

Grassmann identity: $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\cdot\boldsymbol{b})$

$(\boldsymbol{a}\times\boldsymbol{b})\times\boldsymbol{c} = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{a}(\boldsymbol{b}\cdot\boldsymbol{c})$

Lagrange identity: $(\boldsymbol{a}\times\boldsymbol{b})(\boldsymbol{c}\times\boldsymbol{d}) = (\boldsymbol{b}\boldsymbol{d})(\boldsymbol{a}\boldsymbol{c}) - (\boldsymbol{a}\boldsymbol{d})(\boldsymbol{b}\boldsymbol{c})$

Useful relationship: $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) + \boldsymbol{b}\times(\boldsymbol{c}\times\boldsymbol{a}) + \boldsymbol{c}\times(\boldsymbol{a}\times\boldsymbol{b}) = 0$

---

**Differential operators**

$$\nabla = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3}\right) \quad ; \quad \Delta = \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} + \frac{\partial^2}{\partial x_3^2}$$

**Action of $\nabla$ on scalar and vector fields $V(\boldsymbol{x})$ and $\boldsymbol{a}(\boldsymbol{x})$**

$\nabla\cdot(V\boldsymbol{a}) = V(\nabla\cdot\boldsymbol{a}) + (\nabla\cdot V)\boldsymbol{a}$

$\nabla\times(V\boldsymbol{a}) = V(\nabla\times\boldsymbol{a}) + (\nabla\cdot V)\times\boldsymbol{a}$

$\nabla\cdot(\boldsymbol{a}\times\boldsymbol{b}) = \boldsymbol{b}\cdot(\nabla\times\boldsymbol{a}) - \boldsymbol{a}\cdot(\nabla\times\boldsymbol{b})$

$\nabla\times(\boldsymbol{a}\times\boldsymbol{b}) = (\boldsymbol{b}\cdot\nabla)\cdot\boldsymbol{a} - (\boldsymbol{a}\cdot\nabla)\cdot\boldsymbol{b} + \boldsymbol{a}\cdot(\nabla\cdot\boldsymbol{b}) - \boldsymbol{b}\cdot(\nabla\cdot\boldsymbol{a})$

$\nabla\cdot(\boldsymbol{a}\cdot\boldsymbol{b}) = (\boldsymbol{b}\cdot\nabla)\cdot\boldsymbol{a} + (\boldsymbol{a}\cdot\nabla)\cdot\boldsymbol{b} + \boldsymbol{a}\times(\nabla\times\boldsymbol{b}) + \boldsymbol{b}\times(\nabla\times\boldsymbol{a})$

**Repeated application of $\nabla$ to scalar and vector fields $V(\boldsymbol{x})$ and $\boldsymbol{a}(\boldsymbol{x})$**

$\nabla\cdot(\nabla\cdot V) = \Delta V$ (a scalar field)

$\nabla\cdot(\nabla\cdot\boldsymbol{a}) = \nabla\cdot W$ with scalar $W = \nabla\cdot\boldsymbol{a}$ (a vector field)

$(\nabla\cdot\nabla)\cdot\boldsymbol{a} = \Delta\boldsymbol{a}$ (a vector field)

$\nabla\times(\nabla\times\boldsymbol{a}) = \nabla\cdot(\nabla\cdot\boldsymbol{a}) - \Delta\boldsymbol{a}$ (a vector field)

$\nabla\times(\nabla\cdot V) = 0$

$\nabla\cdot(\nabla\times\boldsymbol{a}) = 0$

As we have already observed earlier, the vanishing of the divergence of a vector field $\boldsymbol{a}$, i.e., $\nabla\boldsymbol{a} = 0$, tells us that the vector field $\boldsymbol{a}$ has neither sources nor sinks within the corresponding volume where this condition holds. From eq. (2.35), we also know that in this case, $\boldsymbol{a}$ can always be expressed as the curl of another vector field $\boldsymbol{b}$, i.e., $\boldsymbol{a} = \nabla\times\boldsymbol{b}$. A combination we have not yet considered is the gradient of a divergence, namely $\nabla(\nabla\boldsymbol{a})$. As is easily seen, the result of this operation is a vector: the gradient of the scalar $W = \nabla\boldsymbol{a}$. When working with the nabla operator, one must always bear in mind that this vector takes the first partial derivative of any expression to its right. Otherwise, its vector components behave exactly like the components of any ordinary vector with respect to the expression preceding it on the left. Caution is therefore required whenever the position of the $\nabla$ operator changes within an equation. In case of doubt, it is always best to write out an equation component by component, as this makes it easier to see what actually happens during such a rearrangement. Tab. 2.1 gives a summary of the key relationships which we have encountered in this brief overview of vectors, vector fields, and vector calculus.

The relationships summarized in tab. 2.1 apply within a Cartesian coordinate system. In practice, however, problem-solving is often simplified by taking advantage of the specific symmetry present in a given situation, rather than simply proceeding with calculations in Cartesian coordinates. Consequently, we frequently require the relationships from tab. 2.1 in polar, spherical, or cylindrical coordinates rather than Cartesian ones. Appendix A provides a brief overview of how to convert between coordinate systems, as well as a derivation of the expressions for differential operators in curvilinear orthogonal coordinate systems.

## 2.3 Vector Fields and Integral Theorems

Already in volume 1 of this physics course, we saw that integrating along the path of a particle traversing a force field which acts on that particle tells us something about the work required to transport that particle from one point in the force field to another point. In the previous section, we saw that the flux of a vector field through surfaces bounding it, or respectively the divergence of a vector field within a specific region, can provide valuable information about a vector field. Here, we turn our attention to integral theorems concerning vector fields. These often offer a simpler path to a solution and a more intuitive understanding of the underlying physics than the differential equations provided by vector analysis. We begin with the line integral, often also referred to as a curve integral.

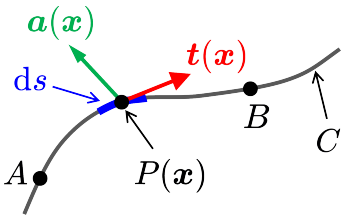



**Fig. 2.7**

To this end, we consider the curve $C$ sketched in fig. 2.7, extending between points $A$ and $B$. Let the vector field along $C$ have the value $\boldsymbol{a}(\boldsymbol{x})$ at any arbitrary point $P(\boldsymbol{x})$ so that $\boldsymbol{a}(\boldsymbol{x})$ is defined everywhere along the curve $C$. Then let $\mathrm{d}s$ be the differential line element at this point $P(\boldsymbol{x})$, and let $\boldsymbol{t}(\boldsymbol{x})$ be the tangential component at that point in the direction of increasing arc length, i.e., in the direction from $A$ to $B$. The component of $\boldsymbol{a}(\boldsymbol{x})$ along $C$ at the point $P(\boldsymbol{x})$ is then given by $\boldsymbol{a}(\boldsymbol{x})\cdot\boldsymbol{t}(\boldsymbol{x})$. If we now sum the components of $\boldsymbol{a}(\boldsymbol{x})$ along the curve $C$ over all points between $A$ and $B$, we obtain:

$$\int\limits_{\substack{C\\ A\text{ bis }B}} \big(\boldsymbol{a}(\boldsymbol{x})\cdot\boldsymbol{t}(\boldsymbol{x})\big)\,\mathrm{d}s \;=\; \int\limits_{\substack{C\\ A\text{ bis }B}} \boldsymbol{a}(\boldsymbol{x})\,\mathrm{d}\boldsymbol{s} \qquad \text{where} \qquad \mathrm{d}\boldsymbol{s} \;=\; \boldsymbol{t}(\boldsymbol{x})\cdot\mathrm{d}s \tag{2.36}$$

Next, we consider the surface integral. Let $S$ be a surface in three-dimensional space and let $\hat{\boldsymbol{n}}(\boldsymbol{x})$ be the unit vector perpendicular to the surface $S$ at every point. This so-called surface normal provides us, by means of

$$\hat{\boldsymbol{n}}(\boldsymbol{x})\,\mathrm{d}\sigma \;\equiv\; \mathrm{d}\boldsymbol{\sigma}$$

with a directed surface element (analogous to the directed line element $\mathrm{d}\boldsymbol{s} = \mathrm{d}s\cdot\boldsymbol{t}(\boldsymbol{x})$ of the curve integral above). In the line integral of eq. (2.36), we are interested in the component of a vector field $\mathbf{a}(\mathbf{x})$ along a given curve, whereas in the surface integral, it is the component in the direction of the surface's orientation. Let $\boldsymbol{a}(\boldsymbol{x})$ be a vector field defined on $S$. To determine the flux of $\boldsymbol{a}(\boldsymbol{x})$ through the surface $S$, we must sum the normal components of $\boldsymbol{a}(\boldsymbol{x})$ over the surface; that is:

$$\int_S \big[\boldsymbol{a}(\boldsymbol{x})\cdot\hat{\boldsymbol{n}}(\boldsymbol{x})\big]\,\mathrm{d}\sigma \;\equiv\; \int_S \boldsymbol{a}(\boldsymbol{x})\,\mathrm{d}\boldsymbol{\sigma} \tag{2.37}$$

Determining the mathematical expression for the directed line element $\mathrm{d}\boldsymbol{s}$ in eq. (2.36) is usually not difficult. Similarly, the directed surface element $\mathrm{d}\boldsymbol{\sigma}$ in eq. (2.37) can often be determined quite easily for many relevant surfaces. Consider, for example, the surface of a sphere of radius $R$ centered at the origin, as sketched in fig. 2.8. The coordinates $x_1, x_2, x_3$ of the point $P$, expressed in spherical coordinates, are:

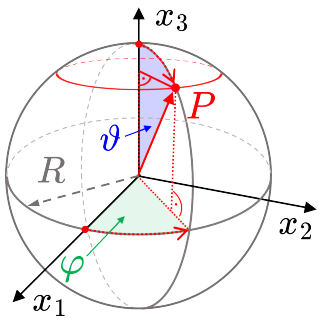



**Fig. 2.8**

$$x_1 \;=\; R\sin\vartheta\cos\varphi \quad ; \quad x_2 \;=\; R\sin\vartheta\sin\varphi \quad ; \quad x_3 \;=\; R\cos\vartheta$$

A parametrization of the spherical surface $S$ for a fixed radius $R$ is thus the set of all points $P$ at a distance $R$ from the origin:

$$S(\vartheta,\varphi) \;=\; R\cdot\begin{pmatrix}\sin\vartheta\cos\varphi\\ \sin\vartheta\sin\varphi\\ \cos\vartheta\end{pmatrix} \qquad \text{with} \qquad 0\le\vartheta\le\pi \quad , \quad 0\le\varphi<2\pi \tag{2.38}$$

The surface element $\mathrm{d}\sigma$ for $S(\vartheta,\varphi)$ can be easily determined by considering the corresponding line elements lying on the two circles that pass through the point $P$. The small red circle in fig. 2.8, which lies parallel to the $x_1x_2$-plane, has a radius of $R\sin\vartheta$, and the differential angular element of the circle is $\mathrm{d}\varphi$. Thus, the differential length element on this circle is $R\sin\vartheta\,\mathrm{d}\varphi$. The differential length element perpendicular to this length element, also passing through point $P$, lies on the circle traced by $\vartheta$ and is given by $R\,\mathrm{d}\vartheta$. The differential area element $\mathrm{d}\sigma$ is the product of these two differential line elements, hence:

$$\mathrm{d}\sigma \;=\; R^2\sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \tag{2.39}$$

The unit normal vector is already given in eq. (2.37); we simply need to set $R = 1$ there. Multiplying eq. (2.39) by $\hat{\boldsymbol{n}}$ yields the directed surface element $\mathrm{d}\boldsymbol{\sigma}$ of the spherical surface. Deriving the directed surface element of an arbitrary surface is not always as straightforward as it is for a sphere. However, there is a helpful rule for determining the directed surface element in the general case. Let $\boldsymbol{r}(u,v)$ be the parametrization of a surface in terms of the coordinates $u$ and $v$. Then, the unit normal vector of this surface is given by:

$$\hat{\boldsymbol{n}} \;=\; \frac{\boldsymbol{r}_u\times\boldsymbol{r}_v}{|\boldsymbol{r}_u\times\boldsymbol{r}_v|} \tag{2.40}$$

Here, $\boldsymbol{r}_u$ and $\boldsymbol{r}_v$ are the partial derivatives of $\boldsymbol{r}(u,v)$ with respect to the coordinates $u$ and $v$. The formula for the differential area element $\mathrm{d}\sigma$ is:

$$\mathrm{d}\sigma \;=\; \sqrt{E\cdot G-F^2}\,\mathrm{d}u\,\mathrm{d}v \quad \text{with} \quad E \;=\; \boldsymbol{r}_u^2 \;;\; G \;=\; \boldsymbol{r}_v^2 \;;\; F \;=\; \boldsymbol{r}_u\cdot\boldsymbol{r}_v \tag{2.41}$$

Using the example of the spherical surface $S(\vartheta,\varphi)$ discussed above, one can easily verify that equations eq. (2.40) and eq. (2.41) yield exactly the result obtained earlier for $\boldsymbol{n}$ and $\mathrm{d}\sigma$. If we now consider the vector field $\boldsymbol{a}(\boldsymbol{r}) = \boldsymbol{r}$ on the spherical surface, that is

$$\boldsymbol{a} \;=\; R\begin{pmatrix}\sin\vartheta\cos\varphi\\ \sin\vartheta\sin\varphi\\ \cos\vartheta\end{pmatrix} \;=\; R\cdot\hat{\boldsymbol{n}}$$

and calculate the flux of this vector field through the spherical surface $S(\vartheta, \varphi)$, then we obtain:

$$\int_S \boldsymbol{a}\,\mathrm{d}\boldsymbol{\sigma} \;=\; \int_S R\cdot \underbrace{\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{n}}}_{1}\cdot R^2 \sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \;=\; 4\pi R^3$$

Now that we know how to calculate the flux through a surface, we can turn to Gauss's theorem. Let $S$ be an arbitrary closed surface in space, let $\Omega$ be the volume enclosed by $S$ and let $\boldsymbol{v}(\boldsymbol{x})$ be a function that is continuously differentiable within $\Omega$ and is continuous on $S$. Furthermore, let $\hat{\boldsymbol{n}}_S$, as sketched in fig. 2.9, be the outward-pointing surface normal of $S$. Then the following integral theorem (Gauss's theorem) holds:

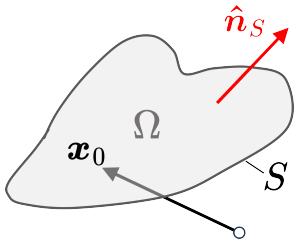



**Fig. 2.9**

$$\int_\Omega \nabla\boldsymbol{v}(\boldsymbol{x})\,\mathrm{d}^3x \;=\; \int_S \boldsymbol{v}(\boldsymbol{x})\cdot\hat{\boldsymbol{n}}_S\,\mathrm{d}\sigma \tag{2.42}$$

Here, $\mathrm{d}^3x$ is, of course, the differential volume element in $\Omega$. If we now shrink this volume $\Omega$, then in the limit $\Omega \to 0$, equation eq. (2.42) becomes:

$$\nabla\boldsymbol{v}(\boldsymbol{x}) \;=\; \lim_{\Omega\to 0}\frac{1}{\Omega}\int_S \boldsymbol{v}(\boldsymbol{x})\cdot\hat{\boldsymbol{n}}_S\,\mathrm{d}\sigma \tag{2.43}$$

This integral definition of the divergence operation clarifies that the divergence of a vector field at any point in the field is a measure of the flux of the vector field. Two other useful integral theorems in electrodynamics are the so-called first and second Green's identity. Let a vector field $\boldsymbol{v}(\boldsymbol{x})$ be given that can be written as

$$\boldsymbol{v}(\boldsymbol{x}) \;=\; U(\boldsymbol{x})\nabla V(\boldsymbol{x})$$

where $U(\boldsymbol{x})$ is continuously differentiable in $\Omega$, and $V(\boldsymbol{x})$ is twice continuously differentiable in $\Omega$. Using eq. (2.30a), the divergence of $\boldsymbol{v}(\boldsymbol{x})$ can be written as:

$$\nabla\cdot\boldsymbol{v} \;=\; U\Delta V + \nabla U\nabla V$$

Substituting this into eq. (2.42) and rearranging slightly yields:

$$\int_\Omega U\Delta V\,\mathrm{d}^3x \;=\; \int_S U(\boldsymbol{n}_s\cdot\nabla V)\,\mathrm{d}\sigma - \int_\Omega \nabla U\cdot\nabla V\,\mathrm{d}^3x \tag{2.44}$$

Eq. (2.44) is Green's first identity. If one swaps $U$ and $V$ in eq. (2.44) and then subtracts the original eq. (2.44) from the result, one obtains the relation:

$$\int_\Omega (U\Delta V - V\Delta U)\,\mathrm{d}^3x \;=\; \int_S (U\nabla V - V\nabla U)\hat{\boldsymbol{n}}_S\,\mathrm{d}\sigma \tag{2.45}$$

Eq. (2.45) is Green's second identity. Finally, we consider Stokes' theorem. As illustrated in fig. 2.10, let there be a surface $S$ in three-dimensional space bounded by a closed curve $C$. Let $\mathrm{d}\boldsymbol{s}$ be the differential arc length element along $C$, and let $\hat{\boldsymbol{n}}_S$ be the surface normal vector of $S$. The sense of direction in which $\mathrm{d}s$ traverses the boundary curve $C$ and the direction in which $\hat{\boldsymbol{n}}_S$ points form a right-handed system. Again, let a vector field $\boldsymbol{v}(\mathbf{x})$ be given that is continuously differentiable on $S$ and continuous on $C$. If $\boldsymbol{t}(\boldsymbol{x})$ is the unit vector in the direction of the tangent to $C$, then Stokes' theorem holds:

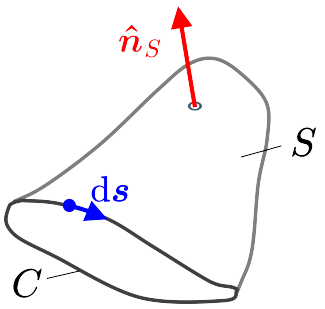


**Fig. 2.10**

$$\int_S \left[\nabla \times \boldsymbol{v}(\boldsymbol{x}) \cdot \underbrace{\hat{\boldsymbol{n}}_S}_{\mathrm{d}\boldsymbol{\sigma}}\right] \mathrm{d}\sigma \;=\; \oint_C \left[\boldsymbol{v}(\boldsymbol{x}) \cdot \underbrace{\boldsymbol{t}(\boldsymbol{x})}_{\mathrm{d}\boldsymbol{s}}\right] \mathrm{d}s \tag{2.46}$$

If we shrink this area $S$, then eq. (2.46) becomes, in the limit $S \to 0$:

$$\hat{\boldsymbol{n}}_S \cdot [\nabla \times \boldsymbol{v}(\boldsymbol{x})] \;=\; \lim_{S \to 0} \frac{1}{S} \oint_C \boldsymbol{v}(\boldsymbol{x}) \cdot \boldsymbol{t}(\boldsymbol{x})\, \mathrm{d}s \tag{2.47}$$

The classical Stokes' theorem in eq. (2.46) is merely a special case of the generalized Stokes' theorem. Here, we present another example of the latter which we will use in a later chapter, namely:

$$\int_V (\nabla \times \boldsymbol{F})\, \mathrm{d}^3x \;=\; \int_S (\hat{\boldsymbol{n}}_S \times \boldsymbol{F})\, \mathrm{d}^2x \tag{2.48}$$

The classical Stokes' theorem in eq. (2.46) relates the operation $\nabla \times \boldsymbol{v}(\boldsymbol{x})$ on a surface $S$ to an operation on the boundary $C$ of the surface. Eq. (2.48) relates the operation $\nabla \times \boldsymbol{F}$ within a volume $V$ to an operation on the boundary $S$ of that volume. To prove eq. (2.48), we consider:

$$\nabla \times \boldsymbol{F} \;=\; \begin{pmatrix} \partial_y F_z - \partial_z F_y \\ \partial_z F_x - \partial_x F_z \\ \partial_x F_y - \partial_y F_x \end{pmatrix} \;=\; \begin{pmatrix} \nabla \boldsymbol{G}_x \\ \nabla \boldsymbol{G}_y \\ \nabla \boldsymbol{G}_z \end{pmatrix} \tag{2.49}$$

For the second equals sign in eq. (2.49), the following must hold:

$$\boldsymbol{G}_x \;=\; \begin{pmatrix} 0 \\ F_z \\ -F_y \end{pmatrix} \quad ; \quad \boldsymbol{G}_y \;=\; \begin{pmatrix} -F_z \\ 0 \\ F_x \end{pmatrix} \quad ; \quad \boldsymbol{G}_z \;=\; \begin{pmatrix} F_y \\ -F_x \\ 0 \end{pmatrix} \tag{2.50}$$

Using eq. (2.49) and eq. (2.50), we can now rewrite the left-hand side of eq. (2.48) and apply Gauss's theorem to each vector component. This yields:

$$\int_V (\nabla \times \boldsymbol{F})\,\mathrm{d}^3x = \int_V \left[(\nabla \boldsymbol{G}_x)\hat{\boldsymbol{i}} + (\nabla \boldsymbol{G}_y)\hat{\boldsymbol{j}} + (\nabla \boldsymbol{G}_z)\hat{\boldsymbol{k}}\right]\mathrm{d}^3x$$
$$= \int_S \left[(\boldsymbol{G}_x\hat{\boldsymbol{n}}_S)\hat{\boldsymbol{i}} + (\boldsymbol{G}_y\hat{\boldsymbol{n}}_S)\hat{\boldsymbol{j}} + (\boldsymbol{G}_z\hat{\boldsymbol{n}}_S)\hat{\boldsymbol{k}}\right]\mathrm{d}^2x \tag{2.51}$$

With $\boldsymbol{G}_x$, $\boldsymbol{G}_y$, and $\boldsymbol{G}_z$ from eq. (2.50) as well as $\hat{\boldsymbol{n}}_S = (n_x, n_y, n_z)$, we have:

$$\boldsymbol{G}_x\hat{\boldsymbol{n}}_S = F_z n_x - F_y n_z \quad ; \quad \boldsymbol{G}_y\hat{\boldsymbol{n}}_S = -F_z n_x + F_x n_z \quad ; \quad \boldsymbol{G}_z\hat{\boldsymbol{n}}_S = F_y n_x - F_x n_y$$

If we substitute this into the integrand on the right-hand side of eq. (2.51), it becomes:

$$(\boldsymbol{G}_x\hat{\boldsymbol{n}}_S)\hat{\boldsymbol{i}} + (\boldsymbol{G}_y\hat{\boldsymbol{n}}_S)\hat{\boldsymbol{j}} + (\boldsymbol{G}_z\hat{\boldsymbol{n}}_S)\hat{\boldsymbol{k}} = \begin{pmatrix} F_z n_x - F_y n_z \\ -F_z n_x + F_x n_z \\ F_y n_x - F_x n_y \end{pmatrix} = \hat{\boldsymbol{n}}_S \times \boldsymbol{F} \tag{2.52}$$

Thus, the right-hand side of eq. (2.51) is indeed the surface integral of $\hat{\boldsymbol{n}}_S \times \boldsymbol{F}$, which proves eq. (2.48). With eq. (2.46) and eq. (2.48), we are now familiar with Stokes' theorem in two and three dimensions.

### 2.3.1 Vector Field of a Conservative Force

We consider the vector field of a conservative force $\boldsymbol{F}$. From volume 1, section 4.1.2, we know that a conservative force $\boldsymbol{F}$ is always the gradient of a potential $\phi$, that is, $\boldsymbol{F} = -\nabla\phi$ holds. Together with eq. (2.34), this implies:

$$\nabla \times \boldsymbol{F} = -\nabla \times \nabla\phi = 0$$

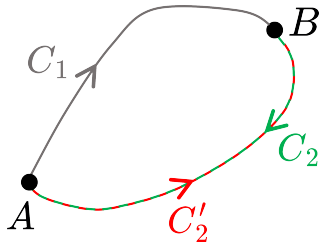



**Fig. 2.11**

Consider the closed path $C = C_1 + C_2$ shown in fig. 2.11, consisting of the two segments $C_1$ from $A$ to $B$ and $C_2$ from $B$ to $A$, as well as the segment $C_2'$ from $A$ to $B$, which runs in the direction opposite to $C_2$. It follows from Stokes' theorem that:

$$\oint_C \boldsymbol{F}\,\mathrm{d}\boldsymbol{s} = 0 = \int_A^B_{C_1} + \int_B^A_{C_2} \quad \Rightarrow \quad \int_A^B_{C_1} = -\int_B^A_{C_2} = \int_A^B_{C_2'}$$

As we know, the line integral of $\boldsymbol{F}\mathrm{d}\boldsymbol{s}$ represents the work done along a path within a force field $\boldsymbol{F}$. Using Stokes' theorem, we have just demonstrated that, for a conservative force field, this work is independent of the path and depends solely on the starting and ending points.

### 2.3.2 Sources and Vortices of Vector Fields I

In this section, we consider the calculation of a vector field $\boldsymbol{v}(\boldsymbol{x})$ from its sources and vortices. Let $\rho(\boldsymbol{x})$ and $\boldsymbol{c}(\boldsymbol{x})$ be two functions defined in three-dimensional space and confined to a finite region, satisfying the following conditions:

$$\nabla \boldsymbol{v}(\boldsymbol{x}) \;=\; 4\pi\rho(\boldsymbol{x}) \tag{2.53a}$$

$$\nabla \times \boldsymbol{v}(\boldsymbol{x}) \;=\; 4\pi\boldsymbol{c}(\boldsymbol{x}) \tag{2.53b}$$

From eq. (2.53b) and eq. (2.34) follows the equation:

$$\nabla\big[\nabla \times \boldsymbol{v}(\boldsymbol{x})\big] \;=\; 0 \;=\; 4\pi\nabla\boldsymbol{c}(\boldsymbol{x}) \;=\; 0 \tag{2.54}$$

We now assert that solutions $\boldsymbol{v}(\boldsymbol{x})$ of eq. (2.54) that decay to zero as $x \to \infty$ at least as $1/x^2$ are unique. To prove this assertion, we assume there are two solutions, $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. Since both solutions satisfy eq. (2.54), the difference between these two solutions, that is

$$\boldsymbol{\omega} \;\coloneqq\; \boldsymbol{v}_1 - \boldsymbol{v}_2$$

must with eq. (2.53a) und eq. (2.53b) satisfy:

$$\nabla\boldsymbol{\omega} \;=\; \nabla \times \boldsymbol{\omega} \;=\; 0$$

However, as we know, $\nabla \times \boldsymbol{\omega} = 0$ implies that $\boldsymbol{\omega}$ is the gradient of a potential, i.e., $\boldsymbol{\omega} = -\nabla\phi$. Since $\nabla\boldsymbol{\omega} = 0$ also holds, it follows for this potential $\phi$ that:

$$\nabla\boldsymbol{\omega} \;=\; -\nabla(\nabla\phi) \;=\; -\Delta\phi \;=\; 0$$

We now make use of Green's first identity from eq. (2.44), but set $U = V = \phi$ in it, thereby obtaining the equation:

$$\int_\Omega \phi \underbrace{\Delta\phi}_{=\,0}\, \mathrm{d}^3x \;=\; 0 \;=\; \int_S \phi(\boldsymbol{n}_S \nabla\phi)\,\mathrm{d}\sigma - \int_\Omega (\nabla\phi)^2\,\mathrm{d}^3x \tag{2.55}$$

Let us consider the first of the two integrals on the right-hand side of this equation in the case $S \to 0$. As we already know from eq. (2.39), the differential surface element $\mathrm{d}\sigma$ scales like $x^2$ in this case. Our assumption for the assertion above was that solutions of eq. (2.54) for $x \to \infty$ scale at least like $1/x^2$. Therefore, with $\boldsymbol{\omega} = \boldsymbol{v}_1 - \boldsymbol{v}_2 = -\nabla\phi$, it must hold that $\nabla\phi$ also scales at least as $1/x^2$ for $x \to \infty$. Accordingly, $\phi$ then scales as $1/x$. Thus, considering the surface integral on the right-hand side of eq. (2.55) for $S \to 0$, we see that

$$\lim_{S\to\infty}\int_S \phi\,(\boldsymbol{n}_S\nabla\phi)\,\mathrm{d}\sigma \;=\; 0$$
$$\qquad\qquad\quad \downarrow \quad\;\; \downarrow \;\; \downarrow$$
$$\qquad\qquad \propto x^{-1} \quad x^{-2} \;\; x^{2}$$

because the integrand scales like $x^{-1}$. However, this implies that the second integral on the right-hand side of eq. (2.55) must also vanish. This is only possible if $\nabla\phi = 0$, meaning $\phi$ must be constant. Given that $\boldsymbol{\omega} = \boldsymbol{v}_1 - \boldsymbol{v}_2 = -\nabla\phi$, this implies that $\boldsymbol{v}_1 = \boldsymbol{v}_2$. Thus, the solution $\boldsymbol{v}$ to eq. (2.54) is indeed unique, provided it scales at least like $1/x^2$ as $x \to \infty$.

## 2.4 The Delta Function

We consider a Gaussian function that becomes increasingly narrow and tall yet maintains a constant area of 1. Fig. 2.12 shows examples of two such Gaussian functions, each with its maximum at $x = a$ on the $x$-axis. The red Gaussian function is significantly narrower and taller than the blue one. Continuing this process while keeping the area under the curve equal to 1 leads, in the limit where the function's width approaches zero and its height approaches infinity to the delta function. Let us first introduce the $\delta$-function in words: The delta function $\delta(x-a)$ is defined such that it approaches zero for $x \neq a$ and infinity for $x = a$, in such a way that the integral of the area under the delta function is always equal to 1. Expressed mathematically:

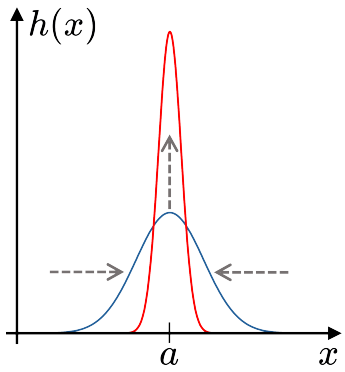


**Fig. 2.12**

$$\delta(x-a) \;:=\; \begin{cases} 0 & \text{if} \quad x \neq a \\ \infty & \text{if} \quad x = a \end{cases} \qquad \text{such that} \qquad \int_{-\infty}^{\infty} \delta(x-a)\mathrm{d}x \;=\; 1 \tag{2.56}$$

From that, it follows immediately that for every function $f(x)$:

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)\,\mathrm{d}x \;=\; f(a) \tag{2.57}$$

Eq. (2.57) defines the $\delta$-function via its integration property. However, the $\delta$-function can also be represented explicitly. One possibility, as already sketched in fig. 2.12, is to represent it as the limit of a sequence of functions. In fig. 2.12, we illustrated this using the Gaussian function. We could, however, just as well have used the resonance function, which is already familiar to us from the discussion of oscillatory motion in the first volume of this physics course. Here, we present only the representation of the $\delta$-function as the

limit of a Gaussian function that becomes increasingly narrow and tall while maintaining a constant area:

$$\delta(x-a) \;=\; \lim_{\epsilon\to 0} \frac{1}{\epsilon\sqrt{\pi}} \cdot \mathrm{e}^{\frac{-(x-a)^2}{\epsilon^2}} \tag{2.58}$$

Another representation of the $\delta$-function describes it as the derivative of the step function known as the Heaviside function:

$$\delta(x-a) \;=\; \frac{\partial}{\partial x}\Theta(x-a) \quad \text{mit} \quad \Theta(x-a) \;=\; \begin{cases} 0 & x < a \\ 1 & x \geq a \end{cases} \tag{2.59}$$

To the step function applies:

$$\int_{-\infty}^{\infty} \Theta(x-a) f(x)\,\mathrm{d}x \;=\; \int_{a}^{\infty} f(x)\,\mathrm{d}x \tag{2.60}$$

For the derivative of the step function to represent the $\delta$-function requires:

$$\int_{-\infty}^{\infty} f(x) \frac{\partial\Theta(x-a)}{\partial x}\,\mathrm{d}x \;=\; f(a) \;=\; \int_{-\infty}^{\infty} f(x)\delta(x-a)\,\mathrm{d}x \tag{2.61}$$

The integral on the left-hand side of this equation is solved using integration by parts:

$$\int_{-\infty}^{\infty} f(x) \frac{\partial\Theta(x-a)}{\partial x}\,\mathrm{d}x \;=\; \Theta(x-a)f(x)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \frac{\partial f(x)}{\partial x}\Theta(x-a)\,\mathrm{d}x$$

Since $-\infty < a < \infty$, the first term on the left-hand side reduces to $f(\infty)$, and the second term becomes, using eq. (2.60),

$$\int_{-\infty}^{\infty} \frac{\partial f(x)}{\partial x}\Theta(x-a)\,\mathrm{d}x \;=\; \int_{a}^{\infty} \frac{\partial f(x)}{\partial x} \;=\; f(\infty) - f(a)$$

and we thus obtain

$$\int_{-\infty}^{\infty} f(x) \frac{\partial\Theta(x-a)}{\partial x}\,\mathrm{d}x \;=\; f(\infty) - \big[f(\infty) - f(a)\big] \;=\; f(a)$$

and thus equation eq. (2.61) is proven. The Heaviside function is therefore indeed the antiderivative of the Dirac delta function. But what is the derivative of the delta function? This derivative is defined by its action on a test function $f(x)$ via:

$$\int_{-\infty}^{\infty} f(x) \frac{\partial\delta(x-a)}{\partial x}\,\mathrm{d}x \;=\; \underbrace{f(x)\delta(x-a)\Big|_{-\infty}^{\infty}}_{=\,0} - \int_{-\infty}^{\infty} \frac{\partial f(x)}{\partial x}\delta(x-a)\,\mathrm{d}x$$

and thus:

$$\int_{-\infty}^{\infty} f(x)\frac{\partial\delta(x-a)}{\partial x}\,\mathrm{d}x \;=\; -\frac{\partial f(x)}{\partial x}\bigg|_{x=a} \tag{2.62}$$

The derivative of the $\delta$-function $\delta(x-a)$ thus picks out the derivative of the function $f(x)$ at the point $x = a$. What happens when the argument of the $\delta$-function is multiplied by a constant $c$? As can be easily verified, the $\delta$-function is highly symmetric, and the relation $\delta(x-a) = \delta(a-x)$ holds. Consequently, in the case where $a = 0$ and a constant $c$ is involved, it follows that $\delta(cx) = \delta(-cx)$, a condition that can only be true if $\delta(cx) = \delta(-cx) = \delta(|c|x)$. As can be readily demonstrated by substituting into eq. (2.56) or eq. (2.57) and using the substitution $y = |c|x$, the following therefore holds:

$$\delta(cx) \;=\; \frac{1}{|c|}\delta x \qquad , \qquad c = \text{constant} \tag{2.63}$$

So far, we have considered the $\delta$-function in only one coordinate. If $\boldsymbol{a}$ is a vector with coordinates $a_1$, $a_2$, and $a_3$, then the extension of the $\delta$-function to three coordinate directions is:

$$\delta^{(3)}(\boldsymbol{x}-\boldsymbol{a}) \;=\; \delta(x_1-a_1)\cdot\delta(x_2-a_2)\cdot\delta(x_3-a_3) \tag{2.64}$$

The reason for examining here the properties of the $\delta$-function in somewhat greater detail is that it allows us to express point-like charges. However, before looking at an example of this, let us turn to another important property of the $\delta$-function, which is the validity of:

$$\Delta\frac{1}{|\boldsymbol{r}|} \;=\; -4\pi\delta^{(3)}(\boldsymbol{r}) \tag{2.65}$$

The fact that the Laplacian of 1 divided by the magnitude of the position vector $\boldsymbol{r}$ is proportional to the delta function at the position $\boldsymbol{r}$ tells us that the result of this operation vanishes everywhere except at $\boldsymbol{r} = 0$. To prove eq. (2.65), we consider this equation in spherical coordinates. In doing so, $|\boldsymbol{r}|$ becomes $r$, and the left-hand side of equation eq. (2.65) becomes (see tab. A.1 in Appendix A):

$$\Delta\frac{1}{|\boldsymbol{r}|} \;=\; \nabla\left[\nabla\frac{1}{r}\right] \;=\; \nabla\left[\left(\frac{\partial}{\partial r}\frac{1}{r}\right)\hat{\boldsymbol{e}}_r\right] \;=\; -\nabla\frac{\hat{\boldsymbol{e}}_r}{r^2}$$

Since the magnitude of $|\boldsymbol{r}|$ does not depend on $\vartheta$ and $\varphi$, the gradient in spherical coordinates reduces here to the component in the direction of $\hat{\boldsymbol{e}}_r$. Therefore, we need only consider the component of the divergence in the direction of $\hat{\boldsymbol{e}}_r$; that is:

$$-\nabla\frac{\hat{\boldsymbol{e}}_r}{r^2} \;=\; \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\cdot\frac{1}{r^2}\right) \;=\; 0 \qquad \text{für} \qquad r \neq 0$$

With that we already know that eq. (2.65) holds for $r \neq 0$. Now, what remains is to consider what happens as $r$ approaches zero. To do this, we choose a test function $f(r)$ and multiply both sides of eq. (2.65) by it. Then, we integrate both sides of the equation over an arbitrary volume containing the origin, i.e., $r = 0$. First, however, we need to examine the following expression:

$$\nabla\left[f(r)\frac{\hat{\boldsymbol{e}}_r}{r^2}\right] = f(r)\nabla\left(\frac{\hat{\boldsymbol{e}}_r}{r^2}\right) + \frac{\hat{\boldsymbol{e}}_r}{r^2}\nabla f(r) = f(r)\nabla\left(\frac{\hat{\boldsymbol{e}}_r}{r^2}\right) + \frac{1}{r^2}\frac{\partial f(r)}{\partial r}$$

We can rewrite this as:

$$f(r)\nabla\left(\frac{\hat{\boldsymbol{e}}_r}{r^2}\right) = \nabla\left[f(r)\frac{\hat{\boldsymbol{e}}_r}{r^2}\right] - \frac{1}{r^2}\frac{\partial f(r)}{\partial r}$$

On the left-hand side of this equation stands the left-hand side of eq. (2.65), multiplied by our test function $f(r)$. We now wish to integrate this over a volume $\Omega$ that contains the origin, that is:

$$\begin{aligned}\int_\Omega f(r)\nabla\left(\frac{\hat{\boldsymbol{e}}_r}{r^2}\right)\mathrm{d}^3r &= \int_\Omega \nabla\left[f(r)\frac{\hat{\boldsymbol{e}}_r}{r^2}\right]\mathrm{d}^3r - \int_\Omega \frac{1}{r^2}\frac{\partial f(r)}{\partial r}\,\mathrm{d}^3r \\ &= \int_S f(r)\frac{\hat{\boldsymbol{e}}_r}{r^2}\hat{\boldsymbol{n}}_S\,\mathrm{d}\sigma - \int_\Omega \frac{1}{r^2}\frac{\partial f(r)}{\partial r}\,\mathrm{d}^3r\end{aligned}$$

Here, between the first and the second line we have applied Gauss's theorem eq. (2.42) to the first integral on the right-hand side. Now, we choose a sphere of radius $R$ as the volume and evaluate the two integrals on the left-hand side for it. We are already familiar with the surface element $\mathrm{d}\sigma$ from eq. (2.39), as well as the corresponding surface unit vector $\hat{\boldsymbol{n}}_S = \hat{\boldsymbol{e}}_r$. The volume element in spherical coordinates is $\mathrm{d}^3r = r^2 \sin\vartheta\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$. This yields:

$$\begin{aligned}\int_\Omega f(r)\nabla\left(\frac{\hat{\boldsymbol{e}}_r}{r^2}\right)\mathrm{d}^3r &= \int_S f(R)\sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \;- \int_\Omega \frac{\partial f(r)}{\partial r}\sin\vartheta\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \\ &= 4\pi f(R) - 4\pi f(R)\Big|_0^R = 4\pi f(0)\end{aligned}$$

Multiplying the left side of eq. (2.65) by a test function $f(r)$ and then integrating over the volume of a sphere with radius $R$ thus yields:

$$\int_\Omega f(r)\Delta\left(\frac{1}{r}\right)\mathrm{d}^3r = -\int_\Omega f(r)\nabla\left(\frac{\hat{\boldsymbol{e}}_r}{r^2}\right)\mathrm{d}^3r = -4\pi f(0) \tag{2.66}$$

In the next step, we multiply the right-hand side of eq. (2.65) by $f(r)$ and integrate over the same sphere volume. For this, we need the expression for the $\delta$-function in spherical coordinates (see Eq. (A.16) in Appendix A):

$$\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0) \;=\; \frac{1}{r^2\sin\vartheta}\delta(r-r_0)\delta(\vartheta-\vartheta_0)\delta(\varphi-\varphi_0)$$

This gives us then:

$$-4\pi\int_\Omega f(r)\delta^{(3)}(\boldsymbol{r})\,\mathrm{d}^3r \;=\; -4\pi\int_0^R\int_0^\pi\int_0^{2\pi} f(r)\delta(r)\delta(\vartheta)\delta(\varphi)\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$$

$$= \; -4\pi\underbrace{\int_0^R f(r)\,\mathrm{d}r}_{=f(0)}\cdot\underbrace{\int_0^\pi\delta(\vartheta)\,\mathrm{d}\vartheta}_{=1}\cdot\underbrace{\int_0^{2\pi}\delta(\varphi)\,\mathrm{d}\varphi}_{=1} \;=\; -4\pi f(0)$$

Multiplication of the right side of eq. (2.65) and integration over the volume of a sphere of radius $R$ whose center lies at the origin thus yields:

$$-4\pi\int_\Omega f(r)\delta^{(3)}(\boldsymbol{r})\,\mathrm{d}^3r \;=\; -4\pi f(0) \tag{2.67}$$

Comparing eq. (2.66) with eq. (2.67) shows that multiplication by $f(r)$ and subsequent integration over a volume containing the origin yields the result $-4\pi f(0)$ for both the left and right sides of eq. (2.65). This proves eq. (2.65). The general form of eq. (2.65), which reduces to eq. (2.65) for $\boldsymbol{r}_0 = 0$, is:

$$\Delta\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_0|} \;=\; -4\pi\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_0) \tag{2.68}$$

### 2.4.1 Sources and Vortices of Vector Fields II

With what we have learned about the $\delta$-function, we can now express point-like charges. Again, we consider a vector field $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$ for which the following shall hold:

$$\nabla\times\boldsymbol{v}_1 \;=\; 0 \tag{2.69a}$$

$$\nabla\boldsymbol{v}_1 \;=\; 4\pi\rho(\boldsymbol{x}) \tag{2.69b}$$

as well as

$$\nabla\boldsymbol{v}_2 \;=\; 0 \tag{2.69c}$$

$$\nabla\times\boldsymbol{v}_2 \;=\; 4\pi\boldsymbol{c}(\boldsymbol{x}) \tag{2.69d}$$

Because of $\nabla\times\nabla\phi = 0$ , eq. (2.69a) is automatically satisfied for

$$\boldsymbol{v}_1 \;=\; -\nabla\phi \tag{2.69e}$$

Consequently, for $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$ the equations already known to us from eq. (2.53a) and eq. (2.53b) apply:

$$\nabla \boldsymbol{v}(\boldsymbol{x}) \;=\; 4\pi\rho(\boldsymbol{x})$$

$$\nabla \times \boldsymbol{v}(\boldsymbol{x}) \;=\; 4\pi \boldsymbol{c}(\boldsymbol{x})$$

Here, $\rho(\boldsymbol{x})$ and $\boldsymbol{c}(\boldsymbol{x})$ are again given functions that only take on finite values. Furthermore, we know from the discussion of eq. (2.54) that solutions $\boldsymbol{v}$ of these equations, which scale with $1/x^2$, are unique. By substituting eq. (2.69e) into eq. (2.69b), we obtain the so-called Poisson equation:

$$-\nabla(\nabla\phi) \;=\; 4\pi\rho \qquad \text{or} \qquad \Delta\phi \;=\; -4\pi\rho \tag{2.70}$$

Now, let a charge distribution $\rho$ be given which consists of $N$ point charges $q_i$and a continuous charge distribution $\widetilde{\rho}(\boldsymbol{r})$, that is:

$$\rho(\boldsymbol{r}) \;=\; \sum_{i=1}^{N} q_i \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}_i) + \widetilde{\rho}(\boldsymbol{r}) \tag{2.71}$$

Here, the fixed numbers $q_i$ give us the magnitude and sign of a charge at location $\boldsymbol{r}_i$. Using eq. (2.68), we know which function gives us a discrete charge distribution expressed by the $\delta$-function when we apply the Laplace operator to it. Therefore, for the potential $\phi$ in eq. (2.70), we make the following approach:

$$\phi \;=\; \sum_{i=1}^{N} \frac{q_i}{|\boldsymbol{r} - \boldsymbol{r}_i|} + \widetilde{\phi}(\boldsymbol{r}) \tag{2.72}$$

The quantity $\widetilde{\phi}(\boldsymbol{r})$ is the potential associated with the continuous charge distribution $\widetilde{\rho}(\boldsymbol{r})$. If we now substitute this potential $\phi$ on the left-hand side of eq. (2.70) and use for the charge distribution on the right-hand side of eq. (2.70) the expression from eq. (2.71), we obtain:

$$\sum_{i=1}^{N} q_i \Delta \frac{1}{|\boldsymbol{r} - \boldsymbol{r_i}|} + \Delta\widetilde{\phi}(\boldsymbol{r}) \;=\; -4\pi \sum_{i=1}^{N} q_i \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}_i) - 4\pi\widetilde{\rho}(\boldsymbol{r})$$

Because of the identity in eq. (2.68), this expression reduces to an equation for the continuous charge distribution $\widetilde{\rho}(\boldsymbol{r})$ and the associated potential $\widetilde{\phi}(\boldsymbol{r})$:

$$\Delta\widetilde{\phi}(\boldsymbol{r}) \;=\; -4\pi\widetilde{\rho}(\boldsymbol{r})$$

To determine $\widetilde{\phi}(\boldsymbol{r})$, we consider Green's second identity in eq. (2.45) for the functions:

$$U \;=\; \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} \qquad \text{and} \qquad V \;=\; \widetilde{\phi}(\boldsymbol{r}')$$

Since $U$ is not differentiable at $\boldsymbol{r}' = \boldsymbol{r}$, this point must be excluded. This exclusion, as sketched in fig. 2.13, results in a new boundary surface $\widetilde{S}$, an internal boundary surface in the volume $\Omega$, which must be considered separately from the external boundary $S$. Substituting the above functions $U$ and $V$ into eq. (2.45), Green's second identity yields:

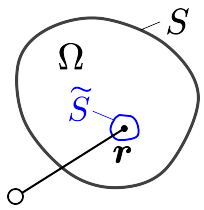


**Fig. 2.13**

$$\int_\Omega \left\{ \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \underbrace{\Delta_{\boldsymbol{r}'}\widetilde{\phi}(\boldsymbol{r}')}_{-4\pi\widetilde{\rho}(\boldsymbol{r})} - \widetilde{\phi}(\boldsymbol{r}') \underbrace{\Delta_{\boldsymbol{r}'}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}}_{-4\pi\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')} \right\} \mathrm{d}^3 r'$$

$$= \lim_{S\to\infty} \int_S \underbrace{\left\{ \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \nabla_{\boldsymbol{r}'}\widetilde{\phi}(\boldsymbol{r}') - \widetilde{\phi}(\boldsymbol{r}') \nabla_{\boldsymbol{r}'} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \right\}}_{\to 0} \hat{\boldsymbol{n}}_S \,\mathrm{d}\sigma$$

$$+ \lim_{\widetilde{S}\to\boldsymbol{r}} \int_{\widetilde{S}} \left\{ \frac{1}{R_0} \frac{\partial\widetilde{\phi}(R_0,\vartheta,\varphi)}{\partial R_0} \hat{\boldsymbol{e}}_{R_0} - \widetilde{\phi}(R_0,\vartheta,\varphi) \underbrace{\frac{\partial}{\partial R_0}\left(\frac{1}{R_0}\right)}_{-R_0^{-2}} \hat{\boldsymbol{e}}_{R_0} \right\} \hat{\boldsymbol{n}}_{\widetilde{S}} \,\mathrm{d}\sigma$$

The surface integral over the outer boundary surface $S$ vanishes for $S \to \infty$ because $\nabla\widetilde{\phi}(r')$ as a unique solution scales like $1/r'^2$ and the two expressions in the integral over $S$ therefore each scale like $1/r'^3$ for $S \to \infty$. In this equation we have written out the contribution of the integral over the inner boundary surface $\widetilde{S}$ in spherical coordinates. The unit vector $\hat{\boldsymbol{n}}_{\widetilde{S}}$ of this inner spherical surface of radius $R_0$ with center $\boldsymbol{r}$ points in the direction of this center, while the unit vector $\hat{\boldsymbol{e}}_{R_0}$ points in the exact opposite direction. Therefore, $\hat{\boldsymbol{e}}_{R_0} \cdot \hat{\boldsymbol{n}}_{\widetilde{S}} = -1$. The contributions in the gradient in the other coordinate directions $\hat{\boldsymbol{e}}_\vartheta$ and $\hat{\boldsymbol{e}}_\varphi$ were not included in the integral because they vanish with $\hat{\boldsymbol{e}}_\vartheta \cdot \hat{\boldsymbol{n}}_{\widetilde{S}} = 0$ and $\hat{\boldsymbol{e}}_\varphi \cdot \hat{\boldsymbol{n}}_{\widetilde{S}} = 0$. For the surface integral over $\tilde{S}$ we obtain with $\mathrm{d}\sigma = R_0^2 \sin\vartheta \,\mathrm{d}\vartheta \,\mathrm{d}\varphi$ ($\widetilde{S} \to \boldsymbol{r}$ means $R_0 \to 0$):

$$\lim_{R_0\to 0} \int_{\widetilde{S}} \{\ldots\} \,\mathrm{d}\sigma = \underbrace{-\int_0^{\pi}\int_0^{2\pi} \frac{\partial\widetilde{\phi}(R_0,\vartheta,\varphi)}{\partial R_0} R_0 \sin\vartheta \,\mathrm{d}\vartheta \,\mathrm{d}\varphi}_{\to 0}$$

$$\underbrace{-\int_0^{\pi}\int_0^{2\pi} \widetilde{\phi}(R_0,\vartheta,\varphi) \sin\vartheta \,\mathrm{d}\vartheta \,\mathrm{d}\varphi}_{4\pi\widetilde{\phi}(\boldsymbol{r})}$$

The first integral over $\widetilde{S}$ therefore vanishes, and since $R_0 = 0$ simply means that we are at the location $\boldsymbol{r}$, the second integral is equal to $4\pi\widetilde{\phi}(\boldsymbol{r})$. The evaluation of Green's second identity with the functions $U$ and $V$ from above thus simplifies to the following equation:

$$-4\pi \int_\Omega \frac{\widetilde{\rho}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' + 4\pi \int_\Omega \widetilde{\phi}(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\,\mathrm{d}^3r' \;=\; -4\pi\widetilde{\phi}(\boldsymbol{r})$$

The integral over the $\delta$-function vanishes because the volume $\Omega$ does not include the singularity at $\boldsymbol{r}$ and thus we obtain the result:

$$\widetilde{\phi}(\boldsymbol{r}) \;=\; \int_\Omega \frac{\widetilde{\rho}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{2.73}$$

We could have achieved this result much faster if we had only assumed that Gauss's theorem and Green's second identity, which derives from it, also apply to the integration of functions with singularities. In that case, the exclusion of the volume $\widetilde{S}$ is unnecessary, and one obtains directly from Green's second identity (the surface integral over $S$ still vanishes):

$$\int_\Omega \left\{ \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}\Delta_{\boldsymbol{r}'}\widetilde{\phi}(\boldsymbol{r}') - \widetilde{\phi}(\boldsymbol{r}')\Delta_{\boldsymbol{r}'}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \right\} \mathrm{d}^3r' \;=\; 0$$

and therefore

$$-4\pi \int_\Omega \frac{\widetilde{\rho}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' + 4\pi \int_\Omega \widetilde{\phi}(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\,\mathrm{d}^3r' \;=\; 0$$

Since the volume $\Omega$ now also contains the singularity at $\boldsymbol{r}$, the integral over the $\delta$-function no longer vanishes, but has the value $\widetilde{\phi}(\boldsymbol{r})$, and we obtain again:

$$\widetilde{\phi}(\boldsymbol{r}) \;=\; \int_\Omega \frac{\widetilde{\rho}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r'$$

This shows us that Gauss's theorem and Green's first and second identity are also applicable when $U$ or $V$ are distributions. Here, it is the function $U$ that can be interpreted as a distribution.

To summarize. With eq. (2.72) and the result of eq. (2.73), the solution of eq. (2.70), i.e., Poisson's equation, is given by:

$$\phi(\boldsymbol{r}) \;=\; \sum_{i=1}^{N} \frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|} + \int_\Omega \frac{\tilde{\rho}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{2.74}$$

This equation tells us that if we know the magnitudes and signs of the charges $q_i$ at positions $\boldsymbol{r}_i$ and the continuous charge distribution $\widetilde{\rho}(\boldsymbol{r}')$, then we can determine the potential $\phi(\boldsymbol{r})$. And with that, we also have the solution for $\boldsymbol{v}_1$ in our example, since it

is equal to eq. (2.69e), the negative gradient of $\phi(\boldsymbol{r})$. What we still need is a solution for $\boldsymbol{v}_2$ that satisfies the equations eq. (2.69c) and eq. (2.69d), that is:

$$\nabla \boldsymbol{v}_2 = 0 \qquad \text{and} \qquad \nabla \times \boldsymbol{v}_2 = 4\pi \boldsymbol{c}$$

As we know from eq. (2.35), $\nabla(\nabla \times \boldsymbol{a}) = 0$ always holds. Therefore, if we set $\boldsymbol{v}_2 = \nabla \times \boldsymbol{A}$, then the first of the two equations (eq. (2.69c)) is always satisfied, and from the second equation for $\boldsymbol{v}_2$ (eq. (2.69d)), we obtain, with the help of eq. (2.33), a determining equation for the vector potential $\boldsymbol{A}$:

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla \boldsymbol{A}) - \Delta \boldsymbol{A} = 4\pi \boldsymbol{c} \tag{2.75}$$

Now we will make the assumption that

$$\boldsymbol{A} = \boldsymbol{A}' + \nabla \Lambda$$

We can safely add the term $\nabla\Lambda$ here because $\nabla \times \nabla\Lambda = 0$ always holds, and thus eq. (2.69c) for $\boldsymbol{v}_2 = \nabla \times \boldsymbol{A}$ is still satisfied. Except for the gradient of an arbitrary scalar field $\Lambda$, the vector potential $\boldsymbol{A}$ is always unique. We now choose $\Lambda$ such that $\nabla \boldsymbol{A} = 0$ always holds, that is:

$$0 = \nabla \boldsymbol{A} = \underbrace{\nabla \boldsymbol{A}'}_{\neq 0} + \underbrace{\nabla(\nabla \Lambda)}_{\Delta \Lambda}$$

$\Delta\Lambda$ can always be chosen appropriately such that $\nabla \boldsymbol{A} = 0$ and thus eq. (2.75) for $\boldsymbol{A}$ simplifies to:

$$\Delta \boldsymbol{A} = -4\pi \boldsymbol{c}$$

We have already solved such an equation, although not for a vector potential $\boldsymbol{A}(\boldsymbol{r})$ but for the scalar potential $\phi(\boldsymbol{r})$ in eq. (2.70). However, we can directly apply the result to the individual vector components of $\boldsymbol{A}$ and thus obtain:

$$\boldsymbol{A}(\boldsymbol{r}) = \int_\Omega \frac{\boldsymbol{c}(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} \, \mathrm{d}^3 r' \tag{2.76}$$

For the vector field $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$, the solution of the equations eq. (2.69a) to eq. (2.69d), we thus obtain

$$\boldsymbol{v}(\boldsymbol{r}) = -\nabla\phi(\boldsymbol{r}) + \nabla \times \boldsymbol{A}(\boldsymbol{r})$$

where $\phi(\boldsymbol{r})$ and $\boldsymbol{A}(\boldsymbol{r})$ are determined by eq. (2.74) and eq. (2.76).

# 3. Fundamentals of Electrostatics

In electrostatics, we consider stationary electric charges or charge distributions and the static electric fields they generate. The interaction between these stationary charges or charge distributions is determined by Coulomb's force law eq. (1.1). From measurements of the Coulomb force $\boldsymbol{F}_{12}$ between two charges $q_1$ and $q_2$ with coordinates $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$, it follows that:

- The force is directly proportional to the charges $q_1$ and $q_2$.
- The force is inversely proportional to $|\boldsymbol{r}_1 - \boldsymbol{r}_2|^2$.
- The force acts in the direction of the line connecting the two charges.
- If the charges $q_1$ and $q_2$ have the same polarity, the force is repulsive, and if they have different polarity, the force is attractive.
- Newton's third law applies, action = reaction, so $\boldsymbol{F}_{12} = -\boldsymbol{F}_{21}$.

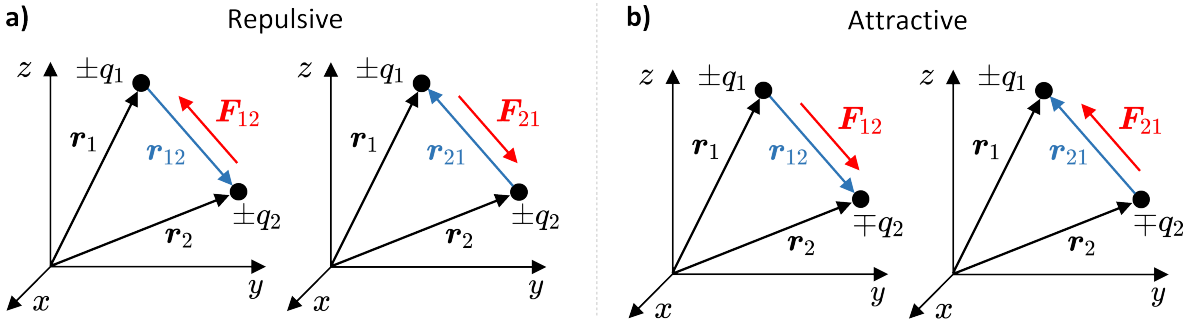



**Fig. 3.1:** Orientations of the distance and force vectors in Coulomb's law: (a) for charges of the same polarity; (b) for charges of opposite polarity.

The conventions illustrated in fig. 3.1 apply for Coulomb's force law: let $\boldsymbol{F}_{12}$ be the force exerted by $q_2$ on charge $q_1$, and $\boldsymbol{r}_{12}$ the separation vector measured from $q_1$ to $q_2$; conversely, let $\boldsymbol{F}_{21}$ be the force exerted by $q_1$ on $q_2$, and $\boldsymbol{r}_{21}$ the separation vector measured from $q_2$ to $q_1$. Then Coulomb's force law is generally:

$$\boldsymbol{F}_{12} = k_e \cdot q_1 \cdot q_2 \cdot \frac{\boldsymbol{r}_1 - \boldsymbol{r}_2}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|^3} = -\boldsymbol{F}_{21} \tag{3.1}$$

The positive proportionality constant $k_e$ depends on the chosen system of units. Here, the SI system is used throughout (see Appendix B) in which the unit of the elementary electric charge is the coulomb, abbreviated C, and in this system

$$k_e = \frac{1}{4\pi\epsilon_0} \tag{3.2}$$

where $\epsilon_0 = 8.8542 \cdot 10^{-12}\,\mathrm{C}^2 \cdot \mathrm{m}^{-2} \cdot \mathrm{N}^{-1}$ is the so-called electric field constant. In the following sections, we will first derive the fundamental equations of electrostatics and then begin to illustrate the practical physical significance of these equations using various examples.

## 3.1 Fundamental Equations of Electrostatics

In this section, we will first familiarize ourselves with the fundamental equations of electrostatics: Gauss's law of electrostatics and Maxwell's equations of electrostatics. These relate the electrostatic field $\boldsymbol{E}(\boldsymbol{r})$ to the electrostatic potential $\phi(\boldsymbol{r})$ and the charge distribution $\rho(\boldsymbol{r})$ associated with $\phi(\mathbf{r})$. We will then examine the changes in potential energy and work done when charges are moved within an electrostatic field, as well as introduce the concept of the electrostatic field energy.

### 3.1.1 The Electrostatic Field

Coulomb's law of electrostatics, eq. (3.1), describing the force between two point charges, is structurally similar to Newton's law which describes the force interaction between two point masses. The difference lies in the fact that there are two types of charges, positive and negative. Therefore, the Coulomb force between two charges can be attractive or repulsive depending on their respective polarity, whereas the Newtonian force between two point masses is always attractive. However, this is not the only difference between these two formally similar laws of nature. Unlike Newton's law, which applies equally to stationary and moving point masses, Coulomb's law is only valid for stationary point charges and not for moving ones. Moreover, in electrostatics, unlike with point masses in Newtonian mechanics, one does not usually speak of the force acting between point charges. Instead, in electrostatics we usually speak of an electric field generated by static charges or a charge distributions at specific locations in space which then exerts a force on a test charge $q_p$. By using such a test charge, which we can place at any desired

position in an electric field, we can then measure, for example, the position-dependent strength of the electric field at location $\boldsymbol{r}$ that is generated by a point charge $q_1$ located at location $\boldsymbol{r}_1$. Hence, we consider with

$$\boldsymbol{F}(\boldsymbol{r}) \;=\; q_p \cdot \boldsymbol{E}(\boldsymbol{r}) \tag{3.3}$$

the force acting on a test charge $q_p$ located at position $\boldsymbol{r}$ exerted by an electrostatic field

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{q_1}{4\pi\epsilon_0} \cdot \frac{\boldsymbol{r} - \boldsymbol{r}_1}{|\boldsymbol{r} - \boldsymbol{r}_1|^3} \tag{3.4}$$

which is generated by a charge $q_1$ located at the position $\boldsymbol{r}_1$. Formally, eq. (3.3) certainly looks like Newton's second law $\boldsymbol{F} = m\boldsymbol{a}$ where $m$ is a point mass and $\boldsymbol{a}$ is the acceleration field acting on it. However, unlike Newton's second law, eq. (3.3) only applies to static and not to moving charges. Like any other electric charge, the test charge $q_p$ also generates its own electrostatic field. Therefore, we must choose this test charge to be so small that the electrostatic field generated by $q_p$ is much weaker than the field we want to measure. In practice, measuring an electric field involves placing the small test charge $q_p$ at a specific point in the electrostatic field, measuring the force on the test charge there, and then determining the electric field strength at that location using eq. (3.3). Then we move the test charge to the next location of interest, measure and evaluate, and repeat this process as often as required to map the electric field to the desired accuracy. However, this only gives us the magnitude of the field strength as a function of location. To determine the direction of the electrostatic field lines, we need to know in which direction the force in eq. (3.3) accelerates the positive test charge $q_p$. At the time of their discovery, electromagnetic fields were something exotic. However, in our modern world, we are surrounded by them almost everywhere. Every electrical device generates them, and there are limits that must be observed when such devices emit electromagnetic radiation. Accordingly, there are a number of modern methods for measuring electric fields that are significantly more precise than simply placing a test charge in a field, as described above.

Eq. (3.4) serves as the definition of the electric field. This electric field can be generated by a single point charge like $q_1$, by a distribution of such discrete point charges, by continuous charge distributions, or by a combination of these different scenarios. If we place a point charge $q$ at the coordinate origin,

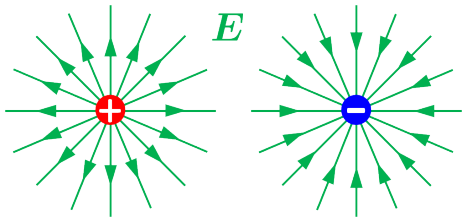



**Fig. 3.2**

i.e., $\boldsymbol{r}_1 = 0$ and $q_1 = q$ in eq. (3.4), then we obtain:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{q}{4\pi\epsilon_0}\frac{\hat{\boldsymbol{r}}}{r^2} \tag{3.5}$$

Fig. 3.2 shows this radially symmetric electrostatic field for a positive and a negative point charge. By convention, the electric field vector always points from a positive to a negative charge. The field strength decreases with the inverse square of distance from the point charge, i.e., $r^{-2}$. Therefore, if one considers the flux of the electric field through a spherical shell with surface area $4\pi r^2$, the flux through such a shell always remains the same, regardless of the value of $r$.

As we already know from the discussion of the Newtonian force in volume 1 of this physics course, whenever several forces act on a given point mass, the respective force vectors add up to a total force acting on this point mass. As illustrated by fig. 3.3 for three point charges, the same applies to the Coulomb force between point charges. The superposition principle applies equally to the Coulomb force and, as eq. (3.3) suggests, also to the corresponding electric fields that generate these forces. Therefore, if $n$ point charges $q_i$ are located at the coordinates $\boldsymbol{r}_i$, then the total field generated at location $\boldsymbol{r}$ by these point charges is given by:

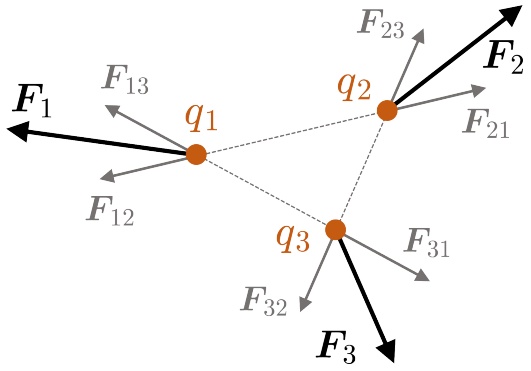



**Fig. 3.3**

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n} q_i \frac{\boldsymbol{r}-\boldsymbol{r}_i}{|\boldsymbol{r}-\boldsymbol{r}_i|^3} \tag{3.6}$$

In the case of a continuous charge distribution $\rho(\boldsymbol{r})$ the charge $\mathrm{d}q = \rho(\boldsymbol{r}')\,\mathrm{d}^3r'$ contained in an infinitesimal volume element $\mathrm{d}^3r'$ generates the electric field

$$\mathrm{d}\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\mathrm{d}q\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} = \frac{1}{4\pi\epsilon_0}\rho(\boldsymbol{r}')\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}\,\mathrm{d}^3r' \tag{3.7}$$

Instead of summing over the contributions of discrete charges as in eq. (3.6), the superposition principle here requires one to integrate over the entire charge distribution $\rho(\boldsymbol{r}')$ present in the volume $\Omega$ in order to obtain the total field generated by this charge distribution, that is:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\int_\Omega \rho(\boldsymbol{r}')\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3}\,\mathrm{d}^3r' \tag{3.8}$$

Now we use Gauss's theorem from eq. (2.42) to determine the flux of the vector field $\boldsymbol{E}(\boldsymbol{r})$, i.e., $\nabla \boldsymbol{E}(\boldsymbol{r})$. Because of

$$\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} = -\boldsymbol{\nabla}_{\boldsymbol{r}} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \boldsymbol{\nabla}_{\boldsymbol{r}'} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \tag{3.9}$$

we can rewrite eq. (3.8) as

$$\boldsymbol{E}(\boldsymbol{r}) = -\frac{1}{4\pi\epsilon_0} \int_\Omega \rho(\boldsymbol{r}') \boldsymbol{\nabla}_{\boldsymbol{r}} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \,\mathrm{d}^3 r' \tag{3.10}$$

This expression for $\boldsymbol{E}(\boldsymbol{r})$ we now insert into Gauss's theorem eq. (2.42) and thereby obtain for the flux of the electric field through the surface $S$ of a volume $\Omega$ in which all charge sources are located:

$$\begin{aligned}
\int_S \boldsymbol{E}(\boldsymbol{r}) \cdot \boldsymbol{n}_s \,\mathrm{d}\sigma &= \int_\Omega \boldsymbol{\nabla}_{\boldsymbol{r}} \cdot \boldsymbol{E}(\boldsymbol{r}) \,\mathrm{d}^3 r \\
&= -\frac{1}{4\pi\epsilon_0} \int_\Omega \boldsymbol{\nabla}_{\boldsymbol{r}} \left[ \int \rho(\boldsymbol{r}') \boldsymbol{\nabla}_{\boldsymbol{r}} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \,\mathrm{d}^3 r' \right] \mathrm{d}^3 r \\
&= -\frac{1}{4\pi\epsilon_0} \int_\Omega \left[ \int \rho(\boldsymbol{r}') \Delta_{\boldsymbol{r}} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \,\mathrm{d}^3 r' \right] \mathrm{d}^3 r
\end{aligned} \tag{3.11}$$

With eq. (2.68) we know that:

$$\Delta_{\boldsymbol{r}} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = -4\pi \delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}') \tag{3.12}$$

Using this relationship, the right side of eq. (3.11) becomes:

$$\int_S \boldsymbol{E}(\boldsymbol{r}) \cdot \boldsymbol{n}_s \,\mathrm{d}\sigma = \frac{1}{\epsilon_0} \int_\Omega \left[ \int \rho(\boldsymbol{r}') \delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}') \,\mathrm{d}^3 r' \right] \mathrm{d}^3 r = \frac{1}{\epsilon_0} \int_\Omega \rho(\boldsymbol{r}) \,\mathrm{d}^3 r \tag{3.13}$$

The application of Gauss's theorem to the flux of an electrostatic field $\boldsymbol{E}(\boldsymbol{r})$ generated by a charge distribution $\rho(\boldsymbol{r})$ located within a volume $\Omega$ which is bounded by the surface $S$, therefore gives us the following equation:

$$\int_S \boldsymbol{E}(\boldsymbol{r}) \cdot \boldsymbol{n}_s \,\mathrm{d}\sigma = \int_\Omega \boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}) \,\mathrm{d}^3 r = \frac{1}{\epsilon_0} \int_\Omega \rho(\boldsymbol{r}) \,\mathrm{d}^3 r \tag{3.14}$$

From eq. (3.14) the central equations of electrostatics follow. Let us begin with Gauss's law of electrostatics, which states:

$$\int_S \boldsymbol{E}(\boldsymbol{r}) \cdot \boldsymbol{n}_s \,\mathrm{d}\sigma = \frac{1}{\epsilon_0} \int_\Omega \rho(\boldsymbol{r}) \,\mathrm{d}^3 r \tag{3.15}$$

If one is not dealing with a continuous charge distribution, but with $n$ discrete individual charges $q_i$, then Gauss's law of electrostatics states:

$$\int_S \boldsymbol{E}(\boldsymbol{r})\cdot\boldsymbol{n}_s \,\mathrm{d}\sigma \;=\; \frac{1}{\epsilon_0}\sum_{i=1}^{n} q_i \tag{3.16}$$

As we will see in the next chapter, Gauss's theorem is a very powerful tool. In many practically important situations in electrostatics, this tool allows us to quickly determine the electric field generated by an electrostatic charge distribution. However, there is more. Eq. (3.14) also provides us with another important result, namely, that the following must always hold true:

$$\int_\Omega \left[\boldsymbol{\nabla}\cdot\boldsymbol{E}(\boldsymbol{r}) - \frac{1}{\epsilon_0}\rho(\boldsymbol{r})\right] \mathrm{d}^3r \;=\; 0 \tag{3.17}$$

This equation applies to any volume, and therefore:

$$\boldsymbol{\nabla}\cdot\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{1}{\epsilon_0}\rho(\boldsymbol{r}) \tag{3.18}$$

Eq. (3.18) is the first Maxwell equation of electrostatics. Sometimes, it is also referred to as the differential form of Gauss's theorem. In deriving eq. (3.18), we learned something else about the electric field $\boldsymbol{E}(\boldsymbol{r})$: eq. (3.10) tells us that $\boldsymbol{E}(\boldsymbol{r})$ is the gradient of a scalar potential $\phi$:

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; -\boldsymbol{\nabla}\phi(\boldsymbol{r}) \tag{3.19}$$

with the electrostatic potential

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{3.20}$$

A first consequence of eq. (3.19) is that the electric field vector is always perpendicular to lines or surfaces for which $\phi(\boldsymbol{r}) = const.$ Or put another way: The electric field component along equipotential lines or equipotential surfaces of the electrostatic potential is always zero, since $\boldsymbol{\nabla}\phi(\boldsymbol{r}) = 0$ always holds there. From eq. (3.18) and eq. (3.19) it further follows that:

$$\Delta\phi(\boldsymbol{r}) \;=\; -\frac{1}{\epsilon_0}\rho(\boldsymbol{r}) \tag{3.21}$$

The electrostatic potential $\phi(\boldsymbol{r})$ thus satisfies the Poisson equation already familiar to us from eq. (2.70). From the discussion of this equation, we know that $\phi(\boldsymbol{r})$ is a unique solution as long as $\phi(|\boldsymbol{r}|)$ scales at least as $r^{-2}$. Eq. (3.19) has a further consequence. With eq. (2.34), we know that the curl of a gradient vanishes, so the following holds for the electric field:

$$\boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}) \;=\; 0 \tag{3.22}$$

Eq. (3.22) is the second Maxwell equation of electrostatics. It states that the electric field is a vector field free of vortices. Unlike the first Maxwell equation, eq. (3.18), which, as we will see, also applies to moving charges, eq. (3.22) is however only valid in electrostatics. For moving electric charges this equation must be modified. Because a vector field whose curl is zero possesses no vortices, it follows that no closed field lines can exist in it. Hence, eq. (3.22) tells us that closed electric field lines do not exist and therefore electric field lines must always originate from and terminate in electric charges. Where field lines begin and end is a matter of definition. In the case of the electric field, the agreed convention is that field lines originate from positive charges and terminate in negative charges. Since the electric force is related to the electric field via $\boldsymbol{F} = q\boldsymbol{E}$, the electric force field is also free of vortices. Therefore, the electric force is a conservative force, meaning that the work done in this force field is independent of the path taken, a fact we will return to later.

### 3.1.2 The Electrostatic Potential

We derived the electrostatic potential $\phi(\boldsymbol{r})$ in eq. (3.20) from a very general perspective. But what is its physical significance? A more fundamental approach can help us to understand this. In the discussion of potentials and fields in volume 1 of this physics course, we learned that the potential energy $U(\boldsymbol{r})$, gained when a point mass moves from a location $\boldsymbol{r}_0$ where the potential is by definition zero to a location $\boldsymbol{r}$ in a gravitational field, is equal to the negative of the work $W(\boldsymbol{r})$ performed during that process. Just as with the motion of a point mass in a gravitational field, the following applies to the displacement of an electric point charge in an electrostatic field:

$$W(\boldsymbol{r}) + U(\boldsymbol{r}) \;=\; 0 \tag{3.23}$$

For the physical quantity of work, the relationship work = force × distance always applies. When dealing with a force field $\boldsymbol{F}(\boldsymbol{r})$, the work performed along a path from $a$ to $b$ is given by the work integral familiar to us from volume 1:

$$W \;=\; \int_a^b \boldsymbol{F}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \tag{3.24}$$

If we bring in an electrical test charge $q$ from infinity to a position $\boldsymbol{r}$ in the field of another electric charge $q_1$ located at $\boldsymbol{r}_1$, then we can express the potential energy $U(\boldsymbol{r})$ of the charge $q$ as the negative of the work done:

$$U(\boldsymbol{r}) \;=\; -\int_{\infty}^{r} \boldsymbol{F}(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}' \;=\; -\frac{q\cdot q_1}{4\pi\epsilon_0}\int_{\infty}^{r} \frac{\boldsymbol{r}'-\boldsymbol{r}_1}{|\boldsymbol{r}'-\boldsymbol{r}_1|^3}\,\mathrm{d}\boldsymbol{r}' \;=\; -q\cdot\int_{\infty}^{r} \boldsymbol{E}(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}' \tag{3.25}$$

Let us now define the electrostatic potential $\phi(\boldsymbol{r})$ at the position $\boldsymbol{r}$ as the capacity to perform the work required to bring a positive unit charge, i.e., $q = 1$, from infinity to the position $\boldsymbol{r}$ within the electric field $\boldsymbol{E}(\boldsymbol{r}')$. Thus we have

$$\phi(\boldsymbol{r}) \;=\; -\int_{\infty}^{r} \boldsymbol{E}(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}' \tag{3.26}$$

from which follows immediately that:

$$U(\boldsymbol{r}) \;=\; q\cdot\phi(\boldsymbol{r}) \tag{3.27}$$

The quantity charge × electrostatic potential represents thus a potential energy, and $\phi$ corresponds to a potential energy divided by an electric charge. If the test charge $q$ is brought from infinity not into the vicinity of a single other charge $q_1$ but rather into the vicinity of a distribution of discrete point charges $q_1 \ldots q_n$, then the potential $\phi(\boldsymbol{r})$ is given by:

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|} \tag{3.28}$$

If one transitions from discrete charges $q_1 \ldots q_n$ to a charge distribution $\rho(\boldsymbol{r}')$, the summation becomes an integration, yielding eq. (3.20) for the electrostatic potential $\phi(\boldsymbol{r})$. If the test charge $q$ is moved within the electrostatic potential from position $\boldsymbol{r}_1$ to position $\boldsymbol{r}_2$, then equations eq. (3.25) and eq. (3.26) imply the following for the change in the potential energy of the charge $q$:

$$U(\boldsymbol{r}_1) - U(\boldsymbol{r}_2) \;=\; q[\phi(\boldsymbol{r}_1) - \phi(\boldsymbol{r}_2)] \tag{3.29}$$

Eq. (3.29) describes the change in the potential energy of an electrical charge moving within an electric field. However, since electrostatics deals only with stationary charges, the term «moving» here implies something other than real movement. Instead of actually moving the charge, one considers its potential energy at a specific point in space and compares it with the potential energy it would possess at another location; or vice versa. Eq. (3.27) tells us that an electric charge $q$ at any point $\boldsymbol{r}$ in an electrostatic field $\boldsymbol{E}(\boldsymbol{r})$ possesses a potential energy $U(\boldsymbol{r}) = q \cdot \phi(\boldsymbol{r})$. As we know from volume 1, the work $W_{12}$ required to move $q$ from $\boldsymbol{r}_1$ to the location $\boldsymbol{r}_2$ is simply the negative of this change in potential energy:

$$W_{12} \;=\; q[\phi(\boldsymbol{r}_2) - \phi(\boldsymbol{r}_1)] \tag{3.30}$$

The change in the potential energy of $q$ in eq. (3.29) and the work done as expressed in eq. (3.30) to move $q$ from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$ depend only on the value of the electrostatic potential $\phi$ at these two locations, and not on the path chosen from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$. In this respect, motion in an electrostatic field resembles the motion in a gravitational field that we examined in volume 1. The reason for this is that the Coulomb force, just like the Newtonian gravitational force, depends solely on the distance $r$ and scales as $r^{-2}$. Any force satisfying these two conditions is classified as a so-called conservative force. For the fields associated with these forces, here specifically for the electrostatic field where $\boldsymbol{F}(\boldsymbol{r}) = q \cdot \boldsymbol{E}(\boldsymbol{r})$, and for the gravitational field where $\boldsymbol{F}(\boldsymbol{r}) = m \cdot \boldsymbol{g}(\boldsymbol{r})$, we already know that these fields are each the gradient of a specific potential: the electrostatic potential in the former case and the gravitational potential in the case of Newtonian gravity. For conservative forces, we already know from volume 1 that the work done by such forces is independent of the chosen path. This follows directly from eq. (3.26) if we consider a potential difference:

$$\phi(\boldsymbol{r}_2) - \phi(\boldsymbol{r}_1) = -\int_{\infty}^{\boldsymbol{r}_2} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} + \int_{\infty}^{\boldsymbol{r}_1} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \tag{3.31}$$

We split the path of bringing the charge $q$ from infinity to $\boldsymbol{r}_2$ into the path of bringing $q$ from infinity to $\boldsymbol{r}_1$ and then from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$; that is:

$$\phi(\boldsymbol{r}_2) - \phi(\boldsymbol{r}_1) = -\left[\int_{\infty}^{\boldsymbol{r}_1} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} + \int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r}\right] + \int_{\infty}^{\boldsymbol{r}_1} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \tag{3.32}$$

This gives us a simple relationship for the potential difference:

$$\phi(\boldsymbol{r}_2) - \phi(\boldsymbol{r}_1) = -\int_{\boldsymbol{r}_1}^{\boldsymbol{r}_2} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \tag{3.33}$$

To determine the change in the potential energy of an electric charge when it is displaced within an electrostatic field, we do not need to know the value of the electrostatic potential $\phi$ at infinity. The change in potential energy associated with moving an electric charge $q$ from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$, as well as the work performed in doing so, is determined solely by the potential difference $\phi(\boldsymbol{r}_2) - \phi(\boldsymbol{r}_1)$ between the two respective positions. We can therefore choose the value of $\phi$ at infinity arbitrarily. However, for practical reasons, we typically stipulate that the electrostatic potential $\phi$ is zero (or takes on another constant value) at infinity, which ensures that the electric field automatically vanishes at infinity. The potential difference $\phi(\boldsymbol{r}_2) - \phi(\boldsymbol{r}_1)$, multiplied by the test charge $q$, corresponds to the work required to move $q$ from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$:

$$W_{12} = -q \int_{\boldsymbol{r_1}}^{\boldsymbol{r_2}} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \tag{3.34}$$

Eq. (3.34) shows us that the work which is required to counter the acting electric forces when moving an electric charge is precisely the negative of the electric field strength integrated along the path from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$. If the component of $\boldsymbol{E}$ along the path from $\boldsymbol{r}_1$ to $\boldsymbol{r_2}$ is parallel to $\mathrm{d}\boldsymbol{r}$, then the integral is positive and the work is negative; however, if the component of $\boldsymbol{E}$ along the path is antiparallel to $\mathrm{d}\boldsymbol{r}$, then the integral is negative and the work is positive. Naturally, whether the work is positive or negative also depends on the sign of the charge $q$. The potential difference in eq. (3.33) is of great practical importance, as it represents the electric voltage $U_{12}$ between the points $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$:[1]

$$U_{12} = \phi(\boldsymbol{r_2}) - \phi(\boldsymbol{r_1}) = -\int_{\boldsymbol{r_1}}^{\boldsymbol{r_2}} \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{r} \tag{3.35}$$

The work required to move the electric charge from $\boldsymbol{r}_1$ to $\boldsymbol{r}_2$ is thus related to the electric voltage by:

$$W_{12} = q{\cdot}U_{12} \tag{3.36}$$

The voltage in eq. (3.35) is a potential difference, a quantity that does not depend on any reference point. In practice, however, voltages must be referenced to something. Per definition, this reference point is the electrical potential of the Earth, the so-called ground potential, which defines the 0-volt potential. Here, we will also frequently assume that the electrical potential vanishes at infinity, i.e., it is zero there as well.

Another important relationship follows from eq. (3.33): namely, that for a closed path around a surface $A$ with boundary $C$ in electrostatics, the following always holds:

$$\oint_C \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{s} = 0 \tag{3.37}$$

With eq. (2.46), Stokes' theorem, it follows from this equation that:

$$\oint_C \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{s} = \int_A \big[\nabla \times \boldsymbol{E}(\boldsymbol{r})\big]\,\mathrm{d}\boldsymbol{\sigma} = 0 \tag{3.38}$$

The surface $A$ can be chosen arbitrarily. It follows, therefore, that $\nabla \times \boldsymbol{E}(\boldsymbol{r}) = 0$ must hold. Eq. (3.37) is the integral form of Maxwell's equation eq. (3.22). Here, however, we have arrived at the result of eq. (3.22) via a different route and, at the same time, learned that force fields with vanishing curl are conservative force fields.

[1] Here, we use the same letter $U$ for the quantities electric voltage and potential energy. However, the context will always make it clear which one we are referring to.

### 3.1.3 Electrostatic Field Energy

In eq. (3.27), we have already encountered the expression for the potential energy of a positive test charge $q$ when it is brought from infinity to a position $\boldsymbol{r}$ in the field of another charge $q_1$ located at position $\boldsymbol{r}_1$:

$$U(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\frac{q\cdot q_1}{|\boldsymbol{r}-\boldsymbol{r}_1|} \;=\; q\cdot\phi_1(\boldsymbol{r}) \tag{3.39}$$

$\phi_1(\boldsymbol{r})$ is the electrostatic potential generated by the electric charge $q_1$ at position $\boldsymbol{r}$. The principle of superposition applies to electrostatic forces and consequently it also applies to the fields or electrostatic potentials generated by electric charges. Therefore, the potential energy of an electric charge $q_i$ at position $\boldsymbol{r}_i$ in the presence of electrostatic potentials generated by charges $q_j$ located at positions $\boldsymbol{r}_j$ is given by:

$$U(\boldsymbol{r}_i) \;=\; \frac{q_i}{4\pi\epsilon_0}\cdot\sum_j \frac{q_j}{|\boldsymbol{r}_i-\boldsymbol{r}_j|} \;=\; q_i\sum_j \phi_j(\boldsymbol{r}_i) \;=\; q_i\phi(\boldsymbol{r}_i) \tag{3.40}$$

Here, $\phi(\boldsymbol{r}_i)$ is the electrostatic potential generated by all other charges at the position $\boldsymbol{r}_i$. The potential energy of the electric charge $q_i$ is determined by summing over two-body interactions; three-body interactions do not exist in electrodynamics, just as they do not exist in mechanics. The analysis applied to $q_i$ naturally applies equally to each of the electric charges $q_j$. We obtain the total potential energy $U$ of all electric charges present by summing over all two-body interactions:

$$U \;=\; \sum_i U(\boldsymbol{r}_i) \;=\; \frac{1}{2}\cdot\frac{1}{4\pi\epsilon_0}\cdot\sum_{i\neq j}\frac{q_i\cdot q_j}{|\boldsymbol{r_i}-\boldsymbol{r}_j|} \;=\; \frac{1}{2}\sum_i q_i\phi(\boldsymbol{r}_i) \tag{3.41}$$

Double counting, which means the counting of both $ij$ and $ji$, is corrected by the factor $1/2$, and since there is no electrostatic self-energy, i.e., there is no interaction of an electric charge with itself, the summation is subject to the restriction $i \neq j$. Upon transitioning from discrete charges $q_i$ to a continuous charge distribution $\rho(\boldsymbol{r})$, the sum in eq. (3.41) becomes an integral

$$\frac{1}{2}\sum_i q_i\phi(\boldsymbol{r}_i) \quad\rightarrow\quad \frac{1}{2}\int_V \rho(\boldsymbol{r})\phi(\boldsymbol{r})\,\mathrm{d}^3r \tag{3.42}$$

over the volume $V$ containing the charge distribution $\rho(\boldsymbol{r})$. Thus eq. (3.41) becomes:

$$U \;=\; \frac{1}{2}\int_V \rho(\boldsymbol{r})\phi(\boldsymbol{r})\,\mathrm{d}^3r \tag{3.43}$$

The interpretation of eq. (3.39) and eq. (3.40) is straightforward. These equations obviously describe the potential energy of an electric test charge in the electrostatic field

of another electric charge or of several other electric charges. However, we must ask ourselves: What, then, do equations eq. (3.41) and eq. (3.43) describe? Evidently, in these equations, the contributions of all charges are either summed up or integrated. The potential energy in these equations no longer refers to any specific electric charge, but rather to something that arises from the interactions of all these charges. So, what kind of potential energy are we talking about here? With what else — if not the potential energy of the discrete electric charges or electric charge distributions themselves — can we possibly associate the potential energy in eq. (3.41) and eq. (3.43)? To understand this, we must remind ourselves that we can express the electric charge distribution $\rho(\boldsymbol{r})$ in eq. (3.43) using eq. (3.18) and eq. (3.19) as:

$$\rho(\boldsymbol{r}) \;=\; \epsilon_0 \cdot \nabla \boldsymbol{E}(\boldsymbol{r}) \;=\; -\epsilon_0 \cdot \Delta\phi(\boldsymbol{r}) \tag{3.44}$$

Substituting this into eq. (3.43), we obtain the following expression for the total potential energy $U$:

$$U \;=\; -\frac{\epsilon_0}{2} \int_V \phi(\boldsymbol{r}) \Delta\phi(\boldsymbol{r}) \, \mathrm{d}^3 r \tag{3.45}$$

What we can immediately deduce from this equation is that the potential energy $U$ is entirely determined by the electrostatic potential $\phi$. Therefore, we can associate the potential energy $U$ with the electrostatic energy linked to the electrostatic potential $\phi$. The electrostatic potential $\phi$ is a scalar field, and we can view $U$ as the potential energy stored in this field. But there is more to this equation and to see that we need to take another step by using the identity

$$\nabla \cdot (\phi \nabla\phi) \;=\; (\nabla\phi) \cdot (\nabla\phi) + \phi \Delta\phi \tag{3.46}$$

to rewrite eq. (3.45) as:

$$U \;=\; \frac{\epsilon_0}{2} \int_V \nabla\phi(\boldsymbol{r}) \cdot \nabla\phi(\boldsymbol{r}) \, \mathrm{d}^3 r - \frac{\epsilon_0}{2} \int_V \nabla \cdot \big[\phi(\boldsymbol{r}) \nabla\phi(\boldsymbol{r})\big] \, \mathrm{d}^3 r \tag{3.47}$$

Let us first consider the second integral in this equation. Using Gauss's integral theorem (eq. (2.42)), this becomes:

$$\int_V \nabla \cdot \big[\phi(\boldsymbol{r}) \nabla\phi(\boldsymbol{r})\big] \, \mathrm{d}^3 r \;=\; \int_S \phi(\boldsymbol{r}) \nabla\phi(\boldsymbol{r}) \, \mathrm{d}\boldsymbol{\sigma} \tag{3.48}$$

We already know that $\phi$, as the unique solution to the Poisson equation, scales as $r^{-1}$, and therefore $\nabla\phi$ scales as $r^{-2}$. Thus, the integrand in the surface integral scales as $r^{-3}$, whereas the infinitesimal surface element $\mathrm{d}\boldsymbol{\sigma}$ scales as $r^2$. Considering a charge distribution where all charges are located within a finite region, the surface integral

consequently vanishes as $S \to \infty$. Only the first integral in eq. (3.47) remains, and with $\boldsymbol{E} = -\nabla\phi$ we obtain for the energy of the electrostatic field the expression:

$$U = \frac{\epsilon_0}{2} \int_V \boldsymbol{E}^2(\boldsymbol{r})\, \mathrm{d}^3 r \tag{3.49}$$

In eq. (3.45), the total stored potential energy $U$ is described entirely by the electrostatic potential $\phi$, and in eq. (3.49), entirely by the electric field $\boldsymbol{E}$. With eq. (3.49), we now know the energy density of the electric field. If $\mathrm{d}V = \mathrm{d}^3 r$ denotes the differential volume element, then the following simple relationship holds for the energy density of the electric field:

$$\frac{\mathrm{d}U}{\mathrm{d}V} = \frac{\epsilon_0 \boldsymbol{E}^2(\boldsymbol{r})}{2} \tag{3.50}$$

Equations eq. (3.41) and eq. (3.50) reflect two different perspectives. Eq. (3.41) interprets $U$ as the potential energy of a system of charged point masses, thereby focusing on the role of the point charges. In contrast, eq. (3.50), the result of evaluating eq. (3.43), focuses on the role of the electric field, which stores the energy $U$ in the form of field energy. In electrostatics, both perspectives are equally valid as they are essentially equivalent. However, this is not the case in electrodynamics, which relies on the concept of the field. The potential energy in eq. (3.43) and eq. (3.50) is the energy of the electric field, expressed in terms of the electrostatic field potential $\phi$ in eq. (3.45) or the electric field strength $\boldsymbol{E}$ in eq. (3.50). In this section, we have referred to $U$ in terms of the stored potential energy which is associated with a given distribution of electric charge. The reason for this is the link between $U$ and the work done, $W$, which has a counterpart in mechanics. However, without electric charges, this energy would not exist. Therefore, it is more appropriate to speak of $U$ as the electrostatic energy associated with a given charge distribution. Because the gravitational force is much weaker than the Coulomb force and the masses involved are usually quite small, the electrostatic energy $U$ is much greater than the potential energy of the electric charge carriers associated with the gravitational force.

When considering the energy of the electrostatic field, we bypassed the problem of self-interaction, that is, the interaction of an electric charge's electrostatic field with the electric charge itself. In eq. (3.41), we excluded such an interaction by imposing the condition $i \neq j$. However, in eq. (3.43), and consequently in eq. (3.49), the integration extends to the location of the charge itself. Therefore, eq. (3.49) includes self-energy contributions. We will revisit this issue in the next chapter when discussing the classical electron radius.

**Tab. 3.1:** Fundamental equations of electrostatics.

**Basic assumption: Stationary electric charges**

$$\rho = \rho(\boldsymbol{r}) \qquad \Rightarrow \qquad \frac{\partial \rho}{\partial t} = 0$$

**Electrostatic potential and electric field strength**

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|} \, \mathrm{d}^3 r' \qquad \text{and} \qquad \boldsymbol{E}(\boldsymbol{r}) = -\boldsymbol{\nabla}\phi(\boldsymbol{r})$$

**Gauss's law of electrostatics**

$$\int_S \boldsymbol{E}(\boldsymbol{r}) \cdot \boldsymbol{n}_s \, \mathrm{d}\sigma = \frac{1}{\epsilon_0} \int_\Omega \rho(\boldsymbol{r}) \, \mathrm{d}^3 r \qquad \text{or} \qquad \int_S \boldsymbol{E}(\boldsymbol{r}) \cdot \boldsymbol{n}_s \, \mathrm{d}\sigma = \frac{1}{\epsilon_0} \sum_{i=1}^{n} q_i$$

**Maxwell's equations of electrostatics**

$$\boldsymbol{\nabla} \cdot \boldsymbol{E}(\boldsymbol{r}) = \frac{1}{\epsilon_0} \rho(\boldsymbol{r}) \qquad \text{and} \qquad \boldsymbol{\nabla} \times \boldsymbol{E}(\boldsymbol{r}) = 0$$

Tab. 3.1 summarizes the fundamental equations of electrostatics derived in this section. A key premise of electrostatics is the absence of moving charges. In electrostatics there are no electric currents. The equations in tab. 3.1 will help us in the next chapter to understand the behavior of stationary, conducting, and non-conducting bodies in electrostatic fields.

At this point, however, it is time to say something about the SI units of the electrical quantities considered so far. We are already familiar with the coulomb as the unit of electric charge from eq. (1.2). However, it is not defined in terms of the elementary charge, as this would be impractical due to the very small electric charge of the electron. The electric charge of 1 coulomb is defined in terms of electric current measured in amperes per unit of time; thus, $1\mathrm{C} = 1\mathrm{A} \cdot \mathrm{s}^{-1}$. According to its definition in eq. (3.2), the electric field strength $\boldsymbol{E}$ has the unit $\mathrm{N} \cdot \mathrm{C}^{-1}$. The electrostatic potential $\phi$, as the gradient of the field $\boldsymbol{E}$, thus has the unit $\mathrm{N} \cdot \mathrm{C}^{-1} \cdot \mathrm{m}^{-1}$. However, due to its great practical importance, the electrical voltage $U$, and consequently $\phi$ as well, has been assigned its own unit, the volt. For the volt, abbreviated as V, the following relations hold: $\mathrm{V} = \mathrm{N} \cdot \mathrm{C}^{-1} \cdot \mathrm{m}^{-1} = \mathrm{J} \cdot \mathrm{C}^{-1}$. However, a look at Appendix B reveals that the volt is defined as $\mathrm{V} = \mathrm{W} \cdot \mathrm{A}^{-1}$, that is, as power divided by electric current. As can be quickly verified, this is by no means contradictory to the relations $\mathrm{W} = \mathrm{J} \cdot \mathrm{s}^{-1}$ and $\mathrm{A} = \mathrm{C} \cdot \mathrm{s}^{-1}$; rather, it simply reflects the

fact that defining the volt in terms of electric power and electric current is of significantly greater practical relevance.

## 3.2 Fields of Bound Charge Distributions

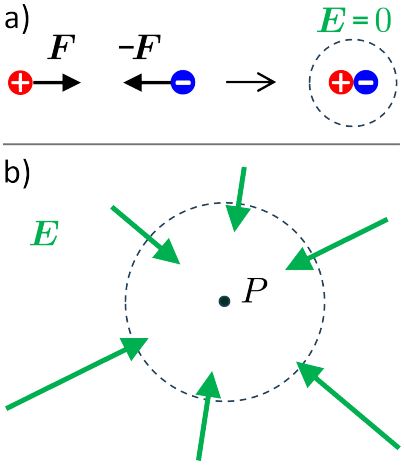



**Fig. 3.4**

In electrostatics, as the name implies, there are no moving charges, all electric charges are at rest. This restriction poses no problem for free electric charges if one considers only a single charge and the field it generates within an otherwise field-free space. A single electrostatic charge in an otherwise force-free space will not move as there is no force acting on it. In that sense, it is thus truly a free, static charge and represents a static «charge distribution» that is in equilibrium. However, this changes as soon as this free charge finds itself in the presence of another charge distribution. Let us assume our solitary free static charge is positive and the additionally introduced charge is negative. Under the influence of the Coulomb force, these electric charges will then move toward each other, as sketched on the left side of fig. 3.4a, and only stop moving once they reach a stable equilibrium, which is the state where they so to speak, stick together, as shown on the right side of fig. 3.4a. In this case, however, we no longer have a charge distribution that produces a measurable electric field. To understand that we only have to place a Gaussian sphere immediately around the two charges that are stuck together, as shown on the right side of fig. 3.4a. This Gaussian sphere encloses a total charge of zero. Outside this Gaussian sphere, the radius of which can be made small enough to just barely enclose both charges completely, there exists therefore no electric field which means that $\boldsymbol{E} = 0$ in that region. This raises an important question: can free static charges in electrostatic fields exist in a state of equilibrium at all? If so, this kind of equilibrium can only be a dynamic one. This means that the forces acting on charges in such equilibrium positions within an electrostatic field must always cancel each other out. In the case of such a hypothetical equilibrium position for a positive test charge in the field of other positive charges, this implies that the $\boldsymbol{E}$-field lines must converge on this equilibrium position from all directions in such a way that, following a small displacement of the test charge from that position, the charge is driven back to this equilibrium position. Fig. 3.4b illustrates a situation in which all field lines in an electric field converge on such a hypothetical equilibrium point $P$. If we imagine

this point $P$ as being surrounded by a Gaussian sphere, there must then exist a non-zero flux of $\boldsymbol{E}$ through the sphere's surface. However, this is a contradiction, because our Gaussian sphere encloses a total charge of zero, meaning the flux of $\boldsymbol{E}$ through it must vanish. Thus, there can be no point $P$ in an electric field toward which field lines converge from all directions.

If we speak of a stable static distribution of free discrete charges, strictly speaking, such a distribution does not exist. Free electric charges will always move under the influence of the Coulomb forces acting between them, that is, unless they are fixed in space in some way or another. Leaving aside the case of a solitary electric charge in an otherwise force-free space, free electric charges do not exist in a state of stable static equilibrium; static charges exist only as bound electric charges. In this context, we can broadly distinguish between two forms of bound electric charges. On the one hand, there are static charge distributions such as those fixed within molecules and on the other hand, there exist quasi freely movable charges in electrical conductors that arrange themselves into a state of static equilibrium in response to an external field. We will turn our attention to conductors that carry a charge and/or are situated in an electrostatic field in the next chapter. Here, we will now begin to examine some fundamental charge configurations, such as those found in a range of simple molecules. Naturally, interatomic distances within molecules can change due to various external forces, which in turn can shift the respective charge distribution in these molecules. However, this generally does not alter the fundamental character of the electric charge distribution present in these molecules, it merely weakens or strengthens its manifestation. The most elementary charge configuration is that of a single electrical charge. The electrostatic potential $\phi$ produced by a single charge located at the coordinate origin scales as $r^{-1}$ in accordance with eq. (3.27). The next elementary charge configuration is that of two charges of opposite polarity, the so-called electric dipole, whose electrostatic potential $\phi$, as we shall see, scales as $r^{-2}$. How do such electric dipoles form?

In molecules such as HCl or CO, one atom is more electronegative than the other and thus attracts electrons more strongly. The result of this is an accumulation of negative charge at the end where the more electronegative atom is located, Cl and O in our examples, and a positive charge at the other end, where the H and C atoms are situated. In such cases where a molecule has a positive and a negative end, it is said to possess a so-called dipole moment. Asymmetric charge distributions, as the ones just described, in which the centers of positive and negative charge do not coincide, are not limited to diatomic molecules. Water, that is, $H_2O$, and ammonia, that is, $NH_3$, also possess a dipole moment.

### 3.2.1 The Electric Dipole

As sketched in fig. 3.5, the electric dipole consists of two point charges, $+q$ and $-q$, separated by a distance $d$. We previously examined such a configuration in the first volume of this physics course in the context of illustrating the field concept in section 4.3. We adopt the corresponding arrangement here but now place a positive electric charge $+q$ on the positive $z$-axis at $z = d/2$ and a negative electric charge $-q$ on the negative $z$-axis. The electrostatic potential of a point charge depends only on the distance from the point charge and does not depend on the direction. Thus, the electrostatic potential $\phi$ of the electric dipole at point $P$, incorporating the contributions of the potentials $\phi_+$ and $\phi_-$ from the positive point charge $+q$ and the negative point charge $-q$, is given by:

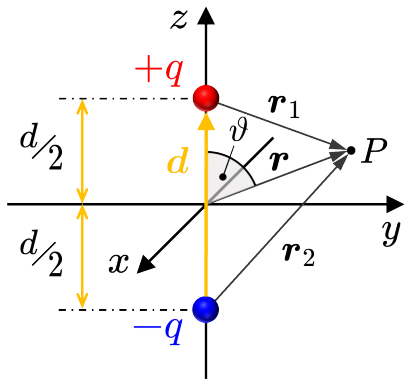



**Fig. 3.5**

$$\phi(r) = \frac{1}{4\pi\epsilon_0}\left[\frac{q}{\sqrt{x^2+y^2+(z-d/2)^2}} + \frac{-q}{\sqrt{x^2+y^2+(z+d/2)^2}}\right] \tag{3.51}$$

Now we will again use the dipole approximation, as in volume 1. For distances $r \gg d$, the so-called far field, we can neglect in the square roots

$$\sqrt{x^2+y^2+\left(z \mp \frac{d}{2}\right)^2} = \sqrt{x^2+y^2+z^2 \mp zd + \frac{d^2}{4}} = r\cdot\sqrt{1 \mp \frac{zd}{r^2} + \frac{d^2}{4r^2}}$$

the term $d^2/4r^2$ and, for $x \ll 1$, use the approximation

$$\frac{1}{\sqrt{1 \mp x}} \approx 1 \pm \frac{x}{2}$$

The result is the so-called far-field approximation for $\phi_+$ and $\phi_-$:

$$\phi_\pm(r) = \frac{1}{4\pi\epsilon_0}\frac{\pm q}{r\cdot\sqrt{1 \mp \frac{zd}{r^2} + \frac{d^2}{4r^2}}} \approx \frac{1}{4\pi\epsilon_0}\frac{\pm q}{r}\cdot\left(1 \pm \frac{zd}{2r^2}\right)$$

This transforms $\phi(r) = \phi_+ + \phi_-$ into:

$$\phi(r) = \frac{1}{4\pi\epsilon_0}\frac{q}{r}\left[\left(1+\frac{zd}{2r^2}\right) - \left(1-\frac{zd}{2r^2}\right)\right] = \frac{1}{4\pi\epsilon_0}\frac{q\cdot d\cdot z}{r^3} \tag{3.52}$$

If $\boldsymbol{d}$ is the displacement vector pointing from the negative charge $-q$ to the positive charge $+q$, as sketched in fig. 3.5, then $\boldsymbol{p} = q\cdot\boldsymbol{d}$ is the so-called dipole moment. Using this dipole moment $\boldsymbol{p}$ and the relation $\cos\vartheta = z/r$, we can express the electrostatic potential of the dipole, independent of its specific orientation within the coordinate system, as:

$$\phi(r) \;=\; \frac{1}{4\pi\epsilon_0}\frac{p\cos\vartheta}{r^2} \;=\; \frac{1}{4\pi\epsilon_0}\frac{\boldsymbol{p}\cdot\boldsymbol{r}}{r^3} \tag{3.53}$$

Based on eq. (3.19), the electric field of the dipole is the negative gradient of eq. (3.53). We now use our small collection of formulas in tab. 2.1 to calculate $\nabla\cdot(\boldsymbol{p}\cdot\boldsymbol{r}/r^3)$ using the formula for $\nabla\cdot(\boldsymbol{a}\cdot\boldsymbol{b})$. Substituting into the respective formula yields:

$$\nabla\cdot\left(\boldsymbol{p}\cdot\frac{\boldsymbol{r}}{r^3}\right) \;=\; \left(\frac{\boldsymbol{r}}{r^3}\cdot\nabla\right)\cdot\boldsymbol{p} + \left(\boldsymbol{p}\cdot\nabla\right)\cdot\frac{\boldsymbol{r}}{r^3} + \boldsymbol{p}\times\left(\nabla\times\frac{\boldsymbol{r}}{r^3}\right) + \frac{\boldsymbol{r}}{r^3}\times\left(\nabla\times\boldsymbol{p}\right) \tag{3.54a}$$

The vector $\boldsymbol{p}$ is constant, and therefore the first and last terms on the left side of this equation vanish. The third term also vanishes because with

$$\frac{\boldsymbol{r}}{r^3} \;=\; -\nabla\frac{1}{r^2}$$

$\boldsymbol{r}/r^3$ is the gradient of a scalar, and from tab. 2.1 we know that for a scalar field $V$, $\nabla\times(\nabla\cdot V) = 0$ holds. Thus, only the second term in eq. (3.54a) is non-zero. To evaluate it, we write out this term:

$$\begin{aligned}
\left(\boldsymbol{p}\cdot\nabla\right)\cdot\frac{\boldsymbol{r}}{r^3} \;&=\; \left(p_x\frac{\partial}{\partial x} + p_y\frac{\partial}{\partial y} + p_z\frac{\partial}{\partial z}\right)\begin{pmatrix} x/r^3 \\ y/r^3 \\ z/r^3 \end{pmatrix} \\
&= \begin{pmatrix} p_x\left(\frac{1}{r^3} - \frac{3x^2}{r^5}\right) - p_y\frac{3xy}{r^5} - p_z\frac{3xz}{r^5} \\ -p_x\frac{3yx}{r^5} + p_y\left(\frac{1}{r^3} - \frac{3y^2}{r^5}\right) - p_z\frac{3yz}{r^5} \\ -p_x\frac{3zx}{r^5} - p_y\frac{3zy}{r^5} + p_z\left(\frac{1}{r^3} - \frac{3z^2}{r^5}\right) \end{pmatrix} \\
&= \frac{\boldsymbol{p}}{r^3} - \frac{3p_x x}{r^5}\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \frac{3p_y y}{r^5}\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \frac{3p_z z}{r^5}\begin{pmatrix} x \\ y \\ z \end{pmatrix} \\
&= \frac{\boldsymbol{p}}{r^3} - \frac{3(\boldsymbol{p}\cdot\boldsymbol{r})\cdot\boldsymbol{r}}{r^5}
\end{aligned} \tag{3.54b}$$

Thus, from eq. (3.53), using $\boldsymbol{E} = -\nabla\phi$, we obtain for the electrostatic field of the electric dipole in the far-field approximation:

$$\boldsymbol{E} \;=\; \frac{1}{4\pi\epsilon_0}\frac{3(\boldsymbol{p}\cdot\boldsymbol{r})\cdot\boldsymbol{r} - \boldsymbol{p}\cdot r^2}{r^5} \tag{3.55}$$

Using $\boldsymbol{p}\cdot\boldsymbol{r} = p\cdot r\cdot\cos\vartheta$, $\cos\vartheta = z/r$ and $\boldsymbol{p} = p\cdot\hat{\boldsymbol{e}}_z$ the field components of $\boldsymbol{E}$ can also be written as:

$$\boldsymbol{E} \;=\; \frac{p}{4\pi\epsilon_0}\left(\frac{3zx}{r^5}\,,\; \frac{3zy}{r^5}\,,\; \frac{3\cos^2\vartheta - 1}{r^3}\right) \tag{3.56}$$

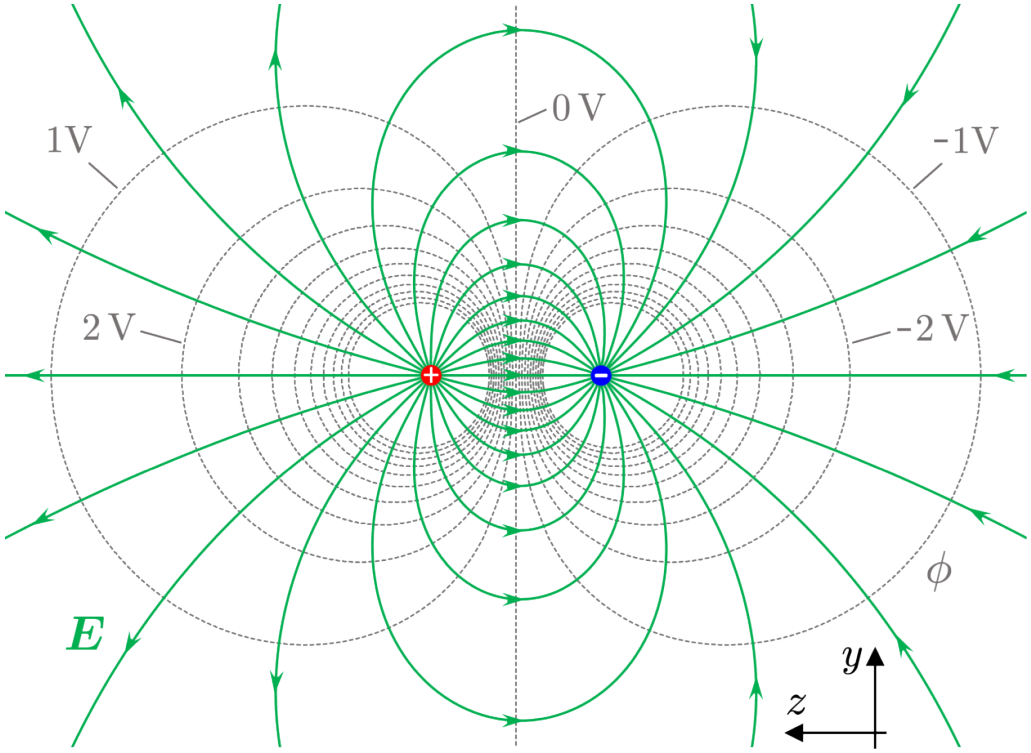



**Fig. 3.6:** Dipole field without dipole approximation: equipotential lines of $\phi$ in 1 V increments from 0 to 10 V, and field lines of $\boldsymbol{E}$. The poles are separated by a distance of $d = 2\,\mathrm{m}$, carrying the equivalent of 10 million positive or negative unit charges $e$.

Thus, the components of $\boldsymbol{E}$ normal and parallel to the dipole axis are:

$$E_\perp = \sqrt{E_x^2 + E_y^2} = \frac{p}{4\pi\epsilon_0}\frac{3\cos\vartheta\sin\vartheta}{r^3} \tag{3.57}$$

and

$$E_\parallel = E_z = \frac{p}{4\pi\epsilon_0}\frac{3\cos^2\vartheta - 1}{r^3} \tag{3.58}$$

Fig. 3.6 shows the exact configuration of the equipotential lines of $\phi$ and the field lines of $\boldsymbol{E}$ in the $yz$-plane, without the dipole approximation. The fields are rotationally symmetric about the dipole axis. The equipotential surfaces of $\phi$ are thus obtained by rotating the equipotential lines of $\phi$ around the dipole axis, in this case, the $z$-axis.
We obtain the field pattern of $\phi$ in the dipole approximation by determining the equipotential lines of $\phi$ in the $yz$-plane using eq. (3.53). To do this, we solve the equation for $y$ and plot pairs of values $(y, z)$ for selected constant values of $\phi$. Fig. 3.7 shows the resulting pattern of the equipotential lines of $\phi$. As can be seen, all equipotential lines converge on the $z$-axis, a situation that is physically impossible, given that each equipotential line correspond to a different potential. In the far field, however, the potential profile agrees very well with the exact solution shown in fig. 3.6.

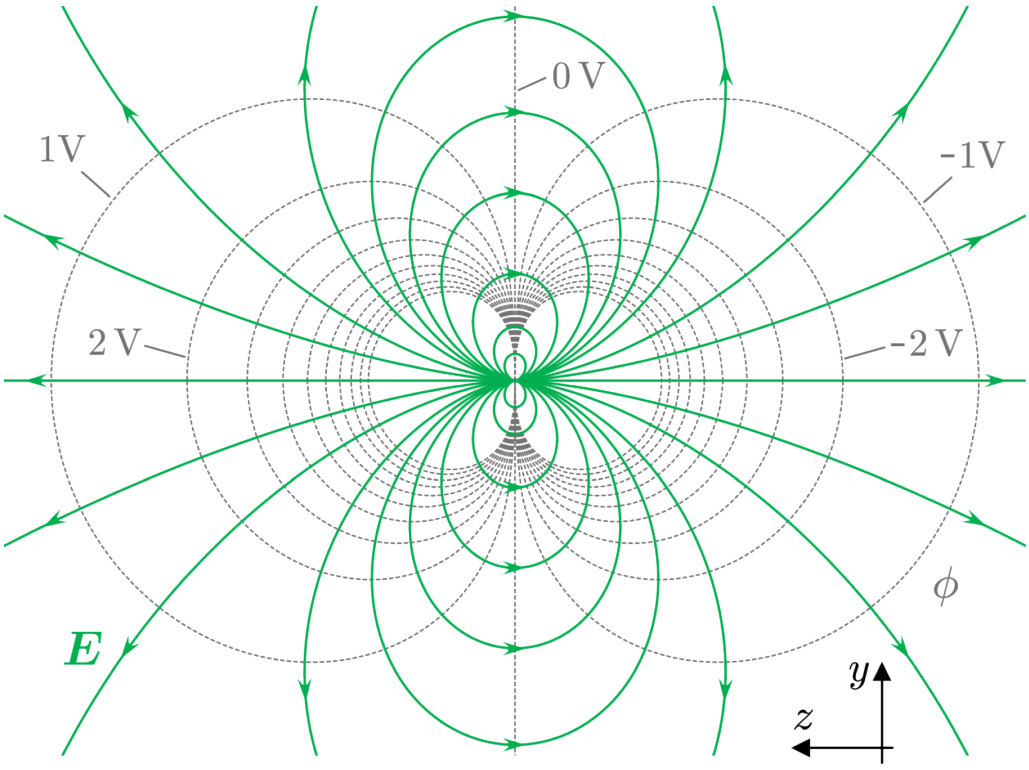



**Fig. 3.7:** Dipole field in dipole approximation: equipotential lines of $\phi$ and field lines of $\boldsymbol{E}$ in the dipole approximation. A comparison with the exact field patterns in fig. 3.6 shows very good agreement in the far field but significant differences in the near field.

To sketch the field lines of $\boldsymbol{E}$ in the dipole approximation, we rewrite eq. (3.55) from Cartesian to polar coordinates. The components of the vector $\boldsymbol{E}$ in polar coordinates we determine by comparing the respective coefficients:

$$\boldsymbol{E} = a_x\hat{\boldsymbol{e}}_i + a_y\hat{\boldsymbol{e}}_j + a_z\hat{\boldsymbol{e}}_k = b_r\hat{\boldsymbol{e}}_r + b_\vartheta\hat{\boldsymbol{e}}_\vartheta + b_\varphi\hat{\boldsymbol{e}}_\varphi$$

For eq. (3.55) we thus obtain in polar coordinates:

$$\boldsymbol{E} = \frac{p}{4\pi\epsilon_0 r^3}\left(2\cos\vartheta\cdot\hat{\boldsymbol{e}}_r + \sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta\right) = E_r\cdot\hat{\boldsymbol{e}}_r + E_\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta \tag{3.59}$$

At any point $\boldsymbol{r}$ on a field line, the vector $\boldsymbol{E}(\boldsymbol{r}) = (E_r, E_\vartheta)$ is the tangent to the field line at that location. The differential components in the directions of $\hat{\boldsymbol{e}}_r$ and $\hat{\boldsymbol{e}}_\vartheta$ are $\mathrm{d}r$ and $r\mathrm{d}\vartheta$. At every point on a field line, the vectors $(E_r, E_\vartheta)$ and $(\mathrm{d}r, r\mathrm{d}\vartheta)$ must point in the same direction. This is only possible if these vectors satisfy the differential equation:

$$\frac{\mathrm{d}r}{r\mathrm{d}\vartheta} = \frac{E_r}{E_\vartheta} \tag{3.60}$$

After rearranging and substituting the values for $E_r$ and $E_\vartheta$ from eq. (3.59), we thus obtain the differential equation

$$\frac{\mathrm{d}r}{r} = \frac{E_r}{E_\vartheta}\mathrm{d}\vartheta = 2\cdot\cot\vartheta\,\mathrm{d}\vartheta \tag{3.61}$$

This equation is not difficult to solve. First, integrate both sides of the equation, that is

$$\int\frac{\mathrm{d}r}{r} = 2\cdot\int\cot\vartheta\,\mathrm{d}\vartheta$$

with the result

$$\ln(r) = 2\cdot\ln(\sin\vartheta) + c = \ln\left(\sin^2\vartheta\right) + c$$

This expression we shape into

$$\ln\left(\frac{r}{\sin^2\vartheta}\right) = c$$

to finally obtain the equation

$$\frac{r}{\sin^2\vartheta} = \mathrm{e}^c = C$$

Thus, as solutions to the differential equation eq. (3.61) for the field lines of $\boldsymbol{E}$ in eq. (3.59), we obtain a family of curves

$$r = C\cdot\sin^2\vartheta \tag{3.62}$$

which differ by the constant parameter $C$. Fig. 3.7 shows a series of these field lines of $\boldsymbol{E}$ for selected values of the parameter $C$. We chose the parameters $C$ such that the resulting field lines illustrate the similarities to and differences from the exact solution shown in fig. 3.6.

What are the differences between the exact solution in Fig. 3.6 and the dipole approximation in Fig. 3.7? For the far-field of $\boldsymbol{E}$ just as for the far-field of $\phi$ we observe good agreement between the exact result in fig. 3.6 and the result of the dipole approximation in fig. 3.7. In the near-field, however, we obtain a completely different result that allows for only one interpretation: the dipole approximation effectively seems to reduce the separation between the two dipole charges to zero. At first glance, this might appears to be a paradoxical result, but it is not. Naturally, the dipole separation $d$ cannot be zero because if that were to happen the positive and negative charges of the dipole would neutralize each other, and the dipole field would vanish. Yet, that does not happen. What occurs instead is something else: the separation of the dipole becomes vanishingly small while the dipole moment $\boldsymbol{p} = q\boldsymbol{d}$ remains constant. Consequently, the dipole approximation yields a dipole field in which the dipole shrinks to the point of vanishing, yet the dipole moment is preserved.

What forces act on an electric dipole, such as the one we have just examined, if it is placed into a homogeneous electric field? To answer this, we consider the sketch in fig. 3.8, which shows an electric dipole in such a homogeneous field. We know the forces acting on the positive charge $q$ and the negative charge $-q$ of the dipole within an electric field $\boldsymbol{E}$. Using eq. (3.2), the total force on the dipole is given by:

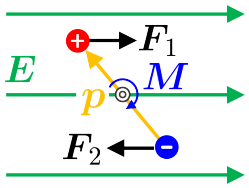


**Fig. 3.8**

$$\boldsymbol{F} = \boldsymbol{F}_1 + \boldsymbol{F}_2 = q\boldsymbol{E} - q\boldsymbol{E} = 0 \tag{3.63}$$

The forces acting on the two electric charges of the dipole thus cancel each other out. Hence, the dipole will move neither in the direction of $\boldsymbol{F}_1$ nor in the direction of $\boldsymbol{F}_2$, nor in any other direction. However, as is clearly evident from the forces $\boldsymbol{F}_1$ and $\boldsymbol{F}_2$ shown in fig. 3.8, these two opposing forces produce a torque $\boldsymbol{M}$ which acts on the dipole, causing it to align itself about its axis of rotation such that the dipole moment $\boldsymbol{p}$ becomes parallel to the electric field vector $\boldsymbol{E}$. If we place the coordinate origin at the axis of rotation and denote the distance between the two charges as $d$, then the positive charge in fig. 3.8 is located at $\boldsymbol{r} = \boldsymbol{d}/2$ and the negative charge at $\boldsymbol{r} = -\boldsymbol{d}/2$. This yields the following expression for the torque $\boldsymbol{M} = \boldsymbol{r} \times \boldsymbol{F}$:

$$\boldsymbol{M} = \frac{\boldsymbol{d}}{2} \times q\boldsymbol{E} + \frac{-\boldsymbol{d}}{2} \times -q\boldsymbol{E} = q\boldsymbol{d} \times \boldsymbol{E} = \boldsymbol{p} \times \boldsymbol{E} \tag{3.64}$$

Now, let us imagine a situation where the electric field is switched off and the dipole finds itself oriented in such a way that its dipole moment $\boldsymbol{p}$ is antiparallel to $\boldsymbol{E}$ when then the uniform electric field is turned on. When we switch on the field, the dipole rotates by 180° until $\boldsymbol{p}$ it lies parallel to $\boldsymbol{E}$. The two states, $\boldsymbol{p}$ antiparallel and parallel to $\boldsymbol{E}$, are associated with two different energy states. For comparison, let us consider a gravity-free space, i.e., a space which Earth's gravity is entirely absent. In such a situation, regardless of where a mass is located in this space, no force will act upon it. If the gravitational field is then switched on, however, this situation changes abruptly. Suddenly, there exist states of higher and lower potential energy, depending on the height at which a mass is located within this space. Evidently, any mass in this space will fall as low as it can in order to minimize its potential energy. An electric dipole in an electric field does exactly the same thing, it minimizes its potential energy: $\boldsymbol{p}$ anti-parallel to $\boldsymbol{E}$ is the state of highest potential energy, and $\boldsymbol{p}$ parallel to $\boldsymbol{E}$ is the state of lowest potential energy for the dipole. When the dipole rotates into the position of lowest potential energy within an electric field. Regardless of the field being homogeneous or inhomogeneous, the dipole's potential energy supplies the energy required for this process. Conversely, if an

external force is applied to rotate the dipole from the position where $\boldsymbol{p}$ is parallel to $\boldsymbol{E}$ into another position, the dipole's potential energy increases by the amount of rotational work performed.

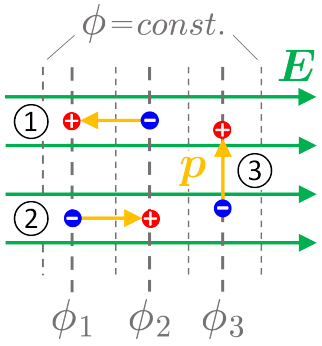


**Fig. 3.9**

Eq. (3.25) describes the change in potential energy of an electric charge moved within an electric field. However, this equation also tells us that a charge $q$ at any point $\boldsymbol{r}$ in an electrostatic field $\boldsymbol{E}(\boldsymbol{r})$ possesses a potential energy $U(\boldsymbol{r}) = q \cdot \phi(\boldsymbol{r})$. Naturally, this applies to both charges of the electric dipole as well. Fig. 3.9 shows an electric dipole in three different orientations relative to the uniform electric field $\boldsymbol{E}$ in which it has been placed. As we know, lines of constant potential $\phi$ run perpendicular to the field lines of $\boldsymbol{E}$. The potential energy of the electric dipole is the sum of the potential energies of the two charges $+q$ and $-q$ at their respective positions in the field $\boldsymbol{E}$. Thus, for the three cases illustrated in fig. 3.9, we obtain the following expression for the potential energy of the dipole:

$$U_{①} = q\phi_1 - q\phi_2 = q(\phi_1 - \phi_2) \tag{3.65a}$$

$$U_{②} = -q\phi_1 + q\phi_2 = -q(\phi_1 - \phi_2) = -U_{①} \tag{3.65b}$$

$$U_{③} = q\phi_3 - q\phi_3 = 0 \tag{3.65c}$$

In case ② in fig. 3.9, where the dipole moment $\boldsymbol{p}$ is parallel to the electric field $\boldsymbol{E}$, the potential energy of the dipole is minimal. The electric field vector always points from higher toward lower potential values. Therefore, $\phi_1 > \phi_2$, and $U_{②}$ is consequently negative. If the dipole moment $\boldsymbol{p}$ is antiparallel to $\boldsymbol{E}$, as in case ①, then the potential energy of the dipole is maximal and it is the negative of $U_{②}$; thus, $U_{①}$ is positive. If the dipole moment $\boldsymbol{p}$ is parallel to a surface $\phi = \text{const}$, as in case ③, then the potential difference is exactly zero and thus $U_{③}$ lies precisely between $U_{①}$ and $U_{②}$. The potential energy of the dipole depends on the change in the electrostatic potential $\phi$ between the positions of the dipole's two charges. In fig. 3.9, we consider the dipole in a homogeneous field $\boldsymbol{E}$. Consequently, the difference in the electrostatic potential $\phi$ is simply the difference between the parallel equipotential lines on which the two charges come to lie in the field. In the more general case of an inhomogeneous electrostatic field, we obtain the potential difference between two points in the scalar field $\phi$ by considering, starting from a given point, the change in potential along the direction of interest. If $\boldsymbol{\xi}$ is this direction, then the change we are interested in is simply the directional derivative

$$\frac{\mathrm{d}\phi}{\mathrm{d}\boldsymbol{\xi}} = \nabla\phi \cdot \boldsymbol{\xi} = \boldsymbol{\xi} \cdot \nabla\phi \tag{3.66}$$

of the scalar field $\phi_0$ in the direction of the vector $\boldsymbol{\xi}$, as we discussed it in section 2.2 (see eq. (2.23)). The direction in which we are interested in the change of $\phi$ is, of course, that of the dipole vector

$$\boldsymbol{d} = \frac{1}{q} \cdot \boldsymbol{p} \tag{3.67}$$

If we replace the potential difference in eq. (3.36) with the directional gradient of $\phi$ in the direction in which the dipole vector $\boldsymbol{d}$ points, we then obtain the potential energy $U$ of the dipole in an electrostatic field $\boldsymbol{E}$:

$$U = q \cdot \nabla\phi \cdot \frac{1}{q} \cdot \boldsymbol{p} = \boldsymbol{p} \cdot \nabla\phi \tag{3.68}$$

Let us examine this equation for the three different cases shown in fig. 3.9. In case $U_{①}$, the dipole moment $\boldsymbol{p}$ points from a location of lower potential $\phi_2$ to a location of higher potential $\phi_1$. Therefore, $\nabla\phi > 0$ and the potential energy $U = U_{max}$ reaches its maximum value. In case $U_{②}$, the dipole moment $\boldsymbol{p}$ points from a location of higher potential $\phi_1$ to a location of lower potential $\phi_2$. Therefore, $\nabla\phi < 0$ and the potential energy $U = U_{min} = -U_{max}$ reaches its minimum value. Finally, in the case of $U_{③}$, the potential is constant in the direction of $\boldsymbol{p}$. Because of that $\nabla\phi = 0$ holds and the potential energy is $U = 0.5 \cdot (U_{max} + U_{min}) = 0$. Using $\boldsymbol{E} = -\nabla\phi$, we can also express the formula in eq. (3.68) for the potential energy $U$ of a dipole with dipole moment $\boldsymbol{p}$ in an electric field $\boldsymbol{E}$ as follows:

$$U = -\boldsymbol{p} \cdot \boldsymbol{E} \tag{3.69}$$

If $\boldsymbol{p}$ and $\boldsymbol{E}$ point in the same direction, the potential energy of the dipole in the field is at its minimum. If $\boldsymbol{p}$ and $\boldsymbol{E}$ point in exactly opposite directions, the potential energy of the dipole in the field is at its maximum. If $\boldsymbol{p}$ and $\boldsymbol{E}$ are perpendicular to each other, the potential energy of the dipole lies exactly between these extreme values. From eq. (3.39), it follows for the force acting on the dipole in a field $\boldsymbol{E}$ (see the corresponding expression in tab. 2.1 with $\boldsymbol{p} = \text{const}$ and $\nabla \times \boldsymbol{E} = 0$):

$$\boldsymbol{F} = -\nabla U = \nabla(\boldsymbol{p} \cdot \boldsymbol{E}) = (\boldsymbol{p} \cdot \nabla)\boldsymbol{E} \tag{3.70}$$

In a homogeneous field, the field strength is constant and in that case $\nabla\boldsymbol{E} = 0$ holds. Therefore, in that case the only torque acting on the dipole arises from the equal and opposite forces exerted on its positive and negative charges. In an inhomogeneous electrostatic field $\boldsymbol{E}$, however, $\nabla\boldsymbol{E} \neq 0$ holds, and the forces acting on the dipole's positive and negative charges which find themselves at locations of different field strength are no

longer equal. Consequently, a dipole in an inhomogeneous field not only experiences a torque but also a force pulling it in the direction of increasing field-strength.

### 3.2.2 The Multipole Expansion of the Potential

The approach of considering the electric potential at a great distance from the potential-generating charges or charge distributions allowed us, in the case of the electric dipole, to obtain a good description of the far-field of this charge configuration in a simple manner. However, the approximation we used to obtain this result is just one example of a more general approach to analyzing potentials and fields of arbitrary charge distributions in the far-field region. The dipole term is merely the linear term in the Taylor expansion of $|\boldsymbol{r}-\boldsymbol{r}'|^{-1}$ in the expression describing the electric potential $\phi$ generated by a charge at position $\boldsymbol{r}'$, which we evaluate at a location $r \gg r'$. In this section, we will examine this so-called multipole expansion of $\phi$ in greater detail.

We begin with a potential $\phi$ generated, as in eq. (3.20), by a charge distribution $\rho(\boldsymbol{r}')$ or, as in eq. (3.28), by a set of discrete electric charges $q_i$ that collectively represent a total charge $Q$. We then place the origin of our coordinate system at the center of charge of the $q_i$ or the charge distribution $\rho$, as sketched in fig. 3.10. These centers of charge $\boldsymbol{r}_{qc}$ for the discrete charges $q_i$ and the continuous charge distribution $\rho(\boldsymbol{r}')$ are given by:

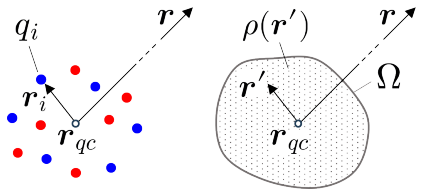



**Fig. 3.10**

$$\boldsymbol{r}_{qc} = \frac{1}{Q}\cdot\sum_{i=1}^{n} q_i\boldsymbol{r}_i \qquad \text{and} \qquad \boldsymbol{r}_{qc} = \frac{1}{Q}\int_{\Omega}\boldsymbol{r}'\rho(\boldsymbol{r}')\,\mathrm{d}^3r' \tag{3.71}$$

Let us consider the discrete charge distribution. For $\boldsymbol{r} \gg \boldsymbol{r}_i$, the electrostatic potential $\phi$ produced by the charges $q_i$ appears, to zeroth order, like the potential of a charge $Q$ located at position $\boldsymbol{r}_{qc}$. Therefore, we can write $\phi$ as:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|} = \frac{1}{4\pi\epsilon_0}\frac{Q}{|\boldsymbol{r}-\boldsymbol{r}_{qc}|} + \mathscr{O}\Big(\frac{1}{r^2}\Big) \tag{3.72}$$

The right-hand side of this equation also applies to continuous charge distributions $\rho$. To determine the higher-order terms in eq. (3.72), we write $1/|\boldsymbol{r}-\boldsymbol{r}_i|$ as:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|} = \frac{1}{\sqrt{r^2-2\boldsymbol{r}\cdot\boldsymbol{r}_i+r_i^2}} = \frac{1}{r}\cdot\frac{1}{\sqrt{1-(2\boldsymbol{r}\cdot\boldsymbol{r}_i-r_i^2)/r^2}} = \frac{1}{r}\cdot\frac{1}{\sqrt{1-\xi}} \tag{3.73}$$

With $r \gg r_i$, we have $\xi \ll 1$, and using the Taylor expansion of the square root for $\xi < 1$, we can write:

$$\frac{1}{\sqrt{1-\xi}} = 1 + \frac{1}{2}\cdot\xi + \frac{3}{8}\cdot\xi^2 + \mathscr{O}(\xi^3)$$

With this Taylor expansion and $\xi = (2\boldsymbol{r}\cdot\boldsymbol{r}_i + r_i^2)/r^2$, eq. (3.73) becomes:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|} = \frac{1}{r} + \frac{1}{2}\cdot\frac{2\boldsymbol{r}\cdot\boldsymbol{r}_i - r_i^2}{r^3} + \frac{3}{8}\cdot\frac{4(\boldsymbol{r}\cdot\boldsymbol{r}_i)^2 - 4(\boldsymbol{r}\cdot\boldsymbol{r}_i)r_i^2 + r_i^4}{r^5} + \mathscr{O}\Big(\frac{1}{r^7}\Big) \tag{3.74}$$

We now group the terms according to powers of $1/r$, explicitly considering only those up to $1/r^3$, and thus obtain:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_i|} = \frac{1}{r} + \frac{\boldsymbol{r}\cdot\boldsymbol{r}_i}{r^3} + \frac{1}{2}\cdot\frac{3(\boldsymbol{r}\cdot\boldsymbol{r}_i)^2 - r^2 r_i^2}{r^5} + \mathscr{O}\Big(\frac{r_i^3}{r^4}\Big) \quad , \quad r_i \ll r \tag{3.75}$$

With that, we can now write the potential $\phi(\boldsymbol{r})$ generated at the position $\boldsymbol{r}$ by $n$ discrete charges $q_i$ located at positions $\boldsymbol{r}_i$ for $r \gg r_i$ as:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|} = \phi_0(\boldsymbol{r}) + \phi_1(\boldsymbol{r}) + \phi_2(\boldsymbol{r}) + \mathscr{O}\Big(\frac{1}{r^4}\Big) \tag{3.76}$$

Here, $\phi_0(\boldsymbol{r})$, $\phi_1(\boldsymbol{r})$, and $\phi_2(\boldsymbol{r})$ are the so-called monopole, dipole, and quadrupole components of the potential $\phi(\boldsymbol{r})$. For these, the following applies:

$$\phi_0(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{1}{r}\cdot\sum_{i=1}^{n} q_i \qquad \text{(Monopol)} \tag{3.77a}$$

$$\phi_1(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{\boldsymbol{r}}{r^3}\cdot\sum_{i=1}^{n} q_i\cdot\boldsymbol{r}_i \qquad \text{(Dipol)} \tag{3.77b}$$

$$\phi_2(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{1}{r^5}\cdot\frac{1}{2}\cdot\sum_{i=1}^{n} q_i\big[3(\boldsymbol{r}\cdot\boldsymbol{r}_i)^2 - r^2 r_i^2\big] \qquad \text{(Quadrupol)} \tag{3.77c}$$

Let us begin with $\phi_0$, the monopole component of the potential. If we again denote the sum of all charges $q_i$ by $Q$, then

$$\phi_0(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\frac{Q}{|\boldsymbol{r}|} \tag{3.78}$$

is nothing other than the Coulomb potential of a point charge $Q$ located at the origin, that is, at the center of charge $\boldsymbol{r}_{qc} = 0$ of the discrete charges $q_i$ (cf. eq. (3.72) for $\boldsymbol{r}_{qc} = 0$). The monopole component of the potential is therefore often referred to as the Coulomb component. The scalar

$$Q = \sum_{i}^{n} q_i \tag{3.79}$$

is the so-called monopole moment of the charge distribution. Next, let us consider the dipole component $\phi_1$ of the potential. Here, we write the sum over the weighted charges

as two sums, one over the positive charges $q_i^+$ and the other over the negative charges $q_i^-$, as follows:[2]

$$\sum_i^n q_i \cdot \boldsymbol{r}_i = \sum_j^l q_j^+ \cdot \boldsymbol{r}_j - \sum_k^m q_k^- \cdot \boldsymbol{r}_k \qquad \text{mit} \qquad l + m = n \tag{3.80}$$

Using eq. (3.71) we can rewrite this equation as:

$$\sum_i^n q_i \cdot \boldsymbol{r}_i = Q \cdot \boldsymbol{r}_{qc} = Q^+ \cdot \boldsymbol{r}_{qc}^+ - Q^- \cdot \boldsymbol{r}_{qc}^- \tag{3.81}$$

Here, $\boldsymbol{r}_{qc}^+$ and $\boldsymbol{r}_{qc}^-$ are the centers of charge of the total positive charge $Q^+$ and the total negative charge $Q^-$, respectively. The center of charge of the total charge $Q = Q^+ - Q^-$ is located at $\boldsymbol{r}_{qc}$. Let us first consider the case where $Q^+ = Q^- = q$ and $\boldsymbol{r}_{qc}^+ \neq \boldsymbol{r}_{qc}^-$; that is, the total positive and negative charges are equal in magnitude, but the positive and negative centers of charge do not coincide. It then follows from eq. (3.81) that:

$$\sum_i^n q_i \cdot \boldsymbol{r}_i = q \cdot (\boldsymbol{r}_{qc}^+ - \boldsymbol{r}_{qc}^-) = q \cdot \boldsymbol{d} = \boldsymbol{p} \tag{3.82}$$

The right-hand side of this equation is the familiar dipole moment, the product of the dipole charge $q$ and the dipole separation vector $\boldsymbol{d} = \boldsymbol{r}_{qc}^+ - \boldsymbol{r}_{qc}^-$, which points from the center of negative charge to the center of positive charge. With eq. (3.82), we have found an expression for the dipole moment of an arbitrary charge distribution:

$$\boldsymbol{p} = \sum_i^n q_i \cdot \boldsymbol{r}_i \tag{3.83}$$

For a continuous charge distribution $\rho(\boldsymbol{r})$ which is enclosed within a volume $\Omega$, we simply need to replace the sum over the weighted discrete charges with an integral over the weighted charge distribution:

$$\boldsymbol{p} = \int_\Omega \rho(\boldsymbol{r}') \cdot \boldsymbol{r}' \, \mathrm{d}^3 r' \tag{3.84}$$

We thus know that $\phi_1(\boldsymbol{r})$ indeed describes the dipole component of the potential $\phi(\boldsymbol{r})$ of an arbitrary charge distribution. Using eq. (3.83), we can write $\phi_1(\boldsymbol{r})$ in the form already familiar to us:

$$\phi_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{\boldsymbol{r} \cdot \boldsymbol{p}}{r^3} \tag{3.85}$$

Whenever $\boldsymbol{r}_{qc}^+ \neq \boldsymbol{r}_{qc}^-$, that is, whenever the centers of positive and negative charges of a charge distribution do not coincide, the distribution possesses a dipole moment. If $\boldsymbol{r}_{qc}^+ = \boldsymbol{r}_{qc}^-$, the respective charge distribution has no dipole moment and represents either

[2] Negative charges are always explicitly written as $-q$ or $-Q$; $q$ and $Q$ themselves are always positive quantities.

a net positive total charge ($Q^+ > Q^-$), a net negative total charge ($Q^+ < Q^-$), or a zero net charge, located at the center of charge $\boldsymbol{r}_{qc}$. Up to this point, we have assumed that the center of charge is located at the coordinate origin. But what happens if this is not the case? To determine this, we shift the origin of the coordinate system by a constant vector $\boldsymbol{a}$, which is equivalent to shifting the position $\boldsymbol{r}_i$ of each individual charge $q_i$ by $-\boldsymbol{a}$, that is:

$$\boldsymbol{p} = \sum_i^n q_i \cdot \boldsymbol{r}_i \longrightarrow \sum_i^n q_i \cdot (\boldsymbol{r}_i - \boldsymbol{a}) = \boldsymbol{p} - Q \cdot \boldsymbol{a} = \boldsymbol{p}' \tag{3.86}$$

If $Q \neq 0$, one can always ensure, by choosing $\boldsymbol{a} = \boldsymbol{p}/Q$, that $\boldsymbol{p}' = 0$ relative to the new coordinate origin. However, if $Q = 0$, the dipole moment $\boldsymbol{p}$ is an intrinsic property of the system under consideration, such as is the case for a polar diatomic molecule like HCl or CO, and remains constant upon a shift of the coordinate system. The third term in the multipole expansion is the quadrupole component $\phi_2$. Unlike the monopole and dipole components, the coordinates $\boldsymbol{r}_i$ of the discrete charges $q_i$ and the point $\boldsymbol{r}$ at which the potential is to be evaluated are not separated in the expression for the quadrupole component given in eq. (3.77c); instead, there are mixed terms. We therefore attempt to separate the point $\boldsymbol{r}$ from the coordinates $\boldsymbol{r}_i$ in order to obtain, as with the monopole and dipole moments, an expression for the quadrupole moment that depends solely on the positions $\boldsymbol{r}_i$ of the charges $q_i$ and on the charges themselves. Since we are already using $i$ as the counting index for the charge distribution, we now use superscripts $k$ and $l$ for the coordinate components of the vectors $\boldsymbol{r}$ and $\boldsymbol{r}_i$. We can thus write the vector product $\boldsymbol{r} \cdot \boldsymbol{r}_i$ and the scalar product $(\boldsymbol{r} \cdot \boldsymbol{r}_i) \cdot (\boldsymbol{r} \cdot \boldsymbol{r}_i)$ as:

$$\boldsymbol{r} \cdot \boldsymbol{r}_i = \sum_{k=1}^3 r^k \cdot r_i^k \tag{3.87a}$$

and

$$(\boldsymbol{r} \cdot \boldsymbol{r}_i)^2 = \sum_{k=1}^3 r^k \cdot r_i^k \cdot \sum_{l=1}^3 r^l \cdot r_i^l = \sum_{k,l=1}^3 r^k \cdot r_i^k \cdot r^l \cdot r_i^l \tag{3.87b}$$

If we now rearrange the terms $r^k, r^l, r_i^k, r_i^l$, insert the charges $q_i$, and sum over the coordinates of the charges while swapping the order of summation over $i$ and $k, l$, we obtain with eq. (3.87b) for the first expression in the summand of the quadrupole term in eq. (3.77c):

$$\sum_i q_i \cdot (\boldsymbol{r} \cdot \boldsymbol{r}_i)^2 = \sum_{k,l=1}^3 r^k \cdot \Big( \sum_i q_i \cdot r_i^k \cdot r_i^l \Big) \cdot r^l \tag{3.88}$$

Using eq. (3.87a), we can simply write the second term in the sum expression of the quadrupole component, $r^2 \cdot r_i^2$, in eq. (3.77c) as follows:

$$\sum_i q_i \cdot r^2 \cdot r_i^2 \;=\; \sum_i q_i \cdot \Big( \sum_{k,l=1}^{3} r^k \cdot r^l \cdot \delta_{rl} \Big) \cdot r_i^2 \;=\; \sum_{k,l=1}^{3} r^k \cdot \Big( \sum_i q_i \cdot r_i^2 \cdot \delta_{kl} \Big) \cdot r^l \tag{3.89}$$

Substituting eq. (3.88) and eq. (3.89) into eq. (3.77c) yields a new equation for the quadrupole component:

$$\phi_2(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{r^5} \cdot \sum_{k,l=1}^{3} r^k \cdot \Big( \frac{1}{2} \sum_i q_i \big[ 3 \cdot r_i^k \cdot r_i^l - r_i^2 \cdot \delta_{kl} \big] \Big) \cdot r^l \tag{3.90}$$

The expression in parentheses depends only on the discrete charges $q_i$ and their respective coordinates $\boldsymbol{r}_i$. The monopole moment is a rank-0 tensor, i.e., the scalar $Q$, the dipole moment is a rank-1 tensor, i.e., the vector $\boldsymbol{p}$, and the quadrupole moment of a charge distribution is a rank-2 tensor, i.e., a matrix $\underline{\underline{\boldsymbol{Q}}}$ with the components:

$$Q_{kl} \;=\; \frac{1}{2} \sum_i q_i \big[ 3 \cdot r_i^k \cdot r_i^l - r_i^2 \cdot \delta_{kl} \big] \tag{3.91}$$

As can easily be verified, for a continuous charge distribution $\rho(\boldsymbol{r}')$, the quadrupole tensor $\underline{\underline{\boldsymbol{Q}}}$, expressed in terms of the dyadic product as we known it from eq. (2.6), is given by

$$\underline{\underline{\boldsymbol{Q}}} \;=\; \frac{1}{2} \cdot \int \rho(\boldsymbol{r}') \big[ 3 \cdot \boldsymbol{r}' \cdot \boldsymbol{r}'^{\,T} - \boldsymbol{r}'^{\,2} \cdot \mathbf{1}_3 \big] \, \mathrm{d}^3 \boldsymbol{r}' \tag{3.92}$$

where $\mathbf{1}_3$ is the $3 \times 3$-unit matrix. With eq. (3.90) and eq. (3.91) we get for the quadrupole component the expression:

$$\phi_2(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{r^5} \cdot \sum_{k,l=1}^{3} r^k \cdot Q_{kl} \cdot r^l \;=\; \frac{1}{4\pi\epsilon_0} \cdot \frac{\boldsymbol{r} \cdot \underline{\underline{\boldsymbol{Q}}} \cdot \boldsymbol{r}}{r^5} \tag{3.93}$$

Here, $\underline{\underline{\boldsymbol{Q}}}$ can represent the quadrupole moment of either a discrete or a continuous charge distribution. As an example of a discrete charge distribution possessing only a quadrupole moment, but neither a monopole nor a dipole moment, we consider the four point charges shown in fig. 3.11: two positive and two negative discrete charges located at the axial intercepts of an ellipse centered at the coordinate origin.

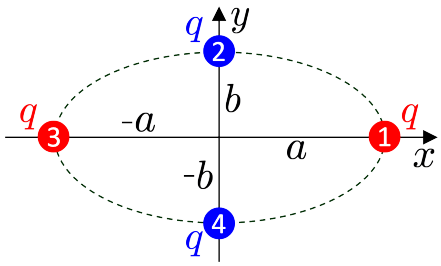



**Fig. 3.11**

Let the length of the ellipse's major axis be $2a$ and that of the minor axis be $2b$. Since the total charge of the four charges is zero, the monopole term of the potential for this charge distribution vanishes; furthermore, because the centers of charge for the positive and negative charges coincide, the dipole moment also vanishes. Consequently, the first non-vanishing moment of this charge distribution must be the quadrupole moment. The

coordinates $\boldsymbol{r}_i$ of these four point charges are:

$$\boldsymbol{r}_1 = a\cdot\hat{\boldsymbol{\imath}} \quad , \quad \boldsymbol{r}_2 = b\cdot\hat{\boldsymbol{\jmath}} \quad , \quad \boldsymbol{r}_3 = -a\cdot\hat{\boldsymbol{\imath}} \quad , \quad \boldsymbol{r}_4 = -b\cdot\hat{\boldsymbol{\jmath}}$$

As is readily apparent, given these four coordinates of the point charges, $\boldsymbol{r}_i^k\cdot\boldsymbol{r}_i^l = 0$ holds for $k \neq l$, and therefore all $Q_{kl}$ with $k \neq l$ vanish. Thus, the quadrupole tensor of this charge arrangement has non-zero values only along the diagonal. When evaluating the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$, we recall that $k, l = 1, 2, 3$ correspond to $x, y, z$. According to eq. (3.91), the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ are:

$$Q_{xx} = \frac{1}{2}\sum_i q_i\left[3x_i^2 - r_i^2\right] = \frac{1}{2}\sum_i q_i\left[2x_i^2 - y_i^2 - z_i^2\right] \tag{3.94a}$$

$$Q_{yy} = \frac{1}{2}\sum_i q_i\left[3y_i^2 - r_i^2\right] = \frac{1}{2}\sum_i q_i\left[2y_i^2 - x_i^2 - z_i^2\right] \tag{3.94b}$$

$$Q_{zz} = \frac{1}{2}\sum_i q_i\left[3z_i^2 - r_i^2\right] = \frac{1}{2}\sum_i q_i\left[2z_i^2 - x_i^2 - y_i^2\right] \tag{3.94c}$$

If we substitute the coordinates and charge values of the four point charges from fig. 3.11 into these equations, we obtain:

$$Q_{xx} = q(2a^2 + b^2) \quad , \quad Q_{yy} = -q(a^2 + 2b^2) \quad , \quad Q_{zz} = q(b^2 - a^2)$$

The quadrupole tensor of the charge distribution in fig. 3.11 is thus:

$$\underline{\underline{\boldsymbol{Q}}} = q\cdot\begin{bmatrix} 2a^2 + b^2 & 0 & 0 \\ 0 & -a^2 - 2b^2 & 0 \\ 0 & 0 & b^2 - a^2 \end{bmatrix} \tag{3.95}$$

We now know what the quadrupole tensor looks like in our example. But what is the meaning of the values on its diagonal, what do they tell us? For a spherically symmetric charge distribution, the following always holds:

$$\sum_i q_i x_i^2 = \sum_i q_i y_i^2 = \sum_i q_i z_i^2$$

A look at the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ as given in eq. (3.94a), eq. (3.94b), and eq. (3.94c) reveals that, for a spherically symmetric charge distribution, these diagonal elements are always zero. The implication of this is simply that the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ describe deviations from a spherically symmetric charge distribution. In our example, the charge distribution shown in Fig. 3.11, all off-diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ were zero. However, if the $Q_{kl}$ terms with $k \neq l$ in eq. (3.91) do not vanish, this can be achieved via a principal axis transformation. This is always possible for a square matrix like $\underline{\underline{\boldsymbol{Q}}}$. In our example, this was not necessary, as we were already working in the principal axis system, where the

axes of the ellipse coincide with the coordinate axes. Once we are in the principal axis system, the diagonal elements describe the deviation of the charge configuration under consideration from a spherically symmetric distribution.

We can directly apply the results of this example of a discrete charge distribution with only a quadrupole moment to a continuous charge distribution in the form of an ellipsoid where the axes of the ellipsoid coincide with the coordinate axes. We are thus back in the principal axis system, and all off-diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ in eq. (3.92) vanish. Consequently, only the diagonal elements of $\boldsymbol{r}' \cdot \boldsymbol{r}'^{T}$ (see eq. (2.6)) remain in eq. (3.92), and for the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ we obtain:

$$Q_{xx} = \frac{1}{2} \cdot \int \rho(\boldsymbol{r}') \left[3x'^2 - r'^2\right] \mathrm{d}^3 r' = \frac{1}{2} \cdot \int \rho(\boldsymbol{r}') \left[2x'^2 - y'^2 - z'^2\right] \mathrm{d}^3 r' \tag{3.96a}$$

$$Q_{yy} = \frac{1}{2} \cdot \int \rho(\boldsymbol{r}') \left[3y'^2 - r'^2\right] \mathrm{d}^3 r' = \frac{1}{2} \cdot \int \rho(\boldsymbol{r}') \left[2y'^2 - x'^2 - z'^2\right] \mathrm{d}^3 r' \tag{3.96b}$$

$$Q_{zz} = \frac{1}{2} \cdot \int \rho(\boldsymbol{r}') \left[3z'^2 - r'^2\right] \mathrm{d}^3 r' = \frac{1}{2} \cdot \int \rho(\boldsymbol{r}') \left[2z'^2 - x'^2 - y'^2\right] \mathrm{d}^3 r' \tag{3.96c}$$

It is not surprising that these equations correspond to those we obtained for the discrete charge distribution in eq. (3.94a), eq. (3.94b), and eq. (3.94c); moreover, we can derive them directly from the latter by replacing the sums over the charges with integrals over the charge distributions. Let us now consider the simple case where the charge density within the ellipsoid sketched in fig. 3.12 is constant. If the total charge of the ellipsoid is $Q$, then the charge density $\rho$ is equal to $Q$ divided by the volume of the ellipsoid, that is, $\rho = Q \cdot (3/4\pi) \cdot (1/abc)$. The diagonal elements for this charge distribution can easily be calculated by integrating over the volume of the ellipsoid. For $Q_{xx}$, for example, one obtains:

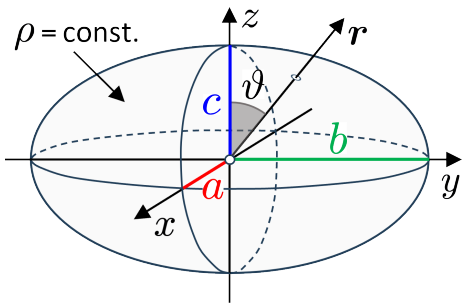



**Fig. 3.12**

$$Q_{xx} = \frac{1}{2} \cdot \frac{3 \cdot Q}{4\pi \cdot abc} \cdot \int_{-a}^{a} \int_{-b}^{b} \int_{-c}^{c} \left[3x'^2 - r'^2\right] \mathrm{d}^3 r' = \frac{Q}{\pi} \cdot (2a^2 - b^2 - c^2)$$

The results for $Q_{yy}$ and $Q_{zz}$ are correspondingly:

$$Q_{yy} = \frac{Q}{\pi} \cdot (2b^2 - a^2 - c^2) \qquad \text{und} \qquad Q_{yy} = \frac{Q}{\pi} \cdot (2c^2 - a^2 - b^2)$$

We now simplify matters by assuming that $b = a$, so that the ellipsoid in fig. 3.12 has the shape of either a cigar ($a > c$) or a disk ($a < c$). In this case, the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ are given by:

$$Q_{xx} = \frac{Q}{\pi}(a^2 - c^2) \quad , \quad Q_{yy} = \frac{Q}{\pi}(a^2 - c^2) \quad , \quad Q_{zz} = -\frac{2Q}{\pi}(a^2 - c^2)$$

Thus, we find that $Q_{xx} = Q_{yy} = -Q_{zz}/2$. Substituting these values for the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ into eq. (3.93), we obtain the following expression for the quadrupole potential of the ellipsoid with constant charge density $\rho$ and total charge $Q$:

$$\phi_2(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q(a^2 - c^2)}{\pi} \cdot \frac{x^2 + y^2 - 2z^2}{r^5} \tag{3.97}$$

Using $\cos\vartheta = z/r$, we can rewrite this equation as:

$$\phi_2(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q(c^2 - a^2)}{\pi} \cdot \frac{3\cos^2\vartheta - 1}{r^3} = \frac{Q_0}{4\pi\epsilon_0} \cdot \frac{3\cos^2\vartheta - 1}{r^3} \tag{3.98}$$

$Q_0$ is the quadrupole moment of our ellipsoid in fig. 3.12, reduced to a single value, for which the following applies:

$$Q_0 = \frac{3Q}{\pi} \cdot (c^2 - a^2) \tag{3.99}$$

$Q_0$ can take positive or negative values, for both positive and negative total charges $Q$, depending on whether $c > a$ or $c < a$. When $c = a$, our ellipsoid becomes a sphere and the quadrupole moment $Q_0$ vanishes, as is required for a spherically symmetric charge distribution. Axially symmetric charge distributions, such as those of ellipsoids, occur frequently in nature. For many atoms, the charge distribution within the atomic nucleus is not spherically symmetric but rather axially symmetric, meaning, it is deformed. Therefore, such atomic nuclei possess an electric quadrupole moment of the kind we have just discussed. To close out this example, let us consider what happens if $c = 0$. In this case, our three-dimensional charge distribution is being reduced to a circular two-dimensional charge distribution of radius $a$ in the $xy$-plane. This two-dimensional charge distribution, however, still possesses a quadrupole moment with the same angular dependence. The reason for that is the rotational invariance of these charge distributions with respect to the $z$-axis. For such charge distributions, the following always holds:

$$\sum_i q_i x_i^2 = \sum_i q_i y_i^2 \quad \text{or} \quad \int \rho(\boldsymbol{r}') x'^2 \, \mathrm{d}^3 r' = \int \rho(\boldsymbol{r}') y'^2 \, \mathrm{d}^3 r'$$

In the next chapter, for instance, we will encounter a ring-shaped charge distribution. Let us assume that the charged ring consists of discrete charges distributed uniformly along a circular ring in the $xy$-plane. In that case, due to rotational invariance with

respect to the $z$-axis, the diagonal elements of $\underline{\underline{\boldsymbol{Q}}}$ satisfy the following:

$$Q_{xx} = \frac{1}{2}\sum_i q_i\left[x_i^2 - z_i^2\right] = Q_{yy} \qquad \text{and} \qquad Q_{zz} = \frac{1}{2}\sum_i q_i\left[2z_i^2 - x_i^2\right]$$

With $Q_{xx} = Q_{yy} = -Q_{zz}/2$ and $Q_0 = Q_{zz}/2$, we can express the quadrupole tensor, analogous to the uniformly charged ellipsoid in fig. 3.12, in terms of a single value $Q_0$, yielding the following result for the quadrupole term of the charged ring:

$$\phi_2(\boldsymbol{r}) = \frac{Q_0}{4\pi\epsilon_0}\cdot\frac{3\cos^2\vartheta - 1}{r^3} \tag{3.100}$$

# 4. Electrostatic Fields

In the second part of the last chapter, we examined the fields of static electric charges in the specific form of bound charges, such as the electric dipole, as they can be found in many molecules or solids. Here, we now will address more generally the electrostatic fields of another kind of bound charge, i.e., macroscopic bodies carrying electric charges bound to them. In doing that, we assume that an electric charge applied to such a macroscopic body distributes itself in such a way that a stable equilibrium is established. Once equilibrium is achieved, we are dealing with macroscopic static charge distributions to which we can apply the laws of electrostatics. In addition to determining the electrostatic potential generated by such charged bodies and understanding the resulting configuration of the $\boldsymbol{E}$-field lines, we will also be interested to learn about the potential energy of such charge-bearing bodies. As we saw in the last section, it follows from the fundamental equations of electrostatics that the electrostatic potential $\phi$ satisfies a Poisson equation:

$$\Delta\phi(\boldsymbol{r}) \;=\; -\frac{1}{\epsilon_0}\rho(\boldsymbol{r}) \tag{4.1}$$

The charge distribution $\rho(\boldsymbol{r})$ can be composed of discrete point charges, a continuous charge distribution, or both. We therefore write in general terms:

$$\rho(\boldsymbol{r}) \;=\; \sum_{i=1}^{N} q_i \delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_i) + \widetilde{\rho}(\boldsymbol{r}) \tag{4.2}$$

Apart from a prefactor arising here due to our choice of the proportionality constant $k_e$ in eq. (3.1), we are already familiar with the solution to eq. (4.1) for such a charge distribution from section 2.3.2 (see eq. (2.73)). It consists of a component arising from discrete point charges $q_i$ (eq. (3.28)) and a component originating from the continuous charge distribution $\widetilde{\rho}(\boldsymbol{r})$ (eq. (3.20)):

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{N}\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|} + \frac{1}{4\pi\epsilon_0}\int\frac{\widetilde{\rho}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{4.3}$$

Put into words, this equation simply tells us that a discrete electric charge $q_i$ located at position $\boldsymbol{r}_i$ contributes to the electrostatic potential at the location with the position vector $\boldsymbol{r}$ the potential

$$\phi_i(r) = \frac{1}{4\pi\epsilon_0}\frac{q_i}{|\boldsymbol{r}-\boldsymbol{r}_i|} \tag{4.4}$$

and the contribution of a continuous charge distribution $\mathrm{d}q = \widetilde{\rho}(\boldsymbol{r}')\,\mathrm{d}^3r'$ at the location $\boldsymbol{r}'$ to the electrostatic potential at $\boldsymbol{r}$ is

$$\mathrm{d}\phi(r) = \frac{1}{4\pi\epsilon_0}\frac{\widetilde{\rho}(\boldsymbol{r})'}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{4.5}$$

Given a charge distribution $\rho(\boldsymbol{r})$ according to eq. (4.3), we can thus determine the electrostatic potential $\phi(\boldsymbol{r})$ by summing up the individual $\phi_i(\boldsymbol{r})$ contributions and integrating $\mathrm{d}\phi(r)$ over the volume containing the charge distribution. Once we know $\phi(\boldsymbol{r})$, we can calculate the electric field $\boldsymbol{E}(\boldsymbol{r})$ by simply taking the gradient of $\phi(\boldsymbol{r})$ according to eq. (3.19). Solving Poisson's equation in eq. (3.21) is one way to determine the electric potential and thus the electric field of a charge distribution. Here, however, we will examine examples where Gauss's law from eq. (3.15) or eq. (3.16) and/or the symmetry of the problem provide a simpler solution path.

## 4.1 Uniformly Charged Solid or Hollow Sphere

A simple application of Gauss's law is the determination of the electric field of a uniformly charged sphere. Specifically, we consider a sphere made of non-conducting material in which the positive charges, summing to a total charge of $+Q$, are uniformly distributed throughout the sphere's volume. As sketched in fig. 4.1, the center of the sphere is located at the origin of the coordinate system. Due to the radial symmetry of this configuration, the electric field must appear the same in all directions $\boldsymbol{r}$. It can therefore depend only on the magnitude of the distance $|\boldsymbol{r}|$ from the coordinate origin, and not on the direction of the vector $\boldsymbol{r}$, that is, $\boldsymbol{E}(\boldsymbol{r}) \to \boldsymbol{E}(r)$. By convention, electric field lines always point away from positive electric charges and toward negative electric charges. For the positive electric charge $Q$ in fig. 4.1, the electric field can therefore only be directed radially outward due to symmetry. As our Gaussian surface $S$ in fig. 4.1, we choose a spherical surface of radius $r > R_0$ that is concentric with the

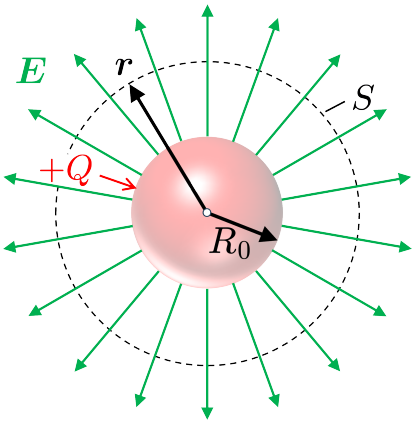



**Fig. 4.1**

uniformly charged sphere of radius $R_0$ through which the field lines originating from the uniformly charged sphere pass through perpendicularly. Applying Gauss's law in this manner, we obtain for the electric flux through the spherical shell $S$ of radius $r$ the expression:

$$\int_S \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{\sigma} \;=\; E(r)\cdot 4\pi r^2 \;=\; \frac{Q}{\epsilon_0} \tag{4.6}$$

Thus, the field strength $E(r)$ at a distance $r > R_0$ from the coordinate origin is:

$$E(r) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{r^2} \tag{4.7}$$

We can choose the radius $r > R_0$ of our Gaussian surface arbitrarily, and given the symmetry of the electric field outside the uniformly charged, non-conducting solid sphere, we can generally write:

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q}{|\boldsymbol{r}|^2}\cdot\hat{\boldsymbol{r}} \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q\cdot\boldsymbol{r}}{|\boldsymbol{r}|^3} \qquad \text{if} \qquad r > R_0 \tag{4.8}$$

However, this is precisely the field that a point charge $Q$ located at the coordinate origin would produce. Thus, we also know the electric potential outside the uniformly charged sphere:

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q}{|\boldsymbol{r}|} \qquad \text{if} \qquad r > R_0 \tag{4.9}$$

Thus, for $r > R_0$, the electric potential and electric field of a uniformly charged sphere of radius $R_0$ behave as if the sphere's entire charge were located at its center. We are already familiar with this result from volume 1, where we encountered it during the discussion of the electric field of a positively charged hollow sphere. There, we also found that the electric field inside a charged hollow sphere vanishes. But what does the field look like inside our non-conducting charged sphere in fig. 4.1? Obviously, for reasons of symmetry, the electric field inside the uniformly charged sphere must also point radially outward. To determine the electric field strength for $r < R_0$, we place our concentric Gaussian spherical surface $S$ inside the uniformly charged sphere. However, this surface now encloses only the fraction $r^3/R_0^3$ of the total charge $Q$ of the sphere, and thus, from Gauss's law, we obtain:

$$\int_S \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{\sigma} \;=\; E(r)\cdot 4\pi r^2 \;=\; \frac{Q}{\epsilon_0}\frac{r^3}{R_0^3} \tag{4.10}$$

Thus, for the electric field strength at a distance $r < R_0$ from the origin inside the uniformly charged, non-conducting sphere, we obtain:

$$E(r) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q\cdot r}{R_0^3} \tag{4.11}$$

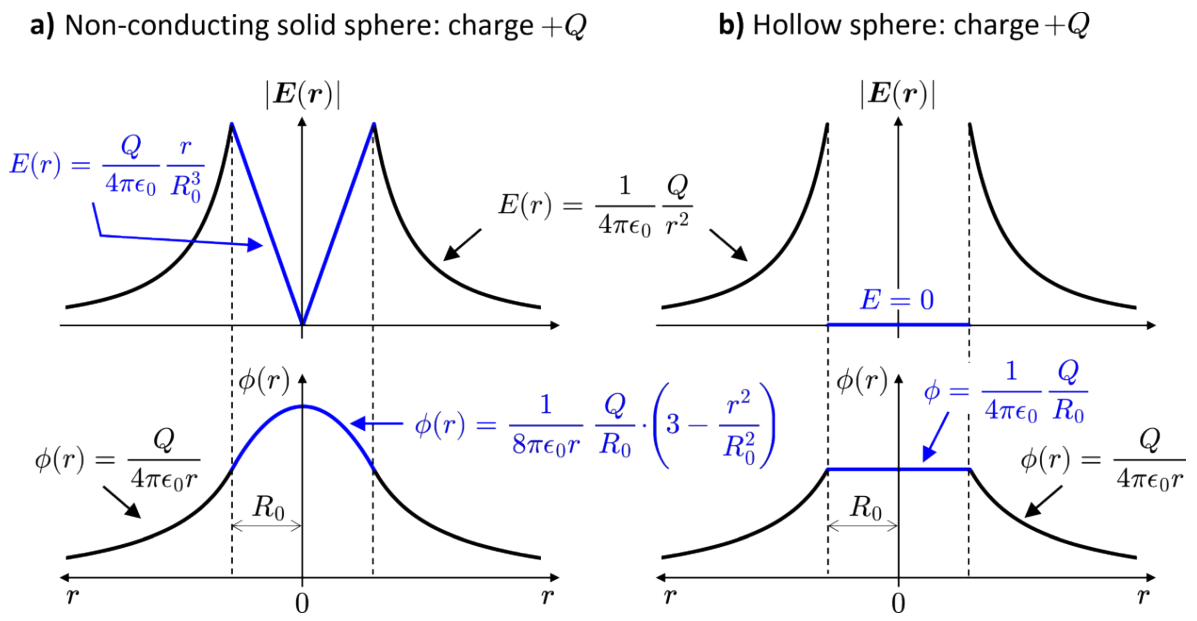



**Fig. 4.2:** Electric field strength $|\boldsymbol{E}(\boldsymbol{r})|$ and electric potential $\phi(r)$ outside and inside a uniformly charged, non-conducting solid sphere (a) and outside and inside a uniformly charged hollow sphere (b), each of radius $R_0$.

Inside the uniformly charged sphere, we can again choose the radius of the Gaussian surface $S$ arbitrarily. Given the symmetry of the electric field within the sphere, the following applies:

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q}{R_0^3}|\boldsymbol{r}|\cdot\hat{\boldsymbol{r}} \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q\cdot\boldsymbol{r}}{R_0^3} \qquad \text{if} \qquad r \le R_0 \tag{4.12}$$

To obtain the corresponding electrostatic potential, we integrate eq. (3.19) inside the uniformly charged sphere between two points, specifically, between $r < R_0$ and $R_0$. We thus make use of eq. (3.33), and since the electric field here depends only on the magnitude of the distance, the calculation is straightforward:

$$\int_r^{R_0} \mathbf{E}(r')\,\mathrm{d}\mathbf{r}' \;=\; -\int_r^{R_0} \mathrm{d}\phi \;=\; \phi(r) - \phi(R_0) \tag{4.13}$$

Substituting eq. (4.12) into the left-hand side and integrating, we obtain:

$$\phi(r) - \phi(R_0) \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q}{R_0^3}\cdot\left(\frac{R_0^2}{2} - \frac{r^2}{2}\right) \tag{4.14}$$

We ensure the continuity of the electrostatic potential $\phi$ at the spherical boundary $r = R_0$ by substituting the corresponding value from eq. (4.9) for $\phi(R_0)$. This yields the following expression for the electrostatic potential inside the sphere:

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{8\pi\epsilon_0}\frac{Q}{R_0}\left(3 - \frac{|\boldsymbol{r}|^2}{R_0^2}\right) \qquad \text{if} \qquad r \le R_0 \tag{4.15}$$

Now we return to the example of the uniformly charged hollow sphere for which we previously derived the electric field and the electrostatic potential in volume 1 in a different context. Outside the hollow sphere, the electric field and the electrostatic potential are identical to what we found here for the uniformly charged solid sphere. Inside the hollow sphere, however, the situation is quite different, since there is no charge there. Because there is no charge inside a hollow sphere, Gauss's law tells us immediately that the electric field there must vanish, and therefore the electrostatic potential inside a uniformly charged hollow sphere is constant. Fig. 4.2 graphically summarizes the results for a uniformly charged solid sphere and a hollow sphere of the same radius $R_0$. Outside $R_0$, the situation is identical in both cases, we obtain the same results for the electric field and the electrostatic potential. Inside the two spheres, however, the situation is completely different.

Next, we ask what changes if, instead of an insulator with an embedded uniform charge distribution, we consider a charged solid metal sphere. The difference is of course that metals are electrical conductors and therefore, other than in insulators, electrons can move freely in metals. If an electric field were to exist inside a solid metallic sphere, similar to the one sketched for the charged insulating sphere in fig. 4.2a, then the electrons would move toward the surface of the sphere in accordance with eq. (3.3), resulting in a situation like the one we encountered with the charged hollow sphere in fig. 4.2b: the field inside the metallic sphere vanishes. Since the surface of the metal sphere is an equipotential surface of $\phi(\boldsymbol{r})$, the tangential component of the electric field vanishes there, and the electric field $\boldsymbol{E}(\boldsymbol{r})$ outside the metallic sphere has only a radial component. Thus, for $r > R_0$, it resembles the fields in fig. 4.2a and fig. 4.2b.

With that we have established that the electrostatic potential at the surface of the sphere, $\phi(R_0)$, is identical for all three cases discussed:

$$\phi(R_0) \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q}{R_0} \tag{4.16}$$

If an additional small charge $\mathrm{d}q$ is brought from infinity onto the surface of the sphere, then in all three cases — a solid sphere of non-conducting material, a hollow sphere, and a solid metal sphere — the same amount of work must be performed, namely:

$$\mathrm{d}W_{R_0} = [\phi(R_0) - \phi(\infty)]\,\mathrm{d}q \tag{4.17}$$

By definition, $\phi(\infty) = 0$, and thus:

$$\mathrm{d}W_{R_0} = \phi(R_0)\,\mathrm{d}q \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q\cdot\,\mathrm{d}q}{R_0} \tag{4.18}$$

We must now distinguish between cases. For the charged solid sphere made of insulating material (Case I), the charges are distributed uniformly throughout the interior of the sphere. However, the situation is different for the charged hollow sphere and the solid metal sphere (Case II), where the electric field vanishes in the interior and all charges reside within an infinitesimal spherical shell or on the surface of the metal sphere. We first consider Case II: the hollow metal sphere and the solid metal sphere. Equation eq. (4.18) tells us how we can easily calculate the work $W$ required to bring the total charge $Q$ from infinity onto the surface of an uncharged sphere of radius $R_0$:

$$W_{R_0}^{\mathrm{II}} = \frac{1}{4\pi\epsilon_0 R_0}\int_0^Q q\,\mathrm{d}q = \frac{Q^2}{8\pi\epsilon_0 R_0} \tag{4.19}$$

We should also obtain this result for the hollow metal sphere and the solid metal sphere using eq. (3.50). With the volume element in spherical coordinates $\mathrm{d}V = 4\pi r^2\mathrm{d}r$, the integration of eq. (3.50) yields the electrostatic field energy $U$:

$$U^{\mathrm{II}} = \frac{\epsilon_0}{2}\int_{R_0}^{\infty} E^2(r)\cdot 4\pi r^2\,\mathrm{d}r \tag{4.20}$$

Since the electric field inside a hollow metal sphere and a solid metal sphere vanishes, the integration runs from $r \geq R_0$ to $r = \infty$. Substituting $E(r)$ for $r \geq R_0$ in the case of the hollow metal sphere and the solid metal sphere yields:

$$U^{\mathrm{II}} = \frac{\epsilon_0}{2}\int_{R_0}^{\infty}\left(\frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\right)^2\cdot 4\pi r^2\,\mathrm{d}r = \frac{Q^2}{8\pi\epsilon_0}\int_{R_0}^{\infty}\frac{1}{r^2}\,\mathrm{d}r \tag{4.21}$$

The integral of $r^{-2}$ is easy to evaluate, yielding the result:

$$U^{\mathrm{II}} = \frac{Q^2}{8\pi\epsilon_0 R_0} \tag{4.22}$$

This is not surprising, as we had expected that using eq. (3.50) would yield the same result we obtained in eq. (4.19): the work $W_{R_0}^{\mathrm{II}}$ required to bring a total charge $Q$ from infinity onto the surface of an uncharged hollow sphere or an uncharged solid metal sphere is equal to the electrostatic field energy $U^{\mathrm{II}}$ possessed by a hollow sphere or a solid metal sphere carrying a total charge $Q$.

Let us now turn to the solid sphere made of insulating material. Here, the charge $Q$ is truly distributed uniformly throughout the volume of the sphere. For any sphere of radius $r \leq R_0$ eq. (4.18) applies, albeit with a modified radius and the fraction of the charge carried by that sphere of radius $r \leq R_0$. The charge density $\rho$ is the total charge $Q$ divided by the volume of the sphere; thus, $\rho = 3Q/4\pi R_0^3$. We now consider a sphere of radius $r \leq R_0$ carrying a charge $q_r$ that is uniformly distributed throughout the sphere.

To expand this sphere into the full sphere of radius $R_0$, we successively add spherical shells with charge $\mathrm{d}q_r$ until $r = R_0$ is reached. For the charge $q_r$ of such a sphere and the charge $\mathrm{d}q_r$ to be added by the next infinitesimal spherical shell, the following holds:

$$q_r = \rho \cdot \frac{4\pi r^3}{3} \qquad \Rightarrow \qquad \mathrm{d}q_r = \rho \cdot 4\pi r^2 \, \mathrm{d}r \tag{4.23}$$

This turns eq. (4.18) into:

$$\mathrm{d}W_r = \frac{1}{4\pi\epsilon_0} \frac{q_r \, \mathrm{d}q_r}{r} \tag{4.24}$$

We now obtain $W_{R_0}^{\mathrm{I}}$ simply by integrating $\mathrm{d}W_r$ from $r = 0$ to $r = R_0$. Substituting $q_r$ and $\mathrm{d}q_r$ into $\mathrm{d}W_r$ and integrating yields:

$$W_{R_0}^{\mathrm{I}} = \frac{4\pi\rho^2}{3\epsilon_0} \int_0^{R_0} r^4 \, \mathrm{d}r = \frac{4\pi\rho^2 R_0^5}{15\epsilon_0} = \frac{3}{5} \cdot \frac{Q^2}{4\pi\epsilon_0 R_0} \tag{4.25}$$

Comparing eq. (4.22) with eq. (4.25) shows that the work required to bring the total charge $Q$ in from infinity and distribute it uniformly within a spherical volume of radius $R_0$ is 20% greater than the work required to charge a hollow sphere or a metallic sphere of radius $R_0$ with a total charge $Q$ in the same manner. In other words, the electrostatic field energy of a hollow metal sphere or a solid metal sphere is exactly 5/6 of the electrostatic field energy of a uniformly charged solid sphere. We can also obtain the result for the uniformly charged solid sphere again using eq. (3.50). However, two contributions to $U$ must be considered here: the contribution $U_i^{\mathrm{I}}$ due to the non-vanishing electric field inside the sphere, and the contribution $U_o^{\mathrm{I}}$ arising from the electric field outside the sphere, that is:

$$U^{\mathrm{I}} = U_i^{\mathrm{I}} + U_o^{\mathrm{I}} \tag{4.26}$$

For $r > R_0$, the electric fields of all three spheres are identical, i.e., $U_o^{\mathrm{I}} \equiv U^{\mathrm{II}}$ from eq. (4.22). For the interior of the non-conducting sphere, we find, using eq. (4.12):

$$U_i^{\mathrm{I}} = \frac{\epsilon_0}{2} \int_0^{R_0} \left( \frac{Q \cdot r}{4\pi\epsilon_0 R_0^3} \right)^2 \cdot 4\pi r^2 \, \mathrm{d}r = \frac{1}{10} \cdot \frac{Q^2}{4\pi\epsilon_0 R_0} \tag{4.27}$$

The field energy of the uniformly charged, non-conducting solid sphere is thus:

$$U^{\mathrm{I}} = \frac{1}{10} \cdot \frac{Q^2}{4\pi\epsilon_0 R_0} + \frac{Q^2}{8\pi\epsilon_0 R_0} = \frac{3}{5} \cdot \frac{Q^2}{4\pi\epsilon_0 R_0} \tag{4.28}$$

As expected, this corresponds to the result from eq. (4.25), where we calculated the work required to bring the charge $Q$ in from infinity and assemble it into a uniformly charged solid sphere.

## 4.2 The Classical Electron Radius

The electrostatic energy of a non-conducting, uniformly charged sphere with arbitrary radius $R$ and charge $Q$ is given by eq. (4.28):

$$U = \frac{3}{5}\frac{Q^2}{4\pi\epsilon_0 R} \tag{4.29}$$

As $R \to 0$, this electrostatic energy diverges. Using eq. (4.22), the electrostatic field energy of a charged metal sphere or hollow sphere of radius $R$ and charge $Q$ is:

$$U = \frac{Q^2}{8\pi\epsilon_0 R} \tag{4.30}$$

This electrostatic energy also diverges as $R \to 0$. In all three cases, a non-conducting uniformly charged sphere, a charged metal sphere, and a charged hollow sphere, the electrostatic energy approaches infinity as $R \to 0$. This presents a problem, because, as far as we know, the electron has no structure and no spatial extent. If we imagine the electron as a spherical charge distribution with radius $\rho$, then the electrostatic energy outside the electron is given by:

$$U_e = \frac{e^2}{8\pi\epsilon_0 \rho} \tag{4.31}$$

The electron has a mass $m_e$ and, via Einstein's $E = m_e c^2$, an energy equivalent. From $U_e = m_e c^2$, we then obtain an equation determining $\rho$:

$$m_e c^2 = \frac{e^2}{8\pi\epsilon_0 \rho} \qquad \Rightarrow \qquad \rho = \frac{1}{8\pi\epsilon_0} \cdot \frac{e^2}{m_e c^2} \tag{4.32}$$

After substituting the numerical values for the corresponding quantities (see Appendix B) in this equation to determine $\rho$ and multiplying by 2, we obtain the so-called classical electron radius $r_e$:

$$r_e = 2 \cdot \rho = 2.82 \cdot 10^{-15}\,\mathrm{m} \tag{4.33}$$

In scattering experiments involving uncharged particles, such as the light photons, off free electrons, an effective electron radius can also be determined from the measured cross-section. This such measured electron radius is indeed of the order of the correct magnitude of $r_e^2$. The classical electron radius $r_e$ corresponds to the size of small atomic nuclei and is therefore important in atomic physics. However, the classical electron radius $r_e$ tells us nothing about the structure of the electron. It is a useful quantity, but the electron is a quantum mechanical object that cannot be described by a classical radius.

## 4.3 Charge Carrying Wire and Coaxial Cable

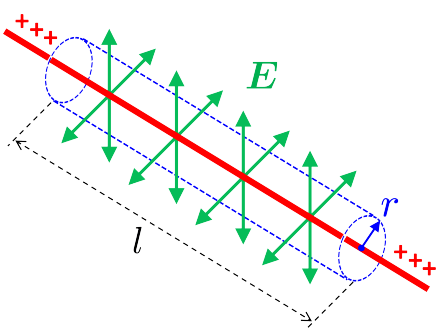



**Fig. 4.3**

We consider a very long and thin wire segment carrying a constant positive electric charge $\lambda$ per unit length. The radius $a$ of the wire is assumed to be much smaller than the length $l$. To allow us to neglect edge effects, the wire segment sketched in fig. 4.3 shall represent merely a section of a much longer wire, with our wire segment located far away from the wire's ends. The wire is a conductor, therefore, the electric field inside the wire must vanish, as does the tangential component of the electric field at the wire's surface. Hence, the electric field vectors are perpendicular to the wire surface and point radially outward. As our Gaussian surface, we now choose, as sketched in fig. 4.3, a cylinder of length $l$ and radius $r$, with the wire running along its axis. Since the electric field vectors are perpendicular to the wire surface and point radially outward, the flux of the electric field through the end faces of the cylinder is zero, and for the flux of the electric field through the cylindrical surface $A$ over the length $l$, we obtain:

$$\int_A \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{\sigma} \;=\; E(r)\cdot 2\pi r\cdot l \tag{4.34}$$

The electric charge enclosed within the Gaussian surface in fig. 4.3 is $l\cdot\lambda$, and thus Gauss's law yields the result:

$$E(r)\cdot 2\pi r\cdot l \;=\; \frac{1}{\epsilon_0}\cdot l\cdot\lambda \tag{4.35}$$

For the magnitude of the electric field strength at a distance $r$ from the wire, we thus obtain:

$$E(r) \;=\; \frac{\lambda}{2\pi\epsilon_0 r} \qquad \text{if} \qquad r > a \tag{4.36}$$

Inside the conducting wire, i.e., for $r < a$, the electric field vanishes. A simple integration of eq. (3.19) using the electric field strength from eq. (4.36) yields the following expression for the electrostatic potential of the wire:

$$\phi(r) \;=\; \frac{\lambda}{2\pi\epsilon_0}\ln|r| + c \qquad \text{if} \qquad r > a \tag{4.37}$$

We determine the integration constant $c$ by requiring that the electrostatic potential must be zero very far from the wire, for example, at a distance $r_0$. We deliberately do not choose $r_0 = \infty$ because $\ln(r_0)$ will become infinite for $r_0 \to \infty$. Although we

are considering only a segment of wire of length $l$ very far from both ends of the wire, in principle, the wire is infinitely long, so to speak. That poses a problem because we cannot determine the potential at infinity for a wire that is itself infinitely long. We must therefore always bear in mind that we are considering only a segment of wire of length $l$, with $l \gg a$, which produces an electrostatic potential $\phi(r)$ that, by definition, vanishes at $r_0$. Since $r$ and $r_0$ are positive by definition, we can omit the absolute value bars, and eq. (4.37) thus becomes:

$$\phi(r) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r}{r_0}\right) \qquad \text{if} \qquad r > a \tag{4.38}$$

The electric potential inside the conducting wire is constant and takes the value that eq. (4.38) assumes at $r = a$:

$$\phi(r) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{a}{r_0}\right) \qquad \text{if} \qquad r \leq a \tag{4.39}$$

With $\phi(r) = 0$ for $r \geq r_0$, the electric field also vanishes for $r \geq r_0$, and for the electrostatic energy of the wire segment of length $l$ (where $l \gg a$), we obtain:

$$U = \frac{\epsilon_0}{2} \int_a^{r_0} \int_0^{2\pi} \int_0^{l} \left(\frac{\lambda}{2\pi\epsilon_0 r}\right)^2 r \,\mathrm{d}r \,\mathrm{d}\varphi \,\mathrm{d}z = \frac{\lambda^2 l}{4\pi\epsilon_0} \ln\left(\frac{r_0}{a}\right) \tag{4.40}$$

As $r_0 \to \infty$, $U$ becomes infinite, which, of course, is impossible. But why is this the case? To determine the work required to bring a charge in from infinity onto the wire, one must know exactly where that charge is being placed. However, that is precisely what we cannot say for a wire that extends infinitely. We simply lack a reference point for where to place this charge on the wire, since it could be located anywhere along the infinitely long wire. To address this problem, we must introduce a boundary, a second cylindrical conductor along whose axis the infinitely long wire runs. As sketched in fig. 4.4, in such an arrangement, referred to as cylindrical capacitor or coaxial cable, the wire is surrounded by a hollow conducting cylinder carrying the same amount of charge but of opposite polarity. In most cases, the space between the inner and outer conductors is filled with an insulator, and the outer conductor is surrounded by an insulating protective sheath. The net charge enclosed by a Gaussian cylindrical surface with the coaxial cable running along its axis is $+\lambda - \lambda = 0$ per unit length. Consequently, the electric field outside the outer conductor must vanish. If we choose for our Gaussian surface a cylinder radius of $r = r_o$ then it will only enclose the positive charge of the

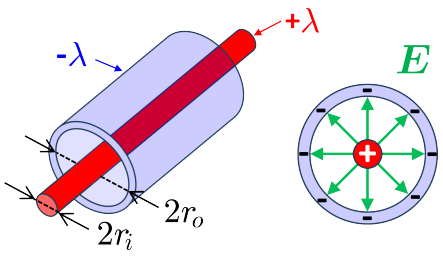



**Fig. 4.4**

inner conductor. In this way, we find that the electric field between the inner and out conductors is given by eq. (4.36). Similarly, the electrostatic potential between the inner and outer conductors is given by eq. (4.37). With the known electric field between the inner and outer conductors we can calculate the electric field energy of a coaxial cable section of length $l$ with the result:

$$U = \frac{\epsilon_0}{2} \int_{r_i}^{r_o} \int_0^{2\pi} \int_0^{l} \left( \frac{\lambda}{2\pi\epsilon_0 r} \right)^2 r \, \mathrm{d}r \, \mathrm{d}\varphi \, \mathrm{d}z = \frac{\lambda^2 l}{4\pi\epsilon_0} \ln\left( \frac{r_o}{r_i} \right) \tag{4.41}$$

We obtain the same result if we consider the work required to deposit the charges. To do this, we assume that a charge $\lambda' \cdot l$ is already present on the electrodes. To deposit an additional charge $\mathrm{d}\lambda' \cdot l$, requires an additional work $\mathrm{d}W$ to be performed which is given by eq. (3.30):

$$\mathrm{W} = [\phi(r_o) - \phi(r_i)] \, \mathrm{d}\lambda' \cdot l \tag{4.42}$$

If we take $\phi(r_i)$ and $\phi(r_o)$ from eq. (4.37) and integrate this equation from $Q = 0$ to $Q = \lambda \cdot l$, we obtain:

$$W = \frac{l}{2\pi\epsilon_0} \ln\left( \frac{r_o}{r_i} \right) \int_0^{\lambda} \lambda' \, \mathrm{d}\lambda' = \frac{\lambda^2 l}{4\pi\epsilon_0} \ln\left( \frac{r_o}{r_i} \right) \tag{4.43}$$

For the electric voltage, that is, the potential difference $U$, between the inner and outer conductors with an applied charge $Q = l \cdot \lambda$, we obtain, using eq. (3.35):

$$U = \phi(r_o) - \phi(r_i) = \frac{\lambda}{2\pi\epsilon_0} \ln\left( \frac{r_o}{r_i} \right) = \frac{2W}{\lambda \cdot l} = \frac{2W}{Q} \tag{4.44}$$

This electric voltage $U$ is constant along the coaxial cable because the inner and outer conductors are at constant electrostatic potentials $\phi(r_i)$ and $\phi(r_o)$. If we consider a section of the inner conductor of length $l$, it carries an electric charge of $Q = l \cdot \lambda$. The ratio of electric charge $Q$ to electric voltage $U$ is the electric capacitance $C$, and for the coaxial cable, this is given by:

$$C = \frac{Q}{U} = \frac{2\pi\epsilon_0 \cdot l \cdot \lambda}{\lambda \cdot \ln\left( r_o / r_i \right)} = \frac{2\pi\epsilon_0 \cdot l}{\ln\left( r_o / r_i \right)} \tag{4.45}$$

As can be seen, $C$ is a purely geometric quantity. It depends solely on the length of the section of coaxial cable under consideration and on the radii $r_i$ and $r_o$ of the inner and outer conductors. The greater this capacitance $C$ is for a given voltage $U$, the more charge such a cylindrical capacitor can store. As we will see later, in addition to geometric factors, there is another factor that significantly influences capacitance: the material that fills the space between the inner and outer conductors. Two further

important relationships follow from eq. (4.44) and eq. (4.45) if we replace the charge $Q$ in eq. (4.44) with the capacitance $C \cdot U$ from eq. (4.45), or the voltage $U$ in eq. (4.45) with the work $W/Q$ from eq. (4.44). This yields the following expression for the energy stored in the cylindrical capacitor:

$$W = \frac{1}{2} C \cdot U^2 \qquad \text{or} \qquad W = \frac{1}{2} \frac{Q^2}{C} \tag{4.46}$$

The relationships in eq. (4.46) between stored energy $W$, voltage $U$, capacitance $C$, and electric charge $Q$ apply not only to the cylindrical capacitor we just discussed but they apply quite generally to all kinds of capacitors. We can, for example, also apply them to the charged spheres we have already examined. But how? What is the electrical capacitance of these spheres, and what is the applied voltage? To answer these questions, we consider a so-called spherical capacitor, that is, a capacitor in which the electrodes consist of a sphere of radius $R_0$ and an enclosing spherical shell of radius $R_1$, as sketched in fig. 4.5. Let the inner sphere carry the electric charge $Q$ and the outer spherical shell the electric charge $-Q$. If we now place a Gaussian surface $S$ in the form of a spherical shell with radius $r$ somewhere between the electrodes of this spherical capacitor, that is, between $R_1 < r < R_2$, then, unsurprisingly, we again obtain eq. (4.7) for the electric field:

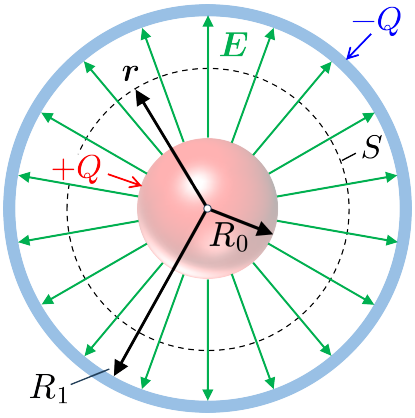



**Fig. 4.5**

$$E(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2} \tag{4.47}$$

When calculating the electric voltage $U$ between the electrodes of the spherical capacitor using eq. (3.33) and eq. (3.35), we choose the path such that the resulting voltage is positive, that is:

$$U = \phi(R_0) - \phi(R_1) = -\int_{R_1}^{R_0} E(r)\,\mathrm{d}r = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{R_0} - \frac{1}{R_1}\right) \tag{4.48}$$

Since $R_1 > R_0$, $U$ is indeed positive. If we now let $R_1$ approach infinity, this expression for the voltage reduces to:

$$U = \frac{1}{4\pi\epsilon_0} \frac{Q}{R_0} \tag{4.49}$$

We could have arrived at this result more quickly: since the electrostatic potential at infinity is zero, the potential $\phi(R_1)$ must approach zero as $R_1 \to \infty$, and the voltage $U$

is being reduced, as shown in eq. (4.49), to the potential $\phi(R_0)$, which we already know from eq. (4.12). However, thanks to this brief detour, we now also know the capacitance of a spherical capacitor:

$$C = \frac{Q}{U} = 4\pi\epsilon_0 \left( \frac{R_0 \cdot R_1}{R_1 - R_0} \right) \tag{4.50}$$

If we let $R_1$ approach infinity in eq. (4.50), we obtain the following expression for the capacitance of our solid and hollow metal spheres:

$$C = 4\pi\epsilon_0 R_0 \tag{4.51}$$

If we then use eq. (4.46) to determine the energy stored in the hollow or solid metal sphere by means of eq. (4.49) and eq. (4.51), we obtain:

$$W = \frac{Q^2}{8\pi\epsilon_0 R_0} \tag{4.52}$$

Unsurprisingly, this result corresponds to the electrostatic energy we previously calculated for the solid metal sphere and the hollow sphere using eq. (4.22). What is the situation for the uniformly charged solid sphere made of insulating material? The electrostatic potential at the surface is identical for all three spheres, and consequently, so is the electric field outside the spheres. Thus, eq. (4.51) also applies to the non-conducting, uniformly charged solid sphere. However, in this case, eq. (4.52) does not correspond to the total electrostatic energy, but only to the component $U^{\mathrm{II}}$ from eq. (4.22).

## 4.4 Infinitely Extended, Planar Surface Charge

As a final example of the application of Gauss's law, we will examine the electrostatic potential and the electric field of the charged planar surface sketched in fig. 4.6. Let us assume it extends infinitely in the $xz$-plane, allowing us to neglect edge effects. A positive electric charge is distributed uniformly across this infinite planar surface. The material of the surface could be a metal or a uniformly charged insulator. In the case of a metal, we know that the tangential component of the electric field must vanish and that, consequently, the electric field vectors on either side of the surface must be perpendicular to it. However, due to symmetry, this must also hold true for an insulator. The electric field therefore has components only in the $+y$- and $-y$-

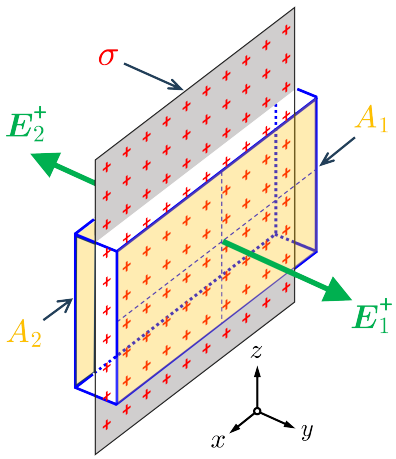



**Fig. 4.6**

directions. If we place a Gaussian surface in the shape of the rectangular box sketched in fig. 4.6 around a portion of the charged surface, the components of the electric field flux through the side faces of the box vanish, leaving only the contributions of the electric field flux through the faces of the box parallel to the $xz$-plane, $A_1$ and $A_2$. If our Gaussian box encloses a total positive charge $Q$, then Gauss's law tells us that the following must hold:

$$E_1^+ \cdot A_1 + E_2^+ \cdot A_2 = \frac{Q}{\epsilon_0} \tag{4.53}$$

The superscript plus sign on the field strength serves to remind us that it originates from a positive charge distribution. We will need this distinction later on. The areas $A_1$ and $A_2$ are equal in size, that is, $A_1 = A_2 = A$, and since $\left|E_1^+\right| = \left|E_2^+\right| = E^+$ also holds, this yields, in terms of the surface charge density $\sigma = Q/A$:

$$E^+ = \frac{1}{2\epsilon_0} \cdot \frac{Q}{A} = \frac{\sigma}{2\epsilon_0} \tag{4.54}$$

The electric field strength $E^+$ is therefore independent of the distance from the charged surface. A constant electric field strength means that the electric potential $\phi$ decreases linearly with the distance from the surface:

$$\phi^+(y) = -\frac{\sigma}{2\epsilon_0} \cdot |y| + \phi^+(0) \tag{4.55}$$

Because the electric field is oriented perpendicular to the $xz$-plane, the equipotential surfaces of $\phi^+$ run parallel to the $xz$-plane. We can now ask ourselves: What happens if, instead of a positively charged surface, we consider a negatively charged one? With this change, the direction of the electric field is reversed, meaning, the electric field vectors point toward a negatively charged surface rather than away from it, and we must also replace $\sigma$ with $-\sigma$ in the above equations. Thus, for an infinitely large, negatively charged surface, we obtain:

$$E^- = \frac{1}{2\epsilon_0} \cdot \frac{-Q}{A} = \frac{-\sigma}{2\epsilon_0} \tag{4.56}$$

and

$$\phi^-(y) = \frac{\sigma}{2\epsilon_0} \cdot |y| + \phi^-(0) \tag{4.57}$$

Taken in isolation, this is not a particularly remarkable result. It becomes interesting when we position the positively and negatively charged surfaces parallel to each other. We do this as shown in fig. 4.7 but no longer use infinitely extended surfaces; instead, we use two thin metal plates where the separation distance $d$ is small compared to the length $l$ of the plates ($d \ll l$). If we add the electric fields generated by the two charged

plates, $E^+$ from the positively charged plate and $E^-$ from the negatively charged plate, we see that they reinforce each other in the space between the metal plates but cancel each other out outside that region. Neglecting edge effects at the capacitor plates, the following applies outside the plates:

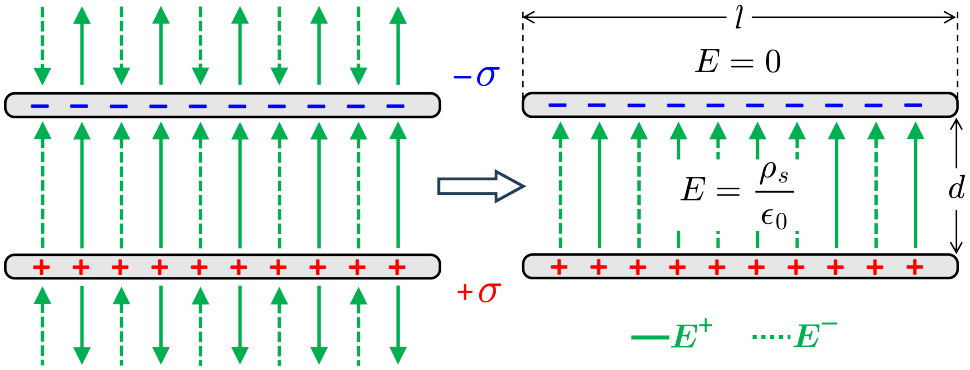



**Fig. 4.7:** The parallel-plate capacitor: Two parallel metallic plates separated by a distance $d \ll l$, with equal charge density but opposite charge polarity.

$$E = \left|E^+\right| - \left|E^-\right| = \frac{\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0} = 0 \tag{4.58}$$

Between the electrodes of this capacitor, the electric field strength is exactly twice the value of eq. (4.54):

$$E = \left|E^+\right| + \left|E^-\right| = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0} \tag{4.59}$$

A potential difference exists between the electrodes of the parallel-plate capacitor, but what is its magnitude? Eq. (4.55) tells us that at a distance $d$ from the positively charged plate, the potential $\phi^+$ is lower than that at the plate by $-d\cdot(\sigma/2\epsilon_0)$. And eq. (4.57) tells us that at a distance $d$ from the negatively charged plate, the potential $\phi^-$ is higher by $d\cdot(\sigma/2\epsilon_0)$ than at the plate itself. Thus, the potential difference $U$ between the two capacitor plates is:

$$U = \frac{\sigma}{\epsilon_0}\cdot d \tag{4.60}$$

This also follows directly from eq. (3.35) if we start with eq. (4.59) and the parameterization $\mathrm{d}s$ of the path between the capacitor plates:

$$U = \phi(0) - \phi(d) = \int_0^d E\cdot \mathrm{d}s = E\cdot d = \frac{\sigma}{\epsilon_0}\cdot d \tag{4.61}$$

If we multiply $U$ by a positive test charge $q$, we obtain with

$$q\cdot U = q\cdot E\cdot d = F\cdot d \tag{4.62}$$

the work required to move this test charge from the negatively charged electrode to the positively charged electrode of the plate capacitor, i.e., against the acting electrostatic force. If we substitute the surface charge density $\sigma = Q/A$ into eq. (4.60), the equation becomes:

$$U = \frac{d}{\epsilon_0 A} \cdot Q = C^{-1} \cdot Q \tag{4.63}$$

Thus, the following applies to the capacitance $C$ of the parallel-plate capacitor:

$$C = \frac{\epsilon_0 A}{d} \tag{4.64}$$

The larger the area $A$ of the capacitor plates and the smaller the distance $d$ between them, the greater the charge $Q$ that a parallel-plate capacitor can store at a given voltage $U$. If we express the energy $W$ stored in the parallel-plate capacitor, as given in eq. (4.46), in terms of the charge $Q$, we obtain, using eq. (4.63) and eq. (4.64):

$$W = \frac{1}{2} C \cdot U^2 = \frac{d}{2\epsilon_0 A} \cdot Q^2 \tag{4.65}$$

Using $U = E \cdot d$, $C = \epsilon_0 \cdot A/d$, and the capacitor volume $V = A \cdot d$, we thus obtain the following expression for the energy density $w = W/V$ in the capacitor:

$$w = \frac{\epsilon_0 \cdot E^2}{2} \tag{4.66}$$

For the force $F$ between the charges $Q$ on the capacitor plates, given $W = F \cdot d$ and $E = \sigma/\epsilon_0 = Q/\epsilon_0 A$, the following applies:

$$F = \frac{1}{2} \cdot Q \cdot E \tag{4.67}$$

Why do we not obtain the equation $F = Q \cdot E$ for the force? Where does the factor 1/2 come from? We can see the reason for this in fig. 4.8. The surface charge layer on the capacitor plates has a finite thickness $\delta$. This means that the electric field strength does not drop abruptly to zero at the surface from the constant value $E$ it has between the capacitor plates. Instead, within the surface charge layer of thickness $\delta$, as sketched in fig. 4.8, it decreases linearly from the value $E$ to zero. That this must be the case becomes apparent when applying Gauss's law of electrostatics to a volume enclosing both plates and their surface charge layers. The effective field strength acting on the surface charge $Q$ of the capacitor plates is therefore, on average, only half as strong as $E$; hence the factor 1/2 in eq. (4.67).

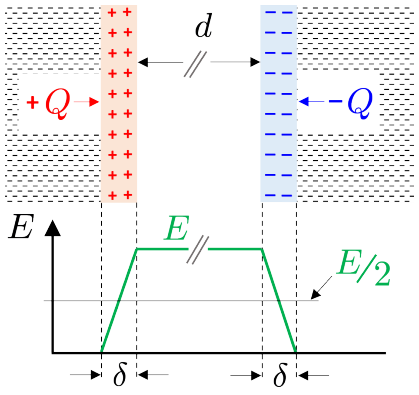



**Fig. 4.8**

## 4.5 Uniformly Charged Ring

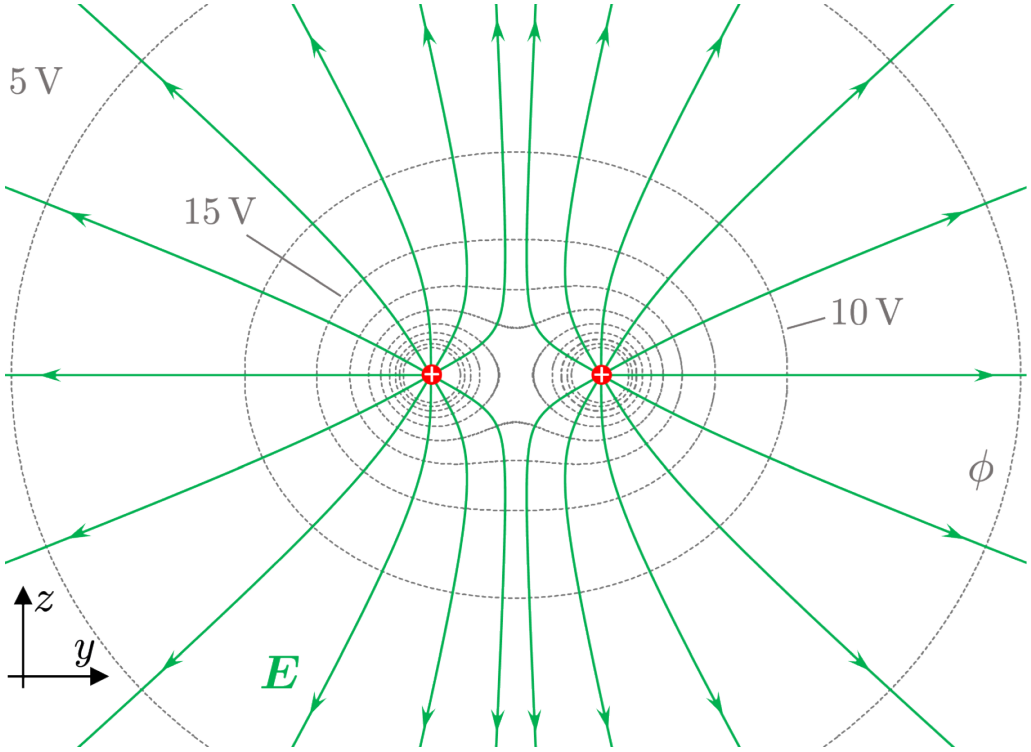



**Fig. 4.9:** Equipotential lines of $\phi$ in 5 V increments from 5 to 50 V and field lines of $\boldsymbol{E}$ for two identical charges, each carrying the equivalent of 10 million positive unit charges $e$ and separated by a distance of $d = 2\,\mathrm{m}$.

We begin with considering the electric potential and electric field produced by a pair of positive charges. To do this, we simply replace the negative charge in the dipole shown in fig. 3.6 with a positive charge of equal magnitude, and we proceed with the calculation of the equipotential lines of $\phi$ and the path of the field lines of $\boldsymbol{E}$ in exactly the same way as we did in section 3.2.1. Here, we choose the $z$-axis as the axis of symmetry and we place the two positive charges on the $y$-axis. The pattern of the equipotential lines of $\phi$ shown in fig. 4.9 is nothing new to us, as we previously encountered it in volume 1 of this physics course when examining the electric potential generated by two hydrogen atoms in an $H_2$ molecule. Here we can see in addition the paths of the $\boldsymbol{E}$-field lines, which intersect the equipotential lines perpendicularly. Of course, instead of two positive charges, we could have used two negative charges of equal magnitude. However, the resulting image would be identical, except that the field lines would point toward the negative charges and we would need to place a minus sign before the potential values. Fig. 4.9 shows fewer field lines than fig. 3.6 because, unlike in fig. 3.6 where they appear every 15°, the $\boldsymbol{E}$-field lines in fig. 4.9 originate from the charge positions at intervals of 30°. Furthermore, field lines

originating from the charges along the line connecting the centers of the two charges and directed toward the respective other charge at $z = 0$ have been omitted. It is evident from fig. 4.9 that the closer a field line of $\boldsymbol{E}$ starts to the line connecting the two charges, the more strongly it curves away from the respective other charge. For $z > 0$, the field lines point in the positive $z$-direction, and for $z < 0$, in the negative $z$-direction. We will address the direction of the field lines that originate at $z = 0$, i.e., parallel to the straight line between the two charge centers, later on. Now, let us simply imagine rotating the two charges shown in fig. 4.9, along with the equipotential lines of $\phi$ and the field lines of $\boldsymbol{E}$, by 180° around the $z$-axis, in such a way that the charges are uniformly smeared out in the process. The result is a ring of charge, and the equipotential surfaces of $\phi$ and the field configuration of $\boldsymbol{E}$ resulting from this rotation are those produced by a uniformly charged ring.

Having now gained an idea of what the electrostatic potential and the electric field of a uniformly charged ring look like, we turn to actually determining the potential and the field of the ring charge sketched in fig. 4.10. In doing that, we are primarily interested in two aspects: the behavior of the electric field along the ring's axis,i.e., the $z$-axis, and what the electric field looks like within the circular disk enclosed by the ring. The thickness of the homogeneously charged ring sketched in fig. 4.10 is to be considered negligibly small compared to the ring radius $R$. For the sake of clarity, we have simply drawn the ring, which carries a total electric charge $Q$, much thicker than it actually is. $M$ is the center of the ring and lies on the $z$-axis. All points $P$ on the ring axis, i.e., the $z$-axis, are equidistant from all segments of the ring; this distance is $s$. Each of the infinitesimal ring segments the ring is made of carries a differential electric charge $\mathrm{d}q$, which sum up to the total charge $Q$ of the ring. The electric potential $\phi$ on the ring axis can thus be easily calculated. All we have to do is to simply sum up all contributions to $\phi$ from these infinitesimal ring segments carrying electric charges $\mathrm{d}q$. Expressing the differential charge $\mathrm{d}q$ in terms of the arc element of the ring where it is located, we obtain:

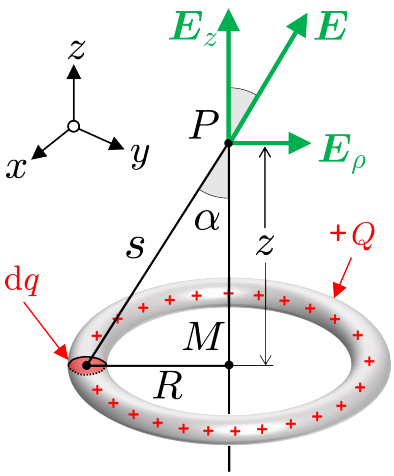



**Fig. 4.10**

$$\mathrm{d}q = \frac{Q}{2\pi R}\cdot R\cdot \mathrm{d}\varphi = \frac{Q}{2\pi}\cdot \mathrm{d}\varphi \tag{4.68}$$

With $s^2 = z^2 + R^2$, the electric potential on the ring axis at a distance $z$ from the center of the ring is thus given by:

$$\phi(z) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{2\pi}\int\limits_0^{2\pi}\frac{\mathrm{d}\varphi}{\sqrt{z^2+R^2}} \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{\sqrt{z^2+R^2}} \tag{4.69}$$

At the location of $M$, i.e., at $x = y = z = 0$, the potential is constant, and thus the electric field strength at $M$ is exactly zero. Here we have the answer to the question of the magnitude of the field strength midway between the two charges shown in fig. 4.9: the electric field at point $M$ is zero; above $M$, the electric field lines point upwards, and below it, they point downwards. The electric field lines converging from the two charges thus bend upwards and downwards and they do so in such a way that the midpoint of the line segment connecting the two charges remains field-free. Far from $M$, in the far-field region, the Taylor expansion of the square root for $z \gg R$ in eq. (4.69) yields the potential:

$$\phi(z) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{z}\left[1-\frac{1}{2}\cdot\frac{R^2}{z^2}+\mathscr{O}\Big(\frac{R^4}{z^4}\Big)\right] \tag{4.70}$$

Thus, from a very large distance, the electric potential of the ring looks like the Coulomb potential of a point charge $Q$. A ring-shaped charge has no dipole moment, and the second term in eq. (4.69) is therefore a quadrupole term. From the multipole expansion of the potential, we are already familiar with the quadrupole term of a ring charge via eq. (3.100). We therefore know, based on eq. (3.78) and eq. (3.100), that the sum of the monopole and quadrupole terms of the potential is:

$$\phi(\boldsymbol{r}) \;=\; \phi_0(\boldsymbol{r})+\phi_2(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\left[\frac{Q}{r}+Q_0\cdot\frac{3\cos^2\vartheta-1}{r^3}\right] \tag{4.71}$$

With eq. (4.71), we know not only the potential on the $z$-axis but also its angular dependence in the far-field approximation. In the last chapter, $Q_0$, the quadrupole moment of the ring charge, was merely a placeholder because we did not know its value. However, we can now determine its value using eq. (4.70) and eq. (4.71). To do so, we set $\vartheta = 0$ and $r = z$ in eq. (4.71) to obtain the exact correspondence of the respective terms in eq. (4.70). A comparison of the respective coefficients then yields:

$$Q_0 \;=\; -\frac{Q\cdot R^2}{4} \tag{4.72}$$

With $Q_0 = -Q_{xx} = -Q_{yy} = Q_{zz}/2$ in eq. (3.100), the quadrupole tensor of the positive ring charge is:

$$\underline{\underline{\boldsymbol{Q}}} \;=\; \frac{Q\cdot R^2}{4}\cdot\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & -2\end{bmatrix} \tag{4.73}$$

For $r \gg R$, the angle-dependent potential of the ring charge in the far-field approximation is thus:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r} \cdot \left[1 - \frac{R^2}{4} \cdot \frac{3\cos^2\vartheta - 1}{r^2} + \mathscr{O}\left(\frac{R^4}{r^4}\right)\right] \quad , \quad r > R \tag{4.74}$$

We can determine the angle-dependent far-field of $\boldsymbol{E}$ from eq. (4.74) by calculating the gradient. However, fig. 4.9 already shows us roughly what this looks like, so we do not learn much new information there. However, what interests us is the strength of the electric field on the $z$-axis and its behavior within the circular disk enclosed by the ring charge. We obtain the electric field on the $z$-axis from eq. (4.69) via $\boldsymbol{E} = -\nabla\phi$:

$$\boldsymbol{E}_z = \frac{1}{4\pi\epsilon_0} \cdot \frac{z \cdot Q}{\left(z^2 + R^2\right)^{3/2}} \cdot \hat{\boldsymbol{k}} \tag{4.75}$$

We thus know that the $z$-component of the electric field at $M$ indeed vanishes. Far from the center $M$, for $z \gg R$, the Taylor expansion of the square root in eq. (4.75) yields the following for the far-field of $\boldsymbol{E}_z$:

$$\boldsymbol{E}_z = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{z^2}\left[1 - \frac{3}{2} \cdot \frac{R^2}{z^2} + \mathscr{O}\left(\frac{R^4}{z^4}\right)\right] \cdot \hat{\boldsymbol{k}} \quad , \quad z > R \tag{4.76}$$

This gives us $\boldsymbol{E}_z$. But what about $\boldsymbol{E}_\rho$? To determine this, we consider an infinitesimally thin slice of the ring carrying a charge $\mathrm{d}q$. For every such $\mathrm{d}q$ on the ring, there is a corresponding $\mathrm{d}q$ on the opposite side that provides a contribution to $\boldsymbol{E}_\rho$ of equal magnitude but exactly opposite direction. For reasons of symmetry alone, it must therefore hold that $\boldsymbol{E}_\rho = 0$ on the cylinder axis. This is not the case for points not lying on the cylinder axis. To determine the behavior of the electric field lines within the circular disk enclosed by the ring, the multipole expansion of the potential for $r > r'$ is of no use to us. However, we can now make use of what we learned in section 3.2.2 regarding the multipole expansion. There, we considered situations where we were interested in the potential at locations $\boldsymbol{r}$ very far from the positions $\boldsymbol{r}_i$ of the charges $q_i$ or a charge distribution $\rho(\boldsymbol{r}')$, that is, the cases where $\boldsymbol{r} \gg \boldsymbol{r}_i$ or $\boldsymbol{r} \gg \boldsymbol{r}'$. Now, let's imagine a charged ring with a very large radius. Conversely, we must ask ourselves which approximation we can use for locations $\boldsymbol{r}$ such that $r \ll R$. In this case, instead of the Taylor expansion of eq. (3.73), we are interested in that of

$$\frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} = \frac{1}{\sqrt{r^2 - 2\boldsymbol{r}\cdot\boldsymbol{r}' + r'^2}} = \frac{1}{r'} \cdot \frac{1}{1 - (2\boldsymbol{r}\cdot\boldsymbol{r}' - r^2)/r'^2} = \frac{1}{r'} \cdot \frac{1}{\sqrt{1-\xi}} \tag{4.77}$$

With $r' \gg r$, we have $\xi \ll 1$, and using the Taylor expansion of the square root for $\xi < 1$, we can write:

$$\frac{1}{\sqrt{1-\xi}} = 1 + \frac{1}{2}\cdot\xi + \frac{3}{8}\cdot\xi^2 + \mathscr{O}(\xi^3)$$

With this Taylor expansion and $\xi = (2\boldsymbol{r}\cdot\boldsymbol{r}' + r^2)/r'^2$, eq. (4.77) becomes:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{1}{r'} + \frac{1}{2}\cdot\frac{2\boldsymbol{r}\cdot\boldsymbol{r}' - r^2}{r'^3} + \frac{3}{8}\cdot\frac{4(\boldsymbol{r}\cdot\boldsymbol{r}')^2 - 4(\boldsymbol{r}\cdot\boldsymbol{r}')r^2 + r^4}{r'^5} + \mathscr{O}\Big(\frac{1}{r'^7}\Big) \tag{4.78}$$

Grouping the terms on the right-hand side according to powers of $1/r'$, while taking into account terms up to $1/r'^3$, yields:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{1}{r'} + \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r'^3} + \frac{1}{2}\cdot\frac{3(\boldsymbol{r}\cdot\boldsymbol{r}')^2 - r^2r'^2}{r'^5} + \mathscr{O}\Big(\frac{r^3}{r'^4}\Big) \quad , \quad r \ll r' \tag{4.79}$$

This approximation for $r \ll r'$ is the mirror image of the approximation for $r_i \ll r$ in eq. (3.75), except that $r$ and $r_i$, or in this case $r'$, have swapped positions. Let us first look at what we obtain for the monopole component $\phi_0$ of the potential in this approximation. With $r' = R$, the magnitude of $\boldsymbol{r}'$ is constant, and with the sum of all charges on the ring equal to $Q$, we obtain:

$$\phi_0(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{R} \tag{4.80}$$

The monopole contribution is thus constant for $r < R$. The dipole term vanishes for our ring charge, and for the quadrupole term in eq. (4.79), we rewrite the numerator to separate $\boldsymbol{r}$ and $\boldsymbol{r}'$, just as we did in section 3.2.2, and obtain the following for the quadrupole contribution $\phi_2(\boldsymbol{r})$ to the potential:

$$\phi_2(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{\boldsymbol{r}\cdot\underline{\underline{\boldsymbol{Q}}}\cdot\boldsymbol{r}}{R^5} \tag{4.81}$$

Thus, the potential of the ring charge in the approximation $r < R$ is:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{1}{R}\cdot\left[Q + \frac{\boldsymbol{r}\cdot\underline{\underline{\boldsymbol{Q}}}\cdot\boldsymbol{r}}{R^4} + \mathscr{O}\Big(\frac{r^4}{R^4}\Big)\right] \quad , \quad r < R \tag{4.82}$$

The quadrupole moment is a property of the ring charge and does not depend on our approximation. Using eq. (4.73), we know the elements of $\underline{\underline{\boldsymbol{Q}}}$. Substituting these into eq. (4.82), we obtain:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{R}\cdot\left[1 - \frac{R^2}{4}\cdot\frac{3z^2 - r^2}{R^4} + \mathscr{O}\Big(\frac{r^4}{R^4}\Big)\right] \quad , \quad r < R \tag{4.83}$$

If $\vartheta$ is again the angle between $\boldsymbol{r}$ and the $z$-axis, i.e., $\cos\vartheta = z/r$, then this becomes:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{R}\cdot\left[1 - \frac{r^2}{4}\cdot\frac{3\cos^2\vartheta - 1}{R^2} + \mathscr{O}\Big(\frac{r^4}{R^4}\Big)\right] \quad , \quad r < R \tag{4.84}$$

A comparison of eq. (4.74) and eq. (4.84) shows that these approximations of the potential for $r > R$ and $r < R$ are symmetric with respect to the interchange of $r$ and $R$. Although neither approximation strictly applies at $r = R$, both yield the same value for the potential at that point. We will discuss what to make of this later. For now,

however, we are interested in the electric field within the circular disk enclosed by the ring charge. We obtain the potential within this disk from eq. (4.84) by setting $\vartheta = \pi/2$. With $\boldsymbol{\rho} = (\rho\cos\varphi, \rho\sin\varphi, 0)$, the potential within this disk is:

$$\phi(\boldsymbol{\rho}) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{R}\cdot\left[1+\frac{\rho^2}{4R^2}+\mathscr{O}\Big(\frac{\rho^4}{R^4}\Big)\right] \qquad , \quad \rho < R \tag{4.85}$$

Thus, using $\boldsymbol{E} = -\nabla\phi$, we obtain the following expression for the electric field $\boldsymbol{E}_\rho$ within the circular disk enclosed by the ring charge:

$$\boldsymbol{E}_\rho \;\approx\; -\frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{2}\cdot\frac{\rho}{R^3}\cdot\hat{\boldsymbol{\rho}} \qquad , \quad \rho < R \tag{4.86}$$

Here, $\hat{\boldsymbol{\rho}} = (\cos\varphi, \sin\varphi, 0)$ is the radial unit vector in the plane of the circular disk enclosed by the ring. The field lines of $\boldsymbol{E}_\rho$, starting from the charge-carrying ring in the plane of this circular disk, thus converge radially toward the center $M$ of the uniformly charged ring, as sketched in Fig. 4.11. However, they never reach this point, as they veer off into the $+z$- or $-z$-direction beforehand. According to eq. (4.86), the electric field strength $|\boldsymbol{E}_\rho|$ decreases linearly from the ring radius to the center $M$, reaching a value of zero at the center. If we were to consider a negative ring charge instead of a positive one, the pattern of the field lines would look exactly as sketched in Fig. 4.11, with the sole difference that the direction of the field is reversed both within the disk and along the ring axis: for a negative ring charge, $\boldsymbol{E}_\rho$ points radially outward from the center $M$ toward the charged ring within the disk, while $\boldsymbol{E}_{+z}$ and $\boldsymbol{E}_{-z}$ point toward the center $M$ along the ring axis. For both positive and negative ring charges, the electric field vanishes at the center $M$; thus, the electric field strength is zero at that point.

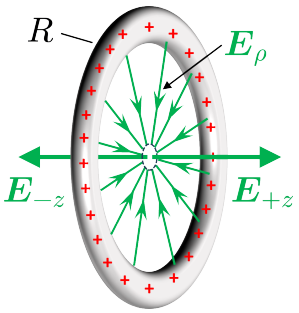



**Fig. 4.11**

At what potential is the charged ring itself? To determine this, we examine the potential in the plane of the ring as a function of the distance from its axis. We already know the behavior of $\phi$ as a function of this distance $\rho$ inside the ring, i.e., for $\rho < R$, from eq. (4.85). We obtain the behavior in the plane of the ring for $\rho > R$ from eq. (4.74) by setting $\vartheta = \pi/2$:

$$\phi(\boldsymbol{\rho}) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{\rho}\cdot\left[1+\frac{R^2}{4\rho^2}+\mathscr{O}\Big(\frac{R^4}{\rho^4}\Big)\right] \qquad , \quad \rho > R \tag{4.87}$$

Eq. (4.86) gives us the path of the field lines for $\rho < R$, and for $\rho > R$ we obtain it via $\boldsymbol{E} = -\nabla\phi$ from eq. (4.87):

$$\boldsymbol{E}(\boldsymbol{\rho}) \;\approx\; \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{\rho^2}\cdot\left[1+\frac{3R^2}{4\rho^2}\right]\cdot\hat{\boldsymbol{\rho}} \qquad , \quad \rho > R \tag{4.88}$$

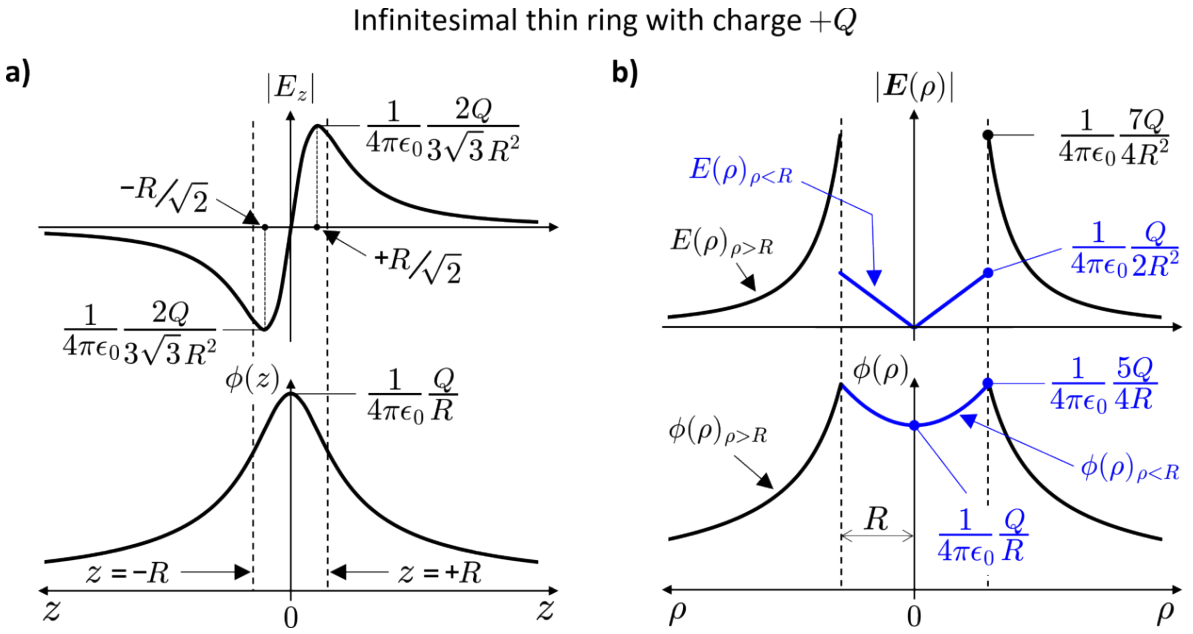



**Fig. 4.12:** Electric field $E_z$ and potential $\phi(z)$ on the axis of a charged ring of radius $R$ (a), as well as the magnitude of the field strength $|\boldsymbol{E}(\boldsymbol{\rho})|$ and potential $\phi(\rho)$ outside and inside the ring in the plane of the ring (b).

Fig. 4.12a shows the variation of the potential $\phi(z)$ and the field strength $E_z$ along the ring axis, which here corresponds to the $z$-axis. The potential on the ring axis is at a maximum at the center of the ring, where the electric field vanishes. $E_z$ changes sign as it passes through zero; the $z$-component of the electric field is positive on the positive $z$-axis and negative on the negative $z$-axis. Fig. 4.12b shows the behavior of the potential $\phi(\rho)$ and the magnitude of the field strength $|\boldsymbol{E}(\boldsymbol{\rho})|$ in the plane of the ring as a function of the distance $\rho$ from the center of the ring, as described by eq. (4.85) and eq. (4.87) for $\phi(\rho)$ and by eq. (4.86) and eq. (4.88) for $|\boldsymbol{E}(\boldsymbol{\rho})|$ for $\rho < R$ and $\rho > R$. Strictly speaking, the approximations we made no longer apply at $\rho = R$. However, the potentials derived for $\rho < R$ and $\rho > 0$ using eq. (4.85) and eq. (4.87) take on the same value at $\rho = R$, providing insight into the behavior of $\phi(\rho)$ and $|\boldsymbol{E}(\boldsymbol{\rho})|$ in the vicinity of $\rho = R$. The potential exhibits a kink at $\rho = R$; it is therefore not differentiable there. The result is that the electric field is discontinuous at $\rho = R$. The behavior of $\phi(\rho)$ and $|\boldsymbol{E}(\boldsymbol{\rho})|$ resembles that of a charged hollow sphere, as shown in fig. 4.2b. If we were to rotate the ring carrying charge $Q$ about an axis passing through its center $M$ within the plane of the ring, we would indeed obtain a charged hollow sphere. With the constant ring potential

$$\phi(R) = \frac{1}{4\pi\epsilon_0} \cdot \frac{5Q}{4R} \tag{4.89}$$

the differential work $\mathrm{d}W$ that must be performed to bring an additional charge $\mathrm{d}q$ onto the ring from infinity is given by:

$$\mathrm{d}W \;=\; [\phi(R) - \phi(\infty)]\mathrm{d}q \tag{4.90}$$

The potential at infinity is zero by definition, and thus:

$$\mathrm{d}W \;=\; \phi(R)\mathrm{d}q \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{5Q}{4R}\cdot\mathrm{d}q \tag{4.91}$$

The total work required to bring the charge on the ring from $Q = 0$ to $Q$ is:

$$W \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{5}{4R}\int_0^Q q\,\mathrm{d}q \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{5}{8}\cdot\frac{Q^2}{R} \tag{4.92}$$

Eq. (4.92) is the result for an infinitesimally thin wire. Such wires do not exist, so we need to examine how this expression for the work required to charge up the ring changes, when we consider a real wire with finite thickness. We know that $W$ in eq. (4.92) is equivalent to the potential energy the fully charged ring possesses. With eq. (4.40) we have already calculated the potential energy for a straight wire segment of length $l$ and charge $\lambda$ per unit length which has a non-vanishing thickness. If our ring of finite thickness is a torus satisfying $R \gg a$, then we can visualize this torus as being composed of short, straight wire segments for which the condition $l \gg a$ still holds. Assembled into a circular ring of length $l = 2\pi R$ and a linear charge density of $\lambda = Q/2\pi R$, eq. (4.40) gives us for the potential energy of the ring of finite thickness the result:

$$U \;=\; \frac{1}{4\pi\epsilon_0}\cdot\left(\frac{Q^2}{4\pi^2R^2}\right)\cdot 2\pi R\cdot\ln\left(\frac{r_0}{a}\right) \;=\; \frac{Q^2}{8\pi^2\epsilon_0 R}\cdot\ln\left(\frac{r_0}{a}\right) \qquad , \quad R \gg a \tag{4.93}$$

This expression for the potential energy of the ring of finite thickness diverges as $a \to 0$. Thus, we cannot expect to derive the expression for the potential energy of the infinitesimally thin ring in eq. (4.92) from eq. (4.93). However, eq. (4.92) and eq. (4.93) exhibit the same $1/R$ dependence and differ ultimately only by a geometric factor. We can use this to gain insight into the magnitude of $r_0$. To do this, one simply needs to set $r_0 = a\cdot\exp(5\pi) = a\cdot 6.64\cdot 10^6$ in eq. (4.93) to see that eq. (4.92) is obtained in this way. Eq. (4.93) applies to a wire made of electrically conducting material. For a wire made of non-conducting material that is homogeneously charged, the interior of the wire also contributes to the potential energy. A circular ring of radius $r \le a$ encloses a fraction $r^2/a^2$ of the charge per unit length $\lambda$ within the wire. Analogous to eq. (4.35), we obtain the field inside the wire using Gauss's law:

$$\int_A \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{\sigma} \;=\; E(r)\cdot 2\pi r\cdot l \;=\; \frac{1}{\epsilon_0}\cdot l\cdot\lambda\cdot\frac{r^2}{a^2} \tag{4.94}$$

and thus

$$E(r) = \frac{\lambda}{2\pi\epsilon_0}\cdot\frac{r}{a^2} \tag{4.95}$$

Using this expression for the electric field inside the wire, we obtain the following for the contribution of the wire's interior to the wire's potential energy:

$$U_I = \frac{\epsilon_0}{2}\int_0^a\int_0^{2\pi}\int_0^l\left(\frac{\lambda}{2\pi\epsilon_0}\frac{r}{a^2}\right)^2 r\,\mathrm{d}r\,\mathrm{d}\varphi\,\mathrm{d}z = \frac{\lambda^2 l}{4\pi\epsilon_0}\ln\left(\frac{r_0}{a}\right) \tag{4.96}$$

For our ring, we again have $l = 2\pi R$ and $\lambda = Q/2\pi R$, and thus we obtain for the contribution of the ring's interior to its potential energy:

$$U_I = \frac{1}{4\pi\epsilon_0}\cdot\left(\frac{Q^2}{4\pi^2R^2}\right)\cdot 2\pi R\cdot\frac{1}{4} = \frac{Q^2}{8\pi^2\epsilon_0 R}\cdot\frac{1}{4} \tag{4.97}$$

The contribution $U_{II}$ to the potential energy of the uniformly charged wire from outside the wire is given, unchanged, by eq. (4.93). Thus we obtain for the total potential energy $U = U_I + U_{II}$ of the uniformly charged wire:

$$U = \frac{Q^2}{8\pi^2\epsilon_0 R}\cdot\left[\frac{1}{4} + \ln\left(\frac{r_0}{a}\right)\right] \quad , \quad R \gg a \tag{4.98}$$

## 4.6 Uniformly Charged Disk

Given what we now know about the electric potential and electric field of a ring carrying an electric charge $Q$, it is only a small step to understand what $\phi$ and $\boldsymbol{E}$ look like for a uniformly charged disk. Fig. 4.13 shows such a uniformly charged disk in the $xy$-plane, with its center $M$ coinciding with the origin of the coordinate system. The disk carries a positive charge $Q$ and has a radius $R$. Again, we first consider the potential at a point $P$ on the $z$-axis. Because of to the same symmetry considerations we discussed when examining the uniformly charged ring, the radial component of the electric field vanishes on the $z$-axis for the uniformly charged disk as well. Now we envision the disk being composed of infinitesimally thin rings. All points on such an infinitesimal thin circular ring of radius $\rho \le R$ within the disk are equidistant from point $P$ on the $z$-axis. Such a circular ring of radius $\rho$ carries the charge:

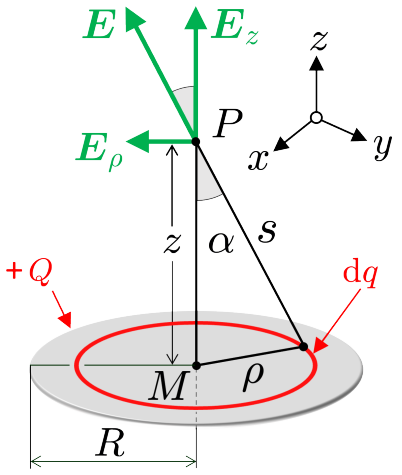



**Fig. 4.13**

$$\mathrm{d}q = \frac{Q}{R^2\pi}\cdot 2\pi\rho\cdot\mathrm{d}\rho = \sigma\cdot 2\pi\rho\cdot\mathrm{d}\rho \tag{4.99}$$

The potential $\mathrm{d}\phi(z)$ generated by such a charge $\mathrm{d}q$ is:

$$\mathrm{d}\phi(z) \;=\; \frac{\sigma}{2\epsilon_0}\cdot\frac{\rho}{|\boldsymbol{s}|}\cdot\mathrm{d}\rho \;=\; \frac{\sigma}{2\epsilon_0}\cdot\frac{\rho\cdot\mathrm{d}\rho}{\sqrt{\rho^2+z^2}} \tag{4.100}$$

To sum the contributions $\mathrm{d}\phi(z)$ from all infinitesimally thin annular rings with $\rho \leq R$, we now integrate this expression from $\rho = 0$ to $\rho = R$ and obtain the potential on the $z$-axis:

$$\phi(z) \;=\; \frac{\sigma}{2\epsilon_0}\cdot\Big[\sqrt{R^2+z^2}-|z|\Big] \tag{4.101}$$

For the electric field in the $z$-direction, we thus obtain, via $\boldsymbol{E}_z = -\partial\phi(z)/\partial z$:

$$E_z \;=\; \frac{\sigma}{2\epsilon_0}\cdot\left[\mathrm{sgn}(z)-\frac{z}{\sqrt{R^2+z^2}}\right]\cdot\hat{\boldsymbol{k}} \quad , \quad \mathrm{sgn}(z) = \begin{cases} +1\,, & z>0 \\ -1\,, & z<0 \end{cases} \tag{4.102}$$

If we let $R$ approach infinity in this equation, thereby considering an infinitely large disk, then eq. (4.102) yields precisely the electric field of the infinitely extended sheet from section 4.4:

$$\boldsymbol{E}_z \;=\; \mathrm{sgn}(z)\cdot\frac{\sigma}{2\epsilon_0}\cdot\hat{\boldsymbol{k}} \tag{4.103}$$

However, the same does not apply to $R\to\infty$ in eq. (4.101), since in this case $\phi(z)\to\infty$ follows. We do, however, obtain the correct result for the potential of an infinitely extended disk by integrating eq. (4.103), namely:

$$\phi(z) \;=\; -\int E(z')\,\mathrm{d}z' \;=\; -\frac{\sigma}{2\epsilon_0}\cdot|z| \tag{4.104}$$

For an infinitely extended, uniformly charged disk, the potential of the disk itself must be constant. However, this is not the case for a disk of finite radius $R$. In this respect, the charged disk differs from an infinitesimally thin ring charge, which is at a constant potential. The reason for the latter is that any charge on the ring, regardless of its position on the ring, sees the same charge configuration surrounding it. The situation is different for a uniformly charged disk. A charge near the center of the disk sees a different charge configuration than one near the edge of the disk. Consequently, the potential of the disk cannot be constant across the disk but will change as one moves from the center of the disk to its edge. What about the electric field? For an infinitesimally thin sheet extending infinitely in the $xy$-plane, the electric field has components only in the $+z$ and $-z$ directions, regardless of whether the sheet consists of a conducting or non-conducting material. If we neglect edge effects then the same applies to our disk. How does this reconcile with the fact that the disk's potential varies along its radius? We know from the discussion of the parallel-plate capacitor that, according to eq. (4.63), the electric voltage between the parallel plates of the capacitor increases as the charge which the

plates carry increases. With eq. (4.61), an increasing voltage implies that the potential difference between the capacitor plates must increase. The situation is similar for our charged disk, except that here the counter-electrode is located at infinity and its potential is, by definition, constantly zero. What changes is the potential between the center and the edge of the disk. This means that annular regions near the center of the capacitor disk are at a higher potential than those near the edge. For circular rings near the center, the density of outgoing field lines directed toward the counter-electrode at infinity is higher than for rings near the edge; however, the electric field always possesses only a $-z$- and a $+z$-component. Thus, there is no radial component of the electric field extending away from the edge of the disk, whether near the disk or within the disk itself. Such a component exists only near the edge of the disk. The variation in the disk potential from the center to the edge arises from the fact that our disk has finite extent. However, this fact itself does not necessitate the existence of a radial component, that is purely an edge effect. For what follows, we will consider the radial component of the electric field in the plane of the disk to be non-existent for $\rho < R$.

Let us now examine what we can determine about the potential on the disk itself. We obtain the potential at the center of the disk, $\phi(0)$, simply by setting $z = 0$ in eq. (4.101):

$$\phi(0) = \frac{\sigma}{2\epsilon_0} \cdot R \tag{4.105}$$

Although this problem exhibits a high degree of symmetry, it is not possible to determine the potential profile along the radius from the center of the disk to its edge as easily as at the center itself. The exception is the edge of the disk, where, with a bit of calculation, we can once again easily determine the potential. To this end, we consider the sketch in fig. 4.14, in which the charge distribution of the disk has been shifted by a distance $R$ in the positive $y$-direction. The $z$-axis, which points out of the plane of the figure in fig. 4.14, thus now passes through a point on the edge of the disk. To obtain the potential at the point $\boldsymbol{r} = 0$, we must integrate over this shifted charge distribution with the constant surface charge density $\sigma = Q/R^2\pi$, specifically over the coordinates of the charge distribution. Since our observation point lies at the origin ($\boldsymbol{r} = 0$), it follows that $|\boldsymbol{r} - \boldsymbol{r}'| \to |\boldsymbol{r}'|$, and $\phi(R)$ is given by:

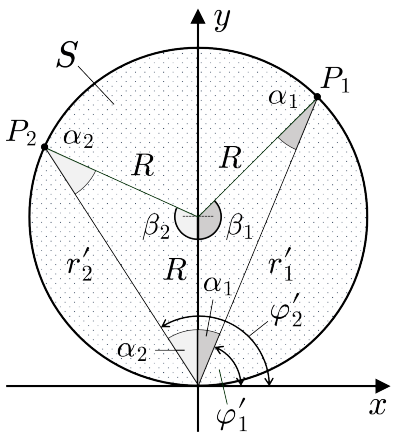



**Fig. 4.14**

$$\phi(\boldsymbol{R}) \;=\; \frac{\sigma}{4\pi\epsilon_0}\cdot\int_S \frac{\mathrm{d}^2 r'}{|\boldsymbol{r}'|} \tag{4.106}$$

With $\mathrm{d}^2 r' = r'\,\mathrm{d}r'\,\mathrm{d}\varphi'$ and $|\boldsymbol{r}'| = r'$, this becomes:

$$\phi(\boldsymbol{R}) \;=\; \frac{\sigma}{4\pi\epsilon_0}\cdot\int_0^{\pi}\int_0^{r(\varphi')} \mathrm{d}r'\,\mathrm{d}\varphi' \;=\; \frac{\sigma}{4\pi\epsilon_0}\cdot\int_0^{\pi} r(\varphi')\,\mathrm{d}\varphi' \tag{4.107}$$

The limits of integration for $\varphi'$ can be easily read from fig. 4.14. However, the upper limit of integration for $r'$ obviously depends on $\varphi'$, so we need to examine this more closely. Let us first consider the point $P_1$ in the right half of the disk in fig. 4.14. For any such point $P_1$ in the right half of the disk, we are dealing with an isosceles triangle, and thus $2\alpha_1 + \beta_1 = 180°$ holds. Since $\alpha_1 = 90° - \varphi_1'$ also holds, it follows immediately that $\beta_1 = 2\varphi_1'$. We now apply the law of sines using the angles $\beta_1$ and $\alpha_1$ and the opposite sides $r_1'$ and $R$:

$$\frac{\sin\beta_1}{r_1'} \;=\; \frac{\sin\alpha_1}{R}$$

If we substitute $\beta_1 = 2\varphi_1'$ and $\alpha_1 = 90° - \varphi_1'$ here, we obtain:

$$\frac{\sin 2\varphi_1'}{r_1'} \;=\; \frac{\sin(\pi/2 - \varphi_1')}{R} \;=\; \frac{\cos\varphi_1'}{R}$$

It follows that

$$r_1' \;=\; 2R\sin\varphi_1'$$

Let us now consider points $P_2$ in the left half of the disk in fig. 4.14. Here, $\alpha_2 = \varphi_2' - 90°$ holds, and for the isosceles triangle, the relation $2\alpha_2 + \beta_2 = 180°$ applies again. It follows that in the left half of the disk, $\beta_2 = 2\pi - 2\varphi_2'$. We again use the law of sines, this time for the angles $\beta_2$ and $\alpha_2$ and the opposite sides $r_2'$ and $R$:

$$\frac{\sin\beta_2}{r_2'} \;=\; \frac{\sin\alpha_2}{R}$$

If we substitute $\beta_2 = 2\pi - 2\varphi_2'$ and $\alpha_2 = \varphi_2' - 90°$ here, we obtain, using $\sin(2\pi - 2\varphi_1') = -\sin 2\varphi_2'$ and $\sin(\varphi_1' - \pi/2) = -\cos\varphi_2'$, the following:

$$-\frac{\sin 2\varphi_2'}{r_2'} \;=\; -\frac{\cos\varphi_2'}{R}$$

It follows:

$$r_2' \;=\; 2R\sin\varphi_2'$$

The magnitude of $\boldsymbol{r}'$ thus exhibits the same angular dependence in both halves of the disk; consequently, $r(\varphi) = 2R\sin\varphi$ in eq. (4.107), and for the potential at the edge of the disk, we obtain:

$$\phi(\boldsymbol{R}) \;=\; \frac{\sigma}{4\pi\epsilon_0}\cdot 2R\cdot \int_0^{\pi} \sin\varphi'\,\mathrm{d}\varphi' \;=\; \frac{\sigma}{2\epsilon_0}\cdot\frac{2R}{\pi} \tag{4.108}$$

In principle, we can also determine the potential for other points between the center and the edge of the disk by shifting the $z$-axis to any location $0 < y_0 < R$ of interest on the disk's radius along the $y$-axis. However, this leads to so-called elliptic integrals, for which analytical solution do not exist. The edge of the disk happens to be a special case because, for $y_0 = R$, the respective elliptic integrals reduce to a simple integral that has an analytical solution. The potential at the center of the disk is greater by a factor of $\pi/2$ than at the edge of the disk. This factor of $\pi/2$ tells us that we are not dealing with a simple linear decrease in potential along the disk radius from $\phi(0)$ to $\phi(R)$. Near the center of the disk, the local charge environment changes less than it does closer towards the edge of the disk. Consequently, the drop in potential near the center will be smaller than it is towards the edge.

Let us now examine the potential produced by the uniformly charged disk for $r < R$ and $r > R$. We begin with $z > R$. To understand the behavior of $\phi(z)$ for $z \gg R$, we rewrite eq. (4.101) as ($\sigma = Q/R^2\pi$):

$$\phi(z) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{2Q}{R^2}\cdot|z|\cdot\left[\sqrt{1+\frac{R^2}{z^2}}-1\right] \tag{4.109}$$

For $z \gg R$, we again use the Taylor expansion of the square root, taking into account terms up to $z^{-4}$:

$$\sqrt{1+\frac{R^2}{z^2}} = 1+\frac{1}{2}\cdot\frac{R^2}{z^2}-\frac{1}{8}\cdot\frac{R^4}{z^4}+\mathscr{O}\Big(\frac{R^8}{z^8}\Big) \tag{4.110}$$

Substituting this into eq. (4.109) gives us for the potential on the z-axis in the far field the expression:

$$\phi(z) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\left[\frac{Q}{|z|}-\frac{1}{4}\cdot\frac{Q\cdot R^2}{|z|^3}+\mathscr{O}\Big(\frac{R^6}{|z|^7}\Big)\right] \tag{4.111}$$

The charged disk possesses the same rotational symmetry as the charged ring. Therefore, we can express the far-field of the charged disk in terms of a monopole term $\phi_0(\boldsymbol{r})$ and a quadrupole term $\phi_2(\boldsymbol{r})$, just as we did for the charged ring in eq. (4.71). If we set there again $\vartheta = 0$ and $r = z$, we obtain for the far-field of the charged disk on the $z$-axis the expression:

$$\phi(z) \;=\; \phi_0(z)+\phi_2(z) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\left[\frac{Q}{z}+Q_0\cdot\frac{2}{z^3}\right] \tag{4.112}$$

Comparing the coefficients of the powers of $z$ in eq. (4.111) and eq. (4.112) shows us that the leading terms are identical and that the following must hold:

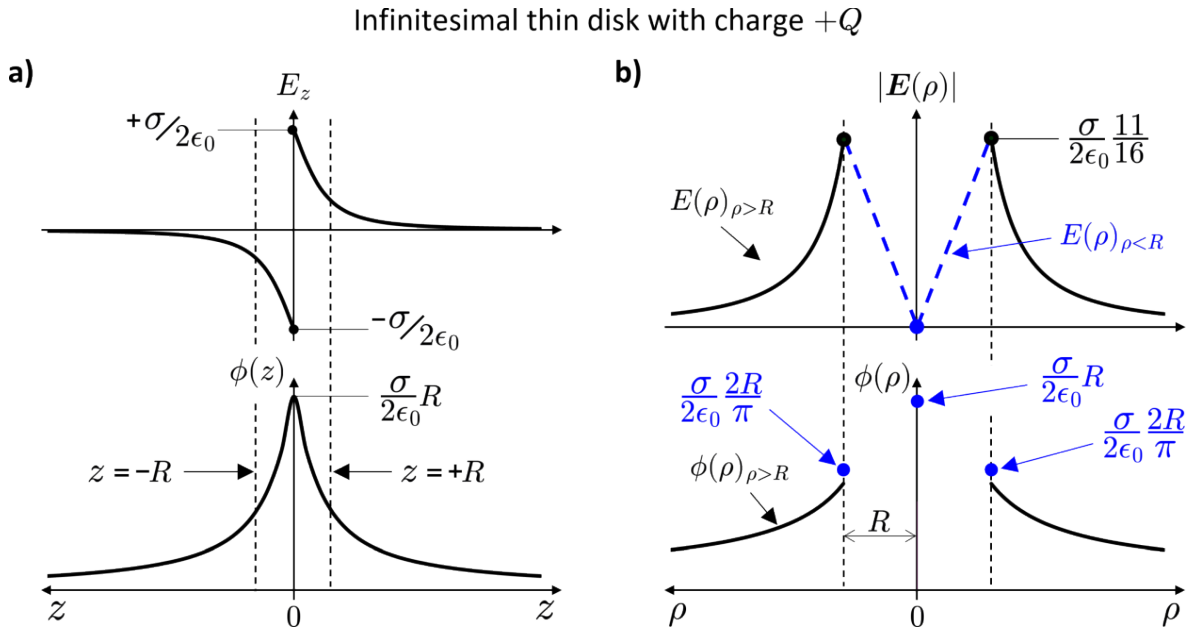



**Fig. 4.15:** Electric field $E_z$ and potential $\phi(z)$ on the axis of a uniformly charged disk of radius $R$ and constant surface charge density $\sigma = Q/R\pi^2$ (a), as well as the magnitude of the field strength $|\boldsymbol{E}(\boldsymbol{\rho})|$ and potential $\phi(\rho)$ outside and inside the disk within the plane of the disk (b).

$$Q_0 = -\frac{Q \cdot R^2}{8} \tag{4.113}$$

so that the quadrupole terms are also identical. Thus, for the charged disk, $Q_0$ is half the value of $Q_0$ for the charge carrying ring in eq. (4.72). The quadrupole tensor for the charged disk is therefore:

$$\underline{\underline{\boldsymbol{Q}}} = \frac{Q \cdot R^2}{8} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \tag{4.114}$$

Substituting the charge $Q_0$ of the charged disk into eq. (4.71), we obtain the angle-dependent potential of the charged disk in the far-field approximation $r \gg R$:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r} \cdot \left[1 - \frac{R^2}{8} \cdot \frac{3\cos^2\vartheta - 1}{r^2} + \mathscr{O}\left(\frac{R^4}{r^4}\right)\right] \quad , \quad r > R \tag{4.115}$$

With this equation, we can now examine the electric potential and the electric field in the plane of the disk. To do this, we set $\vartheta = \pi/2$ in eq. (4.115) and thus obtain, with $\boldsymbol{\rho} = (\rho\cos\varphi, \rho\sin\varphi, 0)$:

$$\phi(\boldsymbol{\rho}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{\rho} \cdot \left[1 + \frac{R^2}{8\rho^2} + \mathscr{O}\left(\frac{R^4}{\rho^4}\right)\right] \quad , \quad \rho > R \tag{4.116}$$

From this potential we obtain for the electric field in the plane of the disk:

$$\boldsymbol{E}(\boldsymbol{\rho}) \approx \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{\rho^2}\cdot\left[1+\frac{3R^2}{8\rho^2}\right]\cdot\hat{\boldsymbol{\rho}} \qquad , \quad \rho > R \tag{4.117}$$

Fig. 4.15a shows the variation of the potential $\phi(z)$ and the field strength $E_z$ along the central axis of the disk, which in our coordinate system corresponds to the $z$-axis. The potential along the disk's axis reaches its maximum at the center of the disk. The magnitude of the $E_z$ component also attains its greatest value there. The electric field directly above and below the infinitesimally thin disk possesses only a $z$-component; as is the case with an infinite planar surface charge, this component points in opposite directions on the two sides of the disk. Fig. 4.15b shows the behavior of the potential $\phi(\rho)$ and the magnitude of the field strength $|\boldsymbol{E}(\boldsymbol{\rho})|$ in the plane of the disk as a function of the distance $\rho$ from the disk's center for $\rho \leq R$. Here, we were able to determine the potential of the disk analytically at only two points: the center of the disk and its edge. However, this is sufficient for us to see that, for $\rho \leq R$, the potential profile of the uniformly charged disk must resemble that of the uniformly charged solid sphere shown in fig. 4.2a. Since the potential of the disk is not constant for $\rho \leq R$ as one moves from the center of the disk to its edge, the radial electric field within the disk cannot vanish. We know that the electric field vector always points from regions of higher potential to regions of lower potential. The potential is highest at the center of the disk and lowest at its edge. This implies that the electric field vector must point radially outward, away from the center. However, this in turn implies that the radial electric field of the disk must vanish at the center.

The radial electric field strength within the disk must therefore follow the same profile as the radial electric field of a uniformly charged solid sphere, as shown in fig. 4.2a. The corresponding pattern of the $\boldsymbol{E}$-field lines, both in the plane of the charged disk (the $xy$-plane) and in the $yz$-plane, is sketched qualitatively in fig. 4.16. At the center of the disk, the electric field has only a $z$-component. Moving from the center of the disk toward the edge causes the radial field component to increase. The radial behavior of the potential and the electric field in the plane of the disk, shown in fig. 4.15b for $\rho \geq R$, is described by eq. (4.116) and eq. (4.117). Unlike the case of the charged ring, the far-field approximation in eq. (4.116) does not yield the same value for the potential at $\rho = R$ as the value determined by

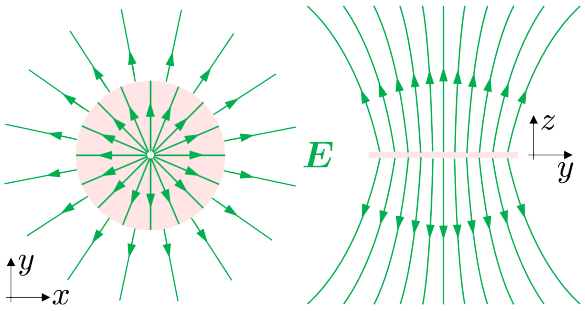



**Fig. 4.16**

shifting the $z$-axis to the edge. The latter was determined without approximation and must therefore be the correct value. The value yielded by the far-field approximation in eq. (4.116) for $\rho = R$ is 11.6% lower than the actual value of the potential at the edge of the disk, $\phi(R)$, as given by eq. (4.108). This should come as no surprise, since, strictly speaking, the far-field approximation is not valid for $\rho = R$. For the electric field, we cannot directly see in fig. 4.15b that the far-field approximation deviates from the correct value at $\rho = R$, because the electric field undergoes a jump at $\rho = R$ anyway. However, we can assume that eq. (4.117) also deviates somewhat from the correct value at $\rho = R$. Conversely, the jump in $E_\rho$ at $\rho = R$ tells us that the potential cannot be differentiable at $\rho = R$, meaning it must have a kink, just as was the case for the uniformly charged solid sphere in fig. 4.2b. Outside the disk, the electric field strength in the plane of the disk falls off as $1/\rho^2$ according to eq. (4.117), whereas inside the disk, it decreases more rapidly.

Finally, it is time to examine the potential energy of the charged disk. We know that the potential of the disk itself is not constant, yet we only know the value of the potential at two points on the disk: at the center and at the edge. However, this suffices, because to calculate the potential energy of an infinitesimally thin disk, we need only know the potential at the edge. Our disk is two-dimensional, and we enlarge it by incrementally adding new rings of charge at the disk's edge. In doing so, we bring this charge, within the plane of the disk, in from zero potential at infinity to the potential at the disk's edge. Using eq. (4.108), we know the potential at the edge of a disk with radius $r$ and constant surface charge density $\sigma$:

$$\phi(r) = \frac{\sigma r}{\pi \epsilon_0} \tag{4.118}$$

For $\sigma$ to remain constant upon the addition of an infinitesimal ring of charge with charge $\mathrm{d}q$, the following condition must hold for the charge brought in from infinity:

$$\mathrm{d}q = 2\pi r \sigma \, \mathrm{d}r \tag{4.119}$$

The work $W$ thus required to assemble a charged disk of radius $R$ and constant surface charge density $\sigma$ in this manner, and with that also its potential energy $U$, is therefore given by:

$$W = U = \int_0^R \frac{\sigma r}{\pi \epsilon_0} 2\pi r \sigma \, \mathrm{d}r = \frac{2\sigma^2}{\epsilon_0} \int_0^R r^2 \, \mathrm{d}r = \frac{2\sigma^2 R^3}{3\epsilon_0} = \frac{2Q^2}{3\pi^2 \epsilon_0 R} \tag{4.120}$$

So far, we have considered a uniformly charged, infinitesimally thin disk with a constant surface charge density $\sigma$. By definition, such a uniformly charged disk must consist of

a non-conducting material, meaning, the charges must be fixed within this material, as their mutual repulsion would otherwise lead to a redistribution of these charges. This charge redistribution is precisely what occurs in a conducting disk of finite radius. As we already know from the example of a solid sphere made of conductive material, charges distribute themselves across the body's surface in such a way that the electric field inside the conductor vanishes. The situation is no different for a circular disk made of electrically conductive material. In the case of our infinitesimally thin disk, this means that charge accumulates at the disk's edge; consequently, the surface charge density of a conductive disk cannot be constant. But what does it look like?

To understand this, let us consider what happens when we compress the upper and lower halves of a hollow sphere of radius $R$, carrying a total charge $Q$, vertically onto the circular disk connecting them. Since the field continues to vanish in the interior during this compression, we obtain a circular disk within which the electric field must be zero. We achieve this compression by projecting the charge located on the upper and lower halves of the hollow sphere onto the circular disk connecting them. As illustrated in fig. 4.17, such a projection maps an arc element $\Delta s$ of the spherical shell onto a radial length element $\Delta s'$ within the circular disk. Using the polar angle $\vartheta$, we deduce the relationship $\Delta s' = \Delta s \cdot \cos\vartheta$ from fig. 4.17. We obtain the arc element $\Delta s$ by noting from fig. 4.17 that $\sin\vartheta = r/R$ and by rewriting the surface element $\mathrm{d}S$ of the hollow sphere of radius $R$ as follows:

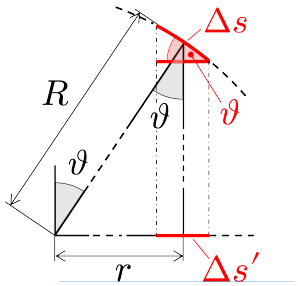



**Fig. 4.17**

$$\mathrm{d}S = 2\pi R^2 \sin\vartheta\,\mathrm{d}\vartheta \;=\; 2\pi R^2 \frac{r}{R}\,\mathrm{d}\vartheta \;=\; 2\pi r \cdot R\,\mathrm{d}\vartheta \;=\; 2\pi r \cdot \Delta s \tag{4.121}$$

Here, we have already utilized axial symmetry and replaced $\mathrm{d}\varphi$ with the factor $2\pi$. On the right-hand side of eq. (4.121), the area element now appears as the product of the circular ring of radius $r$ and the arc element $\Delta s$. The corresponding area element on the circular disk is $\mathrm{d}S' = 2\pi r \cdot \Delta s'$; thus, the following applies to the area element on the sphere and the area element $\mathrm{d}S'$ projected onto the circular disk:

$$\mathrm{d}S' \;=\; \mathrm{d}S \cdot \cos\vartheta \tag{4.122}$$

Given the surface charge density of the sphere, $\sigma_\circ = Q/4\pi R^2$, the charge $Q = \sigma_\circ \mathrm{d}S$ of the surface element $\mathrm{d}S$ of the hollow sphere is located after projection within the surface element $\mathrm{d}S'$ of the circular disk. Thus, the following holds:

$$\sigma(r)\,\mathrm{d}S \;=\; 2\sigma_\circ \mathrm{d}S' \qquad \Rightarrow \qquad \sigma(r) \;=\; \frac{Q}{4\pi R^2 \cos\vartheta} \tag{4.123}$$

The factor of 2 arises because we project the charges from the upper and lower halves of the hollow sphere onto the circular disk. From fig. 4.17, we see that $\cos\vartheta = \sqrt{R^2 - r^2}/R$, and thus obtain the following expression for $\sigma(r)$:

$$\sigma(r) = \frac{Q}{2\pi R\sqrt{R^2 - r^2}} \tag{4.124}$$

We verify this result by integrating $\sigma(r)$ over the disk to see whether we obtain the total charge of the disk:

$$\frac{Q}{2\pi R}\int_0^R \frac{1}{\sqrt{R^2 - r^2}} \cdot 2\pi r\,\mathrm{d}r = -\frac{Q}{R}\sqrt{R^2 - r^2}\Big|_0^R = Q$$

The result of this calculation is, as expected, the total charge $Q$. To obtain the constant potential of the disk, the easiest way is to determine it at the center of the disk, that is, at $r = 0$:

$$\begin{aligned}\phi(0) &= \frac{1}{4\pi\epsilon_0}\int_0^R \frac{Q}{2\pi R\sqrt{R^2 - r'^2}}\frac{2\pi r'\,\mathrm{d}r'}{|\boldsymbol{r}'|} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R}\int_0^R \frac{\mathrm{d}r'}{\sqrt{R^2 - r'^2}} \\ &= \frac{1}{4\pi\epsilon_0}\frac{Q}{R}\cdot\operatorname{asin}\left(\frac{r}{R}\right)\Big|_0^R = \frac{1}{4\pi\epsilon_0}\frac{Q}{R}\cdot\frac{\pi}{2} = \frac{Q}{8\epsilon_0 R}\end{aligned} \tag{4.125}$$

All points on the disk are at this potential. With $\phi(\infty) = 0$, the potential difference between any point on the disk and a point at infinity is therefore $U = \phi(0) - \phi(\infty)$. Using eq. (4.125), we thus obtain the following expression for the capacitance of the infinitesimally thin, charged, and conducting disk of radius $R$:

$$C = \frac{Q}{U} = 8\epsilon_0 R \tag{4.126}$$

We obtain the potential energy of the disk by successively bringing charges from infinity (where $\phi(\infty) = 0$) to the potential of the disk until it carries the charge $Q$. The work done in the process is:

$$W = \frac{1}{8\epsilon_0 R}\int_0^Q q\,\mathrm{d}q = \frac{Q^2}{16\epsilon_0 R} = \frac{Q^2}{2C} \tag{4.127}$$

Comparing this result with that for the uniformly charged disk in eq. (4.92), we see that, for the same charge and radius, the potential energy of the uniformly charged disk is approximately 8% greater than that of the disk made of conducting material. Conversely, this implies that the capacitance of the uniformly charged disk must be lower than that of the conducting disk.

# 5. Conductors in Electrostatic Fields

In the last two chapters, we covered the fundamentals of electrostatics and then examined some simple applications of these basic equations, specifically, Gauss's law for electrostatics. The latter is particularly useful in situations characterized by a high degree of symmetry in the charge distribution and, consequently, in the electrostatic potential $\phi$ and the electrostatic field $\boldsymbol{E}$. However, we often have little choice but to solve Poisson's equation in order to understand the behavior of the potential and the field in certain situations. So far, we have only considered situations in which an electrically charged body is located in an infinitely extended space. But what happens when this is not the case, or, for example, when there are multiple charged bodies in a given finite space, each at a different electrostatic potential $\phi$? Or what happens if we introduce an arbitrary boundary around one of the charge-carrying bodies we have considered so far? Conversely, we must also ask what happens when we place an electrically uncharged body into an electrostatic field. What happens there? How is such a situation handled? In this context, when we speak of a finite volume, we refer to a limited volume that is physically bounded by conductive or non-conductive surfaces that establish specific boundary conditions for the behavior of electrostatic fields at these interfaces. This gives rise to the so-called boundary-value problems for electrostatic fields that must be solved. However, as we shall see, even in those situations there are other approaches, such as the method of images charges, which can frequently help us find solutions to electrostatic problems in a comparatively simple way, without having to resort to solving Poisson's equation directly.

## 5.1 Direct Solution of the Poisson Equation

The field of electrostatics is essentially concerned with solutions to the Poisson equation eq. (4.1) for specific, prescribed charge distributions, each subject to particular boundary conditions. Once the charge distribution is known, the electrostatic potential $\phi(r)$ can

be determined by integration in accordance with eq. (4.3). The electric field $\boldsymbol{E}$, as the gradient of the potential, is then obtained through simple differentiation. We will now examine this procedure in greater detail using the simple example of a spherically symmetric charge distribution.

### 5.1.1 The Radially Symmetric Charge Distribution

We consider a spherically symmetric charge distribution centered at the coordinate origin, i.e., $\rho(\boldsymbol{r}) \to \rho(r)$. While the charge distribution $\rho(r)$ may indeed vary with distance, this variation depends, as sketched in fig. 5.1, solely on the magnitude of the position vector $r = |\boldsymbol{r}|$ and not on the direction of the position vector $\boldsymbol{r}$. It therefore suffices to consider the problem for a given direction of $\boldsymbol{r}$. Conveniently, we align this direction of $\boldsymbol{r}$ with the $z$-axis of our coordinate system, because the angle enclosed by $\boldsymbol{r}$ and $\boldsymbol{r}'$ then corresponds exactly to $\vartheta'$, i.e., the angle in the primed spherical coordinate system $(r', \vartheta', \varphi')$ over which we must integrate. Applying the law of cosines to $|\boldsymbol{r} - \boldsymbol{r}'|$ in eq. (4.3), we therefore have:

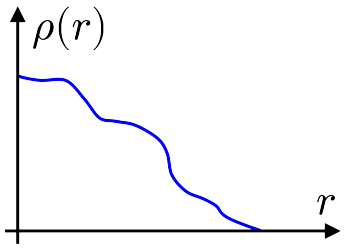


**Fig. 5.1**

$$|\boldsymbol{r} - \boldsymbol{r}'| = \sqrt{r^2 + r'^2 - 2rr'\cos\vartheta'}$$

Substituting into eq. (4.3), we obtain the integral for the electrostatic potential at a distance $r$ from the origin (there are no discrete charges):

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \frac{\rho(r')}{\sqrt{r^2 + r'^2 - 2rr'\cos\vartheta'}} \cdot r'^2 \sin\vartheta' \,\mathrm{d}\varphi' \,\mathrm{d}\vartheta' \,\mathrm{d}r' \tag{5.1}$$

The integrand does not depend on $\varphi'$; thus, the corresponding integration yields a factor of $2\pi$. With the substitution $u = \cos\vartheta'$, $\mathrm{d}u = -\sin\vartheta' \,\mathrm{d}\vartheta'$, and the change of integration limits $\vartheta = 0 \to u = 1$ and $\vartheta = \pi \to u = -1$, we obtain:

$$\phi(r) = -\frac{1}{2\epsilon_0} \int_1^{-1} \int_0^{\infty} \frac{\rho(r')}{\sqrt{r^2 + r'^2 - 2rr'u}} \cdot r'^2 \,\mathrm{d}u \,\mathrm{d}r' \tag{5.2}$$

The integration with respect to $u$ yields:

$$\begin{aligned} \phi(r) &= -\frac{1}{2\epsilon_0} \int_0^{\infty} \left[ -\frac{1}{rr'} \sqrt{r^2 + r'^2 - 2rr'u} \right]_1^{-1} \cdot \rho(r') r'^2 \,\mathrm{d}r' \\ &= \frac{1}{2\epsilon_0 r} \int_0^{\infty} \Big[ (r + r') - |r - r'| \Big] \cdot \rho(r') r' \,\mathrm{d}r' \end{aligned} \tag{5.3}$$

For $r' < r$, the expression in parentheses is equal to $2r'$, and for $r' > r$, it is equal to $2r$. We therefore split the integral into two parts:

$$\phi(r) \;=\; \frac{1}{\epsilon_0 r}\int_0^r \rho(r')\cdot r'^{\,2}\,\mathrm{d}r' + \frac{1}{\epsilon_0}\int_r^\infty \rho(r')\cdot r'\,\mathrm{d}r' \tag{5.4}$$

Apart from the fact that it is radially symmetric, we have made no further assumptions regarding the charge distribution so far. However, eq. (5.4) now shows us that $\rho(r')$ must decay faster than $r'^{-2}$, as the integral would otherwise not converge. If a charge distribution satisfying this condition is substituted into eq. (5.4), then $\phi(r)$ can be calculated via analytical or numerical integration. Here, however, we turn directly to the electric field resulting from eq. (5.4). Since $\phi$ depends only on $r$, the gradient in spherical coordinates (see tab. A.1, Appendix A) simplifies eq. (3.19) to:

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; -\nabla\phi(\boldsymbol{r}) \;=\; -\frac{\boldsymbol{r}}{|\boldsymbol{r}|}\cdot\frac{\partial}{\partial r}\phi(r) \;=\; \hat{\boldsymbol{r}}\cdot\frac{\partial}{\partial r}\phi(r)$$

Substituting $\phi(r)$ and differentiating then gives us:

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{\hat{\boldsymbol{r}}}{\epsilon_0}\cdot\left[\frac{1}{r^2}\int_0^r \rho(r')\cdot r'^{\,2}\,\mathrm{d}r' - \frac{1}{r}\frac{\partial}{\partial r}\int_0^r \rho(r')\cdot r'^{\,2}\,\mathrm{d}r' - \frac{\partial}{\partial r}\int_r^\infty \rho(r')\cdot r'\,\mathrm{d}r'\right] \tag{5.5}$$

Leibniz's integral rule applies to differentiating the limits of integration:

$$\frac{\mathrm{d}}{\mathrm{d}r}\int_{a(r)}^{b(r)} f(r,r')\,\mathrm{d}r' \;=\; f\big(r,b(r)\big)\frac{\partial b(r)}{\partial r} - f\big(r,a(r)\big)\frac{\partial a(r)}{\partial r} + \int_{a(r)}^{b(r)}\frac{\partial}{\partial r}f(r,r')\,\mathrm{d}r' \tag{5.6}$$

The integrands in eq. (5.5) do not depend on $r$, and the last term on the right-hand side of the integral rule therefore drops out. Consequently, we apply in eq. (5.5)

$$\frac{\partial}{\partial r}\int_{a(r)}^{b(r)} f(r,r')\,\mathrm{d}r' \;=\; f\big(r,b(r)\big)\frac{\partial b(r)}{\partial r} - f\big(r,a(r)\big)\frac{\partial a(r)}{\partial r} \tag{5.7}$$

and thus obtain

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{\hat{\boldsymbol{r}}}{\epsilon_0}\cdot\left[\frac{1}{r^2}\int_0^r \rho(r')\cdot r'^{\,2}\,\mathrm{d}r' - \frac{1}{r}\big(\rho(r)\cdot r^2 - 0\big) - \big(0 - \rho(r)\cdot r\big)\right] \tag{5.8}$$

giving us the final result:

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{\hat{\boldsymbol{r}}}{\epsilon_0 r^2}\int_0^r \rho(r')\cdot r'^{\,2}\,\mathrm{d}r' \tag{5.9}$$

As a first example, we now apply eq. (5.4) and eq. (5.9) to the uniformly charged sphere, for which we have already determined the electric field and the electrostatic potential

using Gauss's law. The total charge $Q$ of the sphere is uniformly distributed over the spherical volume $V$, and we describe this charge distribution by:

$$\rho(r) \;=\; \rho_0 \Theta(R-r) \qquad \text{with} \qquad \rho_0 \;=\; \frac{Q}{V} \;=\; \frac{3Q}{4\pi R_0^3}$$

Here, $\Theta(R-r)$ is the step function and $R_0$ is the radius of the sphere. Substituting this charge distribution into eq. (5.4), we obtain for the potential the expression:

$$\phi(r) \;=\; \frac{1}{4\pi\epsilon_0}\frac{3Q}{R_0^3}\left[\frac{1}{r}\int_0^r \Theta(R_0-r')\cdot r'^2\,\mathrm{d}r' + \int_r^\infty \Theta(R_0-r')\cdot r'\,\mathrm{d}r'\right] \tag{5.10}$$

Let us first consider $r < R_0$. In this case, this equation becomes:

$$\phi(r) \;=\; \frac{1}{4\pi\epsilon_0}\frac{3Q}{R_0^3}\left[\frac{1}{r}\int_0^r r'^2\,\mathrm{d}r' + \int_r^{R_0} r'\,\mathrm{d}r'\right] \;=\; \frac{1}{8\pi\epsilon_0}\frac{Q}{R_0}\left(3-\frac{r^2}{R_0^2}\right) \tag{5.11}$$

This, however, is precisely the result expressed by eq. (4.15) that we derived there using Gauss's law of electrostatics. For the case $r > R_0$, one can quickly see that eq. (5.10) implies:

$$\phi(r) \;=\; \frac{1}{4\pi\epsilon_0}\frac{3Q}{R_0^3}\left[\frac{1}{r}\int_0^{R_0} r'^2\,\mathrm{d}r' + 0\right] \;=\; \frac{1}{4\pi\epsilon_0}\frac{Q}{r} \tag{5.12}$$

As expected, this matches the result of eq. (4.9) which we obtained by applying to the problem Gauss's law of electrostatics. We can obtain the corresponding electric field either by calculating the gradient of the potential $\phi(r)$ or by evaluating the integral in eq. (5.9). Just as shown here for the solid sphere, we can calculate the electrostatic potential and the electric field for the hollow sphere using eq. (5.4) and eq. (5.9). When evaluating eq. (5.9), we adopt the following ansatz for the charge distribution of the hollow sphere:

$$\sigma(r) \;=\; \sigma_0 \cdot \delta(R_0-r) \qquad \text{with} \qquad \sigma_0 \;=\; \frac{Q}{4\pi R_0^2}$$

For an infinitely long wire or an infinitely extended charged surface, one can proceed in the same way as was done here for the solid sphere, while making use of the symmetry properties of the respective charge distributions. The examples of electrostatic problems we have addressed so far, using Gauss's law of electrostatics, or, in the case of the spherically symmetric charge distribution, by solving the Poisson equation, share a common feature: the electrostatic fields under consideration are generated by the charged bodies themselves, and there are no other electrostatic fields that need to be taken into account. But what happens if, for example, we place a charged or uncharged body into an electrostatic field?

## 5.2 The Ideal Conductor in an Electrostatic Field

An ideal conductor consists of a material in which charge carriers such as electrons can move without energy loss. Or, put differently, an ideal conductor must consist of a material with zero electrical resistance. In fact, there are materials whose electrical resistance can vanish under certain conditions; however, for various reasons, they are still rarely used. We are interested here in everyday materials, such as certain metals, which, while still possessing some electrical resistance, have such low resistance that we can treat them as ideal conductors. Therefore, when we speak of ideal conductors here, we are referring to metals such as silver, copper, gold, or aluminum. Metals are characterized by the fact that they possess some electrons which are not bound to individual metal atoms but can move freely within the metal as so-called conduction electrons. The free mobility of these conduction electrons within the metal accounts for the high electrical conductivity of metals. When a metal is placed in an electrostatic field, the conduction electrons rapidly redistribute themselves across the metal's surface under the influence of the external field, thereby cancelling out the electric field penetrating the metal. This results in the disappearance of the electric field within the metal and the establishment of a constant electrostatic potential. These electric surface charges are also referred to as electrostatic induction charges, because an external electric field causes them (is inducting them) to move to the surface of the metal. If electric charges are applied to a metal sphere, they likewise distribute themselves across its surface in such a way that the field inside the sphere vanishes. If that were not the case, there would be an electric field inside the sphere, and under the influence of this field, the conduction electrons would rapidly rearrange themselves so that the field inside the sphere vanishes.

As sketched in fig. 5.2, the electric field $\boldsymbol{E}$ vanishes in the interior of a metal body, and the electrostatic potential $\phi$ is constant both in the interior and on the surface of the metal body. The surface of the metallic conductor $M_S$ constitutes always an equipotential surface. If one moves along an equipotential surface, no work is required to do that, because the change in potential energy $\mathrm{d}U$ on this surface is zero by definition. And with $\mathrm{d}W + \mathrm{d}U = 0$, this also implies that no work is performed when a charge is moved along an equipotential surface or an equipotential line. With $\mathrm{d}W = 0$, it follows immediately that:

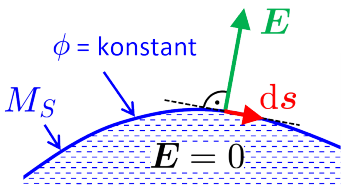



**Fig. 5.2**

$$\mathrm{d}W = \boldsymbol{F}\,\mathrm{d}\boldsymbol{s} = q\boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = 0 \qquad \text{(on equipotential surfaces)} \tag{5.13}$$

The component of $\boldsymbol{E}$ parallel to the path $\mathrm{d}\boldsymbol{s}$ on an equipotential surface or along an equipotential line must therefore vanish. This gives us a first boundary condition for the electric field at surfaces of conductors:

$$\boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{s} \;=\; \boldsymbol{E}(\boldsymbol{r})_{\parallel} \;=\; 0 \qquad \text{if} \qquad \boldsymbol{r} \in M_S \tag{5.14}$$

This means that the field lines of the electric field $\boldsymbol{E}$ intersect equipotential surfaces or equipotential lines perpendicularly. The metal surface $M_S$ in fig. 5.2 and fig. 5.3 is such an equipotential surface and because of that the electric field vector at the metal surface is oriented perpendicular to it everywhere. To determine this normal component of the electric field at the metal surface, we again employ Gauss's law of electrostatics. To do this, as shown in fig. 5.3, we enclose a small section of the metal surface $M_S$ within a so-called Gaussian pillbox, thereby enclosing a charge which scales with the respective surface charge density $\sigma(\boldsymbol{r})$. The end faces of this cylindrical pillbox with volume $V$ and cylinder height $h$ are positioned symmetrically with respect to the metal surface $M_S$, and the pillbox is assumed to be small enough that the unit normal vector $\hat{\boldsymbol{n}}_S$ to the conductor surface $M_S$ is also the normal vector to the end face $A$ of the pillbox. Since the electric field inside the conductor is zero, the flux of $\boldsymbol{E}$ through the boundary surfaces $\partial V$ of the volume $V$ is non-zero only for the top surface $A$. Thus, the right-hand side of Gauss's theorem from eq. (3.15) becomes:

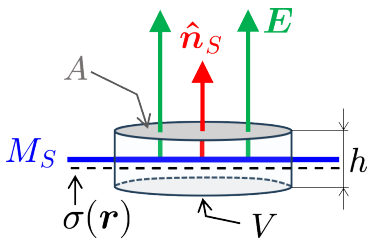


**Fig. 5.3**

$$\int_{\partial V} \boldsymbol{E}(\boldsymbol{r})\cdot\hat{\boldsymbol{n}}_s\,\mathrm{d}\sigma \;=\; \int_{A} \boldsymbol{E}(\boldsymbol{r})\cdot\hat{\boldsymbol{n}}_s\,\mathrm{d}A \tag{5.15}$$

Here, $\mathrm{d}A$ is the area element of the cylinder cap of the Gaussian pillbox, i.e., of the surface $A$. For the left-hand side of Gauss's theorem from eq. (3.15), we obtain, in cylindrical coordinates with the volume element $r\,\mathrm{d}r\,\mathrm{d}\varphi\,\mathrm{d}z = \mathrm{d}A\,\mathrm{d}z$ and with the metal surface at $z = h/2$:

$$\frac{1}{\epsilon_0}\int_V \rho(\boldsymbol{r})\delta(z - h/2)\,\mathrm{d}z\,\mathrm{d}A \;=\; \frac{1}{\epsilon_0}\int_A \sigma(\boldsymbol{r})\,\mathrm{d}A \tag{5.16}$$

In this equation, $\sigma(\boldsymbol{r})$ is the surface charge density, which is non-zero only on the metal surface $M_s$, i.e., for $\boldsymbol{r} \in M_S$. Using the right-hand sides of eq. (5.15) and eq. (5.16), Gauss's theorem becomes:[1]

[1] From this point on, we omit the subscript for $\hat{\boldsymbol{n}}_S$ and simply write $\hat{\boldsymbol{n}}$ for the unit normal vector of the conductor surface.

$$\int_A \boldsymbol{E}(\boldsymbol{r})\cdot\hat{\boldsymbol{n}}\,\mathrm{d}A \;=\; \frac{1}{\epsilon_0}\int_A \sigma(\boldsymbol{r})\,\mathrm{d}A \tag{5.17}$$

It follows immediately from this that the following must hold:

$$\boldsymbol{E}(\boldsymbol{r})\cdot\hat{\boldsymbol{n}} \;=\; E(\boldsymbol{r})_\perp \;=\; \frac{\sigma(\boldsymbol{r})}{\epsilon_0} \qquad \text{if} \qquad \boldsymbol{r}\in M_S \tag{5.18}$$

The boundary conditions eq. (5.14) and eq. (5.18) for the electrostatic field $\boldsymbol{E}$ at conductor surfaces can also be expressed as a boundary condition for the electrostatic potential $\phi(\boldsymbol{r})$. The boundary condition $\phi = \text{const}$ is equivalent to eq. (5.14); indeed, that is how we derived eq. (5.14). And by means of $\boldsymbol{E} = -\nabla\phi$, the left-hand side of eq. (5.18) can be replaced by the directional derivative (see eq. (2.22)) of the scalar field $\phi$ at the conductor surface; thus, instead of the boundary conditions for the electric field $\boldsymbol{E}$ at conductor surfaces, one obtains the boundary conditions for the electrostatic potential $\phi$ at conductor surfaces:

$$\phi(\boldsymbol{r}) \;=\; const. \qquad \text{if} \qquad \boldsymbol{r}\in M_S \tag{5.19a}$$

and

$$\hat{\boldsymbol{n}}\cdot\nabla\phi(\boldsymbol{r}) \;=\; \frac{\partial\phi(\boldsymbol{r})}{\partial\hat{n}} \;=\; -\frac{\sigma(\boldsymbol{r})}{\epsilon_0} \qquad \text{if} \qquad \boldsymbol{r}\in M_S \tag{5.19b}$$

The boundary conditions eq. (5.19a) and eq. (5.19b) are equivalent to eq. (5.14) and eq. (5.18). Which boundary condition is used to solve the Poisson equation depends on what is known. In the case of conductors, this is most often the electrostatic potential, i.e., the applied voltage. If, instead of a single conductor, we consider multiple conductors in a field $\boldsymbol{E}$ generated by a prescribed charge distribution, as illustrated in fig. 5.4 for three metal bodies, then these boundary conditions apply to each of the metal bodies $M_1$, $M_2$, and $M_3$. Their potentials $\phi_1$, $\phi_2$, and $\phi_3$ determine the electric field $\boldsymbol{E}$ at the surfaces $\partial M_1$, $\partial M_2$, and $\partial M_3$. Thus, some of the $\boldsymbol{E}$-field lines originate on the surface of one metal body and terminate on the surface of another, depending on which potential is higher and which is lower. If one considers $N$ such conductors in a field generated by a charge distribution $\rho(\boldsymbol{r})$, then the following holds:

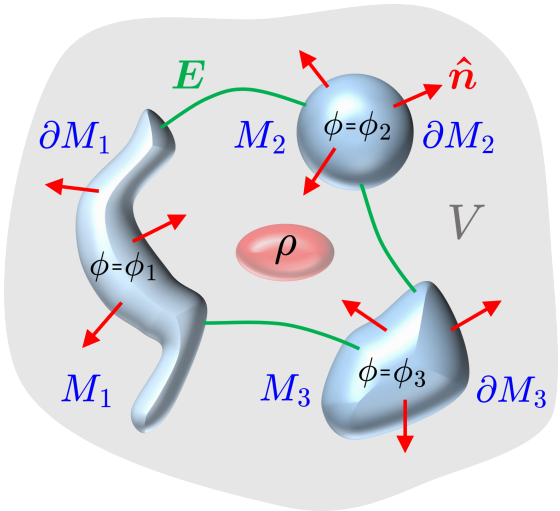



**Fig. 5.4**

$$\Delta\phi(\boldsymbol{r}) \;=\; -\frac{\rho(\boldsymbol{r})}{\epsilon_0} \qquad \text{if} \qquad \boldsymbol{r} \notin M_i \;\;,\;\; i = 1 \ldots N \tag{5.20a}$$

with the boundary conditions

$$\phi(\boldsymbol{r})\big|_{\partial M_i} \;=\; -\phi_i \;=\; const. \qquad \text{if} \qquad \boldsymbol{r} \in \partial M_i \;\;,\;\; i = 1 \ldots N \tag{5.20b}$$

If the surface charge densities $\sigma_i$ on the conductor surfaces are specified rather than the potentials $\phi_i$, then equation eq. (5.19b) applies to the conductors instead of eq. (5.19a) and the Poisson equation is again:

$$\Delta\phi(\boldsymbol{r}) \;=\; -\frac{\rho(\boldsymbol{r})}{\epsilon_0} \qquad \text{if} \qquad \boldsymbol{r} \notin M_i \;\;,\;\; i = 1 \ldots N \tag{5.21a}$$

but now with the boundary conditions

$$\frac{\partial\phi(\boldsymbol{r})}{\partial\hat{n}}\bigg|_{\partial M_i} \;=\; -\frac{\sigma_i(\boldsymbol{r})}{\epsilon_0} \;=\; const. \qquad \text{if} \qquad \boldsymbol{r} \in \partial M_i \;\;,\;\; i = 1 \ldots N \tag{5.21b}$$

Eq. (5.20b) describes the so-called Dirichlet boundary conditions, and eq. (5.21b) describes the von Neumann boundary conditions.

### 5.2.1 Uniqueness of Solutions to the Poisson Equation

In the second chapter (see section 2.3.2 and section 2.4.1), when calculating vector fields from their sources and vortices, we learned that solutions $\phi$ to the Poisson equation are unique provided that $\phi$ decays at least as fast as $r^{-1}$. At that stage, we considered these fields in unbounded space, where it remains true that this asymptotic $r^{-1}$ behavior of $\phi$ guarantees the uniqueness of the solution. However, what does the situation look like if we introduce a boundary, i.e., if the electrostatic potential and electric field produced by a given charge distribution are considered within a bounded volume $V$ containing conductors which themselves possess boundaries, as sketched in fig. 5.4? To answer this question, let us now assume that there are two solutions, $\phi_A$ and $\phi_B$, rather than just one, and that both satisfy eq. (5.20a) or eq. (5.21a). Then, for their difference $\Psi = \phi_A - \phi_B$, it holds that $\Psi = 0$, and it follows that:

$$\Delta\Psi \;=\; 0 \qquad \text{if} \qquad \boldsymbol{r} \in V \tag{5.22a}$$

Thus, $\Psi$ is a solution to the homogeneous version of eq. (5.21a) in $V$. The volume occupied by the conductors $M_i$ in fig. 5.4 is not part of $V$, but the boundaries $\partial M_i$ form an inner boundary $\partial V$ of $V$. For the respective conductor surfaces, $\phi_{A_i} = \phi_{B_i}$ holds, and with the boundary conditions eq. (5.20b) or eq. (5.21b), it follows that:

$$\Psi\big|_{\partial V} \;=\; 0 \qquad \text{or} \qquad \frac{\partial\Psi(\boldsymbol{r})}{\partial\hat{n}}\bigg|_{\partial V} \;=\; 0 \qquad \text{if} \qquad \boldsymbol{r} \in \partial V \tag{5.22b}$$

We now make use of Green's first identity from eq. (2.44) by setting $U = V = \Psi$. This yields:

$$\int_V \Psi \Delta \Psi \, \mathrm{d}^3 r \;=\; -\sum_i \int_{\partial M_i} \Psi (\boldsymbol{n} \cdot \nabla \Psi) \, \mathrm{d}\sigma - \int_V \nabla \Psi \cdot \nabla \Psi \, \mathrm{d}^3 r \tag{5.23}$$

In the derivation of Green's first identity in eq. (2.44), the normal vector pointed outward from the volume under consideration. Here, however, the normal vectors of the boundaries $\partial M_i$ in fig. 5.4 point into the volume under consideration; consequently, the sign preceding the integrals over the conductor boundaries $\partial M_i$ must be negative. Due to eq. (5.22a), the left-hand side of eq. (5.23) vanishes, and due to eq. (5.22b), the surface integrals over the $\partial M_i$ on the right-hand side vanish as well. Thus, eq. (5.23) reduces to the condition:

$$\int_V (\nabla \Psi)^2 \, \mathrm{d}^3 r \;=\; 0 \tag{5.24}$$

Because eq. (5.24) can only be satisfied if $\nabla \Psi = 0$ holds in $V$, it follows that $\Psi$ must be constant. A constant potential difference, however, has no physical consequences. Indeed, given $\boldsymbol{E} = -\nabla \phi$, a constant electrostatic potential has no effect on the resulting electrostatic field. Thus, $\phi_A$ and $\phi_B$ must represent identical physical solutions to the Poisson equation.

### 5.2.2 Separation of Induction Charges

The existence and separation of induction charges can be easily demonstrated, for example, by introducing two additional, smaller metal electrodes into a parallel-plate capacitor in a certain way. This procedure involves placing theses metal electrodes into the parallel-plate capacitor while they are in contact with each other, then separating them while inside the capacitor, and subsequently removing them from the capacitor while they remain separated. Fig. 5.5 graphically brakes out the four steps that lead to the separation of induction charges in this manner. In fig. 5.5a, the first step, the two metal electrodes are located outside the capacitor carrying no charge. They are in contact with each other, meaning they have a conductive connection, and are therefore at the same potential. In fig. 5.5b, the second step, these metal electrodes, while in contact, have been placed into the electric field $\boldsymbol{E}$ of the parallel-plate capacitor. Because these two small metal electrodes are in electrical contact they do behave like a single metal body upon which induction charges form, thereby ensuring that the electric field inside the metal body remains zero. A positive charge layer forms on the surface on one side of

this metal body, and a negative one on the other. In the third step, shown in fig. 5.5c, we then separate the two metal electrodes, which previously formed a single metal body. In doing so, an electric field $\boldsymbol{E}'$ arising between the metal electrodes cancels out the electric field $\boldsymbol{E}$, and the electric field between the separating metal electrodes remains zero. If the metal electrodes are then moved out of the plate capacitor again as shown in fig. 5.5d, the fourth step, the charges on the metal electrodes remain, and an electric field $\boldsymbol{E}' = -\boldsymbol{E}$ exists between them.

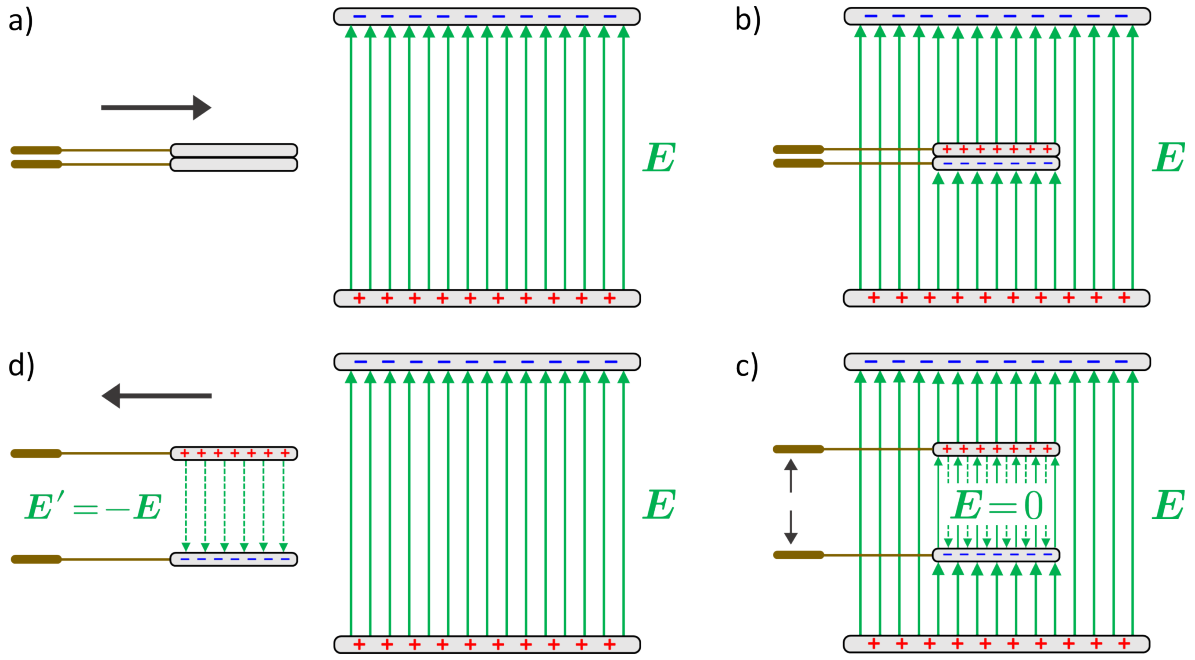



**Fig. 5.5:** a) Two metal electrodes in electrical contact are placed in the field $\boldsymbol{E}$ of a parallel-plate capacitor. b) Induction charges form on the surfaces of the metal electrodes in contact. c) The metal electrodes are separated, and the electric field $\boldsymbol{E}'$ arising between them compensates $\boldsymbol{E}$. d) When the metal electrodes are moved out of the parallel-plate capacitor, the induction charges remain on them, and the field $\boldsymbol{E}'$ exists between the electrodes.

The field $\boldsymbol{E}'$ between the metal electrodes has the same magnitude as $\boldsymbol{E}$ but points in the opposite direction. The situation between the metal electrodes in fig. 5.5c is analogous to fig. 3.8, where the fields outside the capacitor plates cancel each other out and the electric field vanishes. What can we say about the potential of the metal electrodes or the applied electric voltage in fig. 5.5d? We know that the surface charge density $\sigma$ on the metal electrodes must correspond to that on the capacitor plates. If $d_1$ is the separation of the capacitor plates and $d_2$ is the separation of the metal electrodes in fig. 5.5d, then, using eq. (4.58), the following applies to the voltage $U_1$ between the capacitor plates and $U_2$ between the metal electrodes:

$$U_1 = \frac{\sigma}{\epsilon_0} \cdot d_1 \qquad \text{and} \qquad U_2 = \frac{\sigma}{\epsilon_0} \cdot d_2$$

The voltage $U_2$, the potential difference between the metal electrodes, is thus:

$$U_2 = U_1 \cdot \frac{d_2}{d_1}$$

We can only make statements about these potentials themselves if we have a reference point, i.e., if we define a zero potential. If we set the common potential of the metal electrodes shown in fig. 5.5a, i.e., before they are inserted into the parallel-plate capacitor, as the zero potential, then the two metal electrodes in fig. 5.5d are at the potentials $\phi_\pm = \mp\sigma/2\epsilon_0$ (see eq. (4.54) and eq. (4.56)).

### 5.2.3 The Faraday Cage

In electrical conductors, the electrostatic potential $\phi$ is constant; consequently, the electric field $\boldsymbol{E}$ vanishes within electrically conducting bodies. If a cavity is present within an electrical conductor, then $\boldsymbol{E} = 0$ holds inside that cavity as well. Why? To answer this, we consider fig. 5.6, which shows a sketch of a cavity within a conductor. Since $\boldsymbol{E} = 0$ holds inside a conductor, the flux of the electric field through an arbitrary Gaussian surface $S$ enclosing the cavity is zero. While this guarantees that the net charge within the volume enclosed by $S$ is zero, positive and negative charges could still be present on the cavity walls, cancelling each other out while simultaneously generating an electric field inside the cavity. To demonstrate that the electric field inside the cavity truly vanishes, we must make use of a fundamental fact of the nature of electrostatic fields, as expressed in eq. (3.37), which is that they must always be free of vortices. If we choose an arbitrary closed path $C$, then, in conjunction with eq. (3.37), the following holds:

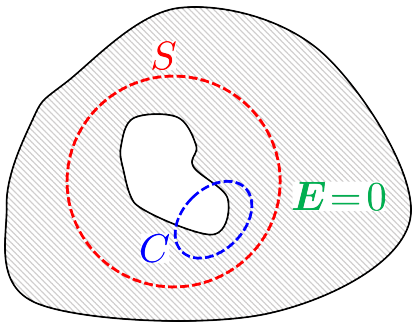



**Fig. 5.6**

$$\oint_C \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = 0 = \underbrace{\int_{C_C} \boldsymbol{E}\,\mathrm{d}\boldsymbol{s}}_{=0} + \int_{C_H} \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} \tag{5.25}$$

On the right-hand side of this equation, we have split the closed path $C$ into the part $C_C$ passing through the conducting body and the part $C_H$ passing through the cavity. Inside the conductor, $\boldsymbol{E} = 0$, leaving only the contribution from the segment $C_H$ that runs through the cavity. For the line integral along the closed curve $C$ to vanish, this quantity, however, must also be zero. However, this is only possible if $\boldsymbol{E}$ is zero within the

cavity along $C_H$ as well. We can choose the closed path $C$ to include arbitrary segments $C_H$ and therefore, $\boldsymbol{E}$ must be zero everywhere within the cavity. Because the electric field vanishes inside the cavities of conductors, metal cages can be used to shield against electric fields. We all frequently use such Faraday cages, for instance, when we get into a car, a tram, a train, or an airplane. In all these cases, these metal cages protect us from hazards such as lightning strikes. Inside them, we are in a field-free space, and the current flows to the ground via the metal outer shell rather than through our bodies. However, the Faraday cage also has disadvantages. One of them is that our phones lose their connection to cell towers while we are in an elevator.

## 5.3 The Principle of Image Charges

For electrostatic problems where despite the symmetry of the setup Gauss's law of electrostatics cannot be easily applied, the method of image charges can often prove helpful. A typical situation of this kind is the determination of the electric field $\boldsymbol{E}$ in the presence of several charged conductors $M_i$. What we know is that the charge distributions on the surfaces of these conductors ($M_i$) will arrange themselves such that their boundaries $\partial M_i$ are equipotential surfaces ($\phi_i = \text{const.}$) for the electric field $\boldsymbol{E}$. We will now consider a few such examples.

### 5.3.1 Point Charge Near a Grounded Metal Plate

If one places a point charge in front of an extended, grounded metal surface, that surface acts as an equipotential surface for the point charge, characterized by $\phi = 0\,\text{V}$. The electric field must vanish within the metal. This shielding of the point charge's electric field within the metal occurs through the movement of conduction electrons in the metal, leading to the formation of induction charges in the metal's surface. In the case of a positive point charge, electrons flow toward the metal surface, which then acquires a net negative charge. However, if a negative point charge is positioned in front of the metal surface, electrons flow away from the surface into the

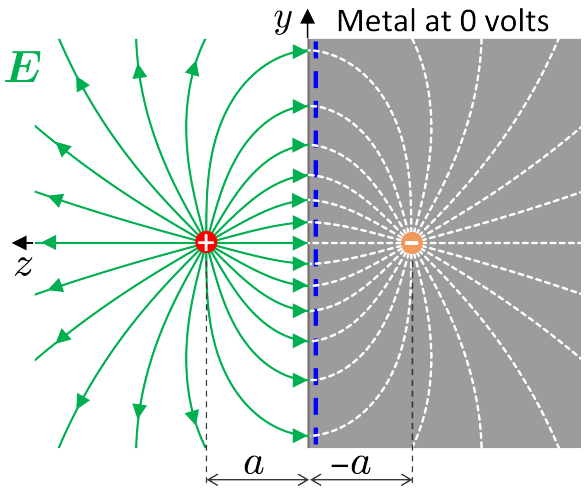



**Fig. 5.7**

interior of the metal, leaving behind a positive charge layer on the surface formed by the stationary positive metal atoms. As sketched in fig. 5.7, we consider a positive point charge in front of a metal body filling the half-space (z < 0). We are already familiar with a situation comparable to fig. 5.7, in which a point charge is located in front of an infinitely extended planar equipotential surface with $\phi = 0\,\mathrm{V}$. In section 3.2.1, we discussed the electrostatic potential and the electric field $\boldsymbol{E}$ of the electric dipole. If we imagine the surface $\phi = 0\,\mathrm{V}$ in fig. 3.6 replaced by a very thin metal plate, then we know the potential $\phi$ and the electric field $\boldsymbol{E}$ for the system consisting of a positive point charge in front of a grounded metal surface. Fig. 5.7 illustrates why the term "mirror charge" or "image charge" is used here. In fig. 5.7, we have replaced the charges within the metal surface, which is at zero potential, with a mirror or image charge - the metal surface acting as the mirror - that carries the negative of the charge which the point charge located in front of the metal surface carries.

Which mathematical problem have we solved here using the method of image charges as just described? For simplicity, we set up our coordinate system as sketched in fig. 5.7. The $xy$-plane coincides with the metal surface, and the positive point charge $q$ is located on the $z$-axis at a distance $\boldsymbol{a} = (0, 0, a)$ from the grounded metal surface. To determine the electrostatic potential $\phi$ in this situation, we must find the solution to the Poisson equation

$$\Delta\phi(\boldsymbol{r}) \;=\; -\frac{q}{\epsilon_0}\delta^{(3)}(\boldsymbol{r}-\boldsymbol{a}) \qquad \text{for} \qquad z > 0 \tag{5.26}$$

satisfying the boundary condition $\phi(x, y, 0) = 0$. From eq. (2.74), we already know the solution to this Poisson equation for $z > 0$, namely

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\frac{q}{|\boldsymbol{r}-\boldsymbol{a}|} + \phi_h$$

In this equation, we introduced on the right-hand side an additional term $\phi_h$ which we require to satisfy the Laplace equation $\Delta\phi_h = 0$, i.e., the homogeneous version of the Poisson equation, at $z = 0$. We choose $\phi_h$ such that $\phi(\boldsymbol{r})$ vanishes at $z = 0$. We achieve this by mentally removing the metal in the right half-space and instead adding the potential of the image charge $-q$. The image charge thus helps us satisfy the boundary condition for Poisson's equation. Since the field of this image charge $-q$ originates in the half-space $z < 0$, the potential resulting from charge $q$ and image charge $-q$ satisfies Poisson's equation eq. (5.26) in the half-space $z > 0$, subject to the boundary condition $\phi(x, y, 0) \;=\; 0$:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{|\boldsymbol{r}-\boldsymbol{a}|} - \frac{q}{|\boldsymbol{r}+\boldsymbol{a}|}\right) \qquad \text{for} \qquad z \geq 0 \tag{5.27}$$

Precisely because the image charge is located in the half-space $z < 0$, its contribution to this potential in the half-space $z > 0$ is a solution to $\Delta\phi(\boldsymbol{r}) = 0$. What can we say about the interaction between the positive point charge and the negative induction charge layer on the metal surface? We know that the electric field must vanish inside the metal body. However, as we have just seen, the presence of this layer of negative induction charges has the same effect on the positive point charge as if a mirror charge $-q$ existed within the metal body. Consequently, the field strength in the left half-space must be identical at every point to that of a dipole field created by a charge $q$ and a mirror charge $-q$. Thus, the force acting between the charge $q$ and the charge layer on the metal surface is the same as that between the poles of a dipole. For the dipole in fig. 5.7 with the unit vector $\hat{\boldsymbol{a}}$ along the dipole axis, this force is:

$$\boldsymbol{F} = -\frac{1}{4\pi\epsilon_0}\frac{q^2}{(2a)^2}\cdot\hat{\boldsymbol{a}} \tag{5.28}$$

This examination of the interaction further clarifies what we are doing when applying the method of images. We replace the material of the conductor with the material in which the point charge which is located outside the conductor resides (usually air or a vacuum). And we replace the induction charges on the conductor's surface with charges located within the volume previously occupied by the conductor, charges that produce the same electrostatic potential at that volume's surface as the induction charges on the conductor's surface.

### 5.3.2 Point Charge Near a Grounded Metal Sphere

Next, we consider the arrangement shown in fig. 5.8 and, for the time being, ignore the image charge $q^*$ already depicted there. The point charge $q$ is located outside a conducting, grounded sphere at a distance $\boldsymbol{r}_1$ from the center of the sphere which is the coordinate origin. Due to the grounding, the conducting sphere is at a potential of $\phi = 0$. In the presence of the point charge $q$, induction charges on the surface of the sphere prevent the electric field generated by this point charge from penetrating the conductor. The spherical surface is an equipotential surface of the electrostatic potential of $q$ for which $\phi = 0$ holds. For this arrangement of a point charge $q$ and a grounded, conducting sphere, Poisson's equation

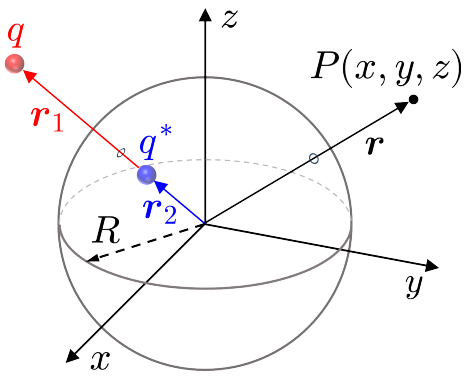



**Fig. 5.8**

$$\Delta\phi(\boldsymbol{r}) \;=\; -\frac{q}{\epsilon_0}\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_1) \qquad \text{for} \qquad |\boldsymbol{r}| > R \tag{5.29a}$$

applies with the boundary condition

$$\phi(\boldsymbol{r}) \;=\; 0 \qquad \text{for} \qquad |\boldsymbol{r}| = R \tag{5.29b}$$

Instead of solving the Poisson equation directly, we again use the method of image charges. We thus replace the conducting sphere with a volume of air or vacuum (depending on the composition of the medium surrounding $q$) of the same size and shape, while using image charges inside the sphere to establish a zero potential at the surface of the now-empty spherical volume. If we attempt to achieve this using only a single image charge, then, for reasons of symmetry, this image charge $q^*$ must be located inside the spherical volume, on the radial vector extending from the origin to the charge $q$, at a distance $r_2 < R$. The electrostatic potential for the arrangement consisting of charge $q$ and image charge $q^*$ shown in fig. 5.8 is thus:

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\left(\frac{q}{|\boldsymbol{r}-\boldsymbol{r}_1|}+\frac{q^*}{|\boldsymbol{r}-\boldsymbol{r}_2|}\right)$$

To satisfy the boundary condition of the Poisson equation, the condition $\phi(\boldsymbol{R}) = 0$ must hold for this potential. The equation determining the charge $q^*$ and position $\boldsymbol{r}_2$ of the image charge is therefore:

$$0 \;=\; \frac{q}{|\boldsymbol{R}-\boldsymbol{r}_1|}+\frac{q^*}{|\boldsymbol{R}-\boldsymbol{r}_2|} \;=\; \frac{q}{\sqrt{R^2+r_1^2-2\boldsymbol{R}\boldsymbol{r}_1}}+\frac{q^*}{\sqrt{R^2+r_2^2-2\boldsymbol{R}\boldsymbol{r}_2}}$$

If $\hat{\boldsymbol{u}}$ is the unit vector in the direction of $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$, and $\hat{\boldsymbol{e}}_r$ is the radial unit vector of the coordinate system, then we can write:

$$\boldsymbol{r}_1 \;=\; r_1\cdot\hat{\boldsymbol{u}} \quad , \quad \boldsymbol{r}_2 \;=\; r_2\cdot\hat{\boldsymbol{u}} \quad , \quad \boldsymbol{R} \;=\; R\cdot\hat{\boldsymbol{e}}_r$$

We use this to rewrite the boundary condition as follows:

$$\begin{aligned} 0 \;&=\; \frac{q}{\sqrt{R^2+r_1^2-2Rr_1\boldsymbol{u}\hat{\boldsymbol{e}}_r}}+\frac{q^*}{\sqrt{R^2+r_2^2-2Rr_2\boldsymbol{u}\hat{\boldsymbol{e}}_r}} \\ &=\; \frac{q}{R\sqrt{1+\left(r_1^2/R^2\right)-2\left(r_1/R\right)\hat{\boldsymbol{u}}\hat{\boldsymbol{e}}_r}}+\frac{q^*}{r_2\sqrt{1+\left(R^2/r_2^2\right)-2\left(R/r_2\right)\hat{\boldsymbol{u}}\hat{\boldsymbol{e}}_r}} \end{aligned}$$

This equation is satisfied for all values of $\hat{\boldsymbol{u}}\cdot\hat{\boldsymbol{e}}_r$ if:

$$\frac{q}{R} \;=\; -\frac{q^*}{r_2} \qquad \text{and} \qquad \frac{r_1}{R} \;=\; \frac{R}{r_2}$$

For the charge and position of the image charge, we thus obtain

$$q^* \;=\; -q\frac{R}{r_1} \qquad \text{and} \qquad \boldsymbol{r}_2 = \frac{R^2}{r_1}\boldsymbol{u} \tag{5.30}$$

and with $\beta = R/r_1$, the solution to the Poisson equation eq. (5.29a) with the boundary condition eq. (5.29b) is the potential:

$$\phi(\boldsymbol{r}) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_1|} - \frac{\beta}{|\boldsymbol{r}-\beta^2\boldsymbol{r}_1|}\right) \qquad \text{with} \qquad \beta < 1 \tag{5.31}$$

This allows us to calculate the electrostatic field $\boldsymbol{E}$ of the arrangement shown in fig. 5.8 using $\boldsymbol{E} = -\nabla\phi$. To visualize this field, we make use of the problem's inherent symmetry. As is evident from fig. 5.8, $\boldsymbol{E}$ must be rotationally symmetric about the axis on which the point charge outside the sphere and the image charge inside the sphere lie. In order to keep things simple, we align this axis with the $z$-axis, position a point charge $q$ at $z = 4\,\mathrm{m}$, and place a metal sphere of radius $R = 2\,\mathrm{m}$ with its center at the coordinate origin. Consequently, the image charge is then located at $z = 1\,\mathrm{m}$, and its value is $q^* = -q/2$. Fig. 5.9 shows the pattern of $\boldsymbol{E}$-field lines originating from the point charge $q$. The electric field lines terminating on the surface of the sphere strike it perpendicularly, and the tangential component of $\boldsymbol{E}$ vanishes there, as must be the case for the surface of a conducting body. Fig. 5.9 also shows how the fictitious field lines continue inside the sphere toward the image charge. If the distance between the point charge and the metal sphere is increased, the image charge moves towards the center of the sphere. As $r_1 \to \infty$, it holds that $r_2 \to 0$ and $q^* \to 0$. If the distance between $q$ and the metal sphere is decreased, the image charge $q^*$ moves towards the point charge from its position within the metal sphere. As $r_1 \to R$, it holds that $r_2 \to R$ and $q^* \to -q$. However, since the point charge and the metal sphere are short-circuited at $r_1 = R$, this scenario does not actually occur; the charge $q$ flows away via the grounded metal sphere, and the electric field collapses.

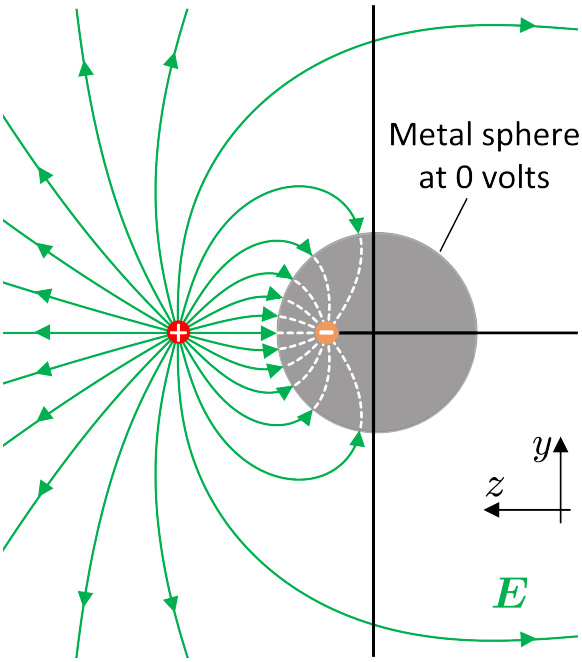



**Fig. 5.9**

As can be clearly seen in fig. 5.9, the field lines of $\boldsymbol{E}$ originating from the point charge $q$, plotted at 15-degree intervals, exhibit radial symmetry in the immediate vicinity of the point charge. However, this radial symmetry of $\boldsymbol{E}$ around the point charge vanishes very quickly on the side facing the sphere. The density of the $\boldsymbol{E}$-field lines terminating on the surface of the sphere is evidently highest where the distance to the point charge

is smallest. Moving away from this point on the sphere's surface causes the field line density to decrease rapidly. The induction charges on the surface of the sphere must therefore be distributed very non-uniformly. From eq. (5.21b), we already know how to determine this surface charge density $\sigma$ given the potential $\phi$. To do this, however, we first rewrite eq. (5.31) as:

$$\phi(\boldsymbol{r}) \;=\; \frac{q}{4\pi\epsilon_0}\left(\frac{1}{\sqrt{r^2+r_1^2-2rr_1\cos\vartheta}} - \frac{\beta}{\sqrt{r^2+\beta^4 r_1^2-2\beta^2 rr_1\cos\vartheta}}\right) \tag{5.32}$$

Here, $\vartheta$ is the angle enclosed by the vectors $\boldsymbol{r}$ and $\boldsymbol{r}_1$, i.e., in fig. 5.9, the angle between $\boldsymbol{r}$ and the $z$-axis. With a brief calculation using eq. (5.21b), we thus obtain the following expression for $\sigma$:

$$\sigma \;=\; -\epsilon_0 \frac{\partial\phi}{\partial r}\bigg|_R \;=\; -\frac{q}{4\pi R^2}\frac{\beta(1-\beta^2)}{\left(1+\beta^2-2\beta\cos\vartheta\right)^{3/2}} \tag{5.33}$$

Fig. 5.10a shows the magnitude of eq. (5.33), normalized to units of $q/4\pi R^2$, for three values of $\beta$ as a function of the angle $\vartheta$. As can be seen, in the situation sketched in fig. 5.9 ($\beta = 1/2$), almost the entire induction charge is located on the side of the sphere facing the point charge $q$. Most of the induction charges are located within a cone defined by $\vartheta \leq \pi/4$. If the distance of the point charge is doubled ($\beta = 1/4$), the induced charges are already distributed much more uniformly over the surface of the sphere. We obtain the total induced charge $Q$ on the spherical surface by integrating $\sigma$ over the spherical shell of radius $R$:

$$Q \;=\; -\frac{q\beta(1-\beta^2)}{4\pi}\cdot\int_0^{2\pi}\int_0^{\pi}\frac{\sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi}{\left(1+\beta^2-2\beta\cos\vartheta\right)^{3/2}} \tag{5.34}$$

The integral over $\varphi$ yields a factor of $2\pi$, and for the integral over $\vartheta$, the value $2/(1-\beta^2)$ is quickly obtained using the substitution $x = \cos\vartheta$. Thus, for $Q$, we obtain, unsurprisingly, exactly the image charge $q^*$:

$$Q \;=\; -\frac{q\beta(1-\beta^2)}{4\pi}\cdot\frac{4\pi}{1-\beta^2} \;=\; -q\beta \;=\; q^* \tag{5.35}$$

To determine the Coulomb force between the point charge and the grounded sphere, we do not need to know the detailed shape of $\sigma$. It suffices to know the position and charge of the image charge $q^*$. We already know both quantities, and thus we know that the Coulomb force between the point charge $q$ and the grounded metal sphere is given by:

$$\boldsymbol{F} \;=\; \frac{q\cdot q^*}{4\pi\epsilon_0}\cdot\frac{\boldsymbol{r}_1-\boldsymbol{r}_2}{|\boldsymbol{r}_1-\boldsymbol{r}_2|^3} \;=\; \frac{q\cdot q^*}{4\pi\epsilon_0}\cdot\frac{1}{|r_1-r_2|^2}\cdot\hat{\boldsymbol{u}} \tag{5.36}$$

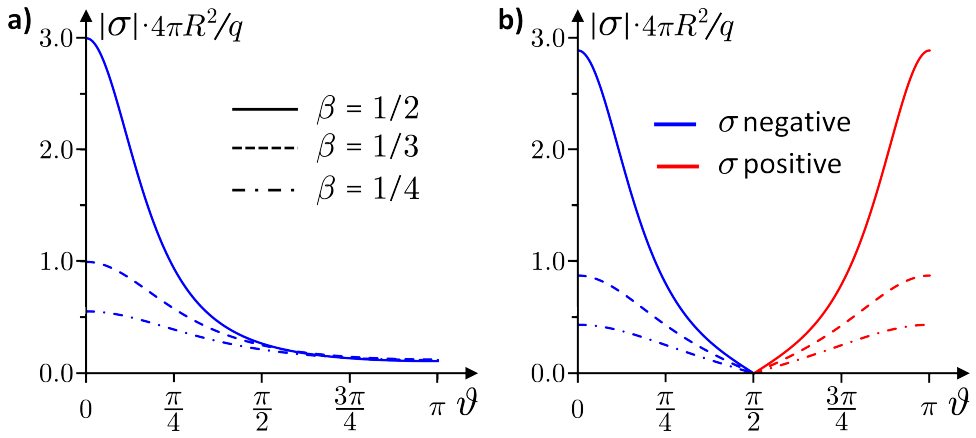



**Fig. 5.10:** a) Normalized magnitude of the surface charge density $\sigma$ for the metal sphere in fig. 5.9. b) Normalized magnitude of the surface charge density $\sigma$ for the metal sphere in fig. 5.11a.

Here, $\hat{\boldsymbol{u}}$ is again the unit vector pointing in the direction of the point charge $q$. If we now substitute the values for $q^*$ and $r_2$, we obtain

$$\boldsymbol{F} = -\frac{q^2\left(R/r_1\right)}{4\pi\epsilon_0} \cdot \frac{1}{\left|r_1 - \left(R^2/r_1\right)\right|^2} \cdot \hat{\boldsymbol{u}} \tag{5.37}$$

or, expressed in terms of $\beta = R/r_1$:

$$\boldsymbol{F} = -\frac{q^2}{4\pi\epsilon_0 R^2} \cdot \frac{\beta^3}{(1-\beta^2)^2} \cdot \hat{\boldsymbol{u}} \qquad \text{with} \qquad \beta < 1 \tag{5.38}$$

As can be easily seen from eq. (5.37), this force scales as $r_1^{-3}$ for $r_1 \to \infty$, i.e., for $\beta \to 0$, and as $(r_1 - R)^{-2}$ for $r_1 \to R$, i.e., for $\beta \to 1$. Instead of the positive charge we used in this example, we could just as well have put a negative charge before the metal sphere, the image charge in the metal sphere would then simply be positive in that case. We will exploit this circumstance, along with the fact that the situation shown in fig. 5.9 is mirror-symmetric about the $y$-axis, in the next example to place the metal sphere in a uniform electrostatic field.

### 5.3.3 Metal Sphere in a Homogeneous Electrostatic Field

If we replace $q$ with $-q$ and $\boldsymbol{r}_1$ with $-\boldsymbol{r}_1$ in eq. (5.29a), while the boundary condition eq. (5.29b) remains unchanged, we obtain the result of eq. (5.31) again from the corresponding Poisson equation, albeit with the signs reversed. Let $\phi_+$ denote the potential of a positive point charge $q$ at position $\boldsymbol{r}$, as we have already determined in eq. (5.31), and $\phi_-$ that of a negative point charge $-q$ at position $-\boldsymbol{r}_1$. The sum of both solutions, $\phi_+$ and $\phi_-$, is then the solution to the Poisson equation:

$$\Delta\phi(\boldsymbol{r}) \;=\; -\frac{q}{\epsilon_0}\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_1) + \frac{q}{\epsilon_0}\delta^{(3)}(\boldsymbol{r}+\boldsymbol{r}_1) \qquad \text{for} \qquad |\boldsymbol{r}| > R$$

with the boundary condition

$$\phi(\boldsymbol{r}) \;=\; 0 \qquad \text{for} \qquad |\boldsymbol{r}| = R$$

Using eq. (5.31), we obtain as the solution $\phi = \phi_+ + \phi_-$ with $\beta < 1$:

$$\phi(\boldsymbol{r}) \;=\; \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r}-\boldsymbol{r}_1|} - \frac{1}{|\boldsymbol{r}+\boldsymbol{r}_1|}\right) - \frac{q\beta}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r}-\beta^2\boldsymbol{r}_1|} - \frac{1}{|\boldsymbol{r}+\beta^2\boldsymbol{r}_1|}\right) \tag{5.39}$$

If we then determine the surface charge density $\sigma$ on the surface of the sphere using eq. (5.21b), we obtain:

$$\sigma \;=\; -\frac{q}{4\pi R^2}\left(\frac{\beta(1-\beta^2)}{\left(1+\beta^2-2\beta\cos\theta\right)^{3/2}} - \frac{\beta(1-\beta^2)}{\left(1+\beta^2+2\beta\cos\theta\right)^{3/2}}\right) \tag{5.40}$$

The first term in eq. (5.40), matching eq. (5.33), originates from the negative image charge, the second term arises from the positive image charge. Analogous to Fig. 5.10a, Fig. 5.10b shows the magnitude of eq. (5.40), normalized to units of $q/4\pi R^2$, as a function of the angle $\vartheta$ for three values of $\beta$. As can be seen, the positive and negative surface charges on the two halves of the spherical shell cancel each other out at $\pi/2$. When calculating the total induced charge $Q$ on the spherical surface, we consider the contributions from the right and left halves of the spherical shell separately, that is:

$$Q = Q_- + Q_+ \;=\; 2\pi R^2\left[\int_0^{\pi/2}\sigma\sin\vartheta\,\mathrm{d}\vartheta + \int_{\pi/2}^{\pi}\sigma\sin\vartheta\,\mathrm{d}\vartheta\right] \tag{5.41}$$

We evaluate the two integrals again using the substitution $\cos\vartheta = x$ and obtain the following result:

$$Q \;=\; -q\left(1 - \frac{1-\beta^2}{\left(1+\beta^2\right)^{1/2}}\right) + q\left(1 - \frac{1-\beta^2}{\left(1+\beta^2\right)^{1/2}}\right) \;=\; 0 \tag{5.42}$$

The magnitudes of $Q_-$ and $Q_+$ are thus exactly equal, and the total charge of the metal sphere is therefore exactly zero. This means that our result for a grounded metal sphere in a uniform electrostatic field, as sketched in fig. 5.11a, applies equally to an isolated, uncharged metal sphere in a uniform electrostatic field. Let us now examine more closely the potential generated by this charge distribution on the surface of the sphere. In eq. (5.39), we have written out the potential $\phi$ in such a way that it can be interpreted as the result of subtracting one dipole potential from another. The first expression, grouped within parentheses, is the potential $\phi_D$ of the dipole formed by the charges $q$ and $-q$ that we introduced:

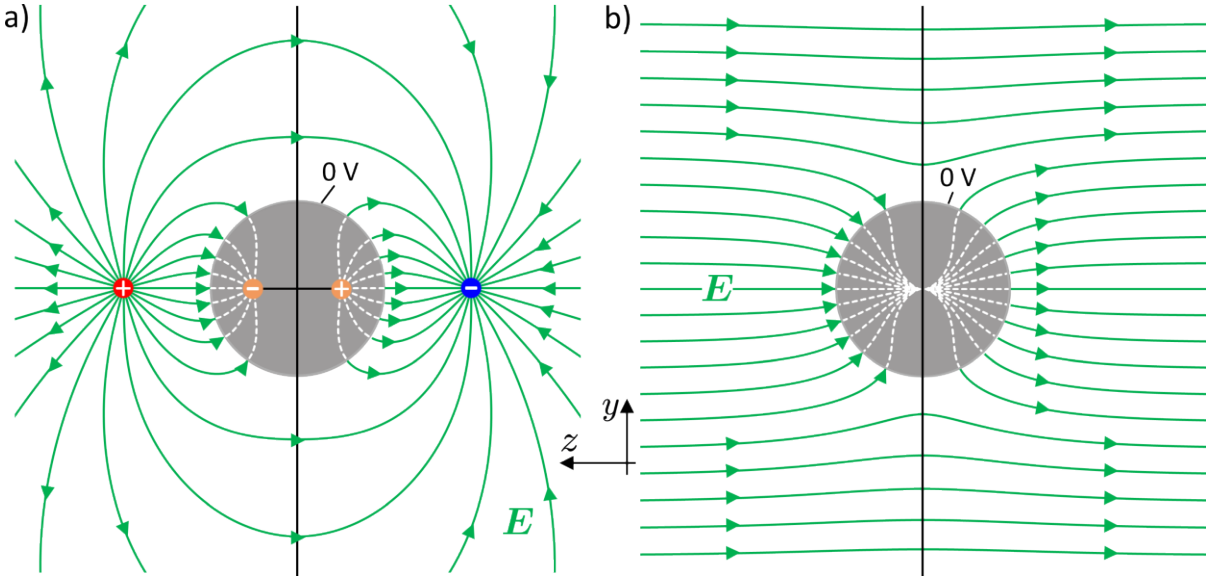



**Fig. 5.11:** a) Grounded metal sphere in a dipole field with positive and negative poles at $y = 0$ and $z = \pm 4\,\mathrm{m}$. b) The poles of the dipole are now at $y = 0$ and $z = \pm 100\,\mathrm{m}$, and the grounded metal sphere ($R = 2\,\mathrm{m}$) is now sitting in a uniform electrostatic field $\boldsymbol{E}$. The cutouts in (a) and (b) are identical, but the field lines in (b) are shown with 30 times higher resolution; otherwise, only the field line running along the $z$-axis from (a) would be visible in (b).

$$\phi_D(\boldsymbol{r}) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r} - \boldsymbol{r}_1|} - \frac{1}{|\boldsymbol{r} + \boldsymbol{r}_1|}\right) \tag{5.43}$$

The second expression enclosed in parentheses on the right-hand side in eq. (5.39), which is subtracted from $\phi_D$, corresponds to another dipole potential $\phi_D^*$ generated by the image charges $q^*$ and $-q^*$:

$$\phi_D^*(\boldsymbol{r}) = \frac{q\beta}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r} - \beta^2\boldsymbol{r}_1|} + \frac{1}{|\boldsymbol{r} + \beta^2\boldsymbol{r}_1|}\right) \tag{5.44}$$

We have already examined the electrostatic potential of a dipole and its electrostatic field in section 3.2.1. We now draw upon this knowledge to take a closer look at $\phi_D$ and $\phi_D^*$. First, however, we consider the electrostatic field associated with eq. (5.39). Fig. 5.11a shows the pattern of the $\boldsymbol{E}$ field lines in the $yz$-plane for a positive charge $q$ at $y = 0$ and $z = 4\,\mathrm{m}$ and a negative charge $q$ at $y = 0$ and $z = -4\,\mathrm{m}$, with a metal sphere of radius $R = 2\,\mathrm{m}$ at the coordinate origin. Except for the additional charge $-q$, the configuration is exactly as shown in fig. 5.9; as in fig. 5.9, the electrostatic field lines, radiating outward from the positive charge and converging on the negative charge, are shown at angular intervals of 15 degrees. Fig. 5.11b shows what happens when we make the dipole separation in $\phi_D$ much larger. In fig. 5.11b, we have increased the separation between the

charges $q$ and $-q$ by a factor of 25 compared to fig. 5.11a. As a result, the electrostatic field strength near the metal sphere becomes much lower, but the electrostatic field also becomes much more uniform. The magnitude of the effect can be gauged by the fact that the first field line of $\boldsymbol{E}$ originating from $q$, which starts at an angle of 15 degrees to the $z$-axis in fig. 5.11a, is no longer visible at all in fig. 5.11b, even though the latter shows exactly the same section of the image. The field lines of $\boldsymbol{E}$ shown in fig. 5.11b originate within an angular segment of $\pm 5$ degrees relative to the $z$-axis from $q$ and $-q$.

We determine the uniform electrostatic field between the two charges $q$ and $-q$ by considering $\phi_D$ for the specific situation shown in fig. 5.11b with a very large separation between the charges. Due to the rotational symmetry of the electrostatic field about the $z$-axis, it suffices to consider $\phi_D$ only in the $yz$-plane. With $\pm\boldsymbol{r_1} = (0, 0, \pm z_1)$, $\phi_D$ in fig. 5.11b is given by:

$$\phi_D(\boldsymbol{r}) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{\sqrt{y^2+(z-z_1)^2}} - \frac{1}{\sqrt{y^2+(z+z_1)^2}}\right) \tag{5.45}$$

We extract $z_1$ from the roots and approximate the remaining roots for $z_1 \to \infty$ by neglecting quadratic terms and then retaining only first-order terms in the Taylor expansion of the roots:

$$\left(\frac{y^2}{z_1^2} + \frac{z^2}{z_1^2} \pm \frac{2z}{z_1} + 1\right)^{-1/2} \approx \left(1 \pm \frac{2z}{z_1}\right)^{-1/2} \approx 1 \mp \frac{z}{z_1}$$

and thus, for $z_1 \to \infty$, it holds that

$$\phi_D(\boldsymbol{r}) = \frac{q}{4\pi\epsilon_0 z_1}\left[1 + \frac{z}{z_1} - \left(1 - \frac{z}{z_1}\right)\right] = \frac{qz}{2\pi\epsilon_0 z_1^2} \tag{5.46}$$

The potential of this dipole thus increases when moving towards the positive charge $q$ and decreases when moving towards the negative charge $-q$, vanishing at $z = 0$, just as we already know from fig. 3.6. The electrostatic field has only one component parallel to the $z$-axis, which is:

$$E_z = -\frac{\partial\phi_D}{\partial z} = -\frac{q}{2\pi\epsilon_0 z_1^2} \tag{5.47}$$

The simple relationship between $\phi_D(\boldsymbol{r})$ and $E_z$ is thus:

$$\phi_D(\boldsymbol{r}) = -z \cdot E_z \tag{5.48}$$

If we were to truly let $z_1$ approach infinity, the electrostatic field $E_z$ would vanish. However, we do not actually do that; we simply make $z_1$ large enough for our approximations to hold. As $z_1$ increases, $E_z$ becomes weaker, but we can compensate for this by increasing the charge $q$ accordingly. However, this is only relevant from a practical standpoint if we

actually intend to test all of this in an experiment, since we would then require a measurable electrostatic field at the sphere's location. From a theoretical perspective, the actual strength of the field $E_z$ makes no difference; all the conclusions we draw here remain valid even for arbitrarily weak fields. While the dipole separation in $\phi_D$ increases as $z_1$ continues to grow, it simultaneously decreases for the image-charge dipole $\phi_D^*$. According to eq. (5.30), this dipole separation is $2R^2/r_1$, which naturally approaches zero as $z_1 \to \infty$. If $r_1$ were to become infinitely large, both the image-charge dipole and $E_z$ would logically vanish. However, we choose $z_1$ such that, firstly, the approximations made earlier in the derivation of $E_z$ remain valid and, secondly, the distance of the image-charge dipole is sufficiently small compared to $R$ for us to apply the dipole far-field approximation from section 3.2.1. With eq. (5.30) and $r_1 = z_1$, the dipole moment $\boldsymbol{p}$ of the image-charge dipole is given by:

$$\boldsymbol{p} = q^* \cdot \frac{2R^2}{z_1}\hat{\boldsymbol{k}} = -q \cdot \frac{R}{z_1} \cdot \frac{2R^2}{z_1}\hat{\boldsymbol{k}} = -q \cdot \frac{2R^3}{z_1^2}\hat{\boldsymbol{k}} \tag{5.49}$$

If we now use eq. (5.47) to replace $q$ with $E_z$ and $z_1$, we obtain:

$$\boldsymbol{p} = 4\pi\epsilon_0 R^3 E_z \hat{\boldsymbol{k}} \tag{5.50}$$

Using eq. (5.50), we have an expression for the dipole moment $\boldsymbol{p}$ of the image-charge dipole induced by the field $E_z$ that depends solely on $E_z$: $\boldsymbol{p}$ grows linearly with $E_z$ and approaches zero linearly as $E_z \to 0$. With eq. (3.53), we can now express the potential of the image-charge dipole from eq. (5.44) in the dipole approximation as:

$$\phi_D^*(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\frac{\boldsymbol{p}\cdot\boldsymbol{r}}{r^3} = \frac{R^3}{r^3} \cdot z \cdot E_z \tag{5.51}$$

Using eq. (5.48) and eq. (5.51), we then can for very large $\boldsymbol{r_1}$, i.e., in this case in fig. 5.11b for very large $z$, write the electrostatic potential of charges and image charges $\phi(\boldsymbol{r}) = \phi_D(\boldsymbol{r}) + \phi_D^*(\boldsymbol{r})$ from eq. (5.39) as:

$$\phi(\boldsymbol{r}) = -z \cdot E_z + \frac{1}{4\pi\epsilon_0}\frac{\boldsymbol{p}\cdot\boldsymbol{r}}{r^3} = \left(\frac{R^3}{r^3} - 1\right) \cdot z \cdot E_z \tag{5.52}$$

For $r = R$, $\phi(\boldsymbol{r})$ vanishes, and therefore eq. (5.52) satisfies the boundary condition eq. (5.29b) for the Poisson equation eq. (5.29a), which requires $\phi$ to be zero at the surface of the metal sphere. From section 3.2.1, we know that

$$\nabla(\boldsymbol{p}\cdot\frac{\boldsymbol{r}}{r^3}) = \frac{r^2\boldsymbol{p} - 3(\boldsymbol{p}\boldsymbol{r})\boldsymbol{r}}{r^5}$$

and thus we obtain for the electric field $\boldsymbol{E}$ from eq. (5.52):

$$\boldsymbol{E}(\boldsymbol{r}) = -\nabla\phi(\boldsymbol{r}) = E_z\hat{\boldsymbol{k}} + \frac{1}{4\pi\epsilon_0}\frac{3(\boldsymbol{p}\boldsymbol{r})\boldsymbol{r} - r^2\boldsymbol{p}}{r^5} \tag{5.53}$$

With eq. (5.52), we now know what happens when a grounded metal sphere is placed in a homogeneous electrostatic field, for example, that of a parallel-plate capacitor. The homogeneous electrostatic field is disturbed, and this disturbance can be described by an additional dipole potential. The pattern of the field lines for the perturbed uniform electrostatic field then appears as sketched in fig. 5.11b. However, we learn something additional here if we determine the surface charge of the metal sphere from eq. (5.52) using eq. (5.21b). With $z = r\cos\vartheta$, we obtain:

$$\sigma \;=\; -\epsilon_0 \frac{\partial\phi}{\partial r}\bigg|_R \;=\; 3\epsilon_0 E_z \cos\vartheta \tag{5.54}$$

The cosine changes sign at $\pi/2$; thus, as shown in fig. 5.11a, we have a situation where one half of the spherical shell carries a negative surface charge and the other a positive one, meaning the metal sphere carries a total charge of zero. We can easily verify that this is the case by once again calculating the total induced charge $Q$ on the spherical surface. In doing so, we again integrate separately over the two halves of the spherical shell of radius $R$ and obtain for $Q = Q_- + Q_+$:

$$Q \;=\; 3\epsilon_0 E_z \cdot 2\pi R^2 \left[ \int_0^{\pi/2} \cos\vartheta \sin\vartheta \,\mathrm{d}\vartheta + \int_{\pi/2}^{\pi} \cos\vartheta \sin\vartheta \,\mathrm{d}\vartheta \right] \tag{5.55}$$

$E_z$ is negative. To make the signs of the charges $Q_-$ and $Q_+$ apparent here, one therefore either substitutes $E_z$ directly from eq. (5.48) or, as is done here, replaces $E_z$ with its negative magnitude. This yields:

$$Q = Q_- + Q_+ \;=\; -6\pi\epsilon_0 R^2 |E_z| \cdot \left( \frac{1}{2} - \frac{1}{2} \right) \;=\; 0 \tag{5.56}$$

Just as for the situation sketched in fig. 5.11a, the condition $Q = 0$ here implies that our result for a grounded metal sphere in a homogeneous electrostatic field, sketched in fig. 5.11b, applies equally to an isolated, uncharged metal sphere in a homogeneous electrostatic field. In general, this also applies to other non-spherical conductors with appropriate symmetry, such as a metal plate inserted parallel to the electrodes of a parallel-plate capacitor into the capacitor (as shown, for example, in fig. 5.5). As long as the total induced charge over the entire surface of the conducting body is zero, meaning the body carries no net charge, the field configuration is identical for both the grounded and the ungrounded conducting body.

## 5.4 Green's Functions

Despite their great practical importance in many areas of physics, Green's functions are usually not introduced until introductory courses on theoretical electrodynamics. However, we have already laid much of the necessary groundwork to address the solution of Poisson's equation using Green's functions within a limited scope; which we will do in this section. The procedure for applying the method of image charges is always the same. In the first step, one determines the solution $\phi_P$ to the Poisson equation for the given charge distribution in free space, without considering the presence of the conducting body. In the second step, a solution is determined for the region inside the conducting body, which itself contains no charges, meaning $\rho(r) = 0$ applies. This solution $\phi_h$ to the homogeneous version of the Poisson equation (i.e., the Laplace equation $\Delta\phi_h = 0$) is chosen such that the sum $\phi_P + \phi_h$ satisfies the boundary condition of the Poisson equation. The approach using Green's functions proceeds in a very similar way but is much more general. The Green's function $G(\boldsymbol{r}, \boldsymbol{r}')$ is defined as the general solution to the Poisson equation

$$\Delta_{\boldsymbol{r}} G(\boldsymbol{r}, \boldsymbol{r}') = -\frac{1}{\epsilon_0}\delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}') \tag{5.57}$$

This solution is given by:

$$G(\boldsymbol{r}, \boldsymbol{r}') = \frac{1}{4\pi\epsilon_0}\frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} + F(\boldsymbol{r}, \boldsymbol{r}') \qquad \text{mit} \qquad \Delta F(\boldsymbol{r}, \boldsymbol{r}') = 0 \tag{5.58}$$

$F(\boldsymbol{r}, \boldsymbol{r}')$ is thus a solution to the Laplace equation. In principle, the procedure described so far corresponds to that of the method of image charges. In eq. (5.57), the right-hand side formally represents a point charge of magnitude 1, and in eq. (5.58), the first term corresponds to the potential of this point charge. This Green's function is thus the specific solution to the Poisson equation for a point charge of magnitude 1. As we saw in the last section, for one, two, or a few point charges, the resulting electrostatic potential can generally be determined using the method of image charges by superimposing the respective individual potentials. However, for more complex charge distributions, this very quickly becomes impractical. We therefore seek a way to determine, in a general manner, the solution $\phi(\boldsymbol{r})$ to a Poisson equation of the form eq. (4.1), that is of

$$\Delta\phi(\boldsymbol{r}) = -\frac{1}{\epsilon_0}\rho(\boldsymbol{r}) \tag{5.59}$$

for an arbitrary, prescribed charge distribution $\rho(\boldsymbol{r})$ and with certain boundary conditions being satisfied. Fortunately, that is precisely what Green's functions enable us to do. How this works becomes clearer when we write down Green's second identity from eq. (2.45)

using a slightly modified notation. In doing so, we consider the two functions $\varphi(\boldsymbol{r}')$ and $\psi(\boldsymbol{r}')$ within a volume $V$ that is bounded by a surface $\partial V$, the unit normal vector of which is $\hat{\boldsymbol{n}}'$. With that Green's second identity becomes

$$\int_\Omega (\varphi \Delta_{\boldsymbol{r}'}\psi - \psi \Delta_{\boldsymbol{r}'}\varphi)\,\mathrm{d}^3r' = \int_{\partial V}\left(\varphi\frac{\partial\psi}{\partial\hat{n}'} - \psi\frac{\partial\varphi}{\partial\hat{n}'}\right)\mathrm{d}\sigma' \tag{5.60}$$

where

$$\frac{\partial\varphi}{\partial\hat{n}'} = \hat{\boldsymbol{n}}'\cdot\nabla_{\boldsymbol{r}'}\varphi \qquad \text{and} \qquad \frac{\partial\psi}{\partial\hat{n}'} = \hat{\boldsymbol{n}}'\cdot\nabla_{\boldsymbol{r}'}\psi$$

are the directional derivatives of $\varphi$ and $\psi$ in the direction of the surface normal $\hat{\boldsymbol{n}}'$ of $\partial V$, pointing outward from the volume $V$. We now set $\varphi = G(\boldsymbol{r},\boldsymbol{r}')$ and $\psi = \phi(\boldsymbol{r}')$ in eq. (5.60) and obtain:

$$\int_V \left[G(\boldsymbol{r},\boldsymbol{r}')\Delta_{\boldsymbol{r}'}\phi(\boldsymbol{r}') - \phi(\boldsymbol{r}')\Delta_{\boldsymbol{r}'}G(\boldsymbol{r},\boldsymbol{r}')\right]\mathrm{d}^3r'$$

$$= \int_{\partial V}\left[G(\boldsymbol{r},\boldsymbol{r}')\frac{\partial\phi(\boldsymbol{r}')}{\partial\hat{n}'} - \phi(\boldsymbol{r}')\frac{\partial G(\boldsymbol{r},\boldsymbol{r}')}{\partial\hat{n}'}\right]\mathrm{d}\sigma'$$

On the left-hand side of this equation, we now substitute the corresponding values from eq. (5.59) and eq. (5.57) for $\phi(\boldsymbol{r}')$ and $\Delta_{\boldsymbol{r}}G(\boldsymbol{r},\boldsymbol{r}')$, yielding:

$$\phi(\boldsymbol{r}) = \int_V G(\boldsymbol{r},\boldsymbol{r}')\rho(\boldsymbol{r}')\,\mathrm{d}^3r' + \epsilon_0\int_{\partial V}\left[G(\boldsymbol{r},\boldsymbol{r}')\frac{\partial\phi(\boldsymbol{r}')}{\partial\hat{n}'} - \phi(\boldsymbol{r}')\frac{\partial G(\boldsymbol{r},\boldsymbol{r}')}{\partial\hat{n}'}\right]\mathrm{d}\sigma' \tag{5.61}$$

The first integral on the right-hand side of this equation describes the contribution to the potential from the charge distribution within the volume $V$, and the second integral describes the value of the potential on the boundary $\partial V$ of this region. Eq. (5.61) provides the solution $\phi(\boldsymbol{r})$ to eq. (5.59) for arbitrary Green's functions, where the boundary conditions of the Poisson equation enter into the determination of $\phi(\boldsymbol{r})$ via the equation's second integral. Since we made no assumptions regarding the value of $\phi$ on the boundary surface $\partial V$ when deriving eq. (5.61), eq. (5.61) is relevant not only for conducting bodies.

At the beginning of our examination of ideal conductors in an electrostatic field, we learned that to solve the Poisson equation, one must know either the potential $\phi(\boldsymbol{r})$ on the boundary of $V$ (eq. (5.20b)) or the directional derivative $\partial\phi(\boldsymbol{r})/\partial n$ on $\partial V$ (eq. (5.21b)) in order for the solution $\phi(\boldsymbol{r})$ to the Poisson equation to be unique. However, regarding the surface integral in eq. (5.61), it appears as though one would need to know both quantities. This is not the case. With a suitable choice of $G(\boldsymbol{r},\boldsymbol{r}')$, given the conditions in eq. (5.20b) or eq. (5.21b), the boundary conditions according to Dirichlet and von Neumann, one of the two terms in the surface integral can be made to vanish or transformed into a physically irrelevant constant.

### Green's Function with Dirichlet Boundary Conditions

For Dirichlet boundary conditions (eq. (5.20b)), the potential $\phi(r)$ is prescribed on $\partial V$. In this case, one chooses for $G(\boldsymbol{r}, \boldsymbol{r}')$:

$$G(\boldsymbol{r}, \boldsymbol{r}') = 0 \qquad \text{if} \qquad \boldsymbol{r} \in \partial V \tag{5.62}$$

As a result, the first term in the surface integral in eq. (5.61) vanishes, and one obtains:

$$\phi(\boldsymbol{r}) = \int_V G(\boldsymbol{r}, \boldsymbol{r}')\rho(\boldsymbol{r}')\,\mathrm{d}^3r' + \epsilon_0 \int_{\partial V} \phi(\boldsymbol{r}') \frac{\partial G(\boldsymbol{r}, \boldsymbol{r}')}{\partial \hat{n}'}\,\mathrm{d}\sigma' \tag{5.63}$$

Another property of Green's functions with Dirichlet boundary conditions is that they are always symmetric; that is, $G(\boldsymbol{r}, \boldsymbol{r}') = G(\boldsymbol{r}', \boldsymbol{r})$ holds (see Appendix A).

### Green's Function with von Neumann Boundary Conditions

In the case of von Neumann boundary conditions (eq. (5.21b)), the directional derivative $\partial\phi(\boldsymbol{r})/\partial n$ is prescribed on $\partial V$. However, here we cannot simply set the directional derivative $\partial G(\boldsymbol{r}, \boldsymbol{r}')/\partial \hat{n}'$ to zero within the surface integral of eq. (5.61) in order to make the term involving $\phi(\boldsymbol{r}')$ vanish. To understand why, we simply need to integrate the left-hand side of eq. (5.57) over the volume $V$ and apply Gauss's theorem. The result is:

$$\int_V \nabla_{\boldsymbol{r}} \big[\nabla_{\boldsymbol{r}} G(\boldsymbol{r}, \boldsymbol{r}')\big]\,\mathrm{d}^3r = \int_{\partial V} \nabla_{\boldsymbol{r}} G(\boldsymbol{r}, \boldsymbol{r}') \cdot \hat{\boldsymbol{n}}\,\mathrm{d}\sigma = \int_{\partial V} \frac{\partial G(\boldsymbol{r}, \boldsymbol{r}')}{\partial \hat{n}}\,\mathrm{d}\sigma$$

If we now perform the same integration for the right-hand side of eq. (5.57), we find that the result is

$$-\frac{1}{\epsilon_0} \int_V \delta^{(3)}(\boldsymbol{r} - \boldsymbol{r}')\,\mathrm{d}^3r = -\frac{1}{\epsilon_0}$$

from which it follows that

$$\int_{\partial V} \frac{\partial G(\boldsymbol{r}, \boldsymbol{r}')}{\partial \hat{n}}\,\mathrm{d}\sigma = -\frac{1}{\epsilon_0} \tag{5.64}$$

Clearly, the directional derivative of $G(\boldsymbol{r}, \boldsymbol{r}')$ cannot possibly vanish. However, by requiring that

$$\frac{\partial G(\boldsymbol{r}, \boldsymbol{r}')}{\partial \hat{n}} = -\frac{1}{\epsilon_0 S} \qquad \text{falls} \qquad \boldsymbol{r} \in \partial V \tag{5.65}$$

holds, where $S$ is the area of $\partial V$, we can then, by using the constant area-weighted potential

$$\langle \phi \rangle_S = \frac{1}{S} \int_{\partial V} \phi(\boldsymbol{r}')\,\mathrm{d}\sigma' \tag{5.66}$$

write eq. (5.61) as:

$$\phi(\boldsymbol{r}) = \int_V G(\boldsymbol{r},\boldsymbol{r}')\rho(\boldsymbol{r}')\,\mathrm{d}^3r' + \epsilon_0\int_{\partial V} G(\boldsymbol{r},\boldsymbol{r}')\frac{\partial\phi(\boldsymbol{r}')}{\partial\hat{n}'}\,\mathrm{d}\sigma' + \langle\phi\rangle_S \tag{5.67}$$

With $\boldsymbol{E} = -\nabla\phi$, the constant term $\langle\phi\rangle_S$ has no influence on the determination of $\boldsymbol{E}$. Unlike the case with Dirichlet boundary conditions, the Green's function is not automatically symmetric under von Neumann boundary conditions.

### 5.4.1 Point Charge in Front of a Grounded Metal Plate

We consider a situation comparable to that shown in fig. 5.7, with the difference that, instead of the entire half-space $z < 0$ being filled with a metal body, there is merely an infinitely extended, thin metal plate lying in the $xy$-plane, with a positive point charge located at a distance $\boldsymbol{r}'$ from it. Thus, our volume $V$ is the gray left half-space ($z > 0$) in fig. 5.12, and $\partial V$ is the surface of the metal plate in the $xy$-plane. Since the potential of the metal plate is specified as $\phi = 0$, we are dealing with a Dirichlet boundary condition. We are already familiar with the Green's function for a point charge of magnitude 1 in free space at $\boldsymbol{r}'$. Using eq. (5.58) and eq. (5.62), the boundary condition for the Green's function is:

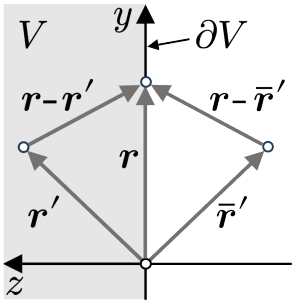


**Fig. 5.12**

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{1}{4\pi\epsilon_0}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} + F(\boldsymbol{r},\boldsymbol{r}') = 0 \qquad \text{if} \qquad \boldsymbol{r}\in\partial V \tag{5.68}$$

The term following the first equals sign in this equation , i.e., the Green's function for a point charge of magnitude 1 in empty space located at $\boldsymbol{r}'$, depends only on the magnitude of the distance between the points $\boldsymbol{r}$ and $\boldsymbol{r}'$ and approaches zero asymptotically as $\boldsymbol{r}\to\infty$ and $\boldsymbol{r}'\to\infty$. This must also hold for $F(\boldsymbol{r},\boldsymbol{r}')$, and $F(\boldsymbol{r},\boldsymbol{r}')$ is therefore likewise the Green's function of a point charge. But where is this point charge located and what charge does it carry? The boundary condition tells us that this location must be at the same distance from all points $\boldsymbol{r}$ on $\partial V$ as the point charge at $\boldsymbol{r}'$. There is only one such location, namely $\bar{\boldsymbol{r}}'$, the reflection of $\boldsymbol{r}'$ across the $xy$-plane. For $G(\boldsymbol{r},\boldsymbol{r}')$ to vanish everywhere on $\partial V$, the point charge at $\bar{\boldsymbol{r}}'$ must be equal to $-1$. Thus, the Green's function satisfying eq. (5.68) is:

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} + \frac{-1}{|\boldsymbol{r}-\bar{\boldsymbol{r}}'|} \tag{5.69}$$

As can be easily verified, $G(\boldsymbol{r},\boldsymbol{r}')$ satisfies eq. (5.57). Written out in coordinates, this Green's function is:

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{1}{4\pi\epsilon_0}\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}} - \frac{1}{4\pi\epsilon_0}\frac{1}{\sqrt{(x-x')^2+(y-y')^2+(z+z')^2}} \tag{5.70}$$

In this form, one immediately sees that $G(\boldsymbol{r},\boldsymbol{r}')$ is symmetric with respect to the interchange of $\boldsymbol{r}$ and $\boldsymbol{r}'$. Let us now consider the specific situation involving a point charge $+q$ at the position $\boldsymbol{a} = (a_1, a_2, a_3)$. The charge distribution $\rho(\boldsymbol{r}')$ that we must substitute into eq. (5.63) is therefore:

$$\rho(\boldsymbol{r}') = q\cdot\delta^{(3)}(\boldsymbol{r}'-\boldsymbol{a}) \tag{5.71}$$

or written out in coordinates:

$$\rho(\boldsymbol{r}') = q\cdot\delta(x'-a_1)\delta(y'-a_2)\delta(z'-a_3) \tag{5.72}$$

We now substitute eq. (5.70) and eq. (5.72) into eq. (5.63). With $\phi(\boldsymbol{r}') = 0$ on $\partial V$, the surface integral in eq. (5.63) vanishes, and the volume integral in eq. (5.63) over the half-space $V$ shown in fig. 5.12, i.e., for $z \geq 0$, is:

$$\int_V G(\boldsymbol{r},\boldsymbol{r}')\rho(\boldsymbol{r}')\,\mathrm{d}^3r' = \frac{1}{4\pi\epsilon_0}\int_{-\infty}^{+\infty}\mathrm{d}x'\int_{-\infty}^{+\infty}\mathrm{d}y'\int_{0}^{+\infty}\mathrm{d}z'\left(\frac{q\cdot\delta(x'-a_1)\delta(y'-a_2)\delta(z'-a_3)}{\sqrt{(x-x')^2+(y-y')^2+(z-z')^2}} - \frac{q\cdot\delta(x'-a_1)\delta(y'-a_2)\delta(z'-a_3)}{\sqrt{(x-x')^2+(y-y')^2+(z+z')^2}}\right)$$

For the sake of simplicity, let us place $q$ on the $z$-axis, i.e., $a_1 = a_2 = 0$ and $a_3 = a$, then eq. (5.63) yields the following for the electrostatic potential:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{\sqrt{x^2+y^2+(z-a)^2}} - \frac{q}{\sqrt{x^2+y^2+(z+a)^2}}\right) \tag{5.73}$$

This is eq. (5.27), which we previously derived in section 5.3.1 for a point charge in front of a grounded metal surface, where the entire half-space $z < 0$ consisted of a metal body. Thus, it makes no difference whether the entire half-space $z < 0$ is filled with metal or whether there is merely a thin metal plate in the $xy$-plane. For the electrostatic field, using $\boldsymbol{E} = -\nabla\phi$, we obtain for $z \geq 0$:

$$\boldsymbol{E} = \frac{1}{4\pi\epsilon_0}\left(\frac{q(\boldsymbol{r}-\boldsymbol{a})}{\left(x^2+y^2+(z-a)^2\right)^{3/2}} - \frac{q(\boldsymbol{r}+\boldsymbol{a})}{\left(x^2+y^2+(z+a)^2\right)^{3/2}}\right) \tag{5.74}$$

where $\boldsymbol{a} = (0,0,a)$ is the coordinate vector of the point charge $q$. The field vector $\boldsymbol{E}$ in the $xy$-plane, i.e., at $z = 0$, is thus:

$$\boldsymbol{E}(x,y,z=0) \;=\; -\frac{1}{4\pi\epsilon_0}\frac{2qa}{\left(x^2+y^2+a^2\right)^{3/2}}\cdot\hat{\boldsymbol{k}} \;=\; E_z\cdot\hat{\boldsymbol{k}} \tag{5.75}$$

In the $xy$-plane, $\boldsymbol{E}$ has only a $z$-component; it is thus perpendicular to the metal plate and points towards the half-space $z<0$, where the negative image charge is located. For the surface charge density, using eq. (5.21b), we find

$$\sigma \;=\; -\epsilon_0\frac{\partial\phi(\mathbf{r})}{\partial\hat{n}_M}\bigg|_{\partial M} \;=\; -\epsilon_0\cdot\hat{\mathbf{n}}_M\cdot\nabla\phi(\mathbf{r})\bigg|_{z=0} \;=\; \epsilon_0\cdot\hat{\mathbf{n}}_{\mathbf{M}}\cdot\hat{\mathbf{k}}\cdot E_z \tag{5.76}$$

Here, $\hat{\boldsymbol{n}}_{\boldsymbol{M}}$ is not the unit normal vector to $\partial V$ but rather the one to the boundary $\partial M$ of the metal plate, which, for $z>0$, points in the same direction as the unit vector $\hat{\boldsymbol{k}}$ of the $z$-coordinate. Thus, we get the following expression for the surface charge density:

$$\sigma \;=\; \epsilon_0 E_z \;=\; -\frac{1}{2\pi}\frac{qa}{\left(x^2+y^2+a^2\right)^{3/2}} \tag{5.77}$$

We obtain the induction charge $Q$ on the metal plate by integrating this surface charge density in the $xy$-plane using polar coordinates ($x^2+y^2=r^2$):

$$Q \;=\; -\frac{qa}{2\pi}\int_0^{2\pi}\!\!\int_0^{\infty}\frac{r}{\left(r^2+a^2\right)^{3/2}}\,\mathrm{d}r\,\mathrm{d}\varphi \;=\; \frac{qa}{\sqrt{r^2+a^2}}\bigg|_0^{\infty} \;=\; -q \tag{5.78}$$

The total induction charge on the metal plate is therefore equal to the image charge $-q$. To determine the force between this induction charge $Q$ and the positive charge in front of the metal plate, we must sum up all contributions

$$\mathrm{d}\boldsymbol{F} \;=\; \mathrm{d}Q\cdot\boldsymbol{E} \;=\; \sigma\boldsymbol{E}\,r\,\mathrm{d}r\,\mathrm{d}\varphi \tag{5.79}$$

For $\boldsymbol{E}$, only the contribution from the positive charge in eq. (5.72) at $z=0$ is to be used here, since otherwise we would be double-counting, i.e., both, the induction charge and the image charge. The field of the positive charge at $z=0$ is:

$$\boldsymbol{E}^{+}_{z=0} \;=\; \frac{1}{4\pi\epsilon_0}\frac{q}{\left(x^2+y^2+a^2\right)^{3/2}}\begin{pmatrix} z\\ y\\ -a\end{pmatrix} \tag{5.80}$$

Thus, upon integration in polar coordinates, we obtain the following expression for the force between the positive point charge and the induction charge:

$$\boldsymbol{F} \;=\; -\frac{qa}{2\pi}\frac{q}{4\pi\epsilon_0}\int_0^{2\pi}\!\!\int_0^{\infty}\begin{pmatrix} r\cos\varphi\\ r\sin\varphi\\ -a\end{pmatrix}\frac{r\,\mathrm{d}r\,\mathrm{d}\varphi}{\left(r^2+a^2\right)^3} \;=\; \frac{q^2a^2}{4\pi\epsilon_0}\int_0^{\infty}\frac{r\,\mathrm{d}r\,\mathrm{d}\varphi}{\left(r^2+a^2\right)^3}\cdot\hat{\boldsymbol{k}} \tag{5.81}$$

Upon angular integration, the $x$- and $y$-components of the force drop out, leaving only a force component in the $z$-direction:

$$\boldsymbol{F} \;=\; \frac{q^2a^2}{4\pi\epsilon_0}\left(-\frac{1}{4}\frac{1}{(r^2+a^2)^2}\bigg|_0^{\infty}\right) \;=\; \frac{1}{4\pi\epsilon_0}\cdot\frac{q^2}{(2a)^2} \tag{5.82}$$

This is exactly the result from eq. (5.28) that we obtained previously by directly calculating the force acting between the positive charge $q$ and the image charge $-q$. Finally, we need to consider the directional derivative of the Green's function on the boundary surface $\partial V$. If we set $x' = 0$, $y' = 0$, and $z' = a$ in eq. (5.70), then $G(\boldsymbol{r}, \boldsymbol{r}')$ differs from the potential $\phi(\boldsymbol{r})$ in eq. (5.73) only by the factor $q$. Thus, we seemingly can obtain the directional derivative of $G(\boldsymbol{r}, \boldsymbol{r}')$ quite simply by substituting $q \cdot G(\boldsymbol{r}, \boldsymbol{r}')$ for $\phi(\boldsymbol{r})$ in eq. (5.76). However, this impression is misleading, because the directional derivative of $\phi(\boldsymbol{r})$ in eq. (5.76) is taken with respect to the boundary of the metal plate (see also the derivation of eq. (5.19b)) and not with respect to the boundary surface of $V$. We must therefore also replace $\hat{\boldsymbol{n}}$ with $-\hat{\boldsymbol{n}}$ in eq. (5.76) to obtain the directional derivative of $G(\boldsymbol{r}, \boldsymbol{r}')$ that we are seeking. Doing so, we obtain:

$$\frac{\partial G(\boldsymbol{r}, \boldsymbol{r}')}{\partial \hat{n}} = \frac{\sigma}{\epsilon_0} = -\frac{1}{2\pi\epsilon_0} \frac{a}{\left(x^2 + y^2 + a^2\right)^{3/2}} \tag{5.83}$$

If we integrate this equation over $\partial V$ and evaluate the right-hand side of the equation in the same way we just did to calculate the total induced charge, the result we obtain is that of eq. (5.64).

### 5.4.2 Point Charge in Front of a Grounded Metal Sphere

As a second example for the Green's functions ansatz, we will revisit the situation of a point charge in front of a grounded metal sphere which we first discussed in section 5.3.2. In this case, we already know that the method of images works. The volume $V$ we are considering here is the region outside the metal sphere, and $\partial V$ is the surface of the sphere. From section 5.3.2 we know where the image charge must be located and we also know the magnitude of its charge. However, let us set aside this knowledge and start from scratch. Fig. 5.13 shows the situation in the plane defined by $\boldsymbol{r}$ and $\boldsymbol{r}'$. The charge $q$ is located at $\boldsymbol{r}'$, and we wish to determine the potential $\phi(\boldsymbol{r})$ for points $\boldsymbol{r}$ outside the metal sphere. We know that, for reasons of symmetry, the image charge must be located somewhere along the radius vector $\boldsymbol{r}'$ inside the sphere, say, at a distance $\bar{r}'$ from the center of the sphere. We therefore make the following ansatz for the function $F(\boldsymbol{r}, \boldsymbol{r}')$:

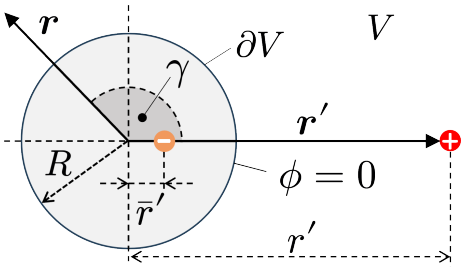



**Fig. 5.13**

$$F(\boldsymbol{r}, \boldsymbol{r}') = \frac{1}{4\pi\epsilon_0} \frac{\alpha}{|\boldsymbol{r} - \bar{\boldsymbol{r}}'|}$$

$F(\boldsymbol{r},\boldsymbol{r}')$ is an $\alpha$-weighted Green's function for a point charge at $\bar{\boldsymbol{r}}'$. We must determine the value of $\alpha$ and the exact position using the boundary condition for the Green's function. Since the sphere is grounded, meaning it is at zero potential, we are once again dealing with a Dirichlet boundary condition. We therefore require, in accordance with eq. (5.62):

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{1}{4\pi\epsilon_0}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} + \frac{1}{4\pi\epsilon_0}\frac{\alpha}{|\boldsymbol{r}-\bar{\boldsymbol{r}}'|} = 0 \qquad \text{if} \qquad \boldsymbol{r}\in\partial V \tag{5.84}$$

Here, $\partial V$ is the spherical surface of radius $R$, and the defining equation for $\alpha$ and $\bar{\boldsymbol{r}}'$ is therefore:

$$\frac{1}{\sqrt{R^2-2Rr'\cos\gamma+r'^2}} = \frac{-\alpha}{\sqrt{R^2-2R\bar{r}'\cos\gamma+\bar{r}'^2}}$$

Since $\boldsymbol{r}'$ and $\bar{\boldsymbol{r}}'$ are collinear, $\cos\gamma$ represents the angle between $\boldsymbol{R}$ and $\boldsymbol{r}'$ as well as the angle between $\boldsymbol{R}$ and $\bar{\boldsymbol{r}}'$. Now, we factor $r'^2$ out of the square root on the left-hand side and $R^2$ on the right-hand side, obtaining

$$\frac{1}{r'\sqrt{1+\left(R^2/r'^2\right)-2\left(R/r'\right)\cos\gamma}} = \frac{-\alpha}{R\sqrt{1+\left(\bar{r}'^2/R^2\right)-2\left(\bar{r}'/R\right)\cos\gamma}}$$

Obviously, this equation is satisfied if

$$\frac{R}{r'} = \frac{\bar{r}'}{R} \quad\Rightarrow\quad \bar{\boldsymbol{r}}' = \frac{R^2}{r'^2}\boldsymbol{r}' \qquad \text{and} \qquad \alpha = -\frac{R}{r'}$$

Thus, the Green's function satisfying eq. (5.84) is:

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{1}{4\pi\epsilon_0}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} - \frac{1}{4\pi\epsilon_0}\frac{R}{r'\left|\boldsymbol{r}-\left(R^2/r'^2\right)\boldsymbol{r}'\right|} \tag{5.85}$$

The symmetry of the problem suggests the use of spherical coordinates, and for this purpose, we write eq. (5.85) as:

$$\begin{aligned} G(\boldsymbol{r},\boldsymbol{r}') = {} & \frac{1}{4\pi\epsilon_0}\frac{1}{\sqrt{r^2+r'^2-2rr'\cos\gamma}} \\ & - \frac{1}{4\pi\epsilon_0}\frac{1}{\sqrt{\left(r^2r'^2/R^2\right)+R^2-2rr'\cos\gamma}} \end{aligned} \tag{5.86}$$

The angle $\gamma$ is a function of the angles $(\vartheta',\varphi')$ specifying the location of the point charge $q$ and the angles $(\vartheta,\varphi)$ specifying the position where we are determining the potential. To obtain $\cos\gamma$ as a function of $(\vartheta',\varphi',\vartheta,\varphi)$, we simply multiply the two unit vectors in the directions of $\boldsymbol{r}$ and $\boldsymbol{r}'$

$$\hat{\boldsymbol{r}}\cdot\hat{\boldsymbol{r}}' \;=\; 1\cdot 1\cdot\cos\gamma \;=\; \begin{pmatrix}\sin\vartheta\cos\varphi\\ \sin\vartheta\sin\varphi\\ \cos\vartheta\end{pmatrix}\cdot\begin{pmatrix}\sin\vartheta'\cos\varphi'\\ \sin\vartheta'\sin\varphi'\\ \cos\vartheta'\end{pmatrix}$$

and thus obtain:

$$\cos\gamma(\vartheta',\varphi',\vartheta,\varphi) \;=\; \cos\vartheta'\cos\vartheta+\sin\vartheta'\sin\vartheta\cos(\varphi'-\varphi) \tag{5.87}$$

Next, we determine $\rho(\boldsymbol{r}')$. For a point charge at the position $\boldsymbol{r}_1 = (r_1,\vartheta_1,\varphi_1)$ with a distance $r_1 > R$ from the center of the sphere, we thus have, using the $\delta$-function in spherical coordinates (see eq. (A.16) in Appendix A):

$$\rho(\boldsymbol{r}') \;=\; \frac{q}{r'^2\sin\vartheta'}\delta(r'-r_1)\delta(\vartheta-\vartheta_1)\delta(\varphi-\varphi_1) \tag{5.88}$$

We now evaluate eq. (5.63) using eq. (5.86) and eq. (5.88). Since $\phi = 0$ on the spherical shell, the surface integral vanishes again, and for the volume integral in eq. (5.63) over the region outside the sphere ($R \geq 0$), we obtain:

$$\int_V G(\boldsymbol{r},\boldsymbol{r}')\rho(\boldsymbol{r}')\,\mathrm{d}^3r' \;=\; \frac{q}{4\pi\epsilon_0}\int_0^{2\pi}\int_0^{\pi}\int_R^{+\infty}\left(\frac{\delta(r'-r_1)\delta(\vartheta-\vartheta_1)\delta(\varphi-\varphi_1)}{\sqrt{r^2+r'^2-2rr'\cos\gamma(\vartheta',\varphi',\vartheta,\varphi)}}\right.$$
$$\left.-\frac{\delta(r'-r_1)\delta(\vartheta-\vartheta_1)\delta(\varphi-\varphi_1)}{\sqrt{\left(r^2r'^2/R^2\right)+R^2-2rr'\cos\gamma(\vartheta',\varphi',\vartheta,\varphi)}}\right)\mathrm{d}r'\,\mathrm{d}\vartheta'\,\mathrm{d}\varphi'$$

The integral is easy to evaluate and with eq. (5.63) we get for $\phi(\boldsymbol{r})$

$$\phi(\boldsymbol{r}) \;=\; \frac{q}{4\pi\epsilon_0}\left(\frac{1}{\sqrt{r^2+r_1^2-2rr_1\cos\gamma}}-\frac{1}{\sqrt{\left(r^2r_1^2/R^2\right)+R^2-2rr_1\cos\gamma}}\right) \tag{5.89}$$

where $\cos\gamma$ is the cosine of the angle between the vector $\boldsymbol{r}_1$ pointing to the point charge $q$ and the vector $\boldsymbol{r}$, i.e., $\cos\gamma(\vartheta_1,\varphi_1,\vartheta,\varphi)$, at which we evaluate the potential. Letting $\beta = R/r_1$, we can rewrite $\phi(\boldsymbol{r})$ as:

$$\phi(\boldsymbol{r}) \;=\; \frac{q}{4\pi\epsilon_0}\left(\frac{1}{\sqrt{r^2+r_1^2-2rr_1\cos\gamma}}-\frac{\beta}{\sqrt{r^2+\beta^4r_1^2-2\beta^2rr_1\cos\gamma}}\right) \tag{5.90}$$

Except for the fact that $\cos\gamma$ appears here in eq. (5.86), whereas $\cos\vartheta$ appears in the corresponding equation eq. (5.32), which we derived using the method of images, the result is identical. The difference is that, at this stage, we have not yet specified a particular orientation for the vector $\boldsymbol{r}_1$ relative to the point charge $q$. However, if we do this and, just as in eq. (5.32), place the point charge $q$ on the positive $z$-axis, then $\cos\vartheta_1 = 1$ and $\sin\vartheta_1 = 0$; and since $\cos\gamma = \cos\vartheta$ holds according to eq. (5.87), we obtain

exactly eq. (5.32). Due to spherical symmetry, however, our result holds for any $\boldsymbol{r}' = \boldsymbol{r}_1$, and we can therefore write the vector form of eq. (5.86) as:

$$\phi(\boldsymbol{r}) \;=\; \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} - \frac{\beta}{|\boldsymbol{r}-\beta^2\boldsymbol{r}'|}\right) \quad , \quad \beta = \frac{R}{|\boldsymbol{r}'|} > 1 \tag{5.91}$$

This, however, is just the result we already derived in eq. (5.31) using the method of image charges. This is no coincidence. The method of image charges is an intuitive approach that corresponds to how one proceeds when applying the Green's function ansatz; and it has the advantage of quickly leading to results when favorable symmetry exists. However, the formal approach using Green's functions is much more generally applicable. For instance, it allows us to consider situations where the prescribed potential on $\partial V$ is not zero but has another constant value, or also cases where the conductor is electrically isolated. If, for example, the metal sphere is not grounded but at a constant potential $\phi_0$, the surface integral in eq. (5.63) no longer vanishes, and we therefore require $\partial G(\boldsymbol{r},\boldsymbol{r}')/\partial\hat{n}'$ for $|\boldsymbol{r}'| = R$.

$$\left.\frac{\partial G(\boldsymbol{r},\boldsymbol{r}')}{\partial\hat{n}'}\right|_{r'=R} \;=\; \hat{\boldsymbol{n}}'\nabla_{\boldsymbol{r}'}G(\boldsymbol{r},\boldsymbol{r}') \;=\; \hat{\boldsymbol{n}}'\cdot\hat{\boldsymbol{r}}'\cdot\frac{1}{4\pi\epsilon_0}\cdot\frac{r^2-R^2}{R\big(r^2+R^2-2rR\cos\gamma\big)^{3/2}}$$

The unit normal vector $\hat{\boldsymbol{n}}'$ of $\partial V$ points outward from the volume $V$ over which the integration is performed. This volume $V$ is the region outside the metal sphere; thus, $\hat{\boldsymbol{n}}'$ points from the spherical shell toward the center of the sphere and is therefore antiparallel to $\hat{\boldsymbol{r}}'$. Consequently, $\hat{\boldsymbol{n}}'\cdot\hat{\boldsymbol{r}}' = -1$, and we obtain the result:

$$\left.\frac{\partial G(\boldsymbol{r},\boldsymbol{r}')}{\partial\hat{n}'}\right|_{r'=R} \;=\; -\frac{1}{4\pi\epsilon_0}\cdot\frac{r^2-R^2}{R\big(r^2+R^2-2rR\cos\gamma\big)^{3/2}} \tag{5.92}$$

The next step is to substitute eq. (5.92), along with the constant potential $\phi(\boldsymbol{r}) = \phi_0$, into the surface integral of eq. (5.63) and then perform the integration. However, before doing so, we simplify matters by placing our point charge $q$ on the $z$-axis, such that $\cos\gamma = \cos\vartheta'$, meaning $\gamma$ becomes identical to the integration angle $\vartheta$. This yields the following expression for the additional potential term arising from the surface integral in eq. (5.63):

$$\phi_{\partial V} \;=\; \frac{\phi_0}{4\pi}\int_0^{2\pi}\!\!\int_0^{\pi}\frac{r^2-R^2}{R\big(r^2+R^2-2rR\cos\vartheta'\big)^{3/2}}R^2\sin\vartheta'\,\mathrm{d}\vartheta'\,\mathrm{d}\varphi' \tag{5.93}$$

Integration over $\varphi'$ yields a factor of $2\pi$, and for the integration over $\vartheta'$ we make the substitution $u = \cos\vartheta'$, with the outcome:

$$\phi_{\partial V} \;=\; \frac{\phi_0 R}{2}(r^2-R^2)\int_{-1}^{1}\frac{\mathrm{d}u}{\big(r^2+R^2-2rRu\big)^{3/2}} \tag{5.94}$$

Evaluating the integral is straightforward, and we obtain the following result:

$$\phi_{\partial V} \;=\; \frac{\phi_0 R}{2}(r^2 - R^2)\cdot\frac{2}{r(r^2-R^2)} \;=\; \phi_0\frac{R}{r} \tag{5.95}$$

If, therefore, the metal sphere is not at potential $\phi(R) = 0$ but rather at a constant potential $\phi(R) = \phi_0$, then eq. (5.91) becomes

$$\phi(\boldsymbol{r}) \;=\; \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} - \frac{\beta}{|\boldsymbol{r}-\beta^2\boldsymbol{r}'|}\right) + \phi_0\frac{R}{|\boldsymbol{r}|} \quad , \quad \beta = \frac{R}{|\boldsymbol{r}'|} > 1 \tag{5.96}$$

The potential $\phi_0$ therefore acts like an additional positive point charge $q_0$ at the center of the metal sphere, carrying a charge of $q_0 = 4\pi\epsilon_0 R\phi_0$, that produces the constant potential $\phi_0$ on the sphere's surface. Using eq. (5.88), we can also determine the potential field for a completely isolated metal sphere. There are no charges inside the metal sphere; the image charge is merely a construct used to generate the surface charge density $\sigma$, which arises from the induced charges that shield the electrostatic field in the interior, in such a way that the prescribed boundary conditions at the sphere's surface are satisfied. For the electrically isolated metal sphere, this implies that the integral of $\sigma$ over $\partial V$ must vanish. The surface charge density corresponding to $\phi_0$ in eq. (5.96) is:

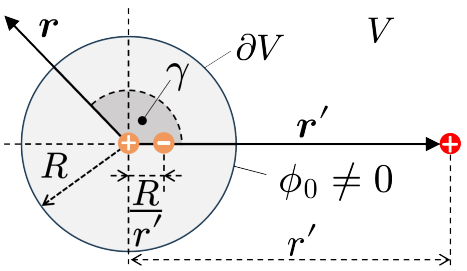



**Fig. 5.14**

$$\begin{aligned}\sigma(\boldsymbol{r}) \;&=\; -\epsilon_0\frac{\partial\phi(\boldsymbol{r})}{\partial\hat{n}}\bigg|_{r=R} \\ &=\; -\epsilon_0\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{R}}\cdot\left(\frac{q}{4\pi\epsilon_0}\frac{r'^2 - R^2}{R\big(r'^2 + R^2 - 2r'R\cos\gamma\big)^{3/2}} + \frac{\phi_0}{R}\right)\end{aligned} \tag{5.97}$$

Here, the unit normal vector $\hat{\boldsymbol{n}}$ points outward from the spherical surface and is therefore parallel to $\hat{\boldsymbol{R}}$; thus, we have $\hat{\boldsymbol{n}}\cdot\hat{\boldsymbol{R}} = 1$. For the integral of $\sigma$ over $\partial V$, the following must hold:

$$\int_0^{2\pi}\int_0^{\pi}\left(\epsilon_0\frac{\phi_0}{R} - \frac{q}{4\pi}\frac{r'^2 - R^2}{R\big(r'^2 + R^2 - 2r'R\cos\gamma\big)^{3/2}}\right)R^2\sin\vartheta\sin\varphi \;=\; 0$$

The first integrand is trivial, and for the second integrand we can adopt the evaluation from eq. (5.93), we simply need to swap $r$ and $r'$, thereby obtaining a simple equation for determining $\phi_0$:

$$4\pi\epsilon_0 R\cdot\phi_0 - q\cdot\frac{R}{r'} \;=\; 0$$

This condition is met for

$$\phi_0 = \frac{1}{4\pi\epsilon_0}\cdot\frac{q}{r'}$$

With that, we now also know the potential for a charge $q$ located outside an electrically isolated metal sphere. Substituting $\phi_0$ into eq. (5.96), we obtain the following for the isolated metal sphere:

$$\phi(\boldsymbol{r}) = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{|\boldsymbol{r}-\boldsymbol{r'}|} - \frac{\beta}{|\boldsymbol{r}-\beta^2\boldsymbol{r'}|} + \frac{R}{|\boldsymbol{r}||\boldsymbol{r'}|}\right) \quad , \quad \beta = \frac{R}{|\boldsymbol{r'}|} > 1 \tag{5.98}$$

Determining the Green's functions for the problems considered here was not difficult; however, the geometry of the problems was of great assistance. Reflecting a point from outside a sphere to its interior constitutes a conformal mapping. Such conformal mappings preserve angles, and the laws of electrostatics are invariant under these angle-preserving transformations. The fact that we found simple analytical solutions for the Green's functions in the examples discussed here is due to this circumstance. This also helped us in the previous section, where we addressed such problems using the intuitive method of image charges. However, the geometry of the problems is often significantly more complex, and the intuitive image-charge method does not suffice. Green's functions offer the advantage of decoupling the problem's geometry from the specific charge distributions and potentials involved. Green's functions can also be determined for less intuitive situations, and when an analytical approach becomes too difficult, numerical methods can be used. Furthermore, there are other ways to find solutions to Poisson's equation, including analytical ones, though discussing them here would go beyond the scope of this text. However, the section on Green's functions in this chapter offers a preview of these solution methods, which are typically covered in advanced electrodynamics courses.

# 6. Electrostatic Fields in Matter

So far, we have considered electrostatic fields in a vacuum or at the surface of conductors. Electrostatic fields cannot exist within conductors because external electrostatic fields are shielded by freely moving charge carriers, the conduction electrons of metals. However, if an electrically non-conducting material is placed in an electrostatic field, one observes that the field passes through it. The term «dielectric», a coined word derived from Ancient Greek roots, was introduced in physics to indicate that something passes through an electrical material designated in this way. These dielectrics are generally non-conductive, or nearly so, and are therefore often referred to as insulators. The essential property of dielectrics is their polarizability. When an electrical conductor is placed in an electrostatic field, induction charges form in its surface layer, thereby shielding the conductor's interior from the external field. When a dielectric is placed in an electrostatic field, polarization charges form in its surface layer. In metals, the free charge carriers are electrons, whereas in gases or electrolytes, the charge carriers can also be ions. Unlike conductors, dielectric materials do not contain freely moving charge carriers, because all electrons within them remain bound to atoms and molecules. If these atoms or molecules in a dielectric material can be polarized, polarization charges will form in the surface layer of the dielectric body if placed in an external electrostatic field. In principle, an electrostatic field penetrating an insulator interacts with all its atoms and molecules, since they consist of positively and negatively charged particles. Hence, in principle, one would therefore have to consider all these microscopic interactions of charge carriers within atoms and molecules with the electrostatic field individually, but that is, of course, not practical. Moreover, that is usually not what is of interest. The fact that there are no freely moving charge carriers in insulators does not mean there are no mobile charge carriers at all; rather, it means they are bound, which restricts their freedom of movement but still allows them to move around their equilibrium position. External electrostatic fields cause a displacement of positive and negative charges from their equilibrium positions within the overall neutral atoms and molecules of an insulator.

The insulator's response to an external electrostatic field is thus the formation of numerous atomic or molecular dipoles. The macroscopic effect resulting from the formation of these many dipoles within the insulator in response to the external field is known as polarization. This polarization itself is associated with an electrostatic field that opposes the external field. Dielectric polarization cannot only occur through the formation of dipoles but also through the reorientation of polar molecules. The latter occurs when molecules possessing a permanent dipole moment, which are initially oriented randomly within the insulator, become aligned due to the external electrostatic field. In this case, the phenomenon is referred to as orientation polarization.
Effects such as polarization and orientation polarization are macroscopic manifestations of phenomena rooted at the microscopic level. Consequently, the macroscopic quantities we use to describe such phenomena result from averaging over microscopic processes. By employing this procedure, which enables a macroscopic description, we assume that the electrostatic Maxwell equations apply to the numerous microscopic interactions between atoms and molecules and that the aggregate effect of these interactions can be described macroscopically using averaged quantities. The electrostatic field induced by polarization or orientation polarization is one such quantity. To this end, we first consider dielectric polarization itself.

## 6.1 Dielectric Polarization

When an insulator is placed in an electrostatic field, the centers of negative and positive charges within the atoms or molecules shift, resulting in the formation of induced atomic or molecular dipole moments. To gain a qualitative understanding of how these induced dipole moments arise, we will consider the simplest conceivable such case: a hydrogen atom in a uniform electrostatic field. Fig. 6.1 shows a hydrogen atom in field-free space and a hydrogen atom sitting in an electrostatic field $\boldsymbol{E}$. Quantum mechanics tells us that we cannot know exactly where the hydrogen atoms electron is located at any given moment, and that we must visualize the electron, as sketched in fig. 6.1, not as a classic particle but as quantum particle, surrounding the hydrogen atom in a kind of cloud. In the simplest case, this electron cloud is treated as a uniformly charged sphere with a radius equal to the Bohr radius $a_0$. In a field-free region, the center of charge of the electron cloud coincides with the position of the hydrogen proton. When a field $\boldsymbol{E}$ is applied, the centers of charge

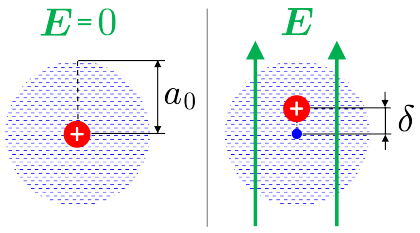



**Fig. 6.1**

of the proton and the electron cloud shift by a distance $\delta$, thereby creating an electric dipole moment

$$\boldsymbol{p} = q \cdot \boldsymbol{\delta} \tag{6.1}$$

where the vector $\boldsymbol{\delta}$ points from the center of negative charge to the center of positive charge. But what is the magnitude of this displacement of the charge centers? The proton and the center of charge of the electron cloud will move apart in an external field until the electrostatic field of the electron cloud at the proton's location cancels out the external field. We are already familiar with the field inside a uniformly charged sphere at a distance $\delta$ from the sphere's center from eq. (4.12). The equilibrium condition for $\delta$ is therefore:

$$\boldsymbol{E} = \frac{q}{4\pi\epsilon_0} \cdot \frac{\boldsymbol{\delta}}{a_0^3} \qquad \Rightarrow \qquad \boldsymbol{\delta} = \frac{4\pi\epsilon_0 a_0^3}{q} \cdot \boldsymbol{E} \tag{6.2}$$

Substituting this equilibrium $\boldsymbol{\delta}$ into eq. (6.1), we obtain:

$$\boldsymbol{p} = \epsilon_0 \cdot 4\pi a_0^3 \cdot \boldsymbol{E} \tag{6.3}$$

The ratio of the atom's induced electric dipole moment to the electric field strength defines the atomic polarizability $\alpha$. Using eq. (6.3), this is:

$$\alpha = \frac{|\boldsymbol{p}|}{|\boldsymbol{E}|} = \epsilon_0 \cdot 4\pi a_0^3 \tag{6.4}$$

In SI units, the atomic polarizability $\alpha$ has the dimension of $\epsilon_0$ times a volume, i.e, its unit of measure is $\mathrm{C^2 \cdot m \cdot N^{-1}}$ or $\mathrm{C \cdot m^2 \cdot V^{-1}}$. In cgs units, $\epsilon_0$ has the value 1 and is dimensionless; consequently, in cgs units, $\alpha$ has the dimension of a volume. In the literature, $\alpha$ is frequently given in units of volume ($\mathrm{Å}^3$), which is, however, incorrect in the SI system. Tab. 6.1 lists measured values for $\alpha/\epsilon_0$ for selected atoms from the first three rows of the periodic table (see tab. A.1 in Appendix B). The polarizability of an atom is greater the easier it is to distort the atom's electron cloud, something that is generally easier with larger atoms. Within a period, atoms with open electron shells are more easily polarized than those with closed shells. Halogens and noble gases are therefore less polarizable than the alkali and alkaline-earth metals in the same row of the periodic table. Given a comparable electron configuration, the more electronegative atom exhibits lower polarizability.

**Tab. 6.1:** Atomic polarizability of selected elements.

| **Element** | H | He | Li | Be | C | N | O | Ne | Na | Al |
|---|---|---|---|---|---|---|---|---|---|---|
| $\boldsymbol{\alpha/\epsilon_0}$ **in [Å$^3$]** | 0.667 | 0.208 | 24.3 | 5.60 | 1.76 | 1.10 | 0.802 | 0.381 | 24.1 | 8.34 |

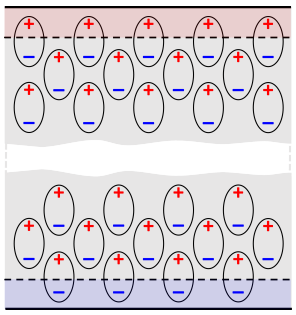


**Fig. 6.2**

The linear relationship between the induced dipole moment and the polarizing electrostatic field in eq. (6.4) holds in many cases, but not for arbitrarily high field strengths $\boldsymbol{E}$. At very high electric field strengths, such as those that can be generated using high-power lasers, $\alpha$ is no longer a constant but becomes a function of the electric field strength itself; consequently, eq. (6.4) no longer applies in this case. However, eq. (6.4) is a very useful relationship for many applications, including the understanding of polarization in an insulator. Let us now consider an insulator composed of atoms or non-polar molecules. Fig. 6.2 schematically illustrates what occurs within the bulk and at the surface of an insulator when it is placed in an external field. Net charges form only at the insulator's interfaces due to induced dipole moments, whereas the interior of the insulator remains neutral. Unlike in metals, where induction charges can flow freely from the bulk of the metal into the metal's boundary layer or from the boundary layer back into the bulk of the metal, surface charges on an insulator are static and limited by the number of induced dipoles located within the respective boundary layer. Therefore, unlike in the case of conductors, one speaks of polarization charges rather than induction charges. Unlike the induction charges in conductors which completely shield electric fields, these polarization charges shield an electric field only partially. Consequently, the electrostatic field penetrates the insulator, and it is precisely this process that gives rise to induced dipoles within the volume of the insulator.

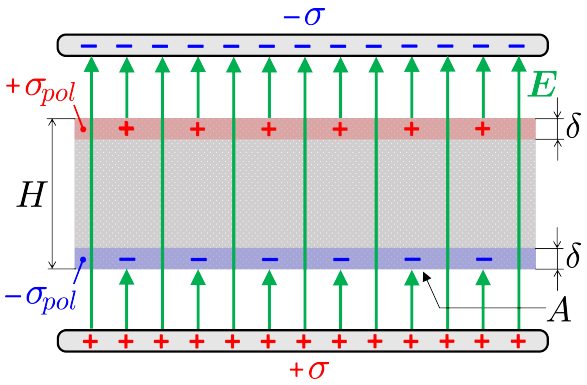



**Fig. 6.3**

To determine the macroscopic polarization of a dielectric body, we consider the simple situation shown in fig. 6.3. Between the electrodes of the illustrated parallel-plate capacitor, there is a cuboid insulator of height $H$, whose faces on the long side , each with an area $A$, are aligned parallel to the capacitor plates. The positive and negative polarization charges in the boundary layers of the insulator facing the electrodes of the parallel-plate capacitor correspond to areas with positive and negative surface charge densities $\pm\sigma_{pol}$. If the insulator contains $n = N/V$ polarizable atoms per unit volume, each associated with a charge $\pm q$ in the corresponding boundary layer, then $\pm\sigma_{pol}$ is given by $n$ multiplied by the volume of the boundary layer $V = A \cdot \delta$ divided by the area $A$, that is

$$\pm\sigma_{pol} \;=\; \pm n\cdot q\cdot\frac{V}{A} \;=\; \pm n\cdot q\cdot\delta \tag{6.5}$$

With eq. (6.1), these surface charge densities are therefore given by

$$\pm\sigma_{pol} \;=\; \pm n\cdot|\boldsymbol{p}| \tag{6.6}$$

Since the quantity $\sigma_{pol}$ does not depend on the volume of the insulator, we can interpret the polarization charges $Q_{pol} = \sigma_{pol}\cdot A$ present in the insulator's boundary layers as the charges of a dipole whose separation distance is simply the height $H$ of the insulator. If we now consider the insulator's dipole moment per unit volume, thereby remaining independent of the insulator's volume $V = A\cdot H$, we obtain the quantity known as dielectric polarization $P$:

$$P \;=\; \frac{1}{V}\cdot Q_{pol}\cdot H \;=\; \frac{1}{V}\cdot\sigma_{pol}\cdot A\cdot H \;=\; \sigma_{pol} \;=\; n\cdot|\boldsymbol{p}| \tag{6.7}$$

The $n$ dipole moments $\boldsymbol{p}$ all point in the same direction. Consequently, the dielectric polarization also points in this direction, and it holds that:

$$\boldsymbol{P} \;=\; n\cdot\boldsymbol{p} \tag{6.8}$$

In the limit of infinitesimal volumes, we replace the particle density $n$ with $\mathrm{d}N/\mathrm{d}V$ and, using $\mathrm{d}\boldsymbol{p} = \boldsymbol{p}\cdot\mathrm{d}N$, obtain the relation

$$\boldsymbol{P} \;=\; \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}V} \tag{6.9}$$

Eq. (6.9) is the defining equation for dielectric polarization, which describes the relative displacement of positive and negative charge carriers within an insulator or, respectively, a dielectric. Although the term «polarization» is frequently used without the adjective «dielectric», this can sometimes be misleading, since the concept of optical polarization also exists alongside dielectric polarization. What can we say about the electric field inside the dielectric? To do this, we consider the dielectric itself as a parallel-plate capacitor with an effective surface charge density $-\sigma_{pol}$. If $E_0 = \sigma/\epsilon_0$ is the electric field strength in the parallel-plate capacitor without the dielectric, then, using eq. (6.7), we obtain the following for the field strength $E$ within the dielectric:

$$E \;=\; \frac{\sigma-\sigma_{pol}}{\epsilon_0} \;=\; \frac{\sigma-P}{\epsilon_0} \;=\; E_0 - \frac{P}{\epsilon_0} \tag{6.10}$$

The electric field strength inside the dielectric is therefore lower by $P/\epsilon_0$ than in a parallel-plate capacitor without a dielectric.

### 6.1.1 Dielectric Displacement

From eq. (6.7), we know that the magnitude of the dielectric polarization, i.e., $|\boldsymbol{P}|$, is precisely the surface charge density $\sigma_{pol}$ arising from the bound charges of the induced dipoles. We shall denote the potential associated with these bound charges as $\phi_{bnd}$. From eq. (3.53), we know the form of the differential contribution $\mathrm{d}\phi_{bnd}(\boldsymbol{r})$ from a microscopic dipole moment $\mathrm{d}\boldsymbol{p}$ at position $\boldsymbol{r}'$ to the potential $\phi_{bnd}(\boldsymbol{r})$:

$$\mathrm{d}\phi_{bnd}(\boldsymbol{r}) \;=\; \frac{\mathrm{d}\boldsymbol{p}(\boldsymbol{r}')}{4\pi\epsilon_0}\cdot\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} \;=\; \frac{\mathrm{d}\boldsymbol{p}(\boldsymbol{r}')}{4\pi\epsilon_0}\cdot\nabla_{\boldsymbol{r}'}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \tag{6.11}$$

If we now rewrite eq. (6.9) to give us

$$\mathrm{d}\boldsymbol{p}(\boldsymbol{r}') \;=\; \boldsymbol{P}(\boldsymbol{r}')\,\mathrm{d}^3r' \tag{6.12}$$

then we can use this relationship to describe the potential $\phi_{bnd}(\boldsymbol{r})$ as an integral over the dielectric polarization within a given volume:

$$\phi_{bnd}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\int\boldsymbol{P}(\boldsymbol{r}')\nabla_{\boldsymbol{r}'}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{6.13}$$

If, in addition to the bound dipole charges of dielectric polarization, there are also free charges satisfying the corresponding equation from tab. 3.1, then the expression for the total electrostatic potential $\phi_{tot}(\boldsymbol{r})$ is given by:

$$\phi_{tot}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\int\left[\frac{\rho_{free}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}+\boldsymbol{P}(\boldsymbol{r}')\nabla_{\boldsymbol{r}'}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}\right]\mathrm{d}^3r' \tag{6.14}$$

Here, $\rho_{free}$ represents the charge density of all free charges, which means it does not include the bound dipole charges contained in $\boldsymbol{P}$. We can now examine how the electrostatic Maxwell equations must change in the presence of insulators. To obtain the first Maxwell equation for this case, we consider the divergence of the electrostatic field generated by eq. (6.14). With $\boldsymbol{E} = -\nabla\phi_{tot}$, we obtain the following expression for $\nabla\boldsymbol{E}$:

$$\nabla\boldsymbol{E}(\boldsymbol{r}) \;=\; -\frac{1}{4\pi\epsilon_0}\int\left[\rho_{free}(\boldsymbol{r}')\Delta_{\boldsymbol{r}}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}+\boldsymbol{P}(\boldsymbol{r}')\nabla_{\boldsymbol{r}'}\left(\Delta_{\boldsymbol{r}}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|}\right)\right]\mathrm{d}^3r' \tag{6.15}$$

Using eq. (2.68), this equation becomes:

$$\nabla\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{1}{\epsilon_0}\int\rho_{free}(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\,\mathrm{d}^3r' + \frac{1}{\epsilon_0}\int\boldsymbol{P}(\boldsymbol{r}')\nabla_{\boldsymbol{r}'}\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\,\mathrm{d}^3r' \tag{6.16}$$

We already know the contribution of the first term on the right-hand side of the equation; it is simply $\rho_{\mathrm{free}}/\epsilon_0$. To evaluate the second integral on the right-hand side of the equation, we first consider the integral of $\nabla_{\boldsymbol{r}'}\left[\boldsymbol{P}(\boldsymbol{r}')\cdot\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\right]$. Since integration and differentiation can be interchanged, the following holds:

$$\int \nabla_{\boldsymbol{r}'}\left[\boldsymbol{P}(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\right]\mathrm{d}^3r' = \nabla_{\boldsymbol{r}'}\int \boldsymbol{P}(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\,\mathrm{d}^3r' = \nabla_{\boldsymbol{r}'}\boldsymbol{P}(\boldsymbol{r}) = 0 \tag{6.17}$$

The integral thus vanishes, and with

$$\nabla_{\boldsymbol{r}'}\left[\boldsymbol{P}(\boldsymbol{r}')\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\right] = \delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\left[\nabla_{\boldsymbol{r}'}\boldsymbol{P}(\boldsymbol{r}')\right] + \boldsymbol{P}(\boldsymbol{r}')\left[\nabla_{\boldsymbol{r}'}\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\right] \tag{6.18}$$

it consequently holds for the second integral on the right-hand side of eq. (6.16) that:

$$\int \boldsymbol{P}(\boldsymbol{r}')\nabla_{\boldsymbol{r}'}\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\,\mathrm{d}^3r' = -\int \delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}')\nabla_{\boldsymbol{r}'}\boldsymbol{P}(\boldsymbol{r}')\,\mathrm{d}^3r' = -\nabla_{\boldsymbol{r}}\boldsymbol{P}(\boldsymbol{r}) \tag{6.19}$$

Substituting this into eq. (6.16), we obtain the equation for the divergence of the electrostatic field in the presence of insulators:

$$\nabla\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{\epsilon_0}\left[\rho_{free}(\boldsymbol{r}) - \nabla\boldsymbol{P}(\boldsymbol{r})\right] \tag{6.20}$$

Before examining this equation in greater detail, we will derive it using a more intuitive approach. Instead of deriving eq. (6.20) from the potential $\phi_{tot}$ generated by the free and bound charges shown in fig. 6.3, we can simply consider an insulator in which the dielectric polarization vector $\boldsymbol{P}$ does not point in the same direction everywhere within the volume of the insulator. To this end, we consider the situation sketched in fig. 6.4, which shows a volume $\Omega$ enclosed by a surface $S$ within an insulator where the direction of $\boldsymbol{P}$ varies. From eq. (6.7), we know that $\boldsymbol{P}$ represents a directed surface charge density. This charge density arises from bound charges and consequently, there can be no flow of free charges out of the volume enclosed by $S$. However, a leakage of charge is still possible. We obtain the magnitude of this polarization charge $\Delta Q_{pol}$ leaking from the volume $\Omega$ by summing the normal component of $\boldsymbol{P}$ over the surface $S$:

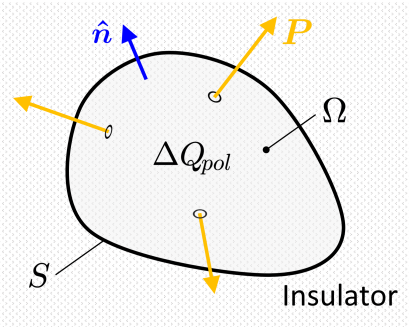



**Fig. 6.4**

$$\Delta Q_{pol} = \oint_S \boldsymbol{P}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{\sigma} = \oint_S \boldsymbol{P}(\boldsymbol{r})\hat{\boldsymbol{n}}\,\mathrm{d}\sigma \tag{6.21}$$

This charge $\Delta Q_{pol}$ leaking out of the volume $\Omega$ is missing from the interior of the volume. If we associate this $\Delta Q_{pol}$ with a polarization charge density $\rho_{pol}(\boldsymbol{r})$, then we can write the missing net charge inside the volume as

$$-\Delta Q_{pol} = \int_\Omega \rho_{pol}(\boldsymbol{r})\,\mathrm{d}^3r \tag{6.22}$$

It then follows from eq. (6.21) and eq. (6.22) that the following must hold:

$$\oint_S \boldsymbol{P}(\boldsymbol{r})\hat{\boldsymbol{n}}\,\mathrm{d}\sigma = -\int_\Omega \rho_{pol}(\boldsymbol{r})\,\mathrm{d}^3r \tag{6.23}$$

A comparison with eq. (3.15) shows that eq. (6.23) is simply a version of Gauss's law of electrostatics, with the difference that the left-hand side of eq. (6.23) features the dielectric polarization $\boldsymbol{P}$ rather than the electric field vector $\boldsymbol{E}$, and that the factor $1/\epsilon_0$ is missing from the right-hand side of eq. (6.23), while a minus sign appears before the integral. If we apply Gauss's theorem eq. (2.42) to the left-hand side of eq. (6.23), then eq. (6.23) becomes

$$\int_\Omega \nabla \boldsymbol{P}(\boldsymbol{r})\,\mathrm{d}^3r \;=\; -\int_\Omega \rho_{pol}(\boldsymbol{r})\,\mathrm{d}^3r \tag{6.24}$$

and therefore it must hold true that

$$\nabla \boldsymbol{P}(\boldsymbol{r}) \;=\; -\rho_{pol}(\boldsymbol{r}) \tag{6.25}$$

The charge distribution $\rho_{pol}(\boldsymbol{r})$ associated with the dielectric polarization $\boldsymbol{P}$ is therefore the negative gradient of $\boldsymbol{P}$. Associated with this charge density $\rho_{pol}(\boldsymbol{r})$ and any potentially present free charge density $\rho_{free}$ is an electrostatic field $\boldsymbol{E}$ which satisfies the following equation:

$$\nabla \boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{1}{\epsilon_0}\big[\rho_{free} + \rho_{pol}\big] \;=\; \frac{1}{\epsilon_0}\big[\rho_{free} - \nabla \boldsymbol{P}(\boldsymbol{r})\big] \tag{6.26}$$

With that, we have derived eq. (6.20) using a different approach. We arrived at eq. (6.20) by considering the surface charge densities generated in the insulator boundary layers by the electrostatic field of the parallel-plate capacitor. However, the derivation of eq. (6.26) did not involve an external field. Instead, we simply considered what happens when the dielectric polarization inside the insulator is direction-dependent. We now rewrite eq. (6.26) as:

$$\nabla\big[\underbrace{\epsilon_0 \boldsymbol{E}(\boldsymbol{r}) + \boldsymbol{P}(\boldsymbol{r})}_{\boldsymbol{D}(\boldsymbol{r})}\big] \;=\; \rho_{free}(\boldsymbol{r}) \tag{6.27}$$

The quantity

$$\boldsymbol{D}(\boldsymbol{r}) \;=\; \epsilon_0 \boldsymbol{E}(\boldsymbol{r}) + \boldsymbol{P}(\boldsymbol{r}) \tag{6.28}$$

whose divergence is determined solely by the free charge carriers is commonly referred to as dielectric displacement or sometimes also as electric flux density. However, the following must still hold true:

$$\nabla \mathbf{E}(\mathbf{r}) \;=\; \frac{1}{\epsilon_0} \cdot \rho_{ges}(\mathbf{r}) \tag{6.29}$$

This allows us to write eq. (6.20) as

$$\rho_{tot}(\boldsymbol{r}) \;=\; \rho_{free}(\boldsymbol{r}) - \nabla \boldsymbol{P}(\boldsymbol{r}) \;=\; \rho_{free}(\boldsymbol{r}) + \rho_{pol}(\boldsymbol{r}) \tag{6.30}$$

Using eq. (6.27) and eq. (6.28), we can write down the first Maxwell equation of electrostatics in the presence of insulators. But what about the second Maxwell equation? $\boldsymbol{E}$ remains the negative gradient of a potential, even though the latter now includes contributions from bound charges. Consequently, the curl of $\boldsymbol{E}$ still vanishes, and the second Maxwell equation of electrostatics holds unchanged. Thus, in the presence of insulators, the Maxwell equations of electrostatics are:

$$\nabla \boldsymbol{D}(\boldsymbol{r}) = \rho_{free}(\boldsymbol{r}) \qquad \text{and} \qquad \nabla \times \boldsymbol{E}(\boldsymbol{r}) = 0 \tag{6.31}$$

If one writes eq. (6.28) as

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{1}{\epsilon_0}\boldsymbol{D}(\boldsymbol{r}) - \frac{1}{\epsilon_0}\boldsymbol{P}(\boldsymbol{r}) \tag{6.32}$$

this illustrates the interpretation of $\boldsymbol{D}/\epsilon_0$ as the field originally produced by free charge carriers, and of $\boldsymbol{P}/\epsilon_0$ as the dielectric polarization induced by this field, which reduces the electrostatic field $\boldsymbol{E}$ in an insulator. The electric field strength $\boldsymbol{E}$ has the unit $\mathrm{N{\cdot}C^{-1}}$ or $\mathrm{V{\cdot}m}$ and $\epsilon_0$ has the unit $\mathrm{C^2{\cdot}m^{-2}{\cdot}N^{-1}}$ or $\mathrm{A{\cdot}s{\cdot}V^{-1}{\cdot}m^{-1}}$. Thus, $\boldsymbol{D}$ and $\boldsymbol{P}$ have the units $\mathrm{C \cdot m^{-2}}$ and $\mathrm{A \cdot s \cdot m^{-2}}$, respectively; they therefore represent surface charge densities, as we were already able to observe for the parallel-plate capacitor example from eq. (6.6) and eq. (6.8) in the case of $\boldsymbol{P}$. As we shall see below, the normal components $D_n = \boldsymbol{D}\cdot\hat{\boldsymbol{n}}$ and $P_n = \boldsymbol{P}\cdot\hat{\boldsymbol{n}}$ of the electric displacement and the electric polarization at surfaces with normal vector $\hat{\boldsymbol{n}}$ correspond to the surface charge density $\sigma$ for free and bound charges, respectively, that is:

$$D_n = \sigma_{free} \qquad \text{and} \qquad P_n = \sigma_{bnd} \tag{6.33}$$

At the beginning of this chapter, we discovered, using a simple model, a linear relationship between the induced dipole moment $\boldsymbol{p}$ of a hydrogen atom and the applied field $\boldsymbol{E}$ that generates this dipole moment. Since $\boldsymbol{P}$ is a dipole density as per eq. (6.9), it is not surprising that such a linear relationship also exists between the electric field $\boldsymbol{E}$ and the dielectric polarization $\boldsymbol{P}$. This generalized form of eq. (6.4) for the linear relationship between $\boldsymbol{E}$ and $\boldsymbol{P}$ is:

$$\boldsymbol{P}(\boldsymbol{r}) = \chi \cdot \epsilon_0 \cdot \boldsymbol{E}(\boldsymbol{r}) \tag{6.34}$$

The proportionality factor $\chi$ in this equation is the so-called dielectric susceptibility. Just as with eq. (6.4), the linear relationship in eq. (6.34) does not hold for arbitrary field strengths. The proportionality factor $\alpha$ in eq. (6.4) is already based on simplifying assumptions but applies only to a comparatively simple hydrogen atom. The dielectric susceptibility $\chi$ in eq. (6.34) is a material property describing the response of the many atoms that make up a solid, a liquid, or a gas.

**Tab. 6.2:** Relative permittivity $\epsilon_r$ of selected substances (RT = room temperature).

| Substance | State and measurement condition | $\epsilon_r$ |
|---|---|---|
| Air | Gas, 0° C, 1 bar | 1.00059 |
| $H_2$ | Gas, 0° C, 1 bar | 1.00026 |
| $H_2O$ | Gas, 110° C, 1 bar / liquid, 20° C | 1.0126 / 80.0 |
| Benzene | Liquid, RT | 2.28 |
| Paraffin | Soft body (wax), RT | 2.1 |
| Paper | Solid, RT | 1.5 - 3 |
| Porcelain | Solid, RT | 4.0 |
| Quartz glas ($SiO_2$) | solid, RT | 3.7 |

Expecting universal validity for a constitutive equation such as eq. (6.34) is therefore even less realistic than for eq. (6.4), which describes the response of a hydrogen atom to an external electric field under simplifying assumptions. Nevertheless, like other such approximations in physics, eq. (6.34) is extremely useful, provided we remain aware of the limits of its validity in every application. A good example of such a highly useful approximation is Hooke's law, which we discussed in the first volume of this physics course. Just as that law ceases to be valid for excessively large displacements, eq. (6.4) and eq. (6.34) lose their validity at excessively high electric field strengths. The utility of eq. (6.34) is demonstrated by the fact that, using this relationship, we can write eq. (6.28) as:

$$\boldsymbol{D}(\boldsymbol{r}) = \epsilon_0(1+\chi)\boldsymbol{E}(\boldsymbol{r}) = \epsilon\boldsymbol{E}(\boldsymbol{r}) \tag{6.35}$$

The dielectric permittivity $\epsilon$, also referred to as dielectric conductivity or the dielectric function, is a material property. Strictly speaking, $\epsilon$ in matter corresponds to $\epsilon_0$ in a vacuum, with the difference that $\epsilon_0$ is a fundamental physical constant, whereas $\epsilon$ is not. Consequently, $\epsilon$ is often expressed as a multiple of $\epsilon_0$, in the form

$$\epsilon = \epsilon_r \cdot \epsilon_0 \qquad \text{with} \qquad \epsilon_r = 1 + \chi \tag{6.36}$$

where $\epsilon_r$ is a material-dependent permittivity value, formerly known as relative permittivity, hence the subscript $r$. Tab. 6.2 lists values of $\epsilon_r$ for several substances. As material-dependent quantities, these values vary with parameters such as pressure or temperature and also depend on the frequency of the electric field. Tab. 6.2 presents $\epsilon_r$ values for electrostatic or low-frequency electric fields, where frequency dependence is either non-existent or negligible. The example of water clearly illustrates that $\epsilon_r$ is a material-dependent property. In its gaseous state, i.e., as water vapor, $H_2O$ exhibits a

significantly lower $\epsilon_r$ value than liquid water. In general, gaseous substances have lower $\epsilon_r$ values than liquid substances. Particularly high $\epsilon_r$ values, up to $10^3 - 10^4$, are exhibited by certain paraelectric or ferroelectric materials, such as $SrTiO_3$ or $BaTiO_3$, which we will discuss in the next chapter. Using eq. (6.31) and eq. (6.35), we can write the first Maxwell equation of electrostatics in dielectrics as

$$\nabla(\epsilon_r \epsilon_0 \boldsymbol{E}(\boldsymbol{r})) = \rho_{free}(\boldsymbol{r}) \tag{6.37}$$

or respectively

$$\nabla\big(\epsilon_r \boldsymbol{E}(\boldsymbol{r})\big) = \frac{1}{\epsilon_0} \cdot \rho_{free}(\boldsymbol{r}) \tag{6.38}$$

Eq. (6.38) looks exactly like the first Maxwell equation of electrostatics in a vacuum, with the difference that the left-hand side features the divergence of $\epsilon_r \boldsymbol{E}$ rather than the divergence of $\boldsymbol{E}$. In the special case where $\epsilon_r = 1$, i.e., in a vacuum, eq. (6.38) becomes the Maxwell equation of electrostatics in a vacuum. For $\epsilon_r \neq 1$, instead of the solutions obtained in a vacuum, we formally obtain the same solutions in dielectrics, but with fields $\boldsymbol{E}$ scaled by a factor $1/\epsilon_r$. For most materials one finds that $\epsilon_r > 1$ holds. This means that the fields in dielectrics are weaker by a factor of $1/\epsilon_r$ than in a vacuum environment. In other words, we can solve electrostatic problems involving dielectric materials by rescaling the corresponding solution for the vacuum case. However, for this simple approach to work, a specific condition must be met: the dielectric materials in question must be isotropic and homogeneous. This means that $\epsilon_r$ must be a scalar, i.e., it must not exhibit any directional dependence.

Let us reconsider the parallel-plate capacitor in fig. 6.3 in light of what we have just learned. Whether or not there is an insulator between the capacitor's electrodes, these electrodes carry the same charge $Q$. However, if the space between the electrodes of the parallel-plate capacitor is filled with a dielectric (Di) rather than air[1], then, for the same free charge $Q$ on the capacitor electrodes, the field $E_{Di}$ between the capacitor plates is weaker by a factor of $1/\epsilon_r$ than $E_{Air}$, the field for the air-filled capacitor. Thus, according to eq. (4.61), the electric voltage $U_{Di} = E_{Di} \cdot d$ is also smaller by this factor. Based on eq. (4.63), it must hold that

$$Q = C_{Air} \cdot U_{Air} = C_{Di} \cdot U_{Di} \tag{6.39}$$

and therefore

$$\frac{C_{Di}}{C_{Air}} = \frac{U_{Air}}{U_{Di}} = \frac{E_{Air}}{E_{Di}} = \epsilon_r \tag{6.40}$$

[1] The value of $\epsilon_r$ for air, at 1.00059, is very close to the vacuum value of 1.

The capacitance $C$ of the capacitor with an insulator between the electrodes is therefore greater by a factor of $\epsilon_r$ than the capacitance of an air-filled capacitor. For a given voltage, more charge can be stored on a capacitor filled with a dielectric having a high $\epsilon_r$ than on one with a dielectric having a lower $\epsilon_r$. We must therefore replace eq. (4.63), the equation for the capacitance of a parallel-plate capacitor, with the more general form for dielectrics

$$C = \frac{Q}{U} = \frac{\epsilon_r \epsilon_0 A}{d} = \frac{\epsilon A}{d} \tag{6.41}$$

where $A$ is again the area of the capacitor plates and $d$ is their separation. Using eq. (4.64) and eq. (6.41), the energy $W$ stored in the capacitor is:

$$W = \frac{1}{2} C \cdot U^2 = \frac{1}{2} Q \cdot U = \frac{1}{2} \cdot \frac{Q^2}{C} \tag{6.42}$$

If one keeps the voltage $U$ constant then a capacitor with a dielectric stores more energy than a capacitor without a dielectric. However, if one keeps the charge $Q$ constant then the energy of a plate capacitor without a dielectric is greater than that of a capacitor with a dielectric between the electrodes. If an insulator is placed between the electrodes of the parallel-plate capacitor, the force between the electrodes is likewise reduced by a factor of $\epsilon_r$, in accordance with eq. (4.67). However, this applies only if the space between the electrodes is completely filled by an insulator. If it is only partially filled with an insulator, the effective capacitance of the capacitor is lower than if it were filled with the insulator. The same applies to the energy stored in the parallel-plate capacitor. In this case, $C$ must be treated as a variable quantity that changes with position within the parallel-plate capacitor.

Tab. 6.3 summarizes the modified electrostatic equations for linear dielectrics which we derived in this chapter. The emphasis here is on the adjective «linear», because in anisotropic or inhomogeneous materials, or at high field strengths, the material properties are generally no longer scalars but tensors. Unlike $\boldsymbol{E}$, $\boldsymbol{P}$ is not a field. According to eq. (6.9), $\boldsymbol{P}$ represents the dipole density, but not the field generated by these dipoles, which is contained in the auxiliary field $\boldsymbol{D}$ rather than in $\boldsymbol{P}$. But what is the physical significance of the field of the dielectric displacement $\boldsymbol{D}$? $\boldsymbol{D}$ is not a directly measurable physical quantity like the quantities $\boldsymbol{E}$ or $\mathbf{P}$, but rather a very useful auxiliary quantity. A comparison of the first Maxwell equation in the presence of dielectric materials using the fields $\boldsymbol{E}$ versus $\boldsymbol{D}$, that is, of

$$\nabla(\epsilon_r \boldsymbol{E}) = \frac{1}{\epsilon_0} \rho_{free}(\boldsymbol{r}) \qquad \text{and} \qquad \nabla \boldsymbol{D} = \rho_{free}(\boldsymbol{r}) \tag{6.43}$$

illustrates why this is the case.

**Tab. 6.3:** Equations of electrostatics in the presence of insulators.

**Free and bound charges**

$$\rho_{tot}(\boldsymbol{r}) = \rho_{free}(\boldsymbol{r}) + \rho_{bnd}(\boldsymbol{r}) \qquad \text{with} \qquad \rho_{bnd}(\boldsymbol{r}) = -\nabla \boldsymbol{P}(\boldsymbol{r})$$

**Maxwell's equations in dielectrics**

$$\boldsymbol{D}(\boldsymbol{r}) = \epsilon_0 \boldsymbol{E}(\boldsymbol{r}) + \boldsymbol{P}(\boldsymbol{r}) \quad ; \quad \nabla \boldsymbol{D}(\boldsymbol{r}) = \rho_{free}(\boldsymbol{r}) \quad ; \quad \nabla \times \boldsymbol{E}(\boldsymbol{r}) = 0$$

**Constitutive equations and material parameters in dielectrics**

$$\boldsymbol{P}(\boldsymbol{r}) = \chi \epsilon_0 \boldsymbol{E}(\boldsymbol{r}) \quad ; \quad \boldsymbol{D}(\boldsymbol{r}) = \epsilon_0(1+\chi)\boldsymbol{E}(\boldsymbol{r}) = \epsilon \boldsymbol{E}(\boldsymbol{r}) \quad ; \quad \epsilon = \epsilon_r \epsilon_0$$

The equation for $\boldsymbol{E}$ contains a material-dependent quantity, namely $\epsilon_r$. This is not the case for the equation for $\boldsymbol{D}$; thus, $\boldsymbol{D}$ is a quantity independent of any specific material properties, whereas $\boldsymbol{E}$ is not. All charges serve as sources or sinks for the electric field $\boldsymbol{E}$, and the electric field acts upon all charges. Only free charges can serve as sources or sinks for the dielectric displacement. Bound charges at dielectric interfaces are not sources or sinks for the electric displacement field $\boldsymbol{D}$, though they certainly are for the electric field $\boldsymbol{E}$ - via $\epsilon_r$. This has consequences for the behavior of $\boldsymbol{D}$ and $\boldsymbol{E}$ at dielectric interfaces.

### 6.1.2 Interfaces of and within Dielectrics

We have already encountered an interface between dielectrics, specifically, the one between the insulator and the air in a parallel-plate capacitor with an insulator between the electrodes. In this case, there was only a normal component of $\boldsymbol{E}$, and we assumed that the field of the parallel-plate capacitor continued in a straight line. However, how do the components of $\boldsymbol{E}$ behave at dielectric interfaces in the general case? To this end, we consider the interface between the two dielectrics ① and ② shown in fig. 6.5. Let the material parameters of the two insulators be $\epsilon_1$ and $\epsilon_2$, and the fields be $\boldsymbol{E}_1$, $\boldsymbol{D}_1$ and $\boldsymbol{E}_2$, $\boldsymbol{D}_2$, respectively. To understand the behavior of the normal components of $\boldsymbol{E}$ and $\boldsymbol{D}$ at the interface, we place a Gaussian pillbox across the interface, just as we did for conductor surfaces in section 5.2. The field quantity associated solely with free charge carriers is the electric displacement $\boldsymbol{D}$. To determine the boundary conditions for $\boldsymbol{D}$ at

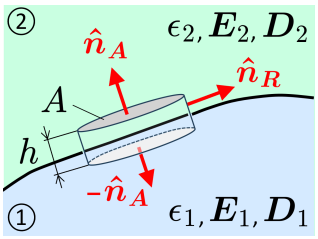



**Fig. 6.5**

the interface between insulator ① and insulator ②, we apply Gauss's law of electrostatics eq. (3.15) to the Gaussian pillbox shown in fig. 6.5. Let this Gaussian pillbox have a volume $V$ enclosed by the surface $\partial V$, the boundary of the box. When applying Gauss's law of electrostatics, we split the surface integral over the boundary $\partial V$ of the Gaussian pillbox into a contribution from the top and bottom of the box and a contribution from the cylindrical side surface. Thus, Gauss's law of electrostatics for the pillbox shown in fig. 6.5 is expressed as follows:

$$\int_{\partial V} \boldsymbol{D}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\sigma \;=\; \int_{A} \boldsymbol{D}\cdot\hat{\boldsymbol{n}}_A\,\mathrm{d}A + \int_{Cyl.} \boldsymbol{D}\cdot\hat{\boldsymbol{n}}_R\cdot R\,\mathrm{d}\varphi\,\mathrm{d}h \;=\; \int_{V} \rho_{free}\,\mathrm{d}V \tag{6.44}$$

In the integral over the cylinder wall, $R$ is the radius of the pillbox, and $h$ is the height of the pillbox, as sketched in fig. 6.5. We first examine the contribution of the cylinder wall in eq. (6.44). To evaluate this integral over the surface of the cylinder wall, we consider the contribution of an annular ring to the integral. All the annular rings that together form the cylinder wall lie in planes parallel to the interface between the dielectrics. Therefore, only components of $\boldsymbol{D}$ that are parallel to the interface between the two insulators can contribute. Let us assume that these parallel components of $\boldsymbol{D}$ form a uniform field across the surface of an annular ring of the can, as illustrated in fig. 6.6. We can always orient our coordinate system such that one of the coordinate axes is parallel to the direction of the homogeneous field $\boldsymbol{D}_\parallel$. If we do this, as shown in fig. 6.6, then the angle $\varphi$ for the line integral along $C$ is, at any point on the circular ring, the angle between the field $\boldsymbol{D}_\parallel$ and the normal vector $\hat{\boldsymbol{n}}_R$ of the circular ring. This yields the following expression for the line integral of $\boldsymbol{D}\cdot\hat{\boldsymbol{n}}_R$ along $C$:

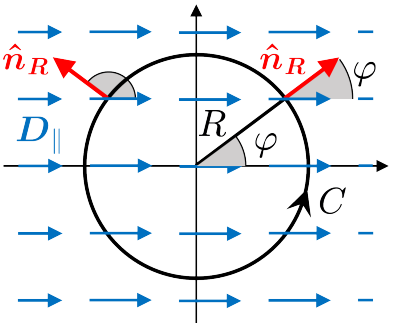



**Fig. 6.6**

$$\oint_C \boldsymbol{D}\cdot\hat{\boldsymbol{n}}_R R\,\mathrm{d}\varphi \;=\; \int_0^{2\pi} D_\parallel\cdot\sin\varphi\cdot R\,\mathrm{d}\varphi \;=\; 0 \tag{6.45}$$

If we make the Gaussian pillbox sufficiently small, then $\boldsymbol{D}_\parallel$ will always be uniform over any annular ring of the pillbox, and the contribution from the cylindrical side surface of the pillbox in eq. (6.44) vanishes. This holds true regardless of whether or not $\boldsymbol{D}_\parallel$ is continuous at the interface. The normal vectors of the end faces $A$ point in exactly opposite directions in insulator ① and insulator ②. Thus, evaluating the terms in eq. (6.44) that originate from the top and bottom of the cylinder yields:

$$\int_A \boldsymbol{D}\cdot\hat{\boldsymbol{n}}_A\,\mathrm{d}A \;=\; \hat{\boldsymbol{n}}_A(\boldsymbol{D}_2 - \boldsymbol{D}_1)\cdot A \;=\; \sigma_{free}\cdot A \tag{6.46}$$

Here, we have evaluated the integral in eq. (6.46) for the case $h \to 0$, that is, when the Gaussian pillbox has shrunk to the circular area lying on the interface between the dielectrics. Thus, $\sigma_{free}$ is the surface charge density of free charges at the dielectric interface. We therefore obtain:

$$\hat{\boldsymbol{n}}_A(\boldsymbol{D}_2 - \boldsymbol{D}_1) = \sigma_{free} \tag{6.47}$$

In insulators, there are no free charge carriers, so $\sigma_{\text{free}} = 0$; consequently, the normal component of $\boldsymbol{D}$ is continuous at the interface between dielectrics, whereas the normal component of $\boldsymbol{E}$, given $\boldsymbol{D} = \epsilon\boldsymbol{E}$ and $\epsilon_1 \neq \epsilon_2$, is discontinuous:

$$\boldsymbol{D}_{2_\perp} = \boldsymbol{D}_{1_\perp} \qquad \text{and} \qquad \boldsymbol{E}_{2_\perp} = \frac{\epsilon_1}{\epsilon_2}\boldsymbol{E}_{1_\perp} \tag{6.48}$$

The reason for this differing behavior of the normal components of $\boldsymbol{E}$ and $\boldsymbol{D}$ is simply the fact that the bound charges at the interface between the two dielectrics do not act as sinks or sources for $\boldsymbol{D}$, whereas they do for $\boldsymbol{E}$. Thus, neither do new field lines of $\boldsymbol{D}$ originate at the interface, nor do incident field lines of $\boldsymbol{D}$ terminate there. Therefore, the normal component of the displacement field $\boldsymbol{D}$ must be continuous at the interface. If field lines of $\boldsymbol{D}$ strike a dielectric perpendicularly, they enter it undisturbed. However, this is not the case for electric field lines. For $\boldsymbol{E}$, the bound charges in the boundary layer do indeed act as sinks for incident field lines or sources of new field lines. Consequently, the normal component of $\boldsymbol{E}$ undergoes a jump across the boundary layer.

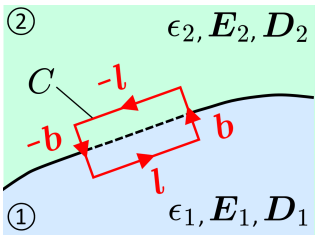



**Fig. 6.7**

Next, we turn to the tangential components of $\boldsymbol{E}$ and $\boldsymbol{D}$. In the case of conductors, we have learned that their free charge carriers shield electric fields, meaning that $\boldsymbol{E} = 0$ always holds within conductors, which is why the tangential component of $\boldsymbol{E}$ always vanishes at conductor surfaces. The situation is different for insulators, as there are no freely moving charge carriers. To understand how they behave at the interface, we use the integral form of the second Maxwell equation of electrostatics, i.e., $\nabla \times \boldsymbol{E} = 0$, which remains valid unchanged in dielectrics:

$$\oint_C \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = 0 \tag{6.49}$$

In doing so, we place the closed path $C$, as shown in fig. 6.7, around the boundary line separating insulator ① from insulator ②. The path segments $\boldsymbol{l}$ and $-\boldsymbol{l}$ lie entirely within insulator ① and insulator ②, respectively, while each of the path segments $\boldsymbol{b}$ and $-\boldsymbol{b}$ lies half within insulator ① and half within insulator ②. Using these path segments, the path integral in eq. (6.49) becomes:

$$\oint_C \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = \boldsymbol{l}\boldsymbol{E}_{1_\parallel} + \frac{\boldsymbol{b}}{2}\boldsymbol{E}_{1_\perp} + \frac{\boldsymbol{b}}{2}\boldsymbol{E}_{2_\perp} - \boldsymbol{l}\boldsymbol{E}_{2_\parallel} - \frac{\boldsymbol{b}}{2}\boldsymbol{E}_{2_\perp} - \frac{\boldsymbol{b}}{2}\boldsymbol{E}_{1_\perp} = 0 \tag{6.50}$$

The contributions from the segments $\boldsymbol{b}$ and $-\boldsymbol{b}$ cancel out, and what remains is:

$$\boldsymbol{l}(\boldsymbol{E}_{1_\parallel} - \boldsymbol{E}_{2_\parallel}) = 0 \tag{6.51}$$

If we now shrink $C$ by letting the short sides $\pm\boldsymbol{b}$ of the rectangular curve approach zero, then eq. (6.51) can only be satisfied if $\boldsymbol{E}_\parallel$ is continuous along the boundary line between insulator ① and insulator ②. Given $\boldsymbol{D} = \epsilon\boldsymbol{E}$, this implies that $\boldsymbol{D}_\parallel$ undergoes a jump, and is thus discontinuous, at the transition from insulator ① to insulator ②. Consequently, the behavior of the parallel components of $\boldsymbol{E}$ and $\boldsymbol{D}$ at the interface between two insulators is as follows:

$$\boldsymbol{E}_{2_\parallel} = \boldsymbol{E}_{1_\parallel} \qquad \text{and} \qquad \boldsymbol{D}_{2_\parallel} = \frac{\epsilon_2}{\epsilon_1}\boldsymbol{D}_{1_\parallel} \tag{6.52}$$

We now know that the normal component of $\boldsymbol{D}$ and the parallel component of $\boldsymbol{E}$ are continuous at the interface between two insulators, whereas the parallel component of $\boldsymbol{D}$ and the normal component of $\boldsymbol{E}$ are discontinuous and undergo a jump at the interface. Let us consider the situation for the case $\epsilon_2 > \epsilon_1$. Fig. 6.8 shows this transition for $\boldsymbol{D}$ and $\boldsymbol{E}$ from a dielectric ① with a smaller $\epsilon$ to a dielectric ② with a larger $\epsilon$. In both insulators, $\boldsymbol{D}$ and $\boldsymbol{E}$ are parallel to each other. The parallel components of $\boldsymbol{E}$ at the interface are identical, but since $\epsilon_2 > \epsilon_1$, it follows from eq. (6.48) that the normal component of $\boldsymbol{E}$ at the interface in dielectric ② is smaller than that in dielectric ①. For $\boldsymbol{D}$, the situation is exactly the reverse: the normal components of $\boldsymbol{D}$ at the interface are identical, but according to eq. (6.52), the parallel component of $\boldsymbol{D}$ at the interface is greater in dielectric ② than in dielectric ①. Thus, using eq. (6.48) and eq. (6.52), the following applies to the angles $\alpha_1$ and $\alpha_2$ in fig. 6.8:

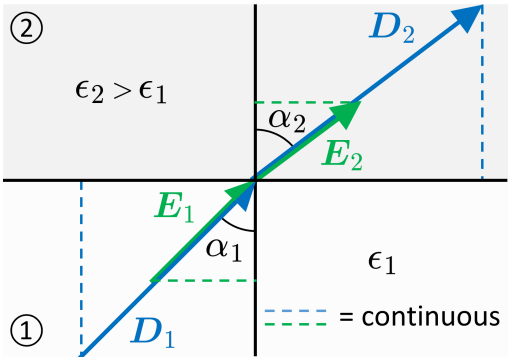



**Fig. 6.8**

$$|\boldsymbol{E}_1|\sin\alpha_1 = |\boldsymbol{E}_2|\sin\alpha_2 \qquad \text{and} \qquad |\boldsymbol{D}_1|\cos\alpha_1 = |\boldsymbol{D}_2|\cos\alpha_2 \tag{6.53}$$

With $\boldsymbol{D}_1 = \epsilon_1\boldsymbol{E}_1$ and $\boldsymbol{D}_2 = \epsilon_1\boldsymbol{E}_2$, the refraction law for the field lines of $\boldsymbol{E}$ and $\boldsymbol{D}$ follows from these equations:

$$\frac{1}{\epsilon_1}\tan\alpha_1 = \frac{1}{\epsilon_2}\tan\alpha_2 \tag{6.54}$$

Unlike with the optical law of refraction, the expression here involves the tangent of the angle rather than the sine, and the reciprocal of the material constant rather than the constant itself. Insulators with a higher or lower $\epsilon$ thus represent electrically denser or thinner media. In contrast to the optical law of refraction, where a light ray is refracted toward the surface normal upon entering an optically denser medium, an electric field line is refracted away from the surface normal upon entering an electrically denser medium. Finally, we make notice of the fact that, according to eq. (6.48), the normal component of $\boldsymbol{E}_2$ in fig. 6.8 decreases as $\epsilon_2$ increases and vanishes completely as $\epsilon_2 \to \infty$. Since $\epsilon_1$ cannot be zero, it follows that $\tan\alpha_1 = 0$ in eq. (6.54), and consequently $\alpha_1 = 0$. Therefore, $\boldsymbol{E}_1$ can only have a normal component. This means that an insulator begins to behave like a conductor as $\epsilon \to \infty$. The fact that the parallel component of $\boldsymbol{D}$ approaches infinity as $\epsilon \to \infty$ is of no consequence, since $\boldsymbol{D}$ is an auxiliary construct rather than a measurable physical field.

## 6.2 Insulators in an Electrostatic Field

### 6.2.1 The Field of a Point Charge in Dielectrics

We consider a point charge $q$ located, as sketched in fig. 6.9, on the positive $z$-axis at a distance $d$ from the $y$-axis within an insulator with permittivity $\epsilon_1$ that fills the positive half-space $z > 0$. The half-space $z < 0$ is filled by an insulator with permittivity $\epsilon_2$. We seek solutions to Maxwell's equations in the half-spaces $z > 0$ and $z < 0$ that satisfy the boundary conditions eq. (6.48) and eq. (6.52). Here, we distinguish these solutions not by indexing the field vectors, but by clearly specifying the half-space in which they apply. With that, Maxwell's equations for the two half-spaces, with the free charge $q$ located in the half-space $z > 0$, are:

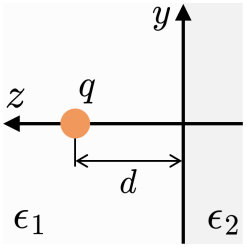


**Fig. 6.9**

$$\begin{aligned} \nabla\boldsymbol{D} &= \epsilon_1\nabla\boldsymbol{E} = q\delta(z-d) \qquad \text{for} \qquad z > 0 \\ \nabla\boldsymbol{D} &= \epsilon_2\nabla\boldsymbol{E} = 0 \qquad \text{for} \qquad z < 0 \\ \nabla\times\boldsymbol{E} &= 0 \qquad \text{in both half-spaces} \end{aligned} \tag{6.55}$$

The boundary conditions for the normal component of $\boldsymbol{D}$ and the parallel component of $\boldsymbol{E}$ at the interface in the $xy$-plane are

$$\lim_{z\to+0}\epsilon_1 E_z = \lim_{z\to-0}\epsilon_2 E_z \qquad \text{and} \qquad \lim_{z\to+0} E_{x,y} = \lim_{z\to-0} E_{x,y} \tag{6.56}$$

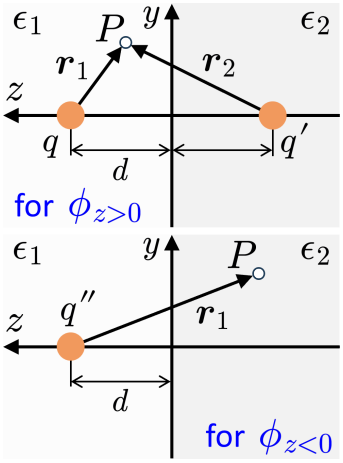



**Fig. 6.10**

Next, we must consider the approach to take regarding the electrostatic potentials in the respective half-spaces that generate the fields we are seeking via $\boldsymbol{E} = -\nabla\phi$. The potential $\phi_{z>0}$, which generates the field in the half-space $z > 0$, must cause the field lines approaching the interface to curve either toward or away from the surface normal of the interface. We already know how to reproduce the field behavior of a point charge in front of an interface, in the case of conductors, by making use of the image charge method. To determine the potential $\phi_{z>0}$ at a point $P$ in the half-space $z > 0$, as sketched in the upper part of fig. 6.10, we therefore choose an approach consisting of a charge $q$ at $z = d$ and an image charge $q'$ at $z = -d$. This image charge $q'$ can be either negative or positive, depending on the direction in which the field lines in the half-space $z > 0$ curve relative to the surface normal. The potential $\phi_{z<0}$, which generates the field in the half-space $z < 0$, can originate only from charges located in the half-space $z > 0$. Furthermore, since there are no free charges in the half-space $z < 0$, the field lines of $\boldsymbol{E}$ must continue in a straight line after entering the half-space $z < 0$, as if they were originating from a point charge in the half-space $z > 0$. This point charge can only be the charge $q$ at $z = d$. However, because $\epsilon_1 \neq \epsilon_2$, the electric field in the half-space $z < 0$ is either stronger or weaker than in the half-space $z > 0$; thus, the field lines of $\boldsymbol{E}$ are either more densely or less densely spaced in the half-space $z < 0$ than in the half-space $z > 0$. The effective charge producing the potential $\phi_{z<0}$ at a point $P$ in the half-space $z < 0$ must therefore be a scaled version of $q$. This scaled charge, denoted as $q''$, as illustrated in the lower part of fig. 6.10, can be greater or smaller than $q$, depending on the relationship between $\epsilon_1$ and $\epsilon_2$. With $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ being the relevant distance vectors, as shown in fig. 6.10, the electrostatic potentials generating the field distribution in the respective half-spaces are consequently given by:

$$\phi_{z>0} = \frac{1}{4\pi\epsilon_1} \cdot \left( \frac{q}{r_1} + \frac{q'}{r_2} \right) \qquad \text{and} \qquad \phi_{z<0} = \frac{1}{4\pi\epsilon_2} \cdot \frac{q''}{r_1} \tag{6.57}$$

For the magnitudes of the distance vectors $\boldsymbol{r}_1 = (x, y, z - d)$ and $\boldsymbol{r}_2 = (x, y, z + d)$ in eq. (6.57), we obtain in cylindrical coordinates ($R^2 = x^2 + y^2$):

$$r_1 = \sqrt{R^2 + (z-d)^2} \qquad \text{and} \qquad r_2 = \sqrt{R^2 + (z+d)^2} \tag{6.58}$$

In cylindrical coordinates, the potentials must therefore satisfy the following boundary conditions in accordance with eq. (6.56):

$$\epsilon_1 \frac{\partial \phi_{z>0}}{\partial z}\bigg|_{z=0} = \epsilon_2 \frac{\partial \phi_{z<0}}{\partial z}\bigg|_{z=0} \qquad \text{and} \qquad \frac{\partial \phi_{z>0}}{\partial R}\bigg|_{z=0} = \frac{\partial \phi_{z<0}}{\partial R}\bigg|_{z=0} \tag{6.59}$$

The second boundary condition is equivalent to $\phi_{z>0}(z=0)=\phi_{z<0}(z=0)$. Hence, we only need to calculate the partial derivatives with respect to the $z$-coordinate, that is:

$$\frac{\partial}{\partial z}\left(\frac{1}{r_1}\right)_{z=0} = -\frac{\partial}{\partial z}\left(\frac{1}{r_2}\right)_{z=0} = \frac{d}{\left(R^2+d^2\right)^{3/2}}$$

Thus, from the boundary condition for the normal component of $\boldsymbol{D}$, we obtain the first governing equation for $q'$ and $q''$:

$$q - q' = q'' \tag{6.60}$$

From $\phi_{z>0}(z=0)=\phi_{z<0}(z=0)$, the boundary condition for the parallel component of $\boldsymbol{E}$, we obtain the following as the second governing equation:

$$\frac{1}{\epsilon_1}(q+q') = \frac{1}{\epsilon_2}q'' \tag{6.61}$$

From eq. (6.60) and eq. (6.61), it follows for the unknown charge $q'$ that

$$q' = -\frac{\epsilon_2-\epsilon_1}{\epsilon_2+\epsilon_1}q \tag{6.62}$$

and for the unknown charge $q''$ that

$$q'' = \frac{2\epsilon_2}{\epsilon_2+\epsilon_1}q \tag{6.63}$$

There are three possible scenarios that we need to examine. In the case $\epsilon_2 < \epsilon_1$, the image charge $q'$ has the same sign as $q$; in the case $\epsilon_2 > \epsilon_1$, the image charge $q'$ has the opposite sign to $q$; and in the case $\epsilon_2 = \epsilon_1$, no image charge exists at all, as is to be expected when both half-spaces are filled with the same insulator. The charge $q''$ always has the same sign as $q$, since it is, after all, a scaled version of $q$. If $\epsilon_2 > \epsilon_1$, then $q'' > q$; if $\epsilon_2 < \epsilon_1$, then $q'' < q$; and if $\epsilon_2 = \epsilon_1$, then, as expected, $q'' = q$. As for the electric displacement field $\boldsymbol{D}$, we have to ask ourselves: Should $q'$ and $q''$ be classified as bound charges or free charges? We introduced the image charge $q'$ to account for the polarization charges in the boundary layer. Therefore, we must regard $q'$ as a bound charge. However, the situation is different for $q''$. This scaled charge produces the same field distribution in the dielectric $\epsilon_2$ as would exist if both dielectrics had the same permittivity $\epsilon_2$. Therefore, $q''$ must be treated as a free charge. With these three charges, which generate the two potentials $\phi_{z>0}$ and $\phi_{z<0}$, we obtain the pattern of the $\boldsymbol{D}$ field lines shown in fig. 6.11a for a positive point charge $q$ in the case where $\epsilon_1 > \epsilon_2$, and the pattern shown in fig. 6.11b for the case where $\epsilon_2 > \epsilon_1$. However, why does fig. 6.11 show the field pattern of $\boldsymbol{D}$ and not that of $\boldsymbol{E}$?

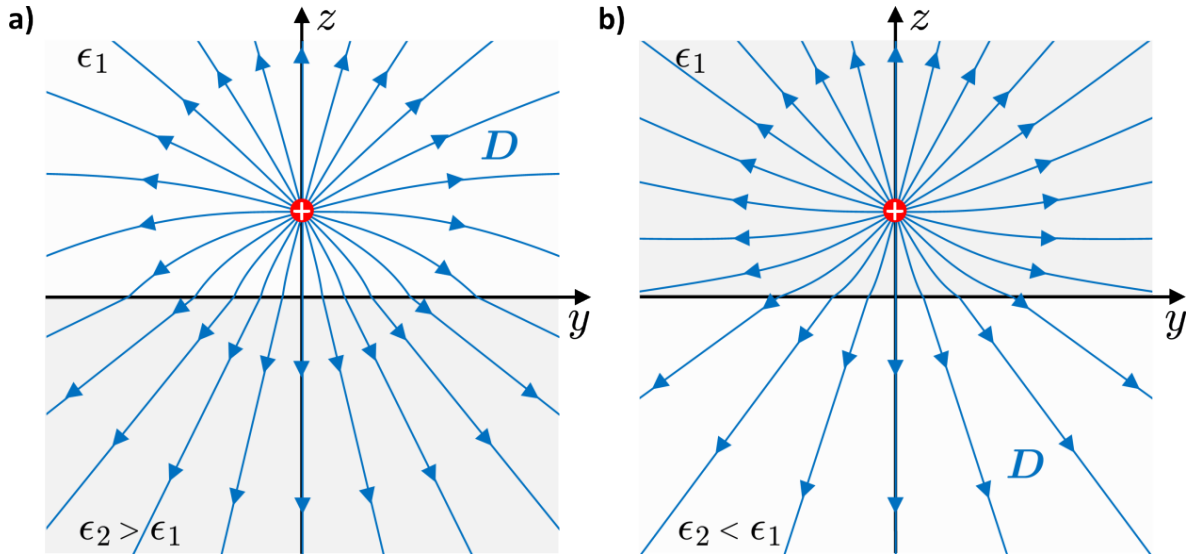



**Fig. 6.11:** Field lines of $\boldsymbol{D}$ for a positive point charge in a dielectric with $\epsilon = \epsilon_1$ adjacent to a dielectric with $\epsilon = \epsilon_2$: (a) $\epsilon_2 = 2 \cdot \epsilon_1$ (b) and $\epsilon_2 = \epsilon_1/2$.

The reason is simple: with $\boldsymbol{D}$, we do not need to worry about field lines terminating or originating at the interface, as would be the case with $\boldsymbol{E}$. Given $\boldsymbol{D} = \epsilon \boldsymbol{E}$, the field configuration of $\boldsymbol{E}$ is identical to that of $\boldsymbol{D}$, but the field strength differs between the two dielectrics. If $\epsilon_2 > \epsilon_1$, the electric field in the half-space $z < 0$ is weaker than in the half-space $z > 0$, and vice versa if $\epsilon_1 > \epsilon_2$. Upon transition into a more electrically dense medium, as shown in fig. 6.11a, the $\boldsymbol{D}$ field lines are refracted away from the normal; conversely, upon transition into a less electrically dense medium, as shown in fig. 6.11b, they are refracted toward the normal. Eq. (6.62) indicates that $q'$ changes sign when transitioning from the case $\epsilon_2 > \epsilon_1$ to $\epsilon_2 < \epsilon_1$. With $q$ positive, $q'$ is negative when $\epsilon_2 > \epsilon_1$ but becomes positive when $\epsilon_2 < \epsilon_1$. Consequently, the sign of the polarization charges at the interface must also change.

In conductors, freely moving charges, known as induction charges, shield external fields and thus define the boundary conditions for $\boldsymbol{E}$ at the conductor surface. At the interface of two insulators, there are no induction charges, but there are polarization charges. The polarization charge density $\rho_{pol}$ is related to the dielectric polarization $\boldsymbol{P}$ according to eq. (6.23), eq. (6.24), and eq. (6.25), respectively. We evaluate the surface integral over $\boldsymbol{P}$ on the left-hand side of eq. (6.23) using the Gaussian pillbox in fig. 6.12. The polarization charge density enclosed in the volume of the pillbox is the surface charge density $\sigma_{pol}$ of the boundary layer multiplied by the area $A$ of the pillbox's base or top, and this does not change as $h \to 0$. Therefore, eq. (6.23) becomes:

$$\hat{\boldsymbol{n}}_A \cdot (\boldsymbol{P}_1 - \boldsymbol{P}_2) \cdot A \;=\; -\sigma_{pol} \cdot A \tag{6.64}$$

Here, $\boldsymbol{P}_1$ denotes the dielectric polarization at the top surface of the Gaussian pillbox inside the insulator with $\epsilon_1$, and $\boldsymbol{P}_2$ denotes the dielectric polarization at the bottom surface of the Gaussian pillbox inside the insulator with $\epsilon_2$. As $h \to 0$, these surfaces converge to the interface separating the two insulators. The area $A$ in eq. (6.64) cancels out, and the surface charge density $\sigma_{pol}$ of the polarization charges at the interface is thus given by:

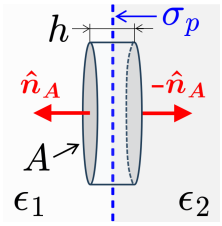



**Fig. 6.12**

$$\sigma_{pol} \;=\; -\hat{\boldsymbol{n}}_A \cdot (\boldsymbol{P}_1 - \boldsymbol{P}_2) \tag{6.65}$$

With eq. (6.28) and eq. (6.35), as well as $\boldsymbol{E} = -\nabla\phi$, we can express $\boldsymbol{P}$ in terms of the potential $\phi$:

$$\boldsymbol{P} \;=\; (\epsilon - \epsilon_0)\boldsymbol{E} \;=\; -(\epsilon - \epsilon_0)\nabla\phi \tag{6.66}$$

However, for the surface charge density $\sigma_{pol}$ of the polarization charges in eq. (6.65), we do not need the quantity $\boldsymbol{P}$ but rather $\hat{\boldsymbol{n}}_A \cdot \boldsymbol{P}$. Therefore, we do not need $\nabla\phi$ but rather the directional derivative $\hat{\boldsymbol{n}}_A \cdot \nabla\phi$. In our example, $\hat{\boldsymbol{n}}_A$ is parallel to the $z$-axis, and consequently $\sigma_{pol}$ satisfies the equation:

$$\sigma_{pol} \;=\; (\epsilon_1 - \epsilon_0) \cdot \left.\frac{\partial \phi_{z>0}}{\partial z}\right|_{z=0} - \;(\epsilon_2 - \epsilon_0) \cdot \left.\frac{\partial \phi_{z<0}}{\partial z}\right|_{z=0} \tag{6.67}$$

We have already determined the derivatives of the potentials $\phi_{z>0}$ and $\phi_{z<0}$ from eq. (6.57), as well as their values at $z = 0$, during the evaluation of the boundary conditions. Substituting these into eq. (6.67) yields:

$$\sigma_{pol} \;=\; \frac{1}{4\pi} \cdot \frac{d}{\left(R^2 + d^2\right)^{3/2}} \cdot \left( \frac{\epsilon_1 - \epsilon_0}{\epsilon_1} \cdot (q - q') - \frac{\epsilon_2 - \epsilon_0}{\epsilon_2} \cdot q'' \right) \tag{6.68}$$

We now express the charges $q'$ and $q''$ in terms of the charge $q$ using eq. (6.62) and eq. (6.63), and, after combining terms, obtain

$$\sigma_{pol} \;=\; \frac{1}{2\pi} \cdot \frac{d \cdot q}{\left(R^2 + d^2\right)^{3/2}} \cdot \frac{\epsilon_2}{\epsilon_2 + \epsilon_1} \cdot \left( \frac{\epsilon_1 - \epsilon_0}{\epsilon_1} - \frac{\epsilon_2 - \epsilon_0}{\epsilon_2} \right) \tag{6.69}$$

and finally, as the end result

$$\sigma_{pol} \;=\; -\frac{1}{2\pi} \cdot \frac{d \cdot q}{\left(R^2 + d^2\right)^{3/2}} \cdot \frac{\epsilon_0(\epsilon_2 - \epsilon_1)}{\epsilon_1(\epsilon_2 + \epsilon_1)} \tag{6.70}$$

We have already seen above, in the discussion of eq. (6.59), the law of refraction for electric field lines, that an insulator begins to behave like a conductor as $\epsilon \to \infty$. Let us consider what becomes of eq. (6.70) in the two limiting cases $\epsilon_2 \gg \epsilon_1$ and $\epsilon_2 \ll \epsilon_1$. As can easily be verified, eq. (6.70) becomes the following in these two limiting cases:

$$\lim_{\substack{\epsilon_2 \gg \epsilon_1 \\ \epsilon_2 \ll \epsilon_1}} \sigma_{pol} = \mp \frac{1}{2\pi} \cdot \frac{d \cdot q}{\left(R^2 + d^2\right)^{3/2}} \cdot \frac{\epsilon_0}{\epsilon_1} = \mp \sigma^*_{pol} \tag{6.71}$$

The dielectric permittivity of a material cannot become arbitrarily small, since the condition $\epsilon \geq \epsilon_0$ must always be satisfied. Nor can it become arbitrarily large. At very high values of $\epsilon$, we are dealing neither with insulators nor yet with conductors, but rather with semiconductors such as $TiO_2$. Fig. 6.13 shows the behavior of $\sigma_{pol}$ across six orders of magnitude of the ratio $\epsilon_1/\epsilon_2$. At the far left of this graph, $\epsilon_2$ is so large that the material effectively acts as a conductor. Indeed, if we let $\epsilon_2$ approach infinity while setting $\epsilon_1 = \epsilon_0$, meaning the point charge $q$ is located in a vacuum environment, then eq. (6.70) yields exactly the result we obtained in section 5.4.1. There, for the induction charge density on a flat metal surface facing a point charge in a vacuum environment, we obtained the result given in eq. (5.77). At the other end of the spectrum, the condition $\epsilon_1 \gg \epsilon_2$ is met only if the point charge is located in a material with a very high $\epsilon$, though not high enough to be considered a conductor, and $\epsilon_2$ takes on the lowest possible value, namely $\epsilon_2 = \epsilon_0$.

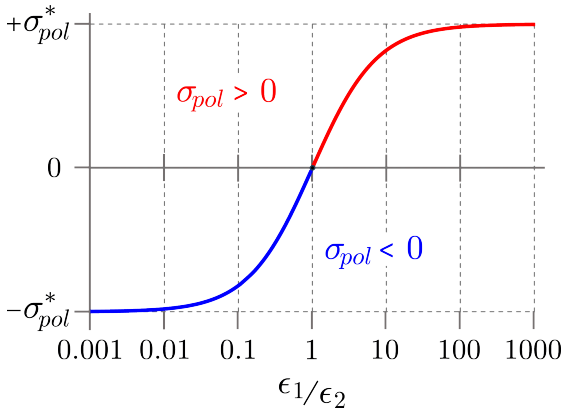



**Fig. 6.13**

Next, we could attempt to determine the electrostatic potential throughout space for a point charge in front of a dielectric sphere. The situation is similar to that of the conducting sphere in an electrostatic field produced by a point charge, which we considered in the last chapter. Again, we could use a modified version of the image charge method, much like we just did for a point charge in front of the planar interface between two dielectrics. However, that does not work in this case. The problem is certainly solvable, but it requires methods that would go beyond the scope of this physics course. In our analysis of conductors in an electric field, we used the point charge in front of a conducting sphere to investigate the field of a conducting sphere in a uniform field (see fig. 5.11). In the case of the dielectric sphere, however, we do not need to take this roundabout approach at all. The method used to determine the potential for a point charge in front of a dielectric sphere is the separation of variables, which, in spherical coordinates, relies on a series expansion of the potential in terms of the so-called Legendre polynomials $P_l(\cos\vartheta)$. In the case of the embedded insulating sphere, this expansion fortunately reduces to the lowest two terms, $P_0 = 1$ and $P_1 = \cos\vartheta$.

### 6.2.2 Insulating Sphere in a Homogeneous Electrostatic Field

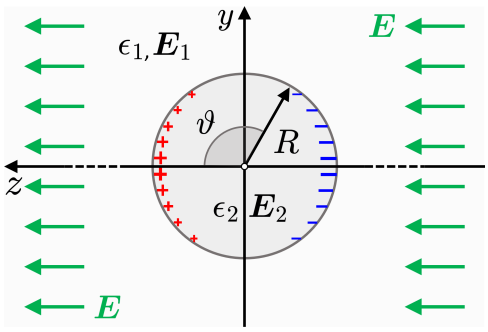



**Fig. 6.14**

For a point charge placed near the interface between two dielectrics, we could proceed in much the same way as we did for a point charge near a flat metal surface in section 5.3.1 or section 5.4.1. However, in the present case, that approach is not as straightforward. We therefore employ a different method to determine the fields around and inside an insulating sphere placed in an external field. To this end, we consider the scenario shown in fig. 6.14, which depicts a relatively small insulating sphere with permittivity $\epsilon = \epsilon_2$ embedded within a much larger insulating medium with permittivity $\epsilon = \epsilon_1$. As illustrated in fig. 6.14, an electrostatic field $\boldsymbol{E}$ exists in the surrounding insulating medium; this field is assumed to be uniform at a great distance from the sphere. The situation shown in fig. 6.14 is analogous to that of a small sphere embedded within the dielectric of a very large parallel-plate capacitor, positioned centrally in the capacitor, far removed from it's electrodes. As shown in fig. 6.14, the electric field of this parallel-plate capacitor gives rise to polarization charges at the interface between the two dielectrics. We can express the electric field $\boldsymbol{E}_1$ in the surrounding insulating medium, far from the insulating sphere, as follows:

$$\boldsymbol{E}_1(\boldsymbol{r}) \;=\; \boldsymbol{E}(\boldsymbol{r}) \;=\; E_1 \cdot \hat{\boldsymbol{k}} \qquad \text{for} \qquad |\boldsymbol{r}| \gg R \tag{6.72}$$

Here, $\hat{\boldsymbol{k}}$ is the unit vector in the $z$-direction. As can be quickly verified by simple differentiation, the corresponding potential generated by this field far from the sphere can only be:

$$\phi_1(\boldsymbol{r}) \;=\; -\boldsymbol{E}_1(\boldsymbol{r}) \cdot \boldsymbol{r} \;=\; -E_1 \cdot r \cdot \cos\vartheta \;=\; -E_1 \cdot z \tag{6.73}$$

This potential is of the general form

$$\phi(\boldsymbol{r}) \;=\; f(r) \cdot \cos\vartheta \tag{6.74}$$

For such potentials, we can easily determine solutions to $\Delta\phi = 0$. In spherical coordinates, $\Delta\phi = 0$ takes the following form in this case:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\Big(r^2 \frac{\partial f(r)}{\partial r}\Big)\cos\vartheta + \frac{1}{r^2 \sin\vartheta}\frac{\partial}{\partial\vartheta}\Big(\sin\vartheta \frac{\partial}{\partial\vartheta}\cos\vartheta\Big) f(r) \;=\; 0 \tag{6.75}$$

Performing the differentiation gives us the differential equation

$$r^2\frac{\partial^2 f(r)}{\partial r^2} + 2r\frac{\partial f(r)}{\partial r} - 2f(r) \;=\; 0 \tag{6.76}$$

Substituting into this equation for $f(r)$ the linear combination

$$f(r) \;=\; a{\cdot}r + b{\cdot}r^{-2} \tag{6.77}$$

produces the result:

$$a{\cdot}\big(\underbrace{-2r+2r}_{=0}\big) + b{\cdot}\big(\underbrace{2{\cdot}r^{-2} - 4{\cdot}r^{-2} + 6{\cdot}r^{-2}}_{=0}\big) \;=\; 0 \tag{6.78}$$

This means that any linear combination with arbitrary $a$ and $b$ in eq. (6.77) is a solution to eq. (6.76), and thus any linear combination of a homogeneous field $f(r) \propto r$ and a dipole field $f(r) \propto r^{-2}$ is also a solution to $\Delta\phi = 0$. How does this help us in determining the electrostatic potentials for the case of an insulating sphere embedded in a dielectric medium? The potential far from the sphere, given by eq. (6.73), is already a solution to $\Delta\phi = 0$. Furthermore, we know that the polarization charges induced by $\boldsymbol{E}_1$ at the dielectric interface generate surface charge densities $\sigma_{pol}$, roughly as sketched in fig. 6.14. There are no charges within the sphere itself. Therefore, the field $\boldsymbol{E}_2$ inside the sphere will run parallel to the $z$-axis, with field lines that are either more or less dense than those of $\boldsymbol{E}_1$, depending on the relationship between $\epsilon_1$ and $\epsilon_2$. The question we need to ask is: what does the external field looks like in the vicinity of the insulating sphere? The polarization charges on the surface of the sphere, shown schematically in fig. 6.14, arise from the displacement of the centers of negative and positive charge within the insulating sphere by a small distance $\boldsymbol{\delta}$. Because the distribution of polarization charges on the surface of the sphere is radially symmetric, we can place the total charge $Q$ of the corresponding dipole $\boldsymbol{p}$ at the center of the insulating sphere, just as we did initially with the hydrogen atom model. The difference is that here we are dealing not with a single atom, but with a great many of them contributing to the total charge $Q = n \cdot q$ of the dipole moment:

$$\boldsymbol{p} \;=\; Q{\cdot}\delta \;=\; \frac{4\pi R^3}{3}{\cdot}n{\cdot}q{\cdot}\boldsymbol{\delta} \tag{6.79}$$

This dipole at the origin does not exist as such in reality. There exists only a dipole density within the sphere which, through the influence of $\boldsymbol{E}_1$, gives rise to polarization charges on the sphere's surface, charges that we express via eq. (6.9) in terms of the macroscopic dielectric polarization:

$$\boldsymbol{p} \;=\; \frac{4\pi R^3}{3}{\cdot}\boldsymbol{P} \tag{6.80}$$

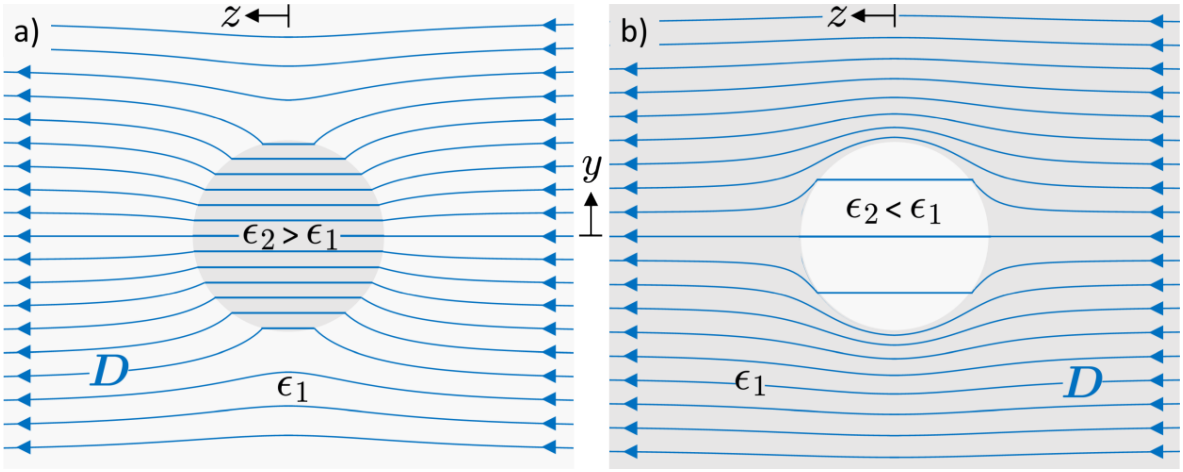



**Fig. 6.15:** Field lines of $\boldsymbol{D}$ for an insulating sphere with dielectric permittivity $\epsilon_2$ embedded in another dielectric with dielectric permittivity $\epsilon_1$, in which a uniform electric field parallel to the $z$-axis exists: (a) $\epsilon_2 > \epsilon_1$ and (b) $\epsilon_2 < \epsilon_1$.

Even though the dipole $\boldsymbol{p} = Q \cdot \boldsymbol{\delta}$ does not actually exist at the center of the insulating sphere, when viewed from a distance, i.e., for $r \geq R \gg \delta$, the electrostatic field appears as a superposition of such a dipole field, according to eq. (3.53), and the uniform field $\boldsymbol{E}_1$. We therefore adopt for the potentials outside and inside the sphere the expressions:

$$\phi_1 = \left( -E_1 \cdot r + \frac{1}{4\pi\epsilon_1} \frac{p}{r^2} \right) \cdot \cos\vartheta \qquad \text{and} \qquad \phi_2 = -E_2 \cdot r \cdot \cos\vartheta \tag{6.81}$$

We determine the parameters $E_2$ and $p$ in these two equations using the boundary conditions at $r = R$ according to eq. (6.52). For the normal and tangential components of the field, these are:

$$\epsilon_1 \frac{\partial \phi_1}{\partial r}\bigg|_{r=R} = \epsilon_2 \frac{\partial \phi_2}{\partial r}\bigg|_{r=R} \qquad \text{and} \qquad \frac{\partial \phi_1}{\partial \vartheta}\bigg|_{r=R} = \frac{\partial \phi_2}{\partial \vartheta}\bigg|_{r=R} \tag{6.82}$$

Equivalent to the boundary condition for the tangential component, we could also have written down the boundary condition $\phi_1(r=R) = \phi_2(r=R)$. From eq. (6.82), we obtain the following equations determining $E_2$ and $p$:

$$\epsilon_1 E_1 + \frac{1}{2\pi} \frac{p}{R^3} = \epsilon_2 E_2 \qquad \text{and} \qquad -E_1 R + \frac{1}{4\pi\epsilon_1} \frac{p}{R^2} = -E_2 R \tag{6.83}$$

Evaluating these equations yields $E_2$ and $p$:

$$E_2 = \frac{3\epsilon_1}{\epsilon_2 + 2\epsilon_1} \cdot E_1 \qquad \text{and} \qquad p = 4\pi R^3 \cdot \epsilon_1 \cdot \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + 2\epsilon_1} \cdot E_1 \tag{6.84}$$

Substituting this into our expression for the potentials $\phi_1$ and $\phi_2$, and using $z = r\cos\vartheta$, we obtain:

$$\phi_1 = \left( -1 + \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + 2\epsilon_1} \frac{R^3}{r^3} \right) \cdot E_1 \cdot z \qquad \text{and} \qquad \phi_2 = -\frac{3\epsilon_1}{\epsilon_2 + 2\epsilon_1} \cdot E_1 \cdot z \tag{6.85}$$

Because of $\boldsymbol{E} = -\nabla\phi$, we thus also know the behavior and strength of the electrostatic fields in the two dielectrics. Fig. 6.15a shows the pattern of the $\boldsymbol{D}$ field lines for $\epsilon_2 > \epsilon_1$, and fig. 6.15b shows it for $\epsilon_2 < \epsilon_1$. The field lines of $\boldsymbol{E}$ follow the same pattern as those of $\boldsymbol{D}$ in fig. 6.15, but according to eq. (6.84) their density decreases inside the dielectric sphere when $\epsilon_2 > \epsilon_1$ and increases inside the sphere when $\epsilon_2 < \epsilon_1$. Using eq. (6.80), we can express the magnitude of the dipole moment $p$ from eq. (6.84) in terms of the dielectric polarization of the sphere:

$$p = \frac{4\pi R^3}{3}\cdot P \qquad \text{with} \qquad P = 3\epsilon_1 \cdot \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + 2\epsilon_1}\cdot E_1 \tag{6.86}$$

In doing so, we assumed that the dipole density $\mathrm{d}p/\mathrm{d}V$ is constant within the volume of the sphere. This yields the following expression for the difference between the homogeneous fields outside and inside the sphere:

$$\boldsymbol{E}_2 - \boldsymbol{E}_1 = \left(\frac{3\epsilon_1}{\epsilon_2 + 2\epsilon_1} - 1\right)\boldsymbol{E}_1 = \frac{\epsilon_1 - \epsilon_2}{\epsilon_2 + 2\epsilon_1}\boldsymbol{E}_1 = -\frac{\boldsymbol{P}}{3\epsilon_1} \tag{6.87}$$

The homogeneous field components $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$ are each parallel to the $z$-axis, and thus the dielectric polarization vector $\boldsymbol{P}$ is also parallel to the $z$-axis.

### 6.2.3 Polarized Dielectric Sphere

We have just seen that the electrostatic field inside a dielectric sphere, situated within a homogeneous electric field in another dielectric, is homogeneous. But what does the polarization charge $\sigma_{pol}$ that produces this homogeneous polarization within the dielectric sphere look like? And what does the dipole moment look like that produces this polarization charge $\sigma_{pol}$ located on the surface of the sphere? To find answers to these questions, we will consider an isolated, uniformly polarized sphere which possesses this polarization we determined in the last section. The polarization vector in this dielectric sphere of radius $R$ lies parallel to the $z$-axis, i.e., $\boldsymbol{P} = P_0 \cdot \hat{\boldsymbol{k}}$. There are no free charges, meaning $\rho_{\text{free}} = 0$ everywhere, and the bound charges are located solely on the surface of the sphere, nowhere else. Eq. (6.20) tells us how the bound charges on the surface are related to the polarization of the sphere. Thus, the charge density in our problem is given by:

$$\rho_{bnd}(r) = 0 \quad \text{for} \quad r > R \qquad \text{and} \qquad \rho_{bnd}(r) = -\nabla\boldsymbol{P} \quad \text{for} \quad r \leq R \tag{6.88}$$

We can express the fact that polarization only exists within the sphere by

$$\boldsymbol{P}(\boldsymbol{r}) = P_0 \cdot \hat{\boldsymbol{k}} \cdot \Theta(r - R) \qquad \text{with} \qquad \Theta(r - R) = \begin{cases} 1 & r \leq R \\ 0 & r > R \end{cases} \tag{6.89}$$

Here, $\Theta(r-R)$ is the step function we already know from eq. (2.59), although its value is 0 here where it is 1 there, and vice versa. Therefore, the step function here is the antiderivative of the negative delta function $-\delta(r-R)$, rather than the positive delta function as was the case for the step function in eq. (2.59), so

$$\frac{\mathrm{d}\Theta(r-R)}{\mathrm{d}r} \;=\; -\delta(r-R) \tag{6.90}$$

Substituting eq. (6.90) into eq. (6.88), we obtain:

$$\rho_{bnd}(r) \;=\; -P_0\big[(\nabla\cdot\hat{\boldsymbol{k}})\cdot\Theta(r-R) + \hat{\boldsymbol{k}}\cdot\nabla\cdot\Theta(r-R)\big] \tag{6.91}$$

Since $\nabla\cdot\hat{\boldsymbol{k}} = 0$, the first term in the square brackets drops out. Furthermore, because we are working in spherical coordinates $\boldsymbol{r} = r\cdot(\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\varphi)$ and there is no dependence on $\vartheta$ or $\varphi$, the gradient reduces to the derivative with respect to $r$; applying eq. (6.90) then transforms eq. (6.91) into:

$$\rho_{bnd}(r) \;=\; -P_0\cdot\hat{\boldsymbol{k}}\cdot\frac{\mathrm{d}\Theta(r-R)}{\mathrm{d}r}\cdot\hat{\boldsymbol{e}}_r \;=\; P_0\cdot\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{e}}_r\cdot\delta(r-R) \tag{6.92}$$

With $\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{e}}_r = \cos\vartheta$ this becomes

$$\rho_{bnd}(r) \;=\; P_0\cos\vartheta\cdot\delta(r-R) \;=\; \sigma_{pol}(\vartheta)\cdot\delta(r-R) \tag{6.93}$$

To determine the polarization of a dielectric sphere with permittivity $\epsilon$ situated in a homogeneous electric field in a vacuum, we need only set $\epsilon_1 = \epsilon_0$ and $\epsilon_2 = \epsilon$ in eq. (6.87). In this case, eq. (6.87) yields:

$$\boldsymbol{P} = -3\epsilon_0\frac{\epsilon_0-\epsilon}{\epsilon+2\epsilon_0}\cdot E_1\cdot\hat{\boldsymbol{k}} \;=\; P_0\cdot\hat{\boldsymbol{k}} \tag{6.94}$$

We thus obtain the following expression for the polarization charge:

$$\sigma_{pol}(\vartheta) \;=\; P_0\cos\vartheta \;=\; 3\epsilon_0\frac{\epsilon-\epsilon_0}{\epsilon+2\epsilon_0}\cdot E_1\cos\vartheta \tag{6.95}$$

Fig. 6.16 shows this cosine distribution of the polarization charge. The north pole of the dielectric sphere, where the normal vector to the sphere's surface in fig. 6.14 points in the positive $z$-direction, is located at $\vartheta = 0$ and carries a positive charge distribution. The south pole of the dielectric sphere, where the normal vector to the sphere's surface in fig. 6.14 points in the negative $z$-direction, is located at $\vartheta = \pi$ and carries a negative charge distribution. The positive charge of the hemisphere ($\vartheta < \pi/2$) and

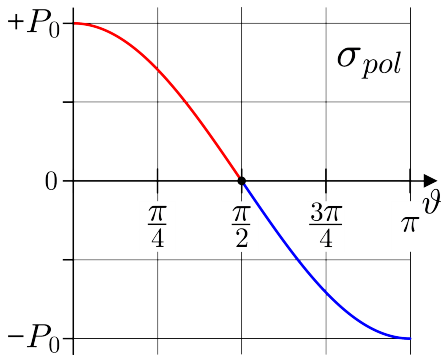



**Fig. 6.16**

the negative charge of the hemisphere ($\vartheta > \pi/2$) cancel each other out, which can easily be verified by a simple integration of eq. (6.95) over the surface of the sphere. We are, in fact, already familiar with the dipole moment of the polarized sphere from eq. (6.80). Here, however, we can see how a homogeneous polarization gives rise to a dipole moment corresponding to a dipole at the center of the dielectric sphere with the strength specified in eq. (6.80). To do this, we simply need to sum all contributions $\boldsymbol{r}\cdot\rho_{bnd}(r)$ to $\boldsymbol{p}$. Using eq. (6.93), we thus obtain the following expression for the dipole moment $\boldsymbol{p}$:

$$\boldsymbol{p} \;=\; \int \boldsymbol{r}\cdot\rho_{bnd}(r)\,\mathrm{d}^3r \;=\; P_0 \int_{-\infty}^{+\infty}\int_{0}^{\pi}\int_{0}^{2\pi} r \begin{pmatrix} \sin\vartheta\sin\varphi \\ \sin\vartheta\cos\varphi \\ \cos\vartheta \end{pmatrix} \cdot\cos\vartheta\cdot\delta(r-R)\cdot\mathrm{d}^3r \tag{6.96}$$

With $\mathrm{d}^3r = r^2\sin\vartheta\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$, integration over $\mathrm{d}r$ yields a factor of $R^3$ due to the $\delta$-function. Upon integration over $\mathrm{d}\varphi$, the $x$- and $y$-components vanish, while the $z$-component yields a factor of $2\pi$. Thus, eq. (6.96) reduces to

$$\boldsymbol{p} \;=\; P_0R^3 2\pi\hat{\boldsymbol{k}}\int_0^{\pi}\cos^2\vartheta\cdot\sin\vartheta\,\mathrm{d}\vartheta \;=\; P_0R^3 2\pi\hat{\boldsymbol{k}}\cdot\left[-\frac{\cos^3\vartheta}{3}\right]_0^{\pi} \tag{6.97}$$

We thus obtain the following result for the dipole moment of the uniformly polarized dielectric sphere:

$$\boldsymbol{p} \;=\; \boldsymbol{k}\cdot P_0\cdot\frac{4\pi R^3}{3} \;=\; \boldsymbol{k}\cdot P_0\cdot V_{Sphere} \tag{6.98}$$

This is exactly the result of eq. (6.80). Thus, outside the sphere, the electric field of an isolated, uniformly polarized dielectric sphere appears identical to that of an electric dipole located at the center of the sphere possessing the dipole moment from eq. (6.98). Using eq. (3.53), the potential outside the polarized sphere is therefore given by

$$\phi_{r\geq R} \;=\; \frac{1}{4\pi\epsilon_0}\frac{p\cos\vartheta}{r^2} \;=\; \frac{1}{4\pi\epsilon_0}\frac{4\pi R^3}{3}\frac{P_0\cos\vartheta}{r^2} \;=\; \frac{P_0}{3\epsilon_0}\frac{R^3}{r^2}\cos\vartheta \tag{6.99}$$

and using $\boldsymbol{E} = -\nabla\phi$, we obtain for the electric field (see eq. (3.59)):

$$\boldsymbol{E}_{r\geq R} \;=\; \frac{P_0}{3\epsilon_0}\frac{R^3}{r^3}\left(2\cos\vartheta\cdot\hat{\boldsymbol{e}}_r + \sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta\right) \;=\; E_r\cdot\hat{\boldsymbol{e}}_r + E_\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta \tag{6.100}$$

With eq. (6.99) and eq. (6.100), we have expressed the electrostatic potential and electric field outside the sphere in terms of the constant polarization $P$. For the validity of the dipole approximation in section 3.2.1, the condition was that the distance of the point $r$, where the potential is determined, must be very large compared to the distance $d$ between the negative and positive charges of the dipole. So why is the potential we just determined for the region outside the polarized sphere also a good approximation near the sphere and right up to its surface? To understand that, we must consider the

magnitude of the dipole length $d$. The idealized dipole at the center of the sphere is very small. Its dimensions correspond to the charge separation that occurs on an atomic scale during polarization. The order of magnitude of $d$ is therefore in the angstrom range or smaller, making $R \gg d$ a very good approximation indeed.

Fig. 6.17 shows the pattern of the field lines of the electrostatic field $\boldsymbol{E}$ outside and inside the polarized dielectric sphere. How did we generate these field line patterns? As for the field line pattern outside the sphere shown in fig. 6.17, it simply results from applying the same procedure to the right-hand side of eq. (6.100) which we previously used to represent the identical right-hand side of eq. (3.59). We have not explicitly derived the field line pattern of $\boldsymbol{E}$ inside the polarized dielectric sphere shown in fig. 6.17, nor is it necessary to do so. To understand why the electric field inside the sphere appears as shown in fig. 6.17, we consider eq. (6.87) and simply imagine "freezing" the polarization induced inside the sphere by the external field $\boldsymbol{E}_1$ and subsequently switching off $\boldsymbol{E}_1$. In our case, $\epsilon_2 = \epsilon$ and $\epsilon_1 = \epsilon_0$; thus, by eliminating $\boldsymbol{E}_1$ while the polarization $\boldsymbol{P}$ remains frozen, we obtain:

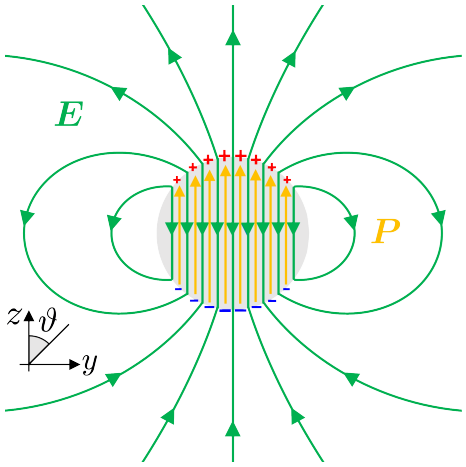



**Fig. 6.17**

$$\boldsymbol{E}_2 - \boldsymbol{E}_1 = -\frac{\boldsymbol{P}}{3\epsilon_0} \qquad \xrightarrow[\text{off}]{\boldsymbol{E}_1} \qquad \boldsymbol{E_2} \;=\; -\frac{\boldsymbol{P}}{\boldsymbol{3\epsilon_0}} \tag{6.101}$$

The electric field inside the dielectric sphere is directed exactly opposite to the polarization, thus pointing in the negative $z$-direction:

$$\boldsymbol{E}(r \leq R) \;=\; -\frac{\boldsymbol{P}}{3\epsilon_0} \;=\; -\frac{P_0}{3\epsilon_0} \cdot \hat{\boldsymbol{k}} \tag{6.102}$$

The corresponding potential is thus:

$$\phi(r \leq R) \;=\; \frac{P_0}{3\epsilon_0} \cdot z \;=\; \frac{P_0}{3\epsilon_0} \cdot r \cos\vartheta \tag{6.103}$$

Fig. 6.17 shows the behavior of the electrostatic field $\boldsymbol{E}$ rather than the field lines of $\boldsymbol{D}$, as is the case in fig. 6.15. The reason for this is that the homogeneous field $\boldsymbol{E}_1$ in fig. 6.15 is generated by free charge carriers, albeit ones located effectively at infinity, whereas in the case of the isolated, homogeneously polarized sphere, there are no free charge carriers anywhere. The polarized charge carriers within the sphere are frozen in place, and the electric field that originally generated them no longer exists, having been switched off after the polarization charges were frozen. Consequently, the boundary conditions regarding

the tangential and normal components of the electric field no longer apply, as the charges that produced the homogeneous field are gone. The remaining boundary condition is that the potentials at the sphere's surface must satisfy $\phi_{r\geq R}(R) = \phi_{r\leq R}(R)$, and this condition is met.

# 7. Dielectric Matter

In the last chapter, we discussed how electrostatic fields change when dielectric materials are introduced into them. In this chapter, we will first examine how the electrostatic field energy in dielectric materials differs from that in a vacuum environment. Subsequently, we will turn our attention to the investigation of the properties of dielectric materials in their various states of matter.

## 7.1 Field Energy in Dielectric Materials

To determine the electrostatic field energy in dielectric materials in the presence of free charge distributions, we recall how we did this in the case of charge distributions in a vacuum environment in section 3.1.3. There we found that with eq. (3.43) the potential energy of a charge distribution is given by:

$$U = \frac{1}{2}\int_V \rho_{free}(\boldsymbol{r})\phi(\boldsymbol{r})\,\mathrm{d}^3r \tag{7.1}$$

In the case of charge distributions in a vacuum environment, we arrived at this result by successively bringing free charges from infinity into the same region while simultaneously summing the potential energies of the associated two-body interactions between these charges. We can do the same in the case of bound charges in dielectrics. But that is not enough, because these bound charges are also linked to dielectric polarization, which did not exist for our charge distribution in a vacuum environment. Therefore, we cannot simply adopt eq. (3.43) for dielectric materials, but we can use it as a starting point. We do this by considering a small change $\delta\rho_{free}$ in the charge density $\rho_{free}$ in eq. (3.43). This small change in charge density is then associated with a change in the potential energy $\delta U$ for which the following holds:

$$\delta U = \int_V \delta\rho_{free}(\boldsymbol{r})\phi(\boldsymbol{r})\,\mathrm{d}^3r \tag{7.2}$$

Here, $\phi(\boldsymbol{r})$ is still the potential of the original distribution of free charge carriers, which now acts on $\delta\rho_{free}$. Eq. (7.2) corresponds to eq. (3.43) for a small change in the charge density $\delta\rho_{free}$ in the presence of a continuous charge distribution that generates the potential $\phi(\boldsymbol{r})$ which acts on $\delta\rho_{free}$. When integrating over this charge distribution, unlike in eq. (7.1), there is no prefactor 1/2 before the integral, which corrected for the double counting of two-body interactions in our derivation of eq. (3.43). This is because there is no double counting, but rather only the influence of $\phi(\boldsymbol{r})$ on $\delta\rho_{free}(\boldsymbol{r})$; therefore, no prefactor of 1/2 appears in eq. (7.2). Via the relationship

$$\nabla(\delta\boldsymbol{D}) = \delta\rho_{free} \tag{7.3}$$

this $\delta\rho_{free}$ is coupled to a small dielectric displacement $\delta\boldsymbol{D}$. Using

$$\nabla(\delta\boldsymbol{D}\cdot\phi) = \nabla(\delta\boldsymbol{D})\cdot\phi + \delta\boldsymbol{D}\cdot(\nabla\cdot\phi) = \delta\rho_{free}\cdot\phi - \delta\boldsymbol{D}\cdot\boldsymbol{E} \tag{7.4}$$

we can write $\delta U$ as:

$$\delta U = \int_V \nabla(\delta\boldsymbol{D}\cdot\phi)\,\mathrm{d}^3r + \int_V \boldsymbol{E}\cdot\delta\boldsymbol{D}\,\mathrm{d}^3r \tag{7.5}$$

We transform the first integral on the right-hand side into a surface integral using Gauss's theorem, which vanishes when we shift the surface to infinity:

$$\int_V \nabla(\delta\boldsymbol{D}\cdot\phi)\,\mathrm{d}^3r = \int_S (\delta\boldsymbol{D}\cdot\phi)\hat{\boldsymbol{n}}_s\,\mathrm{d}\sigma \underset{S\to\infty}{\Longrightarrow} 0 \tag{7.6}$$

Why is this the case? We know that for dielectric materials where the polarization depends linearly on the electric field, $\boldsymbol{D} = \epsilon\boldsymbol{E} = -\epsilon\nabla\phi$ holds true. The potential $\phi$ scales as a unique solution of the Poisson equation like $r^{-1}$, and thus $\boldsymbol{D} = -\epsilon\nabla\phi$ like $r^{-2}$. Hence, the integrand of the surface integral scales like $r^{-3}$ while the surface element $\mathrm{d}\sigma$ only scales like $r^2$. For $V\to\infty$, the surface $S$ shifts to infinity and thus the surface integral vanishes. Let us now look at the second integral on the right-hand side of eq. (7.5). For linear dielectric materials, $\boldsymbol{E}\cdot\delta\boldsymbol{D} = \delta\boldsymbol{E}\cdot\boldsymbol{D}$ also holds, and thus:

$$\delta(\boldsymbol{E}\cdot\boldsymbol{D}) = \delta\boldsymbol{E}\cdot\boldsymbol{D} + \boldsymbol{E}\cdot\delta\boldsymbol{D} = 2\cdot\boldsymbol{E}\cdot\delta\boldsymbol{D} \tag{7.7}$$

We obtain the expression for the potential energy $U$ from the equation for $\delta U$ by increasing the dielectric displacement from 0 to a value $\boldsymbol{D}$, that is

$$\boldsymbol{E}\cdot\delta\boldsymbol{D} \to \int_0^{\boldsymbol{D}} \boldsymbol{E}\cdot\delta\boldsymbol{D} = \frac{1}{2}\int_0^{\boldsymbol{D}} \delta(\boldsymbol{E}\cdot\boldsymbol{D}) = \frac{1}{2}\boldsymbol{E}\cdot\boldsymbol{D} \tag{7.8}$$

Using $\delta U$ from eq. (7.5) we thus obtain the expression:

$$U = \frac{1}{2}\int_V \boldsymbol{E}\cdot\boldsymbol{D}\,\mathrm{d}^3r \tag{7.9}$$

With eq. (3.43) and eq. (7.9) we now have two equations for the potential energy of a charge distribution in a vacuum environment and in a dielectric material that look very different at first glance. However, this is not the case. With

$$\nabla(\phi\cdot\boldsymbol{D}) \;=\; \phi\cdot\nabla\boldsymbol{D} + (\nabla\phi)\cdot\boldsymbol{D} \tag{7.10}$$

and $\boldsymbol{E} = -\nabla\phi$ we can rewrite eq. (7.9) as:

$$U \;=\; -\frac{1}{2}\int_V \nabla\phi\cdot\boldsymbol{D}\,\mathrm{d}^3r \;=\; \frac{1}{2}\int_V \phi\cdot\nabla\boldsymbol{D}\,\mathrm{d}^3r - \frac{1}{2}\int_V \nabla(\phi\cdot\boldsymbol{D})\,\mathrm{d}^3r \tag{7.11}$$

In the first integral on the right-hand side, we now replace the divergence of the dielectric displacement with the free charge distribution $\rho_{free}$ using $\nabla\boldsymbol{D} = \rho_{free}$ and apply Gauss's theorem again to the second integral on the right-hand side. This transforms eq. (7.11) into:

$$U \;=\; \frac{1}{2}\int_V \rho_{free}\cdot\phi\,\mathrm{d}^3r - \frac{1}{2}\int_S (\phi\cdot\boldsymbol{D})\hat{\boldsymbol{n}}_s\,\mathrm{d}\sigma \underset{S\to\infty}{\Longrightarrow} \frac{1}{2}\int_V \rho_{free}\cdot\phi\,\mathrm{d}^3r \tag{7.12}$$

For the same reasons as in eq. (7.6), the surface integral vanishes as $S \to \infty$. Therefore, eq. (7.9) and eq. (3.43) are not different equations. On the contrary, we can transform eq. (7.9) into eq. (3.43). However, there is a difference. In deriving eq. (7.9) we made an assumption that we did not have to make in deriving eq. (3.43) in section 3.1.3, namely, that we are dealing here exclusively with linear dielectric materials for which $\boldsymbol{D} = \epsilon\boldsymbol{E}$ holds. If this is not the case, then eq. (7.7) and eq. (7.9) do not apply and we must calculate the potential energy directly from eq. (7.5) by bringing $\boldsymbol{D}$ from 0 to a value $\boldsymbol{D}$.

### 7.1.1 Field Energy of the Insulator Sphere

As an example of determining the field energy of a dielectric material in a homogeneous electrostatic field, we return to the insulator sphere from section 6.2.2. To calculate the field energy of this dielectric sphere with $\epsilon = \epsilon_2$ embedded in another dielectric material with $\epsilon = \epsilon_1$ in which a homogeneous electrostatic field exists, we first consider the field energy in this other dielectric material without the insulator sphere. Both dielectric materials are linear media, therefore $\boldsymbol{D}_1 = \epsilon_1\boldsymbol{E}_1$ and $\boldsymbol{D}_2 = \epsilon_2\boldsymbol{E}_2$. Using eq. (7.9), we can write the field energy in the dielectric material with $\epsilon = \epsilon_1$:

$$U_1 \;=\; \frac{1}{2}\int_V \boldsymbol{E}_1\cdot\boldsymbol{D}_1\,\mathrm{d}^3r \tag{7.13}$$

If we now introduce our insulator sphere into the dielectric, then the potential energy in the volume now occupied by the insulator sphere is:

$$U_2 \;=\; \frac{1}{2}\int_V \boldsymbol{E}_2\cdot\boldsymbol{D}_2\,\mathrm{d}^3r \tag{7.14}$$

The difference in potential energy $\Delta U = U_2 - U_1$ between the configuration after the introduction of the insulator sphere and the configuration before the introduction of the insulator sphere is therefore:

$$\Delta U \;=\; \frac{1}{2}\int_V \big(\boldsymbol{E}_2\boldsymbol{D}_2 - \boldsymbol{E}_1\boldsymbol{D}_1\big)\,\mathrm{d}^3r \tag{7.15}$$

If we write the potential energy difference $\Delta U$ in this way, we implicitly make the assumption that the dielectric permittivity changes very quickly, but still continuously, during the transition between the region with $\epsilon_1$ and the region with $\epsilon_2$. In its current form, we cannot do much with this equation for the difference in potential energy. But that changes when we learn that for the auxiliary integral

$$I_A \;=\; \int_V \big(\boldsymbol{E}_2 + \boldsymbol{E}_1\big)\big(\boldsymbol{D}_2 - \boldsymbol{D}_1\big)\,\mathrm{d}^3r \tag{7.16}$$

it holds that $I_A = 0$. Let us examine why this auxiliary integral vanishes. Using Maxwell's second equation in electrostatics, $\nabla\times(\boldsymbol{E}_2+\boldsymbol{E}_1) = 0$ must hold, and we can write the sum of the fields as the gradient of a single potential, i.e., $\boldsymbol{E}_2+\boldsymbol{E}_1 = -\nabla\phi$. Together with

$$\nabla\big[\phi(\boldsymbol{D}_2-\boldsymbol{D}_1)\big] \;=\; \phi\cdot\nabla(\boldsymbol{D}_2-\boldsymbol{D}_1) + \nabla\phi\cdot(\boldsymbol{D}_2-\boldsymbol{D}_1) \tag{7.17}$$

our auxiliary integral then becomes:

$$\begin{aligned} I_A \;&=\; -\int_V \nabla\phi\cdot(\boldsymbol{D}_2-\boldsymbol{D}_1)\,\mathrm{d}^3r \\ &=\; -\int_V \nabla\big[\phi\cdot(\boldsymbol{D}_2-\boldsymbol{D}_1)\big]\,\mathrm{d}^3r \;+\; \int_V \phi\cdot\nabla(\boldsymbol{D}_2-\boldsymbol{D}_1)\,\mathrm{d}^3r \\ &=\; \underbrace{-\int_S \big[\phi\cdot(\boldsymbol{D}_2-\boldsymbol{D}_1)\big]\boldsymbol{n}_S\,\mathrm{d}\sigma}_{=\,0} \;+\; \int_V \phi\cdot\nabla(\boldsymbol{D}_2-\boldsymbol{D}_1)\,\mathrm{d}^3r \end{aligned} \tag{7.18}$$

Here we have again applied Gauss's theorem to transform the first volume integral in the second line into a surface integral. This surface integral vanishes for $S\to\infty$ for the same reasons as was the case in eq. (7.6) and eq. (7.12), and we obtain the following result:

$$I_A \;=\; \int_V \phi\cdot\nabla\big(\boldsymbol{D}_2-\boldsymbol{D}_1\big)\,\mathrm{d}^3r \tag{7.19}$$

Embedding the insulator sphere with $\epsilon=\epsilon_2$ in the dielectric material with $\epsilon=\epsilon_1$ has not changed the distribution of the free charges. Therefore, $\nabla\boldsymbol{D}_1 = \rho_{free}$ and $\nabla\boldsymbol{D}_2 = \rho_{free}$

must hold, from which it follows that $\nabla(\boldsymbol{D}_2 - \boldsymbol{D}_1) = 0$ and thus $I_A = 0$. By inserting zero twice, we can now use the auxiliary integral $I_A$ to rewrite the expression for the difference in potential energy in eq. (7.15) as:

$$\begin{aligned}\Delta U &= \frac{1}{2}\int_V \big(\boldsymbol{E}_2\boldsymbol{D}_2 + \underbrace{\boldsymbol{E}_1\boldsymbol{D}_2 - \boldsymbol{E}_1\boldsymbol{D}_2}_{=0} - \boldsymbol{E}_1\boldsymbol{D}_1 + \underbrace{\boldsymbol{E}_2\boldsymbol{D}_1 - \boldsymbol{E}_2\boldsymbol{D}_1}_{=0}\big)\,\mathrm{d}^3r \\ &= \frac{1}{2}\int_V \big(\boldsymbol{E}_2\boldsymbol{D}_1 - \boldsymbol{E}_1\boldsymbol{D}_2\big)\,\mathrm{d}^3r + \underbrace{\frac{1}{2}\int_V \big(\boldsymbol{E}_2 + \boldsymbol{E}_1\big)\big(\boldsymbol{D}_2 - \boldsymbol{D}_1\big)\,\mathrm{d}^3r}_{=I_A=0}\end{aligned} \tag{7.20}$$

If the free charge density $\rho_{free}$ does not change due to the introduction of a dielectric into the electrostatic field generated by $\rho_{free}$, then the following applies to eq. (7.15):

$$\Delta U = \frac{1}{2}\int_V \big(\boldsymbol{E}_2\boldsymbol{D}_2 - \boldsymbol{E}_1\boldsymbol{D}_1\big)\,\mathrm{d}^3r = \frac{1}{2}\int_V \big(\boldsymbol{E}_2\boldsymbol{D}_1 - \boldsymbol{E}_1\boldsymbol{D}_2\big)\,\mathrm{d}^3r \tag{7.21}$$

This difference in potential energy arises from introducing an insulator sphere with $\epsilon = \epsilon_2$ and volume $V_S$ into the dielectric $\epsilon = \epsilon_1$, where it replaces the volume $V_S$. The integration volume in eq. (7.21) is therefore $V_S$ and with $\boldsymbol{D}_1 = \epsilon_1\boldsymbol{E}_1$ and $\boldsymbol{D}_2 = \epsilon_2\boldsymbol{E}_2$ eq. (7.21) becomes:

$$\Delta U = \frac{1}{2}\int_{V_S} (\epsilon_1 - \epsilon_2)\boldsymbol{E}_2\cdot\boldsymbol{E}_1\,\mathrm{d}^3r \tag{7.22}$$

Let us now consider specifically an insulator sphere with dielectric permittivity $\epsilon_2 = \epsilon$ in an electrostatic field in a vacuum environment, i.e., $\epsilon_1 = \epsilon_0$. In this case, the energy difference in eq. (7.22) becomes the potential energy of the polarized dielectric sphere. Using eq. (6.34) and eq. (6.36), we can generally write the relationship between $\boldsymbol{P}$ and $\boldsymbol{E}$ in linear dielectric materials as:

$$\boldsymbol{P} = \chi\epsilon_0\boldsymbol{E} = (\epsilon - \epsilon_0)\boldsymbol{E} \tag{7.23}$$

Using this expression, we obtain for the potential energy of our polarized dielectric sphere in a vacuum field $\boldsymbol{E}_1$:

$$U = -\frac{1}{2}\int_{V_S} (\epsilon - \epsilon_0)\boldsymbol{E}_2\cdot\boldsymbol{E}_1\,\mathrm{d}^3r = -\frac{1}{2}\int_{V_S} \boldsymbol{P}\cdot\boldsymbol{E}_1\,\mathrm{d}^3r \tag{7.24}$$

As long as $\rho_{free}$ remains unchanged by the introduction of a dielectric material into a field, i.e., as long as the free charges are fixed, the following generally applies to the potential energy of dielectrics in a vacuum environment:

$$U = -\frac{1}{2}\int_V \boldsymbol{P}\cdot\boldsymbol{E}\,\mathrm{d}^3r \tag{7.25}$$

The integration is performed here over the volume of the dielectric. The energy density in this dielectric then follows accordingly:

$$\frac{\mathrm{d}U}{\mathrm{d}V} = -\frac{1}{2}\boldsymbol{P}\boldsymbol{E} \tag{7.26}$$

Why is the potential energy in eq. (7.25) negative? The reason is simply that the polarized state of the dielectric, i.e., when the dipoles in the dielectric are aligned with the external field, represents a state of lower energy than one in which the dipoles are randomly oriented in the dielectric. This is exactly analogous to the potential energy of the dipole in eq. (3.69), which is lowest when the dipole is parallel to the field and highest when it is antiparallel to the field. In our dielectric sphere, $\boldsymbol{P}$ is linked to $\boldsymbol{E}_1$ by eq. (6.87). We only need to set $\epsilon_2 = \epsilon$ and $\epsilon_1 = \epsilon_0$ and realize that for $\epsilon > \epsilon_0$ the vector quantities $\boldsymbol{E}_1$ and $\boldsymbol{P}$ point in the same direction. $\boldsymbol{E}_1$ and $\boldsymbol{P}$ are each constant, meaning we can pull both out of the integral in eq. (7.24) and replace the integral with the volume of the sphere. Thus, we obtain the following for the potential energy of our dielectric sphere $DS$ with radius $R$ in the field $\boldsymbol{E}_1$:

$$U_{DS} = -\frac{1}{2}\boldsymbol{P}\cdot\boldsymbol{E}_1\cdot\frac{4\pi R^3}{3} = -\frac{2\pi R^3}{3}\cdot P\cdot E_1 \tag{7.27}$$

### 7.1.2 Force on Dielectric Materials in an Electrostatic Field

Now that we know the potential energy of dielectric materials in electrostatic fields, we can also say something about the force acting on a dielectric material in such a field. From eq. (7.25) we can directly see that for $\epsilon > \epsilon_0$, as the field strength $\boldsymbol{E}_1$ increases, the potential energy becomes increasingly negative, i.e., increasingly smaller. The equation for the electrostatic energy density in a dielectric material in eq. (7.26) tells us the same thing. This simply means that a force acts on a dielectric in an electrostatic field, pulling the dielectric in the direction of increasing field strength. This should not surprise us, as we already know this behavior from section 3.2.1 where we considered the force on the electric dipole in homogeneous and inhomogeneous electrostatic fields. The question is: How can we apply what we learned there about the force acting on the dipole to conductors and insulators?

In conductors, the charge carriers, the conduction electrons of metals, are freely mobile, and under the influence of an external electric field, they will always distribute themselves on the surface of the conductor in such a way that the electric field within the conductor itself vanishes. A homogeneous electric field looks exactly like the field in a parallel-plate capacitor whose electrodes extend to infinity and are sitting far apart from each other, just as we assumed in section 5.3.3. When a conductor is placed in the center of this parallel-plate capacitor, a charge is generated on its surface by induction, shielding the interior from the electric field. The same forces act on the positive surface areas of the

conductor, from which electrons have flowed away, and on the negative surface areas, to which electrons have flowed. And because there are equal numbers of positive and negative induced charges, the net force on the conductor vanishes in a homogeneous electric field. This applies not only to the spherical conductor in section 5.3.3 but to conductors of any shape in a homogeneous field. One can visualize this as follows: part of the conductor forms a positive counterpole to the negative electrode of the parallel-plate capacitor, and another part forms an equally strong negative counterpole to the positive electrode of the parallel-plate capacitor. The forces on these two poles of the conductor always cancel each other out, even in conductors shaped in such a way that the electric field becomes locally very inhomogeneous at their surface. If, for example, we replace one half of the conducting sphere from section 5.3.3 with a cone whose base has the same radius as the sphere, we obtain a conductor end with a tip. The field line density, and thus the field strength, is significantly higher there than on the opposite side of the conductor, which retains a hemispherical shape. The field strength at the pointed end of the conductor can become so high that discharges occur. However, this does not change the fact that the net force acting on the conductor remains zero. This is no different for dielectric bodies in homogeneous electrostatic fields. There, it is not the forces on the positive and negative induction charges that cancel each other out, but rather the forces acting on the positive and negative bound polarization charges. Therefore, the field profiles for the conducting sphere ($\epsilon = \infty$) in the homogeneous electric field in a vacuum environment in fig. 5.11b and those of the dielectric sphere with $\epsilon_2 > \epsilon_1$ in fig. 6.15a are the same. If a dielectric material is placed in a homogeneous field, the state of lowest potential energy is that in which the polarization of the dielectric material points in the same direction as the external field. This is no different than with an electric dipole. Let us consider the force acting on our dielectric sphere in the homogeneous vacuum field $\boldsymbol{E}_1$, i.e., $\epsilon_1 = \epsilon_0$, in fig. 6.15a. Using eq. (7.25) and interchanging integration and differentiation, we can write the force on our insulator sphere as follows:

$$\boldsymbol{F} \;=\; -\nabla U \;=\; \frac{1}{2}\int_{V_S} \nabla(\boldsymbol{P}\boldsymbol{E}_1)\,\mathrm{d}^3r \;=\; \frac{1}{2}\int_{V_S} \big(\boldsymbol{P}\cdot\nabla\boldsymbol{E}_1 + \nabla\boldsymbol{P}\cdot\boldsymbol{E}_1\big)\,\mathrm{d}^3r \tag{7.28}$$

According to the assumption in section 6.2.2, $\boldsymbol{E}_1$ is constant, and therefore the first term vanishes. With eq. (6.25), we know that $\nabla\boldsymbol{P} = -\rho_{bnd}$ holds, and thus we obtain the following expression for the force on the dielectric sphere in the homogeneous field:

$$\boldsymbol{F} \;=\; -\frac{1}{2}\int_{V_S} \rho_{bnd}\cdot\boldsymbol{E}_1\,\mathrm{d}^3r \;=\; -\frac{\boldsymbol{E}_1}{2}\int_{V_S} \rho_{bnd}\,\mathrm{d}^3r \;=\; -\frac{\boldsymbol{E}_1}{2}\cdot 0 \;=\; 0 \tag{7.29}$$

The integral over the bound charge distribution $\rho_{bnd}$ on the surface of the dielectric sphere vanishes because there are an equal number of positive and negative charges,

and therefore the total charge, and thus the integral, is zero. Things change when considering conductors and dielectrics in inhomogeneous fields. For a conductor located in an inhomogeneous electric field, the field strength is greater at some parts of the surface than at others. Regions with higher or lower induction charge density form on the surface, and different forces act on these regions in an inhomogeneous field. This situation is analogous to that of the dipole in an inhomogeneous electric field, which we discussed in section 3.2.1. In an inhomogeneous field, the conductor possesses a temporary dipole moment, meaning it has a positive and a negative end. This leads to two effects. First, a torque acts on the conductor, aligning it so that its potential energy in the field is minimized; second, a resultant force acts on the conductor, pulling it, similar to a dipole, in the direction of increasing field strength. Unlike the constant dipole we considered in section 3.2.1, the dipole moment of the conductor does not remain constant as the conductor moves in the direction of increasing field strength in the inhomogeneous field. This temporary dipole moment itself depends on the local field environment of the conductor, which naturally changes with its movement in the inhomogeneous electrostatic field. The situation is similar for a dielectric body in an inhomogeneous electric field, with the difference that the dipole moment $\boldsymbol{p}$ is associated with the dielectric polarization $\boldsymbol{P}$ and an internal electric field exists in the dielectric that opposes and thereby weakens the external electric field inside the dielectric.

In general, the force acting on a dielectric body can be determined in many practical cases by considering a small displacement of the dielectric body along a selected coordinate $\xi$ within the external field. If, in such a situation, the free charge distribution that generates the external electrostatic field remains unchanged, we can interpret the observed change in the total energy of the system as a change in the potential energy of the dielectric body in the electrostatic field. A simple example of this is the force acting on a dielectric body placed in a parallel-plate capacitor. If a dielectric body is only partially inserted into a parallel-plate capacitor, as sketched in fig. 7.1, then a force is observed acting on the dielectric body, which pulls the dielectric further into the capacitor. The energy stored in a capacitor as given by eq. (4.65) is:

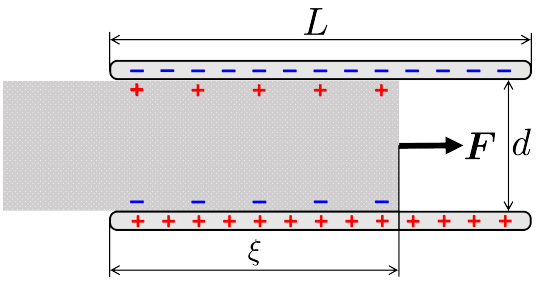



**Fig. 7.1**

$$W = \frac{1}{2} C \cdot U^2 \tag{7.30}$$

Here, $U$ denotes the electrical voltage applied between the capacitor electrodes and $C$ its capacitance including the partially inserted dielectric material. If we now interpret the small change in $W$ which occurs when the capacitor is pulled a small distance further into the capacitor as a change in the potential energy of the dielectric body within the capacitor, then the force acting on the dielectric body, with the voltage $U$ held constant, is given by:

$$F_\xi = \frac{\partial W}{\partial \xi} = \frac{1}{2} U^2 \cdot \frac{\partial C}{\partial \xi} \tag{7.31}$$

With eq. (4.64) and eq. (6.41) we know the capacitance of a parallel-plate capacitor without a dielectric and with a dielectric inserted between the capacitor electrodes. Here we have the situation that the dielectric is only partially inserted. To obtain the total capacitance $C$ of our capacitor when the dielectric is inserted up to a position $\xi$ into the long side $L$ between the electrodes, we must add the two respective capacitances. If $B$ is the short side of the electrodes, then the electrode area $A = L \cdot B$ is given by eq. (4.64) and eq. (6.41)

$$C = \frac{\epsilon_0 B \cdot (L - \xi)}{d} + \frac{\epsilon B \cdot \xi}{d} = \frac{B}{d}\left[(\epsilon - \epsilon_0)\xi + \epsilon_0 L\right] \tag{7.32}$$

This gives us the following for the force in eq. (7.31), which pulls the dielectric into the capacitor when a voltage $U$ is applied:

$$F_\xi = (\epsilon - \epsilon_0) \cdot \frac{U^2 B}{2d} \tag{7.33}$$

## 7.2 Properties of Dielectric Matter

The properties of solid-state dielectrics are based on the properties of the atoms and molecules that constitute the dielectric material. Up to this point, atomic or molecular properties have played no role in our discussion of the dielectric behavior of solids, other than the assumption that the constituents of these dielectric solids are polarizable. Within the solid, we have treated the atomic or molecular dipoles induced by an external electric field as independent of one another. This was possible because the behavior of solid-state dielectrics at the macroscopic level is determined by the surface charges at their outer interfaces. At the microscopic level, however, the dipole interaction between neighboring atoms or molecules in the solid-state dielectric must be taken into account. The situation is simpler for gases and liquids, though even there, one must carefully consider the extent to which dipole interactions can be neglected in each case. With ideal gases, we are on safe ground, since by definition, the atoms or molecules of ideal gases do not interact with one another.

### 7.2.1 Polarizable Gases

Atoms do not possess a permanent dipole moment, but all atoms are polarizable in electric fields, even though some, such as for example the noble gases, are only very weakly polarizable. Most molecules are also polarizable, with the exception of small diatomic molecules like $H_2$, $O_2$, $N_2$, or $F_2$. Most molecules are not only polarizable, but many also possess a permanent dipole moment. Molecules with a permanent dipole moment are referred to as polar molecules, in contrast to non-polar molecules. Before we address polar molecules, let us return to the polarization of atoms, which was already discussed in the previous chapter. There, we considered the situation of a hydrogen atom in an electrostatic field, meaning of an atom subject to a constant force $F = q \cdot E$. But what happens if the force acting on the electron of the hydrogen atom in fig. 6.1 is not constant but rather oscillates with an angular frequency $\omega$? Fortunately, we already know the answer to that question from chapter 10 of volume 1: the result is a forced harmonic oscillation described by:

$$m\ddot{z} \;=\; -m\omega_0^2 z + F(t) \tag{7.34}$$

In this equation, $\omega_0$ is the resonance frequency of the system, in this case, the forced oscillation of the electron in the hydrogen atom, and $F(t) = q \cdot E_z^0 \cos\omega t$ is the periodic force exerted by an electric field in the $z$-direction. As we know from volume 1, the solution to eq. (7.34) is:

$$z(t) \;=\; \frac{q \cdot E_z^0}{m_e(\omega_0^2 - \omega^2)}\big[\cos\omega t - \cos\omega_0 t\big] \;=\; \frac{q}{m_e(\omega_0^2 - \omega^2)} \cdot E_z(\omega, t) \tag{7.35}$$

An oscillating dipole moment $p_z = q \cdot z(t)$ is associated with the oscillating displacement $z(t)$ of the electron according to eq. (6.1), and it follows from eq. (6.4) that:

$$p_z(\omega, t) \;=\; \frac{q^2}{m_e(\omega_0^2 - \omega^2)} \cdot E_z(\omega, t) \;=\; \alpha(\omega) \cdot E_z(\omega, t) \tag{7.36}$$

Evidently, the polarizability $\alpha$ is no longer a constant. Instead, as shown by its behavior in fig. 7.2, it depends on the angular frequency $\omega$ at which the external force, here the electric field $E_z(\omega, t)$, oscillates. For $\omega < \omega_0$, the polarizability $\alpha$ is positive ($\alpha > 0$) and the vectors $\boldsymbol{p}$ and $\boldsymbol{E}$ are therefore parallel, whereas for $\omega > \omega_0$, the polarizability $\alpha$ is negative ($\alpha < 0$) and the vectors $\boldsymbol{p}$ and $\boldsymbol{E}$ are therefore antiparallel. As $\omega$ approaches the resonance frequency $\omega_0$ from below or above, $\alpha$ tends

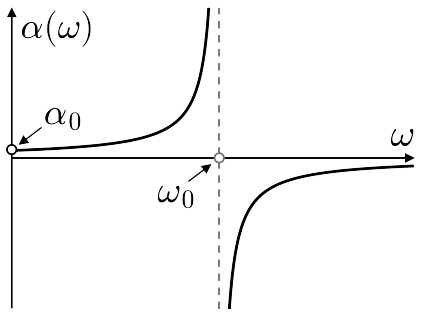



**Fig. 7.2**

to $+\infty$ or $-\infty$, respectively, meaning that the interaction between the polarizable atom and the oscillating field is maximal. However, as can be seen in fig. 7.2, even for values of $|\omega| < \omega_0$ the linear relationship between polarizability and the applied field, i.e., $\alpha = \text{const.}$, which we assumed earlier in our discussion of dielectric polarization, no longer holds. Nevertheless, if we consider only static fields, i.e., $\omega = 0$, this does not matter, and we are dealing exclusively with the static polarizability $\alpha_0$, i.e., the value of $\alpha$ in fig. 7.2 at $\omega = 0$. This value is:

$$\alpha_0 = \frac{q^2}{m_e\omega_0^2} = \frac{e^2}{m_e\omega_0^2} \tag{7.37}$$

If we consider not just one atom but $n$ atoms per unit volume, then the dielectric polarization $\boldsymbol{P}$ is given by eq. (6.8), and using eq. (6.34) we obtain

$$\boldsymbol{P} = n\cdot\boldsymbol{p} = n\cdot\alpha\cdot\boldsymbol{E} = \chi\cdot\epsilon_0\cdot\boldsymbol{E} \tag{7.38}$$

from which it follows:

$$\chi = \frac{n\cdot\alpha}{\epsilon_0} \tag{7.39}$$

This result tells us that the dielectric susceptibility $\chi$ exhibits the same frequency dependence as the atomic polarizability $\alpha$. For ideal gases, the interaction between dipoles within the volume under consideration plays no role, and consequently, eq. (7.39) describes their behavior very well. In denser media, however, the interaction between the dipoles must be taken into account. A special case of a polarizable gas is the case where $\omega_0 = 0$. To understand what this implies, we need to remind ourselves that the resonant frequency always reflects the strength of the coupling within the oscillating system, such as in the harmonic oscillation of a mass $m$ suspended from a spring, where $\omega_0 = \sqrt{k/m}$. Thus, $\omega_0 = 0$ means that the equivalent of the spring constant $k$ in a system is zero. In our case, this implies that the electron is no longer bound to its ion, and consequently, no restoring force exists. Electrons and ions therefore respond independently to the external field; we are no longer dealing with a simple gas, but rather with a plasma for which the following applies:

$$\chi(\omega) = \frac{n\cdot\alpha(\omega_0=0,\omega)}{\epsilon_0} = -\frac{ne^2}{\epsilon_0 m_e\omega^2} = -\frac{\omega_p^2}{\omega^2} \qquad \text{with} \qquad \omega_p^2 = \frac{ne^2}{\epsilon_0 m_e} \tag{7.40}$$

Here, $\omega_p$ is the so-called plasma frequency, and using this, we obtain the following for the dielectric permittivity $\epsilon$ of a plasma via eq. (6.36):

$$\epsilon(\omega) = \epsilon_0\epsilon_r(\omega) = \epsilon_0[1+\chi(\omega)] = \epsilon_0\left[1-\frac{\omega_p^2}{\omega^2}\right] \tag{7.41}$$

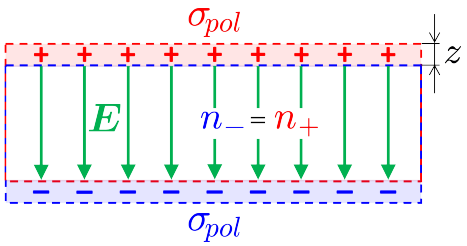



**Fig. 7.3**

At $\omega = \omega_p$ we have $\epsilon_r(\omega) = 0$ and therefore $\epsilon(\omega) = 0$. In this case, $\chi = -1$ must hold, and eq. (7.38) reduces to $\boldsymbol{P} = -\epsilon_0 \boldsymbol{E}$. Alongside matter in the form of gaseous, liquid, or solid bodies, the plasma state is commonly referred to as a fourth state of matter. In the case of a plasma, however, one is dealing not with $n$ polarizable particles per unit volume, but rather with $n = n_-$ negatively charged particles, the electron gas, and $n = n_+$ positively charged particles, the ion gas. If the lighter electrons in the electron gas and the heavier ions in the ion gas, are shifted with respect to each other, positive and negative surface charges $\sigma_{pol} = n \cdot e \cdot z$ arise. As illustrated in fig. 7.3 for a small displacement $z$, this results in the generation of an electric field that opposes the displacement. Fig. 7.3 thus presents a situation analogous to that of a parallel-plate capacitor; using eq. (4.59), the electric field is consequently given by:

$$E_z = \frac{\sigma_{pol}}{\epsilon_0} = \frac{n \cdot e}{\epsilon_0} \cdot z \tag{7.42}$$

With the restoring force $-q \cdot E_z$, the equation of motion describing the plasma oscillation is then:

$$m_e \ddot{z} = -\frac{n \cdot e^2}{\epsilon_0} \cdot z \qquad \text{or} \qquad \ddot{z} + \omega_p^2 = 0 \tag{7.43}$$

This is the equation for a simple harmonic oscillation of the electron gas with the resonance frequency $\omega_0 = \omega_p$, as we already know it from the section on oscillatory motion in volume 1. This resonance frequency depends directly on the electron density, which can vary over many orders of magnitude. Plasma oscillations are of great practical importance because the upper layers of the Earth's atmosphere, starting at an altitude of approximately 60 km above the surface, are ionized by high-energy solar radiation. This ionosphere is essentially a plasma whose density varies with altitude. This has consequences for communication signals on Earth and for transmitting communication signals between Earth and satellites. Plasmas are transparent only to radio signals with frequencies above the plasma frequency. Signals with frequencies below this threshold are reflected back by the ionosphere. The latter explains why it is possible to communicate via shortwave radio ($3 - 30$ MHz) with stations located far beyond the horizon. For communication between ground stations and satellites, which are, after all, located above the ionosphere, signals in the microwave range (4 GHz and higher) are required. We have already spoken several times about the freely moving electron gas in metals, i.e, the conduction electrons. In fact, quantized collective oscillations, referred to as so-called

plasmons, of this electrons gas in a metal relative to the fixed metal ions can be excited using electrons or light. For most of us, the existence of plasma states are also indirectly important for another reason. One modern-day technical application of plasmas is the so-called plasma etching process used in semiconductor manufacturing, which enables the nanostructuring of surfaces with very high precision. Without this technology, there would be no highly integrated electrical circuits with critical dimensions in the nanometer range, in which more than a billion transistors are accommodated on a component the size of a fingernail.

Just as there exist liquids that can conduct electric current via dissolved ions, such as for example non-deionized water, there exist also insulating liquids, i.e., liquids that do not conduct electric currents. These dielectric liquids have long been of great technical importance in applications such as the insulation and cooling of high-performance transformers. More recently, however, they also serve as cooling fluids in data centers, for instance, in so-called immersion cooling. In this application of dielectric liquids, computers are completely submerged in a dielectric fluid to ensure optimal heat transfer. While deionized water is the simplest dielectric fluid, mineral oils are typically used in high-performance applications. Dielectric liquids are characterized by high dielectric strength, i.e., a high resistance to dielectric breakdown. And unlike solid dielectrics, they are self-healing in the event of an electrical breakdown. The properties of dielectric liquids are strongly dependent on their purity and gases or impurities dissolved in them will impact their behavior. Such impurities can arise, for example, when an electrical breakdown occurs. Thus, even though dielectric liquids are self-healing, electrical breakdowns alter their properties.

### 7.2.2 Orientational Polarization

Unlike atoms, molecules can possess a permanent dipole moment. A well-known example of this is the water molecule sketched in fig. 7.4. The bond angle between the hydrogen atoms bonded to the oxygen atom is 104.45°. An oxygen atom is more electronegative than a hydrogen atom. Consequently, the oxygen atom in the water molecule draws the bonding electrons away from the hydrogen atoms and closer to itself. The result is the accumulation of a net negative electric charge $-\delta q$ on the side of the oxygen atom and a net positive electric charge $+\delta q$ on the side of the hydrogen atoms. The center of electric charge of the positive excess charge $+\delta q$ is located at the midpoint of the line connecting the hydrogen

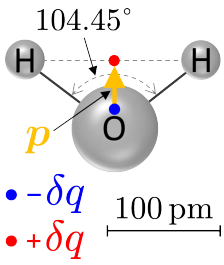



**Fig. 7.4**

atoms, and the center of electric charge of the negative excess charge $-\delta q$ lies within the oxygen atom on the line connecting the oxygen nucleus and the center of electric charge of the positive excess charge. Permanent molecular dipoles are generally much stronger than the induced atomic or molecular dipoles which we have discussed so far. In the dipole-dipole interaction between neighboring polar molecules, the positive end of one dipole attracts the negative end of another. This can lead to a preferred orientation of the dipoles relative to each other. However, this dipole-dipole interaction is counteracted by the thermal energy of the molecules, which promotes a random orientation of the dipoles. If, on the other hand, an external electric field is applied, the dipoles will align themselves along the field lines. We have already examined the behavior of a dipole in an electric field in chapter 3. The three equations - eq. (3.64), eq. (3.69), and eq. (3.70) - that describe this behavior are:

$$\boldsymbol{M} = \boldsymbol{p} \times \boldsymbol{E} \quad , \quad U = -\boldsymbol{p} \cdot \boldsymbol{E} \quad ; \quad \boldsymbol{F} = (\boldsymbol{p} \cdot \nabla)\boldsymbol{E} \tag{7.44}$$

The first equation gives the torque $\boldsymbol{M}$ acting on a dipole in a homogeneous field $\boldsymbol{E}$ when its dipole moment $\boldsymbol{p}$ does not yet point in the same direction as the electric field $\boldsymbol{E}$. In a homogeneous electric field, the dipole aligns itself with the direction of the field, but no further forces act upon it. The situation is different in an inhomogeneous field: there, the dipole is subject not only to a torque but also to a force. The third equation describes this force $\boldsymbol{F}$, which pulls the dipole in the direction of increasing electric field strength. And the second equation gives us the potential energy $U$ of a dipole in an electric field, whether homogeneous or inhomogeneous, which is greatest when $\boldsymbol{p}$ and $\boldsymbol{E}$ are antiparallel and smallest when $\boldsymbol{p}$ is parallel to $\boldsymbol{E}$.

Before considering a gas or liquid composed of polar molecules in an electric field, we will quickly estimate the order of magnitude of the potential energy of a molecular dipole. The dipole moment is the product of the charge and the distance between the positive and negative ends of the dipole. If we assume a typical distance of 1 Å or less and the elementary charge $e$ as the typical charge, then a realistic estimate for the order of magnitude of a molecular dipole moment is $|\boldsymbol{p}| \approx e \cdot 10^{-10}\,\mathrm{C \cdot m} = 1.6 \cdot 10^{-29}\,\mathrm{C \cdot m}$. Tab. 7.1 shows, for several small molecules, that this is indeed the order of magnitude characteristic of the dipole moments of polar molecules.

**Tab. 7.1:** Dipole moments of small polar molecules.

| **Molecule** | CO | HCL | NaCl | CsCl | $SO_2$ | $H_2O$ | $NH_3$ | $CH_3CL$ |
|---|---|---|---|---|---|---|---|---|
| $\boldsymbol{\lvert p \rvert}$ **in** $\mathbf{10^{-30}}$ **[C·m]** | 0.37 | 3.60 | 28.35 | 34.69 | 5.40 | 6.17 | 4.90 | 6.24 |

The permanent dipole moment of a molecule is particularly large in the case of an ionic bond, where one atom completely transfers an electron to another, as is the case with NaCl, for example. The sodium atom has the electron configuration of neon, i.e., it possesses a filled second electron shell, plus an additional single electron in the outermost shell. The chlorine atom, on the other hand, can achieve the electron configuration of Ar by gaining an additional electron. In the ionic NaCl bond, Na transfers its outer single electron to Cl and thus both atoms achieve their goal and acquire the character of a noble gas. Let us now consider the magnitude of the potential energy of a polar molecule using our estimate for the dipole moment. In a homogeneous field with an applied voltage of 1 kV and an electrode separation of 1 m, the electric field strength is $1\mathrm{kV}\cdot\mathrm{m}^{-1}$. If this dipole is oriented parallel to the electric field, it has a potential energy of $U = -10^{-7}\mathrm{eV} = -1.602\cdot 10^{-26}\mathrm{J}$. At room temperature, the thermal energy of a gas particle is around $25\,\mathrm{meV} = 4\cdot 10^{-21}\,\mathrm{J}$, i.e., the thermal energy is much greater than the potential energy of a permanent molecular dipole. High thermal energy implies frequent collisions between gas particles, and the energy exchanged during these collisions is significantly greater than the potential energy of the dipole. Consequently, at room temperature, the orientation of the dipoles in a gas of polar molecules is random.

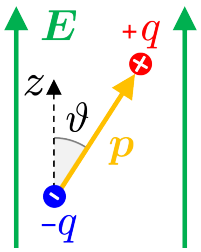


**Fig. 7.5**

Let us now consider the dipole of a polar molecule placed in a homogeneous field, as shown in fig. 7.5. We orient the coordinate system such that the homogeneous field $\boldsymbol{E}$ is parallel to the $z$-axis. Consequently, the angle $\vartheta$ between $\boldsymbol{p}$ and $\boldsymbol{E}$ corresponds to the integration angle $\vartheta$ in spherical coordinates $\boldsymbol{r} = \boldsymbol{r}(r,\vartheta,\varphi)$. According to eq. (7.44), the potential energy of this dipole is given by $U(\vartheta) = -p\,E\cos\vartheta$. In our gas of polar molecules, all molecules have the same dipole moment magnitude $p$ and in the absence of an external force, the corresponding dipole moments point in various directions. In the section on thermal motion in volume 1 of this physics course, we learned that in thermal equilibrium, the number of molecules in a specific internal energy state $E_i$ of the molecule at a given temperature is determined by the factor $\exp(-E_i/k_BT)$. This consideration regarding the internal energy state of a molecule also applies to the potential energy of the dipole associated with a polar molecule. Therefore, the number of polar molecules $n(\vartheta)$ in a solid angle element $\mathrm{d}\omega = \sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$ with dipole energy $U(\vartheta)$ is given by:

$$n(\vartheta) \;=\; n_0\cdot\exp\Big(-\frac{U(\vartheta)}{k_BT}\Big) \;=\; n_0\cdot\exp\Big(\frac{p\,E\cos\vartheta}{k_BT}\Big) \tag{7.45}$$

We still need to determine the proportionality constant $n_0$ we introduced with this equation. We have just established that the thermal energy $k_BT$ at room temperature is much

greater than the potential energy $U$ of the dipole. Therefore, to simplify matters, we can use the Taylor expansion of the exponential function. If we consider only linear terms in the expansion, eq. (7.45) becomes:

$$n(\vartheta) \;=\; n_0 \cdot \left(1 + \frac{p\,E\cos\vartheta}{k_B T}\right) \tag{7.46}$$

To determine $n_0$, we require that the quantity $n(\vartheta)$, when integrated over all angles, must equal the density $n = N/V$ of the polar molecules per unit volume, where $N$ is the total number of molecules in that volume $V$. Thus:

$$n \;=\; \int_0^{2\pi}\!\!\int_0^{\pi} n(\vartheta)\sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \;=\; n_0 \cdot \int_0^{2\pi}\!\!\int_0^{\pi} \left(1 + \frac{p\,E\cos\vartheta}{k_B T}\right)\sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \tag{7.47}$$

The integral of the first term in the parentheses yields the value $4\pi$, while the integral of the second term in the parenthesis vanishes. Hence, preforming the integration gives us the simple expression $n_0 = n/4\pi$ for the relationship between $n_0$ and $n$. Since $0 \leq \vartheta \leq \pi$, $\cos\vartheta$ varies between 1 and $-1$. Eq. (7.45) tells us that there must be far more molecules with the $z$-component of the dipole vector $\boldsymbol{p}$ pointing in the direction of the electric field $\boldsymbol{E}$ than those with it pointing in the opposite direction. The electric field $\boldsymbol{E}$ thus undoubtedly induces a net polarization $\boldsymbol{P}_\parallel$ in the $z$-direction, which we obtain by summing the components $p\cos\vartheta$ of all molecules. Accordingly, the summation over the components $p\sin\vartheta$ of all molecules should yield the magnitude of $\boldsymbol{P}_\perp$, the polarization normal to $\boldsymbol{E}$. Using eq. (6.8), we obtain $\boldsymbol{P}_\parallel$ and $\boldsymbol{P}_\perp$ by integrating the distribution $n(\vartheta)$, weighted by the normal and parallel components of $\boldsymbol{p}$, respectively, over the solid angle $\mathrm{d}\omega = \sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$, that is:

$$\boldsymbol{P}_\parallel \;=\; \hat{\boldsymbol{k}} \cdot \int_0^{2\pi}\!\!\int_0^{\pi} n(\vartheta) p\cos\vartheta\sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \tag{7.48}$$

and

$$\boldsymbol{P}_\perp \;=\; \int_0^{2\pi}\!\!\int_0^{\pi} n(\vartheta) p\sin\vartheta \begin{pmatrix}\cos\varphi\\ \sin\varphi\end{pmatrix}\sin\vartheta\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \tag{7.49}$$

Let us first consider the normal component of $\boldsymbol{P}$ in eq. (7.49). As one can quickly verify, integration of $\cos\varphi$ and of $\sin\varphi$ over the interval $0 \leq \varphi \leq 2\pi$ yields a value of zero. The $x$- and $y$-components of $\boldsymbol{P}$ thus average to zero independently, and consequently $\boldsymbol{P}_\perp = 0$. Hence, the dielectric polarization possesses only a component in the $z$-direction, and therefore $\boldsymbol{P} = \boldsymbol{P}_\parallel$. Substituting eq. (7.46) with $n_0 = 4\pi/n$ into eq. (7.48), we thus obtain the following expression for the dielectric polarization $\boldsymbol{P}$ of our gas of polar molecules:

$$\boldsymbol{P} = \hat{\boldsymbol{k}} \cdot \frac{n}{4\pi} \int_0^{2\pi} \int_0^{\pi} \left(1 + \frac{p\,E\cos\vartheta}{k_B T}\right) p \cos\vartheta \sin\vartheta \,\mathrm{d}\vartheta \,\mathrm{d}\varphi \tag{7.50}$$

Integration with respect to $\varphi$ yields a factor of $2\pi$ and the integration with respect to $\vartheta$ of the first term inside the parentheses, i.e., of $\cos\vartheta \sin\vartheta$, vanishes. The integration with respect to $\vartheta$ of the second term, i.e., $\cos^2\vartheta \sin\vartheta$ is easily solved using the substitution $u = \cos\vartheta$ and $\mathrm{d}u = -\sin\vartheta\,\mathrm{d}\vartheta$, thereby yielding:

$$\boldsymbol{P} = \hat{\boldsymbol{k}} \cdot \frac{n}{2} \frac{p^2 E}{k_B T} \int_{-1}^{1} u^2 \,\mathrm{d}u = \frac{n\,p^2 E}{3 k_B T} \cdot \hat{\boldsymbol{k}} = \frac{n\,p^2}{3 k_B T} \cdot \boldsymbol{E} \tag{7.51}$$

The dielectric polarization $\boldsymbol{P}$ of our gas of polar molecules is directly proportional to the applied field $\boldsymbol{E}$ and inversely proportional to the temperature $T$ of the gas. The higher the temperature, the more collisions occur between the gas molecules. These collisions between the gas molecules destroy the molecular dipole alignment induced by the electric field, and consequently, the dielectric polarization decreases. Using eq. (6.34), the dielectric susceptibility $\chi$ of our gas of polar molecules is:

$$\chi = \frac{\boldsymbol{P}}{\epsilon_0 \boldsymbol{E}} = \frac{n\,p^2}{3 k_B T \epsilon_0} \tag{7.52}$$

In the case of polar molecules, one speaks of paraelectric susceptibility, which decreases with increasing temperature ($T$). This $1/T$ behavior of the susceptibility, also referred to in this context as Curie behavior, is characteristic of gases composed of polar molecules. Gases composed of non-polar molecules do not exhibit this temperature dependence. Fig. 7.6 shows the behavior of the molar dielectric susceptibility as a function of inverse temperature, using the examples of a gas composed of polar molecules and one composed of non-polar molecules. For the gas consisting of polar molecules, we used our earlier estimate of the magnitude of the molecular dipole moment. For the gas consisting of non-polar molecules, no temperature dependence is observed, and the measurement curve runs horizontally. For a gas of polar molecules, the dielectric susceptibility increases linearly with the reciprocal of the temperature, with the steepness of the rise depending on the dipole moment: the larger the dipole moment, the steeper the slope of the measurement curve. In addition to the contribution made by the

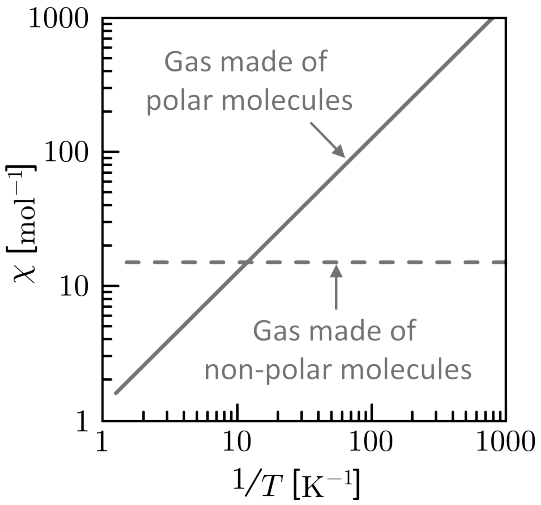



**Fig. 7.6**

permanent dipoles of the polar molecules to the dielectric susceptibility, there remains the much smaller contribution arising from the electric polarizability of atoms, a factor also present in non-polar molecules. In the previous section, we saw that the contribution of electric polarizability to the dielectric susceptibility is frequency-dependent, and thus, naturally, the dielectric susceptibility itself is as well. The same applies to the contribution of the permanent dipoles of polar molecules to the dielectric susceptibility. The only difference is that, in the case of electric polarizability, it is the electrons that can no longer keep up with a rapidly changing electric field, whereas with polar molecules, it is the molecules themselves that cannot reorient quickly enough to align their permanent dipole moment with the changing applied field. In the case of polar molecules, the reorientation of dipole moments is brought about by the oscillating torque exerted on the molecules by the applied electric field. Molecules are far heavier than electrons, and consequently, they are unable to keep pace with a rapidly changing electric field even at much lower frequencies where electrons still have no problem following the field oscillation. Thus, above this cutoff frequency, only the contribution of electric polarizability to the dielectric susceptibility remains, even for gases composed of polar molecules, until this contribution, too, vanishes at even higher frequencies, when even the electrons can no longer keep pace with the oscillating electric field.

### 7.2.3 Dielectric Behavior of Liquids

Up to this point, our analysis has only gone as far as determining how the dielectric polarization $\boldsymbol{P}$ of a medium depends on the applied electric field $\boldsymbol{E}$ via the relation $\boldsymbol{P} = \chi\epsilon_0\boldsymbol{E}$. We must now pay closer attention to the fact that $\boldsymbol{E}$ and $\boldsymbol{P}$ are averaged macroscopic quantities. Gas particles are constantly in motion and are subject, on average, to precisely these quantities. Unlike gases, liquids possess short-range order but lack the long-range order found in solids, where atoms and molecules occupy fixed positions and can only move about their respective equilibrium position. In the case of liquids, we cannot ignore how the local environment influences the electric field acting on individual atoms or molecules within the liquid. Let us consider, for example, a parallel-plate capacitor filled with a liquid dielectric. Eq. (6.10) tells us that the macroscopically homogeneous electric field within the dielectric is given by $E = E_0 + P/\epsilon_0$, where $E_0$ is the electric field strength in the parallel-plate capacitor without the dielectric material. On the atomic length scale, however, the electric field within the dielectric is anything but homogeneous. This becomes immediately apparent when one considers that atoms and molecules consist of positively charged atomic nuclei and negatively charged electrons.

The local electric field varies significantly across the diameter of an atom or molecule, particularly near the nearly point-like atomic nuclei, where the field gradient is especially steep. The electric field is also not constant between neighboring atoms or molecules, though it varies far less than it does within an atom or molecule itself. The question we need to ask is simply: What, then, is the average field acting on any atom or molecule within our dielectric liquid inside the parallel-plate capacitor?

To answer this question, we will examine the average electric field that exists within a small cavity in our dielectric. We can do that, for example, as illustrated in fig. 7.7, in a slot running parallel to the external field $\boldsymbol{E}$ which generates the polarization charges in the first place, and in a slot oriented normal to that field direction. The polarization vector $\boldsymbol{P}$ characterizing the polarization generated in the dielectric lies parallel to the vector $\boldsymbol{E}$ of the electric field producing it. The slot running parallel to the field $\boldsymbol{E}$ shall be sufficiently narrow that any polarization charges, the result of producing the slot, at its upper and lower ends are negligible and play no role. Let us first consider this cavity, whose slot runs parallel to the field $\boldsymbol{E}$ in the dielectric. According to the second Maxwell equation of electrostatics, $\nabla \times \boldsymbol{E} = 0$, the line integral of $\boldsymbol{E}$ along the closed curve $C_1$ bounding $A_1$ must vanish. The segments of $C_1$ that are perpendicular to the field in the dielectric contribute nothing to the line integral, and the segments $\boldsymbol{l}$ inside and outside the cavity that are parallel to the electric field in the dielectric are traversed in opposite directions. It follows that:

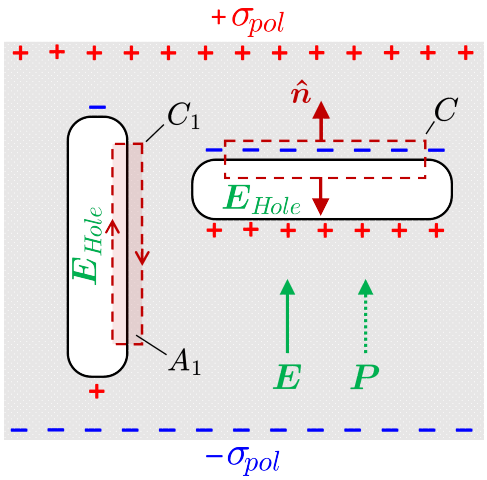



**Fig. 7.7**

$$\int_{A_1} \big[\nabla \times \boldsymbol{E}\big]\, \mathrm{d}\boldsymbol{\sigma} \;=\; \int_{C_1} \boldsymbol{E}\, \mathrm{d}\boldsymbol{s} \;=\; \big(\boldsymbol{E}_{Hole} - \boldsymbol{E}\big)\cdot \boldsymbol{l} \;=\; 0 \tag{7.53}$$

and thus, it must hold that:

$$\boldsymbol{E}_{\boldsymbol{Hole}} \;=\; \boldsymbol{E} \tag{7.54}$$

Let us now consider the cavity whose slot runs perpendicular to the field in the dielectric. The surface charge density on the long sides of this slot corresponds to that of the polarization charges at the dielectric interfaces, albeit with opposite polarity. In this case, the curve $C$ does not represent a path, but rather the rectangular-shaped cross-section of a Gaussian pillbox with height $h$ and base and top areas $A$. This Gaussian pillbox encloses the surface charge $-\sigma_{pol}$ at the top of the slot, and the normal vectors of

the top and bottom surfaces of the pillbox are $\hat{\boldsymbol{n}}$ and $-\hat{\boldsymbol{n}}$, respectively. Considering the boundary $S$ of the volume $V = A \cdot h$, Gauss's law of electrostatics (eq. (3.15)), together with $\sigma_{pol} = P$ from eq. (6.7), yields:

$$\int_S \boldsymbol{E} \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\sigma \;=\; (\boldsymbol{E} - \boldsymbol{E}_{Hole}) \cdot \hat{\boldsymbol{n}} \cdot A \;=\; -\frac{\sigma_{pol}}{\epsilon_0} \cdot A \;=\; -\frac{\boldsymbol{P} \cdot \hat{\boldsymbol{n}}}{\epsilon_0} \cdot A \tag{7.55}$$

The field $\boldsymbol{E}_{Hole}$ in the cavity, which runs perpendicular to $\boldsymbol{E}$ in the dielectric, is thus:

$$\boldsymbol{E}_{Hole} \;=\; \boldsymbol{E} + \frac{\boldsymbol{P}}{\epsilon_0} \;=\; \frac{\boldsymbol{D}}{\epsilon_0} \tag{7.56}$$

We are already encountered this equation in the previous chapter in the form of eq. (6.10) and the situation considered there is analogous to the one here. There, we had considered a dielectric inserted into the vacuum of a parallel-plate capacitor. Here, the cavities shown in fig. 7.7 are regions of vacuum «inserted» into a dielectric that, like a parallel-plate capacitor, bears surface charges, though these are polarization charges. The electric field existing within such a cavity obviously depends on the cavity's geometry. If the cavity is a long vertical slot oriented parallel to the field in the dielectric, the field inside it equals the field $\boldsymbol{E}$ in the dielectric. However, if the cavity is a horizontal transverse slot running perpendicular to the field in the dielectric, the field inside it equals $\boldsymbol{D}/\epsilon_0$. To determine the effective electric field acting on an atom or molecule in a liquid, we must make an assumption regarding the type of cavity in which it is located within the dielectric material. The constituents of a liquid lack long-range order but possess short-range order, and liquids are isotropic. We can therefore assume that an atom or a molecule in a dielectric liquid resides in a sort of spherical cavity. The question we must ask next is therefore: How do we determine the effective field acting on an atom or molecule within such a spherical cavity?

To this end, we consider the situation sketched in fig. 7.8. A dielectric is placed in a homogeneous external electric field $\boldsymbol{E}$, which causes the formation of polarization charges at the interfaces of the dielectric. As we already know, the resulting polarization $\boldsymbol{P}$ must be parallel to the external field $\boldsymbol{E}$. Within this dielectric material, we now consider a spherical volume, represented by the dashed circle in fig. 7.8. Now, let us imagine that we somehow can freeze the polarization within this spherical volume and, so to speak, can remove it, along with the spherical volume itself, from the dielectric material. This leaves behind a spherical hole with its associated polarization charges which is precisely the cavity for which we wish to determine the field $\boldsymbol{E}_{Hole}$. The superposition of the field of the remaining hole, $\boldsymbol{E}_{Hole}$, and the field $\boldsymbol{E}_{Sphere}$ of the cut-out sphere, which arises from the frozen-in polarization, must then yield exactly the field that existed in the dielectric before the sphere was cut out. Thus, the following holds:

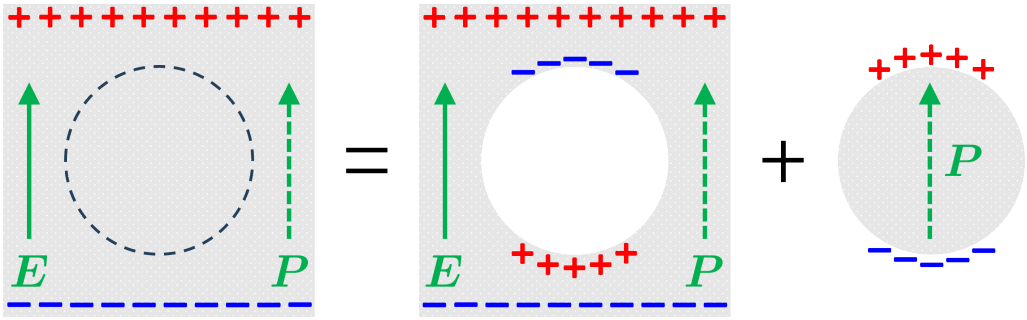



**Fig. 7.8:** The external field $\boldsymbol{E}$ generates polarization charges at the interfaces of the dielectric. This gives rise to a polarization $\boldsymbol{P}$ within the dielectric. The effective electric field within a spherical volume of the dielectric is the superposition of the field of a hollow sphere in the dielectric and the field of a polarized sphere in a vacuum.

$$\boldsymbol{E} = \boldsymbol{E}_{Hole} + \boldsymbol{E}_{Sphere} \tag{7.57}$$

However, we already know the field $\boldsymbol{E}_{Sphere}$ from our discussion of the polarized sphere in section 6.2.3. There we established that

$$\boldsymbol{E}_{Sphere} = -\frac{\boldsymbol{P}}{3\epsilon_0} \tag{7.58}$$

holds. Thus, eq. (7.57) becomes:

$$\boldsymbol{E} = \boldsymbol{E}_{Hole} - \frac{\boldsymbol{P}}{3\epsilon_0} \tag{7.59}$$

The electric field in a spherical cavity inside a dielectric material in which a field $\boldsymbol{E}$ exists is thus given by:

$$\boldsymbol{E}_{Hole} = \boldsymbol{E} + \frac{\boldsymbol{P}}{3\epsilon_0} \tag{7.60}$$

The average electric field acting on an atom or molecule in a dielectric liquid is thus greater than the macroscopic field $\boldsymbol{E}$ in the dielectric by $\boldsymbol{P}/3\epsilon_0$. For the dielectric polarization $\boldsymbol{P}$ and the effective field acting on each atom or molecule in the dielectric liquid, it then follows from eq. (6.4) and eq. (6.8) that:

$$\boldsymbol{P} = n \cdot \boldsymbol{p} = n \cdot \alpha \cdot \boldsymbol{E}_{Hole} = n \cdot \alpha \cdot \left( \boldsymbol{E} + \frac{\boldsymbol{P}}{3\epsilon_0} \right) \tag{7.61}$$

It follows that

$$\boldsymbol{P} = \frac{n \cdot \alpha / \epsilon_0}{1 - \left( n \cdot \alpha / 3\epsilon_0 \right)} \cdot \epsilon_0 \cdot \boldsymbol{E} \tag{7.62}$$

Comparing this expression with eq. (6.34), we see that a non-polar liquid possesses a dielectric susceptibility

$$\chi = \frac{n \cdot \alpha / \epsilon_0}{1 - \left(n \cdot \alpha / 3\epsilon_0\right)} \tag{7.63}$$

and thus the dielectric permittivity

$$\epsilon = \left[1 + \frac{n \cdot \alpha / \epsilon_0}{1 - \left(n \cdot \alpha / 3\epsilon_0\right)}\right] \cdot \epsilon_0 \tag{7.64}$$

Eq. (7.63) is the so-called Clausius-Mossotti equation, which relates the macroscopic quantity $\chi$, and therefore also $\epsilon$, to the microscopic quantity $\alpha$, of the atomic polarizability. For sufficiently low particle densities $n$, such as in dilute gases, we can neglect the factor $n\alpha/3\epsilon_0$ in the denominator and thereby obtain the well-known result $\chi = n\alpha/\epsilon_0$ from eq. (7.39). Solving eq. (7.64) for $n\alpha/3\epsilon_0$ and substituting $\chi = \epsilon_r - 1$ from eq. (6.36), we obtain the equation

$$\frac{n \cdot \alpha}{3\epsilon_0} = \frac{\epsilon_r - 1}{\epsilon_r + 2} \tag{7.65}$$

If we set $n = N_A/V_m$, where $N_A$ is Avogadro's number and $V_m$ is the molar volume, we obtain the version of the Clausius-Mossotti relation useful for gases in the form of an equation for the molar polarization $P_m$:

$$P_m = \frac{N_A \alpha}{3\epsilon_0} = \frac{\epsilon_r - 1}{\epsilon_r + 2} \cdot V_m \tag{7.66}$$

The Clausius-Mossotti equation does not account for the existence of permanent dipole moments but considers exclusively the polarizability of atoms and the associated induced dipole moments. Eq. (7.63) provides a good description of the dielectric behavior of weakly interacting dielectrics materials, which typically include non-polar gases and dilute non-polar liquids. For dielectric materials with strong intermolecular interactions, such as those found in polar liquids like water, the Clausius-Mossotti equation fails. However, we can remedy this shortcoming of the Clausius-Mossotti equation by also taking into account the polarizability due to permanent dipoles. We have already determined the latter with eq. (7.51) and we only need to rewrite this equation as

$$\boldsymbol{P} = \frac{n \cdot p^2}{3k_B T} \cdot \boldsymbol{E} = n \cdot \alpha_{op} \cdot \boldsymbol{E} \tag{7.67}$$

where $p$ is the permanent dipole moment of the molecule under consideration and

$$\alpha_{op} = \frac{p^2}{3k_B T} \tag{7.68}$$

is the polarizability due to orientation polarization. If we then consider the polarizability $\alpha$ appearing in eq. (7.66) as the sum of the atomic polarizability $\alpha_{atom}$ and the orientation polarizability $\alpha_{op}$, that is

$$\alpha = \alpha_{atom} + \alpha_{op} = \alpha_{atom} + \frac{p^2}{3k_B T} \tag{7.69}$$

we obtain a modified form of the Clausius-Mossotti equation for the molar polarization $P_m$, the so-called Debye equation:

$$P_m = \frac{N_A}{3\epsilon_0}\left[\alpha_{atom} + \frac{p^2}{3k_B T}\right] = \frac{\epsilon_r - 1}{\epsilon_r + 2} \cdot V_m \tag{7.70}$$

If one plots this molar polarization $P_m$ as a function of temperature against $1/T$ for polar or non-polar gaseous or liquid dielectrics, one obtains a graph just like the one in fig. 7.6, featuring straight lines whose slopes are determined by the orientation polarizability and whose intercepts on the $P_m$-axis are determined by the atomic polarizability.

### 7.2.4 Dielectric Behavior of Solids

Examining the interaction of electrostatic fields with polar and non-polar gas and liquid particles at the microscopic level has provided us with a description of the dielectric and paraelectric behavior of gaseous and liquid dielectrics at the macroscopic level. Some of what we learned there can also be applied to solids. However, alongside dielectric and paraelectric behavior, solids also exhibit a third form of interaction between matter and electric fields, i.e., the so-called ferroelectric or antiferroelectric behavior. Accordingly, a distinction is made between three types of solids, depending on how a dielectric solid behaves in an external field.

- Solid-state dielectrics: Polarization arises from the atomic polarizability of the atoms within a solid when subjected to an external electric field, specifically, from the displacement of the atoms' negative electron clouds relative to their positive atomic nuclei. In this case, one speaks of so-called electronic polarization, thereby distinguishing it from so-called ionic polarization, which occurs in ionic crystals such as NaCl. The latter involves a displacement of the positive and negative ions relative to each other.

- Paraelectric solids: These solids consist of molecules possessing a permanent dipole moment. However, in the absence of an external field, these dipole moments within the solid are randomly oriented and therefore cancel each other out when averaged over space or time. The situation changes when such a solid is brought into an an external field because the dipole moments align with the electric field, resulting in orientational polarization. Just as we found for gases and liquids in eq. (7.51), this polarization of the solid decreases with the inverse of the temperature.

- Ferroelectric and antiferroelectric solids: These solids possess a polarization below a critical temperature even in the absence of an external field. This so-called spontaneous polarization $P_S$ vanishes when the temperature of the solid reaches the critical temperature $T_c$.

When we speak here of external fields, this refers both to external fields in the sense of electric fields into which a solid is placed, and to internal local electric fields acting upon the atoms and molecules of the solid. This is an important difference compared to gases and liquids, as these local fields do not exist in them because gas and liquid particles are constantly in motion. However, in solids, where atoms and molecules occupy fixed positions, leading to their designation as lattice atoms or lattice molecules, these local fields do indeed exist. For solids, we must therefore add an additional term, $\boldsymbol{E}_{\text{local}}$, to eq. (7.60):

$$\boldsymbol{E}_{Hole} = \boldsymbol{E} + \frac{\boldsymbol{P}}{3\epsilon_0} + \boldsymbol{E}_{local} \tag{7.71}$$

In this modified equation for the field inside the spherical macroscopic cavity, which we have already considered in the case of a liquid dielectric material, $\boldsymbol{E}$ is again the external field into which we introduce now a solid dielectric, $\boldsymbol{P}/3\epsilon_0$ is the mean field in the cavity that the dipoles generate, and $\boldsymbol{E}_{local}$ is the electric field generated by the nearest-neighbor environment of a lattice atom or lattice molecule. The first two terms in eq. (7.71) are thus macroscopically averaged quantities, whereas $\boldsymbol{E}_{local}$ is a microscopic, rapidly varying quantity. To determine it, we must sum the contributions from the dipole fields of the $n$ nearest-neighbor atoms of a lattice atom or molecule. Using eq. (3.55), we can therefore write for $\boldsymbol{E}_{local}$:

$$\boldsymbol{E}_{local} = \frac{1}{4\pi\epsilon_0} \sum_n \frac{3(\boldsymbol{p}\cdot\boldsymbol{r_n})\cdot\boldsymbol{r_n} - \boldsymbol{p}\cdot r_n^2}{r_n^5} \tag{7.72}$$

Here, we have placed the atom or molecule, at whose lattice site we wish to determine the local electric field, at the origin of the coordinate system, and the $\boldsymbol{r}_n$ are the vectors representing the distances from this origin to the neighboring lattice sites of the atom or molecule of interest. We are assuming a monoatomic or monomolecular solid. Thus, all dipole moments within the considered volume of the solid, over which the sum in eq. (7.72) extends, have the same magnitude; furthermore, they all shall point in the same direction. To calculate the

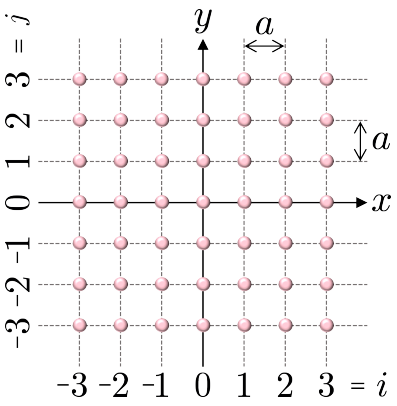



**Fig. 7.9**

strength of the field $\boldsymbol{E}_{local}$ in this case, we must make an assumption regarding the arrangement of the atoms or molecules in the solid under consideration. For this purpose, we choose a so-called simple cubic lattice structure, i.e., an arrangement in which the lattice constituents of the solid are located at the corners of a cube with edge length $a$. The crystal looks the same from every lattice site in the simple cubic lattice structure. A cross-section in the $xy$-plane therefore yields a distribution of nearest neighbors as sketched in fig. 7.9. In Cartesian coordinates, the dipole coordinates at a lattice site in our solid are given by:

$$\boldsymbol{r}_n \;=\; \boldsymbol{r}_{ijk} \;=\; a{\cdot}(i,j,k) \tag{7.73}$$

Due to the symmetry of the lattice, it suffices to consider one component of $\boldsymbol{E}_{\text{local}}$. We choose the $x$-component, for which, after substituting eq. (7.73), we obtain:

$$\boldsymbol{E}_{x_{local}} \;=\; \frac{1}{4\pi\epsilon_0}\frac{1}{a^3}\sum_{ijk}\frac{3(p_x{\cdot}i + p_y{\cdot}j + p_z{\cdot}k){\cdot}i - p_x(i^2+j^2+k^2)}{\left(i^2+j^2+k^2\right)^{5/2}} \tag{7.74}$$

A simple cubic lattice structure possesses inversion symmetry. This means that reflection across any lattice atom acting as an inversion center maps the lattice structure onto itself. For every atom in the crystal lattice, there exists an exactly identical atom on the opposite side of the inversion center. Mathematically, this is reflected in the sum in eq. (7.74) by the fact that, as shown for $i$ and $j$ in fig. 7.9, the indices $i$, $j$, and $k$ run through the same positive and negative values. Consequently, the terms involving the mixed pairs $j{\cdot}i$ and $k{\cdot}i$ within the first set of parentheses in the numerator of the sum in eq. (7.74) cancel out. Furthermore, the squared terms in the second set of parentheses in the numerator are identical, so we can replace $i^2+j^2+k^2$ with $3i^2$ there. This reduces the equation for the $x$-component of $\boldsymbol{E}_{\text{local}}$ to:

$$\boldsymbol{E}_{x_{local}} \;=\; \frac{1}{4\pi\epsilon_0}\frac{1}{a^3}\sum_{ijk}\frac{3p_x{\cdot}i^2 - 3p_x{\cdot}i^2}{\left(3i^2\right)^{5/2}} \;=\; 0 \tag{7.75}$$

Under normal conditions, there is almost no element whose solid form exhibits a simple cubic lattice; polonium is an exception. In the case of ionic crystals such as NaCl, the individual atoms are indeed located at the corners of a cube, but they consist of two different types of atoms; consequently, the lattice structure of an NaCl crystal is not simple cubic but rather face-centered cubic, as shown in fig. 7.10. In this arrangement, $Na^+$ ions occupy the cube's corner points and the centers of its faces. The $Cl^-$ ions are located at the midpoints of the cube's edges, between the $Na^+$ ions, and one $Cl^-$ ion sits at the very center of the cube. Fig. 7.10 shows a cross-section of the NaCl crystal along the $xy$-plane. This two-dimensional structure clearly possesses inversion symmetry.

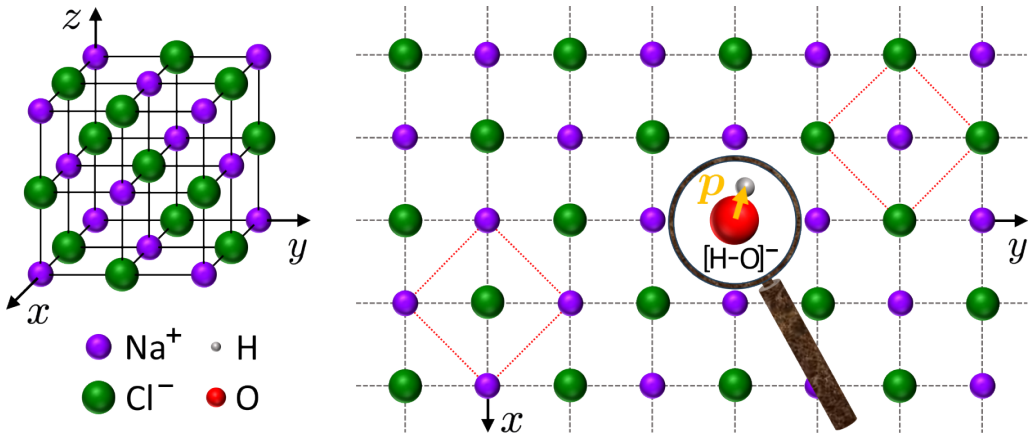



**Fig. 7.10:** The face-centered cubic crystal structure of NaCl. Very few crystals are free of impurities, and the incorporation of molecules such as an $OH^-$ ion can cause an NaCl crystal to exhibit orientational polarization as well.

Because adjacent crystal planes are shifted by one lattice spacing relative to each other, a $Cl^-$ ion sits above and below every $Na^+$ ion in the $xy$-plane, and conversely, a $Na^+$ ion sits above and below every $Cl^-$ ion in the $xy$-plane. Consequently, the NaCl crystal possesses inversion symmetry, and $\boldsymbol{E}_{\text{local}}$ vanishes in a pure NaCl crystal. A perfect crystal with inversion symmetry has no dipole moment. However, most solids are not perfect but contain impurities. A common impurity of this type in NaCl crystals is the hydroxide anion $OH^-$. The hydroxide anion possesses a permanent dipole moment, and when incorporated into a $Cl^-$ vacancy, as shown in fig. 7.10, it is free to rotate. However, even if an NaCl crystal contains many such hydroxide anions in $Cl^-$ vacancies, it still does not possess a permanent dipole moment, because the individual dipoles of the hydroxide anions are randomly oriented. That changes, however, if such an NaCl crystal containing hydroxide anions is placed in an external field as the permanent dipoles will align with the applied field. What is involved here is clearly a case of orientation polarization and due to $OH^-$ impurities the NaCl crystal acquires a dipole moment for as long as the field is applied.

However, there are also so-called polar crystals, i.e., solids that possess a dipole moment even in the absence of an applied electric field. The resulting polarization of the crystal manifests itself through the associated permanent charges that form in the crystal's surface layers. Because these permanent charges are usually neutralized very quickly by other charges present in the environment surrounding the crystal, they are quite difficult to detect. To demonstrate the existence of these surface charges, one can, however, vary the temperature of the crystal or apply pressure to it. If this variation occurs very quickly, the crystal's polarization varies very rapidly as a result and the associated change

in surface charge density can no longer be neutralized quickly enough by charges from the surrounding environment. In the case of a temperature variation, such a change in surface charge density is referred to as the pyroelectric effect, and in the case of a pressure variation, as the piezoelectric effect. A well-known example of the piezoelectric effect is the quartz crystal, which, however, exhibits no polarization in its normal state. Quartz crystals are composed of $SiO_4$ tetrahedra, as shown in fig. 7.11. The fact that the chemical formula for quartz is $SiO_2$ rather than $SiO_4$ is due to the tetrahedra sharing oxygen atoms. Consequently, there are not nominally four oxygen atoms per silicon atom, but only two. Why quartz does not possess a permanent dipole moment in its normal state becomes quickly apparent when considering the symmetry of the $SiO_4$ tetrahedron shown in fig. 7.11. The silicon atom is located at the geometric center of the tetrahedron, specifically at the point that divides the perpendicular dropped from the apex to the base in a 3:1 ratio, causing the $z$-components of the four dipole moments to cancel each other out:

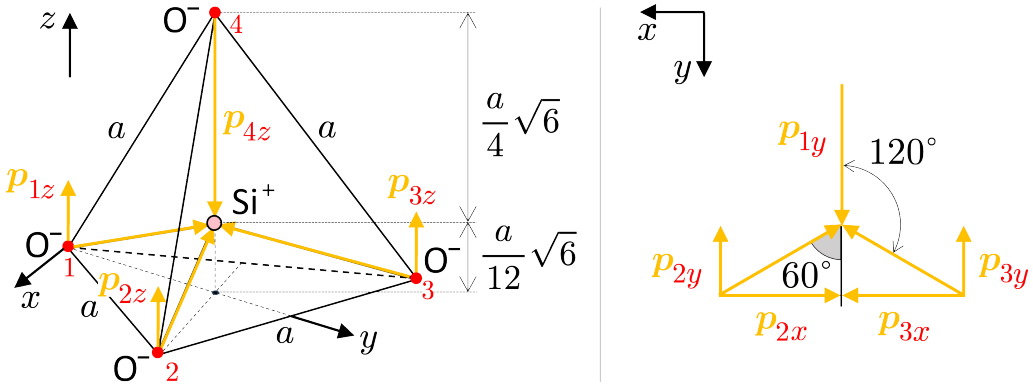



**Fig. 7.11:** In the tetrahedral structure of $SiO_4$, the four dipole moments cancel each other out in all directions, and the total dipole moment of $SiO_4$ is zero.

$$\boldsymbol{p}_{1z} + \boldsymbol{p}_{2z} + \boldsymbol{p}_{3z} = \boldsymbol{p}_{4z} \tag{7.76}$$

The dipole moments of the three oxygen atoms sitting at the corners of the base of the tetrahedron also possess $x$- and $y$-components. However, as is readily apparent from fig. 7.11, the $x$- and $y$-components of the corresponding dipole moments in the $SiO_4$ tetrahedron also cancel each other out due to the 120-degree angle between these three oxygen atoms:

$$\boldsymbol{p}_{1x} = 0 \quad , \quad \boldsymbol{p}_{2x} + \boldsymbol{p}_{3x} = 0 \qquad \text{and} \qquad \boldsymbol{p}_{4x} = 0 \tag{7.77}$$

as well as

$$\boldsymbol{p}_{1y} = \boldsymbol{p}_{2y} + \boldsymbol{p}_{3y} \qquad \text{and} \qquad \boldsymbol{p}_{4y} = 0 \tag{7.78}$$

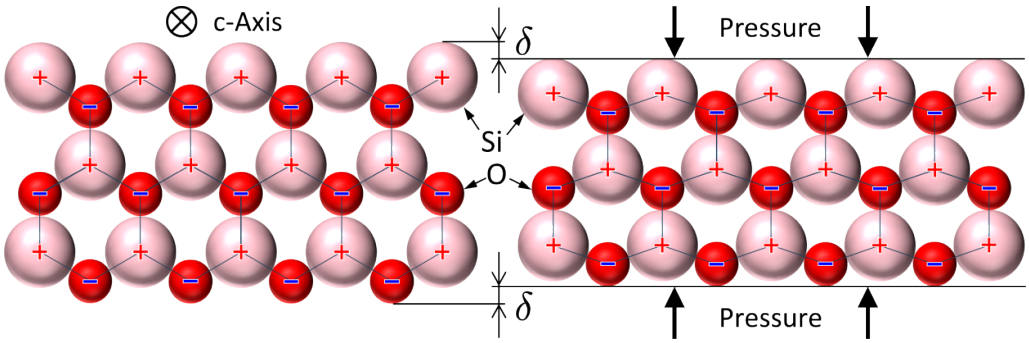



**Fig. 7.12:** View of the $\alpha$-quartz crystal structure along the $c$-axis, showing the hexagonal channels formed by the helical stacking of $SiO_4$ tetrahedra. Under ambient conditions, the dipole moments within a tetrahedron cancel each other out, and the quartz exhibits no net volume dipole moment. Applying pressure can deform the channels, thereby generating a volume dipole moment perpendicular to the $c$-axis.

Hence, overall, the dipole moment of an $SiO_4$ tetrahedron cancels out. Consequently, the quartz crystal, which is composed of these tetrahedra, possesses no permanent dipole moment within its volume. Unlike an NaCl crystal, a quartz crystal lacks a center of symmetry but does possess polar axes. A polar axis refers to a crystallographic direction along which there is no mirror symmetry. What does that mean? Well, it simply means that mirroring one half of the crystal across a plane normal to the polar axis does not yield the crystal structure found on the other side of that plane. In quartz, the so-called $c$-axis is such a polar axis, along which $SiO_4$ tetrahedra are arranged in pairs within helical chains. If one looks into the crystal along the $c$-axis, one will notice that these tetrahedral spirals form hexagonal channels. Fig. 7.12 shows a schematic view into the quartz crystal along this $c$-axis. The hexagons do not represent the bonding structure within the crystal, but rather the open hexagonal channels running along the $c$-axis. If pressure is applied to the crystal perpendicular to this $c$-axis, these hexagonal channels deform, creating a volume dipole moment within the crystal that is oriented perpendicular to the $c$-axis. When the pressure on the crystal is removed, the quartz crystal's volume dipole moment disappears. Different from that, applying pressure to the crystal along the $c$-axis does not generate a volume dipole moment. Polarization of the quartz crystal upon the application of pressure occurs via a mechanical charge displacement. This process also works in reverse. Inducing such a charge displacement via an external electric field results in a mechanical deformation of the crystal. Consequently, applying alternating electric fields allows for the generation of reversible, periodic crystal deformation, thereby enabling a wide range of technical applications. These include, for example, quartz watches, frequency generators of all kinds, industrial sensors and actuators, as well as a wide range of applications in precision mechanics.

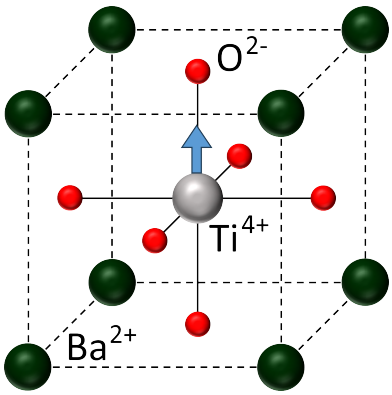



**Fig. 7.13**

Below a critical temperature, ferroelectric materials exhibit spontaneous polarization even in the absence of an applied external electric field and are of great technical importance, for instance, due to their high dielectric permittivity or their use as ferroelectric storage media (FeRAM – Ferroelectric Random Access Memory). One such material is the crystal barium titanate, the cubic unit cell of which is shown in fig. 7.13. In this so-called perovskite structure, a $Ba^{2+}$ ion is located at each of the cube's eight corners, and an $O^{2-}$ ion is situated at the center of each of the six faces. The $Ti^{4+}$ ion is located at the center of the cube, and thus at the center of the octahedron formed by the oxygen ions. Each Ba ion at the corner sites is shared among eight unit cells, and each O ion on the face centers is shared between two unit cells; therefore, the chemical formula of barium titanate is $BaTiO_3$. In the perovskite structure, the centers of charge of the positive and negative ions coincide at the center of the cube: 8 times the charge of 1/8 $Ba^{2+}$ each yields a charge of +2, and 6 times the charge of 1/2 $O^{2-}$ each yields a charge of −6; together with the +4 charge of the $Ti^{4+}$ ion at the center of charge, this results in a net charge of exactly zero. When a $BaTiO_3$ melt cools down, the material first crystallizes into the perovskite structure shown in fig. 7.13. In this structure, as we have just seen, the centers of positive and negative charge coincide. However, at a temperature of approximately 130° C, the $Ti^{4+}$ ion begins to shift away from the center of the oxygen-ion octahedron, as indicated in fig. 7.13. As a result, the centers of charge of the negative and positive ions no longer coincide, and the unit cell now possesses a dipole moment. This displacement of the $Ti^{4+}$ ion is the primary reason for the spontaneous polarization that occurs in the $BaTiO_3$ crystal at a critical temperature of approximately $T_c = 120°\,\mathrm{C}$. Not shown in fig. 7.13 is the fact that the cube also undergoes slight elongation in the direction of the $Ti^{4+}$ ion's movement. The cubic structure shown in fig. 7.13 thus transforms into a so-called tetragonal structure, in which one of the cube's axes, specifically the one aligned with the $Ti^{4+}$ ion's axis of movement, is slightly elongated. During this stretching, the shared center of charge of the Ba and O ions also shifts, though to a significantly lesser extent than the distance the Ti ion moves away from the center of the perovskite structure. The phase transition from the cubic to the tetragonal unit cell is not the only one occurring in $BaTiO_3$. At even lower temperatures, $BaTiO_3$ exhibits two further phase transitions, though these are of no interest to us here. We are far more interested in why spontaneous polarization arises in $BaTiO_3$ in the first place.

To understand this, we need to examine the symmetry of the $BaTiO_3$ perovskite structure. Clearly, the cubic crystal shown in fig. 7.13 possesses inversion symmetry. Every lattice site in the unit cell shown in fig. 7.13 is a center of inversion for the crystal. If the crystal is reflected across one of the crystal planes passing through a lattice site, it appears identical on both sides of the reflection plane when viewed along the surface normal. If one finds it difficult to visualize this inversion symmetry for the entire unit cell shown in fig. 7.13, it suffices to consider the individual sublattices of the Ba, O, and Ti ions separately in order to recognize the inversion symmetry present in each of these sublattices. In eq. (7.74), we can split the sum over all lattice sites into sums over the respective sublattices. Since each of these sublattices possesses inversion symmetry, each of these sums vanishes and thus does the total sum. Therefore, the entire crystal possesses inversion symmetry. $\boldsymbol{E}_{local}$ thus vanishes in the cubic perovskite structure of $BaTiO_3$. Consequently, eq. (7.71) reverts to eq. (7.60), and for $BaTiO_3$ above $T_c$, we can use the Clausius-Mossotti relation from eq. (7.63) for the dielectric susceptibility. $BaTiO_3$ consists of three different atoms; thus, there is no atomic polarizability here, but rather an effective electric polarizability $\alpha$:

$$\chi = \frac{n\cdot\alpha/\epsilon_0}{1-\left(n\cdot\alpha/3\epsilon_0\right)} \qquad \text{or} \qquad \frac{n\cdot\alpha}{\epsilon_0} = \frac{3\chi}{\chi+3} \tag{7.79}$$

Just above the critical temperature $T_c$, $\chi$ in $BaTiO_3$ reaches a value of 1000, and the factor $n\cdot\alpha/\epsilon_0$ is thus 2.991. As the temperature drops further towards $T_c$, the term $n\cdot\alpha/\epsilon_0$ increasingly approaches the value 3; this means the denominator in the Clausius-Mossotti equation tends towards zero, and consequently, $\chi$ tends towards infinity. According to eq. (6.34), even the smallest electric fields can thus generate a very large polarization as $T \to T_c$. Near the critical temperature, we can approximate the behavior of $n\cdot\alpha/\epsilon_0$ by

$$\frac{n\cdot\alpha}{\epsilon_0} = 3 - \beta(T-T_c) + \ldots \tag{7.80}$$

where $\beta$ is a constant. Therefore, near the critical temperature $T_c$, we can write the Clausius-Mossotti relation as:

$$\chi(T) \approx 3\cdot\frac{3-\beta(T-T_c)}{\beta(T-T_c)} \tag{7.81}$$

However, by assumption, $\beta(T-T_c) \ll 1$, and thus eq. (7.81) becomes:

$$\chi(T) \approx \frac{9}{\beta(T-T_c)} \tag{7.82}$$

We thus know the temperature dependence of the dielectric susceptibility of $BaTiO_3$ as $T \to T_c$: $\chi(T)$ approaches infinity as $(T-T_c)^{-1}$.

# 8. Direct Electric Current

In the preceding chapters, we have already considered several different types of electric charges, among them free static charges, bound charge pairs where charge separation can occur through electric polarization in an external electric field, or induction charges that form on the surfaces of conductors in response to external electric fields. Now we turn to our attention to moving electric charges, i.e., electric currents, which can exist, for example, in current carriers such as conductors, semiconductors, liquids, gases, or as streams of charged particles in a vacuum. In this chapter, we consider only direct electric currents. The electric current $I$ is defined as the quantity of charge $\Delta q$ that flows through the cross-sectional area element of the current carrier per unit time $\Delta t$. In the limit $\Delta t \to 0$, this current is defined by:

$$I = \lim_{\Delta t \to 0} \frac{\Delta q}{\Delta t} = \frac{\mathrm{d}q}{\mathrm{d}t} \tag{8.1}$$

The strength of the electric current, measured in the unit ampere with $1\mathrm{A} = 1\mathrm{C}{\cdot}1\mathrm{s}^{-1}$ has been defined since 2019 by the charge of the electron and the time unit second. Before that, it was defined by the force acting between two parallel current-carrying conductors (see chapter 9). As the name suggests, in direct-current circuits, often referred to just as DC-circuits, the current $I$ flows constantly in one direction. However, during the switching on or off process of a direct-current circuit, the electric current naturally changes over time. For this reason, we only consider direct-current circuits in their steady state, i.e., circuits in which a constant current is already flowing. To operate a circuit, a voltage source is needed, such as a battery or a power supply, which, with a constant voltage, determines the direction of current flow through its positive and negative terminals. The polarity of a direct-current voltage source is constant, unlike that of an alternating-current (AC) voltage source. The basic electrical components of an electric circuit are the resistor, the capacitor, and the inductor. We are already familiar with capacitors from the discussion in section 4.4. In this section, we will learn about electrical resistors, and we will deal with current-carrying inductors in the next chapter. In

direct-current circuits, a capacitor represents an open circuit, and an inductor represents a short circuit. One might therefore think that capacitors play no role in direct-current circuits, but that is not correct. When a DC circuit is switched on, a one-time charge displacement occurs, which ensures that, at a given electrical voltage, capacitors in such circuits carry the charge corresponding to their capacitance. An electrical circuit always represents a current loop, i.e., a simple coil, which naturally constitutes an inductor. Therefore, inductors also play a role in direct-current circuits during switching. However, we need not discuss this here, as we are considering only direct-current circuits in which a constant current is already flowing. In this case, ideal inductors, due to their zero electrical resistance, are nothing more than short circuits.

## 8.1 Charge Conservation and Electrical Circuits

Electrical devices of all kinds have become indispensable in our daily lives. In each of these devices, electric current flows according to specific rules in electrical circuits designed to enable the device to fulfill its particular function. Even though, unlike decades ago, these electrical circuits are now often highly miniaturized and the individual components of many of these electrical circuits are sometimes no longer easy to identify, they still function according to the same principles and make use of the same passive electrical components: resistors, capacitors and inductors. We are already familiar with the capacitor, a component that can store electrical charge, from chapter 3. Here we will learn about the properties of electrical resistors, and in chapter 9 we will do the same for inductors. There are of course a number of other important electrical components such as diodes and transistors, without which no digital electrical circuit could exist. However, the electrical resistance $R$ of a conductor, the capacitance $C$ of capacitors and the inductance $L$ of coils are the three basic electronic components with which one can begin to build simple analog electrical circuits.

Fig. 8.1 shows the current flow through an electrical resistance $R$ across which a voltage $U$ is applied. The electrons flow from the negative to the positive terminal of the voltage source $U$ while the electric field points from the positive to the negative terminal of the voltage source. The electrons therefore move in the opposite direction to the electric field. The direction of electron flow in the wire defines the physical current direction, which is opposite to the technical current direction $I$. This difference is historically rooted, as at the time electricity was

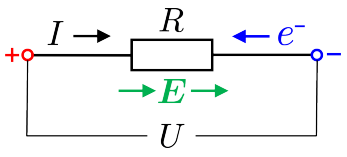



**Fig. 8.1**

discovered, neither atoms nor electrons were known, and it was assumed that the carriers of electric current in a conducting wire were positive charges. The technical current direction $I$ therefore always indicates the direction in which a positive charge would flow in an electrical circuit. The technical and physical current directions are not contradictory, but rather consistent definitions used to analyze electrical circuits. However, it is important to remember that the physical current direction in a metal wire is associated with the transport of electrons, while the technical current direction is not. This becomes important, for example, when we deal with the magnetic fields of moving charges, as we will do in the next chapter; there, the physical direction of the current must always be used.

An electric field is required for electrons to move in a current-carrying wire. But why is there an electric field at all in a current-carrying metal wire? Did we not learn that the electric field vanishes in ideal conductors? That is indeed the case, but only in ideal conductors that offer no resistance to the movement of electrons, and not in conductors with a finite resistance $R$. A potential difference between the ends of a conducting wire can only exist if this wire has a finite resistance. For example, if in fig. 8.1 $R = 0$, then both wire ends would be at the same potential and therefore $U = 0$. However, a real conductor possesses a finite resistance $R$ and because of that a potential gradient exists along the conductor. Consequently, a force $\boldsymbol{F} = q \cdot \boldsymbol{E}$ acts on the freely moving charge carriers in the conductor. To understand how this electric field arises in the conducting wire we can draw on what we learned in the discussion of the charged ring in section 4.5. Why this is the case is illustrated by fig. 8.2 for the simple example of a straight piece of wire over which the potential between the positive and negative ends of the wire drops from $\phi_>$ to $\phi_<$. This potential gradient is reflected in surface charges of the wire, which range from strongly positive at the positive voltage pole to weakly positive, weakly negative, and finally strongly negative at the negative voltage pole. As shown in fig. 8.2, these surface charges can be segmented into ring charges, which generate the electric field in the axis of the conducting wire that we already know from the discussion of ring charges. Between two such positive ring charges, where one carries a slightly larger charge and the other a slightly smaller charge, a net $\boldsymbol{E}$-field is created in the wire axis, pointing from the positive voltage pole to the negative voltage pole. Somewhere along the wire, the polarity of the ring charge changes from positive to negative. However, because the component of the electric field on the ring axis points away from the center of the ring when the ring charge is negative, a net $\boldsymbol{E}$-field component in the direction of the negative voltage pole arises again between two adjacent ring charges carrying slightly unequal amounts of negative charge.

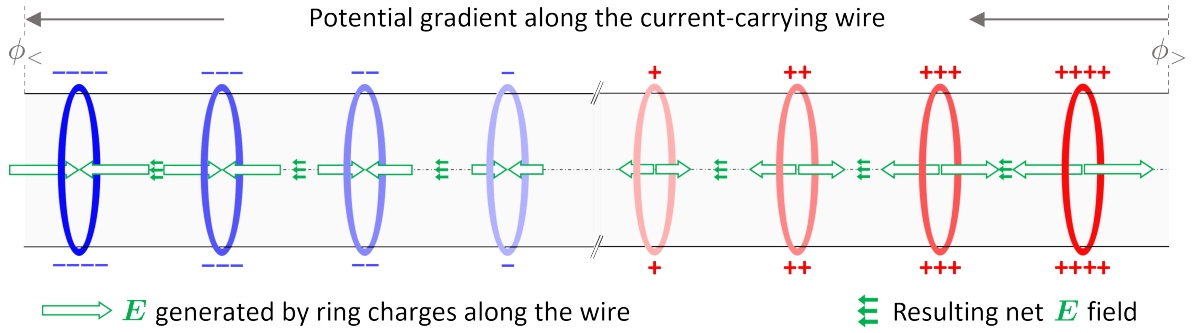



**Fig. 8.2:** Simple ring charge model for the qualitative explanation of the electric field pattern essential for current flow in a current-carrying wire. The electrons move in the net $\boldsymbol{E}$-field, subject to the force $\boldsymbol{F} = -e\boldsymbol{E}$, within the wire segment from $\phi_<$ to $\phi_>$.

A ring charge possesses a radial field in the plane of the ring. For a positive ring charge, this radial field is directed towards the center of the ring, for a negative ring charge, it is directed away from it. At the center of the ring itself, the radial component of the electric field vanishes; therefore, only the net $\boldsymbol{E}$-field along the wire axis acts on the electrons there. Freely moving charge carriers outside the wire axis are therefore subjected to either an electric force radially towards the center of the wire (positive ring charge) or radially towards the edge of the wire (negative ring charge). As we will see in the next chapter, a current-carrying wire also has a radial magnetic field in and around the wire, and the force that this magnetic field has on moving charge carriers in the wire must also be taken into account. It should only be noted here in advance (see fig. 9.6) that this radial magnetic field decreases linearly to zero from the wire's edge to its center. This means that there is no force acting on charge carriers moving along the wire's axis.

### 8.1.1 The Continuity Equation

Consider the conduction electrons in a metal wire with cross-sectional area $A$. Let the conduction electrons within the wire have a uniform density $n_e$ and move with an average speed $|\langle \boldsymbol{v} \rangle|$ from the negative to the positive terminal of the wire. In this case, the current $I$ is given by ($q = e$):

$$I = n_e \cdot A \cdot |\langle \boldsymbol{v} \rangle| \cdot q = j!A \tag{8.2}$$

In our wire, the electric current is simply equal to the magnitude of the current density $j$ multiplied by the cross-sectional area $A$ of the wire. The quantity $|\langle \boldsymbol{v} \rangle|$, the magnitude of the velocity averaged over all charge carriers, is the so-called drift velocity $v_D$ at which the electrons move through the wire when a voltage is applied. At mains voltage, this drift velocity in copper is only about 0.5 mm/s. The typical cross-section of household

electrical wire is $2.5\,\mathrm{mm}^2$. With a current of 1 ampere, $6.24 \times 10^{18}$ electrons per second move through the wire's cross-section, and their density $n_e$ within the wire is $\approx 5 \times 10^{21}$ electrons per $\mathrm{cm}^3$. At room temperature, copper has an atomic density of $\approx 8.4 \cdot 10^{22}$ atoms per $\mathrm{cm}^3$. Thus, on average, each copper atom contributes less than 0.1 electron to the current flowing through the wire. The quantity $j = I/A$ in eq. (8.2) is the current density. In general, $\boldsymbol{j} = \boldsymbol{j}(\boldsymbol{r}, t)$, meaning, the current density $\boldsymbol{j}$ is a vector quantity dependent on position and time. The current density vector points in the direction in which the charges flows, or the charge density $\rho = n_e \cdot q$ moves, and indicates the amount of charge passing through a given area per unit of time:

$$\boldsymbol{j}(\boldsymbol{r}, t) \;=\; n_e \cdot q \cdot \boldsymbol{v}(\boldsymbol{r}, t) \;=\; \rho \cdot \boldsymbol{v}(\boldsymbol{r}, t) \tag{8.3}$$

Eq. (8.3) is the definition of the current density $\boldsymbol{j}$. If we wish to determine not only the electric current flowing through a wire but current flows in the general case, we must consider the net flux of the current density out of an arbitrary volume $V$ with boundary $S$, as sketched in fig. 8.3. As we are already accustomed to, we calculate the flux of a vector quantity such as $\boldsymbol{j}$ through a surface $S$ with surface normal vector $\hat{\boldsymbol{n}}$ by multiplying the scalar product of the vector quantity and the normal vector $\hat{\boldsymbol{n}}$ by the area element $\mathrm{d}\sigma$ and then integrating over the surface. For the electric current $I$ associated with the current density $\boldsymbol{j}(\boldsymbol{r}, t)$, we can thus write:

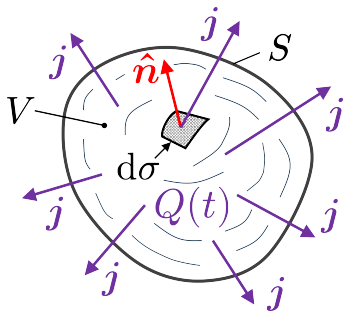



**Fig. 8.3**

$$\mathrm{d}I \;=\; \boldsymbol{j}(\boldsymbol{r}, t) \cdot \hat{\boldsymbol{n}} \cdot \mathrm{d}\sigma \tag{8.4}$$

Let us now consider the volume $V$ bounded by the surface $S$, which, as shown in fig. 8.3, shall enclose a charge quantity $Q$. Electric charge is always a conserved quantity. Its amount within a given volume cannot increase or decrease unless charge flows into the volume from the outside or flows out of it. Even if charges are generated within this volume due to quantum fluctuations, this always occurs in pairs. An electron-positron pair created in this way adds a net charge of zero to the volume. Therefore, the charge within the volume remains conserved. If a charge $\Delta Q$ flows out of the volume $V$ through the closed surface $S$ shown in fig. 8.3 per unit time $\Delta t$, then the charge $Q$ within the closed volume $V$ must decrease accordingly. The charge $\Delta Q$ flowing out through the closed surface $S$ per unit time $\Delta t$ is the electric current $I$, and, using eq. (8.4), this is given by:

$$I \;=\; \oint_S \boldsymbol{j}(\boldsymbol{r}, t) \hat{\boldsymbol{n}} \, \mathrm{d}\sigma \tag{8.5}$$

If $\rho(\boldsymbol{r},t)$ is the charge density in the volume $V$, then the following applies to the decrease in the charge $Q$ within this volume:

$$-\frac{\mathrm{d}Q}{\mathrm{d}t} \;=\; -\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho(\boldsymbol{r},t)\,\mathrm{d}^3r \tag{8.6}$$

The electric current through the boundary $S$ must be equal to the decrease in the charge $Q$ within $V$, and thus the following must hold:

$$\oint_S \boldsymbol{j}(\boldsymbol{r},t)\hat{\boldsymbol{n}}\,\mathrm{d}\sigma \;=\; -\frac{\mathrm{d}}{\mathrm{d}t}\int_V \rho(\boldsymbol{r},t)\,\mathrm{d}^3r \tag{8.7}$$

Eq. (8.7) is the continuity equation of electrodynamics in integral form. To obtain the differential form, we use Gauss's theorem eq. (2.42) to replace the surface integral on the left-hand side of eq. (8.7) with a volume integral over the divergence of the current density $\boldsymbol{j}(\boldsymbol{r},t)$, and move the time derivative on the right-hand side of eq. (8.7) inside the volume integral, yielding the following result:

$$\int_V \left(\nabla\boldsymbol{j}(\boldsymbol{r},t) + \frac{\partial\rho(\boldsymbol{r},t)}{\partial t}\right)\mathrm{d}^3r \;=\; 0 \tag{8.8}$$

This equation is satisfied only if the integrand vanishes. Therefore, it must hold:

$$\nabla\boldsymbol{j}(\boldsymbol{r},t) + \frac{\partial\rho(\boldsymbol{r},t)}{\partial t} \;=\; 0 \tag{8.9}$$

We are already familiar with this type of equation from our study of fluid motion in volume 1 of this physics course. Eq. (8.9) is a continuity equation, in this case, the continuity equation of electrodynamics, which states that the total charge in a closed system always remains constant. Therefore, eq. (8.9) is also referred to as the law of local charge conservation. The integral form eq. (8.7) describes the flux of the current density out of or into a given volume $V$ through its boundary $S$. The differential form eq. (8.9) describes the change in the source strength of the current density at each point $\boldsymbol{r}$ resulting from the change in the charge density at that location $\boldsymbol{r}$. If $\dot{\rho} < 0$ holds for the left-hand side of eq. (8.9), then $\nabla\cdot\boldsymbol{j} > 0$ and the point $\boldsymbol{r}$ in question is a current source. On the other hand, if $\dot{\rho} > 0$ at a point $\boldsymbol{r}$, then $\nabla\cdot\boldsymbol{j} < 0$ and the point is a current sink. There are no electric currents in electrostatics, i.e., in electrostatics, the following holds always and everywhere:

$$\boldsymbol{j} \;=\; 0 \qquad\Rightarrow\qquad \frac{\partial\rho}{\partial t} \;=\; 0 \qquad\Rightarrow\qquad \rho \;=\; \rho(\boldsymbol{r}) \tag{8.10}$$

With a current density of $\boldsymbol{j} = 0$, there is, of course, no current. To deal with electric currents, therefore, we must move beyond electrostatics. We do this by considering stationary currents. stationary currents are currents with constant current density. Thus, applying eq. (8.9), we have:

$$\boldsymbol{j} = \boldsymbol{j}(\boldsymbol{r}) = const. \quad \Rightarrow \quad \nabla \boldsymbol{j} = 0 \quad \Rightarrow \quad \frac{\partial \rho}{\partial t} = 0 \tag{8.11}$$

With $\partial\rho/\partial t = 0$, the charge density remains constant, though now not at $\boldsymbol{j} = 0$, but at $\boldsymbol{j} = \text{const.}$

### 8.1.2 Ohm's Law and Electrical Power

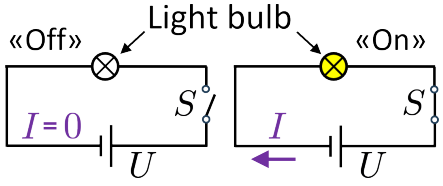



**Fig. 8.4**

Let us look at our metal wire again. This time, however, we cut the wire into two pieces and then connect the two wire segments electrically via an inserted light bulb, as shown in fig. 8.4. In addition, we have installed a switch $S$ in one of the wire segments. We then connect the free wire ends to a voltage source with a voltage $U$ selected to allow the operation of a light bulb. If we then close the switch $S$, the voltage $U$ is applied across the ends of the wire, a current $I$ flows, and the light bulb lights up. If we open switch $S$ again, no current flows and the light bulb stops glowing. For most conductors, such as the wire here, the current $I$ flowing through the conductor at a constant temperature when the switch $S$ is closed is proportional to the voltage $U$:

$$U = R \cdot I \qquad \text{(Ohm's law)} \tag{8.12}$$

Ohm's law, the proportionality between current and voltage expressed by eq. (8.12), is likely familiar even to many non-physicists. The constant of proportionality in eq. (8.12) is the electrical resistance $R$ of wire and light bulb, and the reciprocal of $R$ is the electrical conductance $G$. The unit of electrical resistance $R$ is the ohm, abbreviated as $\Omega$, and the unit of electrical conductance $G$ is the siemens, abbreviated as $S$. With eq. (8.12), the following applies:

$$1\,\Omega = \frac{1\,\text{V}}{1\,\text{A}} \qquad \text{and} \qquad 1\,\text{S} = \frac{1\,\text{A}}{1\,\text{V}} \tag{8.13}$$

For ohmic conductors, resistance does not depend on the current. However, for non-ohmic conductors, such as the light bulb in fig. 8.4, it does depend on the current, because as the current flow increases, the tungsten filament in the bulb heats up, causing its electrical resistance to rise. Our wire is an ohmic conductor, and for it, the resistance $R$ is proportional to the wire's length $L$ and inversely proportional to its cross-sectional area $A$:

$$R = \rho \cdot \frac{L}{A} \tag{8.14}$$

**Tab. 8.1:** Specific resistance $\rho_0$ for selected conductors (C), semiconductors (SC), and non-conductors (NC) at room temperature.

| Material | | $\rho$ [Ω·m] | Material | | $\rho$ [Ω·m] |
|---|---|---|---|---|---|
| Silver | (C) | $1.60 \cdot 10^{-8}$ | Saline solution (10%) | (C) | 0.1 |
| Copper | (C) | $1.71 \cdot 10^{-8}$ | Germanium (pure) | (SC) | 0.46 |
| Gold | (C) | $2.20 \cdot 10^{-8}$ | Silicon (pure) | (SC) | $2 \cdot 10^{3}$ |
| Aluminum | (C) | $2.80 \cdot 10^{-8}$ | NaCl crystal | (NC) | $1.0 \cdot 10^{8}$ |
| Sodium | (C) | $4.20 \cdot 10^{-8}$ | Laminated paper | (NC) | $10^{10} - 10^{12}$ |
| Brass | (C) | $5 \cdot 10^{-8}$ | Porcelain | (NC) | $10^{11} - 10^{12}$ |
| Nickel | (C) | $6.93 \cdot 10^{-8}$ | Diamond | (NC) | $1.0 \cdot 10^{12}$ |
| Iron | (C) | $2 - 1.5 \cdot 10^{-8}$ | Hard rubber | (NC) | $2.0 \cdot 10^{13}$ |
| Graphite (basal plane) | (C) | $2 - 5 \cdot 10^{-6}$ | Quartz glass | (NC) | $10^{16} - 10^{17}$ |

The proportionality constant $\rho$ (not to be confused with charge density) is the specific resistance of our wire, a temperature-dependent material constant. The reciprocal of the specific resistance $\rho$ of our wire is its specific conductance $\sigma$ (not to be confused with surface charge density):

$$\sigma = \frac{1}{\rho} \tag{8.15}$$

According to eq. (8.14), the unit of specific resistance $\rho$ is Ω·m, and that of specific conductance $\sigma$ is correspondingly $\Omega^{-1} \cdot \mathrm{m}^{-1}$. Tab. 8.1 lists values of $\rho$ at room temperature for a range of electrical conductors, semiconductors, and non-conductors. $\rho$ varies by nearly 25 orders of magnitude between very good conductors like silver and very good insulators like quartz glass. Semiconductor materials such as germanium and silicon are somewhere in between conductors and insulators; however, in their pure, i.e., undoped, form, they are effectively non-conductors. The specific resistance of a material is determined not only by the chemical element of which a material is composed but also, to a significant extent, by the specific form in which that element exists. Carbon is a prime example of this. In the form of graphite, carbon is a good electrical conductor along its basal plane, where the carbon atoms are arranged hexagonally. However, perpendicular to this plane, the specific resistance of graphite is three orders of magnitude higher, whereas in the form of diamond, carbon is an insulator. Sodium is a very good conductor, but also a very soft metal with a low melting point of just under 100 C°, which limits its usability. However, potential uses of sodium are even more limited by the fact that it reacts explosively with water, which is why it must be stored in oil. Nevertheless, there are specific technical applications for sodium. Good electrical conductors are generally

also good thermal conductors, as is the case with metals, and the alkali metal sodium is no exception. Furthermore, because liquid sodium hardly slows down fast neutrons, it is used as the coolant of choice in certain nuclear reactors known as fast breeder reactors. Conversely, it is not true that good thermal conductors are also good electrical conductors. Diamond, for example, is an excellent thermal conductor but acts as an insulator. Metals are the best conductors at room temperature. However, there are also so-called superconducting materials whose specific resistance virtually disappears at very low temperatures and/or high pressures, thereby enabling lossless electricity transmission.

If a conductor has finite electrical resistance, energy loss occurs as soon as an electric current flows through the conductor. We can quantify this energy loss by considering the work performed by the electric field. From eq. (3.36), we know that the work done when a charge $q$ traverses a potential difference $U = \phi_2 - \phi_1$ is given by $W = q \cdot U$. A current $I = \mathrm{d}q/\mathrm{d}t$ flows through a conductor, and thus a charge $\mathrm{d}q$ traverses the potential difference $U$ per unit of time $\mathrm{d}t$. The work performed per unit of time, i.e., the power $P$ delivered by the electric field as the charge moves across the electric potential difference $U$, is therefore:

$$\frac{\mathrm{d}W}{\mathrm{d}t} = P = U \cdot I \tag{8.16}$$

For ohmic resistors, voltage and current are related by Ohm's law. The electrical power $P = U \cdot I$ which is being dissipated in an ohmic resistor $R$ in the form of heat is therefore given by:

$$P = R \cdot I^2 \qquad \text{or} \qquad P = \frac{U^2}{R} \tag{8.17}$$

This power loss due to the finite electrical resistance of conductors, referred to as ohmic losses in a circuit, must be continuously replenished, because otherwise, electric current will soon cease to flow in the circuit in question. Therefore, a closed circuit must contain an energy source that compensates for these ohmic losses. This can be a battery storing energy in chemical form. Such a battery will maintain the voltage in the circuit until the chemical energy stored in it is nearly used up at which point the battery is discharged. Alternatively, it can be a voltage generator driven by an internal combustion engine, which first converts the chemical energy of fossil fuel into mechanical energy that is subsequently transformed into electrical energy. And then there are other voltage sources utilizing, for example, the conversion of thermal or light energy into electrical energy. For historical reasons, the source voltage provided by such sources is referred to as electromotive force and is usually abbreviated as *EMF* or often denoted by the symbol $\varepsilon$. However, despite its name, the *EMF* is not a force, but rather the energy per unit

charge that such a voltage source provides to operate a circuit. The *EMF* is measured in the open-circuit state, i.e., without a connected load resistor, as the potential difference between the positive terminal (cathode) and the negative terminal (anode) of the voltage source:

$$EMF = U_\varepsilon = \phi_{cathode} - \phi_{anode} \tag{8.18}$$

In addition to this *EMF* used for voltage generation, i.e., in generator mode, there is also an *EMF* present during motor operation. In an electric motor, the applied *EMF* causes the motor to rotate; thus, electrical energy is converted into mechanical energy.

### 8.1.3 Simple Electrical Circuits

The simplest electrical circuit consists, as sketched in fig. 8.4, of a voltage source and a load, which in the electrical circuit shown in fig. 8.4 is the light bulb. Such a light bulb represents an electrical resistor $R$, the symbol for which is an empty rectangle, as shown in fig. 8.1. However, a load can just as well represent a capacitance $C$ or an inductance $L$, or be a combination of $R$, $C$, and $L$. We have already learned what a capacitance is in chapter 4, and in electrical circuits, a capacitance is represented by two parallel lines, essentially a small, symbolic parallel-plate capacitor. We will not encounter inductors until the next two chapters where will learn a lot about them, but it is worth noting here that the symbol for an inductive component is a filled rectangle. We have already used another symbol, which is that for the circuit element without which no current flows: the voltage source or *EMF* in fig. 8.4. The symbol for a voltage source, like the symbol for a capacitance, consists of two parallel lines, though one is only half the length of the other. The longer of the two lines represents the positive terminal of the voltage source, while the shorter line indicates the negative terminal. This is easy to remember because the minus sign consists of only a single horizontal line, whereas the plus sign is made up of a vertical and a horizontal line which, when combined, form a longer line than the minus sign. Fig. 8.5 shows these important circuit symbols for the electrical components resistor $R$, capacitor $C$, inductor $L$, and for the voltage source *EMF*. Even in small circuits, there are usually multiple resistors, capacitors, or inductors present, each of them designated by the respective letter and an accompanying index, such as $R_1$, $R_2$, $R_3$, etc. For reasons of redundancy, some circuits may also feature more than one *EMF* source, which are likewise numbered sequentially.

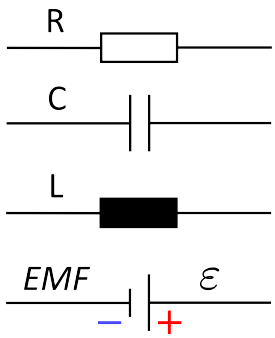



**Fig. 8.5**

Let us now consider the simple circuit shown in fig. 8.6, which consists of an *EMF* with a terminal voltage $U_\varepsilon$ and two resistors, $R_1$ and $R_2$, which are connected in series. According to Ohm's law from eq. (8.12), with the current $I$ flowing through both resistors, voltage drops of $U_1 = R_1 I$ and $U_2 = R_2 I$ must occur across these two resistors. The terminal voltage $U_\varepsilon$ and the voltages $U_1$ and $U_2$ have opposite signs. The reason for this is that $U_\varepsilon$ is the driving force that moves the charges through the circuit, whereas the voltage drops $U_1$ and $U_2$ are the result of the ohmic resistance with which $R_1$ and $R_2$ oppose this very flow of charges. However, the fact that $U_\varepsilon$ must have the opposite sign to $U_1$ and $U_2$ also follows directly from the fundamental equations of electrostatics. The circuit in fig. 8.6 represents a closed path $C$ enclosing a surface $S$, and in this case, we know that, based on eq. (3.35) and eq. (3.37), the following must hold:

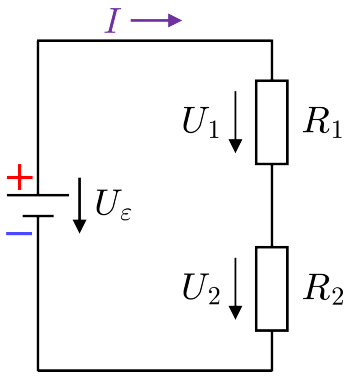



**Fig. 8.6**

$$\oint_C \boldsymbol{E}(\boldsymbol{r})\,\mathrm{d}\boldsymbol{s} \;=\; 0 \;=\; U_1 + U_2 + U_\varepsilon \tag{8.19}$$

This means that the terminal voltage of the *EMF* must have a sign opposite to that of the voltages $U_1$ and $U_2$. The generalization of eq. (8.19) is the so-called loop rule of Kirchhoff, sometimes also referred to as Kirchhoff's mesh rule, which states that for a closed circuit, the sum of all voltages across the circuit components must vanish, with the terminal voltage $U_\varepsilon$ of the *EMF* always counted as negative. Expressed mathematically, for a closed circuit with $n$ components and thus $U_n$ voltages this means that the following always applies:

$$\sum_n U_n \;=\; 0 \qquad \text{(Kirchhoff's loop rule)} \tag{8.20}$$

In addition to Kirchhoff's loop rule, there is a second, no less important rule, which expresses the conservation of charge for stationary currents in a closed circuit. This Kirchhoff rule states that at a current junction in an electrical circuit, also called an electrical circuit node, the sum of the currents flowing into and out of the junction must be zero. Fig. 8.7 illustrates this for the simplest case of a junction. By definition, currents flowing into the junction are counted as positive and currents flowing out of it as negative. Thus, for the example of the three currents in fig. 8.7, the equation $I_1 - I_2 - I_3 = 0$ applies. Generalized, this so-called junction rule states that the following applies to the currents entering and leaving a junction:

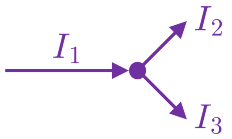



**Fig. 8.7**

$$\sum_n I_n = 0 \qquad \text{(Kirchhoff's junction rule)} \tag{8.21}$$

These two rules, referred to as Kirchhoff's laws, relate individual voltages and currents within a closed electrical circuit. They enable the analysis of electrical circuits that are far more complex than the simple examples shown here. Fig. 8.8 illustrates an elementary application of Kirchhoff's rules. Fig. 8.8a shows the simplest possible electrical circuit in which a voltage drop $U$ occurs across a single resistor $R$ due to a current $I$ flowing through the resistor. In fig. 8.8b, $R$ has been replaced by three resistors $R_1$, $R_2$, and $R_3$ connected in series, and in fig. 8.8c by the same three resistors but now connected in parallel. In a series connection, the same current $I$ flows through all three resistors; thus, Kirchhoff's loop rule implies ($U_\varepsilon = U$):

$$U_\varepsilon = U_1 + U_2 + U_3 = (R_1 + R_2 + R_3)\cdot I = R\cdot I \tag{8.22}$$

In a series connection, the individual resistances therefore add up, and for a series connection of $n$ resistors $R_n$, the total resistance $R$ is therefore given by:

$$R = \sum_n R_n \qquad \text{in a series connection} \tag{8.23}$$

For the resistors connected in parallel in fig. 8.8c, the same voltage $U_\varepsilon = U_1 = U_2 = U_3$ drops across all three resistors. For the respective currents flowing through these resistors Kirchhoff's junction law therefore implies ($U_\varepsilon = U$):

$$\frac{U_\varepsilon}{R} = I = I_1 + I_2 + I_3 = U_\varepsilon\cdot\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) \tag{8.24}$$

In a parallel connection, the reciprocals of the individual resistances add up to the reciprocal of the total resistance. Consequently, the total resistance is smaller than the smallest of the individual resistances. For the parallel connection of $n$ resistors $R_n$, the total resistance $R$ is thus given by:

$$\frac{1}{R} = \sum_n \frac{1}{R_n} \qquad \text{in a parallel connection} \tag{8.25}$$

So much for the series connection and parallel connection of resistors. But what about capacitors? What is the total capacitance of capacitors connected in series or connected in parallel? In the case of resistors, we utilized the voltage characteristic based on Ohm's Law, which describes the linear relationship $U = R\cdot I$ between current and voltage for ohmic resistors. From section 4.4, we know that the voltage of a capacitor is related to its capacitance and to the charge it carries through $U = Q/C$. Consequently, the behavior of capacitors in series connection or capacitors in parallel connection is the exact opposite of that of ohmic resistors.

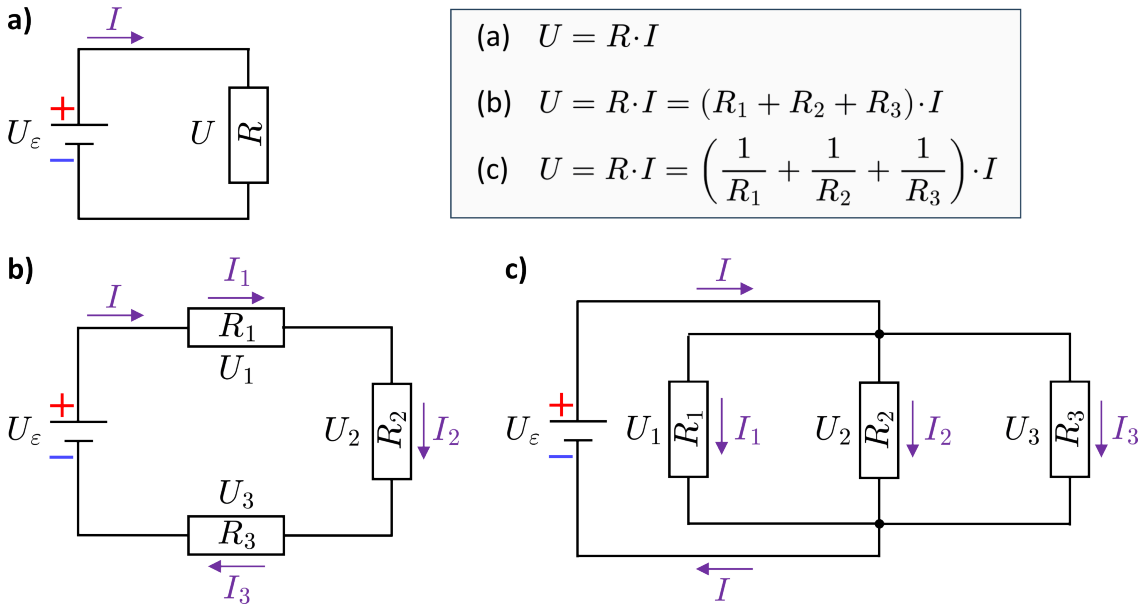


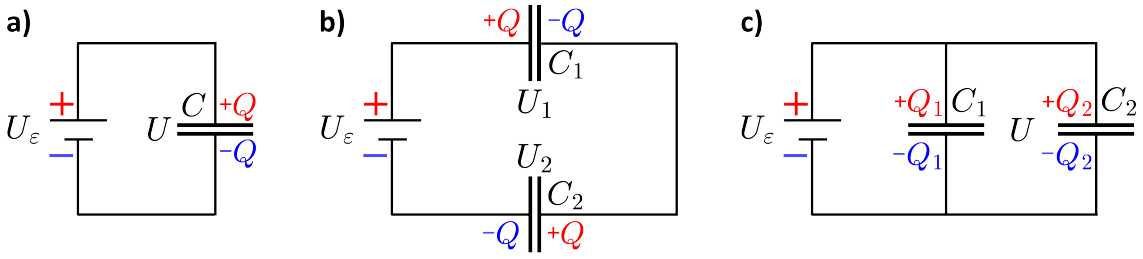

**Fig. 8.8:** a) Simple electrical circuit with a single resistor $R$. b) Same electrical circuit as in (a) with $R$ replaced by three resistors $R_1$, $R_2$, and $R_3$ connected in series. c) Same electrical circuit as in (a) with $R$ again replaced by the resistors $R_1$, $R_2$, and $R_3$, but connected in parallel.

Let us consider the series connection of capacitors $C_1$ and $C_2$ as shown in fig. 8.9b where the respective voltages $U_1$ and $U_2$ add up to the total voltage $U$. The electrodes of $C_1$ and $C_2$ facing each other, as well as the electrodes facing away from each other, each carry charges of equal magnitude and opposite sign. This situation results from a charge redistribution that occurs when the circuit is switched on. When the electrical circuit is switched on, the electrodes of $C_1$ and $C_2$ that are directly connected to the voltage source terminals are brought to the potential of the respective voltage pole, thereby acquiring the corresponding positive charge $+Q$ or negative charge $-Q$. This results, however, in a charge displacement between the facing electrodes of the two capacitors, which consequently must carry the charges $-Q$ and $+Q$, respectively. For a series connection, the following therefore applies:

$$\frac{Q}{C} = U = U_1 + U_2 = Q\left(\frac{1}{C_1} + \frac{1}{C_2}\right) \tag{8.26}$$

The reciprocals of the individual capacitances add up to the reciprocal of the total capacitance, which is consequently smaller than the smallest of the individual capacitances. For a series connection of $n$ capacitances $C_n$, the total capacitance $C$ is thus given by:

$$\frac{1}{C} = \sum_n \frac{1}{C_n} \qquad \text{in a series connection} \tag{8.27}$$




$$\frac{Q}{C} = U \qquad U_1 + U_2 = Q \cdot \left(\frac{1}{C_1} + \frac{1}{C_2}\right) \qquad Q_1 + Q_2 = U \cdot (C_1 + C_2)$$

**Fig. 8.9:** a) Electrical Circuit with a capacitance $C$. b) Same electrical circuit as in (a) with $C$ replaced by two capacitances $C_1$ and $C_2$ connected in series. c) Same electrical circuit as in (a) with $C$ again replaced by $C_1$ and $C_2$, but connected in parallel.

When capacitors are connected in parallel, as shown in fig. 8.9c for two capacitors with capacitances $C_1$ and $C_2$, the same voltage is applied across the electrodes of each, and the charges $Q_1$ and $Q_2$ of the capacitors add up:

$$Q \;=\; Q_1 + Q_2 \;=\; C_1 \cdot U + C_2 \cdot U \;=\; C \cdot U \tag{8.28}$$

Thus, when connected in parallel, the capacitances of the individual capacitors add up to the total capacitance. For a parallel connection of $n$ capacitors $C_n$, the total capacitance $C$ is therefore given by:

$$C \;=\; \sum_n C_n \qquad \text{in a parallel connection} \tag{8.29}$$

In this chapter, we are not considering inductors, i.e., coils, because they act as a short circuit in direct-current circuits. However, we will see later that coils behave exactly like electrical resistors when connected in series or parallel. Let us now take a closer look at our voltage source.

According to eq. (8.16), the power dissipated in an electrical resistor during direct-current operation is given by $P = U \cdot I$. This applies to any ohmic conductor, including the wires connecting the various electrical components within a circuit. Only a negligible fraction of this dissipated power serves to move the electric charges which carry the electric current through the conductor. As we will see in the next section, the reason for this is simply that the speed at which charge carriers move through an electrical conductor is very low. However, an electrical circuit does not exist in isolation; it typically serves a specific purpose, such as keeping our smartphones running. This requires a corresponding amount of electrical power, which must also be supplied by the voltage source. In the case of a smartphone this power source is of course their battery. As we all know, we have to recharge the latter regularly. If we remove the battery from our smartphone and

measure the voltage across its positive and negative terminals, we measure what is known as the open-circuit voltage ($U_0$). During operation, i.e., when the battery is installed and powers the smartphone, we would measure a voltage lower than $U_0$ between the battery terminals.

This behavior of a voltage source, where the open-circuit voltage $U_0$, i.e., when no load is connected to the source, is higher than the terminal voltage $U$ under load, i.e., when a load is connected to the source, can be described using the internal resistance $R_i$ of the voltage source. In the case of our smartphone battery, $R_i$ is the resistance of the electrolyte, for lithium batteries, typically in the range of a few mΩ. Fig. 8.10 shows the equivalent circuit diagram for a voltage source with internal resistance $R_i$ and a connected load resistance $R$ representing the consumer. Without a connected load resistance $R$, we measure the open-circuit voltage $U_{ab} = U_0$. However, if a load $R$ is connected, then we measure the terminal voltage:

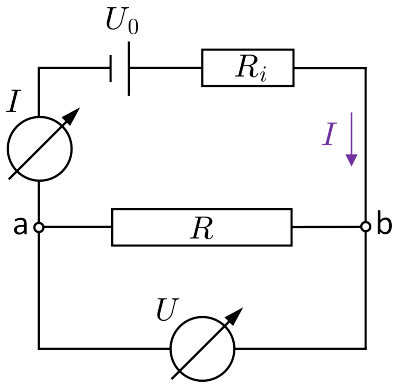



**Fig. 8.10**

$$U_{ab} = U_0 - R_i \cdot I \tag{8.30}$$

For a given load $R$, this terminal voltage decreases linearly with the current drawn, and the more sharply it does so, the higher the internal resistance $R_i$ of the voltage source. According to eq. (8.30), the maximum current $I_{max}$ that can be drawn from a voltage source is:

$$I_{max} = \frac{U_0}{R_i} \tag{8.31}$$

In practice, however, this is not possible, because the terminal voltage ($U$) drops to zero, meaning the load must be short-circuited, in order to draw this current. Internal resistance $R_i$ and load resistance $R$ are connected in series. Consequently, the same current $I$ flows through both resistors. Applying the loop rule yields:

$$U_0 = R_i \cdot I + R \cdot I \tag{8.32}$$

Thus, the current $I$ of the voltage source under load conditions is:

$$I = \frac{U_0}{R_i + R} \tag{8.33}$$

Using eq. (8.30) and eq. (8.33), the power dissipated in the load resistor $R$ is therefore:

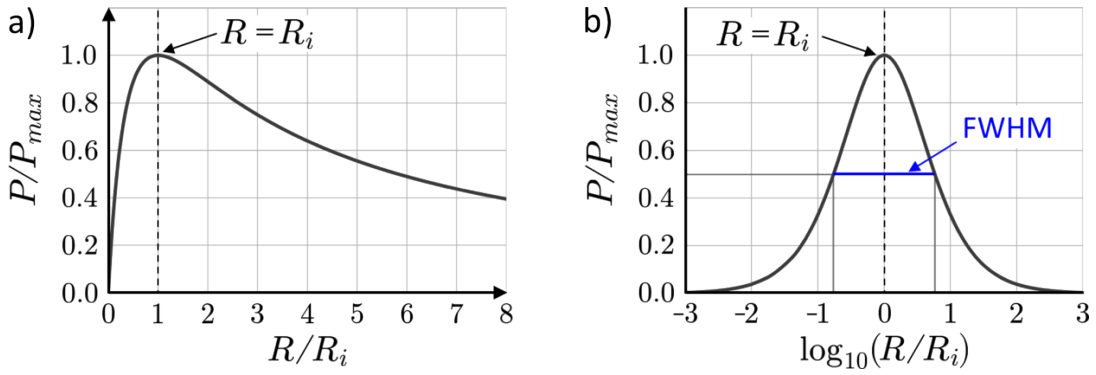



**Fig. 8.11:** Power extraction from a voltage source. The two graphs show the ratio of extracted power to maximum extractable power $P/P_{max}$ as a function of the ratio of load resistance $R$ to the internal resistance $R_i$ of the voltage source, plotted against $R/R_i$ (a) and against $\log_{10}(R/R_i)$ (b).

$$P = U \cdot I = \left(U_0 - R_i \cdot \frac{U_0}{R_i + R}\right) \cdot \frac{U_0}{R_i + R} = \frac{U_0^2 R}{(R_i + R)^2} \tag{8.34}$$

The maximum power that a load with resistance $R$ can draw from a voltage source is determined by $\mathrm{d}P(R)/\mathrm{d}R = 0$. Solving this equation yields $R = R_i$, and for the maximum extractable power, one obtains

$$P_{max} = \frac{U_0^2}{4R_i} \tag{8.35}$$

The maximum power that can be drawn from a voltage source is therefore limited by its internal resistance $R_i$. Fig. 8.11a shows the power drawn from a voltage source, expressed as a fraction of the maximum extractable power, as a function of the ratio between the load resistance $R$ and the internal resistance $R_i$ of the voltage source. $R = 0$ corresponds to a short-circuited voltage source, i.e., the terminal voltage $U$ has collapsed. If we omit the value $R = 0$, we can plot the extracted power against the common logarithm of $R/R_i$, as shown in fig. 8.11b. It becomes evident that $P(R)$ is symmetric with respect to $\log_{10}(R/R_i)$. For $P(R)$, we obtain the same values for $R/R_i = 0.1$, $R/R_i = 0.01$, and $R/R_i = 0.001$ as for $R/R_i = 10$, $R/R_i = 100$, and $R/R_i = 1000$. The full width at half maximum (FWHM) of the distribution in fig. 8.11b is slightly more than 1.5 decades, or, in terms of fig. 8.11a, the range of approximately $0.17 \leq R/R_i \leq 5.83$. To extract maximum power from a direct-current voltage source such as a battery, the internal resistance of the voltage source must be matched to the load resistance. However, this means that, given eq. (8.32) and eq. (8.30), $U_0 = 2R_i I$ and $U = R_i I$ hold true, and thus $U = U_0/2$. In the case of matching the resistances for maximum power transfer, also called impedance matching, the voltage across the load resistor therefore drops to half of the open-circuit voltage $U_0$. This means that half of the energy supplied by the direct-

current voltage source is dissipated as heat in the internal resistance $R_i$, and the voltage source operates with an efficiency of only 50%. Consequently, impedance matching plays a role primarily in fields such as measurement technology or telecommunications, where the objective is to measure or amplify weak signals.

## 8.2 Charge Transport in Electrical Conductors

Ohm's law is not a universal law of nature but rather an experimental finding described by Simon Ohm in 1826, and that is how we introduced it here via eq. (8.12). There was no microscopic explanation for Ohm's law in Simon Ohm's time. The electron was not discovered until 1897, i.e., decades after the formulation of Ohm's law. However, after this discovery, it did not take long for the Drude model of specific conductance in metals to provide an initial classical explanation. In this section, we examine aspects of charge transport in ohmic and non-ohmic conductors. We begin with electron conduction in metals, then look at electron-hole conduction in semiconductors, and finally consider ionic conduction in electrolytes.

### 8.2.1 The Drude Model of Conductivity in Metals

In 1900, Paul Drude proposed a model of electrical conductivity that treats conduction electrons in a metal like gas particles scattering off one another and off the heavy metal ions bound to their lattice sites. Today we known that the scattering of electrons by one another, i.e., electron-electron scattering within the "electron conduction gas," so to speak, plays only a minor role, and that the simple picture of conduction electrons scattering off lattice ions describes reality very inadequately. However, it is, in fact, scattering processes that determine electrical conductivity in metals, albeit processes somewhat more complex than simple scattering by lattice ions. In our discussion of thermal motion in volume 1 of this physics course, we encountered the so-called mean free path $l$, i.e., the distance gas particles travel on average in an ideal gas between two successive scattering processes, and we determined the mean time $\tau$ that elapses between two such scattering events. In the absence of an external electric field, the motion of the electrons in our electron gas is randomly distributed, meaning that just as many electrons move in any given direction as in the exactly opposite direction, and the mean velocity of the electrons $\langle \boldsymbol{v} \rangle$ is therefore zero. The average of this electron velocity squared, .i.e., $\sqrt{\langle \boldsymbol{v}^2 \rangle}$, as we know it from the chapter on thermal motion in volume 1 of this physics course, also called the mean thermal velocity, is by no means zero, but proportional to the

square root of the temperature of the electron gas. If an electrostatic field $\boldsymbol{E}$ is applied, a conduction electron experiences, on average, a change in momentum of the following magnitude between two scattering events:

$$m_e \Delta v \;=\; F \cdot \tau \;=\; e \cdot E \cdot \tau \tag{8.36}$$

The momentum of a conduction electron before a scattering process and its momentum after that process are completely uncorrelated. For instance, the electron might have been scattered forwards or backwards in the preceding scattering event, but after the event, it «retains no memory» of this scattering process. Effectively, the electron starts anew after each scattering process, as if no prior scattering events had occurred. The result is that, averaged over time, the momentum changes of an electron caused by the scattering processes cancel each other out. This implies, however, that $\Delta v$ in eq. (8.36) must be precisely the drift velocity $v_D$ already mentioned in connection with eq. (8.2). Thus, according to eq. (8.36), the drift velocity $v_D$ is proportional to the electric field strength $E$; that is

$$v_D \;=\; \mu \cdot E \qquad \text{with} \qquad \mu \;=\; \frac{e \cdot \tau}{m_e} \tag{8.37}$$

where $\mu$ is the so-called electron mobility. For the current density, setting $\Delta v = v_D$, we can write:

$$j \;=\; n_e \cdot e \cdot v_D \;=\; n_e \cdot e \cdot \mu \cdot E \;=\; \sigma \cdot E \tag{8.38}$$

or, respectively in vector form

$$\boldsymbol{j} \;=\; \sigma \cdot \boldsymbol{E} \tag{8.39}$$

This result of our microscopic analysis shows: The current density $\boldsymbol{j}$ is proportional to the applied electric field $\boldsymbol{E}$, and the specific conductance $\sigma$ is the constant of proportionality. In deriving eq. (8.39), which is frequently referred to as the differential form of Ohm's law, we made a crucial assumption: The mean time $\tau$ elapsing between two scattering processes does not depend on $\boldsymbol{E}$. This assumption is justified if the time of flight between two scattering processes is very short, i.e., if $v_D \ll \sqrt{\langle \boldsymbol{v}^2 \rangle}$ holds. If that is the case, $\tau$ is determined by $\sqrt{\langle \boldsymbol{v}^2 \rangle}$ rather than by $v_D$. This condition is met except at very high electric field strengths. However, there is another constraint to which $\boldsymbol{E}$ is subject and which we need to keep in mind: the field strength $E$ must not become so large that ionization processes occur which would alter the charge density $n_e$ in eq. (8.38). Using $\Delta v = v_D$ we can write eq. (8.36) as

$$e \cdot E - \frac{m_e v_D}{\tau} \;=\; 0 \tag{8.40}$$

and looking at this equation recognize that $-m_e v_D/\tau$ is a velocity-dependent force which opposes the electric force acting on the conduction electrons. We have already encountered such a braking, velocity-dependent force in the form of a frictional force in volume 1 of this physics course. And indeed, we can interpret the force that opposes the electric force acting on the electron in eq. (8.40) as a frictional force. When discussing frictional forces, we introduced the mobility of uncharged gas particles in the atmosphere as the Stokes frictional force divided by their velocity, i.e., $\mu = F_R^\eta/v$, thereby defining it somewhat differently from the electron mobility $\mu = E/v_D$ used here.

As tab. 8.1 shows, metals are the best conductors under standard conditions. By measuring the specific conductance $\sigma$ and using eq. (8.37) and eq. (8.38), the mobility $\mu$ of the conduction electrons in the metal can be determined, and thus the mean collision time $\tau$ of the electrons between two scattering processes:

$$\tau = \frac{\sigma \cdot m_e}{n_e \cdot e^2} \tag{8.41}$$

For a simple estimate of the order of magnitude of $\tau$, we assume that each atom in the metal contributes one conduction electron. For a metal like copper, this assumption yields an electron density of $n_e = 8.5 \cdot 10^{28}\,\mathrm{m}^{-3}$. According to tab. 8.1, copper has a specific conductance of $\sigma = 1/\rho = 5.85 \cdot 10^7\,\Omega^{-1} \cdot \mathrm{m}^{-1}$ at room temperature, and we thus obtain a value of $\tau = 2.44 \cdot 10^{-14}\,\mathrm{s}$ for the mean collision time in copper at room temperature. This is a typical value for metals in which the mean collision time lies in the range $\tau = 10^{-14} - 10^{-15}\,\mathrm{s}$. The Drude model treats conduction electrons like an ideal gas. Consequently, the thermal velocity $v_{th}$ of the electrons in this model is given by the equipartition theorem:[1]

$$\frac{m_e \cdot v_{th}^2}{2} = \frac{3}{2} \cdot k_B T \tag{8.42}$$

At room temperature, the thermal velocity of the electron gas is $v_{th} \approx 10^5\,\mathrm{m/s}$. Consequently, the mean free path $l = v_{th}\tau$, i.e., the average distance a conduction electron travels between two scattering events in metals, lies in the range of 1–10 Å. The interatomic distance in metals is $1 - 4\,\text{Å}$, suggesting that the primary scattering process in metals is the scattering of conduction electrons by lattice atoms. What Drude could not have known around 1900 was that the classical estimate of electron velocity in metals, based on the kinetic theory of gases, yields a value an order of magnitude lower than the actual speed at which conduction electrons move within the metal at room temperature. We now know that the average speed $\sqrt{\langle \boldsymbol{v}^2 \rangle}$ at which conduction electrons move within a metal depends only very weakly on temperature. The situation is different for the mean

[1] See the section on the Maxwell-Boltzmann distribution in chapter 9 of volume 1 of this physics course.

collision time $\tau$. In impure metals, $\tau$ is determined by scattering off impurity atoms and is significantly lower than in pure metals. In the latter case, $\tau$ is temperature-dependent and, at very low temperatures, can be several orders of magnitude higher than the value of $\tau$ at room temperature. Correspondingly, the mean free path of conduction electrons in pure metals at low temperatures is orders of magnitude greater than the value at room temperature.

Do conduction electrons in metals move relatively fast or relatively slowly? To understand this, let us return to our earlier example: copper. Using eq. (8.38), we can express the mobility $\mu$ of the conduction electrons in terms of the specific conductance $\sigma$ or the specific resistance $\rho$:

$$\mu = \frac{\sigma}{n_e \cdot e} = \frac{1}{n_e \cdot e \cdot \rho} \tag{8.43}$$

If we use the values for $n_e$ and $\rho$ from above, we obtain an electron mobility of $\mu = 4.29 \cdot 10^{-3}\,\mathrm{m^2 \cdot V^{-1} \cdot s^{-1}}$ for copper. We will use again as an example a typical household electrical wire with a cross-section of $2.5\,\mathrm{mm^2}$ carrying a current of a few amperes. This means, we are dealing with a current density of a few $A/\mathrm{mm^2}$. Given the specific conductance $\sigma = 5.85 \cdot 10^7\,\Omega^{-1} \cdot \mathrm{m}^{-1}$ and a current of $4\,\mathrm{A}$, the electric field strength in our copper wire is $E = j\sigma = 6.84 \cdot 10^{-2}\,\mathrm{V \cdot m^{-1}}$. Using eq. (8.37), we thus obtain a drift velocity in copper of $v_D = \mu \cdot E = 2.93 \cdot 10^{-4}\,\mathrm{m/s}$, i.e., only a mere $0.03\,\mathrm{mm/s}$. If the speed at which data can travel through the global information networks of our modern world depended on the drift velocity of conduction electrons in metals like copper, the Internet certainly would not exist as we know it today. However, that is not the case, because the signal speed at which data is transmitted corresponds to the propagation speed of the electric field within the respective medium; in fiber-optic cables, that is about 2/3 of the speed of light.

The Drude model provides a half-classical explanation for Ohm's law. It employs classical kinetic gas theory to the electron gas of the conduction electrons to describe an essentially quantum-mechanical phenomenon: the motion of conduction electrons in metals. The form of Ohm's law provided by eq. (8.39) relies on assumptions regarding the interactions among the conduction electrons themselves and between the conduction electrons and the lattice atoms of a metal. We know today that these assumptions are incorrect, yet the Drude model still provides a qualitatively correct explanation for Ohm's law in the form of eq. (8.39). Quantitatively, however, the Drude model reproduces the temperature-dependent behavior of the specific conductance of metals very poorly, and in certain temperature ranges, the model simply fails.

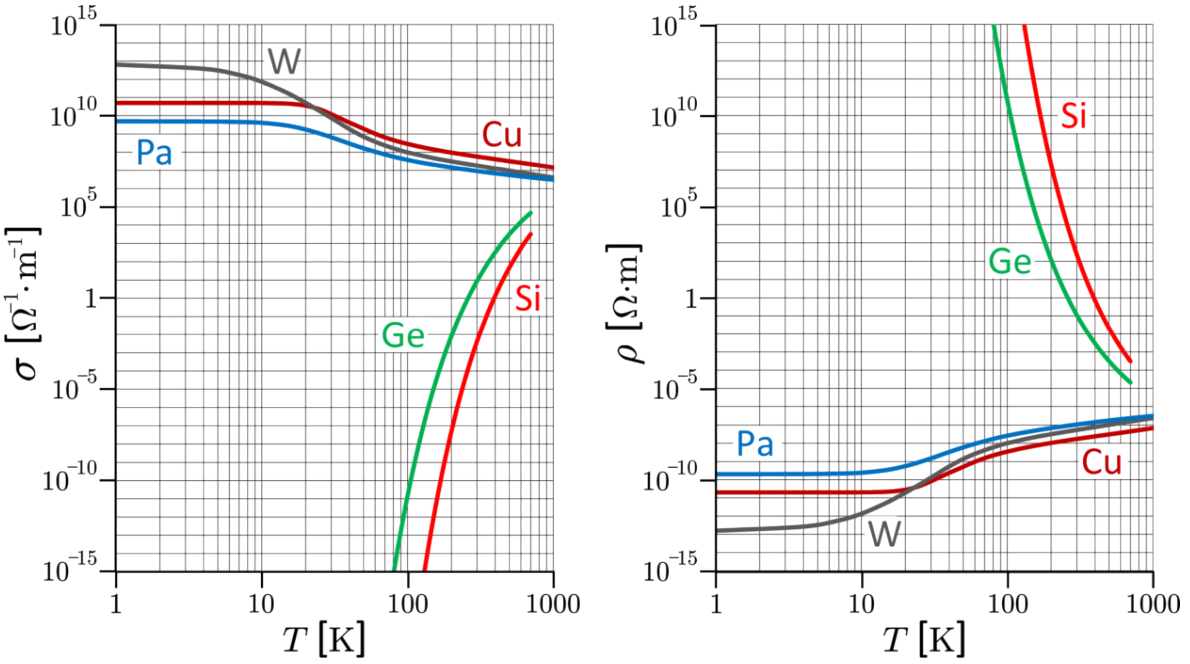



**Fig. 8.12:** Temperature dependence of the specific conductance $\sigma$ (a) and the specific resistance $\rho$ (b) for the metals copper, palladium, and tungsten, as well as the semiconductors silicon and germanium. The curves for the metals are based on experimental data, whereas those for the semiconductors rely on a model (see text).[2,3]

Fig. 8.12a and fig. 8.12b show the behavior of the specific conductance $\sigma$ and the specific resistance $\rho$, respectively, as a function of temperature for three selected metals and two semiconductor materials. At higher temperatures, $\sigma$ and $\rho$ exhibit nearly linear behavior for metals. In this range, specific conductance and specific resistance can be described by two simple equations:

$$\rho = \rho_0[1 + \alpha_0(T - T_0)] \qquad \text{and} \qquad \sigma = \sigma_0/[1 + \alpha_0(T - T_0)] \tag{8.44}$$

Here, $\rho_0$ and $\sigma_0$ are the known measured values of $\rho$ and $\sigma$ at temperature $T_0$, and $\alpha_0$ is the temperature coefficient measured at $T_0$. Eq. (8.44) can then be used to calculate $\rho$ and $\sigma$ in the vicinity of $T_0$. As the temperature of a metal decreases, $\rho$ decreases and $\sigma$ increases. However, below a certain temperature which depends on the particular metal, this decrease or, respectively increase, can no longer be described linearly. At very low temperatures, which in turn depend on the specific material, $\rho$ and $\sigma$ in metals eventually become nearly constant. As we know today, the conduction electrons in a metal do not lose their energy through direct scattering off lattice atoms or through electron-electron

[2] R.A. Matula, J. Phys. Chem. Ref. Data 8, 1147–1298 (1979); https://doi.org/10.1063/1.555614
[3] P. D. Desai, T. K. Chu, H. M. James, and C. Y. Ho, J. Phys. Chem. Ref. Data 13, 1069–1096 (1984); https://doi.org/10.1063/1.555723

scattering, but rather through scattering off collective quantized lattice vibrations which are known as phonons, or off defects and foreign atoms in the crystal lattice, and through relaxation processes from a higher to a lower energy state accompanied by the emission of light quanta. In a perfect crystal, there are no lattice defects or foreign atoms, and at 0 K, all collective lattice vibrations, i.e., all phonons, are frozen out and the electrons themselves are in their lowest energy state. At 0 K, therefore, conduction electrons in metals should no longer lose energy. Consequently, a metals specific resistance should be zero and its specific conductance infinite. Why is this not the case? The reason for this is, firstly, that it is impossible to reach the absolute zero temperature (0 K) and, secondly, that there are no crystals that are 100% defect-free. Most metals, therefore, such as those shown in fig. 8.12, possess a finite residual resistance even at very low temperatures.

However, this does not apply to all metals, as there are some that behave quite differently at very low temperatures. In these so-called superconductors, the specific resistance drops by many orders of magnitude below a critical temperature known as the transition temperature ($T_c$). Superconductors are materials whose specific resistance below $T_c$ is at least $10^{17}$ times lower than that of copper. Superconductivity was first observed in mercury by Kamerlingh Onnes in 1911. As the reproduction of the original measurement data in fig. 8.13 shows, the resistance of Hg changes abruptly at 4.2 K, rising from an immeasurably small value – at the time, immeasurably small meant less than $10^{-6}\,\Omega$ – to over $0.1\,\Omega$ within an interval of 0.01 K. Since then, a number of other superconducting materials have been discovered. Many of these are still metals or metal compounds, but for none of these materials does the transition temperature to superconductivity currently exceed that of $Nb_3Ge$ (23 K) which is 23 K at normal pressure. Such low temperatures can only be achieved using helium cooling. However, helium-cooling is complex and costly, and these superconductors are therefore unattractive for broader technical applications outside of research. In 1986, almost eight decades after Kamerlingh Onnes first observed superconductivity in Hg, the first non-metallic superconductors were discovered. Among them are various ceramic materials that have transition temperatures above 77 K and can therefore be brought into their superconducting state using liquid nitrogen. The

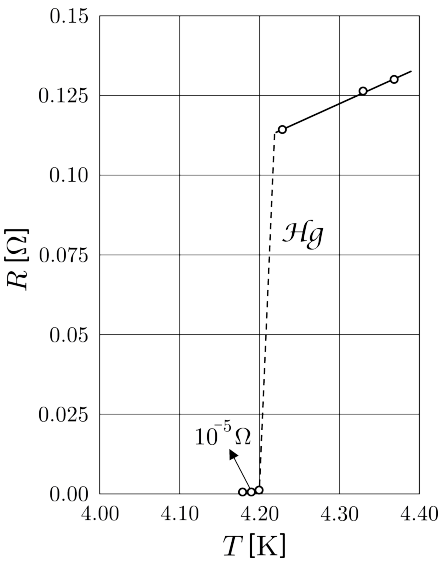



**Fig. 8.13**

record holders are for now copper oxide compounds, so-called cuprates, currently with a maximum transition temperature of 134 K. Some materials exhibit a significantly higher transition temperature under high pressure, though this fact also severely limits their practical applicability.

### 8.2.2 Electron-Hole Conduction in Semiconductors

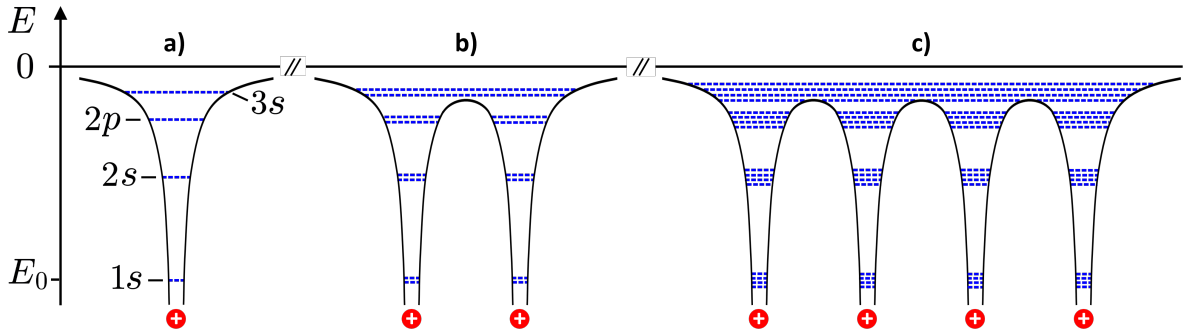



**Fig. 8.14:** Electronic energy levels in the Coulomb potential of an atom (a), in the potential wells of two neighboring atoms (b), and in the potential wells of four atoms arranged in a row (c).

Strictly speaking, Ohm's Law, the linear relationship between electric voltage and electric current expressed in eq. (8.12), often referred to as a linear current-voltage characteristic, applies only to metallic conductors at a constant temperature. However, as most people know from personal experience, electrical devices heat up during operation, and an electrical cable carrying a high current warms up. Consequently, the electrical resistance $R$ is no longer constant but depends on the current flowing through a conductor. Elemental semiconductors such as silicon and germanium, sometimes referred to as semimetals because they possess properties of both metals and non-metals, are non-ohmic conductors by nature.

With respect to electron-hole conduction in semiconductors, we can only target a qualitative understanding of the physics involved as this topic goes significantly beyond the scope of what can be covered here. We begin by looking at what happens when individual atoms are brought together in a solid. Let us consider fig. 8.14a, which schematically shows the electron energy levels within the Coulomb potential of a single atom. The quantum state with the lowest energy level is the 1s state, followed by the 2s, 2p, and 3s states. Two electrons with opposite spins can occupy each of the spherically symmetrical s-orbitals. The dumbbell-shaped p-orbitals are oriented orthogonally to one another, with their respective lobes pointing in the positive and negative directions of the coordinate axes. Again, two electrons with opposite spins can likewise occupy each of these $p_x$-, $p_y$-

and $p_z$-orbitals. The fact that no two electrons in an atom occupy the same quantum state is a consequence of the Pauli principle. This principle states, in general, that no two electrons can ever occupy the same quantum state. Consequently, if two atoms are brought close enough together, as shown in fig. 8.14b, for the corresponding electron states of the individual atoms to overlap, these states must split so that each electron continues to occupy its own distinct quantum state. If four atoms are brought together as shown in fig. 8.14c, the energy levels of the individual atoms split into four energy levels within the combined system. In a solid containing $N$ atoms, this splitting results in $N$ energy levels; since $N$ is very large, these discrete, split energy levels, sketched in fig. 8.14c for four atoms, evolve into energy bands.

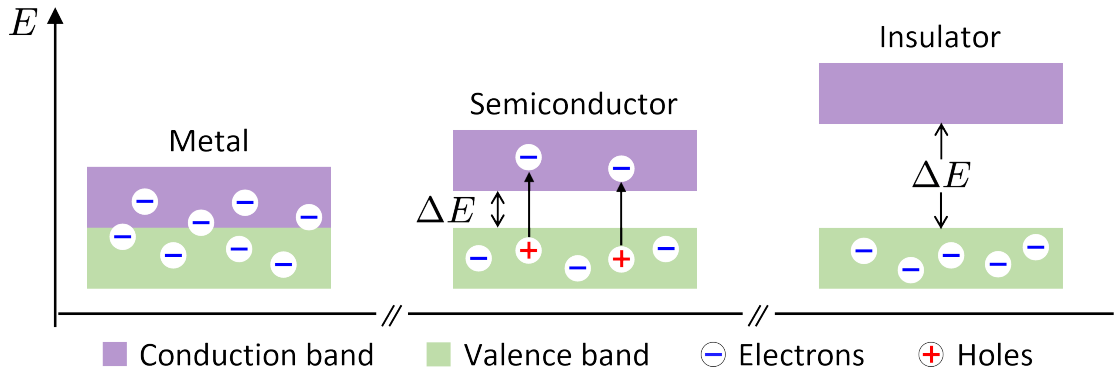



**Fig. 8.15:** Conduction band and valence band in metals, semiconductors, and insulators.

The highest occupied energy band of this type in a solid at absolute zero is the so-called valence band, and the lowest unoccupied energy band is referred to as the conduction band. Fig. 8.15 illustrates the relative energy positions of the valence and conduction bands in metals, semiconductors, and insulators, respectively. In metals, the valence and conduction bands overlap which allows electrons to move freely from the valence band into unoccupied states in the conduction band; alternatively, if the valence band of the metal is only partially filled, electrons can also transition into unoccupied states within the valence band itself. In semiconductors and insulators, the valence band is always completely filled, there are no empty states for electrons to move into. Consequently, electrical conduction within the valence band is impossible. However, in a semiconductor, the energy gap $\Delta E$ is small enough for electrons to transition thermally into the conduction band. With each such transition, the electron in the conduction band and the remaining hole in the valence band contribute to electrical conduction. This fact explains why the temperature dependence of specific conductance and specific resistance in fig. 8.12a and fig. 8.12b for the semiconductors silicon and germanium differs completely from that of metals. For semiconductors, specific conductance increases with temperature, while specific resistance

decreases as temperature rises. As with the three metals examined, the graphs for Si and Ge in fig. 8.12a and fig. 8.12b reflect the behavior of very pure semiconductors. In this case, $\sigma$ and $\rho$ in semiconductors are determined by so-called intrinsic conduction, where the following applies to the charge densities of both charge carriers, negative electrons $n$ and positive electron vacancies $p$, known as holes:

$$n_i = p_i \propto T^{3/2} \exp(E_g/2k_BT) \tag{8.45}$$

Here, $n_i$ and $p_i$ are the respective intrinsic charge carrier densities for electrons and holes in the intrinsic conduction regime, and $E_g$ is the energy gap between the valence and conduction bands of the semiconductor. Fig. 8.12a and fig. 8.12b show the temperature dependence of $\sigma$ and $\rho$ for this model of intrinsic conduction in semiconductors. Only at higher temperatures do electrons in the valence band possess sufficient energy to overcome the energy gap to the conduction band, thereby increasing $n$ by one electron in the conduction band and $p$ by one hole in the valence band. In doped semiconductors, temperature behavior is largely determined by the dopant atoms, making them significantly better conductors at low temperatures than the intrinsic semiconductors considered here. The implantation of $N_D$ electron donors per unit volume, such as phosphorus in silicon, is referred to as $n$-type doping, and the electron charge carrier density is approximately $n \approx N_D$. Accordingly, the implantation of $N_A$ electron acceptors per unit volume, such as boron in silicon, is referred to as $p$-doping, and the hole carrier density is approximately given by $p \approx N_A$. Regardless of the doping level, at a constant temperature, the product of the electron charge density and the hole charge density satisfies the following condition:

$$n \cdot p = n_i^2 \tag{8.46}$$

As the temperature decreases, the intrinsic charge carrier densities $n_i$ and $p_i$ fall according to eq. (8.45). At absolute zero temperature, $n_i = p_i = 0$ holds, and based on eq. (8.46), even a doped semiconductor becomes an insulator. Semiconductors and insulators are alike in that, in both materials, all states within the valence band are filled. A completely filled valence band means that no current can flow within it, because, when an electric field is applied, there are no available states in the valence band for electrons to move through continuously. In semiconductors, the energy gap $\Delta E$ is small enough to allow electrons to jump into the conduction band as the temperature rises. Following each such jump, both the negative electron in the conduction band and the vacancy it leaves behind in the valence band, the positively charged hole, can contribute to electrical conduction. If a voltage $U$ is applied, electrons in the semiconductor can also transition non-thermally

from the energy band to the conduction band when the condition $e \cdot U \geq \Delta E$ is met. This does not work for insulators, because in these materials $\Delta E$ is much larger than in semiconductors, and the voltages required to overcome such a large $\Delta E$ result in structural destruction of the insulator.

### 8.2.3 Ionic Conduction in Electrolytes

Electrolytes are solid or liquid substances in which ions are the charge carriers of electric currents. The simplest electrolyte, and one with which almost everyone is likely familiar, is a common saline solution; in other words, the salt in our soup. If we use distilled water as the solvent to avoid impurities and dissolve NaCl in it, we obtain a saline solution containing $Na^+$ ions and $Cl^-$ ions that serve as charge carriers for conducting electricity. If two metal plates, connected to the positive and negative terminals of a voltage source, are then immersed in the saline solution at a distance $l$ from each other, as sketched in fig. 8.16, then a voltage drop $U$ occurs across the electrolyte between these electrodes. Under the influence of the associated electric field $E = U/l$, the cations, in this case, the $Na^+$ ions, move towards the negative cathode, and the anions, in this case, the $Cl^-$ ions, move towards the positive anode. At the cathode, the cation is neutralized by an electron from the cathode, while conversely, the anion releases an electron at the anode. Thus, cations are reduced at the cathode, and anions are oxidized at the anode. Just as is the case for conduction electrons in metals, ions in electrolytes are subject to a frictional force proportional to their velocity. Consequently, the drift velocity of cations and anions in an electrolyte, like that of conduction electrons in a metal, is proportional to the electric field:

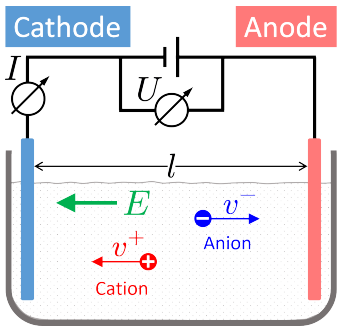



**Fig. 8.16**

$$v^+ = \mu^+ \cdot E \qquad \text{and} \qquad v^- = -\mu^- \cdot E \tag{8.47}$$

Here, $\mu^+$ and $\mu^-$ are the mobilities of the cations and anions, respectively, in the electrolyte. The minus sign for the anions in eq. (8.47) arises because, unlike the cations, they move against the direction of the field $\boldsymbol{E}$. In the case of a saline solution, we are dealing with monovalent ions, but this is not always the case. If $Z_+$ and $Z_-$ denote the valencies of the cations and anions present in the electrolyte, and $n_+$ and $n_-$ their respective particle densities, then for the current densities $j_+$ and $j_-$ in the electrolyte, we obtain with eq. (8.47):

$$j_+ \;=\; Z_+ \cdot n_+ \cdot e \cdot v^+ \;=\; Z_+ \cdot n_+ \cdot e \cdot \mu^+ \cdot E \tag{8.48a}$$

and

$$j_- \;=\; Z_- \cdot n_- \cdot (-e) \cdot v^- \;=\; Z_- \cdot n_- \cdot e \cdot \mu^- \cdot E \tag{8.48b}$$

The total current in the electrolyte is therefore:

$$j \;=\; j_+ + j_- \;=\; (Z_+ \cdot n_+ \cdot \mu^+ + Z_- \cdot n_- \cdot \mu^-) \cdot e \cdot E \tag{8.49}$$

For the conductivity of the electrolyte $\sigma = j/E$, one thus obtains:

$$\sigma \;=\; (Z_+ \cdot n_+ \cdot \mu^+ + Z_- \cdot n_- \cdot \mu^-) \cdot e \tag{8.50}$$

If the surface areas of the metal electrodes in fig. 8.16 are equal to $A$, then the current flowing through the electrolyte at an applied voltage of $U = E \cdot l$ is:

$$I = j \cdot A \;=\; (Z_+ \cdot n_+ \cdot \mu^+ + Z_- n_- \cdot \mu^-) \cdot e \cdot \frac{U}{l} \cdot A \tag{8.51}$$

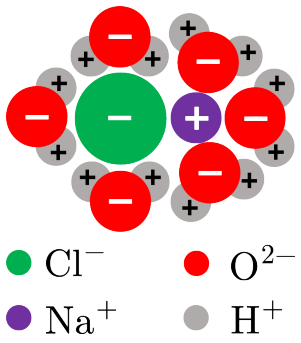


**Fig. 8.17**

In this discussion we have taken for granted that NaCl dissolves in water. Of course, we know from experience that this is the case. But the fact that a current $I$ flows when a voltage $U$ is applied between two metal plates immersed in a saline solution serves as definite proof of this. But why does NaCl dissolve in water even at room temperature? The explanation lies in the special property of water that we discussed in the last chapter. $H_2O$ is a polar molecule, meaning it possesses a permanent dipole moment, and in water, this dipole moment is significantly larger than in the gas phase due to hydrogen bonding. Technically speaking, NaCl is not a polar molecule, yet it behaves like one. The NaCl bond is formed by the transfer of an electron from the Na atom to the Cl atom, whereby both achieve a noble gas configuration (Ne for $Na^+$ and Ar for $Cl^-$). Thus, unlike with the covalent bonds in $H_2O$, the electrons in the NaCl molecule are not shared but transferred. The resulting positive $Na^+$ ion and negative $Cl^-$ ion attract each other electrostatically, thereby forming not a covalent bond but an ionic bond. As shown schematically in fig. 8.17, polar $H_2O$ molecules arrange themselves around an NaCl molecule in such a way that the $Na^+$ ion is surrounded by the negative ends of the $H_2O$ molecules and the $Cl^-$ ion by the positive ends of the $H_2O$ molecules. As a result, opposing forces act on the $Cl^-$ end and the $Na^+$ end of the NaCl molecule, pulling it apart. If these forces exceed the electrostatic force holding the NaCl molecule together, dissociation occurs. Water has a very high dielectric constant of $\epsilon = \epsilon_r \cdot \epsilon_0 \approx 80$ (see tab. 6.2). However, this means that the Coulomb force of

electrostatic attraction between $Na^+$ and $Cl^-$ in water is only about 1/80 of its value in an atmospheric environment. The ionic bond of the NaCl molecule is thus significantly weaker in water than in air, making it much easier for the water molecules in fig. 8.17 to separate the $Cl^-$ ion from the $Na^+$ ion. The Coulomb potential generated by a charge $Q$ in the dielectric medium water is:

$$\phi(r) = \frac{1}{\epsilon_r}\frac{1}{4\pi\epsilon_0}\frac{Q}{r} \tag{8.52}$$

If a charge $q$ is brought from infinity to a distance $r_0$ from the charge $Q$, with the entire system situated in water, then this charge $q$ possesses the potential energy (by definition, $\phi(\infty) = 0$):

$$E_{pot}(r) = q[\phi(r_0) - \phi(\infty)] = \frac{1}{\epsilon_r}\frac{1}{4\pi\epsilon_0}\frac{q{\cdot}Q}{r} \tag{8.53}$$

Let $Q = -e$ be the charge of the $Cl^-$ ion and $q = +e$ that of the $Na^+$ ion. Then the potential energy, i.e., the ionic bond energy of NaCl, at a bond distance of $r_0 = 2.36\,\text{Å}$ is given by:

$$E_{pot}(r_0) = -\frac{8.99{\cdot}10^9\,\text{Nm}^2\text{C}^{-2}}{80}\frac{(1.60{\cdot}10^{-19})^2\,\text{C}^2}{2.36{\cdot}10^{-10}\,\text{m}} = 1.22{\cdot}10^{-20}\,\text{J} = 76.1\,\text{meV}$$

From chapter 9 in volume 1, we know that a diatomic molecule has 6 degrees of freedom: 3 translational, 2 rotational, and 1 vibrational degree of freedom. Each of these degrees of freedom is associated with an energy equivalent of $k_BT/2$. At room temperature, i.e., at 300 K, $k_BT \approx 26\text{meV}$. The thermal energy of NaCl at room temperature and $E_{pot}$ from eq. (8.53) are of the same order of magnitude. Therefore, at room temperature no additional energy is required to dissociate NaCl in water.

The flow of an electric current in an electrolyte, i.e., the flow of positively and negatively charged ions, is always accompanied by the transport of matter, whereby cations are deposited at the cathode and anions at the anode. According to Faraday's law of electrolysis, the amount of substance deposited at the respective electrodes is proportional to the electric charge flowing through the electrolyte:

$$\frac{Q_F}{1\,\text{mol}} = e{\cdot}N_A = F \tag{8.54}$$

Here, $N_A$ is the Avogadro constant, $F = 96485\,\text{C}{\cdot}\text{mol}^{-1}$ is the Faraday constant, and $Q_F$ is the charge required to electrolytically deposit one mole of a monovalent ion. If one considers a substance with valence $Z$, then $e$ in eq. (8.54) must be replaced by $Z{\cdot}e$. One mole of a substance contains exactly $N_A$ atoms or molecules of that substance. The amount of charge $Q_F/N_A$ required per atom of a substance for electrolytic deposition is therefore, according to eq. (8.54):

$$\frac{Q_F}{N_A} = Z \cdot e = Z \cdot \frac{F}{N_A} \tag{8.55}$$

The charge $q$ required to deposit one atom of a substance is always an integer multiple of the elementary charge $e$, and the same applies to the ratio $F/N_A$. By measuring the amount of substance in $n$ mol of a $Z$-valent ion deposited at an electrode while simultaneously measuring the necessary quantity of charge $Q$, one can thus determine the Avogadro constant $N_A$ using the relationship $F = Q/n$.

Our discussion of ionic conduction in electrolytes has treated them as ohmic conductors, even though this was not explicitly stated. The prerequisite for ohmic current conduction is a constant electrical resistance. The latter follows from the proportionality between the drift velocity of the charge carriers, $\boldsymbol{v}_D$, and the electric field, $\boldsymbol{E}$, as expressed by eq. (8.37) for electron conduction in metals and eq. (8.47) for ionic conduction in electrolytes. Many electrolytes are indeed very good ohmic conductors. However, the mobility of ions in electrolytes is about three orders of magnitude lower than that of electrons in metals. Furthermore, ionic conduction in electrolytes involves an additional factor not present in electronic conduction in metals: the concentration of charge carriers within the electrolyte. In many electrolytes, electrical conductivity is a linear function of the concentration of the dissolved substance over several orders of magnitude. In the case of NaCl, this holds true for saline solutions over a range of salt concentrations spanning slightly more than four orders of magnitude. At high concentrations, however, ion mobility decreases; the linear relationship between electrical conductivity and electrolyte concentration breaks down, and ionic conduction within the electrolyte ceases to be ohmic.

# 9. Fundamentals of Magnetostatics

In exploring electrostatics, our starting point was the stationary electric point charge, or, more generally, stationary charge distributions, whose electrostatic potentials $\phi$ and electrostatic fields $\boldsymbol{E}$ we subsequently examined. However, free magnetic charges do not exist. The sources of static magnetic fields, i.e., of time-independent magnetic fields, are stationary electric currents. These current sources of the magnetic field, in the form of infinitesimal current loops, give rise to a magnetic dipole; magnetic monopoles do not exist. In magnetostatics, the magnetic north and south poles of such a magnetic dipole correspond to the positive and negative electric point charges in electrostatics. If a permanent magnet, such as a bar magnet, is split into two parts, each of these parts again possesses a magnetic north pole and a magnetic south pole. If one repeats the process and splits these two parts in turn, the same thing happens. A permanent magnet always remains a magnet, no matter how often one divides it. The magnetic properties of materials can be traced back to electric currents at the atomic and molecular levels. For the magnetic field, the fundamental quantity is the moving electric charge, that means, electric currents.

## 9.1 The Lorentz Force

The fundamental equations of magnetostatics describe the interaction between stationary electric currents and the magnetic fields they generate. We are familiar with the definition of the electric current from eq. (8.1), and we introduced the concept of stationary currents in the previous chapter in section 8.1. However, before turning to the magnetic fields generated by electric currents and the interaction between electric currents and magnetic fields, let us consider the force exerted by a magnetic field on a point charge $q$. Just as static electric point charges generate electrostatic potentials $\phi$ and electrostatic fields $\boldsymbol{E}$, which in turn exert a force on other static electric point charges, stationary electric currents generate magnetic potentials $\boldsymbol{A}$ and magnetic fields $\boldsymbol{B}$, which in turn exert

a force on moving electric charges. According to eq. (3.2), the only force acting on a stationary point charge $q$ in an electrostatic field $\boldsymbol{E}$ is $\boldsymbol{F} = q \cdot \boldsymbol{E}$. A magnetic field $\boldsymbol{B}$ exerts no force whatsoever on an identical stationary point charge $q$ located within it. Only when the point charge $q$ moves with a velocity $\boldsymbol{v}$ in the magnetic field $\boldsymbol{B}$, i.e., if it essentially becomes the equivalent of a localized electric current, does $\boldsymbol{B}$ exert a force on $q$, the so-called Lorentz force:

$$\boldsymbol{F}_L = q \cdot (\boldsymbol{v} \times \boldsymbol{B}) \tag{9.1}$$

This Lorentz force acts on the point charge $q$ perpendicular to the directions of $\boldsymbol{v}$ and $\boldsymbol{B}$. For a positive point charge $q$, the direction of the Lorentz force is determined by the right-hand rule. To apply this, the thumb, index finger, and middle finger of the right hand are positioned at 90-degree angles to one another, an arrangement that is possible in only one way. If the thumb is aligned with the direction of motion of the positive point charge $q$ and the index finger with the direction of the magnetic field $\boldsymbol{B}$, then the middle finger indicates the direction of the Lorentz force. For a negative charge, the force acts in the direction exactly opposite to that indicated by the middle finger. In the example shown in fig. 9.1, the thumb points in the $y$-direction, the index finger in the $z$-direction, and the middle finger in the $x$-direction. For a negative point charge, the force in fig. 9.1 would act in the negative $x$-direction. The combined force exerted by an electrostatic field $\boldsymbol{E}$ and a static magnetic field $\boldsymbol{B}$ on a point charge $q$ moving with a velocity $\boldsymbol{v}$ in these fields is:

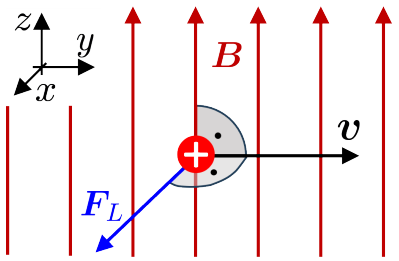



**Fig. 9.1**

$$\boldsymbol{F} = q \cdot (\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) = \boldsymbol{F}_C + \boldsymbol{F}_L \tag{9.2}$$

The total force acting on the point charge $q$ is the sum of the Coulomb force $\boldsymbol{F}_C$ and the Lorentz force $\boldsymbol{F}_L$. At times, this total force is referred to, in a broader sense, as the Lorentz force. However, here we will stick to the definition in the narrower sense, where $\boldsymbol{F}_L$ describes only the force exerted by the magnetic field on the point charge $q$ and does not include the electrostatic force. In the past, one could easily demonstrate the Lorentz force in the living room by holding a strong magnet close to the television and thereby causing image distortion. The force exerted by the magnet on the electrons in the television's cathode-ray tube affected their trajectories, thereby distorting the television image. With modern television sets that do not operate using electron beams, this is no longer possible.

### 9.1.1 Force on a Current-Carrying Wire

Eq. (9.1) describes the experimental finding regarding the force exerted by a magnetic field $\boldsymbol{B}$ on a single moving charge $q$, for example, the force exerted on an electron moving through a magnetic field. As illustrated for a single electron in fig. 9.2, this force $\boldsymbol{F}_L$ acts in the same way on each individual conduction electron in a current-carrying wire situated in a magnetic field. As we learned in the last chapter, these conduction electrons move through a current-carrying wire with a drift velocity $v_D$. The direction of $v_D$, i.e., the vector $\boldsymbol{v}_D$, is determined by the orientation of the wire, with the electrons flowing from the end of the wire which is at a negative potential to the end that is at a more positive potential. However, in a current-carrying wire, it is not just a single electron that moves from a negative potential towards a more positive potential, but many. We must therefore sum the forces acting on these many charges $q$. We can do this by mentally isolating a small section of the wire with volume $\Delta V = A \cdot \Delta L$, as illustrated in fig. 9.2, where $A$ is the cross-sectional area of the wire and $\Delta L$ is the length of the wire section under consideration. If the electron density $n_e$ within the wire is uniform, the total force $\Delta \boldsymbol{F}$ acting on the charges in this wire section is then given by:

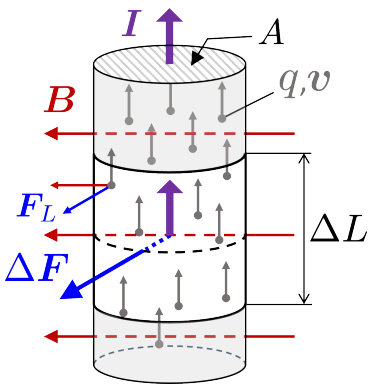



**Fig. 9.2**

$$\Delta \boldsymbol{F} = n_e \cdot \Delta V \cdot q \cdot \boldsymbol{v}_D \times \boldsymbol{B} = (n_e \cdot q \cdot \boldsymbol{v}_D \times \boldsymbol{B}) \cdot A \cdot \Delta L = (\boldsymbol{j} \times \boldsymbol{B}) \cdot A \cdot \Delta L \tag{9.3}$$

Here, in the final step, we have once again made use of eq. (8.3). Using eq. (8.4), we can express $\boldsymbol{j} \cdot A$ as a current vector, thereby obtaining the following expression for the Lorentz force $\Delta \boldsymbol{F}$ acting on the wire segment of length $\Delta L$:

$$\Delta \boldsymbol{F} = (\boldsymbol{I} \times \boldsymbol{B}) \cdot \Delta L \tag{9.4}$$

Thus, a force per unit length acts on the current-carrying wire, corresponding to the cross product $\boldsymbol{I} \times \boldsymbol{B}$. This force depends only on the direction and magnitude of the current flow $\boldsymbol{I}$ and on the direction and magnitude of the magnetic field $\boldsymbol{B}$. It does not matter what kind of particles carry the electric current or what electric charge they possess. Only the magnitude and direction of the current are relevant to the force exerted on the wire in a given magnetic field. If we reverse the polarity of the potential applied along the wire, and let the current flow in the opposite direction, the force also points in the exact opposite direction.

## 9.2 Ampère's Law

A current-carrying wire is subject not only to the Lorentz force exerted by an external magnetic field. The wire itself is also the source of a magnetic field. This can be easily demonstrated by stretching two wires parallel to each other in a region free of external magnetic fields, without them touching, and passing the same current through both. The result is that the two wires attract each other. Conversely, if currents of equal magnitude flow through the two wires in opposite directions, they repel each other. Fig. 9.3 illustrates these situations: the two wires attract each other when the current flows in parallel directions, and they repel each other when the current flows in antiparallel directions. The direction of the force is perpendicular to the direction of the current in each case. Thus, the direction of the generated magnetic field and the direction of the force must span a plane that is perpendicular to the respective wire. Magnetic charges, i.e., magnetic monopoles, which could serve as sources or sinks of the field, do not exist. Therefore, magnetic field lines must always form closed loops and consequently the divergence of the magnetic field is zero, that is

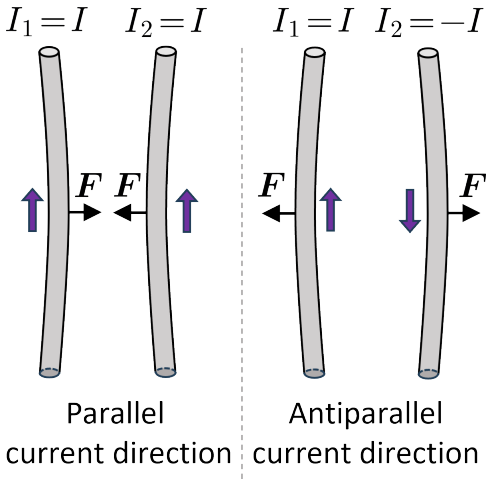



**Fig. 9.3**

$$\nabla \boldsymbol{B} = 0 \tag{9.5}$$

Eq. (9.5) is the first Maxwell equation of magnetostatics in differential form. We obtain the integral form of eq. (9.5) by considering the magnetic flux $\Phi$ of the magnetic field through an open surface $A$. This is given by:

$$\Phi = \int_A \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} \tag{9.6}$$

Now consider a closed surface $S$ enclosing a volume $\Omega$. Then, using eq. (9.5), it follows from Gauss's theorem for the magnetic flux $\Phi$:

$$\Phi = \int_S \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} = \int_V \nabla \boldsymbol{B}\,\mathrm{d}^3 r = 0 \tag{9.7}$$

The integral form of the first Maxwell equation of magnetostatics is thus

$$\int_S \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} = 0 \tag{9.8}$$

With eq. (2.35), it follows immediately from $\nabla \boldsymbol{B} = 0$ that $\boldsymbol{B}(\boldsymbol{r})$ must be the curl of a vector potential $\boldsymbol{A}(\boldsymbol{r})$, that is

$$\boldsymbol{B}(\boldsymbol{r}) = \nabla \times \boldsymbol{A}(\boldsymbol{r}) \tag{9.9}$$

From eq. (9.6) and eq. (9.9), using Stokes' theorem, it follows for the magnetic flux through an open surface $A$:

$$\Phi = \int_A \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} = \int_A (\nabla \times \boldsymbol{A})\,\mathrm{d}\boldsymbol{\sigma} = \oint_C \boldsymbol{A}\,\mathrm{d}s \tag{9.10}$$

where the closed curve $C$ is the boundary of the open surface $A$. Eq. (9.10) tells us that the magnetic flux through an open surface depends only on the value of the vector potential $\boldsymbol{A}$ at the boundary of that open surface. If the open surface $A$ transforms into a closed surface $S$ enclosing a volume, the boundary $C$ vanishes and the line integral becomes zero. Thus, we again obtain $\Phi = 0$, the result from eq. (9.7). The net magnetic flux out of a volume enclosed by a surface $S$ must always be zero. As already derived above, the field lines of the magnetic field generated by a current-carrying wire, as shown in fig. 9.4, can only form concentric circles around the wire, since the magnetic field is source- and sink-free. It was Christian Oersted who discovered in 1820 that a wire carrying an electric current alters the orientation of a nearby compass needle. After a detailed investigation, he explained this effect as follows: The current-carrying wire generates a magnetic field around itself, the strength of which is proportional to the current $I$ and inversely proportional to the distance $r$ from the wire's axis. Today, we express this principle discovered by Oersted as

$$B = \frac{\mu_0}{2\pi} \cdot \frac{I}{r} \tag{9.11}$$

where the quantity $\mu_0$ is the magnetic field constant, which is related to the electric field constant $\epsilon_0$ via the speed of light $c$:

$$c^2 = \frac{1}{\mu_0 \epsilon_0} \tag{9.12}$$

The field lines of the magnetic field generated by the current $I$ are concentric circles. If we express the current as a vector, then, as can be easily verified by considering magnitudes, the vector form of eq. (9.11) is:

$$\boldsymbol{B} = \frac{\mu_0}{2\pi} \cdot \frac{\boldsymbol{I} \times \boldsymbol{r}}{r^2} \tag{9.13}$$

The direction of the magnetic field generated by the wire depends on the direction of the current. It can easily be determined using the right-hand rule: if one makes a thumbs-up

sign using the right hand, pointing the thumb in the direction of the current flow, then the other four fingers, curled toward the palm, indicate the direction of the magnetic field lines.

Fig. 9.4 illustrates this for two wires with opposite current directions. In one case, the direction of the concentric magnetic field is counter-clockwise, while in the other, it is clockwise. Just as with the Coulomb force, the principle of superposition applies to the Lorentz force. If the two wires shown in fig. 9.4 are therefore brought close together, the magnetic field between them intensifies because the field lines point in the same direction. The magnetic field between the wires is stronger than outside them, resulting in a net repulsive force between the wires. However, if the current in the wires flows in the same direction, the magnetic field lines between the wires run in opposite directions; as a result, the fields between the wires partially cancel each other out, creating a field there that is weaker than the fields on the outer sides of the wires. Consequently, this results in a net attractive force between the two parallel wires.

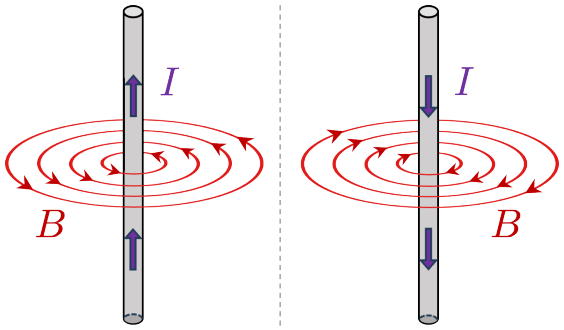



**Fig. 9.4**

The Maxwell equations of electrostatics (see tab. 3.1) tell us that the electrostatic field $\boldsymbol{E}$ has sources and sinks and that it is vortex-free. Regarding the magnetostatic field $\boldsymbol{B}$, we so far only know that magnetic field lines are closed curves and that, consequently, magnetic fields have neither sources nor sinks. But what about vortices: is the magnetic field also vortex-free? To determine this, we can either examine the line integral of the magnetic field along a closed path, i.e., its circulation, or calculate the curl of the magnetic field. The circulation of the magnetic field produced by a current-carrying wire in eq. (9.11) is easily calculated by choosing a closed circle of radius $r$ around the wire as the path of integration:

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} = \frac{\mu_0 I}{2\pi}\int_0^{2\pi}\frac{1}{r}\cdot r\,\mathrm{d}\varphi = \frac{\mu_0 I}{2\pi}\cdot 2\pi = \mu_0 I \tag{9.14}$$

For any arbitrary closed path $C$, this line integral depends only on the current $I$ enclosed by $C$, multiplied by $\mu_0$. If the path $C$ does not enclose a current-carrying conductor, the integral is zero. The equation

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} = \mu_0 I \tag{9.15}$$

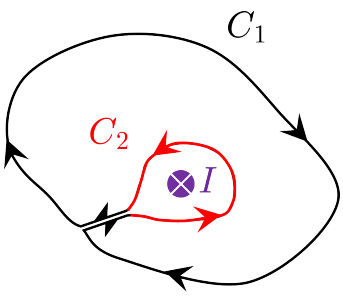



**Fig. 9.5**

is known as Ampère's law, named after André-Marie Ampère, who was the first to formulate it. It states that the circulation of the magnetic field along a closed curve $C$ is proportional to the electric current enclosed by that curve. Let us consider a path integral $C$ consisting of two segments, $C_1$ and $C_2$, which, as sketched in fig. 9.5, shall closely loop around a conductor carrying a current $I$ but without actually enclosing the conductor itself. The line integral eq. (9.15) is therefore zero for the path $C = C_1 + C_2$, since it does not enclose the current-carrying conductor. If the bottleneck connecting $C_1$ and $C_2$ closes, the line integrals over the corresponding segments of the bottleneck cancel each other out, and the path segments $C_1$ and $C_2$ separate. Once $C_1$ and $C_2$ are separated, each of these closed curves encloses the current-carrying conductor. However, they no longer share the same sense of circulation, as $C_2$ is traversed counter-clockwise and $C_1$ clockwise. Expressing in formulas what we have just described in words, this means:

$$\oint_{C_1+C_2} \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} \;=\; 0 \;=\; \oint_{C_1} \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} + \oint_{C_2} \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} \;=\; \oint_{C_1} \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} - \oint_{-C_2} \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma}$$

It follows:

$$\oint_{C_1} \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} \;=\; \oint_{-C_2} \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} \;=\; \mu_0 I$$

$C_1$ and $-C_2$ have the same sense of circulation and this equation therefore tells us that, regardless of how we choose the closed curve $C$ in eq. (9.15), the result will always be $\mu_0 I$ as long as it encloses the current $I$, and zero otherwise. By using eq. (8.5) to replace the current $I$ with the current density $\boldsymbol{j}$, we can write Ampère's law from eq. (9.15) in an even more general form as:

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} \;=\; \mu_0 \oint_A \boldsymbol{j}\,\mathrm{d}\boldsymbol{\sigma} \tag{9.16}$$

Using Stokes' theorem from eq. (2.46), we know that we can write the following for the left-hand side of this equation:

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} \;=\; \int_A (\nabla \times \boldsymbol{B})\,\mathrm{d}\boldsymbol{\sigma} \tag{9.17}$$

The right-hand sides of eq. (9.16) and eq. (9.17) must consequently be identical, and thus it must hold that:

$$\int_A \Big[(\nabla \times \boldsymbol{B}) - \mu_0 \boldsymbol{j}\Big]\,\mathrm{d}\boldsymbol{\sigma} \;=\; 0 \tag{9.18}$$

For eq. (9.18) to be satisfied, the integrand must vanish. With $\mu_0 = 1/c^2\epsilon_0$, it must therefore hold that:

$$c^2 \nabla \times \boldsymbol{B} = \frac{\boldsymbol{j}}{\epsilon_0} \tag{9.19}$$

Eq. (9.19) is the second Maxwell equation of magnetostatics. As we have seen, it follows directly from Ampère's law via Stokes' theorem.

### 9.2.1 Magnetic Field Inside a Current-Carrying Conductor

Using what we just have learned, we can determine the magnetic field inside a conductor using Ampère's law by placing the closed path $C$ from eq. (9.15) inside the conductor. Let us consider a cylindrical conductor of radius $a$ through which a current $I_0$ flows with a constant current density across the conductor's cross-section. The current enclosed by a circular integration path of radius $r \leq a$ inside the conductor is then equal to $I_0 r^2/a^2$, and eq. (9.15) yields the following expression for the corresponding magnetic field:

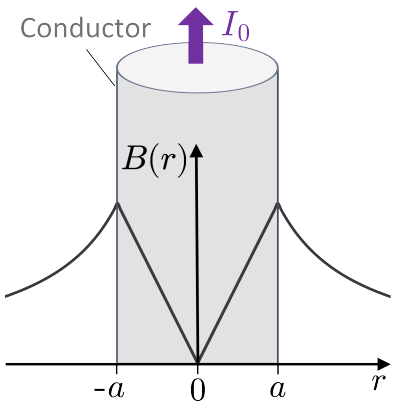



**Fig. 9.6**

$$B(r) \cdot 2\pi r = \mu_0 \cdot I_0 \cdot \frac{r^2}{a^2}$$

It follows:

$$B(r) = \frac{\mu_0 I_0}{2\pi a^2} \cdot r \tag{9.20}$$

Fig. 9.6 shows a cross-sectional sketch of the magnetic field profile inside and outside the conductor. At the conductor's surface, the value corresponds to eq. (9.11) for $r = a$. From there, the magnetic field inside the conductor decreases linearly until it vanishes at the center. Outside the conductor, the magnetic field falls off as $r^{-1}$ in accordance with eq. (9.11). We have written the magnetic field here as a scalar, but it is, of course, a vector quantity. We know that the magnetic field must be perpendicular both to the direction of the current and to the radius vector $\boldsymbol{r}$. With eq. (9.14) we can write the magnetic field vector inside and outside the current-carrying conductor in fig. 9.6 as

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{2\pi} \cdot \frac{(\boldsymbol{I}_0 \times \boldsymbol{r})}{a^2} \qquad \text{in the conductor} \tag{9.21}$$

and

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{2\pi} \cdot \frac{(\boldsymbol{I}_0 \times \boldsymbol{r})}{r^2} \qquad \text{outside the conductor} \tag{9.22}$$

In section 4.3, we determined the electric field of a coaxial cable with static charges of opposite polarity on the inner and outer conductors. There, we learned that the electric field outside the coaxial cable vanishes. Stationary currents flowing in opposite directions in the inner and outer conductors of the coaxial cable correspond to moving positive and negative charge densities on the inner and outer conductors, respectively. Consequently, the electric field outside the coaxial cable remains zero. Typically, as shown in fig. 9.7, a current $I_0$ flows through the inner conductor, while an equal and opposite current $-I_0$ flows through the outer conductor. The current enclosed by a circular path integral running outside the outer conductor is therefore $I_0 - I_0 = 0$ and, according to eq. (9.14), the magnetic field outside the coaxial cable is zero. An integration path running between the inner and outer conductors encloses only the current $I_0$ of the inner conductor, and the magnetic field between the inner and outer conductors of the coaxial cable is given by eq. (9.14) with $I = I_0$. However, what can Ampère's law tell us about the magnetic field within the outer conductor? To this end, we consider the integration path $r$ with $r_{a_1} \leq r \leq r_{a_2}$ shown in fig. 9.7. Evidently, for this integration path the magnetic field is determined by the contribution from the current in the inner conductor and the portion of the outer conductor's current enclosed within the radius $r$:

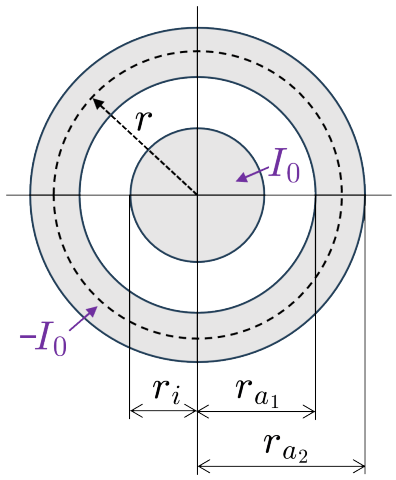



**Fig. 9.7**

$$B(r) = \mu_0 I_0 - \mu_0 I_0 \cdot \frac{\pi r^2 - \pi r_{a_1}^2}{\pi r_{a_2}^2 - \pi r_{a_1}^2} \quad , \quad r_{a_1} \leq r \leq r_{a_2}$$

We can further summarize this as:

$$B(r) = \mu_0 I_0 \cdot \frac{r_{a_2}^2 - r^2}{r_{a_2}^2 - r_{a_1}^2} \quad , \quad r_{a_1} \leq r \leq r_{a_2} \tag{9.23}$$

Using eq. (9.20), eq. (9.11), and eq. (9.23), we know the complete profile of the magnetic field strength within the coaxial cable from the inside out. Fig. 9.8 illustrates this profile for a hypothetical coaxial cable in which $r_i$, $r_{a_1}$, and $r_{a_2}$ have been chosen to be equidistant for visualization purposes. Within the outer conductor, as $r$ increases, the fraction of $-I_0$ that contributes towards compensating the magnetic field generated by the current $I_0$ rises,

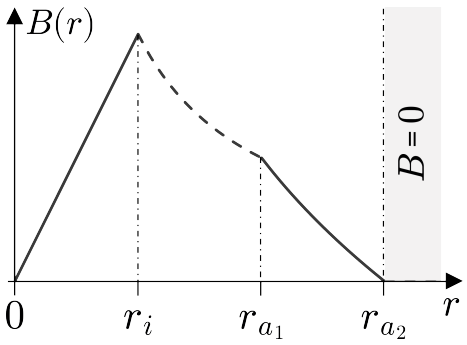



**Fig. 9.8**

until, at $r = r_{a_2}$, the entire current $-I_0$ contributes and the magnetic field strength drops to zero. What can we say about the $\boldsymbol{B}$ field lines within the coaxial cable? Naturally, they must run radially around the axis of the coaxial cable. But in which direction? The direction of the $\boldsymbol{B}$ field lines is determined by the direction of the electric current in the coaxial cable. Up to $r = r_{a_1}$, this is the current vector $\boldsymbol{I}_0$ in the inner conductor. For $r > r_{a_1}$, one begins adding components of the current vector $-\boldsymbol{I}_0$ to $\boldsymbol{I}_0$, yet the total current vector continues to point in the direction of $\boldsymbol{I}_0$; this continues until, at $r = r_{a_2}$, the current vectors of the inner and outer conductors cancel each other out, resulting in the zero vector. The orientation of the $\boldsymbol{B}$ field lines throughout the coaxial cable is thus determined by the direction of the current in the inner conductor.

### 9.2.2 Magnetic Field of a Current-Carrying Coil

We can easily calculate the magnetic field inside a current-carrying coil whose length is much greater than its diameter. For a single circular current loop, we know that the magnetic field inside the loop must pass through the plane of the loop, perpendicular to it. Just as two current-carrying wires running parallel to each other attract one another when the current flows in the same direction, two current loops in which the current flows in the same direction also attract each other. If we place an identical second current-carrying ring over the first one such that the planes of the rings lie parallel to each other, the rings attract one another, and the magnetic field passing perpendicularly through both ring planes is stronger than that of a single ring. In a magnetic field coil, also known as an inductor, such as the one sketched in fig. 9.9, many such rings are stacked on top of one another, with the current flowing in the same direction in all of them. Such an inductor can therefore generate a much stronger magnetic field inside itself that runs parallel to the inductor axis. If the length of the inductor is much greater than its diameter, we can neglect the magnetic field $\boldsymbol{B}_a$ in the region outside the inductor compared to the magnetic field $\boldsymbol{B}_i$ inside it. If we now calculate the line integral of the magnetic field along a path that encloses $N$ turns on the left side of the inductor, as shown in fig. 9.9, we first observe that the segments of the path inside the inductor, which run perpendicular to the internal magnetic field $\boldsymbol{B}_i$, make no contribution. As for the magnetic field outside the inductor, $\boldsymbol{B}_a$, we can neglect it because with $\boldsymbol{B}_a \approx 0$

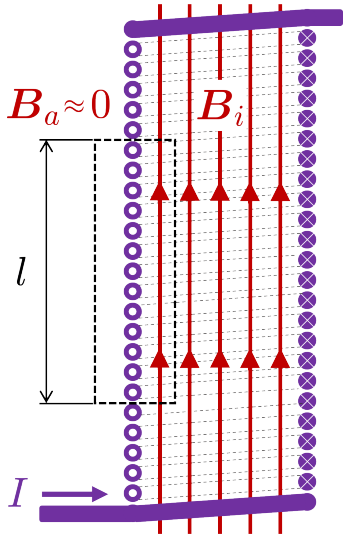


**Fig. 9.9**

it does not contribute to the path integral. Thus, only the contribution from the path integral along the length $l$ of the integration path inside the inductor remains. Through each of the $N$ turns of the inductor that are included in the integration path, the current $I$ flows, and thus it follows from Ampère's law that

$$B_i \cdot l = \mu_0 \cdot N \cdot I \qquad \text{or} \qquad B_i = \mu_0 \cdot n \cdot I \tag{9.24}$$

where $n = N/l$ is the number of turns per unit length. Since the integration paths running perpendicular to the magnetic field make no contribution, we can shift the section of length $l$ anywhere into the volume of the inductor without changing the result. This implies that the magnetic field $B_i$ inside the inductor is uniform, meaning, the magnetic flux $\Phi$ is the same everywhere within the inductor.

## 9.3 The Biot-Savart Law

We begin by looking back at the end of the second chapter, where in section 2.4.1 we derived eq. (2.75), the governing equation for a vector potential such as $\boldsymbol{A}$. We can now apply that result here. Using eq. (9.9) and eq. (9.19), the governing equation for the vector potential $\boldsymbol{A}$ of the magnetic field is:

$$c^2 \nabla \times (\nabla \times \boldsymbol{A}) = \frac{\boldsymbol{j}}{\epsilon_0} \tag{9.25}$$

If we evaluate the left-hand side of eq. (9.25) using the corresponding formula in tab. 2.1, we obtain:

$$\nabla(\nabla \boldsymbol{A}) - \Delta \boldsymbol{A} = \frac{\boldsymbol{j}}{\epsilon_0 c^2} \tag{9.26}$$

However, this is precisely the defining equation eq. (2.75) that we discussed earlier in section 2.4.1. There, we saw that the vector potential is unique only up to the gradient of an arbitrary scalar field $\Lambda$. The reason for this is that $\nabla \times \nabla \Lambda = 0$ always holds. Thus, we can write $\boldsymbol{A}$ as:

$$\boldsymbol{A} = \boldsymbol{A}' + \nabla \Lambda \tag{9.27}$$

We can therefore always freely choose $\nabla \Lambda$, because it drops out as soon as we take the curl of it. We encountered something similar with the potential $\phi$ of the electrostatic field. There, we utilized this freedom to set the value of $\phi$ to zero at infinity. In the context of magnetostatics, we utilize the freedom we have in defining $\boldsymbol{A}$ to impose the condition $\nabla \boldsymbol{A} = 0$, just as we did already in section 2.4.1. If we do this and choose $\nabla \Lambda$ in eq. (9.27) such that $\nabla \boldsymbol{A} = 0$ holds (the so-called Coulomb gauge), then eq. (9.26) becomes:

$$\Delta \boldsymbol{A} \;=\; -\frac{\boldsymbol{j}}{\epsilon_0 c^2} \tag{9.28}$$

Eq. (9.28) is the magnetostatic counterpart to the Poisson equation eq. (3.21) of electrostatics, with the difference that here we are dealing with a vector equation. However, the equations for the components of $\boldsymbol{A}$ and $\boldsymbol{j}$

$$\Delta A_x \;=\; -\frac{j_x}{\epsilon_0 c^2} \quad , \quad \Delta A_y \;=\; -\frac{j_y}{\epsilon_0 c^2} \quad , \quad \Delta A_z \;=\; -\frac{j_z}{\epsilon_0 c^2}$$

are exact mathematical counterparts to eq. (3.21). A comparison of eq. (9.28) and eq. (3.21) shows that the roles played by electric charge distributions in electrostatics and by current density in magnetostatics correspond directly to one another. For the electrostatic potential, it was eq. (2.74) which in the form of eq. (4.3) provided the general solution to the Poisson equation eq. (4.1). Correspondingly, eq. (2.76) gives the general solution to eq. (9.28):

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2} \int \frac{\boldsymbol{j}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\, \mathrm{d}^3 r' \tag{9.29}$$

Since there is no «point current», the right-hand side of eq. (9.29) contains only one term, unlike eq. (4.3). Once we know $\boldsymbol{A}$, we can determine the magnetic field via $\boldsymbol{B} = \nabla \times \boldsymbol{A}$. Using eq. (9.29), we thus obtain:

$$\boldsymbol{B}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2} \int \left( \nabla_{\boldsymbol{r}} \times \frac{\boldsymbol{j}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} \right) \mathrm{d}^3 r' \tag{9.30}$$

Using our collection of formulas from tab. 2.1, we can rewrite the expression in parentheses within the integral as:

$$\begin{aligned} \nabla_{\boldsymbol{r}} \times \frac{\boldsymbol{j}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} \;&=\; \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \Big[ \underbrace{\nabla_{\boldsymbol{r}} \times \boldsymbol{j}(\boldsymbol{r}')}_{=\,0} \Big] + \left( \nabla_{\boldsymbol{r}} \frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} \right) \times \boldsymbol{j}(\boldsymbol{r}') \\ &=\; -\frac{\boldsymbol{r}-\boldsymbol{r}'}{|\boldsymbol{r}-\boldsymbol{r}'|^3} \times \boldsymbol{j}(\boldsymbol{r}') \;=\; \frac{\boldsymbol{j}(\boldsymbol{r}') \times (\boldsymbol{r}-\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|^3} \end{aligned}$$

Substituting this into eq. (9.30), we obtain for $\boldsymbol{B}(\boldsymbol{r})$ the expression:

$$\boldsymbol{B}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2} \int \frac{\boldsymbol{j}(\boldsymbol{r}') \times (\boldsymbol{r}-\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|^3}\, \mathrm{d}^3 r' \tag{9.31}$$

Except for the factor $c^{-2}$ and the fact that the numerator contains the vector product of $\boldsymbol{j}(\boldsymbol{r}')$ and the separation vector $\boldsymbol{r}-\boldsymbol{r}'$ rather than the product of the charge density $\rho(\boldsymbol{r}')$ and the separation vector $\boldsymbol{r}-\boldsymbol{r}'$, equation eq. (9.31) has exactly the same structure as the corresponding equation for the electrostatic field $\boldsymbol{E}(\boldsymbol{r})$ in eq. (3.8). Just as the electrostatic field $\boldsymbol{E}(\boldsymbol{r})$ at a specific point in space is obtained by integrating over the

volume containing the respective charge density distribution $\rho(\boldsymbol{r}')$, the magnetic field $\boldsymbol{B}(\boldsymbol{r})$ at a specific point in space is obtained by integrating over the volume containing the respective current density distribution $\boldsymbol{j}(\boldsymbol{r}')$. Using eq. (9.31), we can determine the magnetic field produced by an arbitrary current density distribution $\boldsymbol{j}$. It makes no difference whether this involves, for example, a stream of charged particles of arbitrary geometry or current-carrying wires. However, the latter are often of interest, particularly thin current-carrying wires.

With that in mind, let us consider the thin wire in fig. 9.10 and the short wire segment shown in the enlarged view. Using the notation from the enlarged wire section in fig. 9.10, we can express the wire volume over which we must integrate in eq. (9.31) as

$$\mathrm{d}^3r' = \mathrm{d}\boldsymbol{a}\cdot\mathrm{d}\boldsymbol{r}'$$

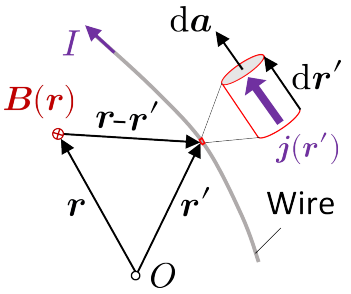



**Fig. 9.10**

where $\mathrm{d}\boldsymbol{a}$ is the directed area element of the wire cross-section and $\mathrm{d}\boldsymbol{r}'$ is the differential arc element along the wire. With the constant current $I = \boldsymbol{j}\cdot\mathrm{d}\boldsymbol{a}$, eq. (9.31) thus becomes:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{I}{4\pi\epsilon_0 c^2}\int_L \frac{\mathrm{d}\boldsymbol{r}'\times(\boldsymbol{r}-\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|^3} \tag{9.32}$$

In eq. (9.32), the integration path $L$ follows the segment of the wire of interest, whose contribution to the magnetic field at the point $\boldsymbol{r}$ is to be calculated. Eq. (9.32) is the Biot-Savart law, which allows one to calculate the contribution of any segment of current-carrying wire to the magnetic field at a specific position.

### 9.3.1 Straight, Infinitely Long, Current-Carrying Wire

We consider a straight, infinitely long wire running along the $z$-axis. Thus, $\boldsymbol{r}' = z'\hat{\boldsymbol{e}}_z$ and $\mathrm{d}\boldsymbol{r}' = \hat{\boldsymbol{e}}_z\mathrm{d}z'$ and so it follows from eq. (9.32) in cylindrical coordinates $r = (R\cos\varphi, R\sin\varphi, z)$ that:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{I}{4\pi\epsilon_0 c^2}\int_{-\infty}^{+\infty}\frac{\hat{\boldsymbol{e}}_z\,\mathrm{d}z'\times(\boldsymbol{r}-z'\hat{\boldsymbol{e}}_z)}{|\boldsymbol{r}-z'\hat{\boldsymbol{e}}_z|^3} = \frac{I\hat{\boldsymbol{e}}_\varphi}{4\pi\epsilon_0 c^2}\int_{-\infty}^{+\infty}\frac{R\,\mathrm{d}z'}{\left[R^2+(z-z')^2\right]^{3/2}}$$

Using the substitution $u = (z-z')/R$, partial fraction decomposition of the integrand, and subsequent integration by parts of the second integrand, this becomes:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{I\hat{\boldsymbol{e}}_\varphi}{4\pi\epsilon_0 c^2}\frac{1}{R}\int_{-\infty}^{+\infty}\frac{\mathrm{d}u}{\left[1+u^2\right]^{3/2}} = \frac{I\hat{\boldsymbol{e}}_\varphi}{4\pi\epsilon_0 c^2}\frac{1}{R}\int_{-\infty}^{+\infty}\left(\frac{1}{\sqrt{1+u^2}} - \frac{u^2}{\left[1+u^2\right]^{3/2}}\right)\mathrm{d}u$$

$$= \frac{I\hat{\boldsymbol{e}}_\varphi}{4\pi\epsilon_0 c^2}\frac{1}{R}\left(\int_{-\infty}^{+\infty}\frac{\mathrm{d}u}{\sqrt{1+u^2}} + u\cdot\frac{1}{\sqrt{1+u^2}}\bigg|_{-\infty}^{+\infty} - \int_{-\infty}^{+\infty}\frac{\mathrm{d}u}{\sqrt{1+u^2}}\right)$$

Of the three terms in the parentheses, the two integrals cancel each other out, and the expression simplifies to:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{I\hat{\boldsymbol{e}}_\varphi}{4\pi\epsilon_0 c^2}\frac{1}{R}\cdot\left(\frac{u}{\sqrt{1+u^2}}\bigg|_{-\infty}^{+\infty}\right)$$

To determine the expression in the parentheses, we must explicitly consider the limiting cases for $-\infty$ and $+\infty$:

$$\frac{u}{\sqrt{1+u^2}}\bigg|_{-\infty}^{+\infty} = \lim_{u\to\infty}\frac{u}{\sqrt{1+u^2}} - \lim_{u\to-\infty}\frac{u}{\sqrt{1+u^2}}$$

$$= \lim_{u\to\infty}\frac{u}{|u|\sqrt{1/u^2+1}} - \lim_{u\to-\infty}\frac{u}{|u|\sqrt{1/u^2+1}}$$

$$= \lim_{u\to\infty}\left(\frac{u}{|u|\sqrt{1/u^2+1}} - \frac{-u}{|u|\sqrt{1/u^2+1}}\right)$$

$$= \lim_{u\to\infty}\left(\frac{1}{\sqrt{1/u^2+1}} - \frac{-1}{\sqrt{1/u^2+1}}\right) = 1-(-1) = 2$$

Thus, the Biot-Savart law yields the following for the straight, infinitely long wire:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{I\hat{\boldsymbol{e}}_\varphi}{2\pi\epsilon_0 c^2}\frac{1}{R} = \frac{1}{2\pi\epsilon_0 c^2}\frac{I}{R}\begin{pmatrix}-\sin\varphi\\ \cos\varphi\\ 0\end{pmatrix} \tag{9.33}$$

Eq. (9.20) already provided us with the magnitude of this magnetic field. However, with eq. (9.33), we now have not only the magnitude of the magnetic field strength but the magnetic field itself as a vector quantity. As eq. (9.33) shows, the magnetic field is indeed radially symmetric: the magnetic field lines are concentric circles around the current-carrying wire, and the magnetic field vector points in the direction of the tangent to the circle, $\hat{\boldsymbol{e}}_\varphi$. In deriving eq. (9.33), we tacitly assumed that the wire has no radial extent, meaning, the wire is infinitely thin. We made the same assumption in the derivation of eq. (9.8); there, too, the wire has no radial extent. However, we corrected this in section 9.2.1, where we examined the magnetic field inside a current-carrying wire. In doing so, we established via eq. (9.20) that the magnetic field inside the wire takes on

the value given by eq. (9.11) at its boundary. Naturally, the same applies to eq. (9.33), which, in place of eq. (9.20), describes the magnetic field outside a long wire with a finite radial extent $a$. For the magnetic field inside the wire, we can then also write, using eq. (9.33):

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{I\hat{\boldsymbol{e}}_\varphi}{2\pi\epsilon_0 c^2}\frac{R}{a^2} = \frac{1}{2\pi\epsilon_0 c^2}\frac{I\cdot R}{a^2}\begin{pmatrix}-\sin\varphi\\ \cos\varphi\\ 0\end{pmatrix} \tag{9.34}$$

### 9.3.2 Force Between Two Current-Carrying Wires

We begin by writing the expression from eq. (9.4), which gives us the force $\Delta\boldsymbol{F}$ acting on a segment $\Delta L$ of a current-carrying wire located in a magnetic field $\boldsymbol{B}$, more generally as:

$$\mathrm{d}\boldsymbol{F} = I\cdot\mathrm{d}\boldsymbol{r}\times\boldsymbol{B}(\boldsymbol{r}) \tag{9.35}$$

It follows that for the total force $\boldsymbol{F}$ exerted on a conductor, through which a current $I_1$ flows and whose path in the magnetic field $\boldsymbol{B}(r)$ is described by the curve $L_1$, we can write the following:

$$\boldsymbol{F} = I_1\int_{L_1}\mathrm{d}\boldsymbol{r}\times\boldsymbol{B}(\boldsymbol{r}) \tag{9.36}$$

If the magnetic field $\boldsymbol{B}(\boldsymbol{r})$ is generated by a second current-carrying conductor through which the current $I_2$ flows along a path $L_2$, then we can use the Biot-Savart law to replace the magnetic field $\boldsymbol{B}(\boldsymbol{r})$ in this force equation with eq. (9.33) and obtain:

$$\boldsymbol{F} = \frac{I_1 I_2}{4\pi\epsilon_0 c^2}\int_{L_1}\int_{L_2}\mathrm{d}\boldsymbol{r}\times\frac{\mathrm{d}\boldsymbol{r}'\times(\boldsymbol{r}-\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|^3} \tag{9.37}$$

Eq. (9.37) applies quite generally and shows us, in the case of two parallel conductors, that the force is attractive if $I_1$ and $I_2$ have the same sign, meaning the current flows in the same direction, and repulsive if the current in the conductors flows in opposite directions, meaning $I_1$ and $I_2$ have different signs. Let us consider the special case of two infinitely long, parallel conductors. Let one conductor, say, $L_1$, lie along the $z$-axis, and the other conductor, $L_2$, lie parallel to it at a radial distance $a$. Thus, we have (again using cylindrical coordinates):

$$\boldsymbol{r} = z\hat{\boldsymbol{e}}_z \quad \text{and} \quad \mathrm{d}\boldsymbol{r} = \hat{\boldsymbol{e}}_z\mathrm{d}z \qquad \text{for} \qquad L_1$$

as well as

$$\boldsymbol{r}' = z'\hat{\boldsymbol{e}}_z + a\hat{\boldsymbol{e}}_R \quad \text{and} \quad \mathrm{d}\boldsymbol{r}' = \hat{\boldsymbol{e}}_z\mathrm{d}z' \qquad \text{for} \qquad L_2$$

Substituting this into eq. (9.37) yields the following for $\boldsymbol{F}$:

$$\begin{aligned}\boldsymbol{F} &= \frac{I_1 I_2}{4\pi\epsilon_0 c^2}\int_{L_1}\int_{L_2}\hat{\boldsymbol{e}}_z \mathrm{d}z \times \frac{\hat{\boldsymbol{e}}_z \mathrm{d}z' \times \left[(z-z')\hat{\boldsymbol{e}}_z - a\hat{\boldsymbol{e}}_R\right]}{\left|(z-z')\hat{\boldsymbol{e}}_z - a\hat{\boldsymbol{e}}_R\right|^3} \\ &= \frac{I_1 I_2}{4\pi\epsilon_0 c^2}\int_{L_1}\int_{L_2}\frac{\hat{\boldsymbol{e}}_z \mathrm{d}z \times -\hat{\boldsymbol{e}}_\varphi a \mathrm{d}z'}{\left|(z-z')\hat{\boldsymbol{e}}_z - a\hat{\boldsymbol{e}}_R\right|^3} = \frac{I_1 I_2}{4\pi\epsilon_0 c^2}\int_{L_1}\int_{L_2}\frac{-\hat{\boldsymbol{e}}_R a \mathrm{d}z\mathrm{d}z'}{\left[a^2+(z-z')^2\right]^{3/2}}\end{aligned}$$

We are already familiar with the integral over $\mathrm{d}z'$, but before we can use the result, we must consider the limits of integration. We certainly cannot consider infinitely long wires, because the force must remain finite. We therefore can only consider the force between two current-carrying parallel wires over a finite length $l$. In this case, the corresponding force is then:

$$\begin{aligned}\boldsymbol{F} &= -\frac{I_1 I_2}{4\pi\epsilon_0 c^2}\cdot\hat{\boldsymbol{e}}_R\int_0^l\int_0^l\frac{a\mathrm{d}z\mathrm{d}z'}{\left[a^2+(z-z')^2\right]^{3/2}} \\ &= -\frac{I_1 I_2}{4\pi\epsilon_0 c^2}\frac{\hat{\boldsymbol{e}}_R}{a}\int_0^l\left(\frac{z-z'}{\sqrt{a^2+(z-z')^2}}\Bigg|_{z'=0}^{z'=l}\right)\mathrm{d}z \\ &= -\frac{I_1 I_2}{4\pi\epsilon_0 c^2}\frac{\hat{\boldsymbol{e}}_R}{a}\int_0^l\left(\frac{z-l}{\sqrt{a^2+(z-l)^2}}-\frac{z}{\sqrt{a^2+z^2}}\right)\mathrm{d}z\end{aligned}$$

Here, for the integration with respect to $\mathrm{d}z'$, we utilized the integral evaluation from section 9.3.1. The integral with respect to $\mathrm{d}z$ is straightforward to evaluate, and we obtain the following result for the force between two parallel current-carrying conductors:

$$\boldsymbol{F} = \frac{I_1 I_2}{2\pi\epsilon_0 c^2}\frac{\hat{\boldsymbol{e}}_R}{a}\left(\sqrt{a^2+l^2}-a\right) \tag{9.38}$$

Let us now consider the arrangement of the two wires for which $l \gg a$ holds. We can then write the following for the expression in parentheses:

$$l\sqrt{1+\frac{a^2}{l^2}}-a \approx l\left(1+\frac{a^2}{l^2}\right)-a = l\left(1-\frac{a}{l}+\frac{a^2}{l^2}\right)$$

If we neglect all terms of order $a/l$ and higher, we obtain the following result for the force between two current-carrying parallel wires in the case where $l \gg a$:

$$\boldsymbol{F} = \frac{I_1 I_2}{2\pi\epsilon_0 c^2}\frac{l}{a}\cdot\hat{\boldsymbol{e}}_R = \frac{I_1 I_2}{2\pi\epsilon_0 c^2}\frac{l}{a}\cdot\begin{pmatrix}\cos\varphi\\ \sin\varphi\\ 0\end{pmatrix} \tag{9.39}$$

Now let us consider a current-carrying wire and a single moving charge. With eq. (9.4) we know the force acting on a charge $q$ moving in a magnetic field. If this magnetic field

is that of a current-carrying wire and the charge $q$ moves parallel to it, then, according to eq. (9.13), the force on this charge is given by:

$$\boldsymbol{F} = q\cdot(\boldsymbol{v}\times\boldsymbol{B}) = \frac{q\mu_0}{2\pi r^2}\cdot\big[\boldsymbol{v}\times(\boldsymbol{I}\times\boldsymbol{r})\big] \tag{9.40}$$

Since $\boldsymbol{v}$ is either parallel or antiparallel to $\boldsymbol{I}$ and orthogonal to $\boldsymbol{r}$, this expression simplifies to:

$$\boldsymbol{F} = \mp\frac{q\mu_0 vI}{2\pi r}\cdot\hat{\boldsymbol{r}} \tag{9.41}$$

If we interpret the expression $I_2\cdot l = \Delta q\cdot l/\Delta t$ in eq. (9.39) as a charge $\Delta q$ moving at a velocity $v = l/\Delta t$, and also replace $1/\epsilon_0 c^2$ with $\mu_0$, then we obtain exactly eq. (9.41). To clarify: We can make this comparison between eq. (9.39) and eq. (9.41) so easily here only because, in deriving eq. (9.39), we implicitly assumed that the wires are infinitesimally thin, meaning that their diameters are negligible compared to their separation $a$. If that were not the case, we would also have to take into account the magnetic fields inside the wires, and consequently, we could only draw this comparison between eq. (9.39) and eq. (9.41) to a limited extent.

### 9.3.3 Magnetic Field of a Current Loop

In section 9.2.2, we examined the magnetic field of a current-carrying inductor. Now, we apply the Biot-Savart law to a single current loop of such an inductor. To this end, however, we consider, as illustrated in fig. 9.11, a truly closed circular current loop lying in the $xy$-plane and centered at the origin. The current $I$ is to flow in the indicated direction through this current loop of radius $a$. The geometry of the problem suggests using cylindrical coordinates $\boldsymbol{r} = (R\cos\varphi, R\sin\varphi, z)$ to describe the current loop. Doing this, we obtain for the directed differential arc element along the current loop in eq. (9.32), the following expression:

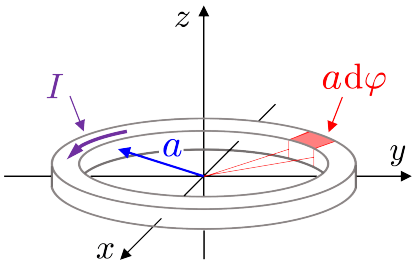



**Fig. 9.11**

$$\mathrm{d}\boldsymbol{r}' = \hat{\boldsymbol{e}}_\varphi a\,\mathrm{d}\varphi = \begin{pmatrix}-\sin\varphi\\ \cos\varphi\\ 0\end{pmatrix} a\,\mathrm{d}\varphi$$

For the distance between the point $\boldsymbol{r}$ where we wish to determine the magnetic field and the location $\mathbf{r}'$ of the current loop contributing to it in eq. (9.32), we have:

$$\boldsymbol{r}-\boldsymbol{r}' = \big(x - a\cos\varphi, y - a\sin\varphi, z\big)$$

For the denominator in eq. (9.32), a brief calculation thus yields:

$$|\boldsymbol{r} - \boldsymbol{r}'|^3 = (r^2 + a^2 - 2ax\cos\varphi - 2ay\sin\varphi)^{3/2}$$

Instead of the integral form of the Biot-Savart law in eq. (9.32), we use its differential form here:

$$\mathrm{d}\boldsymbol{B}(\boldsymbol{r}) = \frac{I}{4\pi\epsilon_0 c^2}\frac{\mathrm{d}\boldsymbol{r}' \times (\boldsymbol{r} - \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^3} \tag{9.42}$$

After substituting the parameters for the current loop into eq. (9.42) and evaluating the cross product, we obtain for the magnetic field of the current loop:

$$\mathrm{d}\boldsymbol{B}(\boldsymbol{r}) = \frac{Ia}{4\pi\epsilon_0 c^2}\frac{\big(z\cos\varphi, z\sin\varphi, a - x\cos\varphi - y\sin\varphi\big)}{(r^2 + a^2 - 2ax\cos\varphi - 2ay\sin\varphi)^{3/2}} \cdot \mathrm{d}\varphi \tag{9.43}$$

Eq. (9.43) gives us the differential contribution of a line element along the current loop to the magnetic field at position $\boldsymbol{r}$. To obtain $\boldsymbol{B}(\boldsymbol{r})$, we must sum the contributions of all line elements along the current loop, i.e., we must integrate from $\varphi = 0$ to $\varphi = 2\pi$. In this case, however, this leads to elliptic integrals for all three coordinates in eq. (9.43) that do not have an analytical solution. We can, however, say something about the magnetic field of the current loop on the $z$-axis. In this case, $x = 0$ and $y = 0$, and eq. (9.43) simplifies to:

$$\mathrm{d}\boldsymbol{B}(\boldsymbol{r}) = \frac{Ia}{4\pi\epsilon_0 c^2}\frac{\big(z\cos\varphi, z\sin\varphi, a\big)}{(z^2 + a^2)^{3/2}} \cdot \mathrm{d}\varphi \tag{9.44}$$

Here, the corresponding integrals over $\mathrm{d}\varphi$ are easily evaluated, with the result that the integrals over the $x$- and $y$-components vanish. Integration over the $z$-component yields:

$$\boldsymbol{B}(z) = \frac{Ia^2\pi}{2\pi\epsilon_0 c^2}\frac{1}{(z^2 + a^2)^{3/2}} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{9.45}$$

In this equation we have not cancelled out the factor $\pi$ because $Ia^2\pi$ is an important quantity. The product of the current $I$ and the area $a^2\pi$ enclosed by the current loop is the so-called magnetic moment $m$ of the current loop, a topic we will examine in greater detail in the next chapter. For now, we use the magnetic moment simply to rewrite eq. (9.45) as:

$$\boldsymbol{B}(z) = \frac{m}{2\pi\epsilon_0 c^2}\frac{\hat{\boldsymbol{e}}_z}{(z^2 + a^2)^{3/2}} = \frac{1}{2\pi\epsilon_0 c^2}\frac{\boldsymbol{m}}{(z^2 + a^2)^{3/2}} \tag{9.46}$$

In the plane of the current loop, the magnetic field has only a component in the $z$-direction. This is easily seen by setting $z = 0$ in eq. (9.43). Thus, the tangents to the

field lines of $\boldsymbol{B}$ are all parallel within the plane of the current loop. Eq. (9.46) tells us that this cannot hold for $z \neq 0$, because the magnitude of $\boldsymbol{B}(z)$ decreases as $z^{-3}$ along the $z$-axis. The density of the field lines passing through the current loop therefore decreases with increasing distance from the plane of the loop along the $z$-axis. This is only possible if these field lines curve radially outward, away from the $z$-axis, as the distance from the loop increases.

Since we know that magnetic field lines must be closed curves, the magnetic field of the current loop must qualitatively resemble the sketch shown in fig. 9.12, where the path of the $\boldsymbol{B}$ field lines can be well described by elliptical curves. In section 9.2.2, we calculated the magnetic field inside an inductor. Using eq. (9.46), we can now calculate the magnetic field of a current-carrying hollow cylinder oriented along the $z$-axis, assuming that its internal magnetic field should correspond to that produced by stacked current loops. To this end, let us take another look at fig. 9.11. The figure shows a sketch of a current loop with finite dimensions. This is simply because a current loop with zero dimensions, which is what we have actually been considering here, is difficult to draw. The current path $\mathrm{d}\boldsymbol{r}'$ along which we have summed the contributions to the Biot-Savart law is a differential current path running along the unit vector $\hat{\boldsymbol{e}}_\varphi$. It has no radial extent and no extent in the $z$-direction; we are dealing with a current loop made of an infinitesimal thin wire. We can therefore regard eq. (9.46) multiplied by $\mathrm{d}z$ as the differential contribution of a current-carrying cylinder element. If we integrate eq. (9.46) with respect to $z$ from $z = -h/2$ to $z = +h/2$, where $h$ is the height of the current-carrying cylinder, we should obtain a constant value for the magnetic field inside the cylinder for the case $h \gg a$. Why do we expect this? Quite simply, because the magnetic field inside such a current-carrying cylinder should correspond to the magnetic field inside a coil, and we know from eq. (9.24) that the latter is constant. Instead of the current $I$ of a single current loop, we assume for the current cylinder that it carries the equivalent current of $N$ loops distributed along the $z$-axis per unit length $L$. With that we obtain for the magnetic field $\boldsymbol{B}_i$ inside the current cylinder of height $h$ with the linear current density $I = N/L = nI$ (here we again explicitly substitute $m = Ia^2\pi$ for $m$):

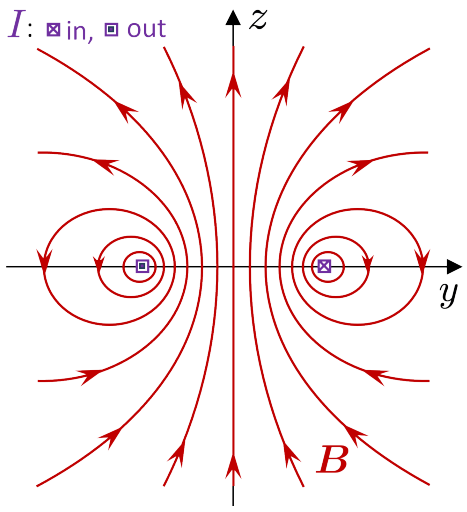



**Fig. 9.12**

$$\boldsymbol{B}_i = \frac{nIa^2\hat{\boldsymbol{e}}_z}{2\epsilon_0 c^2} \int_{-h/2}^{+h/2} \frac{\mathrm{d}z}{(z^2+a^2)^{3/2}} = \frac{nIa^2\hat{\boldsymbol{e}}_z}{2\epsilon_0 c^2} \cdot \frac{z}{\sqrt{z^2+a^2}}\Bigg|_{-h/2}^{h/2} \tag{9.47}$$

Substituting the limits of integration yields the result:

$$\boldsymbol{B}_i = \frac{nIa^2\hat{\boldsymbol{e}}_z}{2\epsilon_0 c^2} \cdot \frac{h}{a^2\sqrt{h^2/4+a^2}} = \frac{nI\hat{\boldsymbol{e}}_z}{2\epsilon_0 c^2} \cdot \frac{1}{\sqrt{1/4+a^2/h^2}} \tag{9.48}$$

In the limit $h \gg a$, the magnitude of the magnetic field inside the current-carrying cylinder is therefore:

$$B_i = \frac{nI}{\epsilon_0 c^2} = \mu_0 \cdot n \cdot I \tag{9.49}$$

This corresponds exactly to eq. (9.24) in section 9.2.2, our result for the magnetic field of a coil, where we had likewise assumed that its length is much greater than the coil's diameter. The magnetic fields inside a coil and an infinitely long current-carrying cylinder are thus identical.

## 9.4 Magnetic Fields and Special Relativity

As far as we know today, all magnetic fields are generated by moving charges. This applies both to magnetic fields produced by electric currents as well as to all types of permanent magnets, whose magnetism is based on the motion of electric charges at the atomic level. As we shall see in the next chapter, a magnetic moment is associated with the motion of such charges at the atomic level. This serves as a measure of the strength of the magnetic field generated by the moving charges. If these atomic magnetic moments within a solid are not randomly distributed but instead possess a preferred orientation, the body itself exhibits a macroscopic magnetic moment, in other words, it is a magnet. The only exception known to us where the source of a magnetic moment, and thus of a magnetic field, is not moving charges is the intrinsic angular momentum of elementary particles. An example of this is the electron spin, which has a purely quantum-mechanical origin and is just as fundamental an intrinsic property of the electron as its charge or mass, which likewise cannot be derived from an internal substructure of the electron. An electron's intrinsic rotation about its axis is frequently used as an analogy for electron spin. However, electron spin is not that; rather, it is an intrinsic property of the electron. If we set aside the electron spin and the intrinsic angular momentum of other elementary particles, we can conclude the following: without moving charges, there would be no magnetic fields. But how, for example, does an observer who moves along with the electron, that is, who, in a sense, flies through a magnetic field together with it, explain

the force acting on that electron? We will address this questions and similar ones in this section.

### 9.4.1 Moving Charges in a Conducting Wire

Here, we need to first recall our discussion of special relativity from the first volume of this physics course. There, we established that the measured length of an object depends on the relative motion between the object and the observer making the measurement. If both are at rest, the observer measures the object's rest length $l_0$. However, if the object and the observer are moving relative to each other, the observer measures a shorter length $l$ in the direction of motion $\boldsymbol{v}$, satisfying the following equation:

$$l = l_0\sqrt{1 - v^2/c^2} = l_0 \cdot \gamma^{-1} \tag{9.50}$$

Thus, an object moving with a velocity $v$ appears to a stationary observer to be shortened by a factor of $\gamma^{-1}$ in the direction of motion. Let us now consider, as sketched in fig. 9.13, a charge $q$ in an inertial frame $S'$ that moves parallel to the conductor in the inertial frame $S$ at a velocity $v$, which shall be the same velocity $v$ at which the electrons in the wire move when a voltage is applied. For the current-carrying wire to be electrically neutral, the linear charge densities $\lambda_+$ and $\lambda_-$ along the wire axis in frame $S$ must satisfy the condition $\lambda_+ = -\lambda_-$. The conduction electrons, which move at velocity $v$ relative to an observer in frame $S$, are at rest relative to an observer in frame $S'$ moving at velocity $v$. For the lattice atoms, the situation is exactly the reverse: they are at rest for the observer in $S$ and move with velocity $-v$ for the observer in $S'$. Therefore, for the observer in $S$, the linear charge density of the electrons $\lambda_-$ that he or she measures is already a contracted charge density; for the observer in $S'$, we must correct for this contraction accordingly. The situation is different for the linear charge density of the lattice atoms, which is contracted for the observer in $S'$. The observer in $S'$ thus measures: first, a contracted linear charge density of the lattice atoms $\lambda'_+ = \lambda_+ \cdot \gamma$, and second, an uncontracted linear charge density of the conduction electrons $\lambda'_- = \lambda_- \cdot \gamma^{-1}$. Taken together, the following applies to the linear charge densities in $S'$:

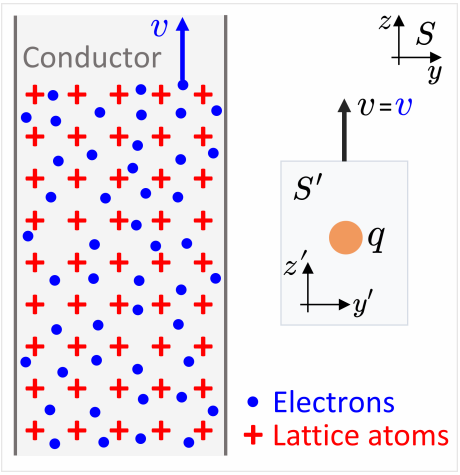



**Fig. 9.13**

$$\lambda'_+ = \lambda_+\gamma \qquad \text{and} \qquad \lambda'_- = \lambda_-\gamma^{-1} \tag{9.51}$$

For the observer in $S'$, the conductor is therefore no longer neutral but exhibits a positive linear charge density $\lambda' = \lambda'_+ - \lambda'_-$ satisfying the following:

$$\lambda' \;=\; \lambda_+\left(\gamma + \frac{1}{\gamma}\right) \;=\; \lambda_+\left(\frac{1}{\sqrt{1 - v^2/c^2}} - \sqrt{1 - v^2/c^2}\right) \;=\; \lambda_+\gamma\frac{v^2}{c^2} \tag{9.52}$$

From eq. (4.36), we know that outside the wire, this positive linear charge density produces an electrostatic field for the observer in $S'$ of the form:

$$E'(r) \;=\; \frac{\lambda'}{2\pi\epsilon_0 r} \;=\; \gamma\frac{q\lambda_+ v^2}{2\pi\epsilon_0 c^2 r} \tag{9.53}$$

From the perspective of an observer in $S'$, the lattice atoms of the wire, moving with velocity $-v$, do indeed generate a magnetic field. However, since $q$ is at rest in $S'$, this magnetic field cannot exert a force on $q$. For the observer in $S'$, the force $F'$ acting on $q$ is solely that produced by the electrostatic field $E'$, that is:

$$F' \;=\; qE' \;=\; \gamma\frac{q\lambda_+ v^2}{2\pi\epsilon_0 c^2 r} \tag{9.54}$$

We know that $E'$ is radially symmetric, and with $c = 1/\sqrt{\epsilon_0\mu_0}$ and the electric current $I = v\lambda_+$, we can write the vector form of this equation as:

$$\boldsymbol{F}' \;=\; q\boldsymbol{E}' \;=\; \gamma\frac{\mu_0 q v I}{2\pi r}\cdot\hat{\boldsymbol{r}} \tag{9.55}$$

With eq. (9.55), we now know the spatial component of the four-force vector in $S'$. From the discussion of special relativity in the first volume of this physics course, we know that the spatial component of the four-force vector transforms as $\gamma\boldsymbol{F}$, and we thus obtain the following for the force $\boldsymbol{F}$ in $S$:

$$\boldsymbol{F} \;=\; \gamma^{-1}\boldsymbol{F}' \;=\; \frac{\mu_0 q v I}{2\pi r}\cdot\hat{\boldsymbol{r}} \tag{9.56}$$

In $S$, a current-carrying conductor generates a magnetic field but no electric field, and according to eq. (9.2), the force exerted on a charge $q$ moving with velocity $\boldsymbol{v}$ in this magnetic field is given by:

$$\boldsymbol{F} \;=\; q(\boldsymbol{v}\times\boldsymbol{B})$$

For an observer in $S'$, however, this is a purely electrostatic force which is described in $S$ by eq. (9.56). Therefore, the following must hold:

$$q(\boldsymbol{v}\times\boldsymbol{B}) \;=\; \frac{\mu_0 q v I}{2\pi r}\cdot\hat{\boldsymbol{r}} \tag{9.57}$$

In our case, the charge $q$ moves parallel to the wire, and therefore $\boldsymbol{v}\times\boldsymbol{B} \;=\; vB\cdot\hat{\boldsymbol{r}}$ and thus eq. (9.57) becomes:

$$qvB\cdot\hat{\boldsymbol{r}} = \frac{\mu_0 qvI}{2\pi r}\cdot\hat{\boldsymbol{r}} \tag{9.58}$$

From eq. (9.58), we can immediately see that the magnetic field of a current-carrying conductor is therefore given by:

$$B = \frac{\mu_0 I}{2\pi r} \tag{9.59}$$

### 9.4.2 The Moving Plate Capacitor

As a further example of the relativistic connection between electric and magnetic fields, we once again consider the electric field of a parallel-plate capacitor, which we have already examined in section 4.4. Let the parallel-plate capacitor be located in an inertial frame $S$, with its long side $l_0$ parallel to the $y$-axis and its short side $b$ parallel to the $x$-axis of $S$. Fig. 9.14 shows a perspective side view of this parallel-plate capacitor, whose plates lie parallel to the $xy$-plane and which is assumed to be at rest in frame $S$. Furthermore, let the capacitor carry a surface charge density $\sigma$. Thus, according to eq. (4.58), the electric field of the capacitor in frame $S$ is:

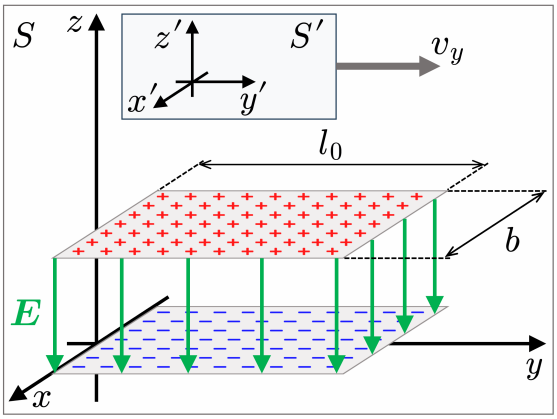



**Fig. 9.14**

$$E_z = \frac{\sigma}{\epsilon_0} \tag{9.60}$$

For an observer in an inertial frame $S'$ moving to the right parallel to the $y$-axis at a velocity $v_y$, the dimension that an observer in $S$ measures as $l_0$ is shortened in accordance with eq. (9.50). As a result, the linear charge density in the $y$-direction increases by a corresponding factor for the observer in $S'$, and they measure an increased surface charge density $\sigma'$ in $S'$ that corresponds to an electric field:

$$E'_z = \frac{\sigma'}{\epsilon_0} = \frac{\sigma}{\epsilon_0}\frac{1}{\sqrt{1 - v_y^2/c^2}} = \gamma\cdot E_z \tag{9.61}$$

This specific result is an example of the following more general rule: any electric field $\boldsymbol{E}$, produced in an inertial frame $S$ by an arbitrary charge distribution at rest in $S$, has the following components for a stationary observer in an inertial frame $S'$ moving at velocity $v$ relative to the inertial frame $S$:

$$E'_{\parallel} = E_{\parallel} \qquad \text{and} \qquad E'_{\perp} = \gamma \cdot E_{\perp} \tag{9.62}$$

where the parallel and normal components here denote the components of $\boldsymbol{E}$ parallel and normal to the direction of the velocity $\boldsymbol{v}$. Eq. (9.61) tells us that the electric field $\boldsymbol{E'}$ measured by the observer in $S'$ is greater than the one measured by the observer in $S$. From eq. (4.67), we know that for the observer in $S$, an electrostatic force $F_E$ acts between the capacitor plates, given by:

$$F_E = \frac{1}{2} \cdot Q \cdot E_z \tag{9.63}$$

Here, $\pm Q$ is the total charge on the respective capacitor plates. In the theory of relativity, electric charge is an invariant quantity. Moreover, since $E'_z > E_z$ holds due to eq. (9.61), this implies that, from the perspective of the observer in $S'$, a greater force $F'_E > F_E$ would act between the capacitor plates, compared to what an observer in $S$ would measure. However, this cannot be the case. Therefore, there must be another force acting for the observer in $S'$ such that the net force in $S'$ is the same as in $S$. And such a force does exist. For the observer in $S'$, after all, more changes than just the component of the electrostatic field normal to the direction of motion. Viewed from the perspective of a stationary observer in $S'$, the capacitor plates move with a relative velocity $-v_y$ and thus constitute moving charges, i.e., electric currents. Since $b$ is perpendicular to the direction of motion, and thus has the same length in both $S$ and $S'$, the observer in $S'$ calculates a current of positive and negative charges of:

$$I_y = \pm \sigma' \cdot b \cdot v_y \tag{9.64}$$

Here, the current flowing in the positive $y$-direction is generated by the negatively charged capacitor plate, and the current flowing in the negative $y$-direction is generated by the positively charged capacitor plate. For the observer in $S'$, the two capacitor plates therefore represent two electrical currents of equal magnitude flowing in opposite directions. From section 9.3.2, we know that the force acting between two parallel current-carrying wires is repulsive when the respective electrical currents flow in opposite directions. Since the same applies to current-carrying plates we have thus identified the repulsive force that ensures that the same force acts between the capacitor plates for observers in both $S$ and $S'$, even though the observer in $S'$ measures a greater electrostatic force. Because of the antiparallel electrical currents $\pm I_y$ present for the observer in $S'$, there exists an additional repulsive magnetostatic force $F'_M$ between the capacitor plates, a force that does not exist for the observer in $S$. Consequently, the force acting on the capacitor plates in $S$ and $S'$ is given by:

$$F_E \;=\; F'_E + F'_M \tag{9.65}$$

This magnetostatic force, which exists only for the observer in $S'$, must therefore be given by:

$$F'_M \;=\; F_E - F'_E \;=\; \frac{1}{2}Q(1-\gamma)E_z \tag{9.66}$$

In the case of low velocities, i.e., $v_y \ll c$, the expression for this magnetic force simplifies to:

$$F'_M \;=\; -\frac{Q}{4}\cdot\frac{v_y^2}{c^2}\cdot E_z \tag{9.67}$$

On the right-hand side of this equation, we have a scaled charge multiplied by a field component of $\boldsymbol{E}$. Therefore, $F'_M$ seemingly must resemble an electrostatic force. However, this is not the case; rather, it is a magnetic force. In order to understand this, we must examine more closely the magnetic field to which an observer in $S'$ attributes this force. When calculating the forces between parallel current-carrying wires in section 9.3.2, we assumed the wires to be infinitesimally thin, and in addition, that their length was much greater than the distance between them. Consequently, we cannot simply apply those results to our capacitor plates. However, in qualitative terms, we have a good idea of what the magnetic field around the capacitor plates must look like, specifically, it must look roughly as sketched in fig. 9.15. In fig. 9.15, the negative $y$-axis points out of the image, that is, in the direction of the current flow that the observer in $S'$ associates with the positively charged plate. Just as with a thin wire, the magnetic field lines around a ribbon conductor, which is what the capacitor plates represent, must form closed loops. However, these loops are not circles as is the case for a current-carrying wire. Furthermore, the right-hand rule tells us that for the positive plate, they must run counter-clockwise, as sketched in fig. 9.15, while for the negative plate, they must run clockwise. As a result, the fields between the capacitor plates add up, making the magnetic field between the plates stronger than in the region outside. This then gives rise to the repulsive force, just as with the parallel wires in section 9.3.2. The sketch in fig. 9.15 shows us that, for the observer in $S'$, the magnetic field between the plates points in the positive $x$-direction, meaning it has only a $B'_x$ component and no $B'_y$ or $B'_z$ components. Can we say anything meaningful about the magnitude of $B'_x$? Yes, we can, provided we assume that the capacitor plates are

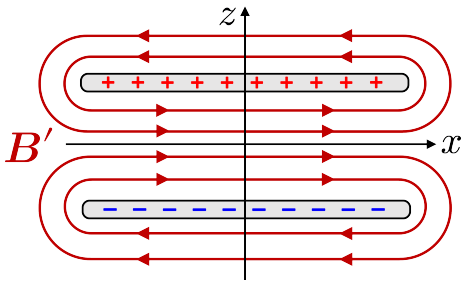



**Fig. 9.15**

infinitesimally thin. That presupposed, we then choose a specific rectangular integration path to evaluate eq. (9.15), running in the $\pm x$-direction directly above and below the respective capacitor plate. In this way, we can then neglect the segments of the integration path oriented in the $\pm z$-direction and thereby obtain the following equation for the magnetic field component $B_x'^{+}$ generated by the positive plate (the integration path is traversed counter-clockwise in both cases):

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} \;\rightarrow\; B_x'^{+}b + (-B_x'^{+})(-b) \;=\; \mu_0 I_y \quad\Rightarrow\quad B_x'^{+} \;=\; \frac{\mu_0 I_y}{2{\cdot}b} \tag{9.68}$$

and for the magnetic field component $B_x'^{-}$ generated by the negatively charged plate:

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} \;\rightarrow\; -B_x'^{-}b + B_x'^{-}(-b) \;=\; \mu_0 I_y \quad\Rightarrow\quad B_x'^{-} \;=\; -\frac{\mu_0 I_y}{2{\cdot}b} \tag{9.69}$$

If we sum the contributions from both capacitor plates and substitute the expression from eq. (9.64) for the current $I_y$, using the respective expressions for the negatively and positively charged plates, we obtain the magnetic field $B_x'$ between the capacitor plates as it exists for the observer in $S'$:

$$B_x' \;=\; B_x'^{+} + B_x'^{-} \;=\; \frac{\mu_0}{2{\cdot}b}\left[\sigma'{\cdot}b{\cdot}v_y - (-\sigma'{\cdot}b{\cdot}v_y)\right] \;=\; \mu_0{\cdot}\sigma'{\cdot}v_y \tag{9.70}$$

With the relationship $c^2 = 1/\mu_0\epsilon_0$ and eq. (9.61) we can express $B_x'$ in terms of $E_z'$ as

$$B_x' \;=\; E_z'{\cdot}\frac{v_y}{c^2} \tag{9.71}$$

or through $E_z$ as

$$B_x' \;=\; \frac{E_z}{\sqrt{1 - v_y^2/c^2}}{\cdot}\frac{v_y}{c^2} \tag{9.72}$$

For $v_y \ll c$, this expression simplifies to:

$$B_x' \;=\; E_z{\cdot}\frac{v_y}{c^2} \tag{9.73}$$

Using this equation, we can now replace $E_z$ in eq. (9.67), the equation describing the magnetic force in $S'$ for $v_y \ll c$, with the low-velocity approximation for $B_x'$. This yields the following expression for $F_M'$:

$$F_M' \;=\; -\frac{Q}{4}{\cdot}v_y{\cdot}B_x' \tag{9.74}$$

We have now described the magnetic force acting on the electrodes of the plate capacitor, as seen by an observer in $S'$, in terms of the magnetic field that exists in $S'$ but not in $S$. And unlike in eq. (9.67), where $F_M'$ resembles an electrostatic force, eq. (9.74) now

clearly describes a Lorentz force. We can see this even more clearly if we write eq. (9.74) in vector form, i.e., as $\boldsymbol{F} = q(\boldsymbol{v} \times \boldsymbol{B})$:

$$\begin{pmatrix} 0 \\ 0 \\ F'_M \end{pmatrix} = -\frac{Q}{4} \cdot \begin{pmatrix} 0 \\ -v_y \\ 0 \end{pmatrix} \times \begin{pmatrix} B'_x \\ 0 \\ 0 \end{pmatrix} \tag{9.75}$$

For the observer in $S'$, the capacitor plates move to the left with a velocity $-v_y$, hence the minus sign in the velocity vector.

### 9.4.3 Point Charge in Motion

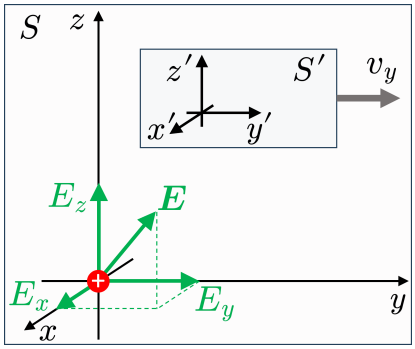



**Fig. 9.16**

As a final example illustrating the relativistic relationship between electric fields and magnetic fields, we consider a stationary point charge in an inertial frame $S$ from the perspective of an observer who finds her- or himself in a second inertial frame $S'$ that moves to the right in the $yz$-plane of $S$ at a velocity $v_y$ parallel to the $y$-axis. We therefore have the same situation as in the previous example, except that here we consider a point charge rather than a parallel-plate capacitor. With eq. (3.5), we know the electrostatic field of the point charge shown in fig. 9.16, which is located at the origin of the coordinate system in $S$ where it is at rest:

$$\boldsymbol{E}(\boldsymbol{r}) = \frac{q}{4\pi\epsilon_0} \cdot \frac{x \cdot \hat{\boldsymbol{\imath}} + y \cdot \hat{\boldsymbol{\jmath}} + z \cdot \hat{\boldsymbol{k}}}{\left[x^2 + y^2 + z^2\right]^{3/2}} = E_x \cdot \hat{\boldsymbol{\imath}} + E_y \cdot \hat{\boldsymbol{\jmath}} + E_z \cdot \hat{\boldsymbol{k}} \tag{9.76}$$

With eq. (9.62), we know how the vector components of $\boldsymbol{E}(\boldsymbol{r})$ that are parallel and normal to $v_y$ transform for a stationary observer in $S'$. We simply need to multiply the components perpendicular to the direction of motion, in this case, $E_x$ and $E_z$, by $\gamma$, and leave the parallel components, here, $E_y$, unchanged. This yields the following in $S'$:

$$\boldsymbol{E}'(\boldsymbol{r}) = \gamma \cdot E_x \cdot \hat{\boldsymbol{\imath}} + E_y \cdot \hat{\boldsymbol{\jmath}} + \gamma \cdot E_z \cdot \hat{\boldsymbol{k}} \tag{9.77}$$

Now we only need to express the vector components $E_x$, $E_y$, and $E_z$, which appear in eq. (9.76) as functions of the coordinates $x$, $y$, and $z$ of $S$, in terms of the coordinates $x'$, $y'$, and $z'$ of $S'$. We learned how this Lorentz transformation of the coordinates from $\boldsymbol{r}$ to $\boldsymbol{r}'$ works in chapter 12 of volume 1 of this physics course, where special relativity was covered. $S'$ moves at velocity $v_y$ in the positive $y$-direction relative to $S$, and thus the Lorentz transformation is given by:

$$x = x' \quad ; \quad y = \gamma(y' + \beta c t') \quad ; \quad z = z' \quad ; \quad t = \left(t + \frac{\beta x'}{c}\right) \tag{9.78}$$

Here, of course, we again have $\beta = v_y/c$ and $\gamma^{-1} = \sqrt{1-\beta^2}$. To keep the calculation simple, we choose $t' = t'_0 = 0$ for the time of the Lorentz transformation, which we can do without loss of generality. We are primarily interested in the qualitative result here. However, if the field at a later time $t' > 0$ becomes of interest, the result for the observer in $S'$ is obtained simply by replacing $x'_0$, the position of the charge $q$, in the equation for $t' > 0$ that we are about to derive, with a new position $x'_1 = x'_0 - v_y t'_1$. For $t' = t'_0 = 0$, eq. (9.76) becomes:

$$\boldsymbol{E}'(\boldsymbol{r}') \;=\; \frac{q}{4\pi\epsilon_0} \cdot \frac{\gamma x' \cdot \hat{\boldsymbol{i}} + \gamma y' \cdot \hat{\boldsymbol{j}} + \gamma z' \cdot \hat{\boldsymbol{k}}}{\left[x'^2 + \gamma^2 y'^2 + z'^2\right]^{3/2}} \tag{9.79}$$

With $\rho'^2 = x'^2 + z'^2$, we obtain for the squared magnitude of $\boldsymbol{E}'(\boldsymbol{r}')$:

$$|\boldsymbol{E}'(\boldsymbol{r}')|^2 \;=\; \left(\frac{q}{4\pi\epsilon_0}\right)^2 \cdot \frac{\gamma^2\rho'^2 + \gamma^2 y'^2}{\left[\rho'^2 + \gamma^2 y'^2\right]^3} \;=\; E'^2_{\perp} + E'^2_{\parallel} \tag{9.80}$$

Here, we rewrite the expression in the parentheses in the denominator as follows:

$$\rho'^2 + \gamma^2 y'^2 \;=\; \frac{\rho'^2(1-\beta^2) + y'^2}{1-\beta^2} \;=\; \frac{r'^2 - \beta^2\rho'^2}{1-\beta^2} \;=\; \frac{r'^2\left(1 - \beta^2\sin^2\theta'\right)}{1-\beta^2}$$

In this equation, $\theta'$ is the angle between the vector $\boldsymbol{\rho}' = (x', z')$ and the vector $\boldsymbol{r}'$. We also rewrite the expression in the numerator:

$$\gamma^2\rho'^2 + \gamma^2 y'^2 \;=\; \gamma^2 \boldsymbol{r}'^2 = \boldsymbol{r}'^2 \frac{1}{1-\beta^2}$$

Substituting the rewritten denominator and numerator into eq. (9.80), we obtain

$$|\boldsymbol{E}'(\boldsymbol{r}')|^2 \;=\; \left(\frac{q}{4\pi\epsilon_0}\right)^2 \cdot \frac{(1-\beta^2)^2}{r'^4\left[1 - \beta^2\sin^2\theta'\right]^3} \tag{9.81}$$

and thus

$$E(r') \;=\; \frac{q}{4\pi\epsilon_0 r'^2} \cdot \frac{1-\beta^2}{\left[1 - \beta^2\sin^2\theta'\right]^{3/2}} \;=\; \frac{q}{4\pi\epsilon_0 r'^2} \cdot f(\theta', \beta) \tag{9.82}$$

What this result tells us is simply that the observer in $S'$ sees the electrostatic field of a point charge $q$, modified by a function $f(\theta', \beta)$. In the case where $S'$ does not move relative to $S$, i.e., $v_y = 0$, we have $f(\theta', 0) = 1$, and the observer in $S'$ sees exactly the same electrostatic field of a stationary point charge $q$ as the observer in $S$. To better understand what happens when $v_y \neq 0$, we consider the function $f(\theta', \beta)$ for $x' = 0$, thereby restricting ourselves to the observer's $y'z'$-plane in $S'$, where $y' = r'\sin\theta'$ and $r'^2 = y'^2 + z'^2$. Fig. 9.17b shows the corresponding behavior of $f(\theta', \beta)$ in the range $0 \leq \theta' \leq \pi/2$ for various values of $\beta$.

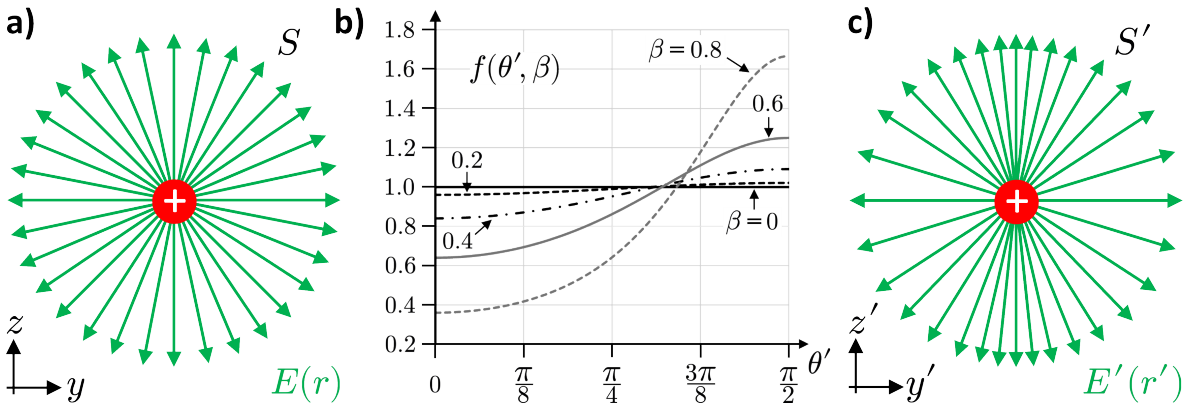



**Fig. 9.17:** (a) Electrostatic field $\boldsymbol{E}$ of the stationary point charge in $S$. (b) The function $f(\theta', \beta)$ with $\sin\theta' = y'/r'$, which modifies the field of the point charge at rest in $S$, as given in eq. (9.82), for a moving observer in $S'$. (c) Electrostatic field of the point charge at rest in $S$ as seen by the moving observer in $S'$.

For $\beta = 0$, the function is constant and equal to one. As $\beta$ increases, the function drops below the value of one at small angles and rises above the value of one at large angles. The angle $\theta'_0$ at which the function takes the value one, i.e., where the electric field strength exactly matches the value a stationary point charge would produce at these coordinates $\boldsymbol{r}'_0 = (y'_0, z'_0)$, shifts toward larger angles as $\beta$ increases. For angles $\theta < \theta'_0$, the observer in $S'$ measures a higher electric field strength than that of a stationary charge, whereas for $\theta > \theta'_0$, the observer in $S'$ measures a lower electric field strength than a stationary charge would produce. For an observer in $S$, the electrostatic field is isotropic, as shown in fig. 9.17a. The equipotential lines of this field are spherical shells with the charge at the center. For an observer in $S'$, the electric field of the point charge is no longer isotropic. Instead, the electric field strength normal to the direction of motion of $S'$ is greater than the field strength in the direction of motion of $S'$. The higher the velocity at which $S'$ moves relative to $S$, the greater this difference between the normal and parallel components of the electric field in $S'$ becomes. An observer in $S'$ would therefore sketch the field lines of $\boldsymbol{E}$ in the $y'z'$-plane within a circle around the point charge $q$ as shown in fig. 9.17c.

The charge at rest in $S$ constitutes an electric current for the observer in $S'$. Thus, the observer in $S'$ sees not only a different electrostatic field but also a magnetic field that does not exist in $S$. For an observer in $S'$, a point charge moving along a straight line represents the limiting case of a current-carrying conductor in which, however, the current consists of only a single charge flowing through. We can therefore expect this magnetic field in $S'$ to be similar to that of a current-carrying wire, with the difference that it exists only locally around the moving charge. However, we will not go into further detail on

this here. Instead, we want to point out something else of greater relevance. As can be easily verified, the circulation of $\boldsymbol{E}'$ in $S'$ no longer vanishes due to the function $f(\theta', \beta)$ in eq. (9.82). For the electrostatic field of the point charge in $S$, given the electrostatic field from eq. (9.76) the following holds in accordance with eq. (3.22), i.e., the second Maxwell equation:

$$\nabla \times \boldsymbol{E}(\boldsymbol{r}) = 0 \tag{9.83}$$

However, this is not the case for the electrostatic field $\boldsymbol{E}'(\boldsymbol{r}')$ in $S'$, since there, using the field from eq. (9.82), one obtains:

$$\nabla_{\boldsymbol{r}'} \times \boldsymbol{E}'(\boldsymbol{r}') \neq 0 \tag{9.84}$$

This result shows us unequivocally that the relation $\nabla \times \boldsymbol{E} = 0$ does not hold for moving charges. For precisely this reason, eq. (9.83) is referred to as the second Maxwell equation of electrostatics. However, let us pause here for a moment and reflect on what we have just demonstrated in this section: electric and magnetic fields are linked by the theory of special relativity. What appears to a stationary observer in an inertial frame $S$ as a purely magnetic or purely electric effect will manifest itself as both to an observer at rest in an inertial frame $S'$ which moves relative to $S$. Even when only considering electrostatics and magnetostatics, i.e., electric charges at rest and stationary electrical currents, a closer look reveals that electromagnetic theory is a relativistic theory. For instance, to derive eq. (9.59), we made use of nothing else but what special relativity tells us about inertial frames moving relative to each other. The fact that a fundamental physical constant such as the speed of light $c$ links electric and magnetic fields via the relationship $c = 1/\sqrt{\epsilon_0 \mu_0}$ demonstrates just how closely the theory of relativity and electrodynamics are interconnected.

Tab. 9.1 summarizes the fundamental equations of magnetostatics that we have addressed in this chapter. A key premise of magnetostatics is the existence of only stationary currents and the absence of both current sources and sinks, that is, $\nabla \boldsymbol{j}(\boldsymbol{r}) = 0$, from which the conservation of charge follows via the continuity equation eq. (8.9). A comparison with tab. 3.1, the fundamental equations of electrostatics, reveals the symmetry between the equations of magnetostatics and electrostatics. Before we examine the magnetic vector potential in greater detail in the next chapter, it is time to discuss the SI units of the magnetic quantities considered here so far. We begin with electric current, which we already know is measured in amperes $\mathrm{A} = \mathrm{C \cdot s^{-1}}$, i.e., in electric charge per unit of time. The ampere is one of the seven base units (see Appendix B) of the SI system from which all other units are derived.

**Tab. 9.1:** Fundamental equations of magnetostatics.

**Basic assumption: Stationary currents**

$$\boldsymbol{j} = \boldsymbol{j}(\boldsymbol{r}) \qquad \text{and} \qquad \operatorname{div}\boldsymbol{j} \,=\, 0 \qquad \Rightarrow \qquad \frac{\partial \rho}{\partial t} \,=\, 0$$

**Magnetostatic vector potential and magnetic field strength**

$$\boldsymbol{A}(\boldsymbol{r}) \,=\, \frac{1}{4\pi\epsilon_0 c^2} \int \frac{\boldsymbol{j}(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\, \mathrm{d}^3 r' \qquad \text{and} \qquad \boldsymbol{B}(\boldsymbol{r}) \,=\, \boldsymbol{\nabla} \times \boldsymbol{A}(\boldsymbol{r})$$

**Ampère's Law and the Biot-Savart Law**

$$\oint_C \boldsymbol{B}(\boldsymbol{r})\, \mathrm{d}\boldsymbol{s} \,=\, \mu_0 I \qquad \text{and} \qquad \boldsymbol{B}(\boldsymbol{r}) \,=\, \frac{I}{4\pi\epsilon_0 c^2} \int_C \frac{\mathrm{d}\boldsymbol{r}' \times (\boldsymbol{r} - \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^3}$$

**Maxwell's equations of magnetostatics**

$$\boldsymbol{\nabla} \cdot \boldsymbol{B}(\boldsymbol{r}) \,=\, 0 \qquad \text{and} \qquad c^2 \nabla \times \boldsymbol{B}(\boldsymbol{r}) \,=\, \frac{\boldsymbol{j}(\boldsymbol{r})}{\epsilon_0} \qquad \text{with} \qquad c^2 \,=\, \frac{1}{\mu_0 \epsilon_0}$$

Until 2019, the definition of the ampere was based on the force between two parallel, infinitely long current-carrying conductors. An ampere was defined as the constant current flowing through two parallel, straight conductors of negligible cross-sectional area separated by a distance of one meter in a vacuum, such that a force of $2 \cdot 10^{-7}$ newtons per meter of length acts between them. Since 2019, the ampere is no longer defined via a force measurement, but rather via the elementary charge $e$, or more precisely, by the number of such elementary charges that produce a current of $1\,\mathrm{C{\cdot}s^{-1}}$. We have previously referred to $\boldsymbol{B}$ as the magnetic field or magnetic field strength, but strictly speaking, we should refer to it as the magnetic flux density $\boldsymbol{B}$. $\boldsymbol{B}$ is a measure of the density of the magnetic flux lines, which, depending on this flux density, naturally implies a locally higher or lower magnetic field strength. According to eq. (9.6), the magnetic flux is the integrated magnetic flux density $\boldsymbol{B}$ over a given surface. The SI unit of magnetic flux $\Phi$ is the weber, with $1\mathrm{Wb} = \mathrm{V{\cdot}s}$, a unit derived from the law of magnetic induction, which we will discuss in the next chapter. Consequently, the unit of magnetic flux density $\boldsymbol{B}$, the tesla, is $\mathrm{T} = \mathrm{Wb{\cdot}m^{-2}}$. The magnetic vector potential $\boldsymbol{A}$ has no unit of its own but given its defining equation $\boldsymbol{B} = \nabla \times \boldsymbol{A}$, its unit must naturally be $\mathrm{T{\cdot}m}$.

# 10. Magnetic Moment and Electromagnetic Induction

In the last chapter, we learned that magnetic fields are created by the movement of electric charges. The fact that magnetic fields can be generated using electric currents is utilized in many technical applications; electromagnets are arguably the best-known example. However, alongside these magnetic fields generated by human technology, there are also naturally occurring magnets. Their magnetism also arises from electric currents, though not from the macroscopic electric currents flowing through electrical conductors that we discussed in the last chapter using various examples. The magnetic fields generated by naturally occurring magnets originate in microscopic electric currents on the atomic length scale, which give rise to atomic magnetic moments. To understand the behavior of bodies with magnetic properties in magnetic fields, we must first examine the magnetostatic vector potential $\boldsymbol{A}$ more closely.

## 10.1 Vector Potential and Magnetic Moment

There are no magnetic charges. Or to put it another way, there are no magnetic monopoles. Elementary electric charges are simply electric monopoles, and there exists no counterpart to this in magnetostatics. The lowest-order interaction in electrostatics is that between two electric charges, i.e., the interaction between two electric monopoles, as described by Coulomb's law. In contrast, the lowest-order interaction in magnetostatics is that between two magnetic dipoles. To understand this better, let us consider the so-called multipole expansion of $1/|\boldsymbol{r} - \boldsymbol{r}'|$, the expression appearing in the denominator of the defining equations for the electrostatic scalar potential $\phi$, eq. (3.20), and the magnetostatic vector potential $\boldsymbol{A}$, eq. (9.29). To this end, just as we did in section 3.2.2, we consider the Taylor expansion for a quantity $0 \leq \xi < 1$:

$$\frac{1}{\sqrt{1-\xi}} = 1 + \frac{1}{2}\xi + \frac{1\cdot 3}{2\cdot 4}\cdot\xi^2 + \ldots + \frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\cdot\xi^3 + \ldots \tag{10.1}$$

The electric charge densities $\rho(\boldsymbol{r}')$ and current densities $\boldsymbol{j}(\boldsymbol{r}')$, over which the integration in eq. (3.20) and eq. (9.29), respectively, must be performed, are localized, say, within a localization radius $R'$. Therefore, to a first approximation, only positions $\boldsymbol{r}'$ with a magnitude $r' < R'$ contribute to the integrals in eq. (3.20) and eq. (9.29). For $r > R'$, it follows that $0 \leq r'/r < 1$, and we can apply eq. (10.1). To do this, we first rewrite $1/|\boldsymbol{r}-\boldsymbol{r}'|$ as:

$$\begin{aligned}\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} &= \frac{1}{\sqrt{(\boldsymbol{r}-\boldsymbol{r}')^2}} = \frac{1}{\sqrt{r^2 - 2\boldsymbol{r}\cdot\boldsymbol{r}' + r'^2}} \\ &= \frac{1}{r}\left(1 - \frac{2\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2} + \frac{r'^2}{r^2}\right)^{-1/2} = \frac{1}{r}\left(1 - \underbrace{\left[\frac{2\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2} - \frac{r'^2}{r^2}\right]}_{=\xi}\right)^{-1/2}\end{aligned}$$

For this $\xi$, we now expand $1/|\boldsymbol{r}-\boldsymbol{r}'|$ using eq. (10.1), retaining only terms up to $\xi^2$. This yields:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{1}{r}\left(1 + \frac{1}{2}\left[\frac{2\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2} - \frac{r'^2}{r^2}\right] + \frac{3}{8}\left[\frac{2\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2} - \frac{r'^2}{r^2}\right]^2\right) + \ldots \tag{10.2}$$

After expanding the right-hand side of the equation and grouping terms by powers of $r'/r$, we obtain:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{1}{r}\left(1 + \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^2} + \frac{3(\boldsymbol{r}\cdot\boldsymbol{r}')^2 - r^2 r'^2}{2r^4} - 3\frac{4(\boldsymbol{r}\cdot\boldsymbol{r}')r'^2 - r'^4}{8r^4}\right) + \ldots \tag{10.3}$$

We are interested here only in the terms containing $r'/r$ or lower powers, i.e., all terms up to the order of $r^{-2}$. Thus, eq. (10.3) becomes:

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} = \frac{1}{r} + \frac{\boldsymbol{r}\cdot\boldsymbol{r}'}{r^3} + \mathscr{O}\left(r^{-3}\right) \tag{10.4}$$

Substituting this into eq. (3.20), we obtain for the electrostatic potential:

$$\phi(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0}\left(\frac{1}{r}\int\rho(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}' + \frac{\boldsymbol{r}}{r^3}\int\boldsymbol{r}'\rho(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}' + \mathscr{O}\left(r^{-3}\right)\right) \tag{10.5}$$

In this equation, we identify the integral over the charge density $\rho(\boldsymbol{r}')$ as the total charge $q$ that generates the potential $\phi$, that is

$$q = \int\rho(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}' \tag{10.6}$$

and the electric dipole moment $\boldsymbol{p}$ as the integral over $\boldsymbol{r}'\rho(\boldsymbol{r}')$

$$\boldsymbol{p} = \int\boldsymbol{r}'\rho(\boldsymbol{r}')\,\mathrm{d}\boldsymbol{r}' \tag{10.7}$$

This gives us the following expression for the electrostatic potential in the far field:

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\frac{q}{r} + \frac{1}{4\pi\epsilon_0}\frac{\boldsymbol{p}\cdot\boldsymbol{r}}{r^3} + \mathscr{O}\big(r^{-3}\big) \tag{10.8}$$

Using $\boldsymbol{E} = -\nabla\phi$, we obtain for the electrostatic field in far-field approximation:

$$\boldsymbol{E}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0}\frac{q\cdot\boldsymbol{r}}{r^3} + \frac{1}{4\pi\epsilon_0}\frac{3(\boldsymbol{p}\cdot\boldsymbol{r})\boldsymbol{r} - \boldsymbol{p}\cdot r^2}{r^5} + \mathscr{O}\big(r^{-4}\big) \tag{10.9}$$

So far, we have not learned anything truly new, as we are already familiar with the monopole and dipole terms of the multipole expansion for $\phi$ and $\boldsymbol{E}$ in eq. (10.8) and eq. (10.9) from chapter 3. This changes when we consider the same derivation for the vector potential $\boldsymbol{A}$. Substituting eq. (10.4) into eq. (9.29) yields:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\left(\frac{1}{r}\int \boldsymbol{j}(\boldsymbol{r}')\,\mathrm{d}^3r' + \frac{1}{r^3}\int (\boldsymbol{r}\boldsymbol{r}')\boldsymbol{j}(\boldsymbol{r}')\,\mathrm{d}^3r' + \mathscr{O}\big(r^{-3}\big)\right) \tag{10.10}$$

Eq. (10.10) contains two explicitly written-out terms as well as higher-order terms. We have retained the latter to preserve the equality. However, for our present considerations, they are of no particular interest. Analogous to our analysis of eq. (10.5), we identify a monopole term in eq. (10.10), that is

$$\boldsymbol{A}_M(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\cdot\frac{1}{r}\int \boldsymbol{j}(\boldsymbol{r}')\,\mathrm{d}^3r' \tag{10.11}$$

and a dipole term, that is

$$\boldsymbol{A}_D(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\cdot\frac{1}{r^3}\int (\boldsymbol{r}\boldsymbol{r}')\boldsymbol{j}(\boldsymbol{r}')\,\mathrm{d}^3r' \tag{10.12}$$

Let us first consider the monopole term $\boldsymbol{A}_M$. Magnetic charges do not exist, so, unlike for the electrostatic case in eq. (10.6), we cannot simply identify the integral in eq. (10.11) with something like a «magnetic charge» that generates the potential $\boldsymbol{A}$. To be able to make sense of the monopole term $\boldsymbol{A}_M$ in eq. (10.11), we will use a trick that exploits the fact that only stationary currents exist in magnetostatics, meaning that $\nabla\boldsymbol{j} = 0$ holds. For such a stationary current $\boldsymbol{j}$, which vanishes outside a sufficiently large volume $V$, and thus also on its surface $S$, the following relationship holds for two scalar fields $f(\boldsymbol{r})$ and $g(\boldsymbol{r})$:

$$\int_V \big[f(\boldsymbol{r})\boldsymbol{j}(\boldsymbol{r})\nabla g(\boldsymbol{r}) + g(\boldsymbol{r})\boldsymbol{j}(\boldsymbol{r})\nabla f(\boldsymbol{r})\big]\,\mathrm{d}^3r \;=\; 0 \tag{10.13}$$

To verify the validity of this relationship, we use the collection of formulas in tab. 2.1 to establish that, given $\nabla\boldsymbol{j} = 0$, the following holds (for the sake of brevity, we do not explicitly write out the $\boldsymbol{r}$-dependence of $g$, $f$, and $\boldsymbol{j}$):

$$\nabla\big[gf\boldsymbol{j}\big] \;=\; (gf)\nabla\boldsymbol{j} + \boldsymbol{j}\nabla(gf) \;=\; f\boldsymbol{j}\nabla g + g\boldsymbol{j}\nabla f \tag{10.14}$$

Using this equation and Gauss's theorem, we can transform the left-hand side of eq. (10.13) into:

$$\int_V \big[f\boldsymbol{j}\nabla g + g\boldsymbol{j}\nabla f\big]\,\mathrm{d}^3r \;=\; \int_V \nabla[gf\boldsymbol{j}]\,\mathrm{d}^3r \;=\; \int_S \big[gf\boldsymbol{j}\big]\,\mathrm{d}^2r \tag{10.15}$$

If we now let $r$ approach infinity, the surface $S$ shifts to infinity, where $\boldsymbol{j} = 0$, which means that:

$$\lim_{r\to\infty}\int_{S(r)}\big[g(\boldsymbol{r})f(\boldsymbol{r})\boldsymbol{j}(\boldsymbol{r})\big]\,\mathrm{d}^2r \;\underset{r\to\infty \Rightarrow \boldsymbol{j}\to 0}{\overset{\downarrow}{=}}\; 0 \tag{10.16}$$

We have thus proven the validity of eq. (10.13) under the aforementioned conditions. How does this help us determine the integral in eq. (10.11)? To this end, we consider the case $f(\boldsymbol{r}) = 1$ and $g(\boldsymbol{r}) = x_i$ for eq. (10.13), where $x_1 = x$, $x_2 = y$, and $x_3 = z$. In this case, eq. (10.13) becomes, using unit vectors $\hat{\boldsymbol{e}}_i$ for $x_i$:

$$\int_V \big[\boldsymbol{j}(\boldsymbol{r})\hat{\boldsymbol{e}}_i + x_i\,\boldsymbol{j}(\boldsymbol{r})\cdot 0\big]\,\mathrm{d}^3r \;=\; \int_V j_i(\boldsymbol{r})\,\mathrm{d}^3r \;=\; 0 \tag{10.17}$$

With these three equations for the components of $\boldsymbol{j}$

$$\int_V j_x(\boldsymbol{r})\,\mathrm{d}^3r \;=\; 0 \quad , \quad \int_V j_y(\boldsymbol{r})\,\mathrm{d}^3r \;=\; 0 \quad , \quad \int_V j_z(\boldsymbol{r})\,\mathrm{d}^3r \;=\; 0 \tag{10.18}$$

applies, of course

$$\int_V \boldsymbol{j}(\boldsymbol{r})\,\mathrm{d}^3r \;=\; 0 \tag{10.19}$$

and thus eq. (10.11), the monopole term $\boldsymbol{A}_M$ in the expansion of $\boldsymbol{A}$, becomes:

$$\boldsymbol{A}_M(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\cdot\frac{1}{r}\int \boldsymbol{j}(\boldsymbol{r}')\,\mathrm{d}^3r' \;=\; 0 \tag{10.20}$$

The monopole term $\boldsymbol{A}_M$ of the magnetic vector potential is therefore zero. Consequently, there is no counterpart in magnetostatics to the electric charge found in electrostatics. Electric charges exist, but magnetic charges do not. The lowest non-vanishing term in the series expansion of $\boldsymbol{A}$ is the dipole term $\boldsymbol{A}_D$ in eq. (10.12). We will examine this term next, again making use of eq. (10.13). To analyze the integral in eq. (10.12), we set $f(\boldsymbol{r}) = x_i$ and $g(\boldsymbol{r}) = x_j$ in eq. (10.13). The quantities $x_i$ and $x_j$, with $i, j = 1, 2, 3$, again represent the components of the Cartesian coordinate vector $\boldsymbol{r} = (x, y, z)$. Thus, eq. (10.13) becomes:

$$\int_V \left[x_i \boldsymbol{j}(\boldsymbol{r})\hat{\boldsymbol{e}}_j + x_j\, \boldsymbol{j}(\boldsymbol{r})\hat{\boldsymbol{e}}_i\right] \mathrm{d}^3r \;=\; \int_V \left[x_i j_j(\boldsymbol{r}) + x_j j_i(\boldsymbol{r})\right] \mathrm{d}^3r \;=\; 0 \tag{10.21}$$

It therefore holds that

$$\int_V x_i j_j(\boldsymbol{r})\, \mathrm{d}^3r \;=\; -\int_V x_j j_i(\boldsymbol{r})\, \mathrm{d}^3r \tag{10.22}$$

From eq. (10.22) it follows directly for $i = j$ that

$$\int_V \big(\boldsymbol{r}\cdot\boldsymbol{j}(\boldsymbol{r})\big)\, \mathrm{d}^3r \;=\; 0 \tag{10.23}$$

Furthermore, we can show using eq. (10.22) that

$$\int_V (\boldsymbol{r}\boldsymbol{r}')\boldsymbol{j}(\boldsymbol{r}')\, \mathrm{d}^3r' \;=\; \frac{1}{2}\int_V \Big[\big(\boldsymbol{r}\boldsymbol{r}'\big)\boldsymbol{j}(\boldsymbol{r}') - \big(\boldsymbol{r}\boldsymbol{j}(\boldsymbol{r}')\big)\boldsymbol{r}'\Big]\, \mathrm{d}^3r' \tag{10.24}$$

holds, because $\boldsymbol{r}$ is a constant vector in this integral. To prove this, we simply write out the vector components of the integral on the right-hand side of eq. (10.24) and then transform the expression using eq. (10.22):

$$\begin{aligned}
&\frac{1}{2}\int_V \Big[\big(\boldsymbol{r}\boldsymbol{r}'\big)\boldsymbol{j}(\boldsymbol{r}') - \big(\boldsymbol{r}\boldsymbol{j}(\boldsymbol{r}')\big)\boldsymbol{r}'\Big]\, \mathrm{d}^3r' \\
&\quad= \frac{1}{2}\sum_{i,j}\int_V \Big[\big(x_i x_i'\big) j_j(\boldsymbol{r}') - \big(x_i j_i(\boldsymbol{r}')\big) x_j'\Big]\, \mathrm{d}^3r' \\
&\quad= \frac{1}{2}\sum_{i,j} x_i \int_V \left[x_i' j_j(\boldsymbol{r}') - j_i(\boldsymbol{r}') x_j'\right] \mathrm{d}^3r' \qquad \text{(now use eq. (10.22))} \\
&\quad= \frac{1}{2}\sum_{i,j} x_i \int_V \left[x_i' j_j(\boldsymbol{r}') + j_j(\boldsymbol{r}') x_i'\right] \mathrm{d}^3r' \\
&\quad= \frac{1}{2}\sum_{i,j} \int_V 2\big(x_i x_i'\big) j_j(\boldsymbol{r}')\, \mathrm{d}^3r' \;=\; \int_V (\boldsymbol{r}\boldsymbol{r}')\boldsymbol{j}(\boldsymbol{r}')\, \mathrm{d}^3r'
\end{aligned}$$

This proves eq. (10.24). We now rewrite the integrand on the right-hand side of eq. (10.24) using the Grassmann identity (see tab. 2.1) and obtain:

$$\int_V (\boldsymbol{r}\boldsymbol{r}')\boldsymbol{j}(\boldsymbol{r}')\, \mathrm{d}^3r' \;=\; \frac{1}{2}\int_V \Big[\big(\boldsymbol{r}' \times \boldsymbol{j}(\boldsymbol{r}')\big) \times \boldsymbol{r}\Big]\, \mathrm{d}^3r' \tag{10.25}$$

With that, we have almost achieved our objective. It remains to define the magnetic dipole moment $\boldsymbol{m}$ as follows:

$$\boldsymbol{m} \;=\; \frac{1}{2}\int_V \left[\boldsymbol{r}' \times \boldsymbol{j}(\boldsymbol{r}')\right] \mathrm{d}^3\boldsymbol{r}' \tag{10.26}$$

The magnetic dipole moment $\boldsymbol{m}$ is the counterpart to the electric dipole moment $\boldsymbol{p}$ in eq. (10.7). The latter is defined as the integral of the product of $\boldsymbol{r}'$ and $\rho(\boldsymbol{r}')$, and the former as the integral of the cross product of $\boldsymbol{r}'$ and $\boldsymbol{j}(\boldsymbol{r}')$. Thus, the current density $\boldsymbol{j}$ plays a role in magnetostatics comparable to that of the charge distribution $\rho$ in electrostatics. Using $\boldsymbol{m}$, we can now write the dipole term $\boldsymbol{A}_D$ from eq. (10.12) as:

$$\boldsymbol{A}_D(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\frac{\boldsymbol{m}\times\boldsymbol{r}}{r^3} \tag{10.27}$$

Substituting this expression into eq. (10.10), we obtain the following for the magnetic vector potential $\boldsymbol{A}$ in the far field:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\frac{\boldsymbol{m}\times\boldsymbol{r}}{r^3} + \mathscr{O}\big(r^{-3}\big) \tag{10.28}$$

We obtain the magnetic field generated by this vector potential via $\boldsymbol{B} = \nabla\times\boldsymbol{A}$. We therefore need to evaluate the expression $\nabla\times(\boldsymbol{m}\times\boldsymbol{r}/r^3)$. To do this, we again use the collection of formulas in tab. 2.1, substituting $\boldsymbol{a}=\boldsymbol{m}$ and $\boldsymbol{b}=\boldsymbol{r}/r^3$ into the formula for $\nabla\times(\boldsymbol{a}\times\boldsymbol{b})$. This yields:

$$\nabla\times\Big(\boldsymbol{m}\times\frac{\boldsymbol{r}}{r^3}\Big) \;=\; \Big(\frac{\boldsymbol{r}}{r^3}\cdot\nabla\Big)\cdot\boldsymbol{m} - \big(\boldsymbol{m}\cdot\nabla\big)\cdot\frac{\boldsymbol{r}}{r^3} + \boldsymbol{m}\cdot\Big(\nabla\cdot\frac{\boldsymbol{r}}{r^3}\Big) - \frac{\boldsymbol{r}}{r^3}\big(\nabla\cdot\boldsymbol{m}\big) \tag{10.29}$$

We now evaluate this expression in Cartesian coordinates. Since $\boldsymbol{m}$ is a constant vector, the first and last terms on the left-hand side of the equation drop out. The third term also vanishes because we know that, according to eq. (2.68):

$$\nabla\cdot\frac{\boldsymbol{r}}{r^3} \;=\; -\nabla\cdot\nabla\cdot\frac{1}{|\boldsymbol{r}|} \;=\; -\Delta\frac{1}{|\boldsymbol{r}|} \;=\; 4\pi\delta^{(3)}(\boldsymbol{r})$$

Therefore, only the second term of eq. (10.29) remains. For this term, we obtain in Cartesian coordinates:

$$\big(\boldsymbol{m}\cdot\nabla\big)\cdot\frac{\boldsymbol{r}}{r^3} \;=\; \Big(m_x\frac{\partial}{\partial x} + m_y\frac{\partial}{\partial y} + m_z\frac{\partial}{\partial z}\Big)\begin{pmatrix} x/r^3 \\ y/r^3 \\ z/r^3 \end{pmatrix}$$

We have already evaluated such an expression, specifically in eq. (3.54b), when calculating the electrostatic field of the charge dipole. The only difference is that the components of the magnetic dipole moment $\boldsymbol{m}$ appear here, rather than those of the electric dipole moment $\boldsymbol{p}$. We can therefore adopt the result of evaluating the corresponding term in eq. (3.54b) by simply replacing the electric dipole moment there with the magnetic dipole moment $\boldsymbol{m}$. The result is:

$$\big(\boldsymbol{m}\cdot\nabla\big)\cdot\frac{\boldsymbol{r}}{r^3} \;=\; \frac{3(\boldsymbol{m}\cdot\boldsymbol{r})\cdot\boldsymbol{r} - \boldsymbol{m}\cdot r^2}{r^5}$$

This gives us the following expression for the magnetic field produced by $\boldsymbol{A}$ in eq. (10.28):

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0 c^2}\frac{3(\boldsymbol{m}\cdot\boldsymbol{r})\cdot\boldsymbol{r} - \boldsymbol{m}\cdot r^2}{r^5} + \mathscr{O}\left(r^{-4}\right) \tag{10.30}$$

A comparison of eq. (10.30) with eq. (10.9) shows that the magnetic field $\boldsymbol{B}(\boldsymbol{r})$ of the magnetic dipole and the electrostatic field $\boldsymbol{E}(\boldsymbol{r})$ of the electric dipole possess exactly the same structure. The source of the electrostatic dipole field consists of two electric charges of opposite polarity, whereas the source of the magnetic dipole field is the magnetic moment $\boldsymbol{m}$ generated by a current density. Just as the electric dipole in electrostatics, the magnetic dipole is of particular importance in magnetostatics. We shall examine it in greater detail in the next section.

### 10.1.1 The Magnetic Dipole

Before turning our attention to the magnetic dipole, we take a look back at the current-carrying inductor which we examined in section 9.2.2. There, we determined the field inside an infinitely long inductor, i.e., in practical terms, an inductor whose length is far greater than its diameter. In section 9.3.3, we then saw that for a current-carrying hollow cylinder whose height is far greater than its diameter, the field inside the cylinder is the same as that of a current-carrying inductor. When determining the magnetic field inside the inductor, we assumed that the field outside the coil could be neglected; similarly for the hollow cylinder, where we made no assumptions at all regarding the magnetic field outside it. However, we can make a qualitative statement regarding the external field of the inductor and the hollow cylinder, as well as the nature of the field at their respective ends. Although magnetic charges do not exist, if they did, the field produced by a separated magnetic north and south pole would look like the one shown in fig. 10.1a. The pattern of the $\boldsymbol{B}$ field lines in fig. 10.1a is exactly the same as that of the electrostatic dipole in fig. 3.6. This is hardly surprising, given that we have just seen that the dipole terms of the electrostatic field $\boldsymbol{E}$ in eq. (10.9) and the magnetostatic field $\boldsymbol{B}$ in eq. (10.30) share the same structure. Although magnetic charges do not exist, there are indeed magnetic fields that, in some respects, resemble the field that would be produced by non-existent, separate magnetic poles. An example of this is the field of long bar magnets or horseshoe magnets, where the magnetic field at the ends of these permanent magnets closely resembles the field that would be generated by two separate magnetic poles. Let us imagine that the two separated magnetic poles in fig. 10.1a expand, in some magical way, towards the opposite pole until they form a single bar magnet, as shown in fig. 10.1b.

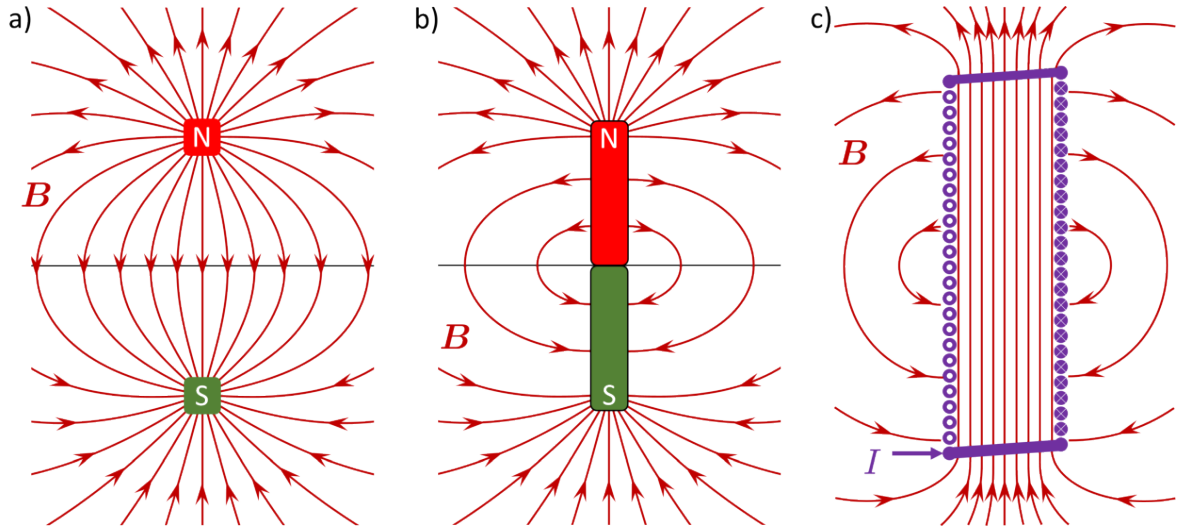



**Fig. 10.1:** Magnetic field of separated magnetic poles, which do not actually exist (a), magnetic field of a bar magnet (b), and magnetic field of a current-carrying inductor (c). At the ends, the magnetic field of the bar magnet resembles that of separated poles, and the magnetic field of a current-carrying inductor resembles that of a bar magnet.

In this process, the short field lines running between the magnetic poles in fig. 10.1a, are pushed outward and carried along as the poles expand toward each other, thereby forming the weak magnetic field along the long side of the bar magnet sketched in fig. 10.1b. Visualizing this transformation of non-existent magnetic monopoles into a bar magnet is not strictly in line with physical principles. However, since the magnetic monopoles in fig. 10.1a do not exist, we are not violating any laws of physics. Still, this approach may help to visualize how the short field lines between the poles in fig. 10.1a transform into the field lines along the long side of the bar magnet shown in fig. 10.1b.
The fields of the bar magnet in fig. 10.1b and the current-carrying inductor in fig. 10.1c resemble each other, even though they arise from completely different causes. Magnetic fields always originate from moving charges. However, in the case of the inductor, the field is generated by macroscopic current loops, whereas for the bar magnet, the macroscopic magnetic field arises from the alignment of atomic magnetic moments produced by the motion of electrons at the atomic level. Magnetic field lines are always closed loops. This is not the case in fig. 10.1a; there, much like in an electrostatic field, the field lines originate at the positive charge (the north pole) and terminate at the negative charge (the south pole). Since magnetic monopoles do not exist, this does not constitute a contradiction. But what is the situation with the bar magnet in fig. 10.1b or the inductor in fig. 10.1c? The field lines of the bar magnet can only form closed loops if the lines run from the south pole to the north pole inside the magnet itself. Thus, the field within the bar magnet resembles the field inside the current-carrying inductor. And what

about the field lines along the sides of the bar magnet and the inductor? The magnetic field lines of $\boldsymbol{B}$ close in this region as well, running back from the south pole to the north pole inside the magnet. However, the magnetic field along the sides of the bar magnet and the inductor is much weaker than at the ends. Consequently, we could neglect it when determining the field inside the inductor. Moreover, the segments that close the field lines along the sides, inside the bar magnet and the inductor, run parallel to the edges from the south pole to the north pole and contribute virtually nothing to the field within the bar magnet or the coil.

With eq. (10.28), we have shown that there are no magnetic charges and that the lowest-order term in the far-field expansion of $\boldsymbol{A}$ is a magnetic dipole. As mentioned at the beginning of the chapter, infinitesimal current loops are the sources of such magnetic dipoles. In the following, we will examine in greater detail the magnetic field of such a magnetic dipole produced by a current loop. In doing so, we want to largely set aside what we learned in the previous section and pretend we have no idea what the vector potential $\boldsymbol{A}$ might look like. Just as we did with the electric dipole in section 3.2.1, we want to work our way step-by-step toward a solution to the problem. To this end, we consider the current-carrying ring sketched in fig. 10.2, through which a current $I$ flows. Conveniently, we treat the problem in cylindrical coordinates, as shown in fig. 10.2, with $\boldsymbol{r} = (R\cos\varphi, R\sin\varphi, z)$ and $\boldsymbol{r}' = (R'\cos\varphi', R'\sin\varphi', z')$. The idealized current loop in fig. 10.2 is assumed to have a radius $a$ and no thickness, and the direction of the current is that of the unit vector $\hat{\boldsymbol{e}}_{\varphi'} = (-\sin\varphi', \cos\varphi', 0)$. Thus, the following must hold for the current density $\boldsymbol{j}(\boldsymbol{r}')$:

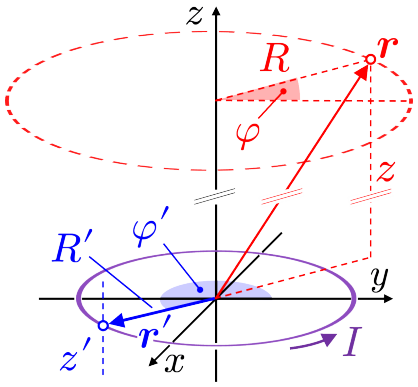



**Fig. 10.2**

$$\boldsymbol{j}(\boldsymbol{r}') = I\delta(R'-a)\delta(z')\hat{\boldsymbol{e}}_{\varphi'} \tag{10.31}$$

We now use eq. (9.29) to determine the magnetic vector potential $\boldsymbol{A}$ generated at an arbitrary point $\boldsymbol{r}$ by the current density flowing in our current loop. Using the cylindrical coordinates in fig. 10.2, $\mathrm{d}^3r' = R'\,\mathrm{d}R'\,\mathrm{d}\varphi'\,\mathrm{d}z'$, and the current density from eq. (10.31), eq. (9.29) becomes:

$$\boldsymbol{A}(\boldsymbol{r}) = \frac{I}{4\pi\epsilon_0 c^2}\int_0^{\infty} R'\,\mathrm{d}R' \int_{-\infty}^{+\infty} \mathrm{d}z' \int_0^{2\pi} \frac{\delta(R'-a)\delta(z')\hat{\boldsymbol{e}}_{\varphi'}}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}\varphi' \tag{10.32}$$

The integrations over $\mathrm{d}R'$ and $\mathrm{d}z'$ are simple because of the $\delta$ functions and with

$$|\boldsymbol{r}-\boldsymbol{r}'| \;=\; \sqrt{(R\cos\varphi - R'\cos\varphi')^2 + (R\sin\varphi - R'\sin\varphi')^2 + (z-z')^2}$$

we obtain, after performing the integrations, the expression:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{Ia}{4\pi\epsilon_0 c^2}\int_0^{2\pi} \frac{\hat{\boldsymbol{e}}_{\varphi'}\,\mathrm{d}\varphi'}{\sqrt{(R\cos\varphi - a\cos\varphi')^2 + (R\sin\varphi - a\sin\varphi')^2 + z^2}} \tag{10.33}$$

If the denominator is expanded, then, with $r^2 = R^2 + z^2$, this equation becomes:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{Ia}{4\pi\epsilon_0 c^2}\int_0^{2\pi} \frac{\hat{\boldsymbol{e}}_{\varphi'}\,\mathrm{d}\varphi'}{\sqrt{r^2 + a^2 - 2Ra\cos(\varphi'-\varphi)}} \tag{10.34}$$

Now we factor $r^2$ out of the square root in the denominator and apply the same far-field approximation for $r \gg a$ that we used for the electric dipole in section 3.2.1: We neglect terms in $a^2/r^2$

$$r\sqrt{1 + \frac{a^2}{r^2} - \frac{2Ra}{r^2}\cos(\varphi'-\varphi)} \;\approx\; r\sqrt{1 - \frac{2Ra}{r^2}\cos(\varphi'-\varphi)}$$

and apply

$$\left(1 \mp x\right)^{-1/2} \;\approx\; 1 \pm \frac{x}{2}$$

With this approximation, we obtain the following expression for $\boldsymbol{A}$:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{Ia}{4\pi\epsilon_0 c^2 r}\int_0^{2\pi} \begin{pmatrix} -\sin\varphi' \\ \cos\varphi' \\ 0 \end{pmatrix} \left(1 + \frac{Ra}{r^2}\cos(\varphi'-\varphi)\right) \mathrm{d}\varphi' \tag{10.35}$$

With $A_z = 0$, we therefore only need to evaluate the integrals for $A_x$ and $A_y$. This is straightforward but involves a bit of writing, yielding the following result for $\boldsymbol{A}$:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{Ia^2\pi}{4\pi\epsilon_0 c^2}\frac{R}{r^3}\hat{\boldsymbol{e}}_\varphi \;=\; \frac{Ia^2\pi}{4\pi\epsilon_0 c^2}\frac{1}{r^3}\begin{pmatrix} -R\sin\varphi \\ R\cos\varphi \\ 0 \end{pmatrix} \;=\; \frac{Ia^2\pi}{4\pi\epsilon_0 c^2}\frac{1}{r^3}\begin{pmatrix} -y \\ x \\ 0 \end{pmatrix} \tag{10.36}$$

A comparison of eq. (10.36) with the corresponding expression for the electrostatic dipole in eq. (3.52) shows that these equations have the same structure, with the difference that $\boldsymbol{A}$ is a vector field and the electrostatic potential $\phi$ is a scalar field. The potential $\phi$ of the electrostatic dipole scales as $z/r^3$, and the vector potential $\boldsymbol{A}$ of the magnetic dipole scales as $R/r^3$. The product of dipole charge $q$ and dipole separation $d$ in eq. (3.52) corresponds in eq. (10.36) to the product of current $I$ and the area $a^2\pi$ of the current loop. The dipole moment $p = q\cdot d$ in electrostatics has a counterpart in the magnetic

dipole moment $m = I \cdot a^2\pi$ in magnetostatics. The dipole moment of the electrostatic dipole is a vector quantity pointing from the negative charge to the positive charge of the dipole. In the case of the magnetic dipole, there is no direction between charges, but there is a direction in which the current flows in the current loop shown in fig. 10.2. The magnetic dipole moment vector $\boldsymbol{m}$ with magnitude $m = I \cdot a^2\pi$ is therefore the directed normal vector to the area enclosed by the current loop. The right-hand rule applies here: the curled fingers of the right hand indicate the direction of the current, and the extended thumb indicates the direction of $\boldsymbol{m}$. If the current flows counter-clockwise as shown in fig. 10.2, the magnetic moment $\boldsymbol{m}$ points in the positive $z$-direction. The next question is how we can rewrite eq. (10.36) to obtain a counterpart to the electrostatic dipole equation eq. (3.53) in which the dipole vector $\boldsymbol{p}$ appears in a scalar product with the position vector $\boldsymbol{r}$. However, eq. (10.36) is already a vector equation where $\boldsymbol{A}$ is oriented everywhere along the unit vector $\hat{\boldsymbol{e}}_\varphi$. We must therefore look for a cross product of $\boldsymbol{m}$ with another vector that yields a vector field $\boldsymbol{A}$ having only a component in the direction of $\hat{\boldsymbol{e}}_\varphi$ for any given $z$-coordinate. As can be easily verified, the vector in question is simply $\boldsymbol{r} = (R\cos\varphi, R\sin\varphi, z)$. Thus, by analogy with eq. (3.53), we can write our equation for the vector potential $\boldsymbol{A}$ as:

$$\boldsymbol{A}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0 c^2}\frac{\boldsymbol{m}\times\boldsymbol{r}}{r^3} \qquad \text{with} \qquad \boldsymbol{m} = I\cdot a^2\pi\cdot\hat{\boldsymbol{e}}_z \tag{10.37}$$

We have thus derived eq. (10.27) by considering a vector potential generated by a ring current $I$ in a current loop of diameter $a$. In fig. 10.2, we selected a specific orientation for the current loop and the direction of the current, and consequently for the magnetic moment $\boldsymbol{m}$. However, eq. (10.37) applies equally to any other orientations, i.e., magnetic dipole moments that can point in arbitrary directions $\mathbf{m} = (m_x, m_y, m_z)$. Apart from the factor $c^{-2}$, eq. (10.37) has the same structure as eq. (3.53), with the only difference being that eq. (10.37) involves the cross product of the magnetic dipole moment and the radius vector, whereas eq. (3.53) involves the scalar product of the electrostatic dipole moment and the radius vector. As we saw in the last section, eq. (10.37) yields the magnetic field via $\boldsymbol{B} = \nabla\times\boldsymbol{A}$:

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{1}{4\pi\epsilon_0 c^2}\frac{3(\boldsymbol{m}\cdot\boldsymbol{r})\cdot\boldsymbol{r} - \boldsymbol{m}\cdot r^2}{r^5} \tag{10.38}$$

To better understand what happens to the magnetic field of the current loop in the dipole approximation, let us examine eq. (10.38) in cylindrical coordinates $\boldsymbol{r} = (R\cos\varphi, R\sin\varphi, z)$. With $\boldsymbol{m} = (0, 0, m)$, we obtain:

$$\boldsymbol{B}(R,\varphi,z) = \frac{1}{4\pi\epsilon_0 c^2}\frac{3mzR\cdot\hat{\boldsymbol{e}}_R + m(2z^2 - R^2)\cdot\hat{\boldsymbol{e}}_z}{\left(R^2 + z^2\right)^{5/2}} \tag{10.39}$$

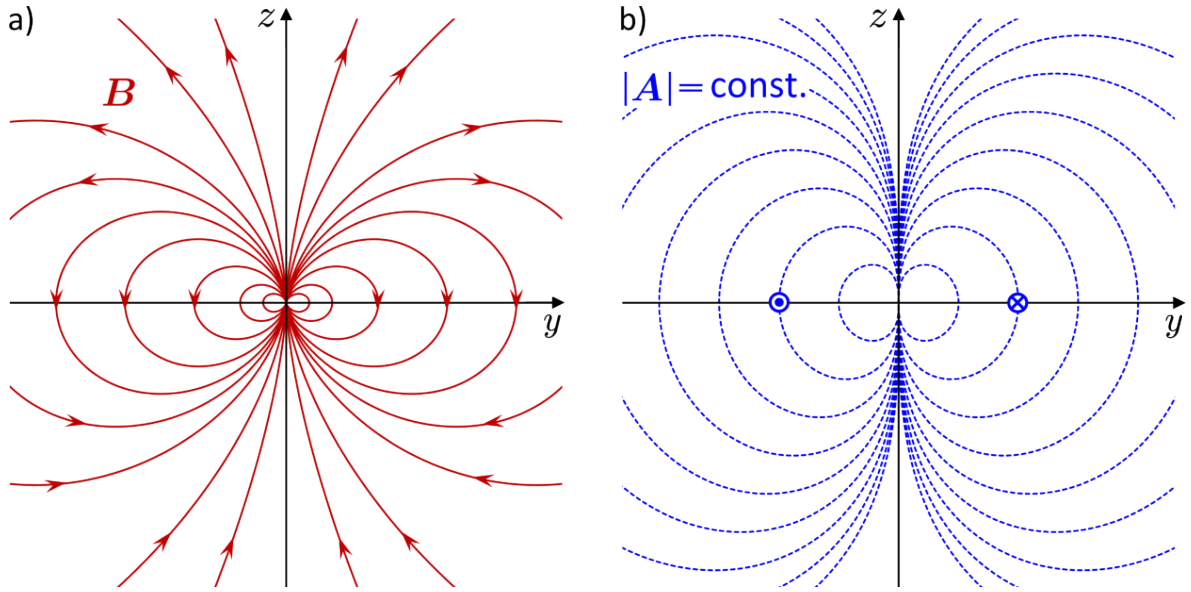



**Fig. 10.3:** Magnetic field lines $\boldsymbol{B}$ (a) and equipotential lines of the vector potential $\boldsymbol{A}$ (b) in the $yz$-plane of the magnetic dipole. The circle with the cross and the circle with the dot indicate the direction of circulation for $\boldsymbol{A}$: at the circle with the cross, the field vector of $\boldsymbol{A}$ enters the $yz$-plane, and at the circle with the dot, it emerges from it.

For $R = 0$, we thus obtain the following for the magnetic field along the $z$-axis:

$$\boldsymbol{B}(z) = \frac{m}{2\pi\epsilon_0 c^2}\frac{\hat{\boldsymbol{e}}_z}{z^3} \tag{10.40}$$

We can compare this result with the one from section 9.3.3, where, without using the dipole approximation, we determined the magnetic field along the axis of a current loop of radius $a$, obtaining the result (see eq. (9.46)):

$$\boldsymbol{B}(z) = \frac{m}{2\pi\epsilon_0 c^2}\frac{\hat{\boldsymbol{e}}_z}{(z^2+a^2)^{3/2}}$$

If we let $a$ approach zero in this equation, while simultaneously keeping the dipole moment $m = Ia^2\pi$ constant, we obtain the result from eq. (10.40), which we determined using the dipole approximation. Just as with the electrostatic dipole, where, in the dipole approximation, the dipole separation approaches zero while the electric dipole moment $\boldsymbol{p}$ remains constant, so too with the magnetic dipole, the radius of the current loop approaches zero in the dipole approximation while the magnetic dipole moment $\boldsymbol{m}$ remains constant. Since the structure of the dipole field in the dipole approximation is identical for the electric dipole, eq. (3.55), and the magnetic dipole, eq. (10.38), we can directly transfer the field line profile of $\boldsymbol{E}$ in fig. 3.7 (see derivation of eq. (3.62)) to the field line profile of $\boldsymbol{B}$. The field lines of $\boldsymbol{B}$ of the magnetic dipole shown in fig. 10.3a are therefore a family of curves which satisfy the equation

$$r = C \cdot \sin^2 \vartheta \tag{10.41}$$

for constant parameters $C$. For the vector potential $\boldsymbol{A}$, we determine the path of equipotential lines in the $yz$-plane by first rewriting eq. (10.37) in polar coordinates, again with $\boldsymbol{m} = (0, 0, m)$:

$$\boldsymbol{A}(r, \vartheta, \varphi) = \frac{m}{4\pi\epsilon_0} \cdot \frac{\sin \vartheta}{r^2} \cdot \hat{\boldsymbol{e}}_\varphi \tag{10.42}$$

We then set $\varphi = 0$ to obtain the value of $\boldsymbol{A}$ in the $yz$-plane and require that the magnitude of $\boldsymbol{A}$ in the $yz$-plane be constant, that is:

$$|\boldsymbol{A}(r, \vartheta, 0)| = \frac{m}{4\pi\epsilon_0} \cdot \frac{\sin \vartheta}{r^2} = C \tag{10.43}$$

This condition is satisfied for the family of curves described by:

$$r^2 = \frac{m}{4\pi\epsilon_0} \frac{\sin \vartheta}{C} \tag{10.44}$$

Fig. 10.3b shows the paths of these curves along which the magnitude of the vector potential $\boldsymbol{A}$ is constant. The vector $\boldsymbol{A}$ has only a component in the direction of $\hat{\boldsymbol{e}}_\varphi$. As shown in fig. 10.3b, $\boldsymbol{A}$ thus points counter-clockwise into the $yz$-plane at the circle marked with a cross, and clockwise out of the $yz$-plane at the circle marked with a dot. The pattern of the $\boldsymbol{B}$ field lines shown in fig. 10.3a, as well as the equipotential lines of the magnitude of $\boldsymbol{A}$ in the $yz$-plane in fig. 10.3b, exhibit rotational symmetry with respect to both the $z$-axis and the $y$-axis. One can readily visualize the three-dimensional configuration of $\boldsymbol{B}$ and the equipotential surfaces of the magnitude of $\boldsymbol{A}$ by rotating these patterns around the coordinate axes.

### 10.1.2 Force on a Magnetic Dipole

We start with the Lorentz force. We consider many moving charges $q_i$ that are moving in the same direction along a current path in a magnetic field $\boldsymbol{B}$. The total force acting on these charges is then:

$$\boldsymbol{F} = \sum_i q_i \left[\boldsymbol{v}_i \times \boldsymbol{B}(\boldsymbol{r_i})\right] \tag{10.45}$$

In the transition from discrete charges to the continuum, that is

$$\sum_i q_i \boldsymbol{v}_i \longrightarrow \int_V \boldsymbol{j}(\boldsymbol{r})\, \mathrm{d}^3 r$$

eq. (10.45) becomes

$$\boldsymbol{F} = \int_V \left[\boldsymbol{j}(\boldsymbol{r}) \times \boldsymbol{B}(\boldsymbol{r})\right] \mathrm{d}^3 r \tag{10.46}$$

For an infinitesimal current loop acting as the source of a magnetic dipole located at $\boldsymbol{a} = 0$, $\boldsymbol{j} \neq 0$ applies only within a very small region around that location. We can therefore assume that $\boldsymbol{B}$ changes very little over the region where $\boldsymbol{j} \neq 0$. In other words, we can approximate $\boldsymbol{B}$ in this region using the first terms of its Taylor expansion about $\boldsymbol{a}$ (see eq. (A.26) in Appendix A), that is:

$$\boldsymbol{B}(\boldsymbol{r}) \;=\; \boldsymbol{B}(\boldsymbol{a}) + \big[(\boldsymbol{r}-\boldsymbol{a})\cdot\nabla_{\boldsymbol{r}'}\big]\boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=a} \;+\; \dots \tag{10.47}$$

Substituting eq. (10.47) with $\boldsymbol{a} = 0$ into eq. (10.46), we obtain:

$$\boldsymbol{F} \;=\; \underbrace{\int_V \boldsymbol{j}(\boldsymbol{r})\,\mathrm{d}^3r}_{=\,0} \times \boldsymbol{B}(0) + \int_V \big(\boldsymbol{r}\cdot\nabla_{\boldsymbol{r}'}\big)\big[\boldsymbol{j}(\boldsymbol{r}) \times \boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0}\big]\,\mathrm{d}^3r \;+\; \dots \tag{10.48}$$

From eq. (10.19), we know that the first integral vanishes. In the second integral on the right-hand side, $\nabla_{\boldsymbol{r}'}$ acts only on $\boldsymbol{B}(\boldsymbol{r}')$, and we can therefore rewrite this integral as follows:

$$\int_V \big(\boldsymbol{r}\cdot\nabla_{\boldsymbol{r}'}\big)\big[\boldsymbol{j}(\boldsymbol{r}) \times \boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0}\big]\,\mathrm{d}^3r \;=\; \int_V \big(\boldsymbol{r}\cdot\nabla_{\boldsymbol{r}'}\big)\boldsymbol{j}(\boldsymbol{r})\,\mathrm{d}^3r \times \boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0} \tag{10.49}$$

When integrating over $\mathrm{d}^3r$, the gradient vector $\nabla_{\boldsymbol{r}'}$ is a constant term, and we can therefore apply to this integral eq. (10.24), yielding:

$$\int_V \big(\boldsymbol{r}\cdot\nabla_{\boldsymbol{r}'}\big)\boldsymbol{j}(\boldsymbol{r})\,\mathrm{d}^3r \;=\; \frac{1}{2}\int_V \Big[\big(\boldsymbol{r}\nabla_{\boldsymbol{r}'}\big)\boldsymbol{j}(\boldsymbol{r}') - \big(\boldsymbol{r}\boldsymbol{j}(\boldsymbol{r}')\big)\nabla_{\boldsymbol{r}'}\Big]\,\mathrm{d}^3r \tag{10.50}$$

We rewrite the integrand on the right-hand side using the Grassmann identity (see tab. 2.1):

$$\int_V \big(\boldsymbol{r}\cdot\nabla_{\boldsymbol{r}'}\big)\boldsymbol{j}(\boldsymbol{r})\,\mathrm{d}^3r \;=\; \frac{1}{2}\int_V \Big[\big(\nabla_{\boldsymbol{r}'} \times \boldsymbol{j}(\boldsymbol{r})\big) \times \boldsymbol{r}\Big]\,\mathrm{d}^3r \tag{10.51}$$

Moving the gradient vector $\nabla_{\boldsymbol{r}'}$ in front of the integral we obtain with eq. (10.26):

$$\int_V \big(\boldsymbol{r}\cdot\nabla_{\boldsymbol{r}'}\big)\boldsymbol{j}(\boldsymbol{r})\,\mathrm{d}^3r \;=\; \nabla_{\boldsymbol{r}'} \times \frac{1}{2}\int_V \big(\boldsymbol{j}(\boldsymbol{r}) \times \boldsymbol{r}\big)\,\mathrm{d}^3r \;=\; -\nabla_{\boldsymbol{r}'} \times \boldsymbol{m} \;=\; \boldsymbol{m} \times \nabla_{\boldsymbol{r}'} \tag{10.52}$$

If we now substitute this result for the second integral into eq. (10.48), we obtain the following expression for the force acting on an infinitesimal current loop with magnetic moment $\boldsymbol{m}$ in a magnetic field $\boldsymbol{B}$:

$$\boldsymbol{F} \;=\; \big(\boldsymbol{m} \times \nabla_{\boldsymbol{r}'}\big) \times \boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0} \tag{10.53}$$

The magnetic moment $\boldsymbol{m}$ is a constant vector, and since $\boldsymbol{B}(\boldsymbol{r}')$ is effectively constant around $\boldsymbol{r}' = 0$, we obtain, using the Grassmann identity (see tab. 2.1):

$$\big(\boldsymbol{m}\times\nabla_{\boldsymbol{r}'}\big)\times\boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0} \;=\; \nabla_{\boldsymbol{r}'}\Big[\boldsymbol{m}\cdot\boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0}\Big] - \boldsymbol{m}\Big[\underbrace{\nabla_{\boldsymbol{r}'}\cdot\boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0}}_{=\,0}\Big]$$

Hence, the force on the infinitesimal current loop at the position $\boldsymbol{r}' = 0$ is:

$$\boldsymbol{F} \;=\; \nabla_{\boldsymbol{r}'}\Big[\boldsymbol{m}\cdot\boldsymbol{B}(\boldsymbol{r}')\big|_{\boldsymbol{r}'=0}\Big] \tag{10.54}$$

Of course, we could have placed our infinitesimal current loop at any other location instead of at $\boldsymbol{r}' = 0$. Therefore, at an arbitrary location $\boldsymbol{r}$, the force acting on an infinitesimal current loop with magnetic moment $\boldsymbol{m}$ is given by:

$$\boldsymbol{F}(\boldsymbol{r}) \;=\; \nabla\big[\boldsymbol{m}\cdot\boldsymbol{B}(\boldsymbol{r})\big] \tag{10.55}$$

A comparison with the corresponding expression in eq. (3.70) shows that the expressions for the potential energy of the electric and the magnetic dipole have exactly the same form, except that eq. (3.70) features the dipole moment $\boldsymbol{p}$, whereas eq. (10.55) features the magnetic moment $\boldsymbol{m}$. Since $\boldsymbol{F} = -\nabla U$, eq. (10.55) provides the expression for the potential energy $U$ of a magnetic dipole in a magnetic field:

$$U \;=\; -\boldsymbol{m}\cdot\boldsymbol{B}(\boldsymbol{r}) \tag{10.56}$$

The expressions for the potential energy of an electric dipole in an electrostatic field $\boldsymbol{E}$ and the potential energy of a magnetic dipole in a magnetic field $\boldsymbol{B}$ are analogous. It appears that we need only replace the electric dipole moment $\boldsymbol{p}$ with the magnetic dipole moment $\boldsymbol{m}$ and $\boldsymbol{E}$ with $\boldsymbol{B}$ in the equations of electrostatics to obtain the corresponding equation for magnetostatics. Just as with the electric dipole, the potential energy of the magnetic dipole is minimal when the dipole moment $\boldsymbol{m}$ is parallel to $\boldsymbol{B}$ and maximal when $\boldsymbol{m}$ and $\boldsymbol{B}$ are antiparallel.

Next, we consider the torque $\boldsymbol{M}_{\bigcirc}$, which acts on a magnetic dipole $\boldsymbol{m}$ that is placed in a constant and homogeneous magnetic field $\boldsymbol{B}$.[1] It is reasonable to assume that the torque on a magnetic dipole in a magnetic field $\boldsymbol{B}$ has the same structure as the torque on an electric dipole in an electrostatic field $\boldsymbol{E}$, as given in eq. (3.64). The total torque on a set of charges $q_i$ at positions $\boldsymbol{r}_i$ moving with velocities $\boldsymbol{v}_i$ in a magnetic field $\boldsymbol{B}$ is given by:

$$\boldsymbol{M}_{\bigcirc} \;=\; \sum_i \boldsymbol{r}_i \times \big[q_i\boldsymbol{v}_i \times \boldsymbol{B}\big] \tag{10.57}$$

For many charges moving in the same direction, the sum in this equation transitions into a continuum, and eq. (10.57) becomes:

[1] For greater clarity and to avoid confusion with the magnetization $\boldsymbol{M}$, we write the torque here as $\boldsymbol{M}_{\bigcirc}$.

$$\boldsymbol{M}_{\bigcirc} = \int_V \boldsymbol{r} \times \big[\boldsymbol{j}(\boldsymbol{r}) \times \boldsymbol{B}\big]\, \mathrm{d}^3 r = \int_V \Big[\big(\boldsymbol{r}\boldsymbol{B}\big)\boldsymbol{j}(\boldsymbol{r}) - \big(\boldsymbol{r}\boldsymbol{j}(\boldsymbol{r})\big)\boldsymbol{B}\Big]\, \mathrm{d}^3 r \tag{10.58}$$

Here, in the second step, we once again used the Grassmann identity to transform the integrand. Since $\boldsymbol{B}$ is constant, we know from eq. (10.23) that the second term in the integrand makes no contribution to the torque, because the corresponding integral vanishes. For the remaining term, we can again apply eq. (10.24) due to the constant vector $\boldsymbol{B}$ and obtain:

$$\boldsymbol{M}_{\bigcirc} = \int_V \big(\boldsymbol{r}\boldsymbol{B}\big)\boldsymbol{j}(\boldsymbol{r})\, \mathrm{d}^3 r = \frac{1}{2}\int_V \Big[\big(\boldsymbol{r}\boldsymbol{B}\big)\boldsymbol{j}(\boldsymbol{r}) - \big(\boldsymbol{j}(\boldsymbol{r})\boldsymbol{B}\big)\boldsymbol{r}\Big]\, \mathrm{d}^3 r \tag{10.59}$$

We rewrite the integrand on the right-hand side using the Grassmann identity, then pull the constant $\boldsymbol{B}$-field out of the integral, and thereby obtain:

$$\boldsymbol{M}_{\bigcirc} = \frac{1}{2}\int_V \big(\boldsymbol{r} \times \boldsymbol{j}(\boldsymbol{r})\big)\, \mathrm{d}^3 r \times \boldsymbol{B} \tag{10.60}$$

From eq. (10.26), we know that the integral term is precisely the magnetic moment $\boldsymbol{m}$. Therefore, the torque on a magnetic dipole in a constant magnetic field is:

$$\boldsymbol{M}_{\bigcirc} = \boldsymbol{m} \times \boldsymbol{B} \tag{10.61}$$

This result shows that our assumption was correct: We indeed can obtain the torque on a magnetic dipole in a constant magnetic field $\boldsymbol{B}$ by simply replacing the electric dipole moment $\boldsymbol{p}$ with the magnetic dipole moment $\boldsymbol{m}$ and the electrostatic field $\boldsymbol{E}$ with a magnetic field $\boldsymbol{B}$ in the equation for the torque on an electric dipole in a constant electrostatic field $\boldsymbol{E}$ (eq. (3.64)). As we know, a torque is always associated with an angular momentum. This naturally also includes the torque on the magnetic dipole we have just considered. To understand the relationship between the magnetic moment $\boldsymbol{m}$ and the angular momentum $\boldsymbol{L}$ associated with the torque $\boldsymbol{M}_{\bigcirc}$ in eq. (10.61), we write the current density $\boldsymbol{j}(\boldsymbol{r}')$ as the sum of $N$ point charges, each with charge $q$, moving at time $t$ at position $\boldsymbol{r}_i(t)$ with velocity $\boldsymbol{v}_i(t)$, that is:

$$\boldsymbol{j}(\boldsymbol{r}') = q\sum_i^N \boldsymbol{v}_i \delta^{(3)}(\boldsymbol{r}' - \boldsymbol{r}_i) \tag{10.62}$$

For the magnetic moment associated with this current density, we obtain:

$$\boldsymbol{m} = \frac{1}{2}\int_V \big(\boldsymbol{r}' \times \boldsymbol{j}(\boldsymbol{r}')\big)\, \mathrm{d}^3 r' = \frac{1}{2}\int_V \Big[\boldsymbol{r}' \times q\sum_i^N \boldsymbol{v}_i \delta^{(3)}(\boldsymbol{r}' - \boldsymbol{r}_i)\Big]\, \mathrm{d}^3 r' \tag{10.63}$$

Due to the $\delta$-function, the evaluation of the integral yields contributions only for $\boldsymbol{r}' = \boldsymbol{r}_i$, and since all point charges are assumed to have the same mass $m_q$, the expression for the dipole moment simplifies to:

$$\boldsymbol{m} = \frac{q}{2}\sum_i^N(\boldsymbol{r}_i \times \boldsymbol{v}_i) = \frac{q}{2m_q}\sum_i^N(\boldsymbol{r}_i \times m_q\boldsymbol{v}_i) = \frac{q}{2m_q}\sum_i^N \boldsymbol{L}_i \tag{10.64}$$

$\boldsymbol{L}_i$ is the angular momentum of a point charge $q$ at position $\boldsymbol{r}_i(t)$ which moves with a velocity $\boldsymbol{v}_i(t)$ at time $t$. This equation applies to any number $N$ of charges $q$ and thus also to a single charge $q$. A single charge $q$, such as an electron or an ion moving in a circular orbit, is thus associated with a magnetic moment. Moreover, as we know from volume 1, section 5.1, an angular momentum is also associated with linear motion, which via the equation

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} \tag{10.65}$$

depends on the position vector $\boldsymbol{r}$ of the observer relative to the moving charge and on the momentum $\boldsymbol{p} = m_q\boldsymbol{v}$ of the charge. Thus, whether on a large or small scale, we must always bear in mind that moving charges are generally associated with angular momentum; unless the position vector $\boldsymbol{r}$ is parallel to the charge's momentum $\boldsymbol{p}$, in which case $\boldsymbol{L}$ vanishes. However, this occurs only if the charge is moving radially away from or toward the observer's position, or if the observer is moving along with the charge. The total angular momentum $\boldsymbol{L}$ of all charges producing the current density, and thus the magnetic moment $\boldsymbol{m}$ in eq. (10.64), is the sum of all $\boldsymbol{L}_i$. The relationship between the magnetic moment of a dipole and its total angular momentum is therefore:

$$\boldsymbol{m} = \frac{q}{2m_q}\boldsymbol{L} \tag{10.66}$$

The ratio of the magnitude of the magnetic moment to the magnitude of the orbital angular momentum is the so-called gyromagnetic ratio $\gamma$:

$$\gamma = \frac{|\boldsymbol{m}|}{|\boldsymbol{L}|} = \frac{|q|}{2m_q} \tag{10.67}$$

From eq. (10.61), we know that a torque $\boldsymbol{M}_\bigcirc$ acts on a magnetic moment $\boldsymbol{m}$ in a homogeneous magnetic field $\boldsymbol{B}$, specifically in such a way that the potential energy $U$ of the dipole given in eq. (10.56) is minimized. This means that a magnetic dipole will orient itself such that $\boldsymbol{m}$ is as parallel as possible to $\boldsymbol{B}$. At the same time, however, this dipole possesses an angular momentum $\boldsymbol{L}$ that is parallel to the «axis of rotation» $\boldsymbol{m}$, the axis about which the charges «rotate». In short, we are dealing with a top governed by the equation of motion for a top:

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \boldsymbol{M}_\bigcirc = \boldsymbol{m} \times \boldsymbol{B} = \gamma\boldsymbol{L} \times \boldsymbol{B} = \boldsymbol{L} \times \gamma\boldsymbol{B} = \boldsymbol{L} \times \boldsymbol{\omega}_p \tag{10.68}$$

We already know this equation from volume 1, section 6.3, where we discussed gyroscopic motion. $\boldsymbol{\omega}_p = \gamma\boldsymbol{B}$ is the precession frequency with which the axis of the dipole and also

$\boldsymbol{m}$, as illustrated in fig. 10.4, precesses around the magnetic field vector $\boldsymbol{B}$. Depending on whether the charge $q$ in eq. (10.66) is positive or negative, $\boldsymbol{m}$ and $\boldsymbol{L}$ are either parallel or antiparallel.

Fig. 10.4 illustrates the situation for a negative charge $q$ where $\boldsymbol{m}$ and $\boldsymbol{L}$ point in opposite directions. This so-called Larmor precession occurs not only with charged particles such as electrons or ions but also with neutral particles composed of charged particles whose magnetic moments do not cancel each other out. Larmor precession is a quantum mechanical effect that can be described classically. In the case of the electron, the magnetic moment $\boldsymbol{m_L}$ arising from the electron's orbital angular momentum is given, both classically and quantum mechanically, by:

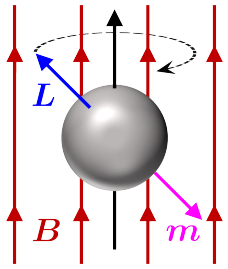


**Fig. 10.4**

$$\boldsymbol{m_L} = -\frac{e}{2m_e}\boldsymbol{L} = -\frac{\mu_B}{\hbar}\boldsymbol{L} \qquad \text{and} \qquad \boldsymbol{m_L} = -g_L\frac{\mu_B}{\hbar}\boldsymbol{L} \tag{10.69}$$

Here, $\mu_B$ is the so-called Bohr magneton, and $g_L$ is a quantum-mechanical quantity. For the electron, $g_L = 1$; that is, the quantum-mechanical and classical results agree. The situation is different for the electron's intrinsic angular momentum $\boldsymbol{S}$, i.e., its so-called spin. The quantum-mechanical result for the magnetic moment $\boldsymbol{m_S}$ arising from the electron spin is:

$$\boldsymbol{m_S} = -g_S\frac{\mu_B}{\hbar}\boldsymbol{S} \approx -2\cdot\frac{\mu_B}{\hbar}\boldsymbol{S} \tag{10.70}$$

With $g_S \approx 2$, the classical and quantum mechanical results thus differ by almost exactly a factor of 2. The corresponding $g$-factors for atoms and other elementary particles are specific to the element or particle being considered. Consequently, atoms can be distinguished based on their precession frequency $f_p = \omega_p/2\pi$ measured in a given magnetic field, provided their magnetic moment is non-zero.

Eq. (3.3) defines the electrostatic field $\boldsymbol{E}$ via the force $\boldsymbol{F}$ it exerts on an electric charge $q$, and eq. (10.61) defines the magnetostatic field $\boldsymbol{B}$ through the torque it exerts on a magnetic dipole $\boldsymbol{m}$. Because magnetic monopoles do not exist, we cannot define the magnetic field $\boldsymbol{B}$ by the force it exerts on a magnetic charge, but rather only by the torque which a magnetic field exerts on a magnetic dipole. Every magnet, whether large or small, is always a dipole. Thus, when we speak of magnetic bodies in magnetic fields, we are always referring, in one way or another, to the behavior of magnetic dipoles in magnetic fields. In principle, any body can be electrically charged, or given an electric charge, by putting positive or negative charges on or in it. Magnetic charges do not

exist. Therefore, it is impossible to place such charges on objects. Some objects can be magnetized, while many cannot. To understand why this is the case, we will look at what lies behind an object's magnetism in the chapter after next.

## 10.2 Faraday's Law of Induction

In 1831, Michael Faraday observed that an electric current is generated in a conducting loop when it moves within a magnetic field or when a nearby magnet is moved. Both of these relative motions result in a time-varying magnetic flux $\Phi$ through the conducting loop, thereby inducing a voltage between the ends of the loop. Fig. 10.5 illustrates Faraday's experimental discovery using the example of a conducting loop $C$ enclosing an area $A$, with the change in magnetic flux through the loop being the result of a bar magnet moved toward or away from the conducting loop. The induced voltage $U_{ind}$ represents an electromotive force $EMF$, or $\varepsilon$, as discussed in section 8.1.2, and is given by:

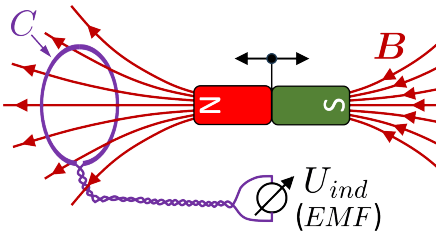



**Fig. 10.5**

$$U_{ind} = \varepsilon = -\frac{\mathrm{d}\Phi(A)}{\mathrm{d}t} \tag{10.71}$$

As shown in fig. 10.5, the leads connecting the contacts on the nearly closed conductor loop $C$ to the voltmeter are twisted together. This ensures that the term $\mathrm{d}\Phi/\mathrm{d}t$ in eq. (10.71), which induces the voltage $U_{ind}$, arises solely from a change in the magnetic flux through the area $A$ of the conductor loop $C$, which is closed except for a very small gap, and that there are no flux changes attributable to the leads connecting to the voltmeter. A change over time in the magnetic flux through the surface $A$ bounded by the curve $C$, as described by eq. (10.71), can, of course, also occur if the loop is situated in a time-varying magnetic field. If, for instance, the bar magnet is replaced by a current-carrying inductor, such as the one shown in fig. 10.1c, and the current flowing through it is varied, then neither the inductor nor the conducting loop needs to move, because $\Phi(A)$ changes in response to the inductor's time-varying magnetic field. We can express the change in magnetic flux in eq. (10.71) using eq. (9.10), and then, using eq. (3.35), we can rewrite $U_{ind}$ as:

$$U_{ind} = \oint_C \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = -\frac{\mathrm{d}}{\mathrm{d}t}\int_A \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} \tag{10.72}$$

We assume that the area $A$ itself is a constant, i.e., it does not change over time in any way, for example by rotating relative to the bar magnet in fig. 10.5 and thus changing the effective cross section exposed to the magnetic flux. Consequently, the change in magnetic flux arises solely from the time-dependent change of the magnetic field $\boldsymbol{B}$ passing through the surface $A$. Thus, from eq. (10.71):

$$\oint_C \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = -\int_A \frac{\partial \boldsymbol{B}}{\partial t}\,\mathrm{d}\boldsymbol{\sigma} \tag{10.73}$$

If $\boldsymbol{B}$ is constant, the right-hand side of this equation is zero, and eq. (10.72) becomes eq. (3.37), the integral form of the second Maxwell equation of electrostatics. Conversely, if $\boldsymbol{B}$ is not constant, the second equation of electrostatics no longer holds, and we must replace it with a modified equation. Here, Stokes' theorem from eq. (2.46) helps us once again, as it allows us to write:

$$\oint_C \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = \int_A [\nabla \times \boldsymbol{E}]\,\mathrm{d}\boldsymbol{\sigma} = -\int_A \frac{\partial \boldsymbol{B}}{\partial t}\,\mathrm{d}\boldsymbol{\sigma} \tag{10.74}$$

It follows directly from this that:

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{10.75}$$

Eq. (10.75) is the second Maxwell equation of electrodynamics and is often referred to as Faraday's law of induction, named after the discoverer of the underlying physical phenomena, though it was Maxwell who first wrote it down in this form. We introduced eq. (10.71) as an experimental finding and subsequently derived Faraday's law of induction, eq. (10.75), from it. To get a better understanding of the physics underlying eq. (10.71), we must examine in detail the effect of the Lorentz force on a wire loop moving through a magnetic field. To do this, however, we must first consider the forces acting on the charge carriers in a stationary current-carrying conductor situated in a magnetic field perpendicular to the direction of the current.

### 10.2.1 The Hall Effect

When in 1879 Edwin Hall discovered the Hall effect named after him, the concept of electricity as a kind of invisible, fluid medium had already been abandoned in favor of models in which moving particles conduct the current; however, the nature of these particles was not yet known, as the electron was not experimentally detected until 1897. With the discovery of the Hall effect, it became clear for the first time that the charge carriers in most metallic conductors had to be negatively charged particles. To understand what happens to the charge carriers in a conductor during the Hall effect, let us

consider fig. 10.6. A conducting plate of thickness $d$, through which a current flows, is situated in a magnetic field. The length $l$ and width $b$ of the plate are assumed to be significantly greater than the plate's thickness. The current flows through the plate in the $y$-direction, and the magnetic field points out of the plane of the figure in the positive $x$-direction. Let the measured current density in fig. 10.6 be given by:

$$\boldsymbol{j}_y = j_y \cdot \hat{\boldsymbol{j}} = q \cdot n \cdot v_D \cdot \hat{\boldsymbol{j}} \tag{10.76}$$

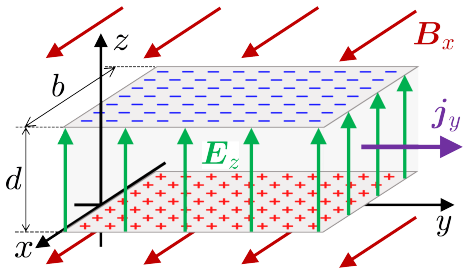



**Fig. 10.6**

Here, $q$ is the electric charge of the charge carriers, $n$ is the charge carrier density, and $v_D$ is the drift velocity of the charge carriers. The Lorentz force acts on these charge carriers $q$ as they move through the plate at the drift velocity $v_D$. Because the magnetic field in fig. 10.6 points out of the plane of the figure, positive charges in the conductor are deflected in the negative $z$-direction, and negative charges in the positive $z$-direction. Regardless of whether the charge carriers $q$ constituting the current are positive or negative, this results in charge separation. This charge separation generates an electric field in the $z$-direction that opposes the Lorentz force. At equilibrium, the Lorentz force and this opposing electric field cancel each other out, and for the situation shown in fig. 10.6, the following applies:

$$q \cdot E_z \cdot \hat{\boldsymbol{k}} = q \cdot v_D \cdot \hat{\boldsymbol{j}} \times B_x \cdot \hat{\boldsymbol{i}} \qquad \Rightarrow \qquad E_z = -v_D \cdot B_x \tag{10.77}$$

If the current flowing in the plate, meaning the current density $\boldsymbol{j}_y$, is kept constant and the fields $E_z$ and $B_x$ are measured, then the magnitude of the drift velocity $v_D$ and its sign can be determined directly from this equation. Given that we know the current density $\boldsymbol{j}_y$, the sign of the charge carriers then follows automatically from eq. (10.77) based on the sign of $v_D$. If the drift velocity $v_D$ is already known, one can substitute $j_y/qn$ for $v_D$ in eq. (10.77) using eq. (10.76), thereby obtaining an equation for determining the charge carrier density $n$:

$$E_z = -\frac{j_y}{q \cdot n} \cdot B_x \tag{10.78}$$

The charge separation in the current-carrying plate, as sketched in fig. 10.6, makes it equivalent to a parallel-plate capacitor with an electrode spacing of $d$. The voltage across this capacitor, i.e., the so-called Hall voltage, is therefore:

$$U_H = E_z \cdot d = \frac{j_y}{q \cdot n} \cdot B_x \cdot d \tag{10.79}$$

The cross-sectional area of the plate through which the current $I$ flows is its thickness $d$ times its width $b$; thus, we can write the current density as $j_y = I/(d \cdot b)$. Substituting this into eq. (10.79), we obtain the following expression for the Hall voltage:

$$U_H = \frac{B_x}{q \cdot n \cdot b} \cdot I = R_H \cdot I \tag{10.80}$$

$R_H$ is the so-called Hall resistance. However, this quantity, which depends on the magnetic field, has nothing to do with ohmic resistance. Instead, it merely expresses the proportionality between the transverse voltage $U_H$ induced by the magnetic field and the current flowing through the plate. By measuring $U_H$ and $I$, one can determine $R_H$ and thus the magnetic field, provided the charge carrier density $n$ is known. Because $n$ in metals is on the order of $10^{22} - 10^{23}\,\mathrm{cm}^{-3}$, the Hall voltages to be measured are very small, even in high magnetic fields. Consequently, the Hall effect only became practically useful for magnetic field measurement with the availability of semiconductors whose intrinsic charge-carrier density is ten orders of magnitude or more lower than that of metals and, thanks to appropriate doping, can be precisely adjusted over a wide range. However, we are not concerned here with magnetic field measurement, but rather with the charge separation demonstrated by the Hall effect in stationary current-carrying conductors situated in a magnetic field.

### 10.2.2 A Relativistic Perspective

The effect of charge separation in conductors can be observed not only in stationary current-carrying conductors but also in conductors, such as a current loop, that does not carry a current but is being moved through a magnetic field. To this end, we first consider a piece of wire being moved through a constant magnetic field, as illustrated in fig. 10.7. The straight wire segment is to move at a constant velocity $\boldsymbol{v}$ perpendicular to the wire axis through a homogeneous magnetic field $\boldsymbol{B}$, which in turn runs perpendicular to $\boldsymbol{v}$. The Lorentz force $\boldsymbol{F}_L$ acts on the free charge carriers $q$ in the wire, i.e., the electrons. The resulting displacement of the charge carriers $q$ in the wire gives rise to an opposing electric field $\boldsymbol{E}$ that counteracts this displacement with the Coulomb force $\boldsymbol{F}_C = q\boldsymbol{E}$. In the state of force equilibrium, the following applies:

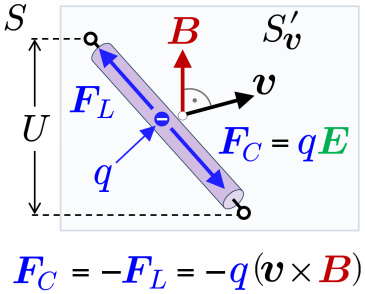



**Fig. 10.7**

$$\boldsymbol{F}_C = q\boldsymbol{E} = -\boldsymbol{F}_L = -q\big(\boldsymbol{v} \times \boldsymbol{B}\big) \tag{10.81}$$

and therefore

$$\boldsymbol{E} = -\boldsymbol{v} \times \boldsymbol{B} \tag{10.82}$$

If one measures the voltage between the ends of the moving wire, one finds that a voltage $U$ exists, given by $U = E \cdot l$. What does the situation look like for an observer moving along with the wire? As we learned in section 9.4, the Lorentz transformation of the $\boldsymbol{E}$ and $\boldsymbol{B}$ fields between a stationary frame $S$ and a frame $S'$ moving relative to it at velocity $\boldsymbol{v}$ yields the following for the field components parallel and perpendicular to the direction of motion $\boldsymbol{v}$:

$$\boldsymbol{E}'_{\parallel} = \boldsymbol{E}_{\parallel} \qquad \text{and} \qquad \boldsymbol{B}'_{\parallel} = \boldsymbol{B}_{\parallel} \tag{10.83}$$

as well as

$$\boldsymbol{E}'_{\perp} = \frac{\boldsymbol{E}_{\perp} + (\boldsymbol{v} \times \boldsymbol{B})_{\perp}}{\sqrt{1 - v^2/c^2}} \qquad \text{and} \qquad \boldsymbol{B}'_{\perp} = \frac{\boldsymbol{B}_{\perp} - (\boldsymbol{v}/c^2 \times \boldsymbol{E})_{\perp}}{\sqrt{1 - v^2/c^2}} \tag{10.84}$$

For the moving observer, therefore, the field components parallel to the direction of motion are the same as in the laboratory frame, whereas the normal components of the fields are not. The $\boldsymbol{E}$-field from eq. (10.82) exists only within the wire, not in the rest of $S$. There, only the fields $\boldsymbol{E}_{\parallel} = 0$ and $\boldsymbol{E}_{\perp} = 0$, as well as $\boldsymbol{B}_{\parallel} = 0$ and $\boldsymbol{B}_{\perp} = \boldsymbol{B}$, exist. Thus, for the co-moving observer applies $\boldsymbol{E}'_{\parallel} = 0$ and $\boldsymbol{B}'_{\parallel} = 0$ as well as

$$\boldsymbol{E}'_{\perp} = \frac{(\boldsymbol{v} \times \boldsymbol{B})_{\perp}}{\sqrt{1 - v^2/c^2}} \qquad \text{and} \qquad \boldsymbol{B}'_{\perp} = \frac{\boldsymbol{B}_{\perp}}{\sqrt{1 - v^2/c^2}} \tag{10.85}$$

For $v \ll c$, the co-moving observer in fig. 10.7 thus sees the fields:

$$\boldsymbol{E}'_{\perp} = (\boldsymbol{v} \times \boldsymbol{B})_{\perp} \qquad \text{and} \qquad \boldsymbol{B}'_{\perp} = \boldsymbol{B}_{\perp} \tag{10.86}$$

Unlike the observer at rest in $S$, however, the co-moving observer sees the electric field $\boldsymbol{E}'$ in $S'$ as existing everywhere, rather than, as the observer in $S$ sees it, as a field $\boldsymbol{E}$ within the wire arising from the charge displacement caused by the Lorentz force and acting in opposition to it. Because the metal wire is at rest in $S'$ and no current flows, for the co-moving observer no electric field exists within the wire. This is ensured by the charge displacement, which cancels out any such field within the wire in the moving system. For an observer in $S'$, the charge displacement within the wire cancels out the field $\boldsymbol{E}'$ inside the wire. Let us now consider what happens when, instead of a straight segment of wire moving through the magnetic field as shown in fig. 10.7, a rectangular wire loop, sketched in fig. 10.8, moves through it. We first examine fig. 10.8a in the laboratory frame $S$. The wire loop moves through a uniform magnetic field in the $x$-direction. With $\boldsymbol{B}_1 = \boldsymbol{B}_2$, the same magnetic field strength acts on the two wire segments of length $a$ that run in the $y$-direction, i.e., perpendicular to the direction of motion $\boldsymbol{v}$.

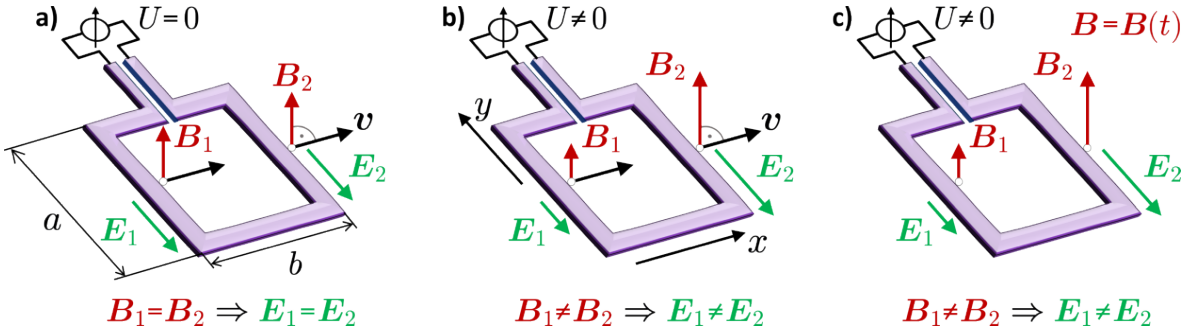



**Fig. 10.8:** (a) A current loop is moved through a homogeneous magnetic field. (b) A current loop is moved through an inhomogeneous magnetic field. (c) The same situation as in (b), but viewed from the perspective of a co-moving observer.

Thus, the Lorentz force acting on the charge carriers in these two wire segments is also equal in magnitude, and consequently, $\boldsymbol{E}_1 = \boldsymbol{E}_2$ holds. In the wire segments of length $b$ that run parallel to the direction of motion $\boldsymbol{v}$ in the $x$-direction, the electric field is zero. With the unit vectors $\hat{\boldsymbol{\imath}} = (1,0,0)$ and $\hat{\boldsymbol{\jmath}} = (0,1,0)$ in the $x$- and $y$-directions, respectively, as well as $\boldsymbol{E}_1 = (0,E_1,0)$ and $\boldsymbol{E}_2 = (0,E_2,0)$, the line integral of $\boldsymbol{E}$ along the wire loop in the counter-clockwise direction becomes:

$$\oint \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = a\cdot\hat{\boldsymbol{\jmath}}\cdot\boldsymbol{E}_2 + b\cdot(-\hat{\boldsymbol{\imath}})\cdot\boldsymbol{0} + a\cdot(-\hat{\boldsymbol{\jmath}})\cdot\boldsymbol{E}_1 + b\cdot\hat{\boldsymbol{\imath}}\cdot\boldsymbol{0} = a\underbrace{(E_2 - E_1)}_{=0} = 0 \qquad (10.87)$$

This is the result of eq. (3.38), telling us that for a wire loop moving through a homogeneous magnetic field, the circulation of the electric field vanishes. Consequently, eq. (3.22), Maxwell's second equation for electrostatics, continues to hold, and the voltmeter in fig. 10.8a indicates a voltage of $U = 0$. This also applies to co-moving observers in $S'$. For them, the wire loop is situated in a homogeneous electric field; consequently, the line integral of $\boldsymbol{E}'$ along the wire loop vanishes for them as well. Let us now examine what changes when the wire loop is moved through an inhomogeneous magnetic field, as sketched in fig. 10.8b. In this case, with $\boldsymbol{B}_1 \neq \boldsymbol{B}_2$, the magnetic field strength acting on the two wire segments of length $a$, which are oriented perpendicular to the direction of motion $\boldsymbol{v}$, is no longer the same. Accordingly, the Lorentz forces acting on the charge carriers in these two wire segments are no longer equal in magnitude, and consequently, neither are the opposing electric fields $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$. The line integral of $\boldsymbol{E}$ along the wire loop therefore yields a non-zero value, and the circulation of the electric field along the wire loop consequently no longer vanishes. However, this contradicts eq. (3.22), the second Maxwell equation of electrostatics. To see how eq. (3.22) must be modified, we first note that a finite voltage $U \neq 0$ is now present across the terminals of the wire loop, which satisfies the following equation:

$$\oint \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} \;=\; a\big(\underbrace{E_2 - E_1}_{\neq 0}\big) \;=\; U \tag{10.88}$$

Because of $\boldsymbol{v} \perp \boldsymbol{B}$ we can also write this voltage with eq. (10.82) as:

$$U \;=\; a{\cdot}(E_2 - E_1) \;=\; -a{\cdot}v{\cdot}(B_2 - B_1) \tag{10.89}$$

If the movement occurs in the $x$-direction as sketched in fig. 10.8b, then we can express $B_2$ in terms of the change in the magnetic field along the $x$-direction over the distance $b$, that is:

$$B_2 \;=\; B_1 + b{\cdot}\frac{\mathrm{d}B}{\mathrm{d}x} \tag{10.90}$$

Substituting this into eq. (10.89), we obtain the following for the voltage:

$$U \;=\; -a{\cdot}v{\cdot}b{\cdot}\frac{\mathrm{d}B}{\mathrm{d}x} \tag{10.91}$$

What does the situation sketched in fig. 10.8b look like for the co-moving observer in $S'$? Viewed from $S$, the wire loop moves through a spatially increasing magnetic field that is constant in time. Viewed from $S'$, the wire loop is at rest in a magnetic field that increases with time. This is the situation shown in fig. 10.8c. Physically, the situations in fig. 10.8b and fig. 10.8c are indistinguishable. The voltage measured by an observer in $S'$ must be the same as that measured by the observer in $S$ according to eq. (10.91). If $v \ll c$, then the lengths of the wire segments $b$ are equal, and the lengths of the wire segments $a$, which lie perpendicular to the direction of motion, are the same in both $S$ and $S'$ regardless. The observer in $S'$ has no knowledge of the velocity $v$. Therefore, with $\boldsymbol{B}_\perp = (0, 0, B) = (0, 0, B') = \boldsymbol{B}'_\perp$ it must hold that:

$$U \;=\; -ab \cdot v\frac{\mathrm{d}B}{\mathrm{d}x} \;=\; -ab \cdot \frac{\mathrm{d}B'}{\mathrm{d}t} \tag{10.92}$$

However, from eq. (10.86), we know that for $v \ll c$ the magnetic fields in $S$ and $S'$ are identical, i.e., $B' = B$ holds, and thus, for the situation in fig. 10.8b, it follows that:

$$\frac{\mathrm{d}B}{\mathrm{d}t} \;=\; v\frac{\mathrm{d}B}{\mathrm{d}x} \tag{10.93}$$

Thus, using eq. (10.88), eq. (10.91), and eq. (10.93), we obtain for the voltage U:

$$U = \oint \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} \;=\; -abv\frac{\mathrm{d}B}{\mathrm{d}x} \;=\; -ab\frac{\mathrm{d}B}{\mathrm{d}t} \tag{10.94}$$

The magnetic flux through the wire loop is given by $\Phi = ab \cdot B$. The magnetic field through which the wire loop moves in the laboratory frame $S$ shown in fig. 10.8b has only a $z$-component. Consequently, the change $\mathrm{d}B/\mathrm{d}t$ occurs perpendicular to the area $ab$ of the wire loop. For the co-moving observer in $S'$ fig. 10.8c, it also holds that the

change $\mathrm{d}B/\mathrm{d}t$ occurs perpendicular to the area $ab$ of the wire loop. In both cases, the rate of change of the magnetic flux is thus given by:

$$\frac{\mathrm{d}\Phi}{\mathrm{d}t} = ab \cdot \frac{\mathrm{d}B}{\mathrm{d}t} \tag{10.95}$$

It then follows immediately from eq. (10.94) and eq. (10.95) that

$$U = \oint \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = -\frac{\mathrm{d}\Phi}{\mathrm{d}t} \tag{10.96}$$

must hold. Using Stokes' theorem and considering very small areas $\mathrm{d}A = ab$ of the wire loop, we can rewrite this equation as:

$$\oint_{\square} \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = \int_{ab} [\nabla \times \boldsymbol{E}]\,\mathrm{d}\boldsymbol{A} = [\nabla \times \boldsymbol{E}] \cdot \mathrm{d}\boldsymbol{A} = \mathrm{d}A \cdot \frac{\mathrm{d}B}{\mathrm{d}t} \tag{10.97}$$

The directed area element $\mathrm{d}\boldsymbol{A}$ points in the same direction as $\boldsymbol{B}$, that is:

$$\mathrm{d}A \cdot \frac{\mathrm{d}B}{\mathrm{d}t} = \mathrm{d}\boldsymbol{A} \cdot \frac{\mathrm{d}\boldsymbol{B}}{\mathrm{d}t} \tag{10.98}$$

Thus, from eq. (10.97) then follows eq. (10.75), which we derived at the beginning from the experimental finding in eq. (10.71), without providing further justification:

$$\nabla \times \boldsymbol{E} = -\frac{\mathrm{d}\boldsymbol{B}}{\mathrm{d}t} \tag{10.99}$$

What is the situation for observers in $S'$ in the case of fig. 10.8c? They observe a time-varying magnetic field $\boldsymbol{B'} = \boldsymbol{B}$ in the $z$-direction and, consequently, a change in magnetic flux in accordance with eq. (10.95). With $\boldsymbol{v} = (v, 0, 0)$ and $\boldsymbol{B} = (0, 0, B)$, eq. (10.86) yields $E' = -v \cdot B$; thus, the counterpart to eq. (10.94) in $S'$ is:

$$\begin{aligned} U &= \oint \boldsymbol{E'}\,\mathrm{d}\boldsymbol{s'} = a(E'_2 - E'_1) \\ &= -a(vB'_2 - vB'_1) = -abv\frac{\mathrm{d}B'}{\mathrm{d}x'} = -ab\frac{\mathrm{d}B'}{\mathrm{d}t'} \end{aligned} \tag{10.100}$$

If we now apply Stokes' theorem for the observer in $S'$, just as we did above for the observer in $S$, then, using the same reasoning we employed for $S$, this equation leads us to the following result for $S'$:

$$\nabla' \times \boldsymbol{E'} = -\frac{\mathrm{d}\boldsymbol{B'}}{\mathrm{d}t'} \tag{10.101}$$

Eq. (10.99) and eq. (10.101) show us that eq. (10.75) holds in both $S$ and $S'$, which means that Faraday's law, that we derived there from the experimental finding expressed by eq. (10.71), holds in both inertial frames. Although the relativistic analysis presented here is subject to limitations, such as the requirement that $v \ll c$, the conclusion remains unaffected: from a physical perspective the situations sketched in fig. 10.8b and fig. 10.8c

are indistinguishable, and the identical voltage measured in both cases is attributable to Faraday's law working in both inertial frames in the same way.

### 10.2.3 Lenz's Law

A current $I$ is associated with the induced voltage described by eq. (10.71), in accordance with Ohm's law:

$$I_{ind} = \frac{U_{ind}}{R} = -\frac{1}{R}\frac{\mathrm{d}\Phi}{\mathrm{d}t} \tag{10.102}$$

But in which direction does this induced current $I_{ind}$ flow? The answer is provided by what is known as Lenz's law, which states that a change in magnetic flux through a conducting loop induces a voltage that drives a current, which gives rise to a magnetic field that opposes the change in magnetic flux through the loop. Lenz's law is a consequence of the conservation of energy, as illustrated by the two cases shown in fig. 10.9. If, as shown in the upper part of fig. 10.9, the north pole of a bar magnet is moved toward a wire loop, the associated change in magnetic flux through the area enclosed by the wire loop induces a current $I_{ind}$ in the loop that flows clockwise when viewed from the bar magnet. This current generates a magnetic field whose north pole points towards the north pole of the bar magnet. However, two magnetic north poles repel each other. Therefore, the magnetic field generated by $I_{ind}$ opposes the motion of the bar magnet, slowing it down. The metal of the wire loop, however, has finite resistance, and the energy lost by the bar magnet as it decelerates corresponds to the heat dissipated by the current flowing through the wire loop. If, on the other hand, the bar magnet is moved away from the conductor loop, again with its north pole facing the loop, as shown in the lower part of fig. 10.9, then the current $I_{ind}$ flows counter-clockwise through the wire loop when viewed from the bar magnet. The magnetic field generated by this current $I_{Ind}$ now has its south pole facing the north pole of the bar magnet as it moves away from the wire loop. The attractive force exerted by this magnetic south pole on the receding north pole of the bar magnet slows the bar magnet down. The energy lost by the bar magnet in the process corresponds again to the heat dissipated by the wire due to ohmic losses resulting from the current flow within it. In short: The current induced by a change in magnetic flux through the wire loop is always directed such that

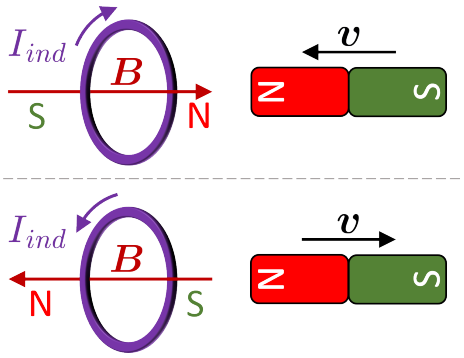



**Fig. 10.9**

the magnetic field it generates opposes the change in the magnetic field that caused it. If the bar magnet in fig. 10.9 does not move at all, i.e., $\boldsymbol{v} = 0$, then the magnetic flux through the wire loop remains constant, and no induced current flows. For an induction current to flow as shown in fig. 10.9, only relative motion between the bar magnet and the wire loop is required. We could therefore also hold the bar magnet stationary and move the wire loop toward or away from it at a velocity $\boldsymbol{v}$ without changing the result.

What is the magnitude of the forces acting on such a wire loop moving in a homogeneous magnetic field? To get an idea of the forces involved, let us consider the situation illustrated in fig. 10.10. For the sake of simplicity, we assume our conductor loop is rectangular and moves in the $xy$-plane along the $x$-direction with a velocity $\boldsymbol{v}$ and it moves out of a uniform magnetic field that fills the half-space $x < 0$ but has only a $z$-component which points out of the plane of the figure in fig. 10.10. First, let us note the obvious: as long as the conductor loop is located entirely within the half-space $x < 0$, no forces act upon it. This is due to the fact that the magnetic field $B_z$ is constant and homogeneous, meaning the magnetic flux through the area of the conductor loop does not change. However, this changes as soon as the conductor loop begins to move out of the magnetic field. At that moment, the magnetic flux through the area of the conductor loop changes, and an induction current begins to flow. In the segments of the conductor loop parallel to the $x$-axis, this current flows in the positive $x$-direction in one of the segments and in the negative $x$-direction in the other. Thus, from eq. (9.41), we know that the force acting on these segments in the $y$-direction must be equal in magnitude and opposite in direction, meaning $\boldsymbol{F}_y + (-\boldsymbol{F}_y) = 0$ must hold, and consequently, no net force in the $y$-direction acts on the conductor loop. If the metal of which the conductor loop is composed has a resistance $R$, then the induction current flowing in the loop at any given time is given by

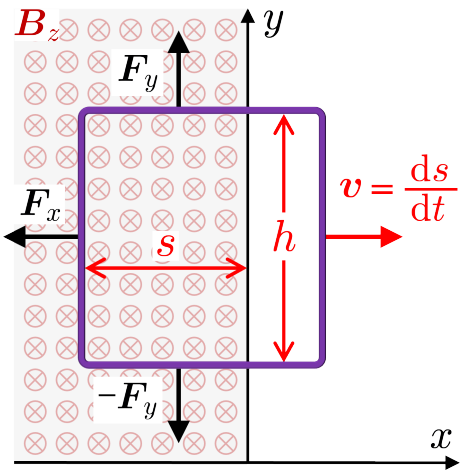



**Fig. 10.10**

$$I = \frac{U}{R} = -\frac{1}{R}\frac{\mathrm{d}\Phi}{\mathrm{d}t} = -\frac{B_z}{R}\frac{\mathrm{d}A(t)}{\mathrm{d}t} \tag{10.103}$$

where $\mathrm{d}A(t)/\mathrm{d}t$ is the rate of change of the area of the conductor loop that is still within the magnetic field, i.e, in the region $x < 0$. At any given time, this area is $A = h \cdot s$ and it decreases at the constant rate of $-\mathrm{d}s/\mathrm{d}t$. Thus, the following applies:

$$\frac{\mathrm{d}A(t)}{\mathrm{d}t} = -h\cdot\frac{\mathrm{d}s}{\mathrm{d}t} = -h\cdot\boldsymbol{v} \tag{10.104}$$

The rate of decrease of the area of the conductor loop remaining within the magnetic field is thus determined by the height $h$ of the loop and the negative of the constant velocity $\boldsymbol{v}$ at which we pull the conductor loop out of the magnetic field. This yields the following expression for the induction current in eq. (10.103):

$$I = \frac{B_z\cdot h\cdot\boldsymbol{v}}{R} \tag{10.105}$$

What can we say about the direction of the current flowing in the conductor loop? The uniform magnetic field points out of the plane of the figure in the positive $z$-direction. According to Lenz's law, the induced current flows in such a way as to oppose the change in magnetic flux through the conductor loop. To achieve this, the current in fig. 10.10 must generate a magnetic field in the $z$-direction that compensates for the loss of magnetic flux through the conductor loop. According to the right-hand rule, this means the current must flow counter-clockwise through the wire loop, as this is the only way the $z$-component of the radial magnetic field generated by the current around the wire points in the correct direction. We can therefore write the current $I$ through the segment that runs parallel to the $y$-axis and lies in the region $x < 0$ as $\boldsymbol{I} = -I\cdot\hat{\boldsymbol{j}}$. If we were to attempt to move the conductor loop, located entirely outside the magnetic field, into that field, a clockwise current would flow through the loop. The $z$-component of the radial magnetic field generated by this current flow through the wire would then be exactly opposite to the $z$-component of the homogeneous magnetic field $B_z$. The force $\boldsymbol{F}_x$ acting on the segment of the conductor loop parallel to the $y$-direction, which is still located within the magnetic field in fig. 10.10 and through which the current from eq. (10.105) flows, is given by:

$$\boldsymbol{F}_x = h\cdot\big[-I\cdot\hat{\boldsymbol{j}}\times B_z\cdot\hat{\boldsymbol{k}}\big] = -h\cdot I\cdot B_z\cdot\hat{\boldsymbol{i}} = -\frac{B_z^2\cdot h^2}{R}\cdot v\cdot\hat{\boldsymbol{i}} \tag{10.106}$$

If we were to attempt to move a conductor loop, located entirely outside the magnetic field shown in fig. 10.10, into that field at a velocity $-\boldsymbol{v}$, it would be opposed by a force $|\boldsymbol{F}_x|$ of equal magnitude.

### 10.2.4 Self-induction

So far, we have considered only changes in magnetic flux resulting either from a conductor loop moving within a magnetic field or from a stationary conducting loop being exposed to a time-varying magnetic field. However, for eq. (10.71), the cause of the change in magnetic flux does not matter. This applies in particular to the case where the conductor

loop itself generates the change in magnetic flux. As previously discussed in section 9.3.3, a current loop generates a magnetic field and if the current in the loop changes, the magnetic field it produces also changes. The result of this is of course a change in the magnetic flux which passes through the conductor loop. As we saw in section 9.3.3, the magnetic field generated by a current loop depends linearly on the generating current and on the geometry of the conductor loop. If the geometry of the conductor loop remains constant, the area enclosed by the conductor loop, through which the magnetic flux passes, also remains unchanged. Thus, a change in magnetic flux is always proportional to the change in current, and using eq. (10.71), we obtain the following expression for the self-induced voltage $U_{ind}$:

$$U_{ind} = -\frac{\mathrm{d}\Phi}{\mathrm{d}t} = -L \cdot \frac{\mathrm{d}I}{\mathrm{d}t} \tag{10.107}$$

The proportionality constant $L$ is the so-called self-inductance of the current loop. $L$ depends on the geometry of the current loop and is measured in henries (H), which expressed through basic units equals $1\,\mathrm{H} = 1\,\mathrm{V \cdot s \cdot A^{-1}}$ or $1\,\mathrm{H} = 1\,\mathrm{Wb \cdot A^{-1}}$. Naturally, when switching an electrical circuit on or off, $\mathrm{d}I/\mathrm{d}t$ is particularly large, and consequently, so is the induced voltage $U_{ind}$. To illustrate this, let us consider the simple circuit shown in fig. 10.11, which consists of a voltage source $U_0$, a current loop with self-inductance $L$, a resistor $R$, and a switch $S$. As long as the switch $S$ is set to «Off», no current flows. If the switch $S$ is flipped to the «On» position, a current $I$ begins to flow. If the current loop were not present in the circuit, the voltage $U_R = U_0$ would immediately drop across resistor $R$, in accordance with Kirchhoff's loop rule, and a current $I_0 = U_0/R$ would immediately flow through $R$. In the circuit containing the current loop, however, the current does not immediately reach the value $I_0$. Instead, it rises to this value much more slowly. The reason for this is that, upon switching on, the flow of current causes a change in the magnetic flux passing through the current loop. According to Lenz's law, this generates an induced voltage $U_{ind}$ that opposes this change in magnetic flux through the current loop, thereby slowing the rise of the current to $I_0$. The question is how we account for the induced voltage in the circuit when applying Kirchhoff's loop rule. When the circuit is switched on, the induced voltage $U_{ind}$ opposes the rise of $U_R$ to the value of $U_0$. The induced voltage thus acts like an additional source voltage that reduces $U_0$. The steady state is reached when $U_{ind}$ has decayed to zero and the constant current $I_0$ is flowing. If the switch in fig. 10.11 is then set to «Off», the result is again an

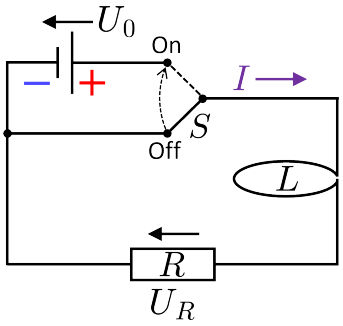



**Fig. 10.11**

induced voltage $U_{ind}$. This time, due to an induction current flowing in the current loop that opposes the change in magnetic flux, thereby slowing the drop in voltage across the resistor from $U_R = U_0$ to $U_R = 0$. Thus, the induced voltage $U_{ind}$ acts once again as an additional source voltage. For Kirchhoff's loop rule, this means:

$$U_0 + U_{ind} = U_R \qquad \text{or} \qquad U_0 - L \cdot \frac{\mathrm{d}I}{\mathrm{d}t} = R \cdot I \tag{10.108}$$

With $I_0 = U_0/R$, we can rewrite this equation as:

$$-\frac{\mathrm{d}I}{I_0 - I} = -\frac{R}{L} \cdot \mathrm{d}t \tag{10.109}$$

Now we integrate both sides of the equation, and since we are evaluating indefinite integrals here, we add the respective constants of integration and combine them into a single constant $c$ for both sides:

$$\ln(I_0 - I) = -\frac{R}{L} \cdot t + c$$

At the moment of switching on at $t = 0$, $I = 0$ must hold, and therefore $c = \ln I_0$. Thus, for the rise in current when the circuit in fig. 10.11 is switched on, we obtain:

$$I = I_0\left(1 - \mathrm{e}^{-R \cdot t/L}\right) \tag{10.110}$$

To determine the current behavior upon switching off at a time $t = t_A$, when the circuit is already in a steady state and the current $I_0 = U_0/R$ is flowing through the circuit, we solve eq. (10.108) for $U_0 = 0$. This leads us to:

$$\frac{\mathrm{d}I}{I} = -\frac{R}{L} \cdot \mathrm{d}t \qquad \Rightarrow \qquad \ln I = -\frac{R}{L} \cdot t + c \tag{10.111}$$

We determine the integration constant $c$ using the boundary condition that the current $I_0$ is flowing at the switch-off time $t_A$. Thus, at $t = t_A$, it must hold that $c = \ln I_0 + R \cdot t_A/L$. If we switch off the circuit, which is in the steady state $U_0 = R \cdot I_0$, the behavior of the current after switching off can be described as follows:

$$I = I_0 \cdot \mathrm{e}^{-R \cdot (t - t_A)/L} \tag{10.112}$$

Fig. 10.12 shows the current waveform for the circuit in fig. 10.11 during the switching-on and switching-off processes, as described by eq. (10.110) and eq. (10.112). For $t < R/L$, the current rises almost linearly from $I = 0$ to approximately $I = I_0/\mathrm{e}$ during switch-on. It then continues to increase but much more slowly, reaching 99% of $I_0$ at the time $t = 4.6 \cdot L/R$. As soon as $I$ reaches the value $I_0$, we are dealing with a discrete circuit. During the switch-off process, the current for $t - t_A < R/L$ initially drops again in a nearly linear fashion and falls to 1% of $I_0$ at $t - t_A = 4.6 \cdot R/L$.

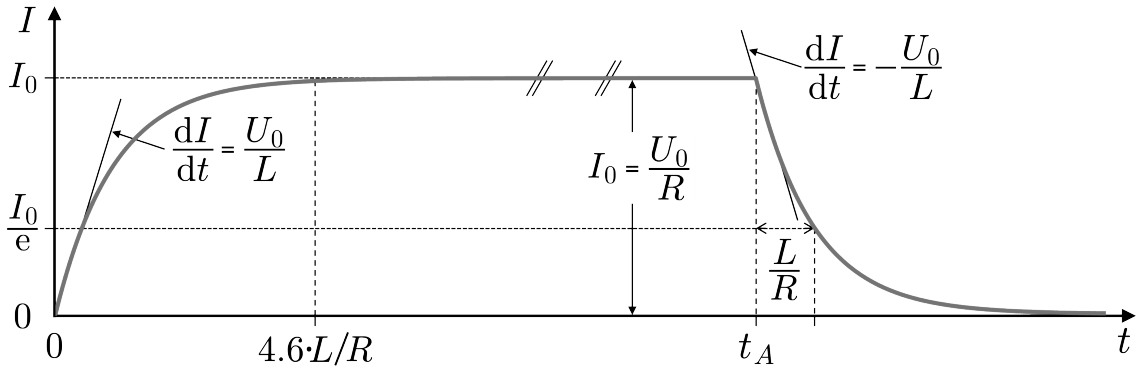



**Fig. 10.12:** Current profile during the switching-on and switching-off processes of the circuit sketched in fig. 10.11, with source voltage $U_0$ and an inductance $L$ and resistor $R$ connected in series.

If we consider not just a single current loop as in fig. 10.11 but many loops wound around a rod, we are dealing with an inductor. From section 9.2.2, we know that the magnetic field inside an inductor is given by eq. (9.24) as $B_i = \mu_0 n I$, where the number of turns $n = N/l$ represents the number of current loops per unit length. If $A$ is the cross-sectional area of the rod around which the current loops are wound, then the magnetic flux passing through each current loop of the inductor is:

$$\Phi = B_i \cdot A = \mu_0 \cdot \frac{A \cdot N}{l} \cdot I \tag{10.113}$$

In each current loop of the inductor, the voltage $U_{ind}$ is induced upon a change in magnetic flux, in accordance with eq. (10.71). For $N$ current loops, the voltage induced in the inductor is therefore:

$$U_{ind} = -N \cdot \frac{\mathrm{d}\Phi}{\mathrm{d}t} = -\mu_0 \cdot \frac{A \cdot N^2}{l} \cdot \frac{\mathrm{d}I}{\mathrm{d}t} \tag{10.114}$$

Thus, the self-inductance $L$ of an inductor with $n = N/l$ current loops per unit length and a constant cross-sectional area $A$ of these current loops is:

$$L = \mu_0 \cdot \frac{A \cdot N^2}{l} = \mu_0 \cdot A \cdot l \cdot \frac{N^2}{l^2} = \mu_0 \cdot V \cdot n^2 \tag{10.115}$$

For an inductor such as the one shown in fig. 9.9, $L$ is therefore proportional to the volume $V$ enclosed by the inductor and it is proportional to the square of the number of turns per unit length, $n = N/l$. To conclude this section on induction, we must address the question of the magnitude of the magnetic energy stored in a current-carrying inductor. If the circuit shown in fig. 10.11 is switched on at $t = 0$ as sketched in fig. 10.12, then the current $I$ flowing through the current loop and the resistor rises, thereby generating a self-induced voltage $U_{ind}$ in the current loop that opposes the flow of charge through the circuit. The voltage source $U_0$ in fig. 10.11 must therefore perform additional work

that would not be required if the current loop $L$ were not present in the circuit. In the equilibrium state, i.e., once the current has reached the value $I = I_0$, this induced voltage has decayed, and the voltage source no longer needs to perform additional work to move charges through the circuit; the magnetic field of the coil is constant in the equilibrium state (with direct current, a conductor loop represents nothing more than a short circuit). The situation is exactly the reverse during the switch-off process. The current loop now performs work to maintain the flow of charge through the circuit. Without the current loop $L$ in fig. 10.11, the current through the resistor would drop immediately to zero upon switching off. With the current loop present, however, a decaying current $I(t)$ flows, as described by eq. (10.112), and power $P(t) = R \cdot I^2(t)$ is consequently dissipated in the resistor $R$ in accordance with Ohm's law. Thus, for the rate of change $\mathrm{d}W/\mathrm{d}t$ of the magnetic energy stored in the current loop during switch-off, the following applies:

$$P(t) = \frac{\mathrm{d}W(t)}{\mathrm{d}t} = R \cdot I_0^2 \cdot \mathrm{e}^{-2 \cdot R \cdot (t - t_A)/L} \tag{10.116}$$

The analysis presented here for the circuit in fig. 10.11 with a single current loop $L$ applies just as well if we replace it with an inductor having an inductance $L$. We obtain the total stored magnetic energy released by such an inductor during the switch-off process by integrating from $t = t_A$ to $t = \infty$:

$$W = R \cdot I_0^2 \cdot \int_{t_A}^{\infty} \mathrm{e}^{-2 \cdot R \cdot (t - t_A)/L} \, \mathrm{d}t = \frac{1}{2} \cdot L \cdot I_0^2 \tag{10.117}$$

We thus know the magnetic field energy stored in an inductor. But what magnetic field strength $B$ is associated with it? Using eq. (9.24), we know the magnetic field strength of the inductor in fig. 9.9 in the equilibrium state:

$$B = \mu_0 \cdot n \cdot I_0 \tag{10.118}$$

Using this equation, we can express the current $I_0$ in fig. 10.11 in terms of the magnetic field strength $B$. If we also replace the self-inductance $L$ of the inductor with the expression in eq. (10.115), we can write the magnetic field energy stored in the inductor as:

$$W = \frac{1}{2} \cdot \mu_0 \cdot V n^2 \cdot \left( \frac{B}{\mu_0 \cdot n} \right)^2 = \frac{1}{2} \cdot V \cdot \frac{B^2}{\mu_0} \tag{10.119}$$

For the energy density $w = W/V$ of the magnetic field in the coil, we thus obtain:

$$w = \frac{B^2}{2 \cdot \mu_0} \tag{10.120}$$

The general formula for the energy density $w = \mathrm{d}U/\mathrm{d}V$ of the magnetic field in a vacuum, which we have determined here for the special case of the magnetic field of an inductor, is given by:

$$\frac{\mathrm{d}U}{\mathrm{d}V} = \frac{1}{2\mu_0}\boldsymbol{B}^2(\boldsymbol{r}) \tag{10.121}$$

With $c^2 = 1/\epsilon_0\mu_0$ we can also write this energy density as:

$$\frac{\mathrm{d}U}{\mathrm{d}V} = \frac{c^2\epsilon_0}{2}\boldsymbol{B}^2(\boldsymbol{r}) = \frac{\epsilon_0}{2}[c\boldsymbol{B}(\boldsymbol{r})]^2 \tag{10.122}$$

A comparison with the energy density of the electric field in eq. (3.50) shows that the two equations have the same form and that one can be obtained from the other by setting $\boldsymbol{E} = c\boldsymbol{B}$.

# 11. Alternating Electric Current

In chapter 8, we dealt with direct electric currents. Here, we now consider electric currents that periodically change their direction, i.e., their polarity. In this case, they are referred to as alternating currents. Much of what we learned about direct-current circuits in chapter 8 can be directly applied to alternating-current circuits, often referred to just as AC-circuits. However, we need to extend some concepts. A direct current has a constant amplitude and does not change over time, except during switching operations. In contrast, the amplitude of alternating currents varies over time, and it possesses a phase. For direct current, an inductor acts merely as a short circuit with a small ohmic resistance; this is not the case for alternating current. In this context, the inductor becomes an important third electrical component, alongside the resistor and the capacitor. An ideal resistor makes no distinction between direct current and alternating current, whereas a capacitor certainly does. However, before we deal with AC circuits, we must first understand how alternating currents are generated in the first place.

## 11.1 The Generation of Alternating Current

An alternating-current generator utilizes Faraday's law of induction to convert mechanical energy into electrical energy. In hydroelectric power plants, this energy comes from either the kinetic energy (run-of-the-river plants) or the potential energy (impoundment plants) of the water. In nuclear power plants, it is water vapor, generated by the heat released during nuclear fission, that provides the energy needed to generate electricity, whereas in gas-fired power plants, burning natural gas supplies this energy. In wind power plants, it is the force of the wind that, via the rotor blades, delivers the energy required to operate an electricity generator. In all these examples, the mechanism of electricity generation is based on the so-called generator principle. Let us once again consider a piece of conducting wire moving within a magnetic field. Fig. 11.1

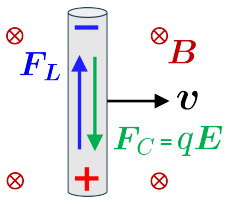


**Fig. 11.1**

illustrates this situation again, a scenario we already examined in section 10.2.2. The Lorentz force $\boldsymbol{F_L}$ acts on the charge carriers in the wire, which is moving at a velocity $\boldsymbol{v}$ perpendicular to the magnetic field $\boldsymbol{B}$. However, only the conduction electrons are free to move within the wire, the ions are bound to their lattice sites. The Lorentz force therefore leads only to a displacement of the conduction electrons, which in turn generates an electric field $\boldsymbol{E}$ in the wire that counteracts the Lorentz force acting on these via the Coulomb force $\boldsymbol{F_C}$. Thus, the following applies:

$$q\boldsymbol{E} + q\boldsymbol{v} \times \boldsymbol{B} = 0 \tag{11.1}$$

With $\boldsymbol{v} \perp \boldsymbol{B}$, it follows that:

$$E = -v \cdot B \tag{11.2}$$

No net force acts on the wire itself as it moves through the magnetic field, because it is electrically neutral. Although the magnetic and electric forces acting on the electrons cancel each other out, the electric field within the straight conducting wire segment of length $l$ is the source of an electromotive force $\varepsilon$:

$$\varepsilon = \int_0^l \boldsymbol{E}\,\mathrm{d}s = -v \cdot B \cdot l \tag{11.3}$$

Eq. (11.3) tells us that by moving an electrical conductor in a magnetic field we can generate a voltage. This physical effect is the principle underlying voltage generation in all alternating-current generators. However, to make this generated voltage technically usable, one must first be able to tap into it. One way to achieve this, for example, is to rotate a conductor loop in a magnetic field. Fig. 11.2 illustrates this using the example of a rectangular conductor loop with area $A$ that rotates at frequency $\omega$ within the field of a permanent magnet about an axis bisecting the loop. This rotation causes the magnetic flux $\Phi$ passing through the conductor loop to change periodically. If the normal vector to the area $A$ of the conductor loop is oriented parallel to the magnetic field $\boldsymbol{B}$, then $\Phi = B \cdot A$ is at its maximum. If we rotate the conductor loop by 90°, the normal vector to $A$ lies perpendicular to the magnetic field $\boldsymbol{B}$, and $\Phi = 0$. If we rotate the conductor loop by another 90°, the normal vector to $A$ is oriented anti-parallel to $\boldsymbol{B}$, and $\Phi = -B \cdot A$. After another rotation by 90°, the normal vector of $A$ is once again perpendicular to $\boldsymbol{B}$, and thus $\Phi = 0$ again. After a final rotation by 90°, we return to the initial state, meaning the normal vector of $A$ is once again oriented parallel to $\boldsymbol{B}$, and

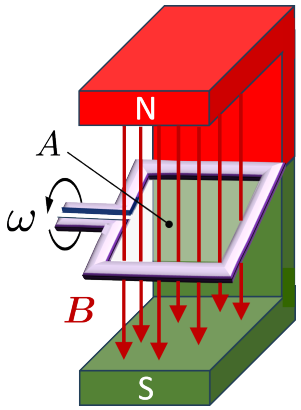



**Fig. 11.2**

thus $\Phi = B \cdot A$. Let $\hat{\boldsymbol{n}}_A$ be the normal vector to the surface $A$. If $\theta$ is the angle between $\hat{\boldsymbol{n}}_A$ and the magnetic field $\boldsymbol{B}$, then, with the directed area $\boldsymbol{A} = \hat{\boldsymbol{n}}_A A$, the magnetic flux $\Phi$ is given by:

$$\Phi \;=\; \boldsymbol{B} \cdot \boldsymbol{A} \;=\; \boldsymbol{B} \cdot \hat{\boldsymbol{n}}_A \cdot A \;=\; B \cdot A \cdot \cos\theta \tag{11.4}$$

If the conductor loop rotates about its axis with an angular velocity $\omega$ as sketched in fig. 11.2, then $\theta = \omega t$ applies, and the time dependence of the magnetic flux through the conducting loop is given by:

$$\Phi(t) \;=\; B \cdot A \cdot \cos\omega t \tag{11.5}$$

For the voltage induced in the conductor loop according to Faraday's law, we thus obtain, following eq. (10.71):

$$U \;=\; -\frac{\mathrm{d}\Phi}{\mathrm{d}t} \;=\; B \cdot A \cdot \omega \cdot \sin\omega t \;=\; U_0 \cdot \sin\omega t \tag{11.6}$$

A rotating-coil generator, the principle of which is illustrated in fig. 11.2, thus produces a sinusoidal voltage with an amplitude $U_0$. The voltage generated in this way is proportional to the strength of the magnetic field and the area enclosed by the conductor loop, and it oscillates with the angular velocity at which the coil rotates. If, instead of a single conductor loop, an inductor with $N$ conductor loops is used, each of these loops experiences the same change in flux, and the amplitude $U_0$ of the generated alternating current in eq. (11.6) increases by a factor of $N$. The more turns $N$ an inductor has, the higher the amplitude of the generated voltage $U_0$. However, rotating an inductor with many turns is impractical, and furthermore, the voltage generated by a rotating inductor is not easily tapped off. A better technical solution is to rotate the magnetic field instead of the inductor. In this setup, several identical inductors, as sketched in fig. 11.3, are each wound around a soft-iron core, which forms the stator. As we shall see later, the elementary magnetic moments in the soft iron align themselves parallel to the field within the inductor, which greatly increases the magnetic flux density inside the inductor, and consequently the change in flux. The permanent magnet of the AC generator is mounted on the shaft of the so-called rotor, which is positioned along the axis of the stator. This rotor is driven by a mechanical force, such as that generated by

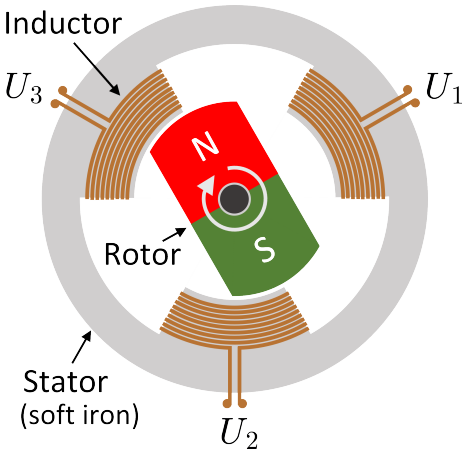



**Fig. 11.3**

water, wind, or a fossil-fuel-powered engine. Since the inductors in fig. 11.3 do not rotate but remain stationary, the generated alternating voltages $U_1$, $U_2$, and $U_3$ can simply be tapped from them.

$$U_1 = U_0 \sin(\omega t + \varphi_1) \ , \ U_2 = U_0 \sin(\omega t + \varphi_2) \ , \ U_3 = U_3 \sin(\omega t + \varphi_3) \tag{11.7}$$

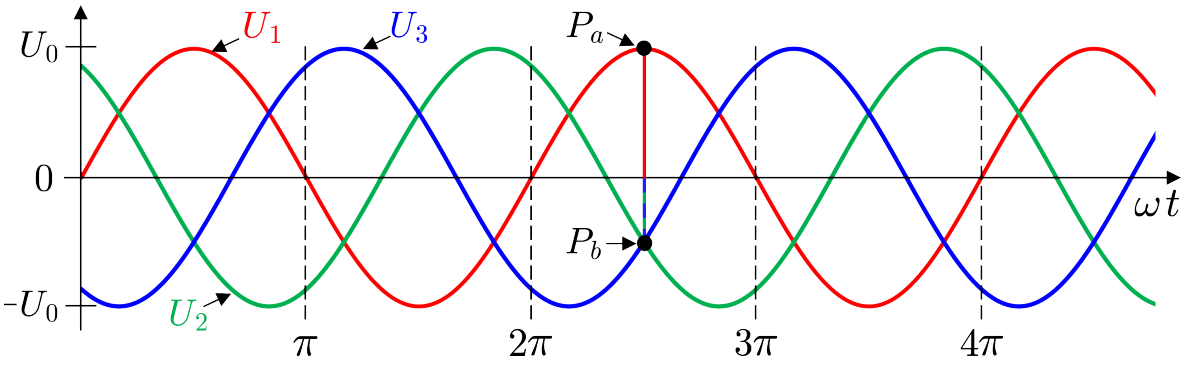



**Fig. 11.4:** Time evolution of the voltages $U_1$, $U_2$, and $U_3$ generated by the AC generator in fig. 11.3: $U_1 = U_0 \sin(\omega t)$, $U_2 = U_0 \sin(\omega t - 2\pi/3)$, $U_3 = U_3 \sin(\omega t - 4\pi/3)$.

To understand the time evolution of these alternating voltages and their relative phase shifts, let us consider what happens during one rotation of the rotor. During a single rotor rotation, each of the three generator inductors shown in fig. 11.3 experiences a change in magnetic flux identical to that experienced by the conductor loop in fig. 11.2 during a full rotation. For instance, if an inductor is positioned opposite the magnet's north pole, then half a rotation later it faces the south pole, and after another half-rotation, it faces the north pole again. Between these two extremes of maximum magnetic flux $\pm\Phi$ passing through the inductor, the magnetic flux passes through zero, i.e, $\Phi = 0$. Consequently, the voltage across each of the three inductors traces a full sine wave cycle during one rotor revolution. Because the inductor axes are positioned at an angle of 120° relative to one another, the sinusoidal waveforms of the three alternating voltages $U_1$, $U_2$, and $U_3$ are phase-shifted by 120° relative to each other. If we set $\varphi_1 = 0$ for $U_1$, then $U_2$ and $U_3$ reach the value of $U_1$ with phase shifts of $\varphi_2 = -120°$ (or $\varphi_2 = -2\pi/3$) and $\varphi_3 = -240°$ (or $\varphi_3 = -4\pi/3$), respectively. Fig. 11.4 shows the time-dependent behavior of this so-called three-phase alternating-current voltage produced by the generator in fig. 11.3. A closer look at the waveforms of the three voltages reveals that the following always holds true:

$$U_1 + U_2 + U_3 \ = \ 0 \tag{11.8}$$

This is best seen in fig. 11.4 by considering a point on the time axis where one of the voltages reaches a maximum. In fig. 11.4, $P_a$ is such a point for $U_1$. The corresponding

point for $U_2$ and $U_3$ on the time axis is $P_b$, where both have the same value, specifically, half the negative value of $U_1$. For alternating currents, it is advantageous to express the time-dependent behavior of voltage and current using complex numbers rather than sine or cosine functions:

$$\underline{U}(t) = \underline{U}_0 \cdot e^{i\omega t} \qquad \text{and} \qquad \underline{I}(t) = \underline{I}_0 \cdot e^{i\omega t} \tag{11.9}$$

$\underline{U}_0$ and $\underline{I}_0$ are complex numbers that do not depend on time. In the case of our three alternating voltages $U_1$, $U_2$, and $U_3$, these complex amplitudes are:

$$\underline{U}_1 = U_{01} \cdot e^{i\varphi_1} \quad , \quad \underline{U}_2 = U_{02} \cdot e^{i\varphi_2} \quad , \quad \underline{U}_3 = U_{03} \cdot e^{i\varphi_3} \tag{11.10}$$

We see the first advantage of describing current and voltage using complex numbers immediately when we examine the complex amplitudes $\underline{U}_1$, $\underline{U}_2$, and $\underline{U}_3$ in the complex plane shown in fig. 11.5. In this so-called phasor representation shown in fig. 11.5, the complex amplitudes are plotted as vectors in the complex plane with $\varphi_1 = 0$, $\varphi_2 = -120°$, and $\varphi_3 = -240°$. As can be easily verified by parallel-shifting the vectors $\underline{U}_2$ and $\underline{U}_3$, as indicated in the inset of fig. 11.5, the three complex amplitude vectors always sum to the zero vector. However, this is precisely what eq. (11.8) states. This shows us that, instead of solving equations, we can perform calculations with complex amplitude vectors in the complex plane, where the real part of the complex solution yields the desired result. We can do this not only with voltages but also with currents. All we need to know for that are the respective complex electrical resistances. But more on that later. Now, using phasor representation, we will familiarize ourselves with the two common configurations of alternating-current circuits: the star connection and the delta connection.

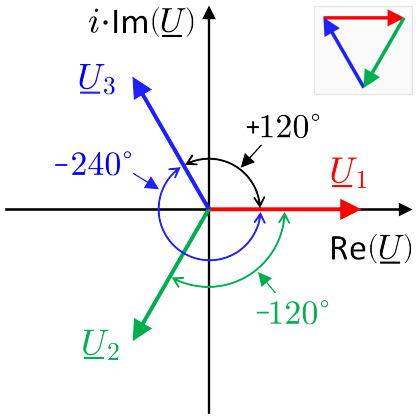



**Fig. 11.5**

Fig. 11.6a shows the star connection, in which the three coils are connected at a single point. A neutral conductor N, which is grounded either directly or via a resistor, may, but need not, be led out from this star point. This is standard in households, as most electrical appliances operate on a single phase, as illustrated for $\underline{U}_{3N}$ in fig. 11.6a. The three coil terminals 1, 2, and 3 are brought out as the so-called phase conductors L1, L2, and L3. The voltage between a phase conductor and the neutral conductor is, as shown in fig. 11.6a for $\underline{U}_{3N}$, always equal to the corresponding inductor voltage, which is also referred to as the phase voltage.

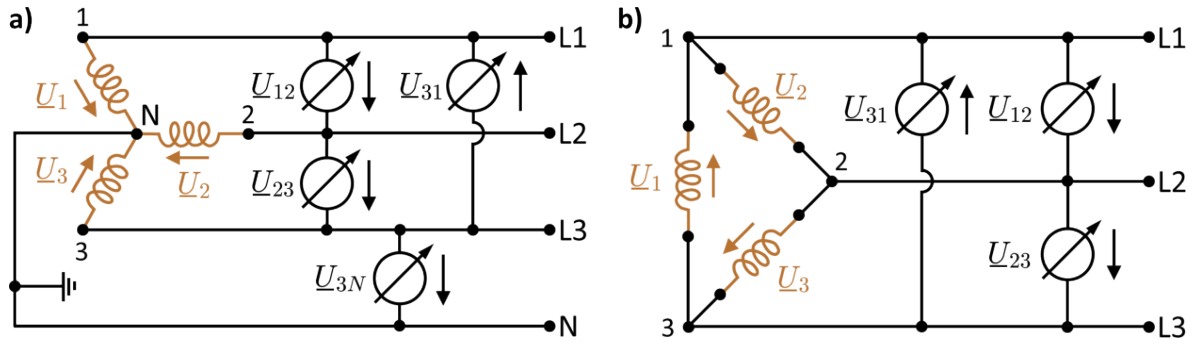



**Fig. 11.6:** Alternating current star connection (a) and delta connection (b).

In a star connection without a neutral conductor, the load resistance is connected between the phase conductors, with the voltages $\underline{U}_{31}$, $\underline{U}_{12}$, and $\underline{U}_{23}$ applied respectively. Regardless of whether a neutral conductor is present, Kirchhoff's current law applies to all currents flowing into and out of the star point:

$$\underline{I}_1 + \underline{I}_2 + \underline{I}_3 = \underline{I}_N \qquad \text{or} \qquad \underline{I}_1 + \underline{I}_2 + \underline{I}_3 = 0 \tag{11.11}$$

If $\underline{I}_N = 0$, the system is said to be operating under a balanced load. If the load is unbalanced, the three phases L1, L2, and L3 are loaded unequally, as is typically the case with many connected loads, and a compensating current flows through the neutral conductor. Naturally, this is not possible without a neutral conductor; consequently, only balanced load operation is feasible in this case, as otherwise electrical equipment could be overloaded and destroyed. To determine the voltage between the phase conductors, we examine the phasor diagram of the star connection shown in fig. 11.7. We have omitted the coordinate system here, as we are concerned solely with the relationship between the phasors themselves, which does not require reference to a coordinate system. The complex phase voltages remain at an angle of 120° to one another and point towards the neutral point N. To determine the line-to-line voltages shown in fig. 11.6a, we simply need to draw the corresponding connecting vectors between points 1, 2, and 3 in fig. 11.7. We can then determine the magnitudes of these complex voltage vectors $\underline{U}_{31}$, $\underline{U}_{12}$, and $\underline{U}_{23}$ using simple trigonometry, as illustrated in fig. 11.7 with the example of $\underline{U}_{31}$. If $\underline{U}_{LL}$ denotes a line-to-line voltage ($\underline{U}_{31}$, $\underline{U}_{12}$, or $\underline{U}_{23}$) and $\underline{U}_{Ph}$ denotes a phase voltage ($\underline{U}_1$, $\underline{U}_2$, or $\underline{U}_3$), then the following applies:

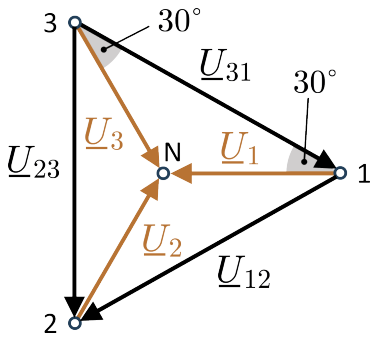



**Fig. 11.7**

$$\underline{U}_{LL} = 2 \cdot \underline{U}_{Ph} \cdot \cos 30° = \underline{U}_{Ph} \cdot \sqrt{3} \tag{11.12}$$

In the delta connection shown in fig. 11.6b, where the inductors are connected in series to form a closed loop, there is no common star point and thus no neutral conductor. As can be seen from fig. 11.6b, in a delta connection the line-to-line voltages are equal to the phase voltages, i.e., $\underline{U}_{LL} = \underline{U}_{Ph}$, and, applying Kirchhoff's loop rule, the following holds:

$$\underline{U}_1 + \underline{U}_2 + \underline{U}_3 = 0 \qquad \text{and} \qquad \underline{U}_{31} + \underline{U}_{12} + \underline{U}_{23} = 0 \tag{11.13}$$

The number of inductors in an alternating-current generator is limited only by its size, but the global standard is a three-phase alternating current. However, the frequency $f = \omega/2\pi$ and peak voltage $U_0$ are not the same everywhere. In North America, in many countries across Central and South America, and in parts of Asia, such as Japan, South Korea, and Taiwan, the grid frequency is $f = 60\,\text{Hz}$. However, in Africa, Europe, many parts of Asia, the southern part of South America, and Australia, the grid frequency is $f = 50\,\text{Hz}$.

## 11.2 Complex Resistances – Impedances

The term impedance is derived from the Latin «impedire», meaning «to inhibit» or «to hinder». Impedance, denoted by the symbol $\underline{Z}$, refers to a form of resistance to alternating current expressed by a complex number that may be frequency-dependent and/or cause a phase shift between current and voltage. The three impedances we consider here are ohmic resistors $R$, capacitors $C$, and inductors $L$. Let us begin with the ohmic resistor $R$. Eq. (8.12), Ohm's law, the linear relationship between current and voltage, also applies to alternating currents. Thus, for time-dependent complex voltages and currents as described by eq. (11.9), the following holds:

$$\underline{U}(t) = R \cdot \underline{I}(t) = \underline{Z}_R \cdot \underline{I}(t) \tag{11.14}$$

The impedance $\underline{Z}_R = R$ of an ohmic resistor $R$ is independent of frequency and causes no phase shift between current and voltage. Next, let us consider the relationship between current and voltage for a capacitance $C$. If $Q$ is the charge on the capacitor, then, using eq. (4.63), the following applies:

$$U = \frac{Q}{C} \qquad \Rightarrow \qquad \frac{\mathrm{d}U}{\mathrm{d}t} = \frac{1}{C} \cdot \frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{1}{C} \cdot I \tag{11.15}$$

If we substitute $\underline{U}(t)$ and $\underline{I}(t)$ from eq. (11.9) here, it follows that:

$$\frac{\mathrm{d}\underline{U}(t)}{\mathrm{d}t} = i\omega \cdot \underline{U}(t) = \frac{1}{C} \cdot \underline{I}(t) \tag{11.16}$$

This gives us for the relationship between current and voltage:

$$\underline{U}(t) = \frac{1}{i\omega C}\cdot\underline{I}(t) = \underline{Z}_C\cdot\underline{I}(t) \tag{11.17}$$

The impedance $\underline{Z}_C = 1/i\omega C$ of a capacitor $C$ is therefore frequency-dependent and results in a phase shift. The higher the angular frequency $\omega$ of the alternating current, the lower the impedance of a capacitance, and vice versa. When transitioning to direct current, $f = \omega/2\pi \to 0$ applies, and consequently, the impedance of a capacitor becomes infinitely large. This is just as it should be, since a capacitor represents an open circuit to direct current. The phase shift between voltage and current caused by a capacitance is $1/i = -i$, i.e., $-90°$. Thus, in eq. (11.17), the voltage lags the current by a phase difference of $\pi/2$.

For the capacitor, we were able to derive the relationship between alternating voltage and alternating current from eq. (4.63). But what does this relationship look like for an inductor? To understand this, let us look at fig. 11.8. The gray box in this circuit represents an inductor in which its ohmic resistance $R_L$ and its inductance $L$ are connected in series. If the box represented only an ohmic resistor, the simple relationship $U_\varepsilon(t) = U_L(t)$ would apply. However, due to the inductance $L$, there is an additional voltage $U_{ind}(t)$ that we must take into account. We know that $U_{ind}(t)$ always opposes $U_L(t)$; therefore, the following must hold:

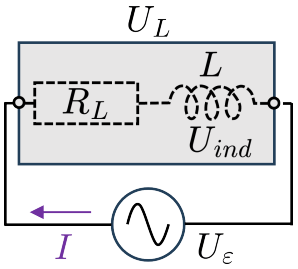



**Fig. 11.8**

$$U_\varepsilon(t) = U_L(t) - U_{ind}(t) \tag{11.18}$$

For the current flowing through the inductor, $I(t) = U_L/R_L$ applies. Using this and $U_{ind}(t) = -L\cdot\mathrm{d}I(t)/\mathrm{d}t$ from eq. (10.107), we can rewrite eq. (11.18) as:

$$U_\varepsilon(t) = I(t)R_L + L\frac{\mathrm{d}I(t)}{\mathrm{d}t} \tag{11.19}$$

If we consider an ideal inductor, then $R_L = 0$, and we thus obtain the following relationship between alternating voltage and alternating current for the inductor:

$$U_\varepsilon(t) = L\frac{\mathrm{d}I(t)}{\mathrm{d}t} \tag{11.20}$$

We apply this to complex voltages and currents and obtain, using eq. (11.9):

$$\underline{U}(t) = L\frac{\mathrm{d}\underline{I}(t)}{\mathrm{d}t} = i\omega L\cdot\underline{I}(t) = \underline{Z}_L\cdot\underline{I}(t) \tag{11.21}$$

The impedance $\underline{Z}_L = i\omega L$ of an inductor $L$ is therefore also frequency-dependent and causes a phase shift between current and voltage. Unlike a capacitor, however, the

alternating-current resistance of an inductor increases with the angular frequency $\omega$ of the alternating current. The reason for this is that at higher frequencies, the current alternates more rapidly, resulting in a greater rate of change of the magnetic flux over time. However, a larger $\mathrm{d}\Phi/\mathrm{d}t$ generates a higher self-induced voltage, which then opposes the flow of current more strongly. An inductance causes a phase shift of 90° between voltage and current. In this case, therefore, it is the current that lags the voltage by a phase difference of $\pi/2$ in eq. (11.21).

Kirchhoff's rules apply to alternating current just as they do to direct current. In the case of direct current, we used them in section 8.1.3 to determine the total resistance or total capacitance for ohmic resistors and capacitors connected in parallel and in series. What does this look like for the parallel or series connection of impedances $\underline{Z}$ in alternating-current circuits? To do this, we first consider the impedances $\underline{Z}_1$, $\underline{Z}_2$, and $\underline{Z}_3$ connected in series as shown in fig. 11.9. The same current $I$ flows through each of the impedances. Kirchhoff's mesh rule eq. (8.20) therefore yields:

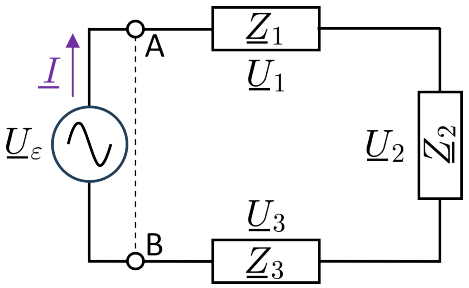



**Fig. 11.9**

$$\underline{U}_\varepsilon = \underline{U}_1 + \underline{U}_2 + \underline{U}_3 = (\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3)\cdot\underline{I} \tag{11.22}$$

If all three impedances in fig. 11.9 are either purely resistive, capacitive, or inductive, the total impedance for a series connection is given by:

$$\underline{Z}_R = R_1 + R_2 + R_3 \tag{11.23a}$$

$$\underline{Z}_C = \frac{1}{i\omega}\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right) \tag{11.23b}$$

$$\underline{Z}_L = i\omega(L_1 + L_2 + L_3) \tag{11.23c}$$

If, in fig. 11.9, $\underline{Z}_1$ is an ohmic resistor $R$, $\underline{Z}_2$ is a capacitor $C$, and $\underline{Z}_3$ is an inductor $L$, then the total impedance of this series connection of impedances is:

$$\underline{Z} = R + \frac{1}{i\omega C} + i\omega L = R + i\left(\omega L - \frac{1}{\omega C}\right) \tag{11.24}$$

Instead of three impedances, we could consider any number of series-connected impedances to the right of terminals A and B in fig. 11.9. Regardless, the total impedance $\underline{Z}_{eff} = \underline{U}_\varepsilon/\underline{I}$ of the circuit would always take the form $\underline{Z}_{eff} = R + iX$.

Next, we consider the parallel-connected impedances $\underline{Z}_1$, $\underline{Z}_2$, and $\underline{Z}_3$ shown in fig. 11.10. According to Kirchhoff's junction rule in eq. (8.21), the three alternating currents flowing through these impedances - $\underline{I}_1$, $\underline{I}_2$, and $\underline{I}_3$ - must sum to the total current $\underline{I}$ supplied by the voltage source $\underline{U}_\varepsilon$. The same voltage drop $\underline{U}$ occurs across each of the three impedances. Thus, the currents through these impedances are given by $\underline{I}_n = \underline{U}/\underline{Z}_n$. For the total impedance of the circuit in fig. 11.10, we thus obtain the equation:

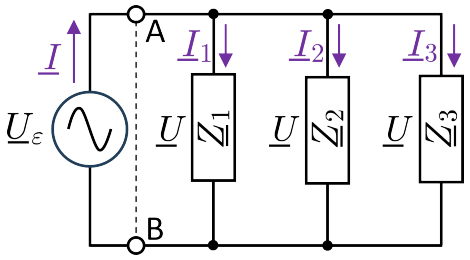



**Fig. 11.10**

$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = \underline{U}\left(\frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3}\right) \tag{11.25}$$

Thus, in a parallel connection, the reciprocals of the individual impedances add up, just as we observed with ohmic resistors and capacitors in direct current-circuits, except that, while a capacitor represents an open circuit for direct currents, here it acts as an alternating-current impedance. If all three impedances in fig. 11.10 are either purely resistive, capacitive, or inductive, the following applies to the total impedance in a parallel connection:

$$\underline{Z}_R = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \tag{11.26a}$$

$$\underline{Z}_C = i\omega(C_1 + C_2 + C_3) \tag{11.26b}$$

$$\underline{Z}_L = \frac{1}{i\omega}\left(\frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}\right) \tag{11.26c}$$

If, in fig. 11.10, $\underline{Z}_1$ is an ohmic resistor $R$, $\underline{Z}_2$ is a capacitor $C$, and $\underline{Z}_3$ is an inductor $L$, then the total impedance of this parallel connection of impedances is:

$$\underline{Z} = \frac{1}{R} + i\omega C + \frac{1}{i\omega L} = \frac{1}{R} + i\left(\omega C - \frac{1}{\omega L}\right) \tag{11.27}$$

As with the series connection of impedances in fig. 11.9, we can also add any number of parallel-connected impedances to the right of terminals A and B in fig. 11.10. Regardless of how many impedances we add, however, the total impedance $\underline{Z}_{eff} = \underline{U}_\varepsilon/\underline{I}$ of the circuit also always takes the form $\underline{Z}_{eff} = R + iX$. This applies not only to simple series or parallel connections of impedances. Rather, any type of circuit connected to an alternating-current voltage source via two terminals A and B can be described by an effective impedance defined by $\underline{Z}_{eff} = \underline{U}_\varepsilon/\underline{I}$.

## 11.3 Alternating Current Power

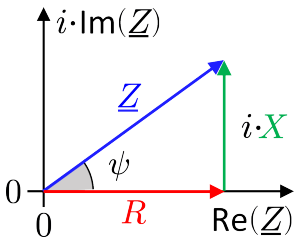



**Fig. 11.11**

In section 8.1.2, we established that the electrical power in a direct-current circuit is given by $P = U \cdot I$ or, alternatively, by $P = R \cdot I^2$ or $P = U^2/R$. In direct-current circuits, this power is dissipated as heat, and we can expect the same to apply to alternating-current operation. But what about the impedances $\underline{Z}_C$ and $\underline{Z}_L$, which, unlike in direct-current circuits, do not represent the equivalent of an open circuit or a short circuit in alternating-current circuits? Obviously, when calculating alternating-current power, we need to take into account the impedances $\underline{Z}_C$ and $\underline{Z}_L$. Fortunately, we have just learned in the previous section what we need to do for that: We must use the total impedance $\underline{Z}$ in the form

$$\underline{Z} = R + iX \tag{11.28}$$

to calculate alternating-current power. As illustrated in fig. 11.11, $\underline{Z}$ represents a phasor in the complex plane. The real part of this phasor is the so-called effective resistance $R_{eff}$, which we have identified here with the ohmic resistance $R$ of the alternating-current circuit, while the imaginary part $X$ consists solely of the circuit's impedances $\underline{Z}_C$ and $\underline{Z}_L$. From fig. 11.11 we read off:

$$\underline{Z} = |\underline{Z}| \cdot e^{i\psi} = \sqrt{R^2 + X^2} \cdot e^{i\psi} \tag{11.29}$$

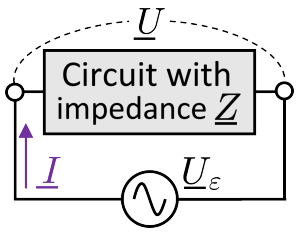



**Fig. 11.12**

Here, the magnitude of $\underline{Z}$, i.e., $|\underline{Z}|$, is the so-called apparent impedance, and the angle $\psi$ in fig. 11.11 describes the phase shift between voltage and current caused by $\underline{Z}$. But between which voltage and which current, exactly? With $\underline{Z}$ being the impedance of our circuit, as shown schematically in fig. 11.12, $\psi$ then must be the phase shift between $\underline{I}$ and $\underline{U}_\varepsilon$. In the case of $\psi = 0$ our circuit acts as a pure ohmic resistor, and the alternating current and alternating voltage are in phase. In the case of $\psi \neq 0$, there are impedances in the circuit, the net effect of which is to produce this effective phase shift between current and voltage. We treat our circuit like an arbitrary resistor across which the voltage $\underline{U}$ drops and through which the current $\underline{I}$ flows. We write this complex current as:

$$\underline{I}(t) = \underline{I}_0 \cdot e^{i\omega t} = I_0 \cdot e^{i\varphi} \cdot e^{i\omega t} \tag{11.30}$$

To simplify matters somewhat, we can choose $\varphi = 0$ here without loss of generality. Using the impedance $\underline{Z}$ from eq. (11.29), we thus obtain the voltage $\underline{U}(t)$ for the circuit shown in fig. 11.12:

$$\underline{U}(t) = \underline{Z} \cdot \underline{I}(t) = |\underline{Z}| \cdot \mathrm{e}^{i\psi} \cdot I_0 \mathrm{e}^{i\omega t} \tag{11.31}$$

The instantaneous value of the time-dependent power $P(t)$ is the product of the real part of $\underline{U}(t)$ and the real part of $\underline{I}(t)$. These are:

$$\mathrm{Re}\left(\underline{I}(t)\right) = I(t) = I_0 \cos \omega t \tag{11.32}$$

and

$$\mathrm{Re}\left(\underline{U}(t)\right) = U(t) = |\underline{Z}| \cdot I_0 \cos(\omega t + \psi) = U_0 \cos(\omega t + \psi) \tag{11.33}$$

This gives us the following for the instantaneous power:

$$P(t) = U(t) \cdot I(t) = U_0 \cdot I_0 \cdot \cos \omega t \cdot \cos(\omega t + \psi) \tag{11.34}$$

To obtain the average or active power $P$ dissipated in the circuit, we must calculate the average value of $P(t)$ over a full oscillation period $T$:

$$P = \langle P(t) \rangle = \frac{1}{T} \int_0^T P(t)\,\mathrm{d}t = \frac{U_0 \cdot I_0}{T} \int_0^T \cos \omega t \cdot \cos(\omega t + \psi)\,\mathrm{d}t \tag{11.35}$$

Using $\cos(\omega t + \psi) = \cos \omega t \cos \psi - \sin \omega t \sin \psi$, this leads us to two integrals that are easy to evaluate:

$$P = \frac{U_0 \cdot I_0}{T} \left[ \cos \psi \cdot \underbrace{\int_0^T \cos^2 \omega t\,\mathrm{d}t}_{= T/2} - \sin \psi \cdot \underbrace{\int_0^T \cos \omega t \cdot \sin \omega t\,\mathrm{d}t}_{= 0} \right] \tag{11.36}$$

If we now define the effective values $U_{eff}$ and $I_{eff}$ of alternating voltage and alternating current as those amplitudes for which the power loss in an ohmic resistor (i.e., $\psi = 0$) exactly corresponds to the power loss in direct-current operation, then the average/active power can be expressed as follows:

$$P = U_{eff} \cdot I_{eff} \cdot \cos \psi = \frac{U_0}{\sqrt{2}} \cdot \frac{I_0}{\sqrt{2}} \cdot \cos \psi \tag{11.37}$$

$U_{\mathrm{eff}}$ and $I_{\mathrm{eff}}$ are related to the time averages of $U^2(t)$ and $I^2(t)$. Since we average these quantities over a full oscillation period, we can simply set the phase shift $\psi$ to zero. In this way we obtain:

$$\langle U^2(t) \rangle = \frac{1}{T} \int_0^T U_0^2 \cos^2 \omega t\,\mathrm{d}t = \frac{U_0^2}{2} = U_{eff}^2 \tag{11.38}$$

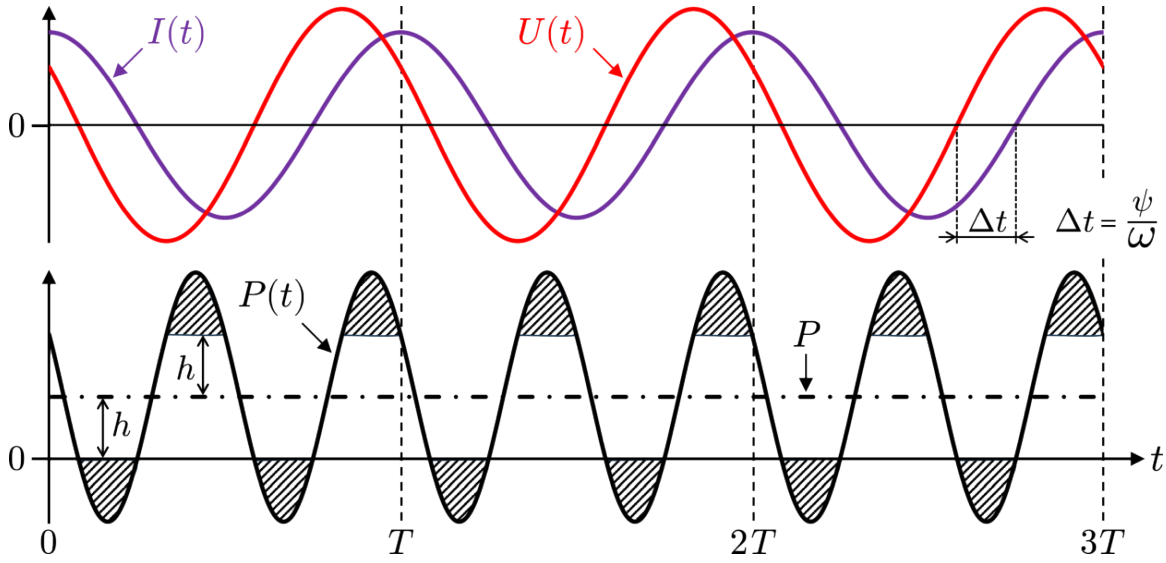



**Fig. 11.13:** Time evolution of alternating current $I(t)$, alternating voltage $U(t)$, and instantaneous power $P(t)$ for an impedance $\underline{Z} = |\underline{Z}|e^{i\psi}$ with $\psi = 60°$. The average/active power $P = \langle P(t) \rangle$ lies exactly between the shaded areas of the $P(t)$ curve. The negative and positive shaded areas below and above the zero line of $P(t)$ cancel each other out and correspond to the reactive power (see text).

and

$$\langle I^2(t) \rangle = \frac{1}{T} \int_0^T I_0^2 \cos^2 \omega t \, \mathrm{d}t = \frac{I_0^2}{2} = I_{eff}^2 \tag{11.39}$$

Calculating the root-mean-square values of voltage and current thus yields exactly the relationships between $U_{\text{eff}}$ and $I_{\text{eff}}$ and the respective amplitudes $U_0$ and $I_0$ as defined in eq. (11.37). The mains voltage used in our households is, depending on the region of the world, either 120 V or 230 V between a phase conductor and the neutral conductor, and 240 V or 400 V between two phase conductors. However, these are the effective voltages in each case. The peak voltages are higher by a factor of $\sqrt{2}$, i.e, 170 V or 325 V for single-phase operation, and 340 V or 566 V for two-phase operation. Fig. 11.13 shows the time evolution of $U(t)$, $I(t)$, and $P(t)$ over three oscillation periods $T$ for a value of $\psi = 60°$, as well as the average/active power $P$. In the case of $\psi = 0$, the grey box in the circuit shown in fig. 11.12 corresponds to a purely ohmic resistor, and the relationship $P = U_{eff} \cdot I_{eff}$ applies. With $U_0 = R \cdot I_0$, the average electrical power dissipated in a purely ohmic resistor is:

$$P = \frac{1}{2} \cdot \frac{U_0^2}{R} \tag{11.40}$$

For the cases $\psi = 90°$ or $\psi = -90°$, it follows that $P = 0$. From eq. (11.17) and eq. (11.21), we know that $\psi = -90°$ implies $\underline{Z} = \underline{Z}_C$ and that $\psi = 90°$ implies $\underline{Z} = \underline{Z}_L$. Thus, if the impedance $\underline{Z}$ in a electrical circuit is purely capacitive or purely inductive, no power is dissipated on average. The shaded areas enclosed by $P(t)$ below the zero line are equal in area to the shaded areas above the zero line that are plotted directly beneath the maxima of the curve $P(t)$. The straight line indicating the average power $P$ lies exactly midway between these areas and is equivalent to the direct-current power equivalent of the unshaded portions of $P(t)$. In the case $P(t) > 0$, the circuit in fig. 11.12 absorbs energy, and in the case $P(t) < 0$, it releases energy. Capacitors and inductors are frequency-dependent alternating-current impedances in which energy is periodically stored by the circuit and then returned to the circuit, without electrical energy being converted into thermal energy as occurs in ohmic resistors. In inductors, this electrical energy is temporarily stored as magnetic field energy, and in capacitors, as electric field energy. If $\psi = 0$, as would be the case in fig. 11.12 for $\underline{Z} = R$, then $P(t) \geq 0$, and the circuit absorbs energy in pulses. We have referred to $P$, the time average of $P(t)$, as average power or active power, since both terms are in common use. Here, we calculated it for sinusoidal currents and voltages, a case where the factor $\sqrt{2}$ applies to the ratio between the effective value and the respective amplitude of current and voltage. However, this is not the case for non-sinusoidal current and voltage waveforms.

Energy storage in circuit elements occurs when $\cos\psi \neq 1$, or, equivalently, $\psi \neq 0$ holds in eq. (11.37). Let us consider an inductor, as shown in fig. 11.14, through which a current $I(t)$ flows. With alternating current, the direction of the magnetic field constantly changes with the direction of the current. Therefore, fig. 11.14 shows only a snapshot at a specific point in time. We now ask how much energy is stored in the coil as a function of time. We could start directly with eq. (10.117), but to deepen our understanding, we will derive eq. (10.117) again here for alternating current using a different approach. To this end, we consider the work required for this change in the inductors energy. The differential work $\mathrm{d}W$ is related to the instantaneous power $P(t)$, and therefore to the current $I(t)$ through the inductor and the voltage drop $U(t)$ across the inductor, by:

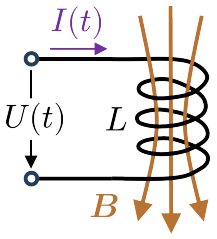


**Fig. 11.14**

$$\mathrm{d}W(t) = P(t)\cdot\mathrm{d}t = U(t)\cdot I(t)\cdot\mathrm{d}t \tag{11.41}$$

From eq. (11.20), we know that for $U(t)$:

$$U(t) = L\cdot\frac{\mathrm{d}I}{\mathrm{d}t} \qquad \Rightarrow \qquad U(t)\cdot\mathrm{d}t = L\cdot\mathrm{d}I(t) \tag{11.42}$$

This turns eq. (11.41) into:

$$\mathrm{d}W(t) \;=\; L\cdot I(t)\cdot \mathrm{d}I(t) \tag{11.43}$$

If we integrate both sides of this equation, we obtain:

$$W(t) \;=\; \frac{1}{2}\cdot L\cdot I^2(t) \tag{11.44}$$

This is the result from eq. (10.117), but now for alternating current rather than direct current. For a sinusoidal current $I(t) = I_0 \sin \omega t$, we thus obtain the following expression for the energy stored in the inductor as a function of time:

$$W(t) \;=\; \frac{1}{2}\cdot L\cdot I_0^2\cdot \sin^2 \omega t \;=\; W_{max}\cdot \sin^2 \omega t \tag{11.45}$$

We can also rewrite this equation as:

$$W(t) \;=\; \frac{1}{2}\cdot W_{max}\cdot \big(1-\cos 2\omega t\big) \;=\; W_{avg}\cdot \big(1-\cos 2\omega t\big) \tag{11.46}$$

Within one cycle of the current, the direction of the current changes. Consequently, the energy stored in the coil oscillates between zero and the maximum value $W_{max}$ at twice the frequency of the sinusoidal current, with an average stored energy of $W_{avg}$. The instantaneous power associated with $W(t)$ is obtained from eq. (11.46) via $P(t) = \mathrm{d}W(t)/\mathrm{d}t$:

$$P(t) \;=\; \omega\cdot W_{max}\cdot \sin 2\omega t \;=\; \omega\cdot L\cdot \frac{I_0^2}{2}\cdot \sin 2\omega t \;=\; \omega\cdot L\cdot I_{eff}\cdot \sin 2\omega t \tag{11.47}$$

For the average/active power $P$, we thus obtain:

$$P \;=\; \frac{1}{T}\int_0^T P(t)\,\mathrm{d}t \;=\; \frac{1}{T}\cdot \omega\cdot L\cdot I_{eff}^2 \underbrace{\int_0^T \sin 2\omega t\,\mathrm{d}t}_{=\,0} \;=\; 0 \tag{11.48}$$

The active power required for the periodic change in the magnetic energy stored in the inductor is therefore zero, just as we would expect from eq. (11.37) for a phase shift of $\psi = 90°$ between the inductor current and the inductor voltage. Let us now consider a capacitor, as shown in fig. 11.15, through which a current $I(t)$ flows. Here, it is the electric field that changes periodically with the direction of the current. In this context, fig. 11.15 shows only a snapshot at a specific point in time. With eq. (11.16), we know the counterpart to eq. (11.42) for the capacitor:

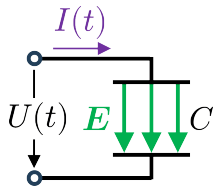


**Fig. 11.15**

$$\frac{\mathrm{d}U(t)}{\mathrm{d}t} \;=\; \frac{1}{C}\cdot I(t) \qquad \Rightarrow \qquad I(t)\cdot \mathrm{d}t \;=\; C\cdot \mathrm{d}U(t) \tag{11.49}$$

We thus obtain for the differential work:

$$\mathrm{d}W(t) \;=\; U(t)\cdot I(t)\cdot \mathrm{d}t \;=\; C\cdot U(t)\cdot \mathrm{d}U(t) \tag{11.50}$$

Integrating both sides of the equation yields:

$$\mathrm{d}W(t) \;=\; U(t)\cdot I(t)\cdot \mathrm{d}t \;=\; C\cdot U(t)\cdot \mathrm{d}U(t) \tag{11.51}$$

With a sinusoidal voltage $U(t) = U_0 \sin \omega t$, we thus obtain, in correspondence with eq. (11.45), the following expression for the time dependence of the electric field energy stored in the capacitor:

$$W(t) \;=\; \frac{1}{2}\cdot C\cdot U_0^2\cdot \sin^2 \omega t \;=\; W_{max}\cdot \sin^2 \omega t \tag{11.52}$$

or, respectively

$$W(t) \;=\; \frac{1}{2}\cdot W_{max}\cdot \big(1-\cos 2\omega t\big) \;=\; W_{avg}\cdot \big(1-\cos 2\omega t\big) \tag{11.53}$$

Just as with the inductor, the energy stored in the capacitor oscillates between zero and the maximum value $W_{max}$ at twice the frequency of the sinusoidal voltage, with an average stored energy of $W_{avg}$. With the difference that, for the capacitor, it is the electric field energy that oscillates, whereas for the inductor, it is the magnetic field energy. Analogous to the inductor, we obtain the following for the instantaneous power associated with $W(t)$:

$$P(t) \;=\; \omega\cdot W_{max}\cdot \sin 2\omega t \;=\; \omega\cdot C\cdot \frac{U_0^2}{2}\cdot \sin 2\omega t \;=\; \omega\cdot C\cdot U_{eff}\cdot \sin 2\omega t \tag{11.54}$$

and for the average/active power $P$:

$$P \;=\; \frac{1}{T}\int_0^T P(t)\,\mathrm{d}t \;=\; \frac{1}{T}\cdot \omega\cdot C\cdot U_{eff}^2 \underbrace{\int_0^T \sin 2\omega t\,\mathrm{d}t}_{=\,0} \;=\; 0 \tag{11.55}$$

As with the inductor, the active power required for the periodic change in the energy stored in the capacitor is also zero. This is, of course, exactly what we would expect based on eq. (11.37) for a phase shift of $\psi = -90°$ between the capacitor voltage and the capacitor current. For the sake of completeness and comparison with inductor and capacitor, we also write down again the instantaneous power and the corresponding active power here for a purely ohmic resistor. To do this, we simply set $\psi = 0$ in eq. (11.34) and eq. (11.35) and obtain the following for $P(t)$ and $P$:

$$P(t) \;=\; U_0\cdot I_0\cdot \cos^2 \omega t \qquad \text{and} \qquad P \;=\; \frac{1}{2}\cdot U_0\cdot I_0 \tag{11.56}$$

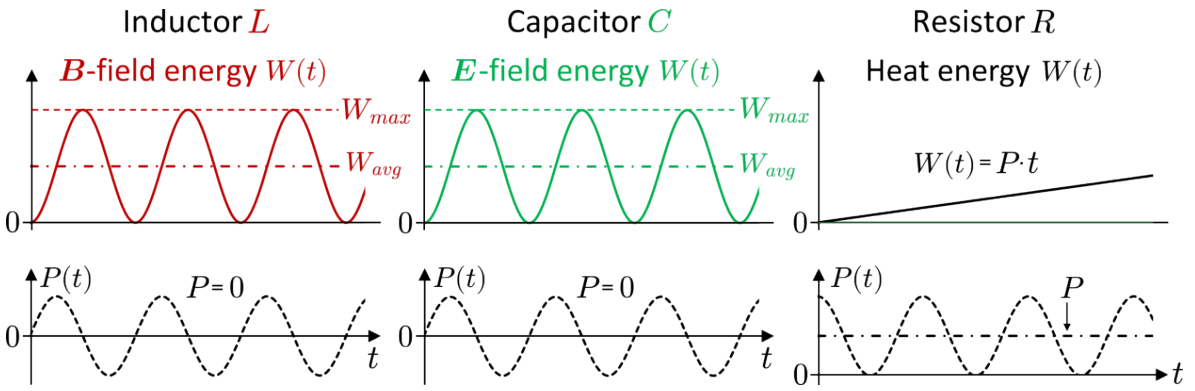



**Fig. 11.16:** Normalized time evolution of instantaneous power $P(t)$, average/active power $P$, and energy balance for an inductor $L$, a capacitor $C$, and an ohmic resistor $R$ under alternating-current operation.

The energy converted into heat within an impedance over time increases linearly with the active power, according to $W = P \cdot t$. For the inductor and the capacitor, this active power is $P = 0$, and no energy is lost through heat dissipation. This is not the case, however, for an ohmic resistor, where $P \neq 0$. Fig. 11.16 compares the energy storage behavior of an inductor and a capacitor with the energy loss due to heat dissipation in an ohmic resistor, alongside the respective waveforms of instantaneous power and active power. No energy losses occur in an ideal inductor or an ideal capacitor. Consequently, neither component performs any net work. However, that does not mean that no work is being done. What is means is that the work required to store energy in an inductor or capacitor is exactly offset by the work the inductor or capacitor perform when returning the stored energy to the circuit.

Thus, work is indeed performed during this periodic energy transfer; however, within the power grid, it simply oscillates between generator and consumer as so-called reactive power. Let us once again look at the expression for active power in eq. (11.37). The quantity $P$ in this equation, the active power, represents the real part of a phasor in the complex plane that has a magnitude of $U_{\text{eff}} \cdot I_{\text{eff}}$. Fig. 11.17 shows this complex phasor, which is referred to as apparent power $\underline{S}$. The imaginary part of $\underline{S}$ is the reactive power, usually denoted by the letter $Q$, which is given by:

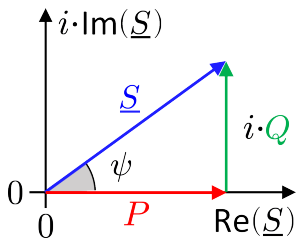



**Fig. 11.17**

$$Q = U_{eff} \cdot I_{eff} \cdot \sin\psi \tag{11.57}$$

In the case of an ohmic resistor, current and voltage are always in phase, meaning $\sin\psi = 0$ and $Q = 0$. This is not the case for an inductor or a capacitor, where current and voltage are phase-shifted by $-90°$ and $90°$, respectively, resulting in $\sin\psi = -1$ or $\sin\psi = 1$. For

the reactive power, this means:

$$Q = -U_{eff} \cdot I_{eff} \quad \text{(Capacitor)} \qquad ; \qquad Q = U_{eff} \cdot I_{eff} \quad \text{(Inductor)} \tag{11.58}$$

The fact that the reactive power of inductors and capacitors has opposite signs is utilized for what is referred to as reactive power compensation in power grids. Its objective is to minimize the phase angle $\psi$ between current and voltage, usually by adding capacitors. For the apparent power $\underline{S}$, the following applies:

$$|\underline{S}|^2 \;=\; P^2 + Q^2 \;=\; U_{eff}^2 \cdot I_{eff}^2 \big( cos^2\psi + \sin^2\psi \big) \;=\; U_{eff}^2 \cdot I_{eff}^2 \tag{11.59}$$

Mains transformers and circuits must be rated for the magnitude of the apparent power $|S| = U_{eff} \cdot I_{eff}$. Why is that so? Even though reactive power is not lost to the grid as dissipated heat, unlike active power, the currents associated with it constitute a real load.

## 11.4 Simple Circuits

With what we have learned so far about the basic electronic components, i.e., resistors, capacitors, and inductors, as well as alternating currents, we can now begin to examine some important circuits. Resistors, capacitors, and inductors are passive components, meaning they do not require an external power supply, such as that needed by transistors or complex integrated components like microprocessors, memory chips, and the like, to perform their respective functions. The functions of passive circuits, i.e., the way any given circuit is designed to operate, can vary widely. Consequently, there are many different circuit modules, each designed to perform a specific function. These modules can, in turn, be incorporated into larger circuits, where they then carry out their respective functions.

### 11.4.1 Four-Terminal Networks

One of the most fundamental and frequently used circuit modules, featuring two input terminals and two output terminals, is schematically illustrated in fig. 11.18a. $U_1(t)$ and $I_1(t)$ are applied at the input side, while $U_2(t)$ and $I_2(t)$ are taken from the output side. Such circuits are referred to as four-terminal or two-port networks because they possess four connection terminals, two at the input port and two at the output port. Simple examples of four-terminal networks are frequency filters, such as the high-pass and low-pass filters shown in fig. 11.18.

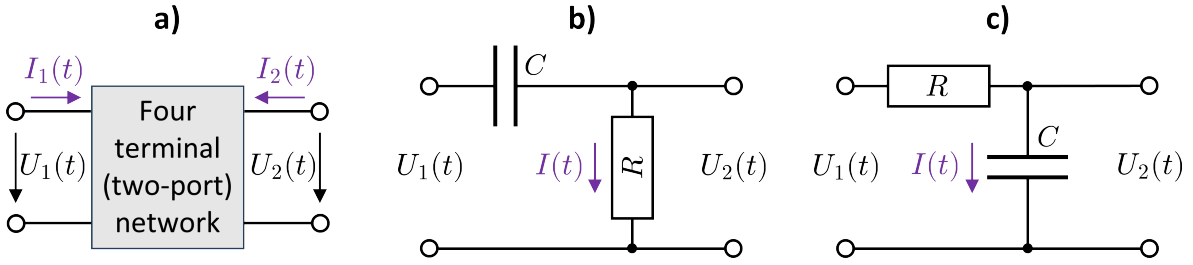



**Fig. 11.18:** Block diagram of a four-terminal network (a), high-pass filter (b), and low-pass filter (c).

Frequency filters are important in electronic signal processing. While frequency filtering is nowadays carried out digitally rather than analog in many applications, analog filtering still offers advantages in real-time or high-frequency applications, for example. Let us first consider the high-pass filter shown in fig. 11.18b. The input voltage $U_1(t)$ is applied to the series-connected impedances $\underline{Z}_C$ and $\underline{Z}_R$, and the output voltage is tapped across the impedance $\underline{Z}_R$. Thus, the following applies to the input and output of this four-terminal network (using complex notation):

$$\underline{U}_1(t) = \underline{Z}\cdot\underline{I}(t) = \left(\frac{1}{i\omega C} + R\right)\cdot\underline{I}(t) \tag{11.60a}$$

$$\underline{U}_2(t) = R\cdot\underline{I}(t) \tag{11.60b}$$

From these two equations, we can eliminate $\underline{I}(t)$ and obtain the following expression for the relationship between the output voltage $\underline{U}_2(t)$ and the input voltage $\underline{U}_1(t)$ of the four-terminal network:

$$\underline{U}_2(t) = \frac{(\omega RC)^2 + i(\omega RC)}{1 + (\omega RC)^2}\cdot\underline{U}_1(t) = g(\omega)\cdot\underline{U}_1(t) \tag{11.61}$$

We write the complex transfer function $g(\omega)$ of the four-terminal network as

$$g(\omega) = |g(\omega)|\cdot e^{i\varphi(\omega)} = \frac{\omega RC}{\sqrt{1 + (\omega RC)^2}}\cdot e^{i\varphi(\omega)} \tag{11.62}$$

As $\omega \to 0$, $g(\omega) \to 0$ ($\omega = 0$ = direct current!), and as $\omega \to \infty$, $|g(\omega)| \to 1$. This four-terminal network is called a high-pass filter because it allows high frequencies to pass with little loss while significantly attenuating low frequencies. The extent to which it reduces them depends on the value of the time constant $R\cdot C$. Fig. 11.19a shows the transfer function $g(\omega)$ of the high-pass filter for several values of $R\cdot C$. Let us now look at the low-pass filter in fig. 11.18c. In this case, the input voltage $U_1(t)$ is applied to the series-connected impedances $\underline{Z}_R$ and $\underline{Z}_C$, and the output voltage is tapped across the impedance $\underline{Z}_C$. Thus, to input and output of this four-terminal network applies:

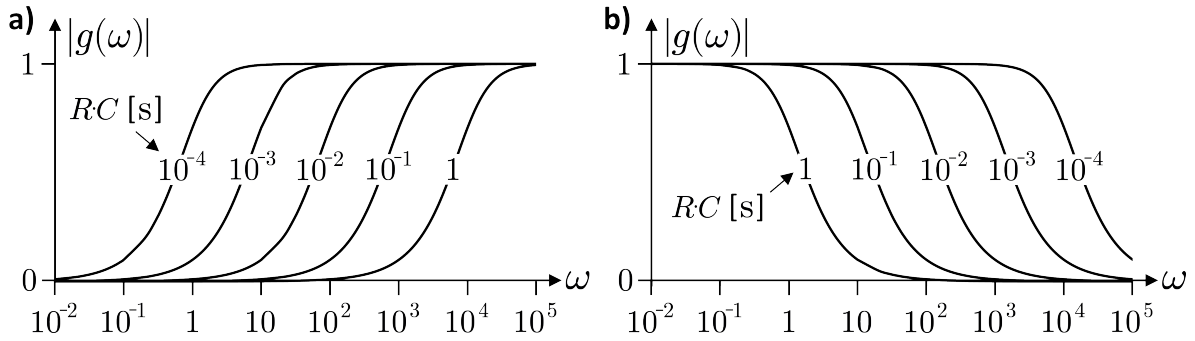



**Fig. 11.19:** Transfer function $|g(\omega)|$ of high-pass filter (a) and low-pass filter (b).

$$\underline{U}_1(t) = \underline{Z}\cdot\underline{I}(t) = \left(R + \frac{1}{i\omega C}\right)\cdot\underline{I}(t) \tag{11.63a}$$

$$\underline{U}_2(t) = \frac{1}{i\omega C}\cdot\underline{I}(t) \tag{11.63b}$$

By eliminating $I(t)$, we obtain the following expression for the relationship between the output voltage $\underline{U}_2(t)$ and the input voltage $\underline{U}_1(t)$ of the four-terminal network:

$$\underline{U}_2(t) = \frac{1 - i(\omega RC)^2}{1 + (\omega RC)^2}\cdot\underline{U}_1(t) = g(\omega)\cdot\underline{U}_1(t) \tag{11.64}$$

Now we write $g(\omega)$ again as

$$g(\omega) = |g(\omega)|\cdot e^{i\varphi(\omega)} = \frac{1}{\sqrt{1 + (\omega RC)^2}}\cdot e^{i\varphi(\omega)} \tag{11.65}$$

For this four-terminal network, therefore, $g(\omega) \to 1$ as $\omega \to 0$ and $g(\omega) \to 0$ as $\omega \to \infty$. This four-terminal network allows low frequencies to pass almost unimpeded while significantly attenuating high frequencies, hence the name low-pass filter. Fig. 11.19b shows the transfer function $g(\omega)$ of this low-pass filter for several values of $R{\cdot}C$.

If a high-pass filter and a low-pass filter with the same resistance $R$ are connected in series, as shown in fig. 11.20a, the resulting four-terminal network is a simple band-pass. Here, we apply Kirchhoff's loop rule to the meshes P-1-2-Q, P-3-4-Q, and R-3-4-S, as well as Kirchhoff's junction rule to junction 1, thereby obtaining the following four equations:

$$\underline{U}_1(t) - \underline{Z}_C\cdot\underline{I}(t) - \underline{Z}_R\cdot\underline{I}_1(t) = 0 \tag{11.66a}$$

$$\underline{U}_1(t) - \underline{Z}_C\cdot\underline{I}(t) - (\underline{Z}_R + Z_C)\cdot\underline{I}_2(t) = 0 \tag{11.66b}$$

$$\underline{U}_2(t) - \underline{Z}_C\cdot\underline{I}_2(t) = 0 \tag{11.66c}$$

$$\underline{I}(t) - \underline{I}_1(t) - \underline{I}_2(t) = 0 \tag{11.66d}$$

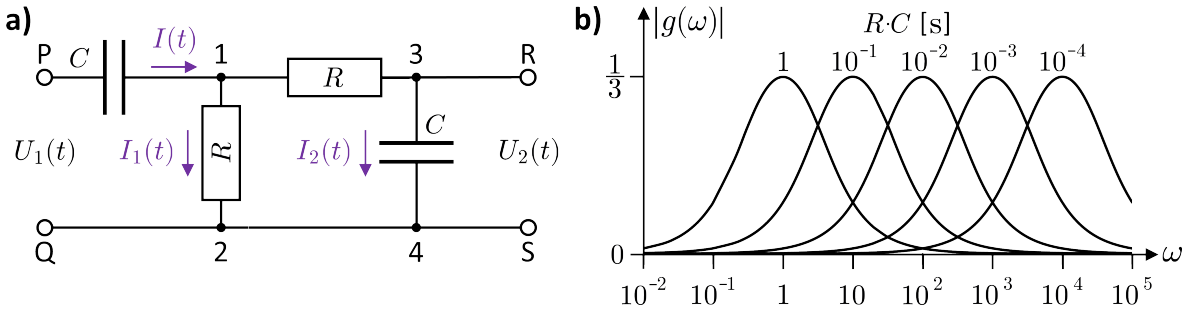



**Fig. 11.20:** Simple band-pass filter (a) and its transfer function $|g(\omega)|$ (b).

To solve these equations, one first uses the fourth equation to replace $\underline{I}(t)$ with $\underline{I}_1(t)$ and $\underline{I}_2(t)$ in the first two equations. Then, $\underline{I}_1(t)$ is eliminated from these two equations, and the resulting expression for $\underline{I}_2(t)$ is substituted into the third equation. This yields the following result:

$$\underline{U}_2(t) \;=\; \frac{\underline{Z}_R \cdot \underline{Z}_C}{\underline{Z}_R^2 + 3\cdot\underline{Z}_R\cdot\underline{Z}_C + \underline{Z}_C^2}\cdot\underline{U}_1(t) \;=\; \frac{-i(\omega RC)}{(\omega RC)^2 - 1 - 3i(\omega RC)}\cdot\underline{U}_1(t) \tag{11.67}$$

The transfer function $g(\omega)$ of the bandpass filter is thus:

$$g(\omega) \;=\; |g(\omega)|\cdot e^{i\varphi(\omega)} \;=\; \frac{\omega RC}{\sqrt{9(\omega RC)^2 + (\omega RC - 1)^2}}\cdot e^{i\varphi(\omega)} \tag{11.68}$$

The magnitude of $g(\omega)$ approaches zero as $\omega \to 0$ and as $\omega \to \infty$ and attains its maximum value $g(\omega) = 1/3$ at $\omega = 1/RC$. This band-pass filter equally attenuates low and high frequencies and allows frequencies in a middle range, determined by the value of $RC$, to pass through. However, this no longer occurs without attenuation, since the amplitude of the output voltage of this frequency filter at $\omega = 1/RC$, where $|g(\omega)|$ reaches its maximum, is only one-third of the input voltage amplitude. Fig. 11.20b shows this behavior of $|g(\omega)|$ for several values of $RC$.

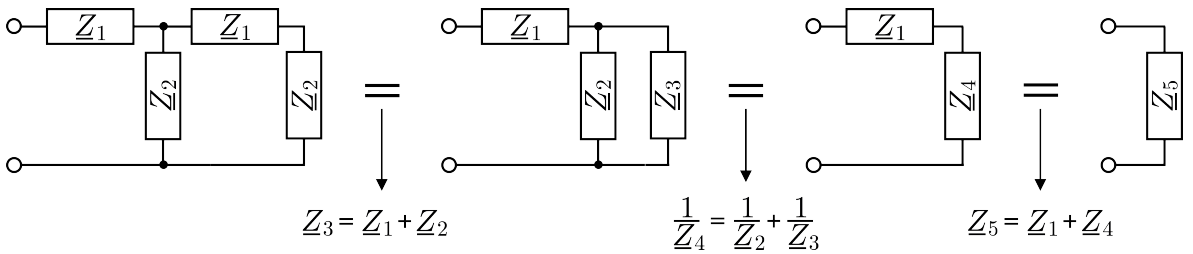



**Fig. 11.21:** Two identical in series four-terminal networks reduced to a single impedance.

The series connection of a high-pass filter and a low-pass filter in fig. 11.20a to form a band-pass filter is an example of a cascaded connection of two-port networks, where the output of one two-port network forms the input of the subsequent one. Here, we will consider a specific type of chain circuit in which a very large number of identical

four-terminal networks are connected in series. This number is to be large enough that adding another four-terminal network does not alter the properties of the chain circuit. Fig. 11.21 illustrates why we require this: it shows how two identical cascaded four-terminal networks can be reduced to a single impedance. This approach can be extended to any number of four-terminal networks, ultimately resulting in a single impedance that represents the characteristics of all the cascaded networks. This impedance is referred to as the characteristic impedance of a cascaded network. This also applies to the case of an infinite number of identical four-terminal networks connected in series. Let $\underline{Z}_0$ be the characteristic impedance of a cascaded network consisting of an infinite number of four-terminal networks of the type shown in fig. 11.21. In fig. 11.22, this is expressed by the equals sign between circuits ① and ②. If we place another such four-terminal network in front of this characteristic impedance $\underline{Z}_0$, we have simply added one more to the infinite number of four-terminal networks in the chain connection. Infinity plus one is still infinity, and the chain connection with the added four-terminal network still possesses the characteristic impedance $\underline{Z}_0$. In fig. 11.22, this is expressed by the equals sign between circuit ② and circuit ③. From this identity of the circuits, it follows immediately that the following must hold:

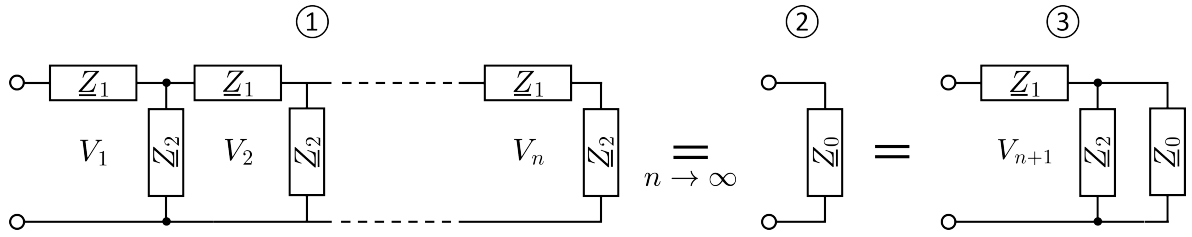



**Fig. 11.22:** The characteristic impedance $\underline{Z}_0$ of a chain connection of an infinite number of identical four-terminal networks does not change when another identical four-terminal network is added.

$$\underline{Z}_0 = \underline{Z}_1 + \frac{\underline{Z}_0 \cdot \underline{Z}_2}{\underline{Z}_0 + \underline{Z}_2} \tag{11.69}$$

Solving for $\underline{Z}_0$, we obtain for the characteristic impedance of the chain connection of identical four-terminal networks with impedances $\underline{Z}_1$ and $\underline{Z}_2$ shown in fig. 11.22:

$$\underline{Z}_0 = \frac{\underline{Z}_1}{2} \pm \sqrt{\frac{\underline{Z}_1^2}{4} + \underline{Z}_1 \cdot \underline{Z}_2} \tag{11.70}$$

In the realm of complex numbers, a quadratic equation always has two solutions. Physical boundary conditions determine which of the two solutions must be selected in a given case. If, for example, $\underline{Z}_1$ and $\underline{Z}_2$ are both ohmic resistors $R$ of equal value, then the square root yields the expression $\sqrt{5} \cdot R/2$. Since $R/2 - \sqrt{5}R/2 < 0$, it follows that $\underline{Z}_0 < 0$ in the case of the negative square root sign. $\underline{Z}_1$ and $\underline{Z}_2$ are purely ohmic resistors, so $\underline{Z}_0$

must also be a purely ohmic resistor. However, this contradicts the condition $\underline{Z}_0 < 0$. A negative square root sign therefore leads to a physically impossible solution, so we must choose the solution with the positive square root sign.

To understand the practical significance of eq. (11.70), let us look at fig. 11.23. It shows a four-terminal network in which $\underline{Z}_1$ consists of a resistor $R$ and an inductor $L$ connected in series, and $\underline{Z}_2$ consists of another resistor, here defined by its conductance $G$, and a capacitor $C$ connected in parallel to the resistor. What fig. 11.23 really depicts is a two-conductor system, such as that found in a coaxial cable with an inner conductor (conductor 1) and a grounded outer conductor (conductor 2). However, it could also represent an antenna cable or a ribbon conductor. Taking the coaxial cable as an example, $R$ is the line resistance and $L$ is the inductance of the inner conductor per unit length $\Delta x$. $G$ is the conductance between the inner and outer conductors, i.e., the conductance of the insulating material between them, and $C$ is the capacitance between the inner and outer conductors; both are again per unit length $\Delta x$. The conductance $G$ represents losses due to imperfect insulation. If we assume a perfectly insulated inner conductor, then $G = 0$ and $1/G = \infty$, from which it follows that $\underline{Z}_2 = -i/\omega C$. If we neglect the ohmic resistance of the inner conductor, then $\underline{Z}_1 = i\omega L$. If we consider not just an infinitesimal conductor segment $\Delta x$ but a long coaxial cable, i.e., a cascade of $n$ four-terminal networks of the type shown in fig. 11.23 with $n \to \infty$, then, according to eq. (11.70), its characteristic impedance $\underline{Z}_0$ is:

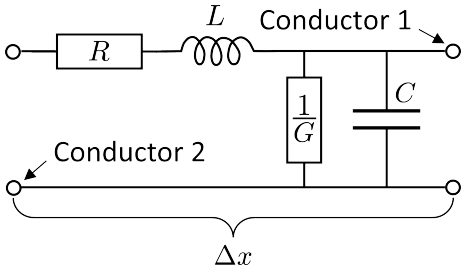



**Fig. 11.23**

$$\underline{Z}_0 = \frac{i\omega L}{2} + \sqrt{\frac{L}{C} - \frac{\omega^2 L^2}{4}} \tag{11.71}$$

Why we chose the positive root sign in eq. (11.70) in this case will become clear a little further down, once we understand the transfer function $g(\omega)$ of such a cascaded circuit with the characteristic impedance $\underline{Z}_0$. To this end, we consider the cascaded network shown in fig. 11.24. The input voltage of the $n$-th two-port network is $U_n$, and its output voltage, which appears across $Z_0$, is $U_{n+1}$. Applying Kirchhoff's loop rule to the $n$-th two-port network yields (again using complex notation):

$$\underline{U}_n - \underline{Z}_1 \cdot \underline{I}_n - \underline{U}_{n+1} = 0 \tag{11.72}$$

The right side of the circuit in fig. 11.24 corresponds to circuit ③ in fig. 11.22 and is thus identical to circuit ② in fig. 11.22. The voltage $U_n$ in fig. 11.24 is therefore applied to the equivalent of $Z_0$, and it follows that $U_n = Z_0 I_n$. Thus, eq. (11.72) becomes:

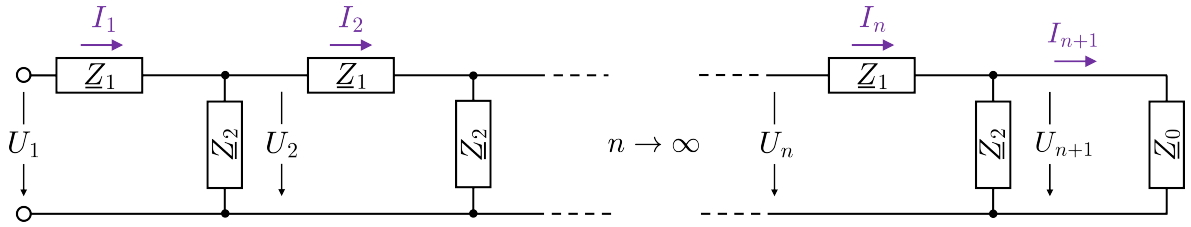



**Fig. 11.24:** Determination of the ratio between output voltages $\underline{U}_{n+1}$ and input voltages $\underline{U}_n$ in a cascade connection of identical four-terminal networks.

$$\underline{U}_n - \underline{Z}_1 \cdot \frac{\underline{U}_n}{\underline{Z}_0} - \underline{U}_{n+1} = 0 \tag{11.73}$$

From eq. (11.73), it follows for the ratio of the voltages $\underline{U}_{n+1}$ and $\underline{U}_n$:

$$\frac{\underline{U}_{n+1}}{\underline{U}_n} = \frac{\underline{Z}_0 - \underline{Z}_1}{\underline{Z}_0} = \gamma \tag{11.74}$$

This voltage ratio $\gamma$ is the propagation constant for a cascade connection of four-terminal networks. With respect to the input voltage $\underline{U}_1$ of the cascaded network in fig. 11.24, the output voltage $\underline{U}_{n+1}$ of the cascaded network is:

$$\underline{U}_{n+1} = \gamma^n \underline{U}_1 = g(\omega) \cdot \underline{U}_1 \tag{11.75}$$

We thus know the transfer function of the cascaded circuit in fig. 11.24 and can determine $|g(\omega)|$ in terms of $|\gamma|$. To this end, we consider the example of the coaxial cable with $\underline{Z}_1 = i\omega L$ and $\underline{Z}_0$ from eq. (11.71). In a passive circuit, the condition $|\gamma| \leq 1$ always holds. This is the physical constraint that determined the choice of the sign of the square root in eq. (11.70). If we were to choose the negative sign, then $|\gamma|$ would exceed 1 for certain frequencies. In the case of the coaxial cable, we obtain the following for $\gamma$:

$$\gamma = \frac{\sqrt{L/C - \omega^2 L^2/4} - i\omega L/2}{\sqrt{L/C - \omega^2 L^2/4} + i\omega L/2} \tag{11.76}$$

We now define the frequency $\omega_0$ by:

$$\frac{L}{C} - \frac{\omega_0^2 L^2}{4} = 0 \qquad \Rightarrow \qquad \omega_0 = \sqrt{\frac{4}{LC}} \tag{11.77}$$

For $\omega \leq \omega_0$, the roots in the numerator and denominator of eq. (11.76) are real, the numerator and denominator are complex conjugates of each other, their magnitudes are identical, and $|\gamma| = 1$ holds. This would also be the case if we used the negative square root sign in eq. (11.70) for eq. (11.71). Thus, for frequencies $\omega \leq \omega_0$, input voltage = output voltage applies for every four-terminal network in the cascade connection, and thus $U_n = U_1$. For $\omega > \omega_0$, the radicands in the numerator and denominator become negative, such that both quantities become purely imaginary and we thus obtain:

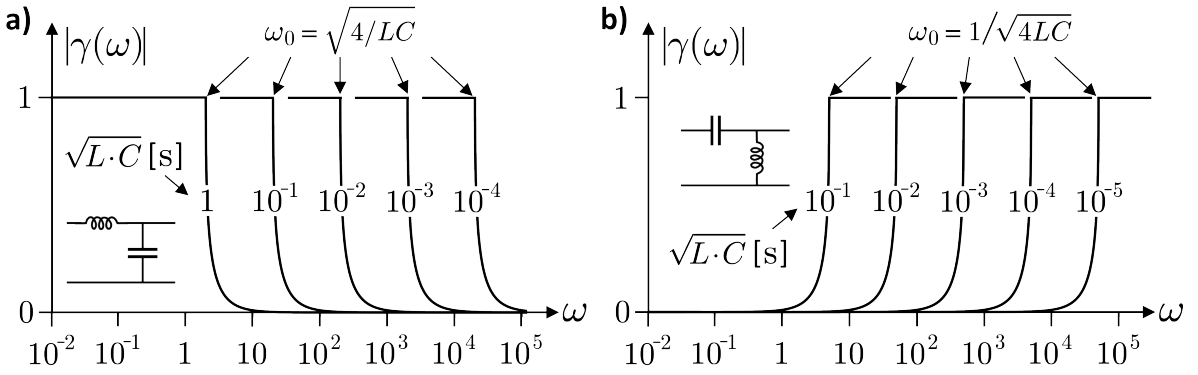



**Fig. 11.25:** Propagation constant $|\gamma(\omega)|$ of cascaded circuits consisting of a low-pass four-terminal network with series inductance $L$ and shunt capacitance $C$ (a) and a high-pass four-terminal network with series capacitance $C$ and shunt inductance $L$ (b).

$$\gamma = \frac{\sqrt{\omega^2 L^2/4 - L/C} - \omega L/2}{\sqrt{\omega^2 L^2/4 - L/C} + \omega L/2} \tag{11.78}$$

The propagation factor $\gamma$ is thus real even for $\omega > \omega_0$, and $|\gamma| < 1$ holds. Had we chosen the positive rather than the negative square root sign in eq. (11.70) for eq. (11.71), then the result would have been $|\gamma| > 1$, a condition that cannot apply to a passive circuit. $|\gamma| < 1$ for the $\gamma$ from eq. (11.78) is thus our physical boundary condition, which determined the choice of sign in eq. (11.70) for the coaxial cable. As $\omega$ increases in eq. (11.78), the numerator decreases and the denominator increases; as $\omega \to \infty$, the numerator approaches zero and the denominator approaches infinity. With increasing frequency $\omega > \omega_0$, the output voltage $U_n$ becomes progressively smaller until it ultimately vanishes with $|\gamma| = 0$. Thus, as shown in fig. 11.25a, our chain connection acts as a frequency filter, specifically, a low-pass filter that allows frequencies $\omega \leq \omega_0$ to pass unimpeded but increasingly attenuates frequencies $\omega > \omega_0$ as $\omega$ rises. If one swaps $L$ and $C$ in the four-terminal network of fig. 11.23, one obtains a chain connection of four-terminal networks with the characteristic impedance:

$$\underline{Z}_0 = -\frac{i}{2\omega C} - \sqrt{\frac{L}{C} - \frac{1}{4\omega^2 C^2}} \tag{11.79}$$

As can easily be verified, the choice of the sign of the square root is again determined by the boundary condition $|\gamma| \leq 1$. However, in this case, one must select the negative sign from eq. (11.70) for eq. (11.79). For the propagation factor $\gamma$ of this circuit, it is found that $\gamma = 1$ holds for $\omega > \omega_0 = 1/\sqrt{4LC}$, and frequencies $\omega \geq \omega_0$ are transmitted without attenuation. For frequencies $\omega < \omega_0$, $\gamma < 1$ holds, and $\gamma \to 0$ as $\omega \to 0$. By swapping $L$ and $C$ in the two-port network of fig. 11.23, one obtains a cascaded network that functions as a high-pass filter, the properties of which are shown in fig. 11.25b for

various values of $LC$. As the product $LC$ approaches zero, $\omega_0$ approaches infinity. For the low-pass filter in fig. 11.25a, in which the capacitor is connected in parallel, this means $\gamma = 1$ for all frequencies, and in this case, our coaxial cable transmits all signal frequencies. The situation is different for the high-pass filter in fig. 11.25b. In this case $\omega_0 = \infty$ means that no signal passes through at all, regardless of the frequency. As the product $LC$ approaches infinity, $\omega_0$ approaches zero. Consequently, a low-pass filter allows nothing to pass through, whereas a high-pass filter allows everything to pass.
The cascaded connections of four-terminal networks considered here are all series connections. However, four-terminal networks can just as well be connected in parallel, series-parallel, or parallel-series configurations. The behavior of four-terminal networks in these configurations is the subject of four-terminal theory in electrical engineering, which describes them as two-dimensional matrices. The four-terminal networks need not consist solely of passive circuit elements; active components, such as transistors, diodes, or entire integrated circuits, can also be incorporated into them.

### 11.4.2 The Electrical Resonant Circuit

An electrical resonant circuit consists of an inductance $L$ and a capacitance $C$, with any losses accounted for by an ohmic resistor $R$ connected in series with the inductance. The inductance $L$ and capacitance $C$ can be connected either in series or in parallel with each other. Here, we consider an oscillating circuit in which $L$ and $C$ are connected in series, as shown in fig. 11.26. The circuit has no voltage source. To set the circuit in motion, we place a charge $Q$ on the capacitor while the switch $S$ is open, for example, by briefly connecting the terminals of a battery to the electrodes of the capacitor $C$. After the clamps connecting the battery terminals with the capacitor electrodes are removed, the capacitor shall carry a charge $Q$, as shown in fig. 11.26. Thus, a voltage $U_C = Q/C$ exists across the capacitor's electrodes. If we now close the switch $S$, a discharge current $I$ flows from the capacitor $C$ through the inductance $L$ and the ohmic resistance $R$. From eq. (11.20), we know that the voltage $U_L = L \cdot \mathrm{d}I/\mathrm{d}t$ is present across the inductor $L$, and that $U_R = R \cdot I$ applies to the ohmic resistor. The voltage across the capacitor's electrodes is determined at any given time by the charge remaining on the capacitor, i.e., $U_C = Q(t)/C$. According to Kirchhoff's loop rule, i.e., the condition $U_L + U_R + U_C = 0$, must hold for the voltages in our circuit. Hence, the following applies to our electrical resonant circuit:

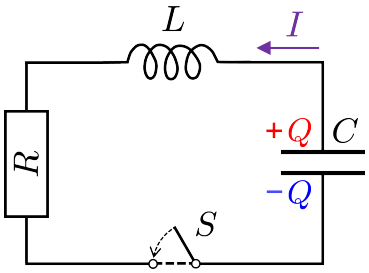


**Fig. 11.26**

$$L \cdot \frac{\mathrm{d}I}{\mathrm{d}t} + R \cdot I + \frac{Q}{C} = 0 \tag{11.80}$$

Differentiating once with respect to time and setting $I = \dot{Q}$ transforms this equation into a second-order homogeneous differential equation for $I(t)$:

$$\ddot{I} + \frac{R}{L} \cdot \dot{I} + \frac{1}{LC} \cdot I = 0 \tag{11.81}$$

This kind of differential equation should look familiar as we already encountered it in chapter 10 of volume 1 of this physics course: eq. (11.81) is the differential equation for a damped harmonic oscillation. Eq. (11.81) can also be derived from the conservation of energy, rather than using Kirchhoff's loop rule for the voltage in the oscillating circuit. This method of derivation has the advantage of leading us to an equation that allows us - unlike eq. (11.81) - to draw a direct analogy to the displacement in mechanical oscillation. To this end, we note that the rate of change of the sum of the magnetic and electric energy plus the power dissipated in the resistor must be zero. Thus, the following must hold:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{L}{2} \cdot I^2 + \frac{1}{2C} \cdot Q^2\right) + R \cdot I^2 = 0 \tag{11.82}$$

If we express the current in this equation in terms of the charge $Q$, i.e., $I = \dot{Q}$, and perform the differentiation on the left-hand side, we obtain:

$$L \cdot \dot{Q} \cdot \ddot{Q} + \frac{1}{C} \cdot Q \cdot \dot{Q} + R \cdot \dot{Q}^2 = 0 \tag{11.83}$$

Here, we can cancel out $\dot{Q}$ and thus obtain a second-order homogeneous differential equation for the charge $Q$:

$$\ddot{Q} + \frac{R}{L} \cdot \dot{Q} + \frac{1}{LC} \cdot Q = 0 \tag{11.84}$$

If one replaces $\dot{Q}$ with $I$ in this equation and differentiates once more with respect to time, one obtains exactly eq. (11.81). However, we will not concern ourselves with eq. (11.81), but rather with eq. (11.84), since it is the charge $Q$, and not the current $I$, that serves as an analogy to mechanical displacement. To this end, we first consider the case $R = 0$, i.e., undamped electrical oscillation:

$$\ddot{Q} + \frac{1}{LC} \cdot Q = 0 \tag{11.85}$$

The general solution for this undamped harmonic oscillation is

$$Q = A \cos \omega_0 t + B \sin \omega_0 t \tag{11.86}$$

and the time derivative of that is

$$\dot{Q} \;=\; I \;=\; -A\omega_0 \sin\omega_0 t + B\omega_0 \cos\omega_0 t \tag{11.87}$$

When the switch $S$ in fig. 11.26 is closed at time $t = 0$, the following applies: $I = 0$ in eq. (11.87), and in eq. (11.86), the charge $Q$ on the capacitor plates is equal to $Q_0$. With these boundary conditions, it follows from eq. (11.86) and eq. (11.87) that $A = Q_0$ and $B = 0$. The oscillation of the charge $Q$ is thus described by:

$$Q(t) \;=\; Q_0 \cos\omega_0 t \tag{11.88}$$

Eq. (11.88) tells us that the charge on the capacitor plates oscillates in the form of an undamped oscillation, periodically changing sign. Substituting this into eq. (11.85), we obtain the following for the resonant frequency $\omega_0$ of the undamped electrical resonant circuit:

$$\omega_0 \;=\; \sqrt{\frac{1}{LC}} \tag{11.89}$$

With $R = 0$, there are no energy losses in the electrical oscillating circuit. From the discussion of undamped mechanical oscillation in volume 1, we know that the system's energy, for instance, in the case of a pendulum, is likewise conserved and oscillates periodically between potential and kinetic energy. Something analogous occurs in an undamped electrical resonant circuit, except that here the system's energy oscillates between the electric field energy of the capacitor and the magnetic field energy of the coil. The magnetic field energy of the coil is given by eq. (10.117):

$$W_L \;=\; \frac{L}{2}\cdot I^2 \;=\; \frac{L}{2}\cdot\dot{Q}^2 \;=\; \frac{L}{2}Q_0^2\omega_0^2 \sin^2\omega_0 t \tag{11.90}$$

The electric field energy of the capacitor is given by eq. (4.65):

$$W_C \;=\; \frac{1}{2}C\cdot U_C^2 \;=\; \frac{1}{2C}\cdot Q^2 \;=\; Q_0^2 \cos^2\omega_0 t \tag{11.91}$$

The magnetic field energy of the coil is at a maximum when the electric field energy of the capacitor is at a minimum, and vice versa. The maximum value that $Q$ can take is $Q = Q_0$, and thus the following applies to the conservation of energy in the undamped electrical oscillating circuit:

$$W_{ges} \;=\; W_L + W_C \;=\; \frac{L}{2}\cdot\dot{Q}^2 + \frac{1}{2C}\cdot Q^2 \;=\; \frac{1}{2C}\cdot Q_0^2 \tag{11.92}$$

If we replace $Q$ in this equation with the mechanical displacement $z$, $L$ with the displaced mass $m$, $1/C$ with the spring constant $k$, and the maximum charge with the maximum amplitude of the mechanical displacement, then eq. (11.92) yields exactly the result for

the undamped mechanical oscillating system: Thus, the magnetic field energy of the coil corresponds to the kinetic energy of the displaced mass, and the potential energy of this mass corresponds to the electric field energy of the capacitor. Next, let us consider the damped electrical oscillation in eq. (11.83). Here, we proceed by drawing an analogy to the damped mechanical oscillation discussed in volume 1:

$$Q = Q_0 \cdot \mathrm{e}^{-\beta t} \cdot \cos \omega t \tag{11.93}$$

where the angular frequency $\omega$ and the damping constant $\beta$ are given by:

$$\omega = \sqrt{\omega_0^2 - \beta^2} \quad \text{with} \quad \omega_0^2 = \frac{1}{LC} \qquad \text{and} \qquad \beta = \frac{R}{2L} \tag{11.94}$$

When the switch $S$ in fig. 11.26 is closed, an alternating current of frequency $\omega$ begins to flow in the circuit, decaying with the time constant $\beta = R/2L$. The energy in the resonant circuit, initially stored in the capacitor, oscillates between the magnetic field energy of the inductor and the electric field energy of the capacitor, but it steadily decreases due to ohmic loss in $R$. We obtain the expression for energy conservation in a damped electrical oscillating circuit by simply replacing $Q_0$ on the right-hand side of eq. (11.92) with $Q_0 \cdot \exp(-\beta t)$:

$$W_{ges} = W_L + W_C = \frac{1}{2C} \cdot Q_0^2 \cdot \mathrm{e}^{-2\beta t} \tag{11.95}$$

A mechanical oscillating system, such as a pendulum, can be excited to oscillate at a frequency different from its resonance frequency $\omega_0$ by the action of an external force that simultaneously supplies the energy required to sustain the oscillation. It should come as no surprise that the same thing can be achieved with an electrical resonant circuit. To do this, we remove the switch from the circuit in fig. 11.26 and instead insert an alternating-current voltage source, as sketched in fig. 11.27. Just as we did for damped forced mechanical oscillations in volume 1, we perform the calculations here using complex numbers. Applying Kirchhoff's loop rule to the circuit in fig. 11.27 therefore yields:

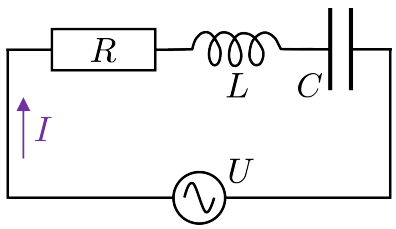



**Fig. 11.27**

$$\underline{U} = \underline{U}_L + \underline{U}_R + \underline{U}_C = \underline{Z}_L \cdot \underline{I} + \underline{Z}_R \cdot \underline{I} + \underline{Z}_C \cdot \underline{I} \tag{11.96}$$

Using eq. (11.9), we write $\underline{U}$ and $\underline{I}$ as:

$$\underline{U} = \underline{U}_0 \cdot \mathrm{e}^{i\omega t} \qquad \text{and} \qquad \underline{I} = \underline{I}_0 \cdot \mathrm{e}^{i\omega t} \tag{11.97}$$

Using this, along with the complex impedances $Z_L = i\omega L$, $Z_R = R$, and $Z_C = -i/\omega C$, eq. (11.96) becomes:

$$i\omega L \cdot \underline{I}_0 + R \cdot \underline{I}_0 + \frac{1}{i\omega C} \cdot \underline{I}_0 = \underline{U}_0 \tag{11.98}$$

This gives us the following expression for the relationship between the complex voltage amplitude $\underline{U}_0$ and the complex current amplitude $\underline{I}_0$:

$$\underline{I}_0 = \frac{\underline{U}_0}{i\omega L + R - i/\omega C} \tag{11.99}$$

Using the quantities

$$\omega_0^2 = \frac{1}{LC} \qquad \text{and} \qquad \frac{1}{\tau} = \frac{R}{L} \tag{11.100}$$

we can cast eq. (11.99) into a form already familiar to us from the discussion of forced mechanical harmonic oscillation in volume 1:

$$\underline{I}_0 = \frac{i\omega \cdot \underline{U}_0}{L\left[(\omega_0^2 - \omega^2) + i(\omega/\tau)\right]} = i\omega \cdot \underline{U}_0 \cdot \underline{R} \tag{11.101}$$

where the resonance function (not to be confused with the real resistance $R$)

$$\underline{R} = \frac{1}{L\left[(\omega_0^2 - \omega^2) + i(\omega/\tau)\right]} = \rho \cdot \mathrm{e}^{i\varphi} \tag{11.102}$$

is a complex quantity with amplitude $\rho$ and phase $\varphi$. Eq. (11.102) corresponds exactly to the expression we obtained for $\underline{R}$ in volume 1 for forced, damped mechanical oscillation, with the difference that the inductance $L$ appears here instead of the displaced mass $m$. We can therefore adopt the results from there directly and obtain the following for $\rho$ and $\varphi$ in eq. (11.102):

$$\rho = \frac{1}{L\sqrt{\left[(\omega_0^2 - \omega^2)^2 + (\omega/\tau)^2\right]}} \tag{11.103}$$

and

$$\cos\varphi = \rho L(\omega_0^2 - \omega^2) \qquad \text{and} \qquad \sin\varphi = -\rho L\frac{\omega}{\tau} \tag{11.104}$$

For the angle $\varphi$, we thus obtain exactly the same result as for the forced damped mechanical oscillation:

$$\tan\varphi = \frac{-(\omega/\tau)}{\omega_0^2 - \omega^2} \tag{11.105}$$

However, the resonance curve for the electrical resonant circuit differs somewhat from that of mechanical oscillation, as we have the additional factor $i\omega$ here in eq. (11.101). For the complex current $\underline{I}$, using eq. (11.97) and eq. (11.101), we obtain:

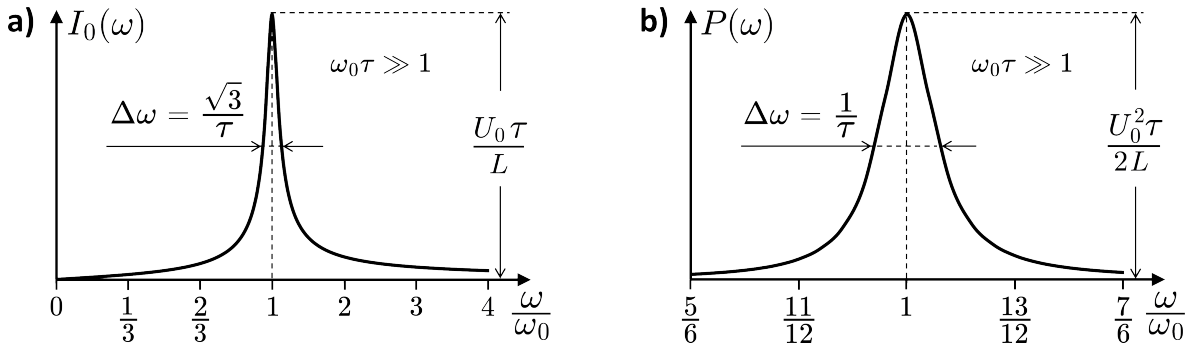



**Fig. 11.28:** Forced harmonic oscillation of the electrical resonant circuit in fig. 11.27: resonance curve (a) and average power dissipated (b).

$$\underline{I} \;=\; \underline{I}_0 \mathrm{e}^{i\omega t} \;=\; i\omega \cdot U_0 \cdot \rho \cdot \mathrm{e}^{i\varphi} \mathrm{e}^{i\omega t} \tag{11.106}$$

We can freely choose the phase of the complex voltage amplitude $\underline{U}_0$ and to ensure the current comes out positive, we set $\underline{U}_0 = -U_0$ and thus obtain:

$$\underline{I} \;=\; -i\omega \cdot U_0 \cdot \rho \big[\cos(\omega t + \varphi) + i \sin(\omega t + \varphi)\big] \tag{11.107}$$

The current $I(t)$ we are looking for is the real part of this equation, that is:

$$I(t) \;=\; \frac{\omega \cdot U_0 \sin(\omega t + \varphi)}{L\sqrt{\big[(\omega_0^2 - \omega^2)^2 + (\omega/\tau)^2\big]}} \;=\; I_0 \sin(\omega t + \varphi) \tag{11.108}$$

Fig. 11.28a shows the behavior of the amplitude $I_0(\omega)$, i.e., the resonance curve of the oscillating circuit. Unlike the case of forced mechanical oscillation, the amplitude here does not approach a finite, non-zero value as $\omega \to 0$; instead, it goes to zero. However, the full width at half maximum is the same in both the mechanical and electrical cases. What about the frequency response of the power dissipated in the resonant circuit? The instantaneous power for the electrical resonant circuit is:

$$P(\omega, t) \;=\; -U_0 \cos(\omega t) \cdot I_0 \sin(\omega t + \varphi) \tag{11.109}$$

If we substitute the expression for $I_0 \sin(\omega t + \varphi)$ from eq. (11.108) here, we obtain the following result for the instantaneous power:

$$P(\omega, t) \;=\; -\omega U_0^2 \rho \cos(\omega t) \big[\sin(\omega t) \cos\varphi + \cos(\omega t) \sin\varphi\big] \tag{11.110}$$

Eq. (11.109) corresponds exactly to the expression we obtained in volume 1 for the instantaneous power $P(\omega, t)$ in the case of a driven, damped mechanical oscillation. If we replace the force $F_0$ and the mass $m$ from that case with the voltage $U_0$ and the inductance $L$ in this case, we can directly adopt the result for the time-averaged value of the instantaneous power $P(\omega) = \langle P(\omega, t) \rangle$:

$$P(\omega) = -\frac{1}{2}U_0^2\omega\sin\varphi = \frac{L\rho^2U_0^2\omega^2}{2\tau} \tag{11.111}$$

If we substitute $\rho$ from eq. (11.103) here, we obtain the average power dissipated in the resonant circuit as a function of $\omega$:

$$P(\omega) = \frac{U_0^2\tau}{2L}\frac{\omega^2/\tau^2}{(\omega_0^2-\omega^2)^2+(\omega/\tau)^2} \tag{11.112}$$

Fig. 11.28b shows this power absorption curve for forced electrical oscillations. It is identical to the power absorption curve for forced mechanical oscillations which we discussed in volume 1. Therefore, we can apply to forced electrical oscillation what we learned there about forced mechanical oscillation. If one considers the energy of the oscillating system, for example, one simply needs to replace the kinetic energy of the oscillating mass with the magnetic energy of the inductance, and the potential energy of the mass with the electric field energy of the capacitor.

## 11.5 Effective Resistance and Ampère's Law

By definition, ohmic resistance $R$ is independent of frequency. However, this does not mean that the same applies generally to effective resistance $R_{eff}$, i.e., the real part of the alternating-current impedance. Furthermore, we must address a gap in the current flow that we have ignored until now: the flow of current through a capacitor. While a capacitor acts as an open circuit for direct current, this is not the case for alternating current. We can easily visualize what the time-varying electric field looks like inside a capacitor through which an alternating current flows. However, what is the situation regarding the alternating magnetic fields that, in accordance with Ampère's law, encircle the supply lines connected to the respective capacitor electrodes? We begin with Ohm's law and then turn to Ampère's law.

### 11.5.1 The Frequency Dependence of Effective Resistance

In section 11.3, we identified the dissipative real part of the alternating-current impedance, the effective resistance $R_{eff}$, with the ohmic resistance $R$. However, not every effective resistance is necessarily a frequency-independent resistance. To understand the transition from a frequency-independent resistance $R$ to a frequency-dependent effective resistance $R_{eff}$, we return to the Drude model of electrical conductivity discussed in section 8.2.1, but this time we consider it for electrons in an alternating electric field. Unlike in a direct-current circuit, however, electrons in an alternating-current circuit do not move in

a single direction but instead oscillate within the alternating electric field. This electron motion represents a forced oscillation, and, due to the resulting collisions, predominantly with lattice atoms, effectively constitutes a damped forced oscillation. From volume 1, we are familiar with the equation of motion for such a forced, damped harmonic oscillation of a particle of mass $m$ about the origin, say, along the $z$-coordinate, under the influence of a periodic force $F(t)$ and a restoring force proportional to the displacement $z$; it is given by:

$$\ddot{z} + \frac{1}{\tau}\dot{z} + \omega_0^2 z \;=\; \frac{F(t)}{m} \tag{11.113}$$

Conduction electrons are free charge carriers and are therefore not subject to any restoring force. Given the periodic force of the oscillating electric field $F(t) = -eE(\omega, t)$ in the $z$-direction, and with $v_z = \dot{z}$, the equation of motion for a conduction electron of mass $m_e$ in an alternating electric field is:

$$m_e\dot{v}_z + \frac{m_e}{\tau}v_z \;=\; -eE_0\mathrm{e}^{i\omega t} \tag{11.114}$$

We are interested here in the velocity of the electrons in an alternating electric field because we can relate it to a current density $\boldsymbol{j}$, and the proportionality factor between that and the electric field $\boldsymbol{E}$, according to $\boldsymbol{j} = \sigma\boldsymbol{E}$, can tell us something about the frequency-dependent conductivity. The velocity of the electron will follow the oscillation of the alternating electric field, so we set $v_z = v_0 e^{i\omega t}$. Substituting this into eq. (11.114) yields:

$$i\omega m_e v_0\mathrm{e}^{i\omega t} + \frac{m_e}{\tau}v_0\mathrm{e}^{i\omega t} \;=\; -eE_0\mathrm{e}^{i\omega t} \tag{11.115}$$

The exponential function cancels out, and we obtain the following expression for the velocity $v_0$:

$$v_0 \;=\; \frac{-e\tau}{m_e(1+i\omega\tau)}\cdot E_0 \tag{11.116}$$

If we equate $v_0$ with the drift velocity of the electrons, just as we did in section 8.2.1, and consider $n$ electrons rather than a single electron, then the current density $j$ of these electrons is given by:

$$j \;=\; n\cdot(-e)\cdot v_0 \;=\; \frac{ne^2\tau}{m_e(1+i\omega\tau)}\cdot E_0 \tag{11.117}$$

We can write this result generally, without restriction, for alternating currents of any frequency as:

$$\boldsymbol{j}(\omega) \;=\; \frac{ne^2\tau}{m_e(1+i\omega\tau)}\cdot\boldsymbol{E}(\omega) \tag{11.118}$$

A comparison with Ohm's law in the form $\boldsymbol{j} = \sigma \boldsymbol{E}$ shows that the frequency-dependent conductivity $\sigma(\omega)$ is given by:

$$\sigma(\omega) \;=\; \frac{ne^2\tau}{m_e}\frac{1}{1+i\omega\tau} \;=\; \frac{\sigma_0}{1+i\omega\tau} \tag{11.119}$$

where $\sigma_0$ is the direct-current conductivity we already know from eq. (8.41). In the limiting case $\omega \to 0$, $\sigma(\omega)$ thus becomes $\sigma_0$, the direct-current conductivity. To see what happens in the case $\omega \to \infty$, we rewrite eq. (11.119) slightly as:

$$\sigma(\omega) \;=\; \frac{\sigma_0}{1+i\omega\tau}\cdot\frac{1-i\omega\tau}{1-i\omega\tau} \;=\; \sigma_0\cdot\frac{1-i\omega\tau}{1+\omega^2\tau^2} \tag{11.120}$$

As $\omega \to \infty$, the real part of $\sigma(\omega)$ approaches zero much faster than the imaginary part; that is:

$$\lim_{\omega\to\infty}\sigma(\omega) \;\approx\; -i\frac{\sigma_0}{\omega\tau} \;=\; -i\frac{ne^2}{m_e\omega} \tag{11.121}$$

What is the significance of this result? The charge carriers in a conductor carrying an alternating current do not behave like free charge carriers, but rather like bound charge carriers, since they locally follow the oscillation of the electric field and do not flow from one end of the conductor to the other. They thus behave like bound charges, or polarization charges, for which the continuity equation eq. (6.25) holds:

$$\nabla\boldsymbol{P} \;=\; -\rho_{bnd} \tag{11.122}$$

If one takes the time derivative of this equation, the bound charge density $\rho_{bnd}$ on the right-hand side becomes the rate of change of this density, which, according to the continuity equation eq. (8.9), is linked to a bound current density $\boldsymbol{j}_{bnd}$ (we can interchange the time and spatial derivatives):

$$\nabla\frac{\partial\boldsymbol{P}}{\partial t} \;=\; -\frac{\partial\rho_{bnd}}{\partial t} \;=\; \nabla\boldsymbol{j}_{bnd} \tag{11.123}$$

We can thus associate the bound current density $\boldsymbol{j}_{bnd}$ with the local oscillation of the electrons in the conductor. The dielectric polarization $\boldsymbol{P}$ is linked to the electric field $\boldsymbol{E}$ via eq. (7.23) and therefore oscillates along with it. Consequently, the following applies to the bound current density:

$$\boldsymbol{j}_{bnd} \;=\; \frac{\partial\boldsymbol{P}}{\partial t} \;=\; i\omega\boldsymbol{P} \tag{11.124}$$

We now equate this bound current density $\boldsymbol{j}_{bnd}$ with the current density $\boldsymbol{j} = \sigma\boldsymbol{E}$ in eq. (11.118). Together with eq. (7.23), it must therefore hold that:

$$i\omega\boldsymbol{P}(\omega) \;=\; \sigma(\omega)\boldsymbol{E}(\omega) \qquad \text{and} \qquad \boldsymbol{P}(\omega) \;=\; (\epsilon-\epsilon_0)\boldsymbol{E}(\omega) \tag{11.125}$$

It follows immediately from these two equations:

$$\sigma(\omega) = i\omega(\epsilon - \epsilon_0) \qquad \text{or} \qquad \epsilon(\omega) = \epsilon_0 - i\frac{\sigma(\omega)}{\omega} \tag{11.126}$$

In section 7.2.1, we found that in an ionized gas, electrons and ions each respond independently to an applied external field oscillating at the angular frequency $\omega$, specifically, as an electron plasma and an ion plasma, respectively. In this case, the dielectric constant of the electron plasma is given by eq. (7.41):

$$\epsilon(\omega) = \epsilon_0\left[1 - \frac{\omega_p^2}{\omega^2}\right] \qquad \text{with} \qquad \omega_p^2 = \frac{ne^2}{\epsilon_0 m_e} \tag{11.127}$$

If we now substitute this into the expression for $\sigma(\omega)$ in eq. (11.126), we obtain:

$$\sigma(\omega) = -i\epsilon_0\frac{\omega_p^2}{\omega} = -i\frac{ne^2}{m_e\omega} \tag{11.128}$$

However, this is precisely the result of eq. (11.121), where we considered the limiting case $\omega \to \infty$. Thus, at high frequencies, the free charge carriers in a conductor behave like a plasma. In doing so, we must distinguish between the cases $\omega < \omega_p$ and $\omega > \omega_p$. For $\omega < \omega_p$, it follows from eq. (11.127) that $\epsilon(\omega) < 0$. Physically, this means that at frequencies below the plasma frequency, the electrons in the conductor can readily follow the oscillation of the electric field. The result is that the electron plasma shields the electric field, and therefore high-frequency currents below the plasma frequency flow not within the volume of the conductor, but only in a thin surface layer. This, in turn, means that for such alternating currents, the effective conductor cross-section through which the current flows is smaller than in the case of direct current, and the effective resistance $R_{eff}$ is therefore greater than the ohmic direct-current resistance $R$. In the case $\omega > \omega_p$, $\epsilon(\omega) > 0$ according to eq. (11.127). At frequencies above the plasma frequency, the electrons can no longer keep pace with the oscillating electric field, and field shielding by the electron plasma becomes increasingly ineffective as the frequency rises. In the limiting case $\omega \to \infty$, $\epsilon(\omega) \to 1$ holds, and with eq. (11.128), the imaginary part of $\sigma(\omega)$ also vanishes and the conductor behaves like a dielectric that is no longer polarizable.

From the discussion in the preceding sections, we know that every conducting wire has not only an effective resistance $R_{eff}$ associated with it per unit length, but also an inductance $L$ and a capacitance $C$. The inductive reactance $\underline{Z}_L = i\omega L$, arising from self-induction linked to the constantly changing magnetic field surrounding an alternating-current-carrying conductor, is evident. As for the capacitive reactance, even when considering only a single conducting wire, it is always at a specific potential relative to its surroundings, specifically, in relation to a defined earth potential. Therefore, a reactance $\underline{Z}_C = C/i\omega$ always exists between the conductor and an earth potential defined in this

way. In such a conductor carrying alternating current, the effective resistance $R_{eff}$ and the inductive reactance $\underline{Z}_L$ increase with rising frequency, while its capacitive reactance $\underline{Z}_C$ steadily decreases. Above the plasma frequency, however, the effective resistance $R_{eff}$ of the conductor, i.e., the real part of the conductor's complex impedance, decreases again, while the imaginary part of the complex impedance rises as the inductive reactance continues to increase. Consequently, at very high frequencies, the capacitive reactance of a conductor is often negligibly small, whereas its inductive reactance is far greater than the conductor's effective resistance.

### 11.5.2 The Extension of Ampère's Law

There is one obvious question regarding the alternating-current circuits we examined in this chapter that we have not yet asked: What happens when an alternating current flows through a conductor segment containing a capacitor? However, we cannot phrase the question that way, because alternating current does not flow, rather, it oscillates, meaning it periodically changes direction. The electrons in the conductor oscillate back and forth at the frequency of the alternating current, meaning they remain more or less in the same place. What flows is energy, not the current itself. The oscillating electrons generate electromagnetic fields that transport electrical and magnetic energy. To illustrate this, let us consider fig. 11.29, which shows the same conductor segment at two different points in time. The conductor segment contains a capacitor through which an alternating current «flows», and the two points in time differ in that the polarity of the capacitor has reversed. From Ampère's law, we know that a magnetic field is associated with a current-carrying wire. To illustrate this, let us consider the hemispherical shell in fig. 11.29, bounded by the curve $C$ and having the surface area $S$. On the left side of fig. 11.29, the current $I$ «flows» through this hemispherical shell, and applying Ampère's law from section 9.2 yields:

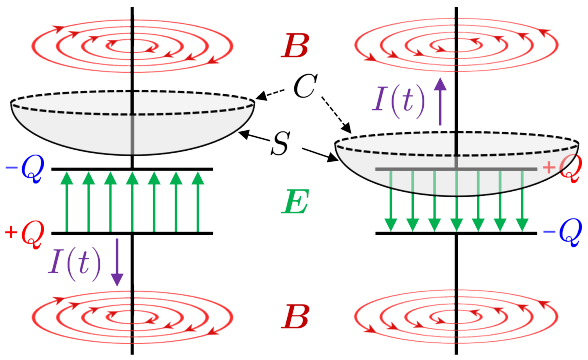



**Fig. 11.29**

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} = \int_S \nabla \times \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} = \mu_0 I(t) \tag{11.129}$$

If we now shift the hemispherical shell slightly downwards, as shown on the right-hand side of fig. 11.29, the electric field of the capacitor passes through the surface $S$, but

no current does. However, as fig. 11.29 shows, the electric field between the capacitor plates is not constant. With the alternating current, the polarity of the capacitor changes periodically, and thus the electric field varies over time. For the relationship between the electric field of the capacitor with plate area $A$ and the charge $Q$ residing on it, it follows from eq. (4.62) and eq. (4.63):

$$E = \frac{Q}{\epsilon_0 A} \tag{11.130}$$

The rate of change of the charge $Q$ is related to a current equivalent $I_D(t)$:

$$I_D(t) = \frac{\partial Q}{\partial t} = \epsilon_0 A \cdot \frac{\partial E}{\partial t} \qquad \Rightarrow \qquad j_D = \epsilon_0 \cdot \frac{\partial E}{\partial t} \tag{11.131}$$

$I_D(t)$ is the so-called displacement current, and $j_D$ is the corresponding displacement current density. What we have found here for the time-varying electric field of the capacitor applies generally to time-varying electric fields: They give rise to a displacement current. If we consider an arbitrary surface $S$ rather than a flat capacitor plate $A$, the relationship between current and current density is given by eq. (8.5):

$$I_D(t) = \int_S \boldsymbol{j}_D \,\mathrm{d}\boldsymbol{\sigma} = \epsilon_0 \cdot \int_S \frac{\partial \boldsymbol{E}}{\partial t} \,\mathrm{d}\boldsymbol{\sigma} = \epsilon_0 \cdot \frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{E} \,\mathrm{d}\boldsymbol{\sigma} \tag{11.132}$$

The final step in eq. (11.132) follows from Leibniz's integral rule (see eq. (5.6)) for constant limits of integration. If we extend the definition of the current in eq. (11.129) to include the displacement current from eq. (11.132), we obtain:

$$\oint_C \boldsymbol{B} \,\mathrm{d}\boldsymbol{s} = \mu_0 \big[ I(t) + I_D(t) \big] = \mu_0 I(t) + \mu_0 \epsilon_0 \cdot \frac{\mathrm{d}}{\mathrm{d}t} \int_S \boldsymbol{E} \,\mathrm{d}\boldsymbol{\sigma} \tag{11.133}$$

Eq. (11.133) is the so-called extension of Ampère's law, also known as the Maxwell-Ampère law or simply as the fourth Maxwell equation of electrodynamics. By accounting for the displacement current from eq. (11.132) in Ampère's law, we now consistently obtain the same result for the magnetic field $\boldsymbol{B}$, regardless of whether we position the hemispherical surface in fig. 11.29 to enclose the current in the wire, as shown on the left in fig. 11.29, or to enclose the displacement current, as shown on the right in fig. 11.29. Eq. (11.133) is the integral form of the fourth Maxwell equation of electrodynamics. The differential form of this equation is obtained by adding the displacement current density to the current density in eq. (9.18), yielding:

$$\int_S \Big[ (\nabla \times \boldsymbol{B}) - \mu_0 (\boldsymbol{j} + \boldsymbol{j}_V) \Big] \,\mathrm{d}\boldsymbol{\sigma} = 0 \tag{11.134}$$

From this, the differential form of the Maxwell-Ampère law follows immediately:

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{j} + \mu_0 \epsilon_0 \cdot \frac{\partial \boldsymbol{E}}{\partial t} \tag{11.135}$$

**Tab. 11.1:** Maxwell's equations of electrodynamics in a vacuum.

**I. Gauss's law for the $\boldsymbol{E}$-field**

$$\int_S \boldsymbol{E}\,\mathrm{d}\boldsymbol{\sigma} = \frac{1}{\epsilon_0}\int_\Omega \rho\,\mathrm{d}^3r \qquad \text{or} \qquad \nabla\boldsymbol{E} = \frac{1}{\epsilon_0}\rho$$

**II. Gauss's law for the $\boldsymbol{B}$ field**

$$\int_S \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} = 0 \qquad \text{or} \qquad \nabla\boldsymbol{B} = 0$$

**III. Faraday's law of induction**

$$\oint_C \boldsymbol{E}\,\mathrm{d}\boldsymbol{s} = -\frac{\partial}{\partial t}\int_A \boldsymbol{B}\,\mathrm{d}\boldsymbol{\sigma} \qquad \text{or} \qquad \nabla\times\boldsymbol{E} = -\frac{\partial\boldsymbol{B}}{\partial t}$$

**IV. Maxwell-Ampère law**

$$\oint_C \boldsymbol{B}\,\mathrm{d}\boldsymbol{s} = \mu_0 I + \mu_0\epsilon_0\frac{\mathrm{d}}{\mathrm{d}t}\int_S \boldsymbol{E}\,\mathrm{d}\boldsymbol{\sigma} \qquad \text{or} \qquad \nabla\times\boldsymbol{B} = \mu_0\boldsymbol{j} + \mu_0\epsilon_0\frac{\partial\boldsymbol{E}}{\partial t}$$

With the Maxwell-Ampère law, we now have the fourth and final Maxwell equation of electrodynamics. Tab. 11.1 summarizes the integral and differential forms of the four Maxwell equations of electrodynamics in a vacuum environment. The first and second Maxwell equations of electrodynamics correspond to the first Maxwell equations of electrostatics and magnetostatics, respectively. The second Maxwell equation of electrostatics becomes the third Maxwell equation of electrodynamics, Faraday's law, and the second Maxwell equation of magnetostatics becomes through the extension of Ampère's Law the fourth Maxwell equation of electrodynamics the Maxwell-Ampère law. In chapter 6 and chapter 7, we saw how these laws for electric fields in dielectric matter must be extended. In the next chapter, we examine how they must be modified when considering magnetic materials.

# 12. Magnetism and Matter

In chapter 9, we examined current-carrying wires in magnetic fields, determined magnetic fields in various situations, such as in and around current-carrying conductors, and learned how magnetic fields are generated. We then examined the magnetic dipole in greater detail in chapter 10, where we were introduced to the magnetic moment associated with a magnetic dipole. From there we then turned our attention to determining the force acting on a magnetic dipole in a magnetic field, as well as its energy within that field. We also learned what Faraday's law tells us about what happens when a conducting loop is moved through a magnetic field, or when the magnetic field in which a stationary conducting loop is located changes over time. Apart from the generation of magnetic fields, these are all examples of how a body, or, more precisely, matter in the form of a magnetic dipole, responds to an external magnetic field. We now turn to the question of what gives certain materials their magnetic properties. We begin with the types of magnetic materials known to us today.

## 12.1 Magnetization

At the atomic level, many materials possess magnetic moments. As discussed in the previous section, these atomic magnetic moments arise from the spin of the electrons ($\boldsymbol{m_S}$) and their orbital motion ($\boldsymbol{m_L}$) around the atomic nucleus. If these atomic magnetic moments within a body are randomly oriented, they cancel each other out, and consequently, the macroscopic body itself possesses no net magnetization. However, if the alignment of a body's magnetic moments is not randomly distributed but instead exhibits a preferred orientation, then the body possesses a resultant magnetization. An externally applied magnetic field can be used to influence the orientation of these atomic magnetic moments in a body. Depending on how the atomic magnetic moments within a body behave in this process, a distinction is made between three main types of magnetic materials: ferromagnetic, paramagnetic, and diamagnetic materials.

- Ferromagnetic materials: These are materials that can be magnetized by bringing them into an external magnetic field and they retain this magnetization even after the external magnetic field is removed. Typical ferromagnetic materials include iron (Latin: ferrum), cobalt, and nickel, or alloys of these metals. Ferromagnetic materials can also possess a permanent magnetic field without their atomic magnetic moments being aligned by external magnetic fields, a phenomenon which is known as spontaneous magnetization.
- Paramagnetic materials: In these materials, the orientation of the atomic magnetic moments also aligns with the direction of an external magnetic field. However, as soon as the external magnetic field is removed, this alignment of the magnetic moments is lost very quickly. Examples of paramagnetic materials include aluminum, magnesium, and platinum.
- Diamagnetic materials: These are materials where the magnetic moments align in a direction opposite to that of an external magnetic field. This results in a repulsive force between the external field and the magnetization induced in the diamagnetic material by that field. If the external field is removed, the magnetization of the diamagnetic material disappears as well. Examples of diamagnetic materials include copper and graphite.

To understand the behavior of such magnetic materials in a magnetic field, we look at an atomic-scale cross section of a ferromagnetic or paramagnetic material which shall be situated in an external homogeneous magnetic field $\boldsymbol{B}_{ext}$ that points in the positive $x$-direction. Each small cube in the cross-section shown in fig. 12.1 represents an atom with a magnetic moment $\boldsymbol{m}_a$ generated by an atomic-scale current. In the external magnetic field $\boldsymbol{B}_{ext}$, the magnetic moments $\boldsymbol{m}_a$ of the atoms align parallel to the external magnetic field, and the atomic currents shown in fig. 12.1 all flow in the same plane. For atoms surrounded on all sides by neighboring atoms, these atomic currents cancel each other out in this case. However, where neighbor atoms are missing, such as at the surface of a ferromagnetic or paramagnetic body, this is not the case. Consequently, at such sites a net surface current $I$ flows locally, which is not compensated by the respective current from a neighboring atom flowing in the

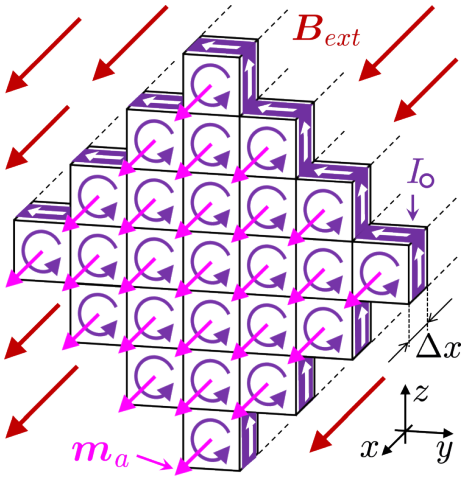



**Fig. 12.1**

opposite direction. These local atomic surface currents are not free currents and therefore cannot combine to form a free ring current on the surface of the body. Nevertheless, they constitute the equivalent of a free current such as that flowing through a single coil of an inductor. Let the dimensions of one of these small cubes representing an atom in fig. 12.1 be $\Delta x$, $\Delta y$, and $\Delta z$. Then, an equivalent current $I_{\bigcirc}$ «flows» along the surface of the ferromagnetic or paramagnetic body in the $yz$-plane within a ring of width $\Delta x$. This equivalent ring current per unit length corresponds to the current $I$ through $n$ turns per unit length in eq. (9.24), and in fig. 12.1 it «flows» counter-clockwise in the $yz$-plane. The magnetic field it generates thus points in the positive $x$-direction, just like $\boldsymbol{B}_{ext}$. The ring current per unit length is $I/\Delta x$, and thus, using eq. (9.24), the total magnetic field $\boldsymbol{B}$ in fig. 12.1 is:

$$\boldsymbol{B} = \boldsymbol{B}_{ext} + \mu_0 \frac{I_{\bigcirc}}{\Delta x} \cdot \hat{\boldsymbol{i}} \tag{12.1}$$

A magnetic moment $\boldsymbol{m}$ is associated with this ring current $I_{\bigcirc}$. According to eq. (10.37), this magnetic moment is the product of the ring current and the directed area enclosed by it. The directed area element shown in fig. 12.1 is $\Delta y \Delta z \cdot \hat{\boldsymbol{i}}$. Thus, we can replace the expression $\mu_0 I \hat{\boldsymbol{i}}/\Delta x$ in eq. (12.1) with:

$$\mu_0 \frac{I_{\bigcirc}}{\Delta x} \cdot \hat{\boldsymbol{i}} = \mu_0 \frac{I_{\bigcirc}}{\Delta x} \cdot \frac{\Delta y \Delta z}{\Delta y \Delta z} \cdot \hat{\boldsymbol{i}} = \mu_0 \frac{\Delta \boldsymbol{m}}{\Delta V} = \mu_0 \boldsymbol{M} \tag{12.2}$$

$\boldsymbol{M}$ is the macroscopic magnetization of the body in fig. 12.1, which arises from the alignment of the atomic magnetic moments $\boldsymbol{m}_a$ of this body in an external magnetic field $\boldsymbol{B}_{ext}$. Just as we defined the macroscopic quantity of dielectric polarization $\boldsymbol{P}$ via eq. (6.9) as the electric dipole moment per unit volume, we now define, by generalizing eq. (12.2), the macroscopic quantity of magnetization $\boldsymbol{M}$ as the magnetic moment per unit volume:

$$\boldsymbol{M} = \frac{\mathrm{d}\boldsymbol{m}}{\mathrm{d}V} \tag{12.3}$$

With that we can now write the expression for the total magnetic field $\boldsymbol{B}$ in eq. (12.1) as the sum of the external magnetic field and the magnetization:

$$\boldsymbol{B} = \boldsymbol{B}_{ext} + \mu_0 \cdot \boldsymbol{M} \tag{12.4}$$

Using eq. (9.19), the second Maxwell equation of magnetostatics, the following applies to the magnetization current density $\boldsymbol{j}_{mag}$ associated with the magnetic field $\mu_0 \boldsymbol{M}$:

$$\boldsymbol{j}_{mag} = \mu_0 \epsilon_0 c^2 \nabla \times \boldsymbol{M} \tag{12.5}$$

Since $\mu_0\epsilon_0 c^2 = 1$ holds, we thus obtain for the magnetization current density:

$$\boldsymbol{j}_{mag} = \nabla \times \boldsymbol{M} \tag{12.6}$$

The magnetization current density $\boldsymbol{j}_{mag}$ is the third kind of current density we encounter. The other two current densities are associated with the free and bound charge densities $\rho_{free}$ and $\rho_{bnd}$, which we discussed in chapter 6 and which each satisfy the continuity equation. For free charges, this is given by eq. (8.9). Non-free charge densities, which we have referred to interchangeably here as bound charges, $\rho_{bound}$, or polarization charges, $\rho_{pol}$, are related to the dielectric polarization $\boldsymbol{P}$ via $\rho_{pol} = -\nabla \boldsymbol{P}$, in accordance with eq. (6.25). For these, the continuity equation is given by:

$$\nabla \boldsymbol{j}_{pol} + \frac{\partial \rho_{pol}}{\partial t} = \nabla\left(\boldsymbol{j}_{pol} - \frac{\partial \boldsymbol{P}}{\partial t}\right) = 0 \qquad \Rightarrow \qquad \boldsymbol{j}_{pol} = \frac{\partial \boldsymbol{P}}{\partial t} \tag{12.7}$$

If we consider a hypothetical material that is not only magnetic but also dielectric, and through which a current of free charge carriers flows, then the total current density, using eq. (12.6) and eq. (12.7), is given by:

$$\boldsymbol{j}_{tot} = \boldsymbol{j}_{free} + \boldsymbol{j}_{pol} + \boldsymbol{j}_{mag} = \boldsymbol{j}_{free} + \frac{\partial \boldsymbol{P}}{\partial t} + \nabla \times \boldsymbol{M} \tag{12.8}$$

Substituting this into the fourth Maxwell equation in tab. 11.1, we obtain:

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{j} + \mu_0\epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} = \mu_0\left(\boldsymbol{j}_{free} + \frac{\partial \boldsymbol{P}}{\partial t} + \nabla \times \boldsymbol{M}\right) + \mu_0\epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \tag{12.9}$$

We now rearrange this to:

$$\nabla \times \left(\boldsymbol{B} - \mu_0 \boldsymbol{M}\right) = \mu_0 \boldsymbol{j}_{free} + \mu_0\epsilon_0 \frac{\partial}{\partial t}\left(\boldsymbol{E} + \frac{\boldsymbol{P}}{\epsilon_0}\right) \tag{12.10}$$

We are already familiar with the expression in parentheses on the right-hand side from chapter 6 as it corresponds to the electric displacement field $\boldsymbol{D}$ divided by $\epsilon_0$. Just as the vector field $\boldsymbol{D}$ is linked only to free charges, we can introduce a vector field $\boldsymbol{H}$ on the left-hand side of eq. (12.10) that is determined solely by free currents and the time variation of $\boldsymbol{D}$:

$$\mu_0 \boldsymbol{H} = \boldsymbol{B} - \mu_0 \boldsymbol{M} \tag{12.11}$$

In the literature, the vector field $\boldsymbol{H}$ is usually referred to as magnetic field strength and the vector field $\boldsymbol{B}$ as magnetic induction. Here, however, we retain our designation of $\boldsymbol{B}$ as the magnetic field strength and refer to $\boldsymbol{H}$ as a vector field or as the magnetization field. A look at eq. (12.4) shows that, for the magnetizing field in fig. 12.1, $\mu_0 \boldsymbol{H} = \boldsymbol{B}_{ext}$ holds true. This is, therefore, an external magnetizing field, such as that generated by a current-carrying inductor inside which the magnetic moments of a soft-iron core align

parallel to $\boldsymbol{H}$. As can further be seen from the Maxwell equations in tab. 11.1, the analogue to the electric field strength $\boldsymbol{E}$ is the field $\boldsymbol{B}$, not the field $\boldsymbol{H}$. With eq. (12.11), eq. (12.10) takes on an elegant form:

$$\nabla \times \boldsymbol{H} = \boldsymbol{j}_{free} + \frac{\partial \boldsymbol{D}}{\partial t} \tag{12.12}$$

At first glance, this equation looks simple. However, this is only the case because all the material properties that make solving it difficult are hidden within the quantities $\boldsymbol{D}$ and $\boldsymbol{H}$. If we consider only time-independent fields, then in magnetostatics, the magnetization field $\boldsymbol{H}$ is the direct counterpart to the electric displacement $\boldsymbol{D}$ in electrostatics. If $\partial \boldsymbol{D}/\partial t$ is zero or negligibly small compared to the free currents, the behavior of magnetic fields in the presence of magnetic materials can be described by the following three equations:

$$\nabla \boldsymbol{B} = 0 \tag{12.13a}$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{j}_{free} \tag{12.13b}$$

$$\mu_0 \boldsymbol{H} = \boldsymbol{B} - \mu_0 \boldsymbol{M} \tag{12.13c}$$

Just as the dielectric susceptibility $\chi$ - a material property we discussed in chapter 7 - is a measure of how easy or difficult it is to polarize a a dielectric material in an external electric field, the magnetic susceptibility $\chi_M$ is a measure of how easy or difficult it is to magnetize a magnetic material in an external magnetic field. In the case of polarization, the relationship between dielectric polarization $\boldsymbol{P}$ and electric field $\boldsymbol{E}$ is linear if the fields are not too strong. Similarly for magnetization, provided again that the fields are not too strong, the relationship between magnetization $\boldsymbol{M}$ and magnetizing field $\boldsymbol{H}$ is also linear, and it holds:

$$\boldsymbol{M} = \chi_M \boldsymbol{H} \tag{12.14}$$

Substituting this relationship into eq. (12.13c) yields:

$$\boldsymbol{B} = \mu_0(1 + \chi_M)\boldsymbol{H} = \mu_0 \mu_r \boldsymbol{H} = \mu \boldsymbol{H} \tag{12.15}$$

The quantity $\mu$, the so-called magnetic permeability, also referred to as magnetic conductivity, is the magnetic counterpart to electric permittivity $\epsilon$. Like the latter, it is a material property that describes the permeability of matter to magnetic fields, just as the physical constant $\mu_0$ determines this for a vacuum environment. And, as in the dielectric case, the material property $\mu$ is frequently expressed as a multiple of the physical constant $\mu_0$:

**Tab. 12.1:** Relative permeability $\mu_r$ of selected substances.

| Substanz | Verhalten im Magnetfeld | $\mu_r$ |
|---|---|---|
| Superconductor | ideally diamagnetic | 0 |
| $H_2O$ | diamagnetic | $\sim 0.999991$ |
| Copper | diamagnetic | $\sim 0.999994$ |
| Gold | diamagnetic | $\sim 0.999998$ |
| Aluminum | paramagnetic | $\sim 1.000022$ |
| Platinum | paramagnetic | $\sim 1.000265$ |
| Stainless steel ($< 85\%$ Fe) | paramagnetic | $\sim 1.003 - 1.05$ |
| Iron ($\sim 98\%$ Fe) | ferromagnetic | $\sim 40 - 1\,800$ |
| Soft-iron (pure Fe)% Fe) | ferromagnetic | $\sim 300 - 10\,000$ |
| Mu-metal (NiFe) | ferromagnetic | $\sim 50\,000 - 500\,000$ |

$$\mu = \mu_r \cdot \mu_0 \qquad \text{mit} \qquad \mu_r = 1 + \chi_M \tag{12.16}$$

The behavior of magnetic materials in a magnetic field, as characterized by eq. (12.15) and eq. (12.16), corresponds formally and exactly to the description of the behavior of dielectric materials in an electric field as given by eq. (6.35) and eq. (6.36). The quantity $\mu_r$ is the so-called material-dependent permeability number, formerly known as relative permeability, hence the subscript $r$. Based on the definition of $\mu_r$ in eq. (12.16), $\mu_r = 1$ in a vacuum environment. If $\mu_r > 1$, then $\chi_M > 0$ and the vectors $\boldsymbol{B}$ and $\boldsymbol{H}$ in eq. (12.15) are parallel and the material in question is paramagnetic. If $0 < \mu_r < 1$, then $\chi_M < 0$ and $\boldsymbol{B}$ and $\boldsymbol{H}$ in eq. (12.15) are antiparallel and this indicates a diamagnetic material. If $\mu_r = 0$, then $\chi_M = -1$ and one then speaks of so-called ideal diamagnetic materials, in which the magnetization $\boldsymbol{M}$ exactly cancels out the magnetizing field $\boldsymbol{H}$, causing the magnetic field within the material to vanish. This is the case, for example, with superconductors, which for reasons we cannot go into here, expel magnetic fields below a critical temperature of the material, thereby making superconductors ideal diamagnets. Tab. 12.1 lists values of $\mu_r$ for several substances. These values do, however, depend on the purity of the materials. For paramagnetic and ferromagnetic materials, $\mu_r$ also depends on temperature, whereas this is virtually not the case for diamagnetic materials. Tab. 12.1 presents $\mu_r$ values for the magnetostatic case or for low-frequency magnetizing fields $\boldsymbol{H}$, where its frequency dependence is absent or negligible. The example of iron in tab. 12.1 demonstrates the extent to which magnetic behavior depends on the purity of the substance. Pure iron, so-called soft iron, exhibits a very high relative permeability.

Steel also consists of approximately 98% iron but has already a significantly lower value of $\mu_r$ than soft iron. In the case of stainless steel, the iron content is less than 85%, and its $\mu_r$ value is again significantly lower than that of steel. Mu-metal, also known as permalloy, has the highest $\mu_r$ value in tab. 12.1. Mu-metal, whose $\mu_r$ value is orders of magnitude higher than that of air (which is practically 1), can be used to effectively shield magnetic fields. If an object is surrounded by Mu-metal, the magnetic field lines remain within the enclosed Mu-metal shielding, due to the metal's high magnetic permeability, and do not penetrate the volume enclosed by the Mu-metal.
What makes a material diamagnetic, paramagnetic, or ferromagnetic? To understand this in detail, one must be able to explain how the magnetic properties of the atoms or molecules that make up a material arise. However, this is not possible within the framework of classical physics as such an endeavor requires the tools of quantum mechanics. In classical physics, one can only reason qualitatively in this regard, while always bearing in mind that there can be no true classical explanation for these phenomena, only classical analogies that convey a qualitative understanding. We begin with a qualitative understanding of diamagnetism.

### 12.1.1 Diamagnetism

In diamagnetic materials, atoms or molecules possess no permanent magnetic moment, or at best only a negligibly small one. The reason for this is that the atoms in question do not have partially filled electron shells, but rather closed shells. However, for atoms with closed shells, the magnetic moments arising from the orbital angular momentum (see eq. (10.69)) of the electrons as well as the magnetic moments arising from electron spin (see eq. (10.70)) cancel each other out for all paired electrons.

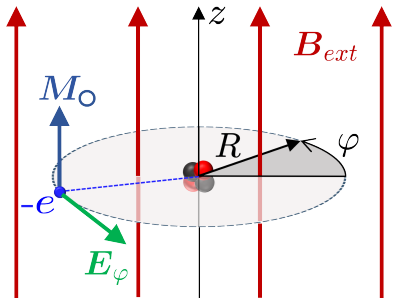


**Fig. 12.2**

When a diamagnetic material is placed in a magnetic field, it possess no atomic magnetic moments that could align with the field. Consequently, a magnetic moment can be generated in a diamagnetic material only through induction. To achieve this, the material must either be moved within a non-uniform static magnetic field or be subjected to a time-varying magnetic field. The latter occurs when a magnetic field is switched on at the location of the diamagnet. To illustrate this, let us consider the electron sketched in fig. 12.2, which moves around its atomic nucleus within the diamagnet along a circular path in a plane normal to the external field $\boldsymbol{B}_{ext} = \mu_0 \boldsymbol{H}$. According to Faraday's law

of induction, a time-varying $\boldsymbol{B}_{ext}(t)$ is associated with an electric field $\boldsymbol{E}$. But what does this electric field look like? To answer this, we consider the situation in cylindrical coordinates. In our setup, the magnetic field has only a $z$-component, and we envision this homogeneous field within our diamagnet being slowly ramped up from zero to a constant value. In this case, Faraday's law of induction, expressed as the second Maxwell equation of electrodynamics from eq. (10.75), therefore takes the following form (we work in cylindrical coordinates):

$$\nabla \times \boldsymbol{E} \;=\; \frac{1}{R}\left[\frac{\partial}{\partial R}(RE_\varphi) - \frac{\partial E_R}{\partial \varphi}\right] \;=\; -\frac{\partial B_{ext}}{\partial t} \tag{12.17}$$

Viewed from our atom, the magnetic field strength in the $xy$-plane, i.e., perpendicular to $\boldsymbol{B}_{ext}$, appears identical in all directions; therefore, $\partial E_R/\partial\varphi = 0$ must hold, and eq. (12.17) thus becomes:

$$\frac{\partial}{\partial R}(RE_\varphi) \;=\; -R\cdot\frac{\partial B_{ext}}{\partial t} \tag{12.18}$$

Since $\boldsymbol{B}_{ext}$ is uniform everywhere inside the diamagnet, $E_R$ cannot depend on $R$ either. Furthermore, since $E_z$ must vanish anyway, the only remaining component of $\boldsymbol{E}$ is the $E_\varphi$ component, i.e., $\boldsymbol{E} = \boldsymbol{E}_\varphi = E_\varphi\cdot\hat{\boldsymbol{e}}_\varphi$, as shown in fig. 12.2. Performing the integration over the $R$-coordinate in eq. (12.18) yields:

$$E_\varphi \;=\; -\frac{R}{2}\cdot\frac{\partial B_{ext}}{\partial t} \;=\; -\frac{R}{2}\cdot\frac{\mathrm{d}B_{ext}}{\mathrm{d}t} \tag{12.19}$$

A torque $\boldsymbol{M}_\mathrm{O}$ is associated with the force $\boldsymbol{F} = -e\boldsymbol{E}_\varphi$ acting on the electron orbiting the atomic nucleus in the $xy$-plane:

$$\boldsymbol{M}_\mathrm{O} \;=\; -e\big(R\cdot\hat{\boldsymbol{e}}_R \times E_\varphi\cdot\hat{\boldsymbol{e}}_\varphi\big) \;=\; -eRE_\varphi\cdot\hat{\boldsymbol{e}}_z \;=\; \frac{eR^2}{2}\frac{\mathrm{d}B_{ext}}{\mathrm{d}t}\cdot\hat{\boldsymbol{e}}_z \tag{12.20}$$

With eq. (10.68), we know that this torque must be equal to the rate of change of the electron's orbital angular momentum $\boldsymbol{L}$, that is:

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} \;=\; \frac{eR^2}{2}\frac{\mathrm{d}B_{ext}}{\mathrm{d}t}\cdot\hat{\boldsymbol{e}}_z \tag{12.21}$$

We integrate this equation from the time $t = 0$, when the field is switched on, to the time $t$, when it reaches the constant value $\boldsymbol{B}_{ext}$, and thus obtain:

$$\boldsymbol{L}(t) - \boldsymbol{L}(0) \;=\; \Delta\boldsymbol{L} \;=\; \frac{eR^2}{2}\cdot B_{ext}\cdot\hat{\boldsymbol{e}}_z \tag{12.22}$$

$\boldsymbol{L}(0)$ is the angular momentum of the electron before $\boldsymbol{B}_{ext}$ is switched on, and $\boldsymbol{L}(t)$ is the angular momentum the electron possesses when the magnetizing field has reached the value $\boldsymbol{B}_{ext}$. $\Delta\boldsymbol{L}$ is thus a purely additional angular momentum of the electron, arising solely from the ramping up of the external magnetic field from zero to the constant value

$B_{ext}$. According to eq. (10.69), a magnetic moment $\Delta\boldsymbol{m}$ is associated with $\Delta\boldsymbol{L}$:

$$\Delta\boldsymbol{m} = \frac{-e}{2m_e}\Delta\boldsymbol{L} = -\frac{e^2R^2}{4m_e}\cdot B_{ext}\cdot\hat{\boldsymbol{e}}_z \tag{12.23}$$

So far, we have said nothing about $R$. What distance $R$ should we use for an electron in the atom? The obvious answer is to use the mean square distance from the axis of rotation. In our case, the axis of rotation is the $z$-axis, and thus the mean squared distance is $R^2 = \langle\rho^2\rangle = \langle x^2\rangle + \langle y^2\rangle$. For a spherically symmetric charge distribution (spherical electron cloud), the following applies:

$$\langle x^2\rangle = \langle y^2\rangle = \langle z^2\rangle = \frac{1}{3}\langle r^2\rangle \qquad \Rightarrow \qquad \langle\rho^2\rangle = \frac{2}{3}\langle r^2\rangle \tag{12.24}$$

Substituting this into eq. (12.23), we thus obtain the magnetic moment of an atom with one electron:

$$\Delta\boldsymbol{m} = -\frac{e^2\langle r^2\rangle}{6m_e}\cdot B_{ext}\cdot\hat{\boldsymbol{e}}_z \tag{12.25}$$

If we are dealing with $n = N/V$ atoms per unit volume with atomic number $Z$, then, using $B_{ext}\cdot\hat{\boldsymbol{e}}_z = \mu_0\boldsymbol{H}$, we obtain the following expression for the magnetization:

$$\boldsymbol{M} = -\frac{n\mu_0 Ze^2\langle r^2\rangle}{6m_e}\boldsymbol{H} \tag{12.26}$$

Comparing eq. (12.26) with eq. (12.14) yields the so-called Langevin equation of diamagnetism for the magnetic susceptibility:

$$\chi_M = -\frac{n\mu_0 Ze^2\langle r^2\rangle}{6m_e} \tag{12.27}$$

Our simple classical model of diamagnetism, known as the Langevin theory of diamagnetism, yields the correct result for the behavior of diamagnets in an external magnetic field, with $\chi_M < 0$. Given $\chi_M < 0$, it follows from eq. (12.16) that $\mu_r < 1$ and thus $\mu < 1$, just as tab. 12.1 indicates for certain diamagnetic materials. The diamagnetic behavior of a material corresponds to what one would expect according to Lenz's law: the induced magnetic field, i.e., the magnetization in eq. (12.26), is directed opposite to the external magnetic field $\boldsymbol{B}_{ext} = \mu_0\boldsymbol{H}$. Diamagnetic materials are thus repelled by magnetic fields. For diamagnets in inhomogeneous magnetic fields, this means that a force acts upon them, moving them toward the region of the weaker magnetic field. The classical Langevin theory of diamagnetism provides a good description of atomic diamagnetism, for instance, in the case of noble gases with their closed electron shells. Gold is also listed as a diamagnetic material in tab. 12.1. However, gold does not possess completely filled electron shells (see tab. A.1 in Anhang B) but rather has a single electron in its outermost $s$-shell. Nevertheless, in the case of gold, as with certain other atoms,

the magnetic moments of the electrons arising from spin or orbital angular momentum cancel each other out, because the latter combine to yield a total angular momentum of zero. The diamagnetism of these atoms cannot be described by the classical Langevin theory without significant limitations. In a solid, specifically, in metallic gold, gold atoms donate their individual electrons to the conduction band. However, the Langevin theory of diamagnetism assumes that all of an atom's electrons remain localized around it and orbit it. This is not the case in metals, and this classical theory therefore cannot explain the diamagnetism of metals.

### 12.1.2 Paramagnetism

The atoms in a paramagnetic material possess a permanent magnetic moment. In the absence of an external magnetic field, these atomic magnetic moments are randomly oriented at room temperature. When a paramagnetic material is placed in a magnetic field $\boldsymbol{B}_{ext}$, the atomic magnetic moments in the material align parallel to the applied magnetic field, as sketched in fig. 12.1. If the magnetic field is removed, the material very quickly loses its magnetization $\boldsymbol{M}$, completely so, such that the orientation of the atomic magnetic moments exhibits a statistical distribution. Thus, the magnetic moments in a paramagnetic material subjected to an external magentic field behave in the same way as permanent dipole moments during orientational polarization (see section 7.2.2). We can therefore use the insights gained from examining orientational polarization to understand paramagnetic behavior. With eq. (10.56), we know the potential energy of an atomic magnetic moment. If $\vartheta$ is the angle between the external magnetic field $\boldsymbol{B}_{ext} = \mu_0 \boldsymbol{H}$ and the magnetic moment $\boldsymbol{m}$, then its potential energy is:

$$U(\vartheta) \;=\; -|\boldsymbol{m}|\cdot\mu_0\cdot|\boldsymbol{H}|\cdot cos\vartheta \tag{12.28}$$

Let there be $n_0$ identical, statistically oriented atomic magnetic moments in our paramagnetic material. For the fraction of these magnetic moments $n(\vartheta)$ at an angle $\vartheta$, given a temperature $T$ and a magnetizing field $\boldsymbol{H}$, we proceed analogously to eq. (7.45) for orientational polarization:

$$n(\vartheta) \;=\; n_0\cdot\exp\Big(-\frac{U(\vartheta)}{k_B T}\Big) \;=\; n_0\cdot\exp\Big(\frac{m\,\mu_0 H\cos\vartheta}{k_B T}\Big) \tag{12.29}$$

If we now proceed in the same way as we did in section 7.2.2 at this stage, then, with $n$ representing the number of magnetic moments per unit volume, we obtain the magnetization $\boldsymbol{M}$ of a paramagnetic body as the paramagnetic analogue to eq. (7.51) for orientation polarization:

$$\boldsymbol{M} \;=\; \frac{n\,m^2}{3k_BT}\cdot\mu_0\boldsymbol{H} \tag{12.30}$$

Using eq. (12.14), we thus obtain the following expression for the magnetic susceptibility $\chi_M$ of paramagnetic substances:

$$\chi_M \;=\; \mu_0\cdot\frac{n\,m^2}{3k_BT} \tag{12.31}$$

Unlike the diamagnetic susceptibility in eq. (12.27), the paramagnetic susceptibility is a function of temperature. As eq. (12.31), the so-called Curie law of paramagnetism, shows, the paramagnetic susceptibility is proportional to $1/T$ and independent of the applied magnetic field $\boldsymbol{B}_{ext} = \mu_0\boldsymbol{H}$. In deriving eq. (12.31), we implicitly assumed that the thermal energy $k_BT$ at room temperature is significantly greater than the potential energy $U$ of the magnetic moment. This means that eq. (12.31) holds only under the approximation $\boldsymbol{m}\mu_0\boldsymbol{H} \ll k_BT$. To see how well this approximation holds, let us consider the magnetic moment of the electron, which we already examined in section 10.1.2. For this purpose, we note here, without further explanation, that in the shell model of the atom, where electron shells are numbered $n = 1, 2, 3, \ldots$, the magnitude of an electron's orbital angular momentum $\boldsymbol{L}$ can only take on specific values calculated according to the following formula:

$$|\boldsymbol{L}| \;=\; \sqrt{l(l+1)}\cdot\hbar \qquad \text{with} \qquad l \;=\; 0, 1, 2, \ldots n-1 \tag{12.32}$$

The lowest non-zero orbital angular momentum of an electron in an atom is thus given by $|\boldsymbol{L}| = \sqrt{2}\cdot\hbar$. Substituting this angular momentum value into eq. (10.69) reveals that the magnetic moment of an electron in an atom is of the order of magnitude of the Bohr magneton $\mu_B$ itself:

$$\mu_B \;=\; \frac{e\hbar}{2m_e} \;=\; 9.27\cdot10^{-24}\,\mathrm{J\cdot T^{-1}} \;=\; 5.79\cdot10^{-2}\,\mathrm{meV\cdot T^{-1}} \tag{12.33}$$

At room temperature, $k_BT \approx 26\,\mathrm{meV}$, thus, magnetic fields significantly stronger than 100 T would be required for the approximation $\boldsymbol{m}\mu_0\boldsymbol{H} \ll k_BT$ to cease to be valid. The strongest stable magnetic fields currently technically feasible are just under 50 T, while magnetic fields exceeding 1000 T can be generated briefly, for durations of $10^{-3}$ to $10^{-6}$ s, via implosion. For even shorter durations ($10^{-23}$ s), it is possible to generate magnetic fields of up to $10^{14}$ T in particle accelerators. These are thus approximately 1000 times stronger than the magnetic fields produced by neutron stars. However, in the case of the magnetization of bodies, we are interested only in stable, technically producible magnetic fields. In this case we can assume that $\boldsymbol{m}\mu_0\boldsymbol{H} \ll k_BT$ is a very good approximation in the vast majority of cases; except in situations involving very low

temperatures. The smaller the thermal energy of the atoms becomes compared to the energy of their magnetic moments, the less thermal motion disrupts the alignment of the magnetic moments by the external magnetic field. Under the right conditions, therefore, all magnetic moments in a paramagnet align completely once a certain point is reached. This means the maximum magnetization, the so-called saturation magnetization, of the paramagnet is reached, and a further increase in the magnetizing field $\boldsymbol{H}$ in eq. (12.31) results in no further increase in the magnetization $\boldsymbol{M}$. At this point, the Curie law ceases to be valid. Eq. (12.31) therefore applies to temperatures and magnetic fields where saturation magnetization does not occur.

In an inhomogeneous magnetic field, a paramagnet moves in the direction of increasing magnetic field strength. Paramagnets are attracted by magnetic fields. This becomes evident when examining the force acting on a magnetic moment $\boldsymbol{m}$. With $\boldsymbol{m} = \text{const}$ and without free currents, i.e., $\boldsymbol{j}_{\text{free}} = 0$ and consequently $\nabla \times \boldsymbol{H} = 0$, eq. (10.55) yields the following expression for this force:

$$\boldsymbol{F} = \nabla(\boldsymbol{m}\cdot\boldsymbol{B}_{ext}) = \mu_0(\boldsymbol{m}\cdot\nabla)\boldsymbol{H} \tag{12.34}$$

Since $\chi_M > 0$, $\boldsymbol{m}$ and $\boldsymbol{H}$ have the same sign, the force on a magnetic moment increases with an increasing magnetizing field $\boldsymbol{H}$.

Interactions between individual magnetic moments played no role in the derivation of the Langevin equation for diamagnetism in eq. (12.27) or in the derivation of the Curie law for paramagnetism in eq. (12.31). However, the situation is different for ferromagnetism as this phenomenon can only be understood if the interactions between the magnetic moments are taken into account.

### 12.1.3 Ferromagnetism

In ferromagnets, local alignments of atomic magnetic moments exists within limited domains even in the absence of an external magnetizing field $\boldsymbol{H}$. These domains, known as Weiss domains or Weiss areas, schematically illustrated in a cross-section of a ferromagnet in fig. 12.3, have linear dimensions in the range of approximately $10 - 1000\,\mu\text{m}$. Consequently, ferromagnets such as iron, nickel, or cobalt typically exhibit what is referred to as spontaneous magnetization, a phenomenon where below a material dependent critical temperature ferromagnets possess a magnetization $\boldsymbol{M}$ and thus their own magnetic field, even in the absence of a magnetizing field $\boldsymbol{H}$. This intrinsic

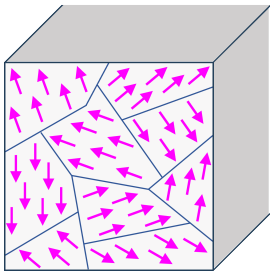


**Fig. 12.3**

magnetic field of a ferromagnet vanishes only if the Weiss domains in the ferromagnetic material are oriented such that their microscopic magnetizations sum to a macroscopic magnetization of zero. If a ferromagnet is placed in a magnetizing field $\boldsymbol{H}$, the walls between the Weiss domains begin to shift. As the magnetizing field $\boldsymbol{H}$ increases, those Weiss domains already aligned parallel to the field grow via wall-shifting at the expense of neighboring domains that are not yet aligned to the external field. Once the magnetizing field $\boldsymbol{H}$ reaches a certain strength, the magnetic moments in those remaining domains not yet aligned parallel to $\boldsymbol{H}$ suddenly flip into alignment with the external field. At this point, all magnetic moments within the ferromagnetic body are aligned parallel to the magnetizing field $\boldsymbol{H}$ and the ferromagnetic body has reached its saturation magnetization. If one then reduces the magnetizing field $\boldsymbol{H}$, however, one does not return to the same initial state; instead, one moves along a so-called hysteresis curve.[1]

When the magnetizing field is switched off, the ferromagnet loses its macroscopic magnetization. However, unlike a paramagnet, it does not lose it completely. A residual magnetization remains due to the existence of Weiss domains in a ferromagent. However, these domains now appear somewhat different from the Weiss domains that existed in the ferromagnet before the magnetizing field $\boldsymbol{H}$ was applied. Weiss domains constitute a kind of magnetic memory for a ferromagnet. Fig. 12.4 shows this memory curve of the ferromagnet, its so-called magnetization curve or hysteresis loop. When examining the curve, we begin with a ferromagnet in which the macroscopic magnetization $\boldsymbol{M}$ shall be zero. This means that the microscopic magnetizations of the Weiss domains of the ferromagnet are statistically oriented in such a way that their respective magnetizations sum to a macroscopic magnetization of zero. We now apply a magnetizing field $\boldsymbol{H}$ directed along the $z$-axis, i.e., $\boldsymbol{H} = H\hat{\boldsymbol{k}}$. If we ramp up this magnetizing field starting from $H = 0$, we trace the so-called initial magnetization curve. Along that curve the magnetization $\boldsymbol{M}$ of the ferromagnet increases as the magnetizing field $\boldsymbol{H}$ grows. Given a sufficiently strong magnetizing field $\boldsymbol{H}$, the ferromagnet reaches its saturation magnetization $\boldsymbol{M}_S$. If the magnetizing field $\boldsymbol{H}$ is then reduced, the magnetization $\boldsymbol{M}$ does not follow the

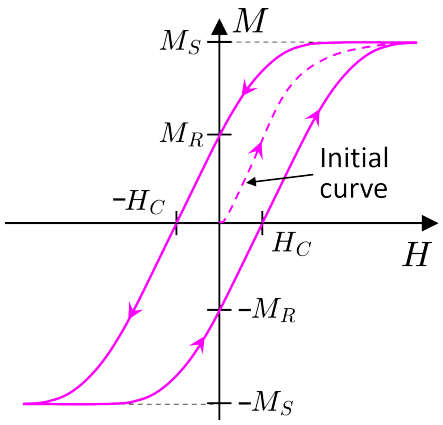



**Fig. 12.4**

[1] The word «hysteresis» is of Ancient Greek origin and derives from terms meaning «later» or «lower». In the present context, it expresses the fact that the magnetization $\boldsymbol{M}$ lags behind the magnetizing field $\boldsymbol{H}$.

path of the initial magnetization curve, but rather that of the left branch of the hysteresis loop. At $H = 0$, the ferromagnet still possesses a macroscopic magnetization $\boldsymbol{M}_R$, known as remanent magnetization. Only by reversing the direction of the magnetizing field, i.e., $\boldsymbol{H} = -H\hat{\boldsymbol{k}}$, does the magnetization $\boldsymbol{M}$ decrease further until it vanishes at $H = -H_C$. $H_C$ is the so-called coercive field strength, which forces the magnetization of the ferromagnet to zero (hence the name: the Latin *coercere* means, among other things, to force or constrain). If $\boldsymbol{H}$ is increased further, the magnetization curve reaches the negative saturation magnetization $-\boldsymbol{M}_S$. If the magnetizing field is now reduced back to zero, the magnetization follows the right-hand branch of the hysteresis loop rather than the left-hand one. At $H = 0$, the ferromagnet exhibits a negative remanent magnetization $-\boldsymbol{M}_R$. Only when the field, now reversed in polarity, i.e., $\boldsymbol{H} = H\hat{\boldsymbol{k}}$, reaches the value $H = H_C$ does the magnetization of the ferromagnet vanish. If the magnetizing field is increased further, saturation magnetization $\boldsymbol{M}_S$ is reached again. Subsequently, if the magnetizing field is reduced again, the magnetization follows the left branch of the hysteresis loop.

So much for the experimental findings. But how can we describe the ferromagnetic behavior, and how can we express the ferromagnetic susceptibility? Clearly, we must take into account the interaction between the individual magnetic moments. We have already seen how this can be done in the case of polarization in section 7.2.3, when determining the electric field $\boldsymbol{E}$ in a dielectric with a polarization $\boldsymbol{P}$. To this end, as shown in fig. 7.8, we cut a sphere out of the dielectric and then considered, on the one hand, the field of the remaining hollow sphere within the dielectric, arising from the polarization charges at the interfaces, and, on the other hand, the polarization field of the sphere that had been cut out. The sum of these two fields was then the effective electric field within the dielectric that we were looking for. We are therefore seeking the magnetostatic counterpart to eq. (7.60). However, we cannot proceed with the magnetization $\boldsymbol{M}$ in the same way as in section 7.2.3, because there are no magnetic charges and, consequently, no magnetic polarization charges. However, we can do something similar. To this end, we first consider the situation sketched in fig. 12.5: a cavity within the magnetic material in the shape of an elongated hole, with its long axis parallel to $\boldsymbol{M}$. We know the magnetic field $\boldsymbol{H}_{Mat}$ in the material and the magnetization $\boldsymbol{M}$, and we seek the magnetic field

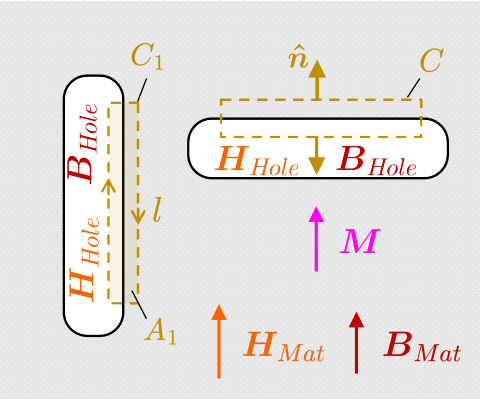



**Fig. 12.5**

$\boldsymbol{H}_{Hole}$ inside the cavity. With $\boldsymbol{j}_{free} = 0$, it follows from eq. (12.13b) that $\nabla \times \boldsymbol{H} = 0$, and the line integral along the closed curve $C_1$ enclosing the area $A_1$ must vanish. The segments of the integration path $C_1$ that lie normal to $\boldsymbol{H}_{Hole}$ and $\boldsymbol{H}_{Mat}$, respectively, contribute nothing to the integral. The segments $\boldsymbol{l}$ of $C$ inside and outside the cavity are traversed parallel to $\boldsymbol{H}_{Hole}$ and antiparallel to $\boldsymbol{H}_{Mat}$, respectively, and thus it holds that:

$$0 = \int_{A_1} \nabla \times \boldsymbol{H} \, \mathrm{d}\boldsymbol{\sigma} = \int_C \boldsymbol{H} \, \mathrm{d}\boldsymbol{s} = \left(\boldsymbol{H}_{Hole} - \boldsymbol{H}_{Mat}\right) \cdot \boldsymbol{l} \tag{12.35}$$

Thus, for the cavity in fig. 12.5 with its long axis parallel to $\boldsymbol{H}_{Mat}$ must hold:

$$\boldsymbol{H}_{Hole} = \boldsymbol{H}_{Mat} \tag{12.36}$$

Since there is no magnetization $\boldsymbol{M}$ within the slot itself, it follows that:

$$\boldsymbol{B}_{Hole} = \boldsymbol{B}_{Mat} - \mu_0 \boldsymbol{M} \tag{12.37}$$

Let us now consider the situation shown in fig. 12.5, involving a cavity whose long axis is oriented normal to the magnetization. However, the curve $C$, which runs half within this cavity and half within the material, is not a path of integration here, but rather the cross-section of a Gaussian pillbox with height $h$ and base and top areas $A$; $\hat{\boldsymbol{n}}$ represents the normal vector to the respective base or top surface $A$. For the surface $S$ enclosing the volume $V = A \cdot h$ of the Gaussian pillbox, the following applies according to the second Maxwell equation:

$$0 = \int_S \boldsymbol{B} \, \mathrm{d}\boldsymbol{\sigma} = \left(\boldsymbol{B}_{Hole} - \boldsymbol{B}_{Mat}\right) \cdot \hat{\boldsymbol{n}} \tag{12.38}$$

Thus, for the cavity in fig. 12.5 with its long axis perpendicular to $\boldsymbol{H}_{Mat}$ must hold:

$$\boldsymbol{B}_{Hole} = \boldsymbol{B}_{Mat} \tag{12.39}$$

Since there is no magnetic field in this cavity, it follows immediately that:

$$\boldsymbol{H}_{Hole} = \boldsymbol{H}_{Mat} + \mu_0 \boldsymbol{M} \tag{12.40}$$

We could have also simply derived the respective expressions for the elongated holes parallel and perpendicular to the magnetization $\boldsymbol{M}$ as shown in fig. 12.5, i.e., eq. (12.36), and eq. (12.40), by drawing an analogy to our analysis of the electric field $\boldsymbol{E}$ and the dielectric polarization $\boldsymbol{P}$ in dielectrics in a similar situation. If, in the corresponding equations for elongated holes parallel and perpendicular to $\boldsymbol{P}$ in section 7.2.3, eq. (7.54) and eq. (7.56), one replaces $\boldsymbol{E}$ with $\boldsymbol{H}/\mu_0$ and $\boldsymbol{P}/\epsilon_0$ with $\boldsymbol{M}$, one obtains exactly eq. (12.36) and eq. (12.40).

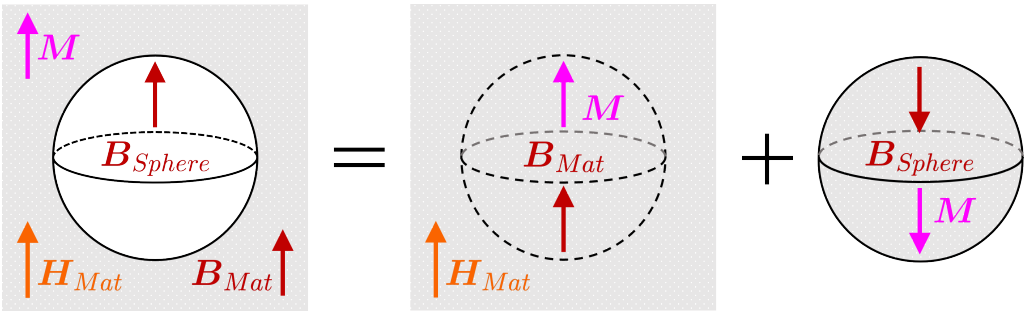



**Fig. 12.6:** The magnetic field $\boldsymbol{B}_{Hole}$ within a spherical cavity in a magnetic material with magnetization $\boldsymbol{M}$ is the superposition of the field $\boldsymbol{B}_{Mat}$ in the magnetic material (without the cavity) and the magnetic field $\boldsymbol{B}_{Sphere}$ inside a sphere of the same size but with opposite magnetization $-\boldsymbol{M}$.

If we make this substitution for the electric field in a hollow sphere within a dielectric, as described by eq. (7.58), then we should obtain the magnetic field in a hollow sphere within a magnetic material with magnetization $M$ as follows:

$$\boldsymbol{E}_{Hole} = \boldsymbol{E}_{Mat} + \frac{\boldsymbol{P}}{3\epsilon_0} \qquad \longrightarrow \qquad \frac{1}{\mu_0}\boldsymbol{H}_{Hole} = \frac{1}{\mu_0}\boldsymbol{H}_{Mat} + \frac{\boldsymbol{M}}{3} \tag{12.41}$$

It follows from this, using eq. (12.13c):

$$\frac{1}{\mu_0}\boldsymbol{B}_{Hole} = \frac{1}{\mu_0}(\boldsymbol{B}_{Mat} - \mu_0\boldsymbol{M}) + \frac{\boldsymbol{M}}{3} \tag{12.42}$$

For the magnetic field in the cavity, we thus obtain:

$$\boldsymbol{B}_{Hole} = \boldsymbol{B}_{Mat} - \frac{2\mu_0}{3}\boldsymbol{M} \tag{12.43}$$

Fig. 12.6 shows the interpretation of this equation. The magnetic field $\boldsymbol{B}_{Hole}$ is the superposition of the magnetic field $\boldsymbol{B}_{Mat}$ without the cavity and the magnetic field $\boldsymbol{B}_{Sphere}$ of a sphere of the same size with magnetization $-\boldsymbol{M}$. A little further on, we will verify in another way that, for the magnetic field inside a homogeneously magnetized sphere with magnetization $\boldsymbol{M}$, the following indeed holds:

$$\boldsymbol{B}_{Sphere} = \frac{2\mu_0}{3}\boldsymbol{M} \tag{12.44}$$

What exactly have we gained by knowing the magnetic field in a spherical cavity within a magnetic material with magnetization $\boldsymbol{M}$? The answer is simple: the same as what we obtained in a similar situation in the case of dielectric polarization. There, the electric field $\boldsymbol{E}_{Hole}$ within such a cavity in the dielectric provided the average electric field acting on an atom or molecule in a body with polarization $\boldsymbol{P}$. Similarly, the magnetic field $\boldsymbol{B}_{Hole}$ within a cavity in a magnet provides the average magnetic field acting on an atomic magnetic moment in a magnetic material with magnetization $\boldsymbol{M}$. Thus, using

$\boldsymbol{B}_{Hole}$, we can account for the interaction between magnetic moments in a ferromagnet. To do this, we simply need to substitute $\boldsymbol{B}_{Hole} = \mu_0 \boldsymbol{H}_{Hole}$ into eq. (12.30), using the expression derived for the magnetization $\boldsymbol{M}$ of a paramagnet as a function of the applied field $\boldsymbol{B}_{ext} = \mu_0 \boldsymbol{H}$, where the magnetic moments do not interact. However, we first write eq. (12.30) and $\boldsymbol{H}_{Hole}$ in a simplified form:

$$\boldsymbol{M} = \frac{C}{T} \cdot \boldsymbol{H}_{Hole} \qquad \text{and} \qquad \boldsymbol{H}_{Hole} = \boldsymbol{H}_{Mat} + \lambda \boldsymbol{M} \tag{12.45}$$

If we substitute $\boldsymbol{H}_{Hole}$ from the right-hand equation into the left-hand equation, we obtain the expression:

$$\boldsymbol{M} = \frac{C}{T} \cdot (\boldsymbol{H}_{Mat} + \lambda \boldsymbol{M}) \tag{12.46}$$

Solving this equation for $\boldsymbol{M}$ yields:

$$\boldsymbol{M} = \frac{C}{T - \lambda C} \cdot \boldsymbol{H}_{Mat} \tag{12.47}$$

Using the Curie temperature $T_c = \lambda C$, we thus obtain the so-called Curie-Weiss law for the magnetic susceptibility of a ferromagnet:

$$\chi_M = \frac{C}{T - T_c} \tag{12.48}$$

For $T \gg T_c$, a ferromagnet behaves exactly like a paramagnet. The reason for this is that, at such high temperatures, the thermal energy is much greater than the energy associated with the interaction between atomic magnetic moments in the ferromagnet. Consequently, the atomic magnetic moments in the ferromagnet behave like non-interacting magnetic moments. As $T_c$ is approached from above, the magnetic susceptibility, governed by the Curie-Weiss law in eq. (12.48), becomes larger as the temperature decreases than it would be according to the Curie law in eq. (12.31) and rises sharply as $T \to T_c$. Nominally, eq. (12.48) diverges at $T = T_c$, but in practice, $\chi_M$ merely assumes very large values at $T = T_c$. $T = T_c$ marks the material's phase transition to ferromagnetic behavior, characterized by the formation of Weiss domains within the ferromagnetic material and the associated spontaneous magnetization. From this point on, the behavior of the ferromagnet below $T = T_c$ is determined by the displacement of the walls between Weiss domains and the rotation of their respective magnetic orientations, rather than by the alignment of magnetic moments constrained by thermal motion, as is the case for $T > T_c$. For $T < T_c$, the relationship between magnetization $\boldsymbol{M}$ and magnetizing field $\boldsymbol{H}$ is no longer described by eq. (12.48), but rather by hysteresis curves as shown in fig. 12.4. For iron, cobalt, and nickel, three of the most important ferromagnetic materials, the Curie temperature is 1043 K, 1400 K, and 627 K, respectively.

## 12.2 Magnetic Fields in Matter

Magnetism in matter is a purely quantum mechanical phenomenon. Nevertheless, its macroscopic effects, as shown in the previous section, can be described classically to a certain extent using Maxwell's equations. Fortunately, for many technical applications, this classical description is entirely sufficient, but we should always be aware that it fails to explain the microscopic causes. Naturally, this applies not only to the magnetic properties within the bulk of a body but also to the properties of magnetic interfaces and surfaces. With that in mind, let us now examine the macroscopic behavior of magnetic fields at magnetic interfaces and surfaces as described by Maxwell's equations. Subsequently, we will turn to the more detailed treatment of the magnetic field of a sphere, as promised in the previous section.

### 12.2.1 Interfaces of and within Magnetic Materials

In section 6.1.2, we examined the behavior of the electric field $\boldsymbol{E}$ and the electric displacement field $\boldsymbol{D}$ at the interfaces of and between dielectrics. We can directly apply what we learned there to the behavior of $\boldsymbol{B}$ and $\boldsymbol{H}$ at interfaces and surfaces of magnetic materials. To this end, we consider the interface between the two magnetic materials ① and ② shown in fig. 12.7. Let the material properties of the two magnetic materials be $\mu_1$ and $\mu_2$, and the corresponding fields be $\boldsymbol{B}_1$ and $\boldsymbol{H}_1$ as well as $\boldsymbol{B}_2$ and $\boldsymbol{H}_2$. To understand the field behavior at the interface, we place a Gaussian pillbox at the interface. Let this Gaussian pillbox have a volume $V$ enclosed by the surface $\partial V$, the boundary of the pillbox. When applying Gauss's theorem, we split the surface integral over the boundary $\partial V$ of the Gaussian pillbox into a contribution from the top and bottom caps ($A$) and a contribution from the cylindrical side surface ($Cyl.$). From $\nabla \cdot \boldsymbol{B} = 0$, the fact that the magnetic field $\boldsymbol{B}$ can have neither sources nor sinks, it follows from Gauss's theorem that:

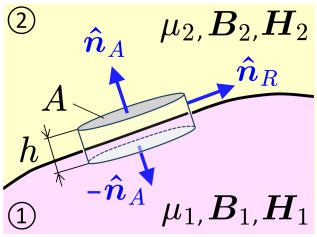



**Fig. 12.7**

$$\int_{\partial V} \boldsymbol{B}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}\sigma \;=\; \int_{A} \boldsymbol{B}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{A}}\,\mathrm{d}A + \int_{Cyl.} \boldsymbol{B}\cdot\hat{\boldsymbol{n}}_{\boldsymbol{R}}\cdot R\,\mathrm{d}\varphi\,\mathrm{d}h \;=\; \int_{V} \nabla\boldsymbol{B}\,\mathrm{d}V \;=\; 0 \tag{12.49}$$

This is the counterpart to eq. (6.44) for $\rho_{free} = 0$, except that $\boldsymbol{B}$ appears here instead of $\boldsymbol{D}$. For the same reasons demonstrated for $\boldsymbol{D}$ at the interface between two insulators in section 6.1.2, the integral over the cylindrical side surface vanishes for $\boldsymbol{B}$ as well. Thus the following must hold:

$$\hat{\boldsymbol{n}}(\boldsymbol{B}_2 - \boldsymbol{B}_1) = 0 \tag{12.50}$$

It follows immediately from this for the normal component of $\boldsymbol{B}$ and, using eq. (12.15), for the normal component of $\boldsymbol{H}$ that:

$$\boldsymbol{B}_{2_\perp} = \boldsymbol{B}_{1_\perp} \qquad \text{and} \qquad \boldsymbol{H}_{2_\perp} = \frac{\mu_1}{\mu_2}\boldsymbol{H}_{1_\perp} \tag{12.51}$$

Thus, the normal component of the magnetic field $\boldsymbol{B}$ is continuous at the interface between two magnetic materials, whereas the normal component of the magnetization field is discontinuous, i.e., it undergoes a jump. How can we understand this behavior? For $\boldsymbol{B}$, this is straightforward. Because of $\nabla\boldsymbol{B} = 0$, there are no sources or sinks for $\boldsymbol{B}$. If we let the height $h$ of the Gaussian pillbox in fig. 12.7 approach zero until the bottom and top surfaces meet at the interface and merge into a single surface, then the number of field lines entering that surface must equal the number of field lines exiting it. However, this only applies if the normal component of $\boldsymbol{B}$ is continuous. The situation is different for the normal component of the magnetic field $\boldsymbol{H}$. The reason for this is that $\boldsymbol{H}$ accounts for the magnetization of the material, and this can change at the interface between material ① and material ② in fig. 12.7. The normal component of $\boldsymbol{H}$ is continuous only if the magnetization is identical in both materials, i.e., if $\mu_1 = \mu_2$ holds.

Next, let us examine the behavior of the tangential components of $\boldsymbol{B}$ and $\boldsymbol{H}$ at the interface between two magnetic materials, or at the interface between a magnetic material and a vacuum environment. For dielectric interfaces in section 6.1.2, we utilized the fact that $\nabla \times \boldsymbol{E} = 0$ holds in electrostatics to show that, in this case, the tangential component of the electric field is continuous at the interface between two dielectrics.

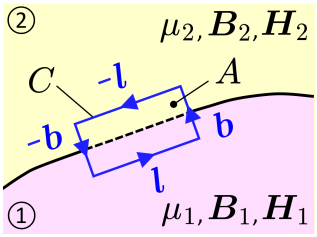



**Fig. 12.8**

Here we proceed analogously but use eq. (12.13b), i.e., $\nabla \times \boldsymbol{H} = \boldsymbol{j}_{free}$, to show that if no free currents flow at the interface between the two magnetic materials or between a magnetic material and vacuum in fig. 12.8, i.e., $\boldsymbol{j}_{free} = 0$, then the tangential component of the magnetization field is continuous at the interface. To do this, we apply Stokes' theorem to the surface $A$ bounded by $C$, along which the boundary line between ① and ② in fig. 12.8 runs:

$$\int_A \nabla \times \boldsymbol{H}\,\mathrm{d}\boldsymbol{A} = \oint_C \boldsymbol{H}\,\mathrm{d}\boldsymbol{s} = 0 \tag{12.52}$$

As shown in fig. 12.8, the closed path $C$ runs in such a way that it intersects the boundary line separating material ① from material ② at its midpoint. The path segments $\boldsymbol{l}$ and $-\boldsymbol{l}$ of $C$ lie entirely within material ① and material ②, respectively, while each of the

path segments $\boldsymbol{b}$ and $-\boldsymbol{b}$ lies half in material ① and half in material ②. Evaluating the line integral in eq. (12.52) using these path segments yields the equation:

$$\oint_C \boldsymbol{H}\,\mathrm{d}\boldsymbol{s} = \boldsymbol{l}\boldsymbol{H}_{1_\parallel} + \frac{\boldsymbol{b}}{2}\boldsymbol{H}_{1_\perp} + \frac{\boldsymbol{b}}{2}\boldsymbol{H}_{2_\perp} - \boldsymbol{l}\boldsymbol{H}_{2_\parallel} - \frac{\boldsymbol{b}}{2}\boldsymbol{H}_{2_\perp} - \frac{\boldsymbol{b}}{2}\boldsymbol{H}_{1_\perp} = 0 \tag{12.53}$$

The contributions from the path segments $\boldsymbol{b}$ and $-\boldsymbol{b}$ cancel out, and the equation reduces to:

$$\boldsymbol{l}(\boldsymbol{H}_{1_\parallel} - \boldsymbol{H}_{2_\parallel}) = 0 \tag{12.54}$$

It follows immediately that the parallel component of $\boldsymbol{H}$ is continuous and, given eq. (12.15), the parallel component of $\boldsymbol{B}$ is discontinuous, i.e., it undergoes a jump:

$$\boldsymbol{H}_{2_\parallel} = \boldsymbol{H}_{1_\parallel} \qquad \text{and} \qquad \boldsymbol{B}_{2_\parallel} = \frac{\mu_2}{\mu_1}\boldsymbol{B}_{1_\parallel} \tag{12.55}$$

If we compare eq. (12.51) and eq. (12.55), which describe the behavior of magnetic fields at magnetic interfaces, with the corresponding equations eq. (6.48) and eq. (6.52) regarding the behavior of electric fields at dielectric interfaces, we see that $\boldsymbol{B}$ behaves like $\boldsymbol{D}$ and $\boldsymbol{H}$ behaves like $\boldsymbol{E}$. As with the dielectric materials in section 6.1.2, we consider only linear media here. This means that $\mu$ in eq. (12.15) is always a scalar quantity. With a scalar $\mu$, the field vectors $\boldsymbol{B}$ and $\boldsymbol{H}$ are always parallel, just as the field vectors $\boldsymbol{E}$ and $\boldsymbol{D}$ are always parallel when $\epsilon$ is a scalar. Fig. 12.9 illustrates the behavior of $\boldsymbol{B}$ and $\boldsymbol{H}$ at a magnetic interface for the case $\mu_2 > \mu_1$. Applying the same considerations we used for dielectric interfaces in section 6.1.2, we can read the law of refraction for magnetic field lines from the drawing in fig. 12.9. With

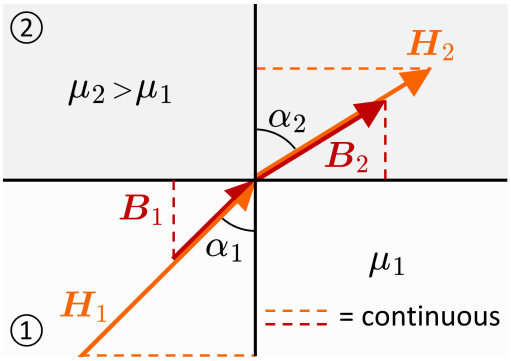



**Fig. 12.9**

$$|\boldsymbol{H}_1|\sin\alpha_1 = |\boldsymbol{H}_2|\sin\alpha_2 \qquad \text{and} \qquad |\boldsymbol{B}_1|\cos\alpha_1 = |\boldsymbol{B}_2|\cos\alpha_2 \tag{12.56}$$

as well as $\boldsymbol{B}_1 = \mu_1\boldsymbol{H}_1$ and $\boldsymbol{B}_2 = \mu_1\boldsymbol{H}_2$ follows:

$$\frac{1}{\mu_1}\tan\alpha_1 = \frac{1}{\mu_2}\tan\alpha_2 \tag{12.57}$$

Just as electric field lines are refracted away from or toward the normal when entering a dielectric material of higher or lower permittivity $\epsilon$, magnetic field lines are refracted away from or toward the normal when entering a magnetic material of higher or lower permeability $\mu$.

### 12.2.2 The Homogeneously Magnetized Sphere

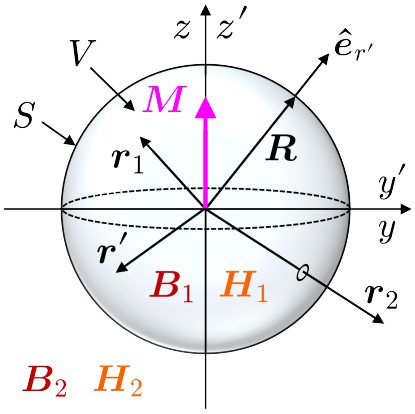



**Fig. 12.10**

We now return to the uniformly magnetized sphere, whose internal magnetic field, described by eq. (12.44), we have already derived earlier in this chapter. Here, we will determine both the field inside a sphere with uniform magnetization $\boldsymbol{M}$ and the corresponding field outside the magnetized sphere. To this end, we consider the sphere of radius $R$ sketched in fig. 12.10, whose magnetization $\boldsymbol{M}$ is parallel to the $z$-axis and whose center coincides with the origin of the coordinate system. The sphere is situated in a vacuum environment, and there is no external magnetic field other than the one generated by the magnetized sphere itself. We examine the magnetic field produced by the magnetization inside the sphere separately from the magnetic field produced outside the sphere. Let us begin with the magnetic field $\boldsymbol{B}_1$ inside the sphere. According to eq. (12.13b), the source of the magnetic field within the sphere must be a magnetization current density $\boldsymbol{j}_{mag}$. However, the magnetization $\boldsymbol{M}$ inside the sphere is constant, and this implies, via eq. (12.6):

$$\boldsymbol{j}_{mag} = \nabla \times \boldsymbol{M} = 0 \tag{12.58}$$

With $\boldsymbol{j}_{mag} = 0$ within the spherical volume $\boldsymbol{V}$, a magnetic field inside the sphere can originate only from magnetization currents in the spherical shell $S$. If we know these surface currents in the spherical shell $S$, we can determine the magnetic field inside the sphere. We obtain these surface currents via the vector potential $\boldsymbol{A}(\boldsymbol{r})$, which is generated by the magnetization $\boldsymbol{M} = \mathrm{d}\boldsymbol{m}/\mathrm{d}V$. To do this, we integrate eq. (10.27) over the volume of the sphere ($\mu_0 = 1/\epsilon_0 c^2$):

$$\boldsymbol{A}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\boldsymbol{M}(\boldsymbol{r}') \times (\boldsymbol{r} - \boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|^3} \, \mathrm{d}^3\boldsymbol{r}' \tag{12.59}$$

With the help of the identity

$$\frac{\boldsymbol{r} - \boldsymbol{r}'}{|\boldsymbol{r} - \boldsymbol{r}'|^3} = \nabla' \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|}$$

we can rewrite eq. (12.59) as:

$$\boldsymbol{A}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \int_V \left[ \boldsymbol{M}(\boldsymbol{r}') \times \nabla' \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} \right] \mathrm{d}^3\boldsymbol{r}' \tag{12.60}$$

Now, using $\boldsymbol{a} = \boldsymbol{M}$ and $V = 1/|\boldsymbol{r} - \boldsymbol{r}'|$, we apply the rule from tab. 2.1:

$$\nabla \times (V\boldsymbol{a}) \;=\; V(\nabla \times \boldsymbol{a}) + (\nabla \cdot V) \times \boldsymbol{a} \;=\; V(\nabla \times \boldsymbol{a}) - \boldsymbol{a} \times (\nabla \cdot V)$$

This allows us to rewrite the integrand in eq. (12.60) as:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{\mu_0}{4\pi}\left[\int_V \frac{\nabla' \times \boldsymbol{M}(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}^3\boldsymbol{r}' - \int_V \nabla' \times \frac{\boldsymbol{M}(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}^3\boldsymbol{r}'\right] \tag{12.61}$$

We can now apply the generalized Stokes' theorem from eq. (2.48) to the second integral in the parentheses, thereby obtaining for the vector potential $\boldsymbol{A}$ the expression:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{\mu_0}{4\pi}\left[\int_V \frac{\nabla' \times \boldsymbol{M}(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}^3\boldsymbol{r}' - \int_S \frac{\hat{\boldsymbol{n}} \times \boldsymbol{M}(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\,\mathrm{d}^2\boldsymbol{r}'\right] \tag{12.62}$$

The volume integral involves, via eq. (12.6), the magnetization current density $\boldsymbol{j}_{max} = \nabla \times \boldsymbol{M}$, which is, however, zero within the interior of the sphere shown in eq. (12.10). Consequently, the volume integral in eq. (12.62) vanishes, leaving only the surface integral containing the currents on the spherical shell $S$ that bounds the spherical volume shown in eq. (12.10). With $\boldsymbol{M} = M \cdot \hat{\boldsymbol{k}}'$ and the normal vector $\hat{\boldsymbol{n}} = \hat{\boldsymbol{e}}_{r'}$ in spherical coordinates, the numerator of the surface integral becomes:

$$\hat{\boldsymbol{n}} \times \boldsymbol{M} \;=\; \hat{\boldsymbol{e}}_{r'} \times M \cdot \hat{\boldsymbol{k}}' \;=\; -M \cdot \sin\vartheta' \cdot \hat{\boldsymbol{e}}'_{\varphi} \tag{12.63}$$

From this point on, we distinguish between the interior of the sphere and the exterior region. In the surface integral, $r'$ is not a variable, instead, $r' = R$ and $d\boldsymbol{r}' = R^2 \sin\vartheta' d\vartheta' d\varphi'$. Consequently, the term $1/|\boldsymbol{r} - \boldsymbol{r}'|$ in the surface integral becomes $1/|\boldsymbol{r} - \boldsymbol{R}|$, and we now replace this expression with the first two terms of the multipole expansion for $r < r' = R$, just as we did in eq. (4.79):

$$\frac{1}{|\boldsymbol{r} - \boldsymbol{R}|} \;\longrightarrow\; \frac{1}{R} + \frac{\boldsymbol{r} \cdot \boldsymbol{R}}{R^3} \qquad \text{for} \qquad r < R \tag{12.64}$$

If we write out the vector product $\boldsymbol{r} \cdot \boldsymbol{R}$ of

$$\boldsymbol{r} = r(\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$$

and

$$\boldsymbol{R} = R(\sin\vartheta'\cos\varphi', \sin\vartheta'\sin\varphi', \cos\vartheta')$$

we obtain the somewhat lengthy expression:

$$\boldsymbol{r} \cdot \boldsymbol{R} \;=\; r \cdot R\big(\sin\vartheta\cos\varphi\sin\vartheta'\cos\varphi' + \sin\vartheta\sin\varphi\sin\vartheta'\sin\varphi' + \cos\vartheta\cos\vartheta'\big) \tag{12.65}$$

After substituting eq. (12.63) and eq. (12.64) into the surface integral in eq. (12.62) and making use of eq. (12.65), we obtain the following expression for the vector potential $\boldsymbol{A}$ after some rearrangement:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{\mu_0 M}{4\pi} \int_0^{2\pi}\!\!\int_0^{\pi} \Big[ \big(R + r\cos\vartheta\cos\vartheta'\big) + r(\sin\vartheta\cos\varphi\sin\vartheta'\cos\varphi' + \sin\vartheta\sin\varphi\sin\vartheta'\sin\varphi') \Big] \hat{\boldsymbol{e}}'_\varphi \sin^2\vartheta' \,\mathrm{d}\vartheta' \,\mathrm{d}\varphi' \qquad , \quad r < R \tag{12.66}$$

We first consider the integral over $\varphi'$ containing the expression $R + r\cos\vartheta\cos\vartheta'$. This expression does not depend on $\varphi'$, so only the integration over $\hat{\boldsymbol{e}}'_\varphi = -\hat{\boldsymbol{i}}\sin\varphi' + \hat{\boldsymbol{j}}\cos\varphi'$ remains. Integrating $\cos\varphi$ or $\sin\varphi$ from 0 to $2\pi$ yields zero, and the term involving $R + r\cos\vartheta\cos\vartheta'$ in the integrand of eq. (12.66) makes no contribution to $\boldsymbol{A}(\boldsymbol{r})$. The remaining integral expression from the $\mathrm{d}\varphi'$-integration of eq. (12.66) is therefore:

$$r \int_0^{2\pi} \Big[ -\hat{\boldsymbol{i}} \big( \sin\vartheta\cos\varphi\sin^3\vartheta'\sin\varphi'\cos\varphi' + \sin\vartheta\sin\varphi\sin^3\vartheta'\sin^2\varphi' \big) + \hat{\boldsymbol{j}} \big( \sin\vartheta\cos\varphi\sin^3\vartheta'\cos^2\varphi' + \sin\vartheta\sin\varphi\sin^3\vartheta'\sin\varphi'\cos\varphi' \big) \Big] \mathrm{d}\varphi'$$

Here, we can immediately omit the mixed terms $\sin\varphi'\cos\varphi' = \sin 2\varphi'/2$ in the integrand because they drop out upon integration. Consequently, the $\varphi'$-integration in eq. (12.66) reduces to:

$$\begin{aligned} & r\sin\vartheta\sin^3\vartheta' \int_0^{2\pi} \big( -\hat{\boldsymbol{i}}\sin\varphi\sin^2\varphi' + \hat{\boldsymbol{j}}\cos\varphi\cos^2\varphi' \big) \,\mathrm{d}\varphi' \\ &= r\sin\vartheta\sin^3\vartheta' \left[ -\hat{\boldsymbol{i}}\sin\varphi\left(\frac{\varphi'}{2} - \frac{1}{4}\sin 2\varphi'\right) + \hat{\boldsymbol{j}}\cos\varphi\left(\frac{\varphi'}{2} + \frac{1}{4}\sin 2\varphi'\right) \right] \Bigg|_0^{2\pi} \\ &= r\sin\vartheta\sin^3\vartheta' \Big[ -\hat{\boldsymbol{i}}\sin\varphi\cdot\pi + \hat{\boldsymbol{j}}\cos\varphi\cdot\pi \Big] \;=\; \pi r\sin\vartheta\sin^3\vartheta'\cdot\hat{\boldsymbol{e}}_\varphi \end{aligned}$$

After substituting this expression into eq. (12.66), we now only need to perform the integration over $\vartheta'$. This yields a factor of $4/3$, and we thus obtain the following result for the vector potential:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{\mu_0 M r\sin\vartheta}{4}\cdot\hat{\boldsymbol{e}}_\varphi \cdot \int_0^{\pi} \sin^3\vartheta' \,\mathrm{d}\vartheta' \;=\; \frac{\mu_0 M r\sin\vartheta}{3}\cdot\hat{\boldsymbol{e}}_\varphi \qquad , \quad r < R \tag{12.67}$$

Hence, the vector potential inside the sphere shown in fig. 12.10 has only an $A_\varphi$ component as $A_r$ and $A_\vartheta$ are both zero. We obtain the magnetic field $\boldsymbol{B}_1$ inside the sphere using $\boldsymbol{B} = \nabla\times\boldsymbol{A}$ (in accordance with eq. (9.9)). For convenience, we perform the calculation in spherical coordinates (see tab. A.1 in Appendix A):

$$\boldsymbol{B}_1 \;=\; \nabla\times\boldsymbol{A} \;=\; \frac{1}{r\sin\vartheta}\frac{\partial}{\partial\vartheta}(\sin\vartheta A_\varphi)\cdot\hat{\boldsymbol{e}}_r - \frac{1}{r}\frac{\partial}{\partial r}(rA_\varphi)\cdot\hat{\boldsymbol{e}}_\vartheta \qquad , \quad r < R \tag{12.68}$$

After evaluating the differentials on the right-hand side of the equation, this becomes:

$$\boldsymbol{B}_1 \;=\; \frac{2\mu_0 M}{3}\big(\cos\vartheta\cdot\hat{\boldsymbol{e}}_r - \sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta\big) \tag{12.69}$$

With $\cos\vartheta\hat{\boldsymbol{e}}_r - \sin\vartheta\hat{\boldsymbol{e}}_\vartheta = \hat{\boldsymbol{k}}$, we thus obtain the final result:

$$\boldsymbol{B}_1 \;=\; \frac{2\mu_0 M}{3}\cdot\hat{\boldsymbol{k}} \;=\; \frac{2\mu_0}{3}\cdot\boldsymbol{M} \tag{12.70}$$

However, this corresponds exactly to the result from eq. (12.44), which we derived there in a different way for the magnetic field $\boldsymbol{B}_{Sphere}$ inside a uniformly magnetized sphere. Associated with the sphere's magnetization $\boldsymbol{M}$ is a magnetization field within the sphere, for which, using eq. (12.13c) and eq. (12.70), the following holds:

$$\mu_0\boldsymbol{H}_1 \;=\; \boldsymbol{B}_1 - \mu_0\boldsymbol{M} \;=\; -\frac{\mu_0}{3}\boldsymbol{M} \tag{12.71}$$

We thus obtain the result in spherical or Cartesian coordinates:

$$\boldsymbol{H}_1 \;=\; -\frac{M}{3}\big(\cos\vartheta\cdot\hat{\boldsymbol{e}}_r - \sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta\big) \;=\; -\frac{M}{3}\cdot\hat{\mathbf{k}} \tag{12.72}$$

The field $\mu_0\boldsymbol{H}_1$ is only half as strong as the field $\boldsymbol{B}_1$. That means, the field-line density of $\mu_0\boldsymbol{H}_1$ inside the sphere is lower by a factor of 2 than that of $\boldsymbol{B}_1$. Furthermore, $\boldsymbol{H}_1$ points in the opposite direction to $\boldsymbol{B}_1$. This is the case not only for the uniformly magnetized sphere considered here but for all bodies possessing their own uniform magnetization. For this reason, $\boldsymbol{H}_1$ is also referred to as the demagnetizing field. This demagnetizing field $\boldsymbol{H}_1$ is not an external field, but rather the field generated by the body's own permanent magnetization.

Next, we examine what eq. (12.62) yields for the vector potential outside the sphere, i.e., for $r > R$. The volume integral still vanishes, and the field outside the sphere arises exclusively from the magnetization currents in the spherical shell $S$. In this case, instead of eq. (12.64), we use the following approximation for $1/|\boldsymbol{r}-\boldsymbol{R}|$ in the surface integral (see also eq. (3.75)):

$$\frac{1}{|\boldsymbol{r}-\boldsymbol{R}|} \;\longrightarrow\; \frac{1}{r} + \frac{\boldsymbol{r}\cdot\boldsymbol{R}}{r^3} \qquad \text{for} \qquad r > R \tag{12.73}$$

Substituting this into the surface integral in eq. (12.62) and using eq. (12.65) then yields, instead of eq. (12.66):

$$\begin{aligned}\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{\mu_0 M}{4\pi}\int_0^{2\pi}\!\!\int_0^{\pi}\Big[\Big(\frac{R^2}{r} + \frac{R^3}{r^2}\cos\vartheta\cos\vartheta'\Big) + \frac{R^3}{r^2}(\sin\vartheta\cos\varphi\sin\vartheta'\cos\varphi' \\ + \sin\vartheta\sin\varphi\sin\vartheta'\sin\varphi')\Big]\hat{\boldsymbol{e}}'_\varphi\sin^2\vartheta'\,\mathrm{d}\vartheta'\,\mathrm{d}\varphi' \qquad , \quad r > R\end{aligned} \tag{12.74}$$

Using exactly the same steps by which we reduced eq. (12.66) to eq. (12.67), we can reduce eq. (12.74) to:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{\mu_0 M R^3 \sin\vartheta}{4r^2}\cdot\hat{\boldsymbol{e}}_\varphi\cdot\int_0^\pi \sin^3\vartheta'\,\mathrm{d}\vartheta' \;=\; \frac{\mu_0 M R^3 \sin\vartheta}{3r^2}\cdot\hat{\boldsymbol{e}}_\varphi \qquad , \quad r > R \tag{12.75}$$

For $r > 0$, the vector potential $\boldsymbol{A}(\boldsymbol{r})$ again has only an $A_\varphi$ component, which, however, now scales not with $r$ but with $r^{-2}$. For the magnetic field $\boldsymbol{B_2}$ outside the sphere, this vector potential yields:

$$\boldsymbol{B}_2 \;=\; \nabla\times\boldsymbol{A} \;=\; \frac{\mu_0 M R^3}{3r^3}(2\cos\vartheta\cdot\hat{\boldsymbol{e}}_r + \sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta) \tag{12.76}$$

We now transform the expression in parentheses using a small trick, which is adding a term that we immediately subtract again, thereby obtaining the identity:

$$2\cos\vartheta\cdot\hat{\boldsymbol{e}}_r + \cos\vartheta\cdot\hat{\boldsymbol{e}}_r - \cos\vartheta\cdot\hat{\boldsymbol{e}}_r + \sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta \;=\; 3\cos\vartheta\cdot\hat{\boldsymbol{e}}_r - \hat{\boldsymbol{k}} \tag{12.77}$$

With $\hat{\boldsymbol{e}}_r = \boldsymbol{r}/r$, eq. (12.76) thus becomes:

$$\boldsymbol{B}_2 \;=\; \frac{\mu_0 R^3}{3}\left(3\cdot\frac{rM\cos\vartheta}{r^4}\cdot\frac{\boldsymbol{r}}{r} - \frac{M}{r^3}\cdot\boldsymbol{k}\right) \tag{12.78}$$

The first fraction inside the parentheses was expanded by $r$ so that the expression $rM\cos\vartheta$ appears in the numerator. Since $\vartheta$ is the angle between $\boldsymbol{M}$ and $\boldsymbol{r}$, the expression $\boldsymbol{M}\cdot\boldsymbol{r}$ now appears in the numerator of the fraction. With $\boldsymbol{M} = (0, 0, M)$, the second term inside the parentheses is the vector $\boldsymbol{M}$ divided by $r^3$. We can thus now write the magnetic field in the region outside the magnetized sphere as:

$$\boldsymbol{B}_2 \;=\; \frac{\mu_0 R^3}{3}\left(\frac{3\cdot(\boldsymbol{M}\cdot\boldsymbol{r})\cdot\boldsymbol{r}}{r^5} - \frac{\boldsymbol{M}}{r^3}\right) \tag{12.79}$$

A comparison of eq. (12.79) with eq. (10.38) shows that the magnetic field $\boldsymbol{B}_2$ outside the sphere is that of a magnetic dipole located at the center of the sphere. We are already familiar with the expression in parentheses in eq. (12.76) from eq. (3.59) or eq. (6.100), which describe the radial and angular dependence of the electric field produced by an electrostatic dipole. Therefore, to visualize the path of the field lines of $\boldsymbol{B}$ outside the sphere, we can apply the same method to the right-hand side of eq. (12.76) that we used earlier to visualize the dipole fields in fig. 3.7 and fig. 6.17. fig. 12.11 shows the path of the $\boldsymbol{B}$ field lines for the magnetized sphere obtained in this way. Magnetic

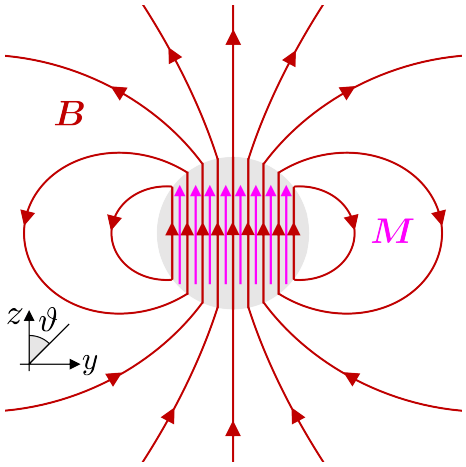



**Fig. 12.11**

field lines in the region outside the sphere, described by eq. (12.76), are closed by the magnetic field lines running parallel to the magnetization $\boldsymbol{M}$ inside the sphere, described by eq. (12.70). Outside the magnetized sphere, the magnetization $\boldsymbol{M}$ is zero. Consequently, $\boldsymbol{B}_2 = \mu_0 \boldsymbol{H}_2$ holds there and we obtain for $\boldsymbol{H}_2$:

$$\boldsymbol{H}_2 \;=\; \frac{MR^3}{3r^3}(2\cos\vartheta\cdot\hat{\boldsymbol{e}}_r + \sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta) \tag{12.80}$$

With eq. (12.69), eq. (12.72), eq. (12.76), and eq. (12.80), we now know the fields $\boldsymbol{B}$ and $\boldsymbol{H}$ inside the sphere ($r < R$) and outside the sphere ($r > R$). Next, we examine the behavior of these fields at the surface of the magnetized sphere, i.e., at $r = R$. To obtain the respective normal components of these fields at $r = R$, we simply need to multiply eq. (12.70) and eq. (12.76) by the unit normal vector $\hat{\boldsymbol{e}}_r$ of the sphere. Doing so yields the following for the $\boldsymbol{B}$ field:

$$\boldsymbol{B}_1(R)_\perp \;=\; \hat{\boldsymbol{e}}_r\cdot\boldsymbol{B}_1 \;=\; \frac{2\mu_0 M}{3}\cos\vartheta \tag{12.81a}$$

and

$$\boldsymbol{B}_2(R)_\perp \;=\; \hat{\boldsymbol{e}}_r\cdot\boldsymbol{B}_2(R) \;=\; \frac{2\mu_0 M}{3}\cos\vartheta \tag{12.81b}$$

Regarding these two equations, it should be noted that $\hat{\boldsymbol{e}}_r$ runs parallel to $\boldsymbol{B}_1$ and $\boldsymbol{B}_2$ at $\vartheta = 0$, and antiparallel to them at $\vartheta = \pi$. As required by eq. (12.51), the normal component of $\boldsymbol{B}$ in the spherical shell is continuous. For the normal components of $\boldsymbol{H}$, we obtain:

$$\boldsymbol{H}_1(R)_\perp \;=\; \hat{\boldsymbol{e}}_r\cdot\boldsymbol{H}_1 \;=\; -\frac{M}{3}\cos\vartheta \tag{12.82a}$$

and

$$\boldsymbol{H}_2(R)_\perp \;=\; \hat{\boldsymbol{e}}_r\cdot\boldsymbol{H}_2(R) \;=\; \frac{2M}{3}\cos\vartheta \tag{12.82b}$$

The normal component of $\boldsymbol{H}$ is therefore not continuous within the spherical shell but undergoes a jump of $M\cos\vartheta$. Let us now consider the tangential components. The $\boldsymbol{B}$ and $\boldsymbol{H}$ fields have no $\varphi$-component, and we obtain their variable tangential components on the spherical surface simply by multiplying the corresponding fields at $r = R$ by the tangential unit vector $\hat{\boldsymbol{e}}_\vartheta$. For the magnetic field, this yields:

$$\boldsymbol{B}_1(R)_\parallel \;=\; \hat{\boldsymbol{e}}_\vartheta\cdot\boldsymbol{B}_1 \;=\; -\frac{2\mu_0 M}{3}\sin\vartheta \tag{12.83a}$$

and

$$\boldsymbol{B}_2(R)_\parallel \;=\; \hat{\boldsymbol{e}}_\vartheta\cdot\boldsymbol{B}_2(R) \;=\; \frac{\mu_0 M}{3}\sin\vartheta \tag{12.83b}$$

The tangential component of $\boldsymbol{B}$ is discontinuous and jumps by $\mu_0 M \sin\vartheta$ at $r = R$. For the tangential component of $\boldsymbol{H}$, we obtain:

$$\boldsymbol{H}_1(R)_\parallel \;=\; \hat{\boldsymbol{e}}_\vartheta \cdot \boldsymbol{H}_1 \;=\; \frac{M}{3}\sin\vartheta \tag{12.84a}$$

and

$$\boldsymbol{H}_2(R)_\parallel \;=\; \hat{\boldsymbol{e}}_\vartheta \cdot \boldsymbol{H}_2(R) \;=\; \frac{M}{3}\sin\vartheta \tag{12.84b}$$

The tangential component of $\boldsymbol{H}$ is continuous across the spherical shell. Thus, the normal and tangential components of $\boldsymbol{B}$ and $\boldsymbol{H}$ behave as expected: $\boldsymbol{B}_\perp$ and $\boldsymbol{H}_\parallel$ are continuous, while $\boldsymbol{B}_\parallel$ and $\boldsymbol{H}_\perp$ undergo a jump at the spherical surface.

### 12.2.3 Magnetic Sphere in a Homogeneous Magnetic Field

In the previous section, we examined the magnetic fields inside and outside a magnetized sphere which are generated by the uniform magnetization $\boldsymbol{M}$ of that sphere. In this section, we consider how these magnetic fields change when, as shown in Fig. 12.12, the magnetized sphere is placed in an external, uniform magnetic field $\boldsymbol{B}_{ext}$ that is oriented parallel to the sphere's magnetization $\boldsymbol{M}$. We have already examined a similar situation earlier in section 6.2.2, where we considered a polarized sphere in an external, uniform electrostatic field aligned parallel to the polarization $\boldsymbol{P}$ of that sphere. Just as in that case, our external field is to extend to infinity in both directions along the $z$-axis here as well. Thus, very far from the sphere, only the field $\boldsymbol{B}_{ext}$ exists, and it is only in the vicinity of the sphere that it is modulated by the sphere's magnetic field. To truly make use of the analogy with section 6.2.2, we need a magnetostatic counterpart to the electrostatic potential $\phi$. And such a counterpart exists here. Since there are no free currents or charge carriers, eq. (12.10) implies that:

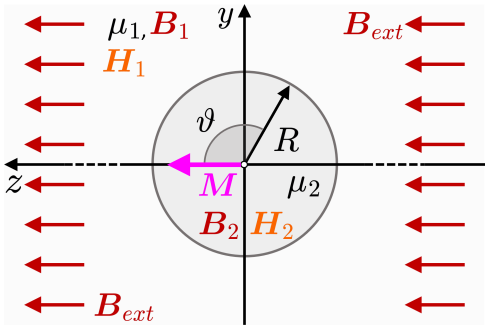



**Fig. 12.12**

$$\nabla \times (\boldsymbol{B} - \mu_0 \boldsymbol{M}) \;=\; \nabla \times \mu_0 \boldsymbol{H} \;=\; 0 \tag{12.85}$$

With this, however, we can write the magnetizing field $\boldsymbol{H}$, analogous to the $\boldsymbol{E}$-field in electrostatics, as the gradient of a potential $\psi$, that is:

$$\boldsymbol{H} \;=\; -\nabla \cdot \psi \tag{12.86}$$

By analogy with the electrostatic potential, we can interpret $\psi$ as a magnetostatic potential. If we consider only linear, isotropic media, then according to eq. (12.15), $\boldsymbol{B} = \mu \boldsymbol{H}$. If these media are also homogeneous, meaning $\mu$ does not depend on position, then for $\boldsymbol{B}$ we have:

$$\nabla \boldsymbol{B} \;=\; -\mu \Delta \psi \;=\; 0 \qquad \Rightarrow \qquad \Delta \psi \;=\; 0 \tag{12.87}$$

Just as the electrostatic potential $\phi$ satisfies Laplace's equation in the absence of free charges, $\psi$ likewise satisfies Laplace's equation. Now, we simply need an equivalent to eq. (6.80) to apply the ansatz used in section 6.2.2 for the electrostatic potential $\phi$ in eq. (6.81) to the magnetostatic potential $\psi$ here. A comparison of eq. (10.38) and eq. (12.79) shows that the equivalent magnetic dipole $\boldsymbol{m}$, which generates the magnetic field of the magnetized sphere, is given by:

$$\boldsymbol{m} \;=\; \frac{4\pi R^3}{3} \boldsymbol{M} \tag{12.88}$$

As a quick comparison shows, eq. (12.88) is the direct counterpart to eq. (6.80). For the potential $\psi$, we therefore choose an ansatz that is the formal equivalent of the electrostatic dipole potential:

$$\psi \;=\; \frac{\boldsymbol{m}\boldsymbol{r}}{4\pi r^3} \;=\; \frac{\boldsymbol{M}\boldsymbol{r} R^3}{3r^3} \;=\; \frac{M R^3}{3r^2} \cdot \cos\vartheta \tag{12.89}$$

As can be quickly verified, the magnetization field $\boldsymbol{H}$ associated with this potential is given by:

$$\boldsymbol{H} \;=\; -\nabla \cdot \psi \;=\; \frac{M R^3}{3r^2} (2\cos\vartheta \cdot \hat{\boldsymbol{e}}_r + \sin\vartheta \cdot \hat{\boldsymbol{e}}_\vartheta) \tag{12.90}$$

However, this is precisely eq. (12.80), and thus we know the generating potential $\psi$ for the magnetic dipole field outside the sphere shown in fig. 12.12. By analogy with eq. (6.81), we can now formulate the expressions for the magnetic potential $\psi$ inside and outside the magnetic sphere shown in fig. 12.12:

$$\psi_1 \;=\; \left( -H_1 \cdot r + \frac{1}{4\pi} \frac{m}{r^2} \right) \cdot \cos\vartheta \qquad \text{and} \qquad \psi_2 \;=\; -H_2 \cdot r \cdot \cos\vartheta \tag{12.91}$$

$H_1$ is the homogeneous field very far from the sphere, possessing only a $z$-component. Or, in other words, for $r \gg R$, $\boldsymbol{B}_{ext} = \mu_1 \boldsymbol{H}_1$ holds. In the case of the electrostatic potential in eq. (6.81), the parameters of this ansatz were $E_2$ and $p$, in eq. (12.91), these parameters are $H_2$ and $m$. Just as in section 6.2.2, we determine these using the boundary conditions for the potential at $r = R$. For $\boldsymbol{H}$, the following must hold at $r = R$:

$$\mu_1 \boldsymbol{H}_{1\perp} \;=\; \mu_2 \boldsymbol{H}_{2\perp} \qquad \text{and} \qquad \boldsymbol{H}_{1\parallel} \;=\; \boldsymbol{H}_{2\parallel}$$

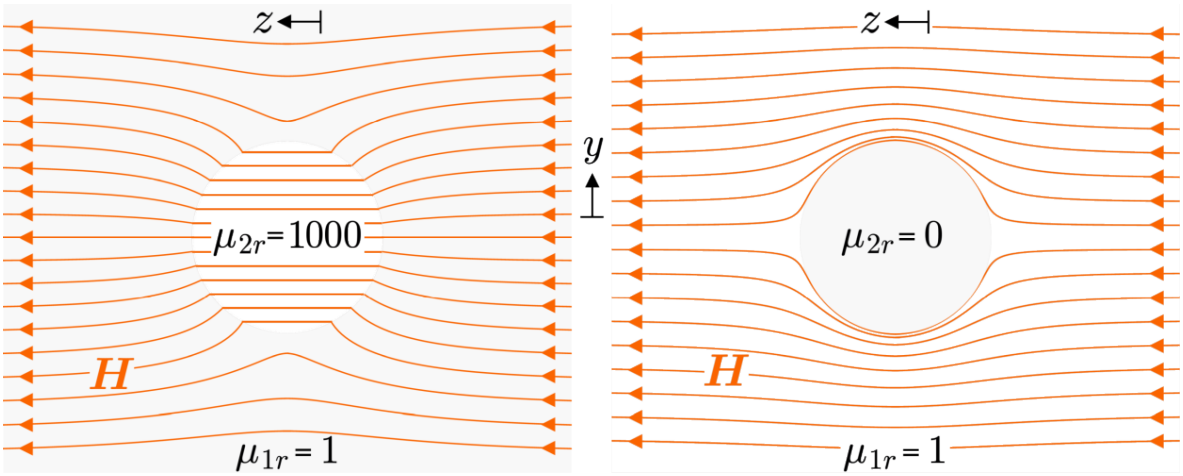



**Fig. 12.13:** Field lines of $\boldsymbol{H}$ for a magnetized sphere of paramagnetic or ferromagnetic material in a homogeneous external magnetic field (a) and for a superconducting sphere (ideal diamagnet) in a homogeneous magnetic field (b).

The corresponding boundary conditions for the normal and tangential components of the scalar field $\psi$ are thus:

$$\mu_1 \frac{\partial \psi_1}{\partial r}\bigg|_{r=R} = \mu_2 \frac{\partial \psi_2}{\partial r}\bigg|_{r=R} \quad \text{and} \quad \frac{\partial \psi_1}{\partial \vartheta}\bigg|_{r=R} = \frac{\partial \psi_2}{\partial \vartheta}\bigg|_{r=R} \tag{12.92}$$

The boundary condition for the tangential component is equivalent to $\psi_1(R) = \psi_2(R)$. From eq. (12.92), we obtain the following equations to determine $H_2$ and $m$:

$$-\mu_1 H_1 - \frac{\mu_1}{2\pi}\frac{m}{R^3} = -\mu_2 H_2 \tag{12.93a}$$

and

$$-H_1 R + \frac{1}{4\pi}\frac{m}{R^2} = -H_2 R \tag{12.93b}$$

Evaluating these equations yields the following for $H_2$ and $m$:

$$H_2 = \frac{3\mu_1}{\mu_2 + 2\mu 1} \cdot H_1 \quad \text{and} \quad m = 4\pi R^3 \cdot \frac{\mu_2 - \mu_1}{\mu_2 + 2\mu_1} \cdot H_1 \tag{12.94}$$

We now substitute this, with $z = r\cos\vartheta$, into our expression for the potentials $\psi_1$ and $\psi_2$ in eq. (12.91) and thus obtain:

$$\psi_1 = \left(-1 + \frac{\mu_2 - \mu_1}{\mu_2 + 2\mu_1}\frac{R^3}{r^3}\right) \cdot H_1 \cdot z \tag{12.95a}$$

and

$$\psi_2 = -\frac{3\mu_1}{\mu_2 + 2\mu_1} \cdot H_1 \cdot z \tag{12.95b}$$

We can now determine the magnetization field $\boldsymbol{H}$ using the relation $\boldsymbol{H} = -\nabla\psi$. When deriving the potential, it makes no difference whether we place an already magnetized

sphere into a magnetic field or place a sphere made of paramagnetic, ferromagnetic, or diamagnetic material into a homogeneous magnetic field. In any case, the result arises from the superposition of the external homogeneous magnetic field and the magnetic field of the magnetized sphere. Fig. 12.13a shows an example in which the permeability of the sphere is greater than that of the surrounding medium, as is the case with a paramagnetic or ferromagnetic material. However, with $\mu_{2r} = 1000$, the material in fig. 12.13a behaves more like a ferromagnet. For the reverse situation, i.e., $\mu_{2r} < \mu_{1r}$, fig. 12.13b illustrates the example of an ideal diamagnet, where $\mu_{1r} = 0$, as is the case with a superconducting sphere. The expulsion of the magnetic field from a superconducting material below a critical temperature arises from causes different from those in a conventional diamagnetic material, yet it is nevertheless encompassed by our model. Moreover, it offers the advantage of being far more illustrative than the merely partial displacement observed in ordinary diamagnets. Using eq. (12.88) and eq. (12.94), we can express the magnetization in terms of the applied field:

$$M = 3\frac{\mu_2 - \mu_1}{\mu_2 + 2\mu_1} H_1 \tag{12.96}$$

For $r \gg R$, $\boldsymbol{B} = \mu_1 \boldsymbol{H}_1$ holds, and using eq. (12.94), we can express the difference between the uniform magnetizing field inside the sphere and the uniform magnetizing field at infinity as:

$$\boldsymbol{H}_2 - \boldsymbol{H}_1 = \left(\frac{3\mu_1}{\mu_2 + 2\mu_1} - 1\right)\boldsymbol{H}_1 = \frac{\mu_1 - \mu_2}{\mu_2 + 2\mu_1}\boldsymbol{H}_1 = -\frac{\boldsymbol{M}}{3} \tag{12.97}$$

If we now switch off the external magnetic field $\boldsymbol{B}_{ext}$, then $\boldsymbol{H}_1 = 0$ applies, and eq. (12.97) becomes:

$$\boldsymbol{H}_2 = -\frac{\boldsymbol{M}}{3} \tag{12.98}$$

However, eq. (12.13c) still applies to the field inside the sphere, and therefore:

$$\mu_0 \boldsymbol{H}_2 = -\frac{\mu_0}{3}\boldsymbol{M} = \boldsymbol{B}_2 - \mu_0 \boldsymbol{M} \tag{12.99}$$

It follows immediately from this for the field $\boldsymbol{B}_2$:

$$\boldsymbol{B}_2 = \frac{2\mu_0}{3}\boldsymbol{M} \tag{12.100}$$

Here, after eq. (12.44) and eq. (12.70), we have thus derived the magnetic field inside a uniformly magnetized sphere in a third way.

# 13. Electromagnetic Waves I

The electromagnetic interaction between charges, between currents, and between charges and currents is fully described by Maxwell's equations. As these equations tell us, however, sources of electric fields are not only charges but also time-varying magnetic fields, and sources of magnetic fields are not only currents but also time-varying electric fields. Even in the absence of charges and currents, the Maxwell equations possess solutions that, in addition to the interactions between charges, between currents, or between charges and currents, have a physical reality of their own. These solutions are space-filling electromagnetic fields. We are all familiar with examples of such electromagnetic fields from our daily experience, ranging from the electromagnetic radiation that reaches us as sunlight and without which we would not exist, to the many technical applications found in areas such as electronic communication or the manufacturing of circuits, where wire diameters and components are measured in nanometers. In this chapter, we consider the electromagnetic wave solutions of Maxwell's equations in a vacuum environment. In the next chapter, we then address the generation of electromagnetic waves and their propagation in matter.

## 13.1 Maxwell's Equations in a Vacuum

In the absence of charges and currents, Maxwell equations from tab. 11.1 are:

$$\nabla \boldsymbol{E} = 0 \tag{13.1a}$$

$$\nabla \boldsymbol{B} = 0 \tag{13.1b}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{13.1c}$$

$$\nabla \times \boldsymbol{B} = \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \tag{13.1d}$$

If we now apply the relation $\nabla \times (\nabla \times \boldsymbol{a}) = \nabla(\nabla \boldsymbol{a}) - \Delta \boldsymbol{a}$ to the curl of $\boldsymbol{E}$ and the curl of $\boldsymbol{B}$, we obtain, given $\nabla \boldsymbol{E} = 0$ and $\nabla \boldsymbol{B} = 0$, the equations:

$$-\Delta \boldsymbol{E} \;=\; -\nabla \times \frac{\partial \boldsymbol{B}}{\partial t} \;=\; -\frac{\partial}{\partial t}(\nabla \times \boldsymbol{B}) \;=\; -\mu_0 \epsilon_0 \frac{\partial^2 \boldsymbol{E}}{\partial t^2} \tag{13.2}$$

and

$$-\Delta \boldsymbol{B} \;=\; \mu_0 \epsilon_0 \nabla \times \frac{\partial \boldsymbol{E}}{\partial t} \;=\; \mu_0 \epsilon_0 \frac{\partial}{\partial t}(\nabla \times \boldsymbol{E}) \;=\; -\mu_0 \epsilon_0 \frac{\partial^2 \boldsymbol{B}}{\partial t^2} \tag{13.3}$$

With $c^2 = 1/\mu_0 \epsilon_0$, these two equations become:

$$\Delta \boldsymbol{E} - \frac{1}{c^2} \frac{\partial^2 \boldsymbol{E}}{\partial t^2} \;=\; 0 \tag{13.4}$$

and

$$\Delta \boldsymbol{B} - \frac{1}{c^2} \frac{\partial^2 \boldsymbol{B}}{\partial t^2} \;=\; 0 \tag{13.5}$$

We are already familiar with this type of equation from the discussion of wave motion in volume 1 of this physics course; they are wave equations. Not only do the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ satisfy wave equations, but so do the corresponding potentials $\phi$ and $\boldsymbol{A}$. To this end, we first consider eq. (13.1c), where we replace $\boldsymbol{B}$ with the corresponding potential $\boldsymbol{B} = \nabla \times \boldsymbol{A}$:

$$\nabla \times \boldsymbol{E} \;=\; -\frac{\partial \boldsymbol{B}}{\partial t} \;=\; -\frac{\partial}{\partial t}(\nabla \times \boldsymbol{A}) \tag{13.6}$$

By interchanging the time derivative and the curl, this becomes:

$$\nabla \times \left( \boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t} \right) \;=\; 0 \tag{13.7}$$

However, this means that there must be a potential $\phi$ for which the following holds:

$$\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t} \;=\; -\nabla \phi \qquad \Rightarrow \qquad \boldsymbol{E} \;=\; -\nabla \phi - \frac{\partial \boldsymbol{A}}{\partial t} \tag{13.8}$$

If we also set $\boldsymbol{B} = \nabla \times \boldsymbol{A}$ in eq. (13.1d) and use eq. (13.8), we obtain:

$$\nabla \times (\nabla \times \boldsymbol{A}) \;=\; \frac{1}{c^2} \frac{\partial}{\partial t} \left( -\nabla \phi - \frac{\partial \boldsymbol{A}}{\partial t} \right) \tag{13.9}$$

Now we again use $\nabla \times (\nabla \times \boldsymbol{a}) = \nabla(\nabla \boldsymbol{a}) - \Delta \boldsymbol{a}$ and move everything to the left-hand side:

$$\Delta \boldsymbol{A} - \frac{1}{c^2} \frac{\partial^2 \boldsymbol{A}}{\partial t^2} - \nabla \left( \nabla \boldsymbol{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} \right) \;=\; 0 \tag{13.10}$$

With $\nabla \boldsymbol{E} = 0$, it also follows from eq. (13.8) that:

$$\Delta \phi + \frac{\partial}{\partial t}(\nabla \boldsymbol{A}) \;=\; 0 \tag{13.11}$$

Eq. (13.10) and eq. (13.11) are coupled differential equations for $\phi$ and $\boldsymbol{A}$. However, we can decouple them by choosing:

$$\nabla \boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0 \tag{13.12}$$

As we learned at the end of chapter 2, the vector potential $\boldsymbol{A}$ is uniquely determined only up to the gradient of a scalar:

$$\boldsymbol{A} = \boldsymbol{A}' + \nabla \Lambda \qquad \Rightarrow \qquad \nabla \times \boldsymbol{A} = \nabla \times \boldsymbol{A}' \tag{13.13}$$

We can therefore choose $\nabla\Lambda$ freely, and in chapter 2 we chose it such that $\nabla \boldsymbol{A} = 0$ holds. This choice of $\nabla\Lambda$ is the so-called Coulomb gauge, which simplifies the Maxwell equation in electrostatics; for instance, eq. (13.11) thereby becomes the Laplace equation. The choice we made in eq. (13.12) is the so-called Lorenz gauge. Substituting eq. (13.12) into eq. (13.10) and eq. (13.11), we obtain the wave equations for the electric potential $\phi$ and the vector potential $\boldsymbol{A}$:

$$\Delta\phi - \frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} = 0 \tag{13.14}$$

and

$$\Delta\boldsymbol{A} - \frac{1}{c^2}\frac{\partial^2 \boldsymbol{A}}{\partial t^2} = 0 \tag{13.15}$$

Two solutions to such wave equations that we encountered in volume 1 are plane waves and spherical waves. Although eq. (13.10) and eq. (13.11), as well as eq. (13.14) and eq. (13.15), are decoupled equations, they are not independent wave equations, because the Maxwell equations from which they are derived remain valid.

### 13.1.1 Monochromatic Plane Waves

Let us consider the wave equations for the electric field $\boldsymbol{E}$ and the magnetic field $\boldsymbol{B}$, eq. (13.4) and eq. (13.5). From volume 1, we know that the ansatz for a plane monochromatic wave with wave vector $\boldsymbol{k}$ and angular frequency $\omega$ must be:[1]

$$\boldsymbol{E}(\boldsymbol{r}, t) = \boldsymbol{E}_0 \mathrm{e}^{i(\boldsymbol{k}\boldsymbol{r} - \omega t)} \tag{13.16}$$

and

$$\boldsymbol{B}(\boldsymbol{r}, t) = \boldsymbol{B}_0 \mathrm{e}^{i(\boldsymbol{k}\boldsymbol{r} - \omega t)} \tag{13.17}$$

Substituting this into eq. (13.4) or eq. (13.5) yields, in each case:

$$-\boldsymbol{k}^2 + \frac{\omega^2}{c^2} = 0 \qquad \Rightarrow \qquad \omega^2 = k^2 c^2 \tag{13.18}$$

[1] Except in the case of complex conjugate quantities, complex quantities are not specifically marked here, as the context makes it clear in each instance whether or not a complex quantity is involved.

For the wavelength $\lambda$ and the wave vector $\boldsymbol{k}$ of the plane wave, as well as for the relationship between its angular frequency $\omega$ and the period $T$ or frequency $f$, the relations already known from volume 1 apply:

$$\lambda = \frac{2\pi}{k} \qquad \text{and} \qquad T = \frac{2\pi}{\omega} = \frac{1}{f} \tag{13.19}$$

The speed of light $c$ is a fundamental physical constant, with $c^2 = 1/\mu_0\epsilon_0$ in a vacuum. All electromagnetic waves propagate in a vacuum at the speed of light $c$, regardless of their frequency or direction of propagation. The wavefronts of a plane wave are planes in space on which the solution to the wave equation has the same value at any given time. Mathematically, this means that the phase of eq. (13.16) or eq. (13.17) must be constant; thus, the following must hold:

$$\boldsymbol{k}\cdot\boldsymbol{r} - \omega t = \varphi_0 = const \tag{13.20}$$

We rewrite this using $\boldsymbol{k} = \hat{\boldsymbol{n}}_k k$, where $\hat{\boldsymbol{n}}_k$ is the unit vector in the direction of $\boldsymbol{k}$, and, with the help of eq. (13.18), obtain:

$$\hat{\boldsymbol{n}}_k\cdot\boldsymbol{r} = \frac{1}{k}(\varphi_0 + \omega t) = \frac{\varphi_0}{k} + ct \tag{13.21}$$

Eq. (13.21) represents the equation of a plane at any given time $t$. As sketched in fig. 13.1, these planes constitute the wavefronts of the electromagnetic wave propagating at the speed of light $c$ in the direction of $\boldsymbol{k}$. The distance between two wavefronts of the propagating plane wave, corresponding to the distance between adjacent wave crests or adjacent wave troughs of the electromagnetic wave, is exactly equal to its wavelength $\lambda$. The boundary conditions that the solutions to eq. (13.4) and eq. (13.5) must satisfy are determined by Maxwell's equations. To see what this entails, we must determine the divergence and curl of the proposed solutions for $\boldsymbol{E}$ and $\boldsymbol{B}$ from eq. (13.16) and eq. (13.17), respectively, and substitute them into the Maxwell equations. This reveals that $\boldsymbol{E}$ and $\boldsymbol{B}$ must satisfy the following equations:

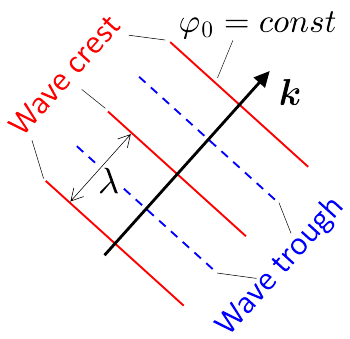



**Fig. 13.1**

$$\nabla\boldsymbol{E} = i\boldsymbol{k}\cdot\boldsymbol{E} \qquad \text{and} \qquad \nabla\boldsymbol{B} = i\boldsymbol{k}\cdot\boldsymbol{B} \tag{13.22}$$

as well as

$$\nabla\times\boldsymbol{E} = i\boldsymbol{k}\times\boldsymbol{E} \qquad \text{and} \qquad \nabla\times\boldsymbol{B} = i\boldsymbol{k}\times\boldsymbol{B} \tag{13.23}$$

From eq. (13.22) and the two Maxwell equations eq. (13.1a) and eq. (13.1b), it follows immediately that the following must hold:

$$\boldsymbol{k}\cdot\boldsymbol{E} = 0 \qquad \text{and} \qquad \boldsymbol{k}\cdot\boldsymbol{B} = 0 \tag{13.24}$$

Both field vectors $\boldsymbol{E}$ and $\boldsymbol{B}$ must therefore be perpendicular to the direction of propagation $\boldsymbol{k}$, meaning they lie in the plane of the propagating wavefront. The relative orientation of $\boldsymbol{E}$ and $\boldsymbol{B}$ within this plane follows from eq. (13.23) and the two Maxwell equations eq. (13.1c) and eq. (13.1d); according to these, the following must hold:

$$\boldsymbol{k}\times\boldsymbol{E} = \omega\boldsymbol{B} \tag{13.25a}$$

$$\boldsymbol{k}\times\boldsymbol{B} = -\frac{\omega}{c^2}\boldsymbol{E} \tag{13.25b}$$

Thus, the field vectors $\boldsymbol{E}$ and $\boldsymbol{B}$ are not only perpendicular to the direction of propagation $\boldsymbol{k}$ of the plane wave but also perpendicular to each other. The combination of the properties $\boldsymbol{E}\perp\boldsymbol{k}$ and $\boldsymbol{B}\perp\boldsymbol{k}$ means that a plane electromagnetic wave is a transverse wave. In addition to this transversality, electromagnetic waves exhibit another important property: phase alignment. To see this, we can consider either eq. (13.25a) or eq. (13.25b), since these two equations can be transformed into one another. To demonstrate this, we substitute the expression for $\boldsymbol{B}$ from eq. (13.25a) into the left-hand side of eq. (13.25b) and using $\omega = kc$ as well as $\boldsymbol{k}\cdot\boldsymbol{E} = 0$, we obtain:

$$\begin{aligned}\boldsymbol{k}\times\boldsymbol{B} &= \frac{1}{\omega}\big[\boldsymbol{k}\times(\boldsymbol{k}\times\boldsymbol{E})\big] \\ &= \frac{1}{\omega}\big[\boldsymbol{k}(\boldsymbol{k}\cdot\boldsymbol{E}) - \boldsymbol{E}(\boldsymbol{k}\cdot\boldsymbol{k})\big] = -\frac{k^2}{\omega}\boldsymbol{E} = -\frac{\omega}{c^2}\boldsymbol{E}\end{aligned}$$

Conversely, if we substitute the expression for $\boldsymbol{E}$ from eq. (13.25b) into the left-hand side of eq. (13.25a), we obtain, with $\omega = kc$ and $\boldsymbol{k}\cdot\boldsymbol{B} = 0$:

$$\begin{aligned}\boldsymbol{k}\times\boldsymbol{E} &= -\frac{c^2}{\omega}\big[\boldsymbol{k}\times(\boldsymbol{k}\times\boldsymbol{B})\big] \\ &= -\frac{c^2}{\omega}\big[\boldsymbol{k}(\boldsymbol{k}\cdot\boldsymbol{B}) - \boldsymbol{B}(\boldsymbol{k}\cdot\boldsymbol{k})\big] = \frac{c^2k^2}{\omega}\boldsymbol{B} = \omega\boldsymbol{B}\end{aligned}$$

Hence, eq. (13.25a) and eq. (13.25b) can be transformed into one another. With $\boldsymbol{E}\perp\boldsymbol{k}$ and $\boldsymbol{B}\perp\boldsymbol{k}$, it follows from these two equations that:

$$|\boldsymbol{k}|\cdot|\boldsymbol{E}| = \omega|\boldsymbol{B}| \qquad \text{and} \qquad |\boldsymbol{k}|\cdot|\boldsymbol{B}| = \frac{\omega}{c^2}|\boldsymbol{E}| \tag{13.26}$$

If we now substitute $\omega = kc$ from eq. (13.18) here, then each of these two equations yields the relationship:

$$|\boldsymbol{E}| = c\cdot|\boldsymbol{B}| \tag{13.27}$$

Except for the factor $c$, $\boldsymbol{E}$ and $\boldsymbol{B}$ have the same magnitude at every point in time and space. In other words: $\boldsymbol{E}$ and $\boldsymbol{B}$ are in phase, meaning they reach their maxima and minima at the same time at every location. Now that we know the solutions to the wave equations eq. (13.4) and eq. (13.5), we can examine one of them. To do this, we consider a monochromatic plane wave propagating in the positive $x$-direction, i.e., $\boldsymbol{k} = k\hat{\boldsymbol{\imath}}$. In this case, the electric field $\boldsymbol{E}$ is given by:

$$\boldsymbol{E} \;=\; \mathrm{Re}\Big[\big(E_{0y}\cdot\hat{\boldsymbol{\jmath}} + E_{0z}\cdot\hat{\boldsymbol{k}}\big)\mathrm{e}^{i(kx-\omega t)}\Big] \tag{13.28}$$

and with eq. (13.25a) and eq. (13.27), we can write for the magnetic field:

$$c\boldsymbol{B} \;=\; \mathrm{Re}\Big[\big(-E_{0z}\cdot\hat{\boldsymbol{\jmath}} + E_{0y}\cdot\hat{\boldsymbol{k}}\big)\mathrm{e}^{i(kx-\omega t)}\Big] \tag{13.29}$$

From eq. (13.28) and eq. (13.29), we see that the plane wave is determined solely by the electric field. However, we could just as well have written eq. (13.28) and eq. (13.29) in such a way that the wave is determined solely by the magnetic field. For the complex amplitudes $E_{0y}$ and $E_{0z}$, we make the following ansatz:

$$E_{0y} \;=\; |E_{0y}|\cdot\mathrm{e}^{i\varphi} \qquad \text{and} \qquad E_{0z} \;=\; |E_{0z}|\cdot\mathrm{e}^{i(\varphi+\delta)} \tag{13.30}$$

Substituting into eq. (13.28) and eq. (13.29), we thus obtain for the electric field:

$$\boldsymbol{E} \;=\; |E_{0y}|\cos(kx-\omega t+\varphi)\cdot\hat{\boldsymbol{\jmath}} + |E_{0z}|\cos(kx-\omega t+\varphi+\delta)\cdot\hat{\boldsymbol{k}} \tag{13.31a}$$

and for the magnetic field:

$$c\boldsymbol{B} \;=\; -|E_{0z}|\cos(kx-\omega t+\varphi+\delta)\cdot\hat{\boldsymbol{\jmath}} + |E_{0y}|\cos(kx-\omega t+\varphi)\cdot\hat{\boldsymbol{k}} \tag{13.31b}$$

### 13.1.2 Polarized Plane Waves

Depending on the value of the phase shift $\delta$, eq. (13.31a) or eq. (13.31b) describes either a linearly polarized, a circularly polarized, or an elliptically polarized electromagnetic wave. Let us examine these three cases:

**Linearly polarized waves:** $\delta = \pm n\pi$ with $n = 0, 1, 2, \ldots$ applies. Linearly polarized waves consist of two partial waves that oscillate perpendicularly to each other in phase ($n$ even, "+") or out of phase ($n$ odd, "−") and thus eq. (13.31a) becomes:

$$\boldsymbol{E} \;=\; \big[|E_{0y}|\cdot\hat{\boldsymbol{\jmath}} \pm |E_{0z}|\cdot\hat{\boldsymbol{k}}\big]\cos(kx-\omega t+\varphi) \tag{13.32a}$$

The amplitude of this wave is independent of position and time, the electric field vector always lies in the same plane of oscillation. The same applies to the magnetic field vector in eq. (13.31b):

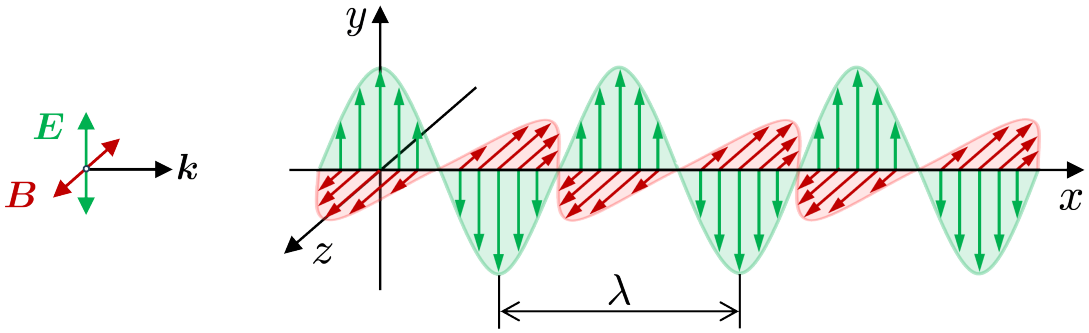



**Fig. 13.2:** Linearly polarized, monochromatic plane wave propagating in the $x$-direction. The ratio $|\boldsymbol{E}| = c|\boldsymbol{B}|$ is not represented to scale here.

$$c\boldsymbol{B} \;=\; \big[\mp|E_{0z}|\cdot\hat{\boldsymbol{j}} + |E_{0y}|\cdot\hat{\boldsymbol{k}}\big]\cos(kx-\omega t+\varphi) \tag{13.32b}$$

The two component waves in eq. (13.32a) and eq. (13.32b) are themselves linearly polarized waves. Any linearly polarized wave can be represented as a superposition of two linearly polarized component waves oscillating perpendicular to each other. Let us select one of these component waves in eq. (13.32a) and eq. (13.32b) by setting $|E_{0z}| = 0$. Then $\boldsymbol{E}$ always lies in the $xy$-plane and $\boldsymbol{B}$ always lies in the $xz$-plane. Fig. 13.2 illustrates the in-phase oscillation of $\boldsymbol{E}$ and $\boldsymbol{B}$ in these planes for the case $\varphi = 0$.

**Circularly polarized waves:** $\delta = \pm(2n+1)\pi/2$ with $n = 0, 1, 2, \ldots$ applies and in addition $|E_{0y}| = |E_{0z}| = E_0$ holds. Circularly polarized waves thus arise from the superposition of two waves of equal amplitude with, for example, a phase shift of $\pm\pi/2$. In this case, eq. (13.31a) becomes:

$$\boldsymbol{E} \;=\; E_0\big[\cos(kx-\omega t+\varphi)\cdot\hat{\boldsymbol{j}} \mp \sin(kx-\omega t+\varphi)\cdot\hat{\boldsymbol{k}}\big] \tag{13.33a}$$

And eq. (13.31b) becomes:

$$c\boldsymbol{B} \;=\; E_0\big[\pm\sin(kx-\omega t+\varphi)\cdot\hat{\boldsymbol{j}} + \cos(kx-\omega t+\varphi)\cdot\hat{\boldsymbol{k}}\big] \tag{13.33b}$$

The upper sign in eq. (13.33a) and eq. (13.33b) applies to $\delta = \pi/2$, and the lower sign applies to $\delta = -\pi/2$. The vector appearing in square brackets in eq. (13.33a) and eq. (13.33b) is the radius vector of the unit circle. At any given fixed location, this radius vector traces out the unit circle over time. The electric field vector $\boldsymbol{E}$ and the magnetic field vector $\boldsymbol{B}$ thus each trace a circle of radius $E_0$ and $E_0/c$, respectively, with angular velocity $\omega$ in the plane perpendicular to the direction of propagation, in this case, the $yz$-plane. Let us first consider $\delta = \pi/2$. When viewing planes parallel to the $yz$-plane from the front, the equations $y = E_0\cos\omega t$ and $z = -E_0\sin\omega t$ describe a clockwise rotation, i.e, a right-handed rotation.

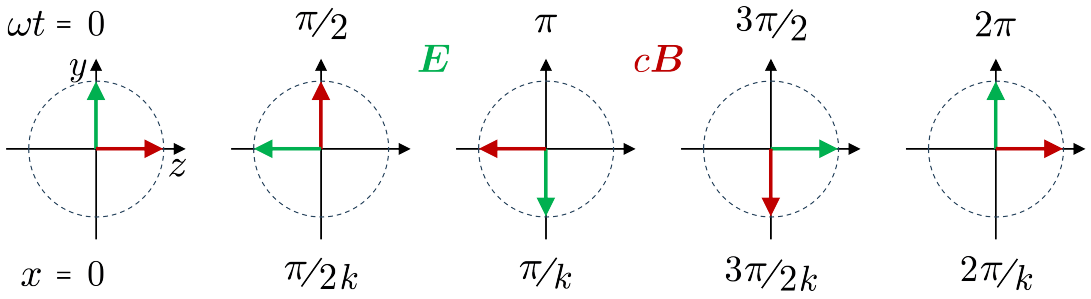



**Fig. 13.3:** View of the plane of oscillation of a left-circularly polarized plane wave, looking along the direction of propagation. Here, the constant phase of the wavefront is set to $\varphi_0 = 0$, and $\varphi = 0$ is chosen in eq. (13.33a) and eq. (13.33b), such that $kx - \omega t = 0$. The field vectors $\boldsymbol{E}$ and $c\boldsymbol{B}$ are perpendicular to each other and trace a full circle as $\omega t$ ranges from 0 to $2\pi$.

When viewing planes parallel to the $yz$-plane from the rear, i.e., looking in the direction of propagation, the vector $(y, z)$ describes a counter-clockwise rotation, or a left-handed rotation. Fig. 13.3 illustrates this counter-clockwise rotation for one circular cycle. For this illustration, $\varphi = 0$ was set in eq. (13.33a) and eq. (13.33b), and $\varphi_0 = 0$ was chosen for eq. (13.20), such that as time progresses, the $x$-coordinate is always given by $x = \omega t/k$. Taking into account the time dependence of $\boldsymbol{E}$, eq. (13.33a) describes a left-handed screw motion. Consequently, in the case of $\delta = \pi/2$, the wave is referred to as a left-circularly polarized wave. The coordinates of $c\boldsymbol{B}$ are $y = E_0 \sin\omega t$ and $z = E_0 \cos\omega t$. This means that, for left-handed rotation, $c\boldsymbol{B}$ lags behind $\boldsymbol{E}$ by a phase of $\omega t = \pi/2$. When viewed in the direction of propagation, for $\delta = -\pi/2$, the vectors $\boldsymbol{E}$ and $c\boldsymbol{B}$ describe a clockwise rotation, i.e., a right-handed rotation or right-handed screw motion. In this case the wave is referred to as a right-circularly polarized wave.

**Elliptically polarized waves:** With the exception of the linearly polarized ($\delta = \pm n\pi$) and the circularly polarized ($\delta = \pm(2n+1)\pi/2$ and $|E_{0y}| = |E_{0z}| = E_0$) waves just discussed, the superposition of two plane waves with any other $\delta$ always results in elliptically polarized waves. Let us look at the example $\delta = \pm\pi/2$ and $|E_{0y}| \neq |E_{0z}|$. In this case, eq. (13.31a) becomes:

$$\boldsymbol{E} = |E_{0y}| \cos(kx - \omega t + \varphi) \cdot \hat{\boldsymbol{j}} \mp |E_{0z}| \sin(kx - \omega t + \varphi) \cdot \hat{\boldsymbol{k}} \tag{13.34a}$$

And eq. (13.31b) becomes:

$$c\boldsymbol{B} = \pm|E_{0z}| \sin(kx - \omega t + \varphi) \cdot \hat{\boldsymbol{j}} + |E_{0y}| \cos(kx - \omega t + \varphi) \cdot \hat{\boldsymbol{k}} \tag{13.34b}$$

With

$$E_y = |E_{0y}| \cos(kx - \omega t + \varphi) \quad \text{and} \quad E_z = \mp|E_{0z}| \sin(kx - \omega t + \varphi)$$

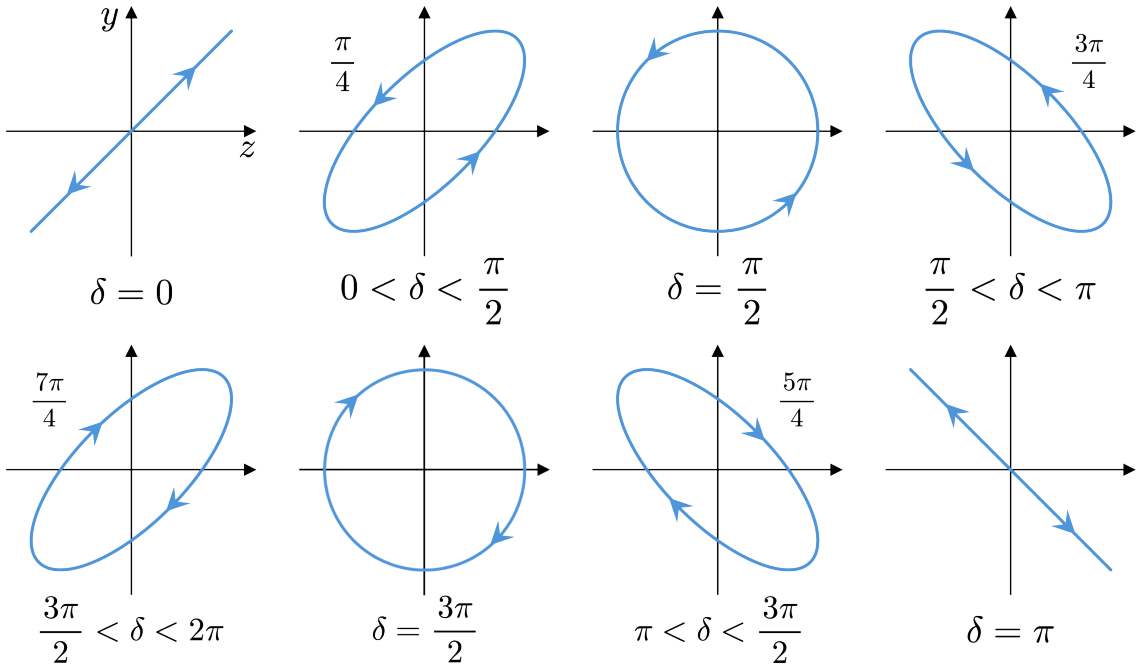



**Fig. 13.4:** Envelope of the motion of the electric/magnetic field vector in the plane of oscillation, depending on the phase difference $\delta$ of the respective orthogonal vector components, for the case where $|E_{0y}| = |E_{0z}| = E_0$. The direction of view into the plane of the figure corresponds to the direction of propagation.

as well as

$$cB_y = \pm|E_{0z}|\sin(kx - \omega t + \varphi) \quad \text{and} \quad cB_z = |E_{0y}|\cos(kx - \omega t + \varphi)$$

the $y$- and $z$-components of these fields each satisfy an ellipse equation

$$\left(\frac{E_y}{|E_{0y}|}\right)^2 + \left(\frac{E_z}{|E_{0z}|}\right)^2 = 1 \quad \text{and} \quad \left(\frac{cB_y}{|E_{0z}|}\right)^2 + \left(\frac{cB_z}{|E_{0y}|}\right)^2 = 1$$

with the principal axes $|E_{0y}|$ and $|E_{0z}|$ or $|E_{0z}|/c$ and $|E_{0y}|/c$, respectively. Thus, for $\delta = \pm\pi/2$ and $|E_{0y}| \neq |E_{0z}|$, the field vectors $\boldsymbol{E}$ and $\boldsymbol{B}$ no longer trace a circular path, but rather an ellipse. Just as with circularly polarized waves, elliptically polarized waves exhibit left-handed rotation ($\delta = \pi/2$) and right-handed rotation ($\delta = -\pi/2$) of the field vectors when viewed in the direction of wave propagation. Accordingly, these are referred to as left-elliptically polarized or right-elliptically polarized waves. That covers the special case where $\delta = \pm\pi/2$. But what is the situation for arbitrary $\delta$? To answer this question, it is best to examine selected graphical representations of eq. (13.31a) and eq. (13.31b).

Fig. 13.4 shows the envelope of the field vector motion of $\boldsymbol{E}$ in eq. (13.31a) and of $c\boldsymbol{B}$ in eq. (13.31b) for arbitrary phase differences $\delta$ between the respective orthogonal vector components. Here, $|E_{0y}| = |E_{0z}| = E_0$ was assumed, such that for $\delta = \pi/2$ and

$\delta = 3\pi/2$, the elliptical polarization becomes circular polarization. For $0 < \delta < \pi/2$ and $\pi/2 < \delta < \pi$, one obtains left-elliptical polarization, as the examples $\delta = \pi/4$ and $\delta = 3\pi/4$ for these ranges demonstrate. And for $\pi < \delta < 3\pi/2$ and $3\pi/2 < \delta < 2\pi$, one obtains right-elliptical polarization, as the examples $\delta = 5\pi/4$ and $\delta = 7\pi/4$ for these ranges illustrate. If $|E_{0y}| \neq |E_{0z}|$, then left- and right-circular polarization at $\delta = \pi/2$ and $\delta = 3\pi/2$ becomes left- and right-elliptical polarization; consequently, the entire range $0 < \delta < \pi$ is left-elliptically polarized, and the entire range $\pi < \delta < 2\pi$ is right-elliptically polarized.

### 13.1.3 Spherical Waves

Spherical waves are spatial waves that propagate uniformly in all directions of space from the point source where such a spherical wave was generated. The wavefronts of such a spherical wave, described by a function $\psi(r,t)$, are concentric spherical shells of constant magnitude and constant phase. The wave equation for a scalar quantity $\psi(\boldsymbol{r},t) = \psi(r,t)$ that depends only on the radial distance $r$ and time $t$, such as the electric potential $\phi(r,t)$ in eq. (13.14), is:

$$\Delta\psi(r,t) - \frac{1}{c^2}\frac{\partial^2\psi(r,t)}{\partial t^2} = 0 \tag{13.35}$$

For convenience, we use the Laplace operator in spherical coordinates $\boldsymbol{r}(r,\vartheta,\varphi)$ here (see tab. A.1 in Appendix A). Since $\psi(r,t)$ depends only on the radial distance $r$ and not on the angles $\vartheta$ and $\varphi$, $\Delta\psi(r,t)$ in spherical coordinates becomes:

$$\Delta\psi(r,t) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi(r,t)}{\partial r}\right) = \frac{\partial^2\psi(r,t)}{\partial r^2} + \frac{2}{r}\frac{\partial\psi(r,t)}{\partial r} \tag{13.36}$$

Now we use the relationship

$$\frac{1}{r}\frac{\mathrm{d}^2}{\mathrm{d}r^2}(r\psi) = \frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(\psi + r\frac{\mathrm{d}\psi}{\mathrm{d}r}\right) = \frac{\mathrm{d}^2\psi}{\mathrm{d}r^2} + \frac{2}{r}\frac{\mathrm{d}\psi}{\mathrm{d}r} \tag{13.37}$$

to replace the right-hand side in eq. (13.36) and thus obtain for $\Delta\psi(r,t)$:

$$\Delta\psi(r,t) = \frac{1}{r}\frac{\partial^2}{\partial r^2}\big[r\psi(r,t)\big] \tag{13.38}$$

If we now substitute this expression into eq. (13.35) and multiply by $r$, the wave equation becomes:

$$\frac{\partial^2}{\partial r^2}\big[r\psi(r,t)\big] - \frac{1}{c^2}\frac{\partial^2\big[r\psi(r,t)\big]}{\partial t^2} = 0 \tag{13.39}$$

With eq. (13.39), we have reduced the three-dimensional wave equation eq. (13.35) for $\psi(r,t)$ to a one-dimensional wave equation for $r\psi(r,t)$. Fortunately, we are already

familiar with the solution to this one-dimensional wave equation from the discussion of wave motion in volume 1. We can therefore directly adopt the solutions $f(x \mp v_x t)$ we found there for the wave traveling in a rope. For that, we simply need to replace the propagation speed of the rope wave, $v_x$, with the speed of light, $c$. Thus, we obtain $r\psi(r,t) = f(r \mp ct)$ as solutions to our wave equation, or rather

$$\psi(r,t) = \frac{f(r \mp ct)}{r} \tag{13.40}$$

In volume 1, $f(x - v_x t)$ was a rope wave traveling to the right, and $f(x + v_x t)$ was one traveling to the left. Here, $f(r - ct)$ is an outgoing spherical wave, which we will denote as $\psi_+(r - ct)$, and $f(r + ct)$ is an incoming spherical wave, for which we write $\psi_-(r + ct)$. The general solution to eq. (13.39) is a linear combination of an outgoing spherical wave and an incoming spherical wave:

$$\psi(r,t) = \frac{1}{r}\big[\psi_+(r - ct) + \psi_-(r + ct)\big] \tag{13.41a}$$

Alternatively, we can also write this with $\omega = kc$ as

$$\psi(r,t) = \frac{1}{r}\big[\psi_+(kr - \omega t) + \psi_-(kr + \omega t)\big] \tag{13.41b}$$

If we substitute the plane-wave solutions already known to us for $\psi_\mp(kr \mp \omega t)$ here, we obtain:

$$\psi(r,t) = \frac{1}{r}\big(A_- \mathrm{e}^{i(kr-\omega t)} + A_+ \mathrm{e}^{i(kr+\omega t)}\big) \tag{13.41c}$$

The surfaces of constant phase of $\psi(r,t)$ are spherical shells, and the amplitude of $\psi(r,t)$ scales with $r^{-1}$. The quantity $\psi(r,t)$ is a scalar function; thus, the considerations just made are directly applicable to the potential $\phi(r,t)$, which, according to eq. (13.14), satisfies such a wave equation. But what about the wave equations for the electric field $\boldsymbol{E}(r,t)$ and the magnetic field $\boldsymbol{B}(r,t)$ in eq. (13.4) and eq. (13.5), as well as the wave equation for the vector potential $\boldsymbol{A}(r,t)$ in eq. (13.15)? As can easily be verified, each of the vector components of these three vector quantities satisfies a wave equation of the type shown in eq. (13.35). Thus, the solutions to the wave equations for the electromagnetic fields are:

$$\boldsymbol{E}(r,t) = \frac{1}{r}\big(\boldsymbol{E}_1 \mathrm{e}^{i(kr-\omega t)} + \boldsymbol{E}_2 \mathrm{e}^{i(kr+\omega t)}\big) \tag{13.42}$$

and

$$\boldsymbol{B}(r,t) = \frac{1}{r}\big(\boldsymbol{B}_1 \mathrm{e}^{i(kr-\omega t)} + \boldsymbol{B}_2 \mathrm{e}^{i(kr+\omega t)}\big) \tag{13.43}$$

So much for the mathematics of the solution to eq. (13.35). Regarding the physics of this solution, we must ask ourselves two questions. First: The amplitude of the outgoing and incoming spherical wave scales as $r^{-1}$, but this becomes problematic at $r = 0$. Second: Spherical waves emanating from a source possess physical reality, a reality of which we encounter numerous examples daily, even if we never truly perceive them consciously. But what about incoming spherical waves? Let us first consider the problem at $r = 0$. This issue arises because we assumed that the source of the spherical waves has no physical extent, i.e., it is an infinitely small point at $r = 0$. However, such sources do not exist in reality. If they did, they would possess an infinitely high energy density. Real sources always have a finite size, however small, even as small as an atom, so the case $r = 0$ never actually occurs. Because the solution to eq. (13.35) consists of a linear combination of outgoing and incoming spherical waves, it is invariant under time reversal. We can swap $t$ with $-t$, and all that happens is that the outgoing spherical wave becomes the incoming one, and vice versa. One of the few examples of incoming spherical waves is the pressure waves generated in a spherical implosion. In such a case, the pressure waves resulting from the explosion of a spherical shell propagate toward the center of the shell, because the shell itself is so heavily confined that the waves can only spread inward toward the center of the exploded shell, rather than outward. Even though such a thing as an incoming spherical wave exists, it often makes no sense from a physical perspective. Consider, for example, a point charge at the origin. If it is set in motion, it generates an outgoing wave. However, simultaneously, an incoming spherical wave would have to be generated at infinity, yet this wave would arrive at the point charge the very moment the outgoing wave began its journey. We do not observe such phenomena. Moving point charges generate only outgoing spherical waves, not incoming ones. If we eliminate the incoming spherical wave from eq. (13.41c), the resulting solution is no longer invariant under time reversal. In volume 1, we encountered a type of physical process in which time-reversal symmetry plays a role: mechanical scattering processes. The situation is no different for quantum mechanical scattering processes, where incoming spherical waves, which represent particles within the framework of wave-particle duality, play a significant role.

If we consider only outgoing spherical waves, then the solutions to the wave equations for the electromagnetic fields $\boldsymbol{E}$ and $\boldsymbol{B}$ are:

$$\boldsymbol{E}(r,t) = \frac{1}{r}\boldsymbol{E}_0 \mathrm{e}^{i(kr-\omega t)} \qquad \text{and} \qquad \boldsymbol{B}(r,t) = \frac{1}{r}\boldsymbol{B}_0 \mathrm{e}^{i(kr-\omega t)} \tag{13.44}$$

The physical electromagnetic wave consists of the respective real parts of the complex waves in eq. (13.44).

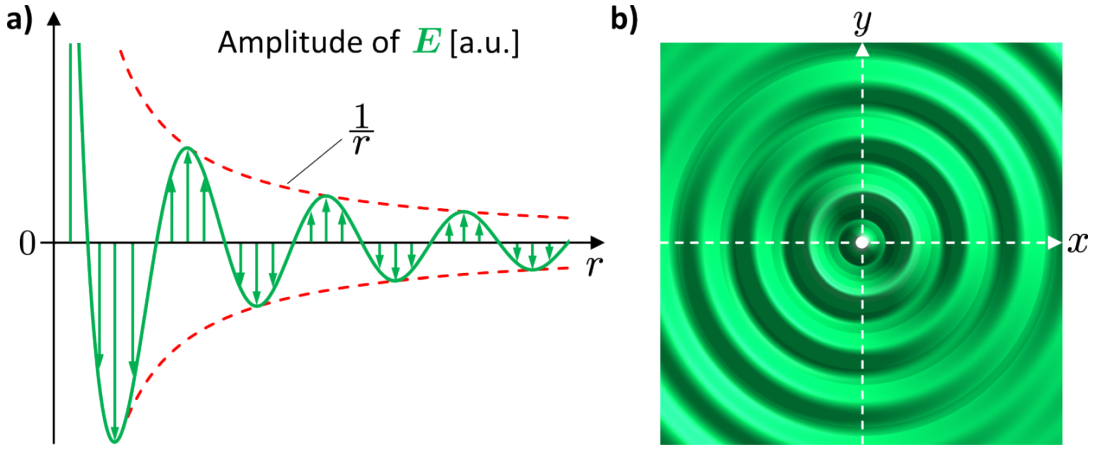



**Fig. 13.5:** Electric field amplitude of a spherical wave as a function of the distance from the wave's origin (a) and contour plot of the outgoing spherical wave in a plane passing through the center of the spherical wave (b).

Fig. 13.5a shows the amplitude of the electric field vector as a function of the distance $r$ from the origin of the spherical wave along an arbitrary coordinate axis. The envelope of this wave decays as $r^{-1}$. Fig. 13.5b shows the contour plot of this outgoing spherical wave in a plane containing the center of the sphere. Having familiarized ourselves with these fields, the question remains as to what type of electromagnetic waves these spherical waves represent. For plane electromagnetic waves, we established that they are transverse electromagnetic waves, i.e., so-called TEM waves. Is this also the case for spherical electromagnetic waves? To this end, we consider the source-free nature of the fields $\boldsymbol{E}$ and $\boldsymbol{B}$, eq. (13.1a) and eq. (13.1b). For a vector field $\boldsymbol{F}$ that, like $\boldsymbol{E}$ and $\boldsymbol{B}$, depends only on the radial distance $r$, the divergence in spherical coordinates is given by (see tab. A.1 in Appendix A):

$$\nabla \boldsymbol{F}(r) \;=\; \frac{1}{r^2}\frac{\partial}{\partial r}\big(r^2 F_r\big) \;=\; \hat{\boldsymbol{e}}_r \cdot \frac{1}{r^2}\frac{\partial}{\partial r}\big(r^2 \boldsymbol{F}\big) \;=\; 0$$

If we now substitute the fields from eq. (13.44) here, we obtain the following boundary conditions:

$$\nabla \boldsymbol{E}(r,t) \;=\; (\boldsymbol{r}\cdot\boldsymbol{E}_0)\frac{(1+ikr)}{r^3}\mathrm{e}^{i(kr-\omega t)} \;=\; 0 \tag{13.45}$$

and

$$\nabla \boldsymbol{B}(r,t) \;=\; (\boldsymbol{r}\cdot\boldsymbol{B}_0)\frac{(1+ikr)}{r^3}\mathrm{e}^{i(kr-\omega t)} \;=\; 0 \tag{13.46}$$

Eq. (13.45) and eq. (13.46) are satisfied for arbitrary $k$ and $r$ only if $\boldsymbol{r}\cdot\boldsymbol{E}_0 = 0$ and $\boldsymbol{r}\cdot\boldsymbol{B}_0 = 0$ hold, and it follows immediately that the following must apply:

$$\boldsymbol{E} \perp \boldsymbol{r} \qquad \text{and} \qquad \boldsymbol{B} \perp \boldsymbol{r} \tag{13.47}$$

Since $\boldsymbol{r}$ is the direction of propagation, meaning $\boldsymbol{r} \parallel \boldsymbol{k}$, eq. (13.47) shows that both $\boldsymbol{E}$ and $\boldsymbol{B}$ must be perpendicular to $\boldsymbol{k}$. Analogous to the approach used for plane waves, one can demonstrate using Maxwell's equations eq. (13.1c) and eq. (13.1d) that the condition $\boldsymbol{E} \perp \boldsymbol{B}$ must also hold. Electromagnetic spherical waves are thus, at least locally, transverse electromagnetic waves, or TEM waves. Why this restriction to local validity? Far from the source, i.e., for very large $r$, the wavefront of a spherical wave increasingly resembles that of a plane wave. Consequently, at a great distance from the source, we can always treat a spherical wave as a TEM wave. However, this does not apply near the source of a spherical wave. There, the fields possess not only a radial component but also angular components, meaning they have longitudinal components. A pure TEM wave exists only when the wavefronts are strictly parallel. For spherical waves, this is the case at a very large distance from the source of the spherical wave, where the radius of curvature of the spherical shells forming the wave fronts is negligible. However, at a short distance from the source of the spherical wave, the curvature of the shell-shaped wavefronts is no longer negligible. To say that a spherical wave is a TEM wave only locally means that it qualifies as a TEM wave only in a region where the radius of curvature of the wavefronts is negligibly small.

### 13.1.4 TE and TM Waves

We consider the superposition of two plane electromagnetic waves that are identical in every respect but do not share the same direction of propagation. We can always orient the coordinate system such that the wave vectors of both plane electromagnetic waves lie within one of the coordinate planes and one of the coordinate axes bisects the angle $2\alpha$ enclosed by the these two wave vectors. In fig. 13.6, we have placed these wave vectors $\boldsymbol{k}_1$ and $\boldsymbol{k}_2$ in the $yz$-plane and selected the $z$-axis as the angle bisector. In the chosen representation, the $x$-axis points out of the plane of the figure. Let us first consider fig. 13.6a. Here, the magnetic field vectors lie perpendicular to the wave vectors within the plane of the figure, and the corresponding electric field vectors therefore point out of the plane of the figure. Fig. 13.6a shows a snapshot, but the field vectors of these plane electromagnetic waves naturally oscillate. In fig. 13.6a, the magnetic field vectors oscillate in the $yz$-plane orthogonal to the respective wave vectors, while the associated electric fields oscillate perpendicular to them along the $x$-coordinate. Since the two waves are identical except for their direction of propagation, meaning they have the same amplitude, frequency, and phase, the observations we make based on the snapshot in fig. 13.6a hold true at any other point in time as well. We can thus write down the fields of the two waves. The $\boldsymbol{E}$-field of the first wave is:

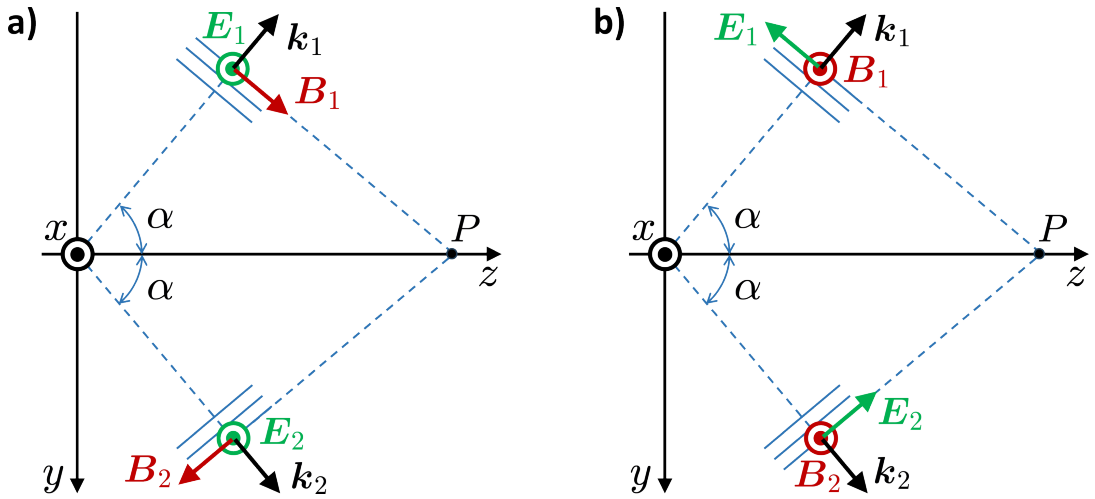



**Fig. 13.6:** Superposition of two identical plane waves with different directions of propagation: (a) the magnetic field vectors lie in the $yz$-plane and the electric field vectors are perpendicular to it; (b) the electric field vectors lie in the $yz$-plane and the magnetic field vectors are perpendicular to it.

$$\boldsymbol{E}_1 = \boldsymbol{E}_0 \cdot \mathrm{e}^{i(\boldsymbol{k}_1 \boldsymbol{r} - \omega t)} = E_{1x} \cdot \hat{\boldsymbol{i}} \qquad \text{with} \qquad E_{1x} = E_0 \cdot \mathrm{e}^{i(\boldsymbol{k}_1 \boldsymbol{r} - \omega t)} \tag{13.48a}$$

The corresponding $\boldsymbol{B}$-field is, with $\boldsymbol{k}_1 = k \cdot \hat{\boldsymbol{e}}_{k_1}$ and $\omega = c \cdot k$, given by eq. (13.25a):

$$\boldsymbol{B}_1 = \frac{1}{\omega} \boldsymbol{k}_1 \times \boldsymbol{E}_1 = \frac{E_{1x}}{c} \left( \hat{\boldsymbol{e}}_{k_1} \times \hat{\boldsymbol{i}} \right) \tag{13.48b}$$

The $\boldsymbol{E}$-field of the second wave is:

$$\boldsymbol{E}_2 = \boldsymbol{E}_0 \cdot \mathrm{e}^{i(\boldsymbol{k}_2 \boldsymbol{r} - \omega t)} = E_{2x} \cdot \hat{\boldsymbol{i}} \qquad \text{with} \qquad E_{2x} = E_0 \cdot \mathrm{e}^{i(\boldsymbol{k}_2 \boldsymbol{r} - \omega t)} \tag{13.49a}$$

The corresponding $\boldsymbol{B}$-field is given by eq. (13.25a), with $\boldsymbol{k}_2 = k \cdot \hat{\boldsymbol{e}}_{k_2}$ and $\omega = c \cdot k$:

$$\boldsymbol{B}_2 = \frac{1}{\omega} \boldsymbol{k}_2 \times \boldsymbol{E}_2 = \frac{E_{2x}}{c} \left( \hat{\boldsymbol{e}}_{k_2} \times \hat{\boldsymbol{i}} \right) \tag{13.49b}$$

The wave vectors $\boldsymbol{k}_1$ and $\boldsymbol{k}_2$ transform into each other upon reflection across the $z$-axis. Decomposed into their $y$- and $z$-components, they are:

$$\boldsymbol{k}_1 = -k_y \cdot \hat{\boldsymbol{j}} + k_z \cdot \hat{\boldsymbol{k}} \qquad \text{and} \qquad \boldsymbol{k}_2 = k_y \cdot \hat{\boldsymbol{j}} + k_z \cdot \hat{\boldsymbol{k}}$$

The cross products in eq. (13.48b) and eq. (13.49b) thus become:

$$\hat{\boldsymbol{e}}_{k_1} \times \hat{\boldsymbol{i}} = \frac{k_z}{k} \cdot \hat{\boldsymbol{j}} + \frac{k_y}{k} \cdot \hat{\boldsymbol{k}} \qquad \text{and} \qquad \hat{\boldsymbol{e}}_{k_2} \times \hat{\boldsymbol{i}} = \frac{k_z}{k} \cdot \hat{\boldsymbol{j}} - \frac{k_y}{k} \cdot \hat{\boldsymbol{k}}$$

With this and the known relationships

$$\cos \beta = \frac{1}{2} \left( \mathrm{e}^{i\beta} + \mathrm{e}^{-i\beta} \right) \qquad \text{and} \qquad \sin \beta = \frac{1}{2i} \left( \mathrm{e}^{i\beta} - \mathrm{e}^{-i\beta} \right)$$

we now have everything needed to write down the superposition of the two plane waves. For the electric field $\boldsymbol{E} = \boldsymbol{E}_1 + \boldsymbol{E}_2$, we thus obtain:

$$\boldsymbol{E} = E_0\left(\mathrm{e}^{-ik_y y} + \mathrm{e}^{ik_y y}\right)\cdot\hat{\boldsymbol{i}}\cdot\mathrm{e}^{i(k_z z-\omega t)} = 2E_0\cos(k_y y)\cdot\hat{\boldsymbol{i}}\cdot\mathrm{e}^{i(k_z z-\omega t)} \tag{13.50a}$$

For the magnetic field $\boldsymbol{B} = \boldsymbol{B}_1 + \boldsymbol{B}_2$, the following applies accordingly:

$$\begin{aligned}\boldsymbol{B} &= \frac{E_0}{ck}\left[\left(\mathrm{e}^{-ik_y y} + \mathrm{e}^{ik_y y}\right)\cdot k_z\cdot\hat{\boldsymbol{j}} + \left(\mathrm{e}^{-ik_y y} - \mathrm{e}^{ik_y y}\right)\cdot k_y\cdot\hat{\boldsymbol{k}}\right]\cdot\mathrm{e}^{i(k_z z-\omega t)} \\ &= \frac{E_0}{ck}\left[2\cos(k_y y)\cdot k_z\cdot\hat{\boldsymbol{j}} - 2i\sin(k_y y)\cdot k_y\cdot\hat{\boldsymbol{k}}\right]\cdot\mathrm{e}^{i(k_z z-\omega t)}\end{aligned} \tag{13.50b}$$

Eq. (13.50a) and eq. (13.50b) describe a plane electric wave and a plane magnetic wave, respectively, both propagating in the positive $z$-direction. The electric field vector oscillates along the $x$-coordinate. Hence, the electric wave in eq. (13.50a) is a purely transverse wave. However, this is not the case for the magnetic wave in eq. (13.50b). In addition to the transverse component in the $y$-direction, the magnetic field vector also has a longitudinal component in the $z$-direction. Consequently, the superposition of the two waves shown in fig. 13.6a does not result in a TEM wave, but rather a TE wave, because the magnetic field contains a longitudinal and a transverse component.

Let us now consider the situation shown in fig. 13.6b. Here, it is not the magnetic field vectors of the two waves that lie in the $yz$-plane, but rather the electric field vectors. Consequently, the magnetic field vectors lie perpendicular to this, along the $x$-direction. Thus, the fields of the two plane waves are described as follows:

$$\boldsymbol{B}_1 = \boldsymbol{B}_0\cdot\mathrm{e}^{i(\boldsymbol{k}_1\boldsymbol{r}-\omega t)} = B_{1x}\cdot\hat{\boldsymbol{i}} \qquad \text{with} \qquad B_{1x} = B_0\cdot\mathrm{e}^{i(\boldsymbol{k}_1\boldsymbol{r}-\omega t)} \tag{13.51a}$$

and with eq. (13.25b)

$$\boldsymbol{E}_1 = -\frac{c^2}{\omega}\,\boldsymbol{k}_1\times\boldsymbol{B}_1 = -cB_{1x}\left(\hat{\boldsymbol{e}}_{k_1}\times\hat{\boldsymbol{i}}\right) \tag{13.51b}$$

as well as

$$\boldsymbol{B}_2 = \boldsymbol{B}_0\cdot\mathrm{e}^{\boldsymbol{k}_2\boldsymbol{r}-\omega t} = B_{2x}\cdot\hat{\boldsymbol{i}} \qquad \text{with} \qquad B_{2x} = B_0\cdot\mathrm{e}^{\boldsymbol{k}_1\boldsymbol{r}-\omega t} \tag{13.52a}$$

and with eq. (13.25b)

$$\boldsymbol{E}_2 = -\frac{c^2}{\omega}\,\boldsymbol{k}_2\times\boldsymbol{B}_2 = -cB_{2x}\left(\hat{\boldsymbol{e}}_{k_2}\times\hat{\boldsymbol{i}}\right) \tag{13.52b}$$

Summing these fields yields the following result for the superposition of the two plane waves shown in fig. 13.6b:

$$\boldsymbol{B} = \boldsymbol{B}_1 + \boldsymbol{B}_2 = 2B_0\cos(k_y y)\cdot\hat{\boldsymbol{i}}\cdot\mathrm{e}^{i(k_z z-\omega t)} \tag{13.53a}$$

and

$$\boldsymbol{E} \;=\; \boldsymbol{E}_1 + \boldsymbol{E}_2 \;=\; -\frac{cB_0}{k}\Big[2\cos(k_y y)\cdot k_z\cdot\hat{\boldsymbol{j}} - 2i\sin(k_y y)\cdot k_y\cdot\hat{\boldsymbol{k}}\Big]\cdot \mathrm{e}^{i(k_z z-\omega t)} \tag{13.53b}$$

As a result of the superposition of the two plane electromagnetic waves in fig. 13.6b, we thus obtain, once again, a plane electric wave and a plane magnetic wave propagating in the positive $z$-direction. Here, we have the magnetic field wave in eq. (13.53a), which is purely transverse in nature, and the electric wave in eq. (13.53b), which exhibits a longitudinal component in addition to its transverse component. In this case, it is referred to as a TM wave.

Let us now consider the points $P$ in fig. 13.6 where the planes of constant phase of the two electromagnetic waves overlap on the $z$-axis. The speed at which this point moves to the right along the $z$-axis depends on the angle $\alpha$. If $\alpha = 0$, both electromagnetic waves move in the same direction, and the point $P$ moves to the right at the phase velocity of the medium in which the waves are propagating. In case that medium is a vacuum, this velocity corresponds to the speed of light. If $\alpha > 0$, it is clear that the point $P$ must move to the right more rapidly the larger $\alpha$ is. And obviously, as $\alpha \to \pi/2$, this phase velocity would have to approach the value of infinity. To understand what this means, we need to examine more closely what the phase velocity of a wave actually refers to. Phase velocity is the speed at which a point of constant phase, such as a wave maximum or minimum, moves within the wave. For a monochromatic plane electromagnetic wave, this phase velocity is equal to the wave's propagation speed. In a vacuum, this corresponds to the speed of light. Thus, for each of the two individual electromagnetic waves shown in fig. 13.6, the following applies:

$$c \;=\; \frac{\omega}{|k|} \;=\; \frac{\omega}{\sqrt{k_y^2 + k_z^2}} \qquad \Rightarrow \qquad k_z \;=\; \sqrt{\frac{\omega^2}{c^2} - k_y^2} \tag{13.54}$$

For the phase velocity of the superposition of these waves, i.e., the speed at which point $P$ in fig. 13.6 moves to the right, the following applies:

$$v_P \;=\; \frac{\omega}{k_z} \;=\; \frac{\omega}{\sqrt{\omega^2/c^2 - k_y^2}} \;=\; \frac{c}{\sqrt{1 - k_y^2 c^2/\omega^2}} \;\geq\; c \tag{13.55}$$

The implication of $\alpha > 0$, i,e., $k_y \neq 0$, is $v_P > c$, i.e., a phase velocity greater than the speed of light. This seemingly contradicts the theory of special relativity. However, this is not the case for a very fundamental reason: the superimposed electromagnetic waves propagating in the positive $z$-direction shown in fig. 13.6 are monochromatic waves with constant amplitude and phase and, as such, do not transmit information. An electromagnetic wave, or for that matter any kind of wave, carries information only if

either its amplitude or its phase exhibits modulation. The simplest example of such modulation is the switching on and off of the wave signal, so-called pulse modulation. Information is embedded in the zeros and ones of such a pulse modulation, or in any other type of wave modulation. For every such modulated wave, the restriction applies that the transmitted signal cannot propagate faster than the speed of light. Energy transmission via a wave is another example of information transmission that cannot occur faster than the speed of light.

## 13.2 Field Energy and Field Momentum

In this section, we turn our attention to the energy and momentum of the electromagnetic field. We are already familiar with the field energies of electric and magnetic fields in their own right. Here, we will see how electromagnetic waves transport energy and that this energy transport always occurs through a combination of electric and magnetic field energy. Furthermore, we will see that, just as with the waves of continuum mechanics discussed in volume 1, a momentum density and a momentum current density are associated with electromagnetic waves.

### 13.2.1 Electromagnetic Energy Density

What energy does an electromagnetic wave transport? Using eq. (3.50) and eq. (10.121), the energy density of the electric and magnetic fields in a vacuum is:

$$w_{el} = \frac{\mathrm{d}U_{el}}{\mathrm{d}V} = \frac{\epsilon_0}{2}\boldsymbol{E}^2 \qquad \text{and} \qquad w_{mag} = \frac{\mathrm{d}U_{mag}}{\mathrm{d}V} = \frac{1}{2\mu_0}\boldsymbol{B}^2$$

The sum of these two energy densities is the electromagnetic energy density $w_{em}$. Let us now consider this energy density for a linearly polarized plane electromagnetic wave propagating in the positive $x$-direction, as described by eq. (13.32a) and eq. (13.32b). We simplify matters by setting $|E_{0z}| = 0$ for the wave polarization in eq. (13.32a) and eq. (13.32b); in addition, we choose $\delta = 0$ for the phase shift and $\varphi = 0$ for the phase. This yields the wave shown in fig. 13.2 for the electric and magnetic fields, a section of which is displayed in

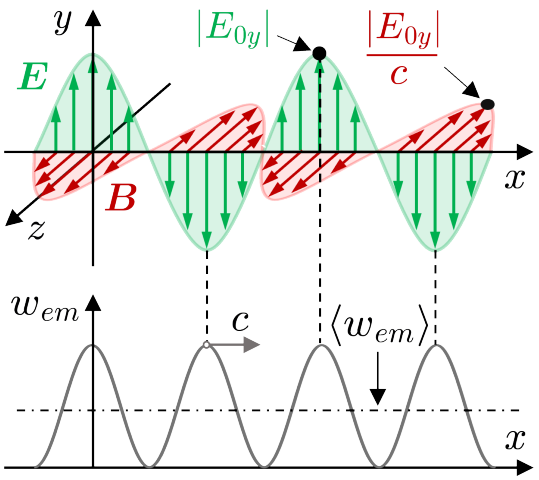



**Fig. 13.7**

the upper part of fig. 13.7. The $\boldsymbol{E}$-field and the $\boldsymbol{B}$-field of this plane electromagnetic wave are given by:

$$\boldsymbol{E} = |E_{0y}| \cos(kx - \omega t) \cdot \hat{\boldsymbol{j}} \qquad \text{and} \qquad c\boldsymbol{B} = |E_{0y}| \cos(kx - \omega t) \cdot \hat{\boldsymbol{k}} \tag{13.56}$$

The energy density $w_{em} = w_{el} + w_{mag}$ of this electromagnetic wave is:

$$w_{em} = \left( \frac{\epsilon_0}{2} E_{0y}^2 + \frac{1}{2\mu_0 c^2} E_{0y}^2 \right) \cos^2(kx - \omega t) \tag{13.57}$$

We can simplify this because with

$$c^2 = \frac{1}{\epsilon_0 \mu_0} \qquad \text{applies} \qquad \frac{1}{2\mu_0 c^2} E_{0y}^2 = \frac{\epsilon_0}{2} E_{0y}^2$$

and therefore, the electric field energy $w_{el}$ of the plane wave and its magnetic field energy $w_{mag}$ are always exactly equal. For $w_{em}$, we thus obtain:

$$w_{em} = \epsilon_0 E_{0y}^2 \cos^2(kx - \omega t) = \epsilon_0 \boldsymbol{E}^2 \tag{13.58}$$

The lower part of fig. 13.7 shows this local variation of $w_{em}$. With

$$\cos^2(kx - \omega t) = \frac{1}{2} + \cos 2(kx - \omega t)$$

this spatial dependence of $w_{em}$ is a cosine function with double the period, oscillating around the mean value 1/2. Thus, for the mean value $\langle w_{em} \rangle$ shown in fig. 13.7, the following applies:

$$\langle w_{em} \rangle = \epsilon_0 \langle \boldsymbol{E}^2 \rangle = \frac{\epsilon_0 E_{0y}^2}{2} \tag{13.59}$$

Every point on the spatial distribution of $w_{em}$ shown in fig. 13.7 moves to the right along this curve at the speed of light $c$. The electromagnetic wave therefore transports energy at the speed of light. If one considers the energy flow through a surface perpendicular to the direction of wave propagation, one finds that an energy flux density $S$ passes through it that satisfies the following condition:

$$S = w_{em} \cdot c = \epsilon_0 \boldsymbol{E}^2 \cdot c \tag{13.60}$$

This energy flux density exhibits the same spatial variation as $\boldsymbol{E}^2$, i.e., it varies according to the square of the cosine, depending on the location along the path of wave propagation where we place the surface through which we observe the energy flow. This raises the question of how energy is transported in a wave. If we place our surface where $|\boldsymbol{E}| = 0$, the corresponding energy flux density is zero according to eq. (13.60) and if we place the surface at a maximum of the wave, it is maximal there. However, this consideration overlooks the fact that the wave is a stationary pattern in which every point within the

wave moves at speed $c$ in the direction of propagation. We must therefore consider the average energy associated with such a point moving within the wave. However, we must not make the mistake of regarding such a point as a material point, for energy transport in an electromagnetic wave occurs without the transport of mass as it takes place via the electromagnetic field. The average energy transported by the plane wave is given by eq. (13.59) as:

$$\langle S\rangle \;=\; \langle w_{em}\rangle\cdot c \;=\; \epsilon_0\langle \boldsymbol{E}^2\rangle\cdot c \;=\; \frac{\epsilon_0 E_{0y}^2}{2}\cdot c \tag{13.61}$$

With eq. (13.56), $|\boldsymbol{E}| = c|\boldsymbol{B}|$ holds, and we can therefore write for $S$ in eq. (13.60):

$$S \;=\; \epsilon_0 c^2|\boldsymbol{E}|\cdot|\boldsymbol{B}| \;=\; \frac{1}{\mu_0}|\boldsymbol{E}|\cdot|\boldsymbol{B}| \tag{13.62}$$

The energy flow occurs in the direction of propagation, i.e., in the direction of the wave vector $\boldsymbol{k}$ of the plane wave. The energy flux is thus a directed quantity $\boldsymbol{S}$, known as the Poynting vector. Since $\boldsymbol{E}$ and $\boldsymbol{B}$ are perpendicular to each other, the expression $|\boldsymbol{E}|\cdot|\boldsymbol{B}|$ is identical to $|\boldsymbol{E}\times\boldsymbol{B}|$. However, according to eq. (13.25a):

$$\boldsymbol{E}\times\boldsymbol{B} \;=\; \frac{1}{\omega}\boldsymbol{E}\times\big(\boldsymbol{k}\times\boldsymbol{E}\big) \;=\; \frac{1}{\omega}\big[\boldsymbol{k}(\boldsymbol{E}\cdot\boldsymbol{E}) - \boldsymbol{E}(\boldsymbol{E}\cdot\boldsymbol{k})\big] \;=\; \frac{\boldsymbol{k}\boldsymbol{E}^2}{\omega}$$

If we now use $|\boldsymbol{E}| = c|\boldsymbol{B}|$ again, it follows that:

$$\boldsymbol{E}\times\boldsymbol{B} \;=\; \frac{\boldsymbol{k}c}{\omega}\cdot(|\boldsymbol{E}|\cdot|\boldsymbol{B}|) = \frac{\boldsymbol{k}}{k}\cdot(|\boldsymbol{E}|\cdot|\boldsymbol{B}|) \;=\; \hat{\boldsymbol{n}}_k\cdot(|\boldsymbol{E}|\cdot|\boldsymbol{B}|)$$

The vector $\hat{\boldsymbol{n}}_k$ is the unit vector in the direction of the wave vector $\boldsymbol{k}$, and thus we can write for the Poynting vector $\boldsymbol{S}$:

$$\boldsymbol{S} \;=\; \frac{1}{\mu_0}\boldsymbol{E}\times\boldsymbol{B} \tag{13.63}$$

Since $\mu_0\boldsymbol{H} = \boldsymbol{B}$ holds in a vacuum, one frequently encounters the notation:

$$\boldsymbol{S} \;=\; \boldsymbol{E}\times\boldsymbol{H} \tag{13.64}$$

Here, we have derived the Poynting vector $\boldsymbol{S}$ for a plane electromagnetic wave. However, eq. (13.63) involves only the electric field $\boldsymbol{E}$ and the magnetic field $\boldsymbol{B}$ and these quantities need not be associated with a plane electromagnetic wave but can be assigned to arbitrary electromagnetic fields. The Poynting vector $\boldsymbol{S}$ describes, in general terms, the transport of energy in electromagnetic fields, for instance, in the case of a resistor $R$ carrying a current, as shown in fig. 13.8. In this situation, the Poynting vector points toward the axis running through the cylindrical

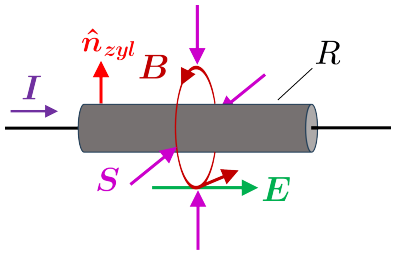



**Fig. 13.8**

resistor. Let us consider the energy flow into this cylindrical resistor, which has length $l$ and radius $r$. The directed area element $\boldsymbol{d\sigma} = \hat{\boldsymbol{n}}_{\text{cyl}}\,\mathrm{d}\sigma$ of the cylinder's lateral surface points outward, i.e., in exactly the opposite direction to $\boldsymbol{S}$. Thus, for the energy flow into the resistor, we obtain:

$$\int_{Zyl} \boldsymbol{S}\cdot\hat{\boldsymbol{n}}_{zyl}\,\mathrm{d}\sigma \;=\; -|\boldsymbol{S}|\cdot 2\pi r\cdot l \;=\; -\frac{1}{\mu_0}(\boldsymbol{E}\cdot\boldsymbol{B})\cdot 2\pi r\cdot l \tag{13.65}$$

For the voltage drop across the resistor and the magnetic field surrounding the current-carrying resistor, the following applies, using eq. (3.35) and eq. (9.11):

$$U \;=\; |\boldsymbol{E}|\cdot l \qquad \text{and} \qquad |\boldsymbol{B}| \;=\; \frac{\mu_0 I}{2\pi r}$$

Substituting this into eq. (13.65), we obtain, using eq. (8.16), the expression:

$$\int_{Zyl} \boldsymbol{S}\cdot\hat{\boldsymbol{n}}_{zyl}\,\mathrm{d}\sigma \;=\; U\cdot I \;=\; P \tag{13.66}$$

The flow of energy through the lateral surface of the cylindrical resistor is exactly equal to the electrical power $P$ dissipated within the resistor. This illustrates that energy transport does not occur within the current-carrying wire itself, but rather via the electromagnetic field. Energy flows from the electromagnetic field into the current-carrying wire, or a resistor $R$, and is dissipated there as heat.

Temporal changes in the density of a quantity are linked to local changes in the flux density of another quantity. This is expressed in continuity equations such as eq. (8.9), which relates the divergence of the electric current density $\boldsymbol{j}$ to the rate of change of charge density within a volume. What, then, is the continuity equation for the energy flux density $\boldsymbol{S}$? To answer this question, we must consider the rate of change of the energy density of the electromagnetic field. Using eq. (3.50) and eq. (10.121), this energy density is:

$$w_{em} \;=\; \frac{\epsilon_0}{2}\boldsymbol{E}^2 + \frac{1}{2\mu_0}\boldsymbol{B}^2 \tag{13.67}$$

The rate of change of this energy density over time is:

$$\frac{\partial w_{em}}{\partial t} \;=\; \epsilon_0 \boldsymbol{E}\frac{\partial \boldsymbol{E}}{\partial t} + \frac{1}{\mu_0}\boldsymbol{B}\frac{\partial \boldsymbol{B}}{\partial t} \tag{13.68}$$

We now use Maxwell's equations eq. (13.1c) and eq. (13.1d) in a vacuum to express the time derivatives of $\boldsymbol{B}$ and $\boldsymbol{E}$ in terms of these, and thereby, with the help of the collection of formulas in tab. 2.1, obtain the following result:

$$\frac{\partial w_{em}}{\partial t} \;=\; \frac{1}{\mu_0}\boldsymbol{E}\big(\nabla\times\boldsymbol{B}\big) - \frac{1}{\mu_0}\boldsymbol{B}\big(\nabla\times\boldsymbol{E}\big) \;=\; -\frac{1}{\mu_0}\nabla\big(\boldsymbol{E}\times\boldsymbol{B}\big) \tag{13.69}$$

Using eq. (13.63), we then obtain the continuity equation for the energy of the electromagnetic field:

$$\frac{\partial w_{em}}{\partial t} + \nabla \boldsymbol{S} \;=\; 0 \tag{13.70}$$

This continuity equation relates only to the conservation of the electromagnetic field energy, not to the conservation of total energy. The latter must also include any electrical charges that interact with the electromagnetic field by absorbing energy from it or transferring energy to it. To account for such interactions between the electromagnetic field and charges, we must use the Ampère-Maxwell law from eq. (11.135) instead of the Maxwell equation eq. (13.1d) to replace the time derivative of $\boldsymbol{E}$ in eq. (13.68). Doing so, we obtain the following equation instead of eq. (13.69):

$$\frac{\partial w_{em}}{\partial t} \;=\; \frac{1}{\mu_0}\boldsymbol{E}\big(\nabla \times \boldsymbol{B}\big) - \boldsymbol{E}\boldsymbol{j} - \frac{1}{\mu_0}\boldsymbol{B}\big(\nabla \times \boldsymbol{E}\big) \tag{13.71}$$

The continuity equation for the conservation of total electromagnetic energy is therefore given by:

$$\frac{\partial w_{em}}{\partial t} + \nabla \boldsymbol{S} \;=\; -\boldsymbol{E}\boldsymbol{j} \tag{13.72}$$

In this continuity equation, also known as «Poynting's theorem», the term $\boldsymbol{E}\boldsymbol{j}$ describes the work done by the electromagnetic field on the existing current density. To see this, we move all terms in eq. (13.72) to the left-hand side and integrate over a volume $V$, thereby obtaining:

$$\frac{\partial}{\partial t}\int_V w_{em}\,\mathrm{d}^3r + \int_V \nabla \boldsymbol{S}\,\mathrm{d}^3r + \int_V \boldsymbol{E}\boldsymbol{j}\,\mathrm{d}^3r \;=\; 0 \tag{13.73}$$

Let us consider these three terms. The integral of the energy density $w_{em}$ is, of course, the field energy $W$ within the volume $V$. Therefore, the first term in eq. (13.73) corresponds to the rate of change of $W$:

$$\frac{\partial}{\partial t}\int_V w_{em}\,\mathrm{d}^3r \;=\; \frac{\partial W}{\partial t} \tag{13.74a}$$

Now consider the second integral in eq. (13.73). By applying Gauss's theorem to this integral, we obtain the energy flux through the closed surface $\partial V$ enclosing the volume $V$, that is:

$$\int_V \nabla \boldsymbol{S}\,\mathrm{d}^3r \;=\; \int_{\partial V} \boldsymbol{S}\hat{\boldsymbol{n}}\,\mathrm{d}^2r \;=\; S_{\partial V} \tag{13.74b}$$

The third integral in eq. (13.73) corresponds to a mechanical work $P_V^{mc}$ performed during the motion of charges within the volume $V$:

$$\int_V \boldsymbol{E}\boldsymbol{j}\,\mathrm{d}^3r \;=\; P_V^{mc} \tag{13.74c}$$

With that, we can now write eq. (13.73) as:

$$\frac{\partial W}{\partial t} + S_{\partial V} + P_V^{mc} = 0 \tag{13.75}$$

Eq. (13.75) describes nothing other than the conservation of energy within the volume $V$. If $S_{\partial V} > 0$, the field energy $W$ in the volume $V$ decreases; the same applies if $P_V^{mc} > 0$. However, if, for example, $S_{\partial V} < 0$, energy flows into the volume $V$ and the field energy $W$ increases. Just as one must account for both electromagnetic field energy and the energy of matter when considering energy conservation in electrodynamics, one must account for both electromagnetic field momentum and the momentum of matter when considering the conservation of momentum. But what is electromagnetic field momentum?

### 13.2.2 Electromagnetic Field Momentum

To determine field momentum associated with an electromagnetic field, we must consider the electromagnetic force density $\boldsymbol{f}_{em}$. We already did something quite similar in volume 1 when considering momentum density and momentum flux density for moving fluids. The electric force density $\boldsymbol{f}_{el}$, the force acting on a charge in an electric field, is given by:

$$\boldsymbol{f}_{el} = \frac{\boldsymbol{F}_{el}}{V} = \frac{q\boldsymbol{E}}{V} = \rho\boldsymbol{E} \tag{13.76}$$

Accordingly, the magnetic force density $\boldsymbol{f}_{mag}$ acting on a charge moving with velocity $\boldsymbol{v}$ in a magnetic field is:

$$\boldsymbol{f}_{mag} = \frac{\boldsymbol{F}_{mag}}{V} = \frac{q}{V}(\boldsymbol{v} \times \boldsymbol{B}) = \rho\boldsymbol{v} \times \boldsymbol{B} = \boldsymbol{j} \times \boldsymbol{B} \tag{13.77}$$

Thus, the electromagnetic force density is $\boldsymbol{f}_{em} = \boldsymbol{f}_{el} + \boldsymbol{f}_{mag}$:

$$\boldsymbol{f}_{em} = \rho\boldsymbol{E} + (\boldsymbol{j} \times \boldsymbol{B}) \tag{13.78}$$

This force density is linked to the electromagnetic field and to the matter present within the field. Let us consider the work per unit time associated with this force density, i.e., its effect on a charge within the field. We obtain this by multiplying eq. (13.78) by the velocity at which the charge in question moves through the electromagnetic field. With $\rho\boldsymbol{v} = \boldsymbol{j}$, this becomes:

$$\boldsymbol{v}\cdot\boldsymbol{f}_{em} = \rho\boldsymbol{v}\cdot\boldsymbol{E} + \boldsymbol{v}\cdot(\boldsymbol{j} \times \boldsymbol{B}) = \boldsymbol{E}\cdot\boldsymbol{j} \tag{13.79}$$

Examining the force density acting on matter in an electromagnetic field thus yields precisely the term in «Poynting's theorem», given in eq. (13.72), that is associated with the work done by the field on the current density within the volume under consideration.

To express eq. (13.78) solely as a function of the fields, we now express the charge density $\rho$ and the current density $\boldsymbol{j}$ in this equation using the first Maxwell equation eq. (3.18) and the Maxwell-Ampère law eq. (11.135), respectively. This yields:

$$\boldsymbol{f}_{em} = \epsilon_0(\nabla \boldsymbol{E})\cdot\boldsymbol{E} + \left(\frac{1}{\mu_0}(\nabla \times \boldsymbol{B}) - \epsilon_0\frac{\partial \boldsymbol{E}}{\partial t}\right) \times \boldsymbol{B} \tag{13.80}$$

Using the third Maxwell equation eq. (13.1c), we can write:

$$\begin{aligned}\frac{\partial}{\partial t}(\boldsymbol{E}\times\boldsymbol{B}) &= \left(\frac{\partial}{\partial t}\boldsymbol{E}\times\boldsymbol{B}\right) + \left(\boldsymbol{E}\times\frac{\partial}{\partial t}\boldsymbol{B}\right)\\ &= \left(\frac{\partial}{\partial t}\boldsymbol{E}\times\boldsymbol{B}\right) - \big[\boldsymbol{E}\times(\nabla\times\boldsymbol{E})\big]\end{aligned} \tag{13.81}$$

Using this result, we can now replace the last term on the right-hand side of eq. (13.80) with:

$$\epsilon_0\frac{\partial}{\partial t}\boldsymbol{E}\times\boldsymbol{B} = \frac{\partial}{\partial t}(\epsilon_0\boldsymbol{E}\times\boldsymbol{B}) + \big[\boldsymbol{E}\times(\nabla\times\epsilon_0\boldsymbol{E})\big] \tag{13.82}$$

If we now add a zero to the right-hand side of eq. (13.80) by adding $(\nabla\boldsymbol{B})\boldsymbol{B}/\mu_0$ and rearrange the equation slightly, eq. (13.80) becomes:

$$\begin{aligned}\boldsymbol{f}_{em} = &-\frac{\partial}{\partial t}(\epsilon_0\boldsymbol{E}\times\boldsymbol{B}) +\\ &+\epsilon_0\Big((\nabla\boldsymbol{E})\cdot\boldsymbol{E} - \big[\boldsymbol{E}\times(\nabla\times\boldsymbol{E})\big]\Big)\\ &+\frac{1}{\mu_0}\Big((\nabla\boldsymbol{B})\boldsymbol{B} - \big[\boldsymbol{B}\times(\nabla\times\boldsymbol{B})\big]\Big)\end{aligned} \tag{13.83}$$

We have thus combined the time derivative on the right-hand side of eq. (13.80) into a single quantity. Since the left-hand side of eq. (13.83) represents the force density of the electromagnetic field, this quantity must be the momentum density of the electromagnetic wave, which we denote by $\boldsymbol{g}$. Based on eq. (13.63), the expression for $\boldsymbol{g}$ is:

$$\boldsymbol{g} = \epsilon_0\boldsymbol{E}\times\boldsymbol{B} = \epsilon_0\mu_0\boldsymbol{S} = \frac{1}{c^2}\boldsymbol{S} \tag{13.84}$$

As can easily be verified, $\boldsymbol{g}$ indeed has the dimension of momentum per unit volume. The momentum density $\boldsymbol{g}$ of the electromagnetic field describes the density of the momentum associated with the electromagnetic field. It tells us how much momentum is present in a specific volume of the electromagnetic field. In the second and third lines on the right-hand side of eq. (13.83), the expressions within the parentheses for the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ are identical. Let us consider this expression for $\boldsymbol{E}$, for example. To do so, we use the vector identity (see Appendix A, eq. (A.18)):

$$\boldsymbol{E}\times(\nabla\times\boldsymbol{E}) \;=\; \frac{1}{2}\nabla(\boldsymbol{E}^2) - (\boldsymbol{E}\cdot\nabla)\boldsymbol{E} \tag{13.85}$$

and thus obtain

$$(\nabla\boldsymbol{E})\cdot\boldsymbol{E} - \boldsymbol{E}\times(\nabla\times\boldsymbol{E}) \;=\; (\nabla\boldsymbol{E})\cdot\boldsymbol{E} - \frac{1}{2}\nabla(\boldsymbol{E}^2) + (\boldsymbol{E}\cdot\nabla)\boldsymbol{E} \tag{13.86}$$

With $(x_1, x_2, x_3) = (x, y, z)$ and $\partial_k = \partial/\partial x_k$, the vector components thereof are:

$$\begin{aligned}
\left[(\nabla\boldsymbol{E})\cdot\boldsymbol{E} - \boldsymbol{E}\times(\nabla\times\boldsymbol{E})\right]_i \;&=\; \sum_{k=1}^{3}\left[(\partial_k E_k)E_i - \frac{1}{2}\partial_i\boldsymbol{E}^2 + (E_k\partial_k)E_i\right] \\
&=\; \sum_{k=1}^{3}\left[\partial_k(E_iE_k) - \frac{1}{2}\partial_k\delta_{ik}\boldsymbol{E}^2\right] \\
&=\; \sum_{k=1}^{3}\frac{\partial}{\partial x_k}\left(E_iE_k - \frac{1}{2}\delta_{ik}\boldsymbol{E}^2\right)
\end{aligned} \tag{13.87}$$

For $(\nabla\boldsymbol{B})\boldsymbol{B} - \big[\boldsymbol{B}\times(\nabla\times\boldsymbol{B})\big]$, we obtain the corresponding expression, and thus we can write for the vector components of the electromagnetic force density $\boldsymbol{f}_{em}$:

$$\boldsymbol{f}_{em\,i} \;=\; -\frac{\partial\boldsymbol{g}_i}{\partial t} + \sum_{k=1}^{3}\frac{\partial}{\partial x_k}\left[\epsilon_0 E_iE_k + \frac{1}{\mu_0}B_iB_k - \frac{1}{2}\left(\epsilon_0\boldsymbol{E}^2 + \frac{1}{\mu_0}\boldsymbol{B}^2\right)\delta_{ik}\right] \tag{13.88}$$

Formally, this expression for the electromagnetic force density corresponds exactly to the expression we derived in the discussion of momentum equilibrium in fluids in volume 1, except that here we have momentum density and momentum current density of the electromagnetic field. The tensor

$$T_{ik} \;=\; \epsilon_0 E_iE_k + \frac{1}{\mu_0}B_iB_k - \frac{1}{2}\left(\epsilon_0\boldsymbol{E}^2 + \frac{1}{\mu_0}\boldsymbol{B}^2\right)\delta_{ik} \tag{13.89}$$

a quantity which describes the momentum flux density of an electromagnetic field, is the so-called Maxwell stress tensor $\underline{\underline{\boldsymbol{T}}}$. From eq. (13.89), we see that $T_{ik} = T_{ki}$ holds, and thus $\underline{\underline{\boldsymbol{T}}}$ is a symmetric second-rank tensor. Let us consider the diagonal elements $T_{ii}$ of $\underline{\underline{\boldsymbol{T}}}$. Using eq. (13.67), we find:

$$T_{ii} \;=\; \epsilon_0 E_i^2 + \frac{1}{\mu_0}B_i^2 - w_{em} \tag{13.90}$$

The diagonal elements thus contain the negative of the total electromagnetic energy density plus the electromagnetic energy density associated with the respective coordinate direction. Consequently, the trace of $\underline{\underline{\boldsymbol{T}}}$ vanishes. Consider, for example, a plane wave propagating in the $z$-direction; then $E_z = 0$ and $B_z = 0$, and the normal vector in the direction of propagation $\boldsymbol{k}$ is $\hat{\boldsymbol{n}} = (0, 0, 1)$. For the component of the momentum flux density in the direction of propagation, we thus obtain:

$$\underline{\underline{\boldsymbol{T}}}\cdot\hat{\boldsymbol{n}} = \begin{bmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{bmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} T_{xz} \\ T_{yz} \\ T_{zz} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -w_{em} \end{pmatrix} \tag{13.91}$$

The momentum flux density $\underline{\underline{\boldsymbol{T}}}$ has the dimensions of momentum per unit area per unit time, equivalent to an energy density. While $\boldsymbol{g}$ tells us the amount of momentum present in a specific volume of the electromagnetic field, $\underline{\underline{\boldsymbol{T}}}$ tells us the amount of momentum flowing out of or flowing into that volume per unit area and per unit time. An electromagnetic force density of $\boldsymbol{f}_{em} = 0$ in any given volume tells us that no force is exerted by electric or magnetic fields on charge carriers within that volume. And that, in turn, can mean only one thing: namely, that no such charge carriers are present in the volume under consideration. In this case, the rate of change of the momentum density $\boldsymbol{g}$ corresponds to the negative gradient of the momentum flux density $\underline{\underline{\boldsymbol{T}}}$. For $\boldsymbol{f}_{em} = 0$, eq. (13.88) reduces to the continuity equation for the momentum of the electromagnetic field in a vacuum:

$$-\frac{\partial \boldsymbol{g}_i}{\partial t} + \sum_{k=1}^{3} \frac{\partial}{\partial x_k} T_{ik} = 0 \qquad \text{or} \qquad (\nabla \underline{\underline{\boldsymbol{T}}})_i - \frac{\partial \boldsymbol{g}_i}{\partial t} = 0 \tag{13.92}$$

In three dimensions, the following applies accordingly:

$$\nabla \underline{\underline{\boldsymbol{T}}} - \frac{\partial \boldsymbol{g}}{\partial t} = 0 \tag{13.93}$$

This equation tells us that an electromagnetic wave carries momentum and that this momentum is a conserved quantity, just as in mechanics. In eq. (13.93), $\nabla \underline{\underline{T}}$ describes a source or sink of momentum flux within the volume under consideration, and $\partial \boldsymbol{g}/\partial t$ describes the change in momentum density within that volume associated with this source or sink. Let us take light as an example. If we switch from the wave model to the particle model, it is, naturally, the massless photons, which we already know from volume 1, that carry the momentum of this electromagnetic wave. When light strikes a surface, it exerts pressure on it through the transfer of momentum. For instance, in the case of the plane electromagnetic wave for which we calculated the momentum flux density in eq. (13.91), to determine the radiation pressure it exerts on a surface normal to its direction of propagation, we simply need to multiply eq. (13.91) by the normal vector of that surface. However, the normal vector is simply $-\hat{\boldsymbol{n}}$. Thus, for the radiation pressure on a surface normal to this incident plane wave, we obtain:

$$p = -\hat{\boldsymbol{n}}\cdot\underline{\underline{\boldsymbol{T}}}\cdot\hat{\boldsymbol{n}} = w_{em} \tag{13.94}$$

# 14. Electromagnetic Waves II

In the previous chapter, we encountered solutions to Maxwell's equations in a vacuum that possess a reality of their own even in the absence of charges or currents: electromagnetic waves. Prior to the formulation of Maxwell's equations, electromagnetic waves were entirely unknown, and it took another two decades before the electromagnetic waves postulated by these equations were successfully demonstrated. In this chapter, we turn our attention to aspects of electromagnetic wave propagation in matter and how electromagnetic waves are generated in the first place.

## 14.1 Electromagnetic Waves in Matter

Here, we restrict our consideration of wave propagation in matter to isotropic and homogeneous materials. The physical properties of the medium are thus uniform throughout, regardless of the spatial direction. In particular, the electric permittivity and magnetic permeability of such a material are constant:

$$\epsilon = \epsilon_0 \epsilon_r \qquad \text{and} \qquad \mu = \mu_0 \mu_r \tag{14.1}$$

For the ratio of the speed of light in matter $c_M$ to the speed of light in a vacuum $c$, the following applies:

$$c_M = \frac{1}{\sqrt{\epsilon\mu}} = \frac{1}{\sqrt{\epsilon_0\mu_0}} \cdot \frac{1}{\sqrt{\epsilon_r\mu_r}} = c \cdot \frac{1}{n} \tag{14.2}$$

This equation defines the refractive index $n$ of a material. In a vacuum, $n = 1$, in the vast majority of materials, $n > 1$, and only in rare cases does $n < 1$ occur. A value of $n < 1$ implies $c_M > c$. However, as we saw in the discussion regarding eq. (13.55), provided no information is transmitted, a phase velocity greater than $c$ does not contradict the principle that nothing can propagate faster than the speed of light $c$. The materials considered here must not possess any free charge carriers or exhibit any magnetization and if any magnetization is present, it must be negligibly small. We therefore require

$\nabla \boldsymbol{D} = 0$, $\boldsymbol{M} = 0$ (or $\boldsymbol{M}$ negligibly small), and consequently $\boldsymbol{j} = 0$. The first three Maxwell equations in such a material are identical to those in a vacuum, and in the fourth Maxwell equation, we simply need to replace $\epsilon_0\mu_0$ with $\epsilon\mu$. In summary, the following Maxwell equations apply here:

$$\nabla \boldsymbol{E} = 0 \tag{14.3a}$$

$$\nabla \boldsymbol{B} = 0 \tag{14.3b}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{14.3c}$$

$$\nabla \times \boldsymbol{B} = \epsilon\mu \frac{\partial \boldsymbol{E}}{\partial t} \tag{14.3d}$$

Since these Maxwell equations in matter differ from those in a vacuum (see eq. (13.1a) to eq. (13.1d)) only by the replacement of $\epsilon_0\mu_0$ with $\epsilon\mu$, we can directly apply the entire discussion regarding wave equations in a vacuum from the previous chapter. We do this simply by replacing $c$ with $c_M$ in the wave equations and their corresponding solutions. As the solution to the wave equations in these materials, we again consider monochromatic plane waves:

$$\boldsymbol{E} = \boldsymbol{E_0} \mathrm{e}^{i(\boldsymbol{kr}-\omega t)} \qquad \text{and} \qquad \boldsymbol{B} = \boldsymbol{B_0} \mathrm{e}^{i(\boldsymbol{kr}-\omega t)} \tag{14.4}$$

For the relationship between $\omega$ and $k$, it follows from the wave equations in the medium:

$$\omega = c_M \cdot |\boldsymbol{k}| \qquad \text{or} \qquad |\boldsymbol{k}| = \frac{\omega}{c} \cdot n \tag{14.5}$$

Instead of eq. (13.59), we obtain the following for the average energy of an electromagnetic wave in matter:

$$\langle w_{em} \rangle = \epsilon \langle \boldsymbol{E}^2 \rangle = \frac{\epsilon E_0^2}{2} = \frac{E_0^2}{2\mu} \cdot \frac{n^2}{c^2} \tag{14.6}$$

It follows for the energy flux density $\boldsymbol{S}$, the momentum density $\boldsymbol{g}$, and the momentum flux density $\underline{\underline{\boldsymbol{T}}}$:

$$\boldsymbol{S} = \frac{1}{\mu} \boldsymbol{E} \times \boldsymbol{B} \qquad \text{or} \qquad \boldsymbol{S} = \frac{E_0^2}{2\mu} \cdot \frac{n}{c} \cdot \frac{\boldsymbol{k}}{|\boldsymbol{k}|} \tag{14.7a}$$

$$\boldsymbol{g} = \epsilon \boldsymbol{E} \times \boldsymbol{B} = \frac{n^2}{c^2} \boldsymbol{S} \tag{14.7b}$$

$$T_{ik} = \epsilon E_i E_k + \frac{1}{\mu} B_i B_k - \frac{1}{2} \left( \epsilon \boldsymbol{E}^2 + \frac{1}{\mu} \boldsymbol{B}^2 \right) \delta_{ik} \tag{14.7c}$$

When discussing the properties of dielectric matter in section 7.2, we already found that quantities such as the polarizability of gases, and consequently their dielectric response

function to an applied external electric field, are frequency-dependent. In low-frequency fields, the dielectric polarization $\boldsymbol{P}$ can follow an alternating electric field $\boldsymbol{E}(\omega)$ almost instantaneously. However, this is no longer the case as the field frequency increases. In the chapter on magnetism in matter, we did not explicitly discuss what the frequency-dependent response of a body possessing magnetic moments to an external alternating magnetic field looks like. However, it is evident that, for instance, the reorientation of magnetic moments and even more so the displacement of Weiss domain walls in a ferromagnetic material cannot follow an alternating magnetic field $\mathbf{B}(\omega)$ arbitrarily fast. This material-specific frequency dependence of the response function to alternating electromagnetic fields is expressed in the frequency dependence of the material parameters $\epsilon$ and $\mu$. Instead of

$$\boldsymbol{D} = \epsilon \boldsymbol{E} \qquad \text{and} \qquad \boldsymbol{B} = \mu \boldsymbol{H} \tag{14.8}$$

we must write in general terms:

$$\boldsymbol{D} = \epsilon(\omega) \boldsymbol{E} \qquad \text{and} \qquad \boldsymbol{B} = \mu(\omega) \boldsymbol{H} \tag{14.9}$$

A frequency-dependent $\epsilon$ and $\mu$ also imply, via eq. (14.2), that the refractive index must be frequency-dependent as well, i.e., $n = n(\omega)$. For a wave packet containing various frequencies, this results in the packet spreading out over time, as component waves of different frequencies propagate at different speeds. Here, however, we are interested in what happens when a monochromatic electromagnetic plane wave strikes an interface between two media.

### 14.1.1 Reflection and Refraction of Light

The behavior of the field vectors $\boldsymbol{E}$ and $\boldsymbol{D}$ at dielectric interfaces in the electrostatic case is already familiar to us from the discussion in section 6.1.2. We also examined the behavior of the field vectors $\boldsymbol{B}$ and $\boldsymbol{H}$ at magnetic interfaces in the magnetostatic case in section 12.2.1. Neither in section 6.1.2 nor in section 12.2.1 did we have to consider electric or magnetic field components reflected at the interface. In these cases, we were dealing only with refraction and not with the reflection of field vectors. The boundary conditions for static electric and magnetic fields discussed in section 6.1.2 and section 12.2.1 remain valid, as the time derivatives in Maxwell's equations do not affect them. Thus, for an electromagnetic wave incident on an interface between two media, the following must hold at the interface:

$$\boldsymbol{E}_{\parallel}, \boldsymbol{D}_{\perp}, \boldsymbol{H}_{\parallel}, \boldsymbol{B}_{\perp} \qquad \text{all continuous}$$

To this end, let us consider the interface between two media ① and ② with material parameters $\epsilon_1$ and $\mu_1$ as well as $\epsilon_2$ and $\mu_2$, as sketched in fig. 14.1. Let $\epsilon_1\mu_1 < \epsilon_2\mu_2$ hold, thus, the propagation speed of an electromagnetic wave in medium ② is lower than in medium ①, i.e., $c_2 < c_1$. Now consider an incident electromagnetic wave whose electric field vector $\boldsymbol{E}_i$ oscillates parallel to the interface, and whose magnetic field vector $\boldsymbol{B}_i$ and wave vector $\boldsymbol{k}_i$ lie in a plane normal to the interface, specifically, the $xy$-plane, which we designate as the $k$-plane. With this specification, $\boldsymbol{E}_i$ oscillates in the $z$-direction. From the discussion of the behavior of electrostatic and magnetostatic fields at interfaces, we know that for $c_2 < c_1$, as shown in fig. 14.1, refraction towards the normal must occur. Therefore, the angle $\theta_t$ that the transmitted wave makes with the normal to the surface must be smaller than $\theta_i$, the angle that the incident wave makes with it. However, we now know that for the ratio of $|\boldsymbol{E}|$ to $|\boldsymbol{B}|$, the following must hold for both the incident and the transmitted wave:

$$\frac{|\boldsymbol{E}_i|}{|\boldsymbol{B}_i|} = c_1 \qquad \text{and} \qquad \frac{|\boldsymbol{E}_t|}{|\boldsymbol{B}_t|} = c_2 \tag{14.10}$$

Since $\boldsymbol{E} = \boldsymbol{E}_\parallel$ holds for the wave in fig. 14.1, we can write the boundary condition $\boldsymbol{E}_{t\parallel} = \boldsymbol{E}_{i\parallel}$ for the tangential component of $\boldsymbol{E}$ at the interface as:

$$c_1|\boldsymbol{B}_i| = c_2|\boldsymbol{B}_t| \qquad \text{or} \qquad \frac{c_1}{c_2} = \sqrt{\frac{\boldsymbol{B}_{t\perp}^2 + \boldsymbol{B}_{t\parallel}^2}{\boldsymbol{B}_{i\perp}^2 + \boldsymbol{B}_{i\parallel}^2}} \tag{14.11}$$

With the boundary conditions for the magnetic field at the interface, $\boldsymbol{B}_{t\perp} = \boldsymbol{B}_{i\perp}$ and $\boldsymbol{B}_{t\parallel} = \boldsymbol{B}_{i\parallel} \cdot \mu_2/\mu_1$, this condition becomes:

$$\frac{c_1}{c_2} = \sqrt{\frac{\epsilon_2\mu_2}{\epsilon_1\mu_1}} = \sqrt{\frac{\mu_1^2\boldsymbol{B}_{i\perp}^2 + \mu_2^2\boldsymbol{B}_{i\parallel}^2}{\mu_1^2\boldsymbol{B}_{i\perp}^2 + \mu_1^2\boldsymbol{B}_{i\parallel}^2}} \tag{14.12}$$

Let us now consider $\theta_i \to 0$, the limiting case of a wave approaching the interface perpendicularly. With $\theta_i \to 0$, $\boldsymbol{B}_{i\perp} \to 0$ also holds, and the equation leads to a contradiction because:

$$\lim_{\theta_i \to 0} \sqrt{\frac{\mu_1^2\boldsymbol{B}_{i\perp}^2 + \mu_2^2\boldsymbol{B}_{i\parallel}^2}{\mu_1^2\boldsymbol{B}_{i\perp}^2 + \mu_1^2\boldsymbol{B}_{i\parallel}^2}} = \frac{\mu_2}{\mu_1} \neq \sqrt{\frac{\epsilon_2\mu_2}{\epsilon_1\mu_1}} \tag{14.13}$$

This contradiction shows that the boundary condition for the electromagnetic field of the plane wave at the interface $A$ cannot be satisfied by an incident and a transmitted wave component alone. This is only possible if one assumes the existence of a third wave, a reflected electromagnetic wave $\boldsymbol{E}_r$. The boundary conditions for the parallel components of the fields $\boldsymbol{E}$ and $\boldsymbol{H}$ at the interface $A$, which has the normal vector $\hat{\boldsymbol{n}}_A$, are therefore as follows (see section 6.1.2 and section 12.2.1):

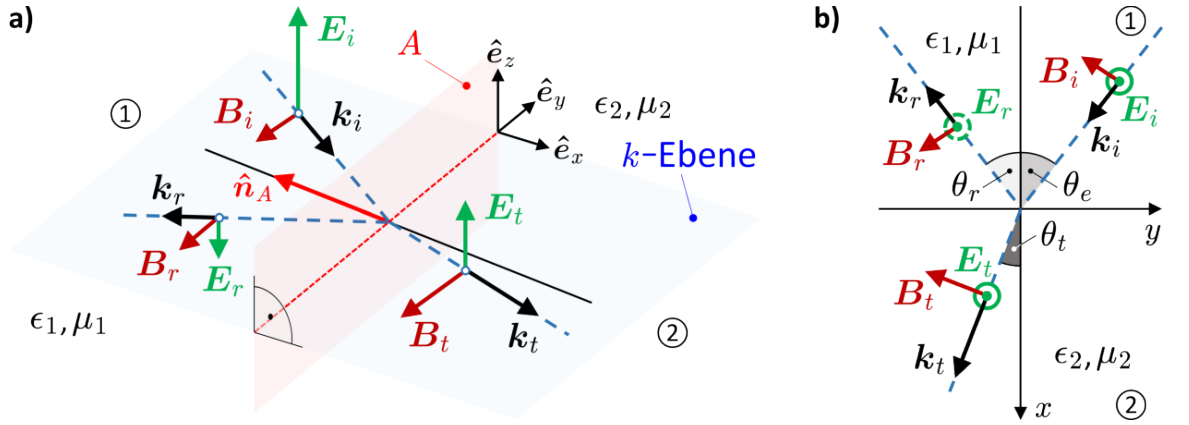



**Fig. 14.1:** Reflection and refraction of a plane electromagnetic wave at an interface $A$ between two media. The plane of incidence ($k$-plane) is perpendicular to the interface $A$ (a), and for $\theta_i$, the angle of incidence, and $\theta_r$, the angle of reflection, $\theta_r = \theta_i$ holds (b).

$$\hat{\boldsymbol{n}}_A \times (\boldsymbol{E}_i + \boldsymbol{E}_r - \boldsymbol{E}_t) = 0 \qquad \text{and} \qquad \hat{\boldsymbol{n}}_A \times (\boldsymbol{H}_i + \boldsymbol{H}_r - \boldsymbol{H}_t) = 0 \tag{14.14}$$

The boundary conditions for the normal components of the fields $\boldsymbol{D}$ and $\boldsymbol{B}$ at the interface $A$ are accordingly:

$$\hat{\boldsymbol{n}}_A \cdot (\boldsymbol{D}_i + \boldsymbol{D}_r - \boldsymbol{D}_t) = 0 \qquad \text{and} \qquad \hat{\boldsymbol{n}}_A \cdot (\boldsymbol{B}_i + \boldsymbol{B}_r - \boldsymbol{B}_t) = 0 \tag{14.15}$$

These boundary conditions for the electromagnetic field components at the interface can only be satisfied if all three partial waves have the same phase at the interface. Such phase matching of the three partial waves is a consequence of the implicit assumption of an ideal interface. Such an ideal interface possesses the same physical properties everywhere, is perfectly flat, and exhibits no roughness whatsoever. Therefore, the point of incidence of the electromagnetic wave on such an ideal interface must not matter. Furthermore, the phases of the incident, transmitted, and reflected waves must match at every point of such a perfectly flat interface $A$ and at every instant in time in order to satisfy this boundary condition of translational invariance at the interface $A$. However, this is only the case if all three component waves exhibit the same time dependence, i.e., possess the same angular frequency $\omega$, and if, at any given time, the following holds:

$$(\boldsymbol{k}_i \cdot \boldsymbol{r})_{\text{in}\,A} = (\boldsymbol{k}_r \cdot \boldsymbol{r})_{\text{in}\,A} = (\boldsymbol{k}_t \cdot \boldsymbol{r})_{\text{in}\,A} \tag{14.16}$$

For convenience, we have placed the coordinate origin at the interface $A$ in fig. 14.1, which is always permissible. The boundary condition of equal phase in the plane $A$ is therefore equivalent to the requirement that the wave vector components parallel to the interface $A$ must be identical at the interface. Thus, the following must hold:

$$\boldsymbol{k}_{i\parallel} = \boldsymbol{k}_{r\parallel} = \boldsymbol{k}_{t\parallel} \tag{14.17}$$

It follows immediately from this that the three wave vectors $\boldsymbol{k}_i$, $\boldsymbol{k}_r$, and $\boldsymbol{k}_t$ lie in a plane, designated as the $k$-plane in fig. 14.1. We see that the assumption of an ideal interface implies translational invariance, and from this follows the coplanarity of the wave vectors. Since the three component waves have the same angular frequency, the relation $\omega = c_M k$ implies:

$$c_1 k_i \;=\; c_1 k_r \;=\; c_2 k_t \tag{14.18}$$

From this, we first see that $k_r = k_i$, and with $k_{i\parallel} = k_i \sin\theta_i$ and $k_{r\parallel} = k_r \sin\theta_r$, we thus obtain the law of reflection from eq. (14.17):

$$k_i \sin\theta_i \;=\; k_r \sin\theta_r \qquad \Rightarrow \qquad \theta_i \;=\; \theta_r \tag{14.19}$$

Furthermore, from eq. (14.18), with $c_1 k_i = c_2 k_t$, we can see that:

$$\frac{k_i}{k_t} \;=\; \frac{c_2}{c_1} \;=\; \frac{n_1}{n_2} \tag{14.20}$$

With $k_{i\parallel} = k_{t\parallel}$ according to eq. (14.17), as well as $k_{i\parallel} = k_i \sin\theta_i$ and $k_{t\parallel} = k_t \sin\theta_t$, the law of refraction follows:

$$1 \;=\; \frac{k_{i\parallel}}{k_{t\parallel}} \;=\; \frac{k_i \sin\theta_i}{k_t \sin\theta_t} \;=\; \frac{n_1 \sin\theta_e}{n_2 \sin\theta_t} \qquad \Rightarrow \qquad n_1 \sin\theta_i \;=\; n_2 \sin\theta_t \tag{14.21}$$

From $\theta_i = \theta_r$ and $k_{i\parallel} = k_{r\parallel}$, it follows that the normal components of the incident and reflected waves must satisfy $k_{i\perp} = k_{r\perp}$. The magnitudes of the normal and parallel components of the incident and reflected waves are thus equal, implying that $\boldsymbol{k}_{r\perp} = -\boldsymbol{k}_{i\perp}$ must hold.

The considerations regarding the reflection and refraction of plane waves made so far apply to media with arbitrary $\epsilon$ and $\mu$. Now, however, we turn to the special case of electromagnetic waves in the visible range, i.e., light that we can see. Most optical materials in the frequency range of visible light are not ferromagnetic, and their relative magnetic permeability $\mu_r$ is close to 1. For the two media ① and ② in fig. 14.1, we can therefore set $\mu_1 = \mu_2 = \mu_0$. It follows from this, with $\boldsymbol{H} = \mu_0^{-1}\boldsymbol{B}$ and the continuity condition eq. (14.14) for the parallel component of $\boldsymbol{H}$ at the interface $A$, that:

$$\hat{\boldsymbol{n}}_A \times (\boldsymbol{H}_i + \boldsymbol{H}_r - \boldsymbol{H}_t) \;=\; 0 \quad \underset{\mu_{1,2}=\mu_0}{\Rightarrow} \quad \hat{\boldsymbol{n}}_A \times (\boldsymbol{B}_i + \boldsymbol{B}_r - \boldsymbol{B}_t) \;=\; 0 \tag{14.22}$$

This boundary condition for the parallel components of $\boldsymbol{B}$ and the continuity of the normal components of $\boldsymbol{B}$ according to eq. (14.15) can only be satisfied simultaneously if:

$$\boldsymbol{B}_i + \boldsymbol{B}_r - \boldsymbol{B}_t \;=\; 0 \tag{14.23}$$

In the case of $\mu_{1,2} = \mu_0$, the magnetic field vectors therefore add up, and the following applies to their magnitudes:

$$B_i + B_r \;=\; B_t \tag{14.24}$$

If we now express the magnitudes of the magnetic field vectors in terms of those of the electric field vectors using $E = c_M B$, we obtain the expression:

$$\frac{n_1}{c} E_i + \frac{n_1}{c} E_r \;=\; \frac{n_2}{c} E_t \tag{14.25}$$

In our case, the electric field vectors oscillate only in the $\pm z$-direction; thus, they have no $x$- or $y$-components. Consequently, the boundary condition for the parallel components of $\boldsymbol{E}$ at the interface $A$ becomes:

$$\hat{\boldsymbol{n}}_A(\boldsymbol{E}_i + \boldsymbol{E}_r - \boldsymbol{E}_t) \;=\; 0 \quad \Rightarrow \quad \hat{\boldsymbol{e}}_z(\boldsymbol{E}_i + \boldsymbol{E}_r - \boldsymbol{E}_t) \;=\; 0 \tag{14.26}$$

Since $\boldsymbol{E}_i$ and $\boldsymbol{E}_t$ are parallel to $\hat{\boldsymbol{e}}_z$, but $\boldsymbol{E}_r$ is antiparallel to $\hat{\boldsymbol{e}}_z$ it follows:

$$E_i - E_r - E_t \;=\; 0 \tag{14.27}$$

We now eliminate $E_t$ from equations eq. (14.25) and eq. (14.27), solve for $E_r$, and thus obtain:

$$E_r \;=\; E_i \frac{n_2 - n_1}{n_2 + n_1} \tag{14.28}$$

For the reflection coefficient $R$, the fraction of the incident wave's electric field amplitude that is reflected at the interface $A$, we thus obtain:

$$R \;=\; \frac{E_r}{E_i} \;=\; \frac{n_2 - n_1}{n_2 + n_1} \tag{14.29}$$

If we eliminate $E_r$ from equations eq. (14.25) and eq. (14.27) and solve for $E_t$, we obtain the transmission coefficient $T$ for the field amplitude:

$$T \;=\; \frac{E_t}{E_i} \;=\; \frac{2n_1}{n_2 + n_1} \;=\; 1 - R \tag{14.30}$$

So much for the balance of field amplitudes at the interface. For the energy balance, we must consider the energy flux into the interface, as well as the energy flux reflected from and transmitted through it. According to eq. (14.7a), the respective energy flux density in the three component waves is:

$$|\boldsymbol{S}_i| \;=\; \frac{E_i^2}{2} \cdot \frac{n_1}{c} \quad ; \quad |\boldsymbol{S}_r| \;=\; \frac{E_r^2}{2} \cdot \frac{n_1}{c} \quad ; \quad |\boldsymbol{S}_t| \;=\; \frac{E_t^2}{2} \cdot \frac{n_2}{c} \tag{14.31}$$

Using eq. (14.31), the ratio of reflected to incident energy is:

$$r \;=\; \frac{|\boldsymbol{S}_r|}{|\boldsymbol{S}_i|} \;=\; \frac{E_r^2}{E_i^2} \;=\; \left(\frac{n_2 - n_1}{n_2 + n_1}\right)^2 = R^2 \tag{14.32}$$

The ratio of transmitted to incident energy is accordingly:

$$t = \frac{|\boldsymbol{S}_t|}{|\boldsymbol{S}_i|} = \frac{n_2}{n_1} \cdot \frac{E_t^2}{E_i^2} = \frac{n_2}{n_1} \cdot \left( \frac{2n_1}{n_2 + n_1} \right)^2 = \frac{n_2}{n_1} \cdot T^2 \tag{14.33}$$

If we add the fractions $r$ and $t$ of the energy flux density, then, due to the conservation of energy, the relation $r + t = R^2 + n_2 T^2 / n_1 = 1$ must hold. Substituting the expressions for $R$ and $T$ from eq. (14.32) and eq. (14.33) indeed yields:

$$r + t = \frac{n_1(n_2 - n_1)^2 + 4n_2 n_1^2}{n_1(n_2 + n_1)^2} = \frac{n_1(n_2^2 - 2n_2 n_1 + n_1^2 + 4n_2 n_1)}{n_1(n_2 + n_1)^2} = 1$$

Here, we have considered the special case in which the plane of incidence, i.e., the $k$-plane in fig. 14.1, is perpendicular to the interface $A$. Furthermore, we assumed a polarized incident wave with its electric field vector oriented perpendicular to the plane of incidence. Furthermore, we have considered only materials with $\mu_r \approx 1$; a valid assumption for the reflection and refraction of light at optical materials. Considering the general case of the reflection and calculation of plane light waves with arbitrary angles of incidence and polarization, one obtains the Fresnel equations, of which the equations listed here are a special case.

### 14.1.2 Electromagnetic Waves in Conductors

In an electromagnetic wave, energy is transported without the transport of mass. For the propagation of an electromagnetic wave in an ohmic conductor, this means that the free charge carriers follow the local oscillation of the electromagnetic field, yet no flow of charge carriers driven by a potential difference occurs. This means that, with regard to an electromagnetic wave propagating within the conductor, there are no free charge carriers in the conductor; thus, $\rho_{\text{free}} = 0$ applies. However, in an ohmic conductor with conductivity $\sigma$, the oscillating electromagnetic field is associated with a local current flow $\boldsymbol{j}_{local}$ at every point of the propagating electromagnetic wave, via the relation $\boldsymbol{j}_{local} = \sigma \boldsymbol{E} \neq 0$. Thus, the Maxwell equations for an electromagnetic wave propagating in a conductor are:

$$\nabla \boldsymbol{E} = \frac{1}{\epsilon} \rho_{free} = 0 \tag{14.34a}$$

$$\nabla \boldsymbol{B} = 0 \tag{14.34b}$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t} \tag{14.34c}$$

$$\nabla \times \boldsymbol{B} = \mu \boldsymbol{j}_{local} + \epsilon\mu \frac{\partial \boldsymbol{E}}{\partial t} = \mu\sigma \boldsymbol{E} + \epsilon\mu \frac{\partial \boldsymbol{E}}{\partial t} \tag{14.34d}$$

We now substitute the plane-wave ansatz from eq. (14.4) for $\boldsymbol{E}$ and $\boldsymbol{B}$ into these equations. It then follows immediately from eq. (14.34a) that:

$$\boldsymbol{k}\cdot\boldsymbol{E} \;=\; 0$$

Accordingly, it follows from eq. (14.34c) and eq. (14.34d):

$$\boldsymbol{B} \;=\; \frac{1}{\omega}(\boldsymbol{k}\times\boldsymbol{E}) \tag{14.35}$$

and

$$i(\boldsymbol{k}\times\boldsymbol{B}) \;=\; \mu\sigma\boldsymbol{E} - i\omega\epsilon\mu\boldsymbol{E} \tag{14.36}$$

Since $\sigma \neq 0$ holds in an ohmic conductor, eq. (14.36) is a complex equation, unlike the situations considered previously. Substituting eq. (14.35) into eq. (14.36), we obtain the following equation for $\boldsymbol{E}$:

$$\frac{i}{\omega}\big[\boldsymbol{k}\times(\boldsymbol{k}\times\boldsymbol{B})\big] \;=\; \frac{i}{\omega}\big[\boldsymbol{k}(\boldsymbol{k}\cdot\boldsymbol{E}) - \boldsymbol{E}(\boldsymbol{k}\cdot\boldsymbol{k})\big] \;=\; -\frac{ik^2}{\omega}\boldsymbol{E} \;=\; \mu\sigma\boldsymbol{E} - i\omega\epsilon\mu\boldsymbol{E}$$

Therefore, the following must hold:

$$\left(\mu\sigma - i\omega\epsilon\mu + \frac{ik^2}{\omega}\right)\boldsymbol{E} \;=\; 0 \tag{14.37}$$

For $\boldsymbol{E} \neq 0$, this is only possible if the expression in the parentheses vanishes, i.e., only if:

$$k^2 \;=\; \epsilon\mu\omega^2\left(1 + i\frac{\sigma}{\epsilon\omega}\right) \tag{14.38}$$

The wavenumber $k$ of an electromagnetic wave propagating in a conductor with $\sigma \neq 0$ is thus a complex quantity. For this, we make the ansatz ($\alpha$ and $\beta$ being real):

$$k = \alpha + i\beta \qquad \text{and thus} \qquad k^2 \;=\; \alpha^2 + \beta^2 + 2i\alpha\beta \tag{14.39}$$

In eq. (14.38), the magnitudes of $\epsilon$, $\mu$, and $\sigma$ appear, i.e., real quantities, and thus it follows from a comparison of eq. (14.38) and eq. (14.39) that:

$$\alpha^2 + \beta^2 \;=\; \epsilon\mu\omega^2 \qquad \text{and} \qquad 2\alpha\beta \;=\; \mu\sigma\omega \tag{14.40}$$

This system of equations with two equations and two real unknowns, $\alpha$ and $\beta$, is easily solved, yielding the result:

$$\alpha^2 \;=\; \frac{\mu\epsilon\omega^2}{2}\left(\sqrt{1 + \left(\frac{\sigma}{\epsilon\omega}\right)^2} + 1\right) \tag{14.41}$$

and

$$\beta^2 = \frac{\mu\epsilon\omega^2}{2}\left(\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}-1\right) \tag{14.42}$$

In the limiting case $\sigma \to 0$, we have $\beta \to 0$ and $\alpha \to \mu\epsilon\omega^2$. Hence, the imaginary part of $k$ approaches zero, and for the real part of $k$, we recover eq. (14.5). In the case $\sigma \neq 0$, an exponentially damped electromagnetic wave propagates within the conductor. The sign for $\alpha$ to be used when taking the square root in eq. (14.41) is determined by the direction of propagation. For a plane wave traveling in the positive $x$-direction, we must choose $\alpha > 0$. In any case, $\beta > 0$ must hold, and thus, using eq. (14.39), we obtain for the electromagnetic wave in a conductor the result

$$\boldsymbol{E} = \boldsymbol{E}_0 \mathrm{e}^{i(kx-\omega t)} \quad \longrightarrow \quad \boldsymbol{E} = \boldsymbol{E}_0 \mathrm{e}^{i(\alpha x-\omega t)} \cdot \mathrm{e}^{-\beta x} \tag{14.43}$$

and

$$\boldsymbol{B} = \boldsymbol{B}_0 \mathrm{e}^{i(kx-\omega t)} \quad \longrightarrow \quad \boldsymbol{B} = \boldsymbol{B}_0 \mathrm{e}^{i(\alpha x-\omega t)} \cdot \mathrm{e}^{-\beta x} \tag{14.44}$$

In section 11.5.1, we saw that the conductivity $\sigma$ of a conductor depends on the frequency of the alternating current flowing through it. For exactly the same reasons, this also applies to the propagation of an electromagnetic wave inside a conductor. It is the relationship between the plasma frequency of the charge carriers in the conductor, namely, the electrons, and the frequency of the electromagnetic wave that determines whether and how an electromagnetic wave can propagate within a conductor. At low frequencies, the imaginary part of the conductivity in eq. (11.120) is negligibly small, and $\sigma$ is therefore purely real. In other words, we are dealing with an ideal conductor in which electrons follow the oscillation of the electromagnetic field without delay, thereby enabling them to shield the field. When an electromagnetic wave traveling through a dielectric like air, strikes such an ideal conductor, part of the wave is reflected and part penetrates the conductor. The penetration depth of the wave into the conductor, the depth at which the field amplitudes within the conductor have decayed to 1/e, is $\beta^{-1}$. Let us examine more closely the frequency-dependent conductivity $\sigma(\omega)$ for conductors, as derived in section 11.5.1. For the conductivity of an ohmic conductor, the following applies according to eq. (11.120):

$$\sigma(\omega) = \frac{\sigma_0}{1+\omega^2\tau^2} - i\frac{\sigma_0 \cdot \omega\tau}{1+\omega^2\tau^2} = \sigma'(\omega) - i\sigma''(\omega) \tag{14.45}$$

where $\sigma_0$ is the conductivity of a conductor according to eq. (8.41), as known from the Drude model for metals (see section 8.2.1):

$$\sigma_0 = \frac{n_e e^2 \tau}{m_e} \tag{14.46}$$

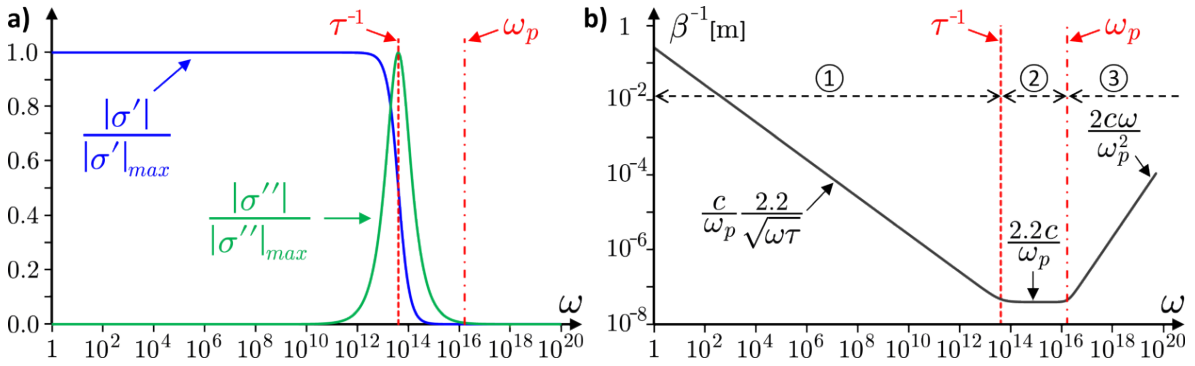



**Fig. 14.2:** Electromagnetic waves in a conductor, using copper as an example. Normalized real part $\sigma'$ and imaginary part $\sigma''$ of the complex conductivity $\sigma(\omega)$ in the metal (a). Penetration depth $\beta^{-1}$ in the metal (b).

Let us consider copper as a concrete example, for which the measured conductivity at room temperature is $\sigma_0 = 5.85 \times 10^7\,\Omega^{-1}\mathrm{m}^{-1}$. In section 8.2.1, assuming that each copper atom in the metal contributes one conduction electron to $n_e$, this gave us a value of $n_e = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$. With that we calculated a value of $\tau = 2.44 \times 10^{-14}\,\mathrm{s}$ for the mean collision time, i.e., the time elapsing between two successive collisions of an electron with lattice atoms. Using these values for $\sigma_0$ and $\tau$, we can now examine the behavior of the real and imaginary parts of $\sigma(\omega)$ for copper, in accordance with eq. (14.46). We do this by plotting the magnitudes of $\sigma'(\omega)$ and $\sigma''(\omega)$, each normalized to its respective maximum value, as functions of $\omega$, as shown in fig. 14.2a. From fig. 14.2a, we see that up to approximately $\omega = 10^{12}$, the real part of $\sigma(\omega)$ dominates and the imaginary part of the conductivity is negligibly small. Beyond this point, however, $|\sigma'(\omega)|$ drops sharply and $|\sigma''(\omega)|$ rises steeply. At $\omega = \tau^{-1}$, the imaginary part $|\sigma''(\omega)|$ reaches its maximum, while the real part $|\sigma'(\omega)|$ has dropped to half its maximum value. Beyond this point, $|\sigma''(\omega)|$ falls off steeply and is effectively zero above $\omega = \omega_p$; $|\sigma'(\omega)|$ has already dropped to zero almost two orders of magnitude below $\omega_p$. We are familiar with the plasma frequency $\omega_p$ from the discussion of polarizable gases in section 7.2.1, and we encountered it again in the chapter on alternating current when examining the frequency dependence of effective resistance in section 11.5.1. In an alternating electric field, electrons in a conductor do not behave like free charge carriers, but rather like bound charge carriers, since they locally follow the oscillation of the electric field but do not flow from one end of the conductor to the other. For $\omega < \omega_p$, the metal electrons can readily follow the oscillations of the alternating electric field: the system comprising stationary positive metal ions and locally oscillating electrons behaves like a polarizable medium. However, the electrons find it increasingly difficult to follow alternating electric fields with $\omega > \omega_p$, and the metal ceases to behave like a polarizable medium.

In chapter 5, we examined conductors in electrostatic fields and learned that a conducting body completely shields electric fields, causing the electric field within the conductor to vanish. We can also express this fact differently: the permittivity $\epsilon$ of a conductor in an electrostatic field is infinitely large. This must be the case because, even in an infinitesimally small electrostatic field, a charge displacement, known as induction charges, occurs within a conductor, compensating the field inside the conductor. In the limiting case where such an electrostatic field approaches zero, i.e., $\boldsymbol{E} \to 0$, this requires $\epsilon \to \infty$ for the charge displacement to occur in accordance with $\boldsymbol{P} = (\epsilon - \epsilon_0)\boldsymbol{E}$. In alternating electric fields, however, the dielectric constant of conductors is, as derived in section 11.5.1, a complex, frequency-dependent quantity:

$$\epsilon(\omega) = \epsilon_0 - i\frac{\sigma(\omega)}{\omega} \tag{14.47}$$

With eq. (14.45) and eq. (14.47), we know the complex quantities $\sigma(\omega)$ and $\epsilon(\omega)$ of a conductor. By substituting the magnitudes of these quantities into eq. (14.42), we can calculate the penetration depth $\beta^{-1}$ of an electromagnetic wave into a conductor. If we do this for our example of copper, we obtain the behavior of $\beta^{-1}$ shown in fig. 14.2b. The behavior of the penetration depth $\beta^{-1}$, as shown in the double-logarithmic plot in fig. 14.2b, differs significantly across the three frequency ranges indicated in the diagram: range ①, where $\beta^{-1}$ decreases linearly with increasing $\omega$ for $\omega < \tau^{-1}$; range ②, where $\beta^{-1}$ is nearly constant for $\tau^{-1} < \omega < \omega_p$; and range ③, where, above the plasma frequency $\omega > \omega_p$, $\beta^{-1}$ increases linearly. In region ①, the behavior of electromagnetic waves incident on a conductor is described by the so-called normal skin effect; in region ②, by the so-called anomalous skin effect; and in region ③, electromagnetic waves can propagate within the conductor. Let us consider the quantity $\sigma/\epsilon\omega$ in eq. (14.42). For the magnitudes of $\sigma(\omega)$ and $\epsilon(\omega)$, it follows from eq. (14.45) and eq. (14.47) that:

$$|\sigma(\omega)|^2 = \frac{\sigma_0^2}{1+\omega^2\tau^2} \qquad \text{and} \qquad |\epsilon(\omega)|^2 = \epsilon_0^2 + \frac{|\sigma(\omega)|^2}{\omega^2} \tag{14.48}$$

Thus, we can write:

$$\left(\frac{\sigma}{\epsilon\omega}\right)^2 = \frac{\sigma_0^2}{\sigma_0^2 + \epsilon_0^2\omega^2(1+\omega^2\tau^2)} = \frac{1}{1 + \dfrac{\omega^2(1+\omega^2\tau^2)}{\omega_p^4\tau^2}} \tag{14.49}$$

In this second step, we have made use of the fact that, via eq. (7.40) and eq. (14.47), the ratio $\epsilon_0/\sigma_0$ can be expressed in terms of $\omega_p$ and $\tau$. To determine the range of $\omega$ values for which the frequency-dependent term in the denominator on the right-hand side of eq. (14.49) can be neglected, we ask when it reaches a specific value $\xi$. To do this, we solve the equation

$$\frac{\omega^4\tau^2+\omega^2}{\omega_p^4\tau^2} = \xi \qquad \text{with the result} \qquad \omega^2 = \frac{1}{2\tau}\left(\sqrt{1+4\omega_p^4\tau^4\cdot\xi}-1\right)$$

If we substitute the values of $\tau$ and $\omega_p$ for copper and require $\xi < 0.01$ for a good approximation, then the condition $\omega < 0.316\cdot\omega_p$ must hold. Hence, in the range from ① up to $\omega = \tau^{-1}$ in fig. 14.2, $\sigma/\epsilon\omega = 1$ is a very good approximation. Since $\xi = 1$ is reached only at $\omega = 0.999998\cdot\omega_p$, it remains a good approximation in range ②. Substituting $\sigma/\epsilon\omega = 1$ into eq. (14.43), we obtain:

$$\beta^2 = \mu\epsilon\omega^2\cdot\frac{\sqrt{2}-1}{2} \tag{14.50}$$

What does the frequency dependence of $\epsilon$ look like in this equation? To answer this, we examine the ratio between the frequency-independent real part and the frequency-dependent imaginary part of $|\epsilon(\omega)|^2$ in eq. (14.48):

$$\frac{\epsilon_0^2\cdot\omega^2}{|\sigma(\omega)|^2} = \frac{\epsilon_0^2}{\sigma_0^2}\cdot\omega^2(1+\omega^2\tau^2) = \frac{\omega^2(1+\omega^2\tau^2)}{\omega_p^4\tau^2} \tag{14.51}$$

We are already familiar with this expression: it is the same dimensionless quantity $\xi$ that we just examined. This tells us immediately that the magnitude of the imaginary part of $\epsilon$ drops to the level of the real part $\epsilon_0$ only once the angular frequency approaches the value $\omega = 0.999998\cdot\omega_p$. From the discussion above, we also know that for $\omega < 0.316\cdot\omega_p$, the imaginary part of $\epsilon$ is still a factor of 100 larger than the real part. In the frequency range ①, we can therefore safely neglect the real part of $\epsilon$, and in frequency range ②, this is also a good approximation up to just below $\omega_p$. Thus, eq. (14.51) becomes the following equation for the frequency ranges ① and ②:

$$\beta^2 = \frac{\mu\sigma_0\omega}{\sqrt{1+\omega^2\tau^2}}\cdot\frac{\sqrt{2}-1}{2} \tag{14.52}$$

For non-magnetic metals such as copper, $\mu = \mu_0$ effectively holds, and with $\sigma_0 = \epsilon_0\omega_p^2\tau$ and $\mu_0\epsilon_0 = c^{-2}$, we can rewrite eq. (14.52) as:

$$\beta^2 = \mu_0\epsilon_0\omega_p^2\cdot\frac{\omega\tau}{\sqrt{1+\omega^2\tau^2}}\cdot\frac{\sqrt{2}-1}{2} = \frac{\omega_p^2}{c^2}\cdot\frac{\omega\tau}{\sqrt{1+\omega^2\tau^2}}\cdot\frac{\sqrt{2}-1}{2} \tag{14.53}$$

Let us consider the fraction containing the frequency term $\omega\tau$ in eq. (14.53) for $\omega\tau < 1$ and $\omega\tau > 1$. For $\omega\tau < 1$, the expansion of this frequency term is:

$$\frac{\omega\tau}{\sqrt{1+\omega^2\tau^2}} = \omega\tau - \frac{1}{2}(\omega\tau)^3 + \ldots \qquad \omega\tau < 1$$

And for $\omega\tau > 1$, this expansion is:

$$\frac{\omega\tau}{\sqrt{1+\omega^2\tau^2}} = 1 - \frac{1}{2}\frac{1}{(\omega\tau)^2} + \ldots \qquad \omega\tau > 1$$

In the zeroth-order approximation, eq. (14.53) thus yields the following expression for the penetration depth $\beta^{-1}$ in the regime $\omega\tau < 1$:

$$\beta^{-1} = \frac{c}{\omega_p}\cdot\frac{1}{\sqrt{\omega\tau}}\cdot\sqrt{\frac{2}{\sqrt{2}-1}} \approx \frac{c}{\omega_p}\cdot\frac{2.2}{\sqrt{\omega\tau}} \qquad \text{in range ①} \tag{14.54}$$

In the regime $\omega\tau > 1$, eq. (14.53) yields the following for the penetration depth $\beta^{-1}$ in the zeroth approximation:

$$\beta^{-1} = \frac{c}{\omega_p}\cdot\sqrt{\frac{2}{\sqrt{2}-1}} \approx 2.2\frac{c}{\omega_p} \qquad \text{in range ②} \tag{14.55}$$

In the frequency range ①, the penetration depth scales with $1/\sqrt{\omega}$, and we obtain a straight line with a negative slope in the double-logarithmic plot shown in fig. 14.2b. In the frequency range ②, the penetration depth is constant. To correctly describe the transition between the two ranges ① and ②, one must take into account the next-higher-order terms in the approximation for the frequency term in eq. (14.53). Let us now consider range ③ where $\omega > \omega_p$. As can be seen from eq. (14.49), for $\omega > \omega_p$ we have $\sigma/\epsilon\omega < 1$ and we can expand the square root in eq. (14.42). Thus, for $\beta^2$, we obtain the following expression based on eq. (14.42) for the regime $\omega > \omega_p$:

$$\beta^2 = \frac{\mu\epsilon\omega^2}{2}\cdot\left[\left(1+\frac{1}{2}\left(\frac{\sigma}{\epsilon\omega}\right)^2+\dots\right)-1\right] \tag{14.56}$$

To a first approximation, we thus obtain for $\beta^2$ in the range ③:

$$\begin{aligned}\beta^2 &= \frac{\mu\epsilon\omega^2}{4}\cdot\left(\frac{\sigma}{\epsilon\omega}\right)^2 = \frac{\mu}{4\epsilon}\cdot\frac{\sigma_0^2}{1+\omega^2\tau^2} \\ &= \frac{\mu}{4\epsilon}\cdot\frac{\epsilon_0^2\omega_p^4\tau^2}{1+\omega^2\tau^2} = \frac{\mu\epsilon_0^2}{4\epsilon}\cdot\frac{\omega_p^4}{\omega^2}\cdot\frac{1}{1+(\omega\tau)^{-2}}\end{aligned} \tag{14.57}$$

Since $\omega\tau > 1$ in range ③, the term $1/(\omega\tau)^2$ becomes very small rapidly as $\omega$ increases. Consequently, in the zeroth-order approximation, we can replace the fraction containing $1/(\omega\tau)^2$ in the denominator with 1. As we have already seen, at $\omega = 0.999998\cdot\omega_p$, the frequency-dependent imaginary part of $\epsilon$ has dropped to the value of the constant real part of $\epsilon$ and approaches zero very rapidly for $\omega > \omega_p$. This means we can replace $\epsilon$ in eq. (14.57) with its real part $\epsilon_0$. Furthermore, as before, we set $\mu = \mu_0$ for the non-magnetic metal copper, thereby obtaining the following for $\beta^2$:

$$\beta^2 = \frac{\mu_0\epsilon_0}{4}\cdot\frac{\omega_p^4}{\omega^2} = \frac{\omega_p^2}{4c^2}\cdot\frac{\omega_p^2}{\omega^2} \tag{14.58}$$

For the penetration depth $\beta^{-1}$, we thus obtain, in the case $\omega > \omega_p$:

$$\beta^{-1} = \frac{2c}{\omega_p}\cdot\frac{\omega}{\omega_p} \qquad \text{in range ③} \tag{14.59}$$

In the frequency range ③, therefore, $\beta^{-1} \propto \omega$ applies, which results in a straight line with a positive slope in the double-logarithmic plot shown in fig. 14.2b. As the frequency increases, the penetration depth $\beta^{-1}$ rises rapidly. Consequently, the metal copper should, in principle, become increasingly transparent to electromagnetic waves with $\omega > \omega_p$. However, this is not observed experimentally; copper remains an opaque metal. One can immediately see why metals cannot truly become transparent for $\omega > \omega_p$ by switching from the wave picture of electromagnetic radiation to the particle picture. The energy of photons of electromagnetic radiation at the plasma frequency of copper is $E = \hbar\omega_p = 57.7\,\mathrm{eV}$. At such energies, interactions with lattice atoms and conduction electrons, essentially absorption and scattering, occur. None of these energy losses are accounted for here. Thus, while in principle, electromagnetic waves can propagate through copper for $\omega > \omega_p$, as derived above, they are rapidly extinguished in the metal by energy losses due to absorption and scattering. The situation changes, however, if, instead of copper, one considers, for example, an alkali metal polished down to a very thin layer. Such very thin alkali metal films are opaque to optical wavelengths but become transparent in the ultraviolet range, i.e., at shorter wavelengths and consequently higher light frequencies; just as our model above predicts. For sodium, this is the case for light with wavelengths $\lambda \lessapprox 210\,\mathrm{nm}$. Consequently, the plasma frequency for this alkali metal is $\omega_p \approx 8.97 \times 10^{15}\,\mathrm{s}^{-1}$.

## 14.2 Generation of Electromagnetic Waves

Every electric charge is surrounded by an electric field, and every moving electric charge generates a magnetic field. Both phenomena were known before James Clerk Maxwell formulated his equations shortly after the middle of the 19th century. These equations provided an explanation for a wide variety of electromagnetic phenomena associated with charges and currents. In the last chapter, we saw that these were not the only solutions to the Maxwell equations. In a vacuum Maxwell's equation have solutions in the form of electromagnetic waves, which possess a reality of their own, independent of electric charges and currents. These electromagnetic waves postulated by the Maxwell equations were first experimentally demonstrated by Heinrich Hertz about twenty years after the Maxwell equations were formulated. Electromagnetic waves are generated whenever electric charges are accelerated or decelerated. Charges moving at a constant velocity (stationary currents) produce magnetic fields but not electromagnetic waves. Alternating currents, however, generate electromagnetic waves because the electric charges oscillate within an alternating field, undergoing periodic acceleration and deceleration in the

process. A rigorous treatment of the generation of electromagnetic waves is not possible within the scope of an introductory course on electrodynamics. The following discussion is therefore limited to a qualitative explanation of key aspects, using the tools we have developed so far.

### 14.2.1 The Oscillating Dipole

When considering the electric fields generated by stationary charges, we need not concern ourselves with any time dependence, as these fields are static. No matter how far we move away from a stationary charge distribution $\rho(\boldsymbol{r}')$, we immediately know the field it produces at any location $\boldsymbol{r}$ in space and time. The simple reason for this is, of course, that the field is invariant; it is an electrostatic field which consistently maintains the same position-dependent value at every point in time and space. If, however, someone were to simply switch off the stationary charge distribution, located at $\boldsymbol{r}'$, and subsequently switch it back on, we at our location $\boldsymbol{r}$, situated at a distance $\Delta r = |\boldsymbol{r} - \boldsymbol{r}'|$ from the stationary charge distribution $\rho(\boldsymbol{r}')$ would only notice such a «switching off or on» event once this event reached our location $\boldsymbol{r}$. Electromagnetic waves, or disturbances such as the «switching on or off» event of a stationary charge distribution, propagate at the speed of light. In air or a vacuum environment, an event occurring at the location of the stationary charge distribution $\rho(\boldsymbol{r}')$ at time $t$ is not registered at location $\boldsymbol{r}$ until a later time, i.e., at $t + \Delta r/c$. In other words, the electric potential we measure at a position $\boldsymbol{r}$ at a time $t$ is always determined by the stationary charge distribution $\rho(\boldsymbol{r}')$ as it existed at an earlier time $t - \Delta r/c$. Hence, observers located at some distance from the sources of such potentials measure so-called retarded potentials, since, due to the finite propagation speed of electromagnetic fields, these always reflect changes in the sources of the potentials, i.e., in charge distributions or currents, with a time delay. From eq. (3.20), we know that the electrostatic potential $\phi(\boldsymbol{r})$ produced at position $\boldsymbol{r}$ by a stationary charge distribution $\rho(\boldsymbol{r}')$ is given by:

$$\phi(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\boldsymbol{r}')}{|\boldsymbol{r} - \boldsymbol{r}'|}\, \mathrm{d}^3 r' \tag{14.60}$$

We obtain the time-dependent electric potential $\phi(\boldsymbol{r}, t)$ generated by a non-stationary charge distribution $\rho(\boldsymbol{r}', t')$ from this equation by taking into account the finite propagation speed of electromagnetic fields. To achieve this we must relate the potential $\phi$ at position $\boldsymbol{r}$ and time $t$ to the state of the charge distribution $\rho(\boldsymbol{r}', t')$ at an earlier time $t' = t - \Delta r/c$:

$$\phi(\boldsymbol{r}, t) \;=\; \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\boldsymbol{r}', t - \Delta r/c)}{|\boldsymbol{r} - \boldsymbol{r}'|}\, \mathrm{d}^3 r' \qquad \text{with} \quad \Delta r = |\boldsymbol{r} - \boldsymbol{r}'| \tag{14.61}$$

We can now apply to the magnetic vector potential the reasoning we just used to derive the equation for the time-dependent potential of a non-stationary charge distribution from the corresponding equation for the electrostatic potential of a stationary charge distribution. From eq. (9.29), we know that the vector potential for steady currents is given by:

$$\boldsymbol{A}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\int \frac{\boldsymbol{j}(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{14.62}$$

The time-dependent vector potential $\boldsymbol{A}(\boldsymbol{r},t)$, generated by a non-stationary current density $\boldsymbol{j}(\boldsymbol{r}',t')$ at an earlier time $t' = t - \Delta r/c$, is then accordingly:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\int \frac{\boldsymbol{j}(\boldsymbol{r}',t-\Delta r/c)}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \qquad \text{with} \quad \Delta r = |\boldsymbol{r}-\boldsymbol{r}'| \tag{14.63}$$

We therefore only need to integrate over the non-stationary charge distribution in eq. (14.61) or the non-stationary current in eq. (14.63) to obtain $\phi(\boldsymbol{r},t)$ and $\boldsymbol{A}(\boldsymbol{r},t)$. However, except in very simple situations, this is not trivial. Nevertheless, we can learn a great deal by considering such a simple example, specifically, that of an oscillating charge distribution. A very small, spherically symmetric, and homogeneous charge distribution $\rho(\boldsymbol{r}')$ is to undergo harmonic oscillation along the $z$-axis about the origin, as sketched in fig. 14.3. For the amplitude of this oscillation, the condition $z_0 \gg |\boldsymbol{r}'|$ is assumed to hold. The coordinates of the center of mass of the charge distribution $\rho(\boldsymbol{r}')$ are:

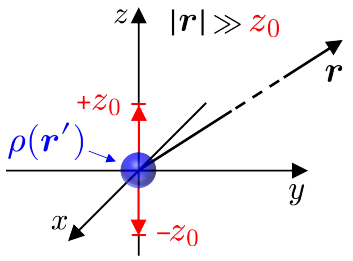



**Fig. 14.3**

$$x' = 0 \quad , \quad y' = 0 \quad \text{and} \quad z'(t') = z_0 \sin\omega t' \tag{14.64}$$

The oscillating charge distribution represents an alternating-current density for which holds:

$$\boldsymbol{j}(\boldsymbol{r}',t') \;=\; \hat{\boldsymbol{k}}\cdot\rho(\boldsymbol{r}')\frac{\partial z'(t')}{\partial t'} \;=\; \hat{\boldsymbol{k}}\cdot\frac{\partial}{\partial t'}\left[\rho(\boldsymbol{r}')z_0\sin\omega t'\right] \tag{14.65}$$

To obtain the vector potential $\boldsymbol{A}(\boldsymbol{r},t)$ in eq. (14.63), we must evaluate the integral in that equation using the current density at time $t' = t - \Delta r/c$. We can move the time derivative outside the integral, and using $\partial/\partial t' = \partial/\partial t$, we thus obtain the following expression for $\boldsymbol{A}(\boldsymbol{r},t)$:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{\hat{\boldsymbol{k}}}{4\pi\epsilon_0 c^2}\frac{\partial}{\partial t}\int \frac{\rho(\boldsymbol{r}')z_0\sin[\omega(t-\Delta r/c)]}{|\boldsymbol{r}-\boldsymbol{r}'|}\,\mathrm{d}^3r' \tag{14.66}$$

We are not interested here in the electromagnetic near-field generated by such an oscillating charge distribution, but rather in the electromagnetic field at locations $|\boldsymbol{r} \gg \boldsymbol{r}'|$

far removed from the charge distribution, the so-called far-field. This allows us to set $\Delta r = r$, which is equivalent to treating the charge distribution as a point charge $q$ with no spatial extent. Without this approximation, we would have to take into account that, due to the finite extent of $\rho(\boldsymbol{r}')$, changes occurring in different parts of the moving charge distribution $\rho(\boldsymbol{r}')$ arrive at our location $\boldsymbol{r}$ at different times. With this approximation, we effectively assume that all changes originating from different parts of the charge distribution $\rho(\boldsymbol{r}')$ reach us simultaneously. Importantly, this implies not only that the charge distribution has practically no spatial extent, but above all that the oscillation of this charge distribution must be slow enough for us to consider a quasi-static situation at any given moment. In other words, the condition $\dot{z}' \ll c$ must hold. Our analysis is therefore valid only as long as we do not need to take relativistic effects into account. If, using this approximation, we move the quantities that no longer depend on $\boldsymbol{r}'$ outside the integral, eq. (14.66) becomes:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{\hat{\boldsymbol{k}}}{4\pi\epsilon_0 c^2}\frac{\partial}{\partial t}\left[\frac{z_0\sin[\omega(t-r/c)]}{r}\int \rho(\boldsymbol{r}')\,\mathrm{d}^3r'\right] \tag{14.67}$$

The integral over the charge distribution is equal to the total charge $q$, and thus:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{\hat{\boldsymbol{k}}}{4\pi\epsilon_0 c^2}\frac{\partial}{\partial t}\frac{q\cdot z_0\sin[\omega(t-r/c)]}{r} \tag{14.68}$$

The quantity $p_0 = q\cdot z_0$ corresponds to an electric dipole moment, as we already know from section 3.2.1, and with $p(t-r/c) = p_0\sin[\omega(t-r/c)]$, this dipole moment represents a charge $q$ oscillating along the $z$-axis:

$$\boldsymbol{p}(t-r/c) = p_0\sin[\omega(t-r/c)]\cdot\hat{\boldsymbol{k}} \tag{14.69}$$

This gives us the following result for the vector potential:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\frac{\dot{p}(t-r/c)}{r}\cdot\hat{\boldsymbol{k}} \tag{14.70}$$

From section 13.1.3, we know that the right-hand side of this equation represents a solution to the three-dimensional wave equation eq. (13.35) for the vector potential $\boldsymbol{A}(\boldsymbol{r},t)$. The vector potential generated by the oscillating charge distribution $\rho(\boldsymbol{r}')$ is thus an outgoing spherical wave. Let us now consider the magnetic field $\boldsymbol{B} = \nabla\times\boldsymbol{A}$ associated with this vector potential. Since we are dealing with a spherical symmetric problem, we switch to spherical coordinates. With $A_x = 0$, $A_y = 0$, and only $A_z \neq 0$, we obtain the following for the components $A_r$, $A_\vartheta$, and $A_\varphi$ of $\boldsymbol{A}(\boldsymbol{r},t)$:

$$A_r \;=\; A_z\cdot\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{e}}_r \;=\; A_z\cdot\cos\vartheta \;=\; \frac{1}{4\pi\epsilon_0 c^2}\frac{\dot{p}(t-r/c)}{r}\cdot\cos\vartheta \tag{14.71a}$$

$$A_\vartheta \;=\; A_z \cdot \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{e}}_\vartheta \;=\; -A_z \cdot \sin\vartheta \;=\; -\frac{1}{4\pi\epsilon_0 c^2}\frac{\dot{p}(t-r/c)}{r}\cdot\sin\vartheta \tag{14.71b}$$

$$A_\varphi \;=\; A_z \cdot \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{e}}_\varphi \;=\; A_z \cdot 0 \;=\; 0 \tag{14.71c}$$

With $A_r$ and $A_\vartheta$ independent of $\varphi$ and $A_\varphi = 0$, the expression $\boldsymbol{B} = \nabla \times \boldsymbol{A}$ simplifies in spherical coordinates to (see tab. A.1 in Appendix A):

$$\boldsymbol{B}(\boldsymbol{r},t) \;=\; \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\vartheta) - \frac{\partial A_r}{\partial \vartheta}\right]\hat{\boldsymbol{e}}_\varphi \;=\; \frac{1}{r}\left[A_\vartheta + r\frac{\partial A_\vartheta}{\partial r} - \frac{\partial A_r}{\partial \vartheta}\right]\hat{\boldsymbol{e}}_\varphi \tag{14.72}$$

From eq. (14.72), we can immediately conclude that the magnetic field lines must be concentric circular rings centered on the $z$-axis, extending within the $xy$-plane. Furthermore, a look at eq. (14.71a) and eq. (14.71b) shows that $\partial A_r/\partial\vartheta = A_\vartheta$ holds, thereby simplifying eq. (14.72) to:

$$\boldsymbol{B}(\boldsymbol{r},t) \;=\; \frac{\partial A_\vartheta}{\partial r}\hat{\boldsymbol{e}}_\varphi \;=\; -\frac{1}{4\pi\epsilon_0 c^2}\frac{\partial}{\partial r}\frac{\dot{p}(t-r/c)}{r}\sin\vartheta\cdot\hat{\boldsymbol{e}}_\varphi \tag{14.73}$$

To evaluate eq. (14.73), we must first consider the spatial derivative of $\dot{p}(t-r/c)$, since we must take into account that $\dot{p}$ is a function of $t' = t - r/c$. With $\partial/\partial t' = \partial/\partial t$, the following applies here:

$$\frac{\partial \dot{p}(t')}{\partial r} \;=\; \frac{\partial \dot{p}(t')}{\partial t'}\frac{\partial t'}{\partial r} \;=\; -\frac{\partial \dot{p}(t-r/c)}{\partial t}\frac{1}{c} \;=\; -\frac{\ddot{p}(t-r/c)}{c} \tag{14.74}$$

We thus obtain the following expression for the magnetic field of the oscillating charge distribution:

$$\boldsymbol{B}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\left[\frac{\dot{p}(t-r/c)}{r^2} + \frac{\ddot{p}(t-r/c)}{cr}\right]\sin\vartheta\cdot\hat{\boldsymbol{e}}_\varphi \tag{14.75}$$

We can write this equation in a more elegant form by making use of the fact that for the cross product $\boldsymbol{p}\times\boldsymbol{r}$:

$$\begin{pmatrix}0\\0\\p\end{pmatrix} \times r\begin{pmatrix}\sin\vartheta\cos\varphi\\ \sin\vartheta\sin\varphi\\ \cos\vartheta\end{pmatrix} \;=\; \begin{pmatrix}-pr\sin\vartheta\sin\varphi\\ pr\sin\vartheta\cos\varphi\\ 0\end{pmatrix} \;=\; p\cdot r\cdot\sin\vartheta\cdot\hat{\boldsymbol{e}}_\varphi \tag{14.76}$$

Using this relationship, we can rewrite eq. (14.75) as:

$$\boldsymbol{B}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\left[\frac{\dot{\boldsymbol{p}}(t-r/c)}{r^3} + \frac{\ddot{\boldsymbol{p}}(t-r/c)}{cr^2}\right]\times\boldsymbol{r} \tag{14.77}$$

The first term in this equation scales as $r^{-2}$ and is proportional to the time derivative of the dipole moment $\boldsymbol{p}$. However, from eq. (14.65), we know that $\dot{\boldsymbol{p}}$ corresponds to a current density. The first term in the numerator of eq. (14.77) is thus the cross product of a current density and the position vector $\boldsymbol{r}$. Earlier, we assumed that $|\boldsymbol{r}| \gg |\boldsymbol{r}'|$, which allowed us to replace $\boldsymbol{r}-\boldsymbol{r}'$ with $\boldsymbol{r}$ and $|\boldsymbol{r}-\boldsymbol{r}'|$ with $r$. Keeping this in mind, let us look

at the differential form of the defining equation eq. (9.31), which led us to the Biot-Savart law in section 9.3:

$$\mathrm{d}\boldsymbol{B}(\boldsymbol{r}) \;=\; \frac{1}{4\pi\epsilon_0 c^2}\frac{\boldsymbol{j}(\boldsymbol{r}')\times(\boldsymbol{r}-\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|^3}\,\mathrm{d}^3r' \tag{14.78}$$

Obviously, the first term in eq. (14.77) formally corresponds to the differential form of eq. (9.31). However, there is a crucial difference: the argument of the current density associated with $\dot{\boldsymbol{p}}$ is the retarded time $t - r/c$, i.e., the earlier instant when a specific change in the charge distribution occurred, rather than the current time $t$ at which a change in the current density takes place in eq. (14.78). Let us assume that $r$ is significantly larger than $r'$, such that the approximation used to derive eq. (14.77) still holds, but that $r/c$ is small enough for us to expand $\boldsymbol{p}(t-r/c)$. In this case, the following applies:

$$\boldsymbol{p}(t-r/c) \;=\; \boldsymbol{p}(t) - \frac{r}{c}\frac{\partial\boldsymbol{p}(t)}{\partial t} + \frac{1}{2}\frac{r^2}{c^2}\frac{\partial^2\boldsymbol{p}(t)}{\partial t^2}\ldots \tag{14.79}$$

If we take into account terms up to the second time derivative of $\boldsymbol{p}$, then in this approximation we obtain for $\dot{\boldsymbol{p}}(t-r/c)$ and $\ddot{\boldsymbol{p}}(t-r/c)$:

$$\dot{\boldsymbol{p}}(t-r/c) \;\approx\; \dot{\boldsymbol{p}}(t) - \frac{r}{c}\ddot{\boldsymbol{p}}(t) \qquad \text{and} \qquad \ddot{\boldsymbol{p}}(t-r/c) \;\approx\; \ddot{\boldsymbol{p}}(t) \tag{14.80}$$

If we substitute this into eq. (14.77), we see that for small values of $r/c$ applies:

$$\boldsymbol{B}(\boldsymbol{r},t) \;\approx\; \frac{1}{4\pi\epsilon_0 c^2}\frac{\dot{\boldsymbol{p}}(t)\times\boldsymbol{r}}{r^3} \tag{14.81}$$

For values of $r$ that are not too large, the magnetic field of the oscillating charge distribution indeed resembles eq. (14.78). However, as the values of $r/c$ from the charge distribution increases, the first term in eq. (14.77) rapidly diminishes, and the second term, scaling as $r^{-1}$, becomes dominant. At large distances from the oscillating charge distribution, the $r^{-2}$ term rapidly becomes negligible compared to the $r^{-1}$ term, and it suffices to consider only the $r^{-1}$ term. This second term in eq. (14.77), which is proportional to the acceleration experienced by the charge distribution during oscillation, represents something new. We have not yet encountered such an expression in a magnetic field.

To determine the electric field of the oscillating charge distribution, we must first know the electric potential $\phi(\boldsymbol{r},t)$. Just as eq. (14.70) is the solution to a wave equation for the vector potential $\boldsymbol{A}(\boldsymbol{r},t)$, the sought-after solution for the electric potential $\phi(\boldsymbol{r},t)$ must satisfy a wave equation of its own. However, this decoupling of the wave equations, which we carried out in section 13.1 for $\phi(\boldsymbol{r},t)$ and $\boldsymbol{A}(\boldsymbol{r},t)$, relied on a prerequisite, namely, that in accordance with eq. (13.12), the following must hold:

$$\nabla \boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0 \tag{14.82}$$

We thus obtain the time derivative of the potential associated with the oscillating charge distribution by calculating the divergence of the known vector potential. With $A_\varphi = 0$, the divergence of $\boldsymbol{A}(\boldsymbol{r}, t)$ in spherical coordinates is (see tab. A.1 in Appendix A):

$$\nabla \boldsymbol{A}(\boldsymbol{r}, t) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 A_r\right) + \frac{1}{r \sin\vartheta}\frac{\partial}{\partial \vartheta}(\sin\vartheta A_\vartheta) \tag{14.83}$$

Using eq. (14.71a) and eq. (14.71b), we have $A_r = A_z \cos\vartheta$ and $A_\vartheta = -A_z \sin\vartheta$, and since $A_z$ depends only on $r$, we can write the divergence of this vector potential as follows:

$$\nabla \boldsymbol{A}(\boldsymbol{r}, t) = \frac{2}{r} A_z \cos\vartheta + \frac{\partial A_z}{\partial r}\cos\vartheta - \frac{2}{r} A_z \cos\vartheta = \frac{\partial A_z}{\partial r}\cos\vartheta \tag{14.84}$$

Substituting $A_z$ from eq. (14.70) here, we obtain, using eq. (14.84), the following expression for the divergence of the vector potential:

$$\nabla \boldsymbol{A}(\boldsymbol{r}, t) = -\frac{1}{4\pi\epsilon_0 c^2}\left[\frac{\dot{p}(t - r/c)}{r^2} + \frac{\ddot{p}(t - r/c)}{cr}\right]\cdot\cos\vartheta \tag{14.85}$$

Using $\boldsymbol{p}\cdot\boldsymbol{r} = p\cdot r\cdot\cos\vartheta$, we can also write this equation as:

$$\nabla \boldsymbol{A}(\boldsymbol{r}, t) = -\frac{1}{4\pi\epsilon_0 c^2}\left[\frac{\dot{\boldsymbol{p}}(t - r/c)}{r^3} + \frac{\ddot{\boldsymbol{p}}(t - r/c)}{cr^2}\right]\cdot\boldsymbol{r} \tag{14.86}$$

According to eq. (14.82), the time derivative of the electric potential $\phi(\boldsymbol{r}, t)$ is thus:

$$\frac{\partial \phi(\boldsymbol{r}, t)}{\partial t} = \frac{1}{4\pi\epsilon_0}\left[\frac{\dot{\boldsymbol{p}}(t - r/c)}{r^3} + \frac{\ddot{\boldsymbol{p}}(t - r/c)}{cr^2}\right]\cdot\boldsymbol{r} \tag{14.87}$$

Up to a constant of integration representing a static potential, which we can always choose to be zero by definition, we obtain $\phi(\boldsymbol{r}, t)$ from this equation by simple integration with respect to time, yielding the result:

$$\phi(\boldsymbol{r}, t) = \frac{1}{4\pi\epsilon_0}\left[\frac{\boldsymbol{p}(t - r/c)}{r^3} + \frac{\dot{\boldsymbol{p}}(t - r/c)}{cr^2}\right]\cdot\boldsymbol{r} \tag{14.88}$$

With eq. (14.85) and eq. (14.88), we now know the vector potential and the electric potential of our oscillating charge distribution. This is all the information which is required to determine the electric field using eq. (13.8), that is:

$$\boldsymbol{E}(\boldsymbol{r}, t) = -\nabla\phi(\boldsymbol{r}, t) - \frac{\partial}{\partial t}\boldsymbol{A}(\boldsymbol{r}, t) \tag{14.89}$$

In the following calculation of the gradient of $\phi(\boldsymbol{r}, t)$ and the time derivative of $\boldsymbol{A}(\boldsymbol{r}, t)$, we are going to omit the retarded argument of $\boldsymbol{p}$, $\dot{\boldsymbol{p}}$, and $\ddot{\boldsymbol{p}}$ for the sake of brevity. However, we must bear in mind that when taking spatial derivatives of these quantities, one must first take their time derivative and then the spatial derivative of the retarded argument. We have already calculated the gradient of a potential very similar to the one

in eq. (14.88) in section 3.2.1, and we will make use of that here. To this end, we write the gradient of eq. (14.88) as:

$$\nabla\phi(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0}\nabla\left(\frac{\left[\boldsymbol{p}+\left({}^{r}/_{c}\right)\dot{\boldsymbol{p}}\right]\cdot\boldsymbol{r}}{r^3}\right) \tag{14.90}$$

Now we define a vector $\tilde{\boldsymbol{p}}$:

$$\tilde{\boldsymbol{p}} \;=\; \boldsymbol{p}+\frac{r}{c}\dot{\boldsymbol{p}} \tag{14.91}$$

In the following, we also require the quantity $(\boldsymbol{r}\nabla)\tilde{\boldsymbol{p}}$ and the curl of this vector, which depends only on $r$. For these, one quickly finds:

$$\left(\boldsymbol{r}\nabla\right)\tilde{\boldsymbol{p}} \;=\; -\frac{r^2}{c^2}\ddot{\boldsymbol{p}} \qquad \text{and} \qquad \nabla\times\tilde{\boldsymbol{p}} \;=\; -\frac{1}{c^2}(\boldsymbol{r}\times\ddot{\boldsymbol{p}}) \tag{14.92}$$

Using the vector $\tilde{\boldsymbol{p}}$ we now write the gradient of $\phi(\boldsymbol{r},t)$ as

$$\nabla\phi(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0}\nabla\left(\frac{\tilde{\boldsymbol{p}}}{r^3}\boldsymbol{r}\right) \tag{14.93}$$

Formally, this expression is identical to the electric potential of the static dipole discussed in section 3.2.1. Here, however, we are no longer dealing with an electrostatic dipole; $\tilde{\boldsymbol{p}}$ represents an oscillating dipole moment. To evaluate the gradient in eq. (14.90), we use equation eq. (3.54a), already considered in section 3.2.1, but now applied to an oscillating dipole moment:

$$\nabla\cdot\left(\tilde{\boldsymbol{p}}\cdot\frac{\boldsymbol{r}}{r^3}\right) \;=\; \left(\frac{\boldsymbol{r}}{r^3}\cdot\nabla\right)\cdot\tilde{\boldsymbol{p}}+\left(\tilde{\boldsymbol{p}}\cdot\nabla\right)\cdot\frac{\boldsymbol{r}}{r^3}+\tilde{\boldsymbol{p}}\times\left(\nabla\times\frac{\boldsymbol{r}}{r^3}\right)+\frac{\boldsymbol{r}}{r^3}\times\left(\nabla\times\tilde{\boldsymbol{p}}\right) \tag{14.94a}$$

As we saw for the static dipole in section 3.2.1, for a constant $\tilde{\boldsymbol{p}}$, all terms on the right-hand side of this equation are zero, except for the second term. The third term is zero even for a non-constant $\tilde{\boldsymbol{p}}$, because $\boldsymbol{r}/r^3$ is the gradient of a potential. However, the remaining terms on the right-hand side of eq. (14.94a) do not vanish for a non-constant $\tilde{\boldsymbol{p}}$; therefore, we must calculate each of them individually. For the first term, using eq. (14.92), we obtain:

$$\left(\frac{\boldsymbol{r}}{r^3}\cdot\nabla\right)\cdot\tilde{\boldsymbol{p}} \;=\; -\frac{\ddot{\boldsymbol{p}}}{c^2 r} \tag{14.94b}$$

We have already calculated the second term in section 3.2.1, we simply need to substitute $\tilde{\boldsymbol{p}}$ for $\boldsymbol{p}$ in eq. (3.54b):

$$\left(\tilde{\boldsymbol{p}}\cdot\nabla\right)\frac{\boldsymbol{r}}{r^3} \;=\; -\frac{3(\tilde{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}-\tilde{\boldsymbol{p}}r^2}{r^5} \tag{14.94c}$$

For the last non-vanishing term on the right-hand side of eq. (14.94a), we obtain, using eq. (14.92):

$$\frac{1}{r^3}\big[\boldsymbol{r}\times(\nabla\times\tilde{\boldsymbol{p}})\big] \;=\; -\frac{1}{c^2r^3}\big[\boldsymbol{r}\times(\boldsymbol{r}\times\ddot{\boldsymbol{p}})\big] \;=\; -\frac{(\ddot{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}}{c^2r^3}+\frac{\ddot{\boldsymbol{p}}}{c^2r} \tag{14.94d}$$

Using eq. (14.94b), eq. (14.94c), and eq. (14.94d), we can now write down the result for eq. (14.94a):

$$\nabla\cdot\Big(\tilde{\boldsymbol{p}}\cdot\frac{\boldsymbol{r}}{r^3}\Big) \;=\; -\frac{(\ddot{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}}{c^2r^3}-\frac{3(\tilde{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}-\tilde{\boldsymbol{p}}r^2}{r^5} \tag{14.94e}$$

We thus obtain the following result for the gradient of the electric potential of our oscillating charge distribution:

$$\nabla\phi(\boldsymbol{r},t) \;=\; -\frac{1}{4\pi\epsilon_0 r^3}\left[\frac{\boldsymbol{r}(\ddot{\boldsymbol{p}}\boldsymbol{r})}{c^2}+\frac{3(\tilde{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}-\tilde{\boldsymbol{p}}r^2}{r^2}\right] \tag{14.95}$$

Let us now turn to the time derivative of the vector potential. In spherical coordinates, the vector potential from eq. (14.70) is:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0c^2}\frac{\dot{p}}{r}\big(\cos\vartheta\cdot\hat{\boldsymbol{e}}_r-\sin\vartheta\cdot\hat{\boldsymbol{e}}_\vartheta\big) \tag{14.96}$$

With $\hat{\boldsymbol{e}}_\vartheta=\hat{\boldsymbol{e}}_\varphi\times\hat{\boldsymbol{e}}_r$ we rewrite this as:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0c^2}\left[\frac{\dot{p}r}{r^2}\cos\vartheta\cdot\hat{\boldsymbol{e}}_r-\frac{\dot{p}r}{r^2}\sin\vartheta\cdot(\hat{\boldsymbol{e}}_\varphi\times\hat{\boldsymbol{e}}_r)\right] \tag{14.97}$$

Using $\boldsymbol{pr}=pr\cos\vartheta$, $\boldsymbol{p}\times\boldsymbol{r}=pr\sin\vartheta\cdot\hat{\boldsymbol{e}}_\varphi$, and $\hat{\boldsymbol{e}}_r=\boldsymbol{r}/r$ this equation becomes:

$$\boldsymbol{A}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0c^2r^3}\big[(\dot{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}-\big((\dot{\boldsymbol{p}}\times\boldsymbol{r})\times\boldsymbol{r}\big)\big] \tag{14.98}$$

The time derivative of this is:

$$\frac{\partial\boldsymbol{A}(\boldsymbol{r},t)}{\partial t} \;=\; \frac{1}{4\pi\epsilon_0c^2r^3}\big[(\ddot{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}-\big((\ddot{\boldsymbol{p}}\times\boldsymbol{r})\times\boldsymbol{r}\big)\big] \tag{14.99}$$

Using eq. (14.95) and eq. (14.99), we thus obtain the expression for $\boldsymbol{E}(\boldsymbol{r},t)$ according to eq. (14.89):

$$\boldsymbol{E}(\boldsymbol{r},t) \;=\; \frac{1}{4\pi\epsilon_0r^3}\left[\frac{3(\tilde{\boldsymbol{p}}\boldsymbol{r})\boldsymbol{r}-\tilde{\boldsymbol{p}}r^2}{r^2}+\frac{1}{c^2}\big((\ddot{\boldsymbol{p}}\times\boldsymbol{r})\times\boldsymbol{r}\big)\right] \tag{14.100}$$

Formally, the first term in eq. (14.100) is identical to eq. (3.55), the expression we obtained for the electric field of a static dipole in section 3.2.1. And indeed, for very small distances from our dipole, $r/c\ll 1$ holds, and with that $\tilde{\boldsymbol{p}}$ reduces to a static dipole moment. However, at greater distances from our oscillating charge distribution, the first term in eq. (14.100) rapidly becomes small, and the second term begins to dominate. If we are interested only in the electric field very far from the oscillating dipole, we can therefore neglect the first term in eq. (14.100). To graphically represent the potentials and fields we have just determined, we slightly rewrite our results from eq. (14.70), eq. (14.77),

eq. (14.88), and eq. (14.100) for $\boldsymbol{A}(\boldsymbol{r},t)$, $\boldsymbol{B}(\boldsymbol{r},t)$, $\phi(\boldsymbol{r},t)$, and $\boldsymbol{E}(\boldsymbol{r},t)$. To do this, we make use of the fact that $\omega = 2\pi f$, $c = \lambda f$, and $2\pi/\lambda = k$ imply:

$$\frac{\omega}{c} = \frac{2\pi}{\lambda} = k \qquad \text{and thus} \qquad \omega\left(t - \frac{r}{c}\right) = \omega t - kr \tag{14.101}$$

Next, we consider the fields at a fixed time, for which we choose $t = 0$. Let us begin with the vector potential and the magnetic field. The vector potential has only a $z$-component, given by eq. (14.70):

$$A_z(r, t = 0) = \frac{p_0}{4\pi\epsilon_0 c}\frac{k \cdot cos(kr)}{r} \tag{14.102}$$

Fig. 14.4a illustrates this behavior of $A_z$ according to eq. (14.102), where the symbols ⊙ and ⊛ indicate that the $A_z$ vector points out of or into the plane of the figure at the respective locations. At the origin, where the oscillating charge distribution is located, $A_z$ is undefined; the symbol at this point indicates that, in the immediate vicinity of the origin, the $A_z$ vector points out of the plane of the figure. For the magnetic field, we obtain from eq. (14.75):

$$\boldsymbol{B}(r, t = 0) = \frac{p_0}{4\pi\epsilon_0 c}\left[\frac{k \cdot cos(kr)}{r^2} + \frac{k^2 \cdot \sin(kr)}{r}\right] \sin\vartheta \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \tag{14.103}$$

If we set $\vartheta = \pi/2$ in this equation, we obtain the pattern of magnetic field lines in the $xy$-plane shown in fig. 14.4b for the time $t = 0$. These field lines are circles centered at the origin, with a direction of circulation that changes with the distance from the origin. The magnetic field lines plotted in fig. 14.4b correspond to the maximum values of $\boldsymbol{B}$ in the respective direction of circulation. As can easily be shown, the expression in square brackets takes on its extreme values at $kr = (n-1)\pi/2$, where $n$ is an integer. Between these maximum values of $\boldsymbol{B}$, the dashed circles in fig. 14.4b indicate where the magnetic field strength drops to zero. This occurs where the expression in the square brackets has its zeros. The first of these zeros lies at $r/\lambda \approx 0.5$, and the subsequent zeros lie increasingly close to multiples of $r/\lambda = 0.5$. In other words, the further away one is from the oscillating charge distribution, the more precisely the distance between two of these zeros corresponds to the condition $k\Delta r = \pi$. The expression in the square brackets depends only on the magnitude of the distance. This means that the factor in eq. (14.103) describes, in each plane parallel to the $xy$-plane, a circular ring belonging to a spherical shell on which this factor has a constant value. The direction of circulation of the magnetic field on the spherical shell remains the same, but due to the $\sin\vartheta$ term, the magnetic field strength decreases with $\sin\vartheta$ for $\vartheta \neq \pi/2$.

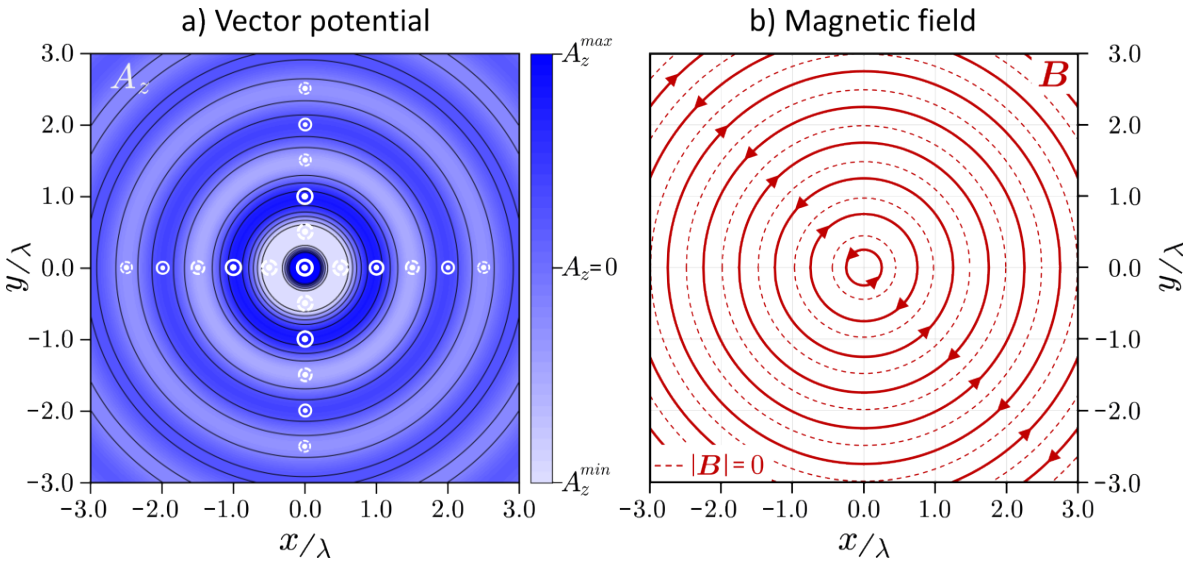



**Fig. 14.4:** Vector potential $\boldsymbol{A}$ (a) and magnetic field $\boldsymbol{B}$ (b) of the oscillating charge distribution in fig. 14.3 at time $t = 0$ according to eq. (14.102) and eq. (14.103) in the $xy$-plane. ($\odot = A_z$ parallel to the $z$-axis; $\otimes = A_z$ antiparallel to the $z$-axis.)

Vector potential and magnetic field lines in fig. 14.4 show the snapshot at time $t = 0$. However, the vector potential of the oscillating charge distribution represents an outgoing spherical wave with angular frequency $\omega$. Therefore, the $A_z$ component oscillates at every location between a maximum negative and a maximum positive value, i.e., the orientation of $A_z$ oscillates over time between being parallel and antiparallel to the $z$-axis. As shown in fig. 14.4b for $t = 0$, this gives rise to magnetic field waves with alternating directions of circulation that propagate radially outward over time in planes normal to $A_z$.

Let us now consider the electric field. For $\phi$, using eq. (14.88) at time $t = 0$, a brief calculation yields the expression:

$$\phi(r, \vartheta, t = 0) \;=\; -\frac{p_0}{4\pi\epsilon_0} \cdot \left[ \frac{\sin(kr)}{r^2} - \frac{k \cdot \cos(kr)}{r} \right] \cdot \cos\vartheta \tag{14.104}$$

Fig. 14.5a shows the contour plot of $\phi$ in the $yz$-plane derived from eq. (14.104) at time $t = 0$. Distances from the origin are again plotted as multiples of the wavelength $\lambda$, i.e., fig. 14.5a depicts the electric potential around the origin in the $yz$-plane over three wavelengths. To represent the electric field, we first write out the vector components of $\boldsymbol{E}$ from eq. (14.100) in spherical coordinates. For the first term in eq. (14.100), the near field, we can simply adopt the results for $E_r$ and $E_\vartheta$ from our discussion of the static dipole, i.e., eq. (3.59) in section 3.2.1. As for the second term in eq. (14.100), the far field, it can be quickly shown that the radial component vanishes, leaving only $E_\vartheta$. Taken together, the result is:

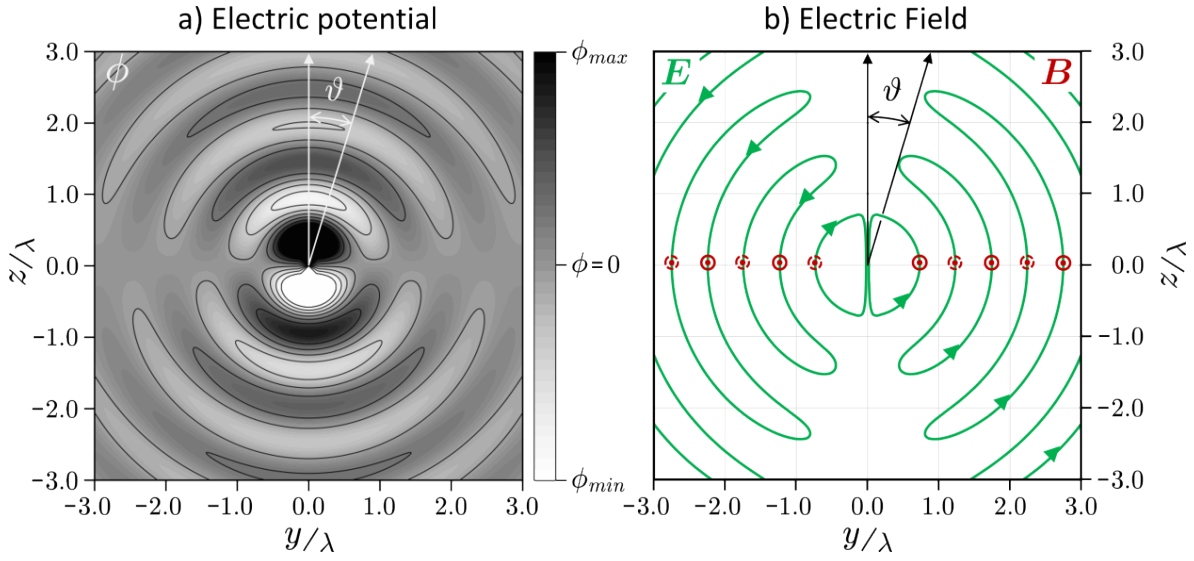



**Fig. 14.5:** Electric potential $\phi$ (a) and electric field $\boldsymbol{E}$ (b) of the oscillating charge distribution in fig. 14.3 at time $t = 0$ according to eq. (14.104) and eq. (14.107) in the $yz$-plane. The potential and electric field are rotationally symmetric with respect to rotations about the $z$-axis. (In b): $\odot = B_x$ parallel to the $x$-axis; $\otimes = B_x$ antiparallel to the $x$-axis; see also fig. 14.4b.)

$$E_r = \frac{2\cos\vartheta}{4\pi\epsilon_0 r^3}\left(p + \frac{r}{c}\dot{p}\right) \qquad \text{and} \qquad E_\vartheta = \frac{\sin\vartheta}{4\pi\epsilon_0 r^3}\left(p + \frac{r}{c}\dot{p} + \frac{r^2}{c^2}\ddot{p}\right) \tag{14.105}$$

After substituting the dipole moment from eq. (14.69), as well as its first and second time derivatives, we obtain the following for $E_r$ and $E_\vartheta$ at time $t = 0$:

$$E_r(r,\vartheta,t=0) = \frac{2\cos\vartheta}{4\pi\epsilon_0 r^3}\Big[-sin(kr) + kr\cos(kr)\Big] \tag{14.106a}$$

$$E_\vartheta(r,\vartheta,t=0) = \frac{\sin\vartheta}{4\pi\epsilon_0 r^3}\Big[-sin(kr) + kr\cos(kr) + (kr)^2\sin(kr)\Big] \tag{14.106b}$$

This gives us with

$$E_y = E_r\sin\vartheta + E_\vartheta\cos\vartheta \qquad \text{and} \qquad E_z = E_r\cos\vartheta - E_\vartheta\sin\vartheta \tag{14.107}$$

the representation of the field lines of $\boldsymbol{E}$ in the $yz$-plane shown in fig. 14.5b. For very small distances $r \ll \lambda$, the first term in eq. (14.100) dominates; consequently, in a snapshot such as the one shown in fig. 14.5b for $t = 0$, the field line pattern around the origin is the same as that encountered in the discussion of the static dipole in section 3.2.1. However, the charge distribution here is not static, instead, it oscillates, thereby constituting an alternating current. As the direction of the current changes periodically, the direction of the associated dipole moment changes periodically as well. As we know, electric field lines originate at the positive end of the dipole moment and terminate at its negative

end. Here, however, the positive and negative ends of the dipole alternate periodically with the direction of the oscillating dipole moment. This causes field lines to periodically detach from the oscillating dipole, thereby making way for new field lines that run in the now-reversed direction. Hence, within a single oscillation of the dipole, two sets of field lines detach, each characterized by an opposite direction of rotation of the field vector. Consequently, a full oscillation period elapses between two sets of detached field lines sharing the same orientation, as shown in fig. 14.5b. In other words, the distance between them corresponds exactly to one wavelength $\lambda$ of the oscillation, as can be seen from fig. 14.5b. From fig. 14.4b, we can transfer the direction of the magnetic field vector on the $y$-axis, i.e. for $x = 0$, to fig. 14.5b. As can be seen, at these points on the $y$-axis, the relationship $\boldsymbol{k} = \boldsymbol{E} \times \boldsymbol{B}$ holds for the propagation direction $\boldsymbol{k}$ of the electromagnetic wave (thumb) and the directions of $\boldsymbol{E}$ (index finger) and $\boldsymbol{B}$ (middle finger), as is required for an electromagnetic wave. For $r \approx \lambda$, the quasistatic electric field, which dominates for $r \ll \lambda$, transitions into an electromagnetic wave of the type described by the second term in eq. (14.100). For distances $r \gg \lambda$, not shown here, the second term in eq. (14.100) dominates, resulting in an electromagnetic spherical wave of the kind we examined in section 13.1.3.

### 14.2.2 Electromagnetic Radiation

We consider the electromagnetic field of the oscillating dipole for $r \gg \lambda$, i.e., in the far-field. We do this by neglecting all terms in eq. (14.75) for $\boldsymbol{B}$ and in eq. (14.100) for $\boldsymbol{E}$ that fall off faster than $r^{-1}$, retaining only those terms that scale as $r^{-1}$. For the electric field, this means that the radial term $E_r$ of $\boldsymbol{E}$ in eq. (14.105) vanishes. Therefor, in the far-field approximation, $\boldsymbol{E}$ possesses only an $E_\vartheta$ component. Furthermore, eq. (14.75) tells us that $\boldsymbol{B}$ has only a $\varphi$ component anyway. Consequently, the cross product of the field vectors $\boldsymbol{E}$ and $\boldsymbol{B}$ reduces to:

$$\boldsymbol{E} \times \boldsymbol{B} \;=\; E_\vartheta \hat{\boldsymbol{e}}_\vartheta \times B_\varphi \hat{\boldsymbol{e}}_\varphi \;=\; E_\vartheta \cdot B_\varphi \cdot \hat{\boldsymbol{e}}_r \tag{14.108}$$

Since we neglect all terms that fall off faster than $r^{-1}$ in the far field, we obtain the following for $E_\vartheta$ and $B_\varphi$ using eq. (14.75) and eq. (14.105):

$$E_\vartheta \;=\; \frac{\sin\vartheta}{4\pi\epsilon_0 r c^2} \cdot \ddot{p} \qquad \text{and} \qquad B_\varphi \;=\; \frac{\sin\vartheta}{4\pi\epsilon_0 r c^3} \cdot \ddot{p} \tag{14.109}$$

For the energy flux density $\boldsymbol{S}$ in the far field, it thus follows from eq. (14.108) and eq. (14.109), as well as with $\ddot{p} = -\omega^2 p_0 \sin(\omega t - kr)$ from eq. (14.69):

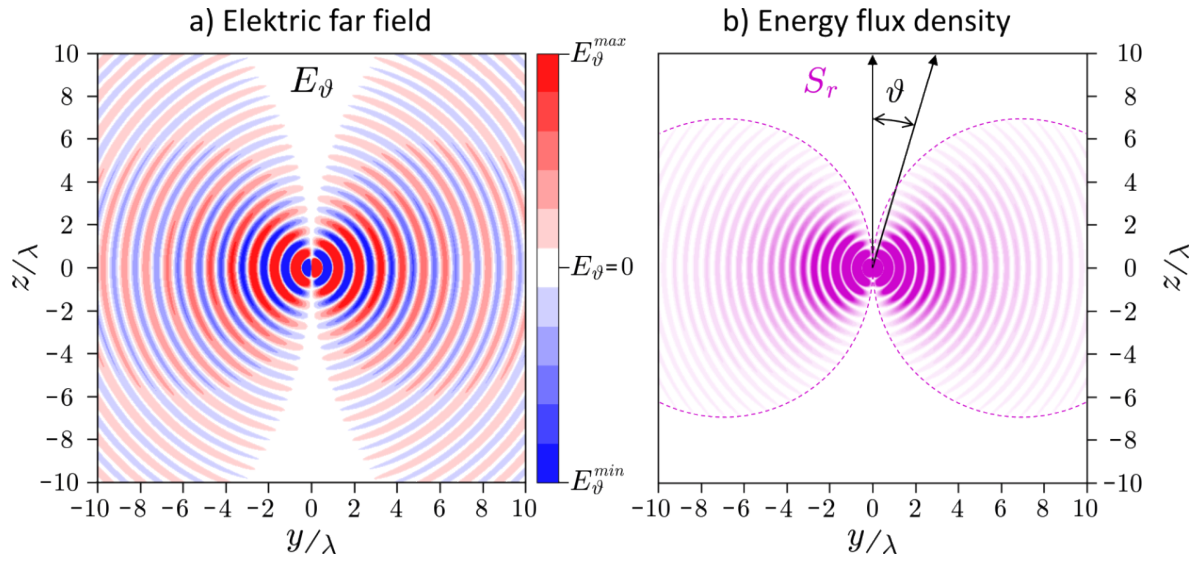



**Fig. 14.6:** Electric far-field ($E_\vartheta$) (a) and corresponding energy flux density ($S_r$) (b) of the oscillating charge distribution from fig. 14.3 at time $t = 0$ in the $yz$-plane, according to eq. (14.109) and eq. (14.111). $E_\vartheta$ and $S_r$ are rotationally symmetric with respect to rotations about the $z$-axis.

$$\boldsymbol{S} = \frac{1}{\mu_0}\boldsymbol{E} \times \boldsymbol{B} = \frac{1}{\mu_0} \cdot \frac{\omega^4 p_0^2}{16\pi^2 \epsilon_0^2 c^5} \cdot \frac{\sin^2\vartheta}{r^2} \cdot \sin^2(\omega t - kr) \cdot \hat{\boldsymbol{e}}_r \tag{14.110}$$

We can simplify this expression further using $\mu_0\epsilon_0 = 1/c^2$:

$$\boldsymbol{S} = \frac{\omega^4 p_0^2}{16\pi^2 \epsilon_0 c^3} \cdot \frac{\sin^2\vartheta}{r^2} \cdot \sin^2(\omega t - kr) \cdot \hat{\boldsymbol{e}}_r = S_r \cdot \hat{\boldsymbol{e}}_r \tag{14.111}$$

As expected, the energy flux density of an electromagnetic spherical wave has only a radial component $S_r(r, \vartheta, t)$. Fig. 14.6 shows $E_\vartheta$, i.e., the electric far field from eq. (14.109), and $S_r$ at time $t = 0$. Because of

$$\sin^2\alpha = \frac{1}{2} - \frac{1}{2}\cos(2\alpha) \qquad (*)$$

the radial component $S_r$ in fig. 14.6b exhibits twice the period of $E_\vartheta$ in fig. 14.6a and, unlike $E_\vartheta$, is always positive. As indicated by the dashed circles in fig. 14.6b, the angular distribution of the energy flux density corresponds to a figure-eight shape, which in this case is composed of two perfect circles. The reason for the latter is the implicit assumption that the dimensions of our oscillating charge distribution are small compared to the wavelength of the electromagnetic radiation. However, if the distance $z_0$ in fig. 14.3 becomes comparable to the radiated wavelength, these circles flatten out along the $z$-coordinate, and the figure-eight shape transforms into a lobe-shaped angular distribution. In most cases, it is not the time evolution of the energy flux density in eq. (14.111) that is of interest, but rather its time average. As can be seen from $(*)$, the time average of

$\sin^2(\omega t - kr)$ is exactly $1/2$. Thus, the time-averaged energy flux density of the radial component $S_r$ is given by:

$$\langle S_r \rangle \;=\; \frac{\omega^4 p_0}{32\pi^2 \epsilon_0 c^3} \cdot \frac{\sin^2 \vartheta}{r^2} \tag{14.112}$$

The fact that $\langle S_r \rangle$ varies as $r^{-2}$ tells us that the total energy flux through any spherical surface of area $4\pi r^2$ enclosing the oscillating charge distribution is constant. However, since $\langle S_r \rangle$ also depends on $\vartheta$, we cannot simply multiply $\langle S_r \rangle$ by the spherical surface area $4\pi r^2$ to obtain the power $P$ radiated into the far field. Instead, we must explicitly integrate $\langle S_r \rangle$ over the spherical shell:

$$P \;=\; \int \langle S_r \rangle r^2 \sin\vartheta \,\mathrm{d}\vartheta \,\mathrm{d}\varphi \;=\; \frac{\omega^4 p_0}{32\pi^2 \epsilon_0 c^3} \cdot \int_0^{2\pi} \int_0^{\pi} \sin^3 \vartheta \,\mathrm{d}\vartheta \,\mathrm{d}\varphi \tag{14.113}$$

Integrating over $\varphi$ yields a factor of $2\pi$, and integrating over $\vartheta$ yields a factor of $4/3$, and we obtain the result:

$$P \;=\; \frac{\omega^4 p_0^2}{12\pi \epsilon_0 c^3} \tag{14.114}$$

The power radiated by an oscillating charge distribution, or an oscillating dipole, therefore scales with the fourth power of the angular frequency $\omega$, or with $\lambda^{-4}$. The electromagnetic radiation emitted by an oscillating charge distribution of the type we have considered here is monochromatic, reflecting the frequency at which the charge distribution or dipole oscillates. It arises from the inherent periodic acceleration and deceleration necessary to keep the charge distribution oscillating at a given frequency. However, any acceleration or deceleration of a charge, not merely the motion of an oscillating charge distribution, results in the emission of electromagnetic radiation. In an X-ray tube, to use a practical example many are likely familiar with, high-energy electrons with energies in the range of $10 - 30\,\mathrm{kV}$ strike a metal plate, typically made of tungsten due to its high atomic number and high melting point. The energy released as the electrons are forced to decelerate in the Coulomb fields of the tungsten atoms is emitted as X-ray radiation across a broad frequency continuum. Another less well-known, yet equally highly relevant example is the acceleration of electrons to near-light speed along a circular path. Motion along a circular path is always accelerated motion, and electrons moving at near-light speed in such a path emit highly intense, collimated electromagnetic radiation tangentially to the circular path within a narrow angular range. This radiation produced by very large circular particle accelerators called synchrotrons, hence the name synchrotron radiation, has become an important tool in many areas of scientific research and technical development.

# Appendix A

This appendix provides a brief overview of common orthogonal coordinate systems, the conversions between them, and the differential operators, such as gradient, divergence, curl, and the Laplace operator, expressed in these systems. It also briefly explains a few other useful relationships.

## Orthogonal Coordinate Systems

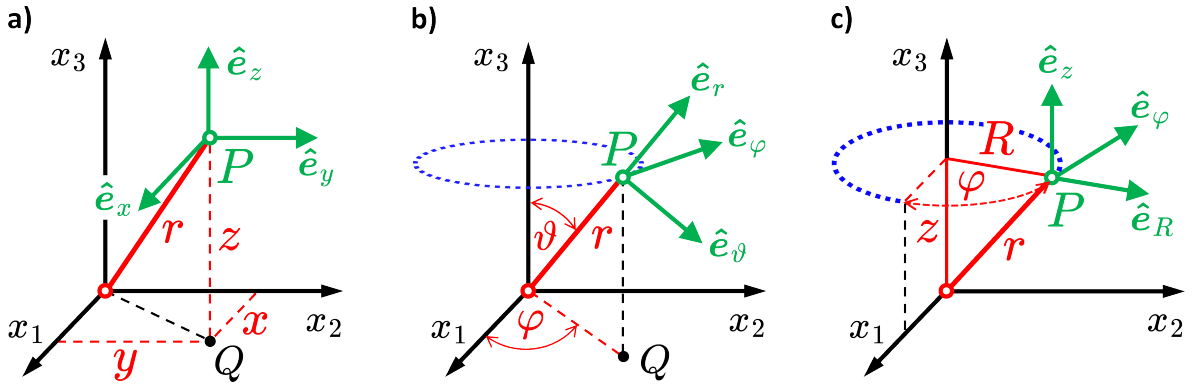



**Fig. A.1:** Nomenclature and unit vectors of three orthogonal coordinate systems: (a) Cartesian coordinates; (b) spherical coordinates; and (c) cylindrical coordinates.

### Cartesian coordinates

In Cartesian coordinates, the position vector of a point in space $\boldsymbol{r}$, as sketched in fig. A.1a, is given by:

$$\boldsymbol{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \tag{A.1}$$

The three unit vectors along the orthogonal coordinate directions $x = x_1$, $y = x_2$, and $z = x_3$ are:

$$\hat{\boldsymbol{\imath}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad ; \quad \hat{\boldsymbol{\jmath}} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad ; \quad \hat{\boldsymbol{k}} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{A.2}$$

### Spherical coordinates

In spherical coordinates, also referred to as spherical polar coordinates, the position vector $\boldsymbol{r}$ of a point in space, as sketched in fig. A.1b, is given by:

$$\boldsymbol{r} = r \cdot \begin{pmatrix} \sin\vartheta\cos\varphi \\ \sin\vartheta\sin\varphi \\ \cos\vartheta \end{pmatrix} \tag{A.3}$$

The unit vectors orthogonal at each point $P(r, \vartheta, \varphi)$ are:

$$\hat{\boldsymbol{e}}_r = \begin{pmatrix} \sin\vartheta\cos\varphi \\ \sin\vartheta\sin\varphi \\ \cos\vartheta \end{pmatrix} \quad ; \quad \hat{\boldsymbol{e}}_\vartheta = \begin{pmatrix} \cos\vartheta\cos\varphi \\ \cos\vartheta\sin\varphi \\ -\sin\vartheta \end{pmatrix} \quad ; \quad \hat{\boldsymbol{e}}_\varphi = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \tag{A.4}$$

**Cylindrical coordinates**

In cylindrical coordinates, the position vector of a point in space $\boldsymbol{r}$, as sketched in fig. A.1c, is given by:

$$\boldsymbol{r} = \begin{pmatrix} R\cos\varphi \\ R\sin\varphi \\ z \end{pmatrix} \tag{A.5}$$

The unit vectors orthogonal at every point $P(R, \varphi, z)$ are:

$$\hat{\boldsymbol{e}}_R = \begin{pmatrix} \cos\varphi \\ \sin\varphi \\ 0 \end{pmatrix} \quad ; \quad \hat{\boldsymbol{e}}_\varphi = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix} \quad ; \quad \hat{\boldsymbol{e}}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{A.6}$$

Anyone who has read carefully will have noticed that, while we speak of unit vector directions in the case of Cartesian coordinates, we do not do so for spherical and cylindrical coordinates. The reason for this is evident in fig. A.1. Only the unit vectors in the Cartesian coordinate system always point in the same direction. In spherical coordinates, all three unit vectors point in different directions at every point, whereas in cylindrical coordinates, two of the three unit vectors point in different directions at every point. Spherical and cylindrical coordinates are therefore referred to as curvilinear coordinate systems. However, just as with rectangular Cartesian coordinates, the unit vectors in curvilinear coordinate systems are orthogonal to each other at every point in space, as can easily be verified by calculation.

Choosing $\vartheta = \pi/2$ in spherical coordinates or $z = 0$ in cylindrical coordinates yields, in both cases, the two-dimensional coordinate representation of polar coordinates in the plane. We can switch between these three coordinate systems by writing out the corresponding mappings between the respective coordinates. For example, when switching from Cartesian coordinates to spherical or cylindrical coordinates, it is important to note that:

$$x = x(r,\vartheta,\varphi) = x(R,\varphi,z)$$
$$y = y(r,\vartheta,\varphi) = y(R,\varphi,z)$$
$$z = z(r,\vartheta,\varphi) = z(R,\varphi,z)$$

We could just as well have written down the mapping for a conversion from spherical coordinates to Cartesian or cylindrical coordinates, or from cylindrical coordinates to Cartesian or polar coordinates. When switching from one triple of orthogonal coordinates to another, the mapping is unique at every point in space. Let us first consider the transformation from Cartesian to spherical coordinates:

$$x = r\sin\vartheta\cos\varphi \quad ; \quad y = r\sin\vartheta\sin\varphi \quad ; \quad z = r\cos\vartheta$$

Thus, for $r$, $\vartheta$, and $\varphi$, the following applies:

$$r = \sqrt{x^2+y^2+z^2} \quad ; \quad \vartheta = \arccos\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right) \quad ; \quad \varphi = \arctan\left(\frac{y}{x}\right)$$

As can be quickly verified, this implies the following for the partial derivatives of $x(r,\vartheta,\varphi)$, $y(r,\vartheta,\varphi)$, and $z(r,\vartheta,\varphi)$:

$$\frac{\partial x}{\partial r} = \sin\vartheta\cos\varphi \quad ; \quad \frac{\partial x}{\partial \vartheta} = r\cos\vartheta\cos\varphi \quad ; \quad \frac{\partial x}{\partial \varphi} = -r\sin\vartheta\sin\varphi \tag{A.7a}$$

$$\frac{\partial y}{\partial r} = \sin\vartheta\sin\varphi \quad ; \quad \frac{\partial y}{\partial \vartheta} = r\cos\vartheta\sin\varphi \quad ; \quad \frac{\partial y}{\partial \varphi} = r\sin\vartheta\cos\varphi \tag{A.7b}$$

$$\frac{\partial z}{\partial r} = \cos\vartheta \quad ; \quad \frac{\partial z}{\partial \vartheta} = -r\sin\vartheta \quad ; \quad \frac{\partial z}{\partial \varphi} = 0 \tag{A.7c}$$

Conversely, for the partial derivatives of $r(x,y,z)$, $\vartheta(x,y,z)$, and $\varphi(x,y,z)$ applies:

$$\frac{\partial r}{\partial x} = \sin\vartheta\cos\varphi \quad ; \quad \frac{\partial r}{\partial y} = \sin\vartheta\sin\varphi \quad ; \quad \frac{\partial r}{\partial z} = \cos\vartheta \tag{A.8a}$$

$$\frac{\partial \vartheta}{\partial x} = \frac{\cos\vartheta\cos\varphi}{r} \quad ; \quad \frac{\partial \vartheta}{\partial y} = \frac{\cos\vartheta\sin\varphi}{r} \quad ; \quad \frac{\partial \vartheta}{\partial z} = -\frac{\sin\vartheta}{r} \tag{A.8b}$$

$$\frac{\partial \varphi}{\partial x} = -\frac{\sin\varphi}{r\sin\vartheta} \quad ; \quad \frac{\partial \varphi}{\partial y} = \frac{\cos\varphi}{r\sin\vartheta} \quad ; \quad \frac{\partial \varphi}{\partial z} = 0 \tag{A.8c}$$

For the transformation from Cartesian coordinates to cylindrical coordinates

$$x = R\cos\varphi \quad ; \quad y = R\sin\varphi \quad ; \quad z = z$$

one requires the corresponding derivatives of $x(R,\varphi,z)$, $y(R,\varphi,z)$, and $z(R,\varphi,z)$, as well as the derivatives of $R(x,y,z)$, $\varphi(x,y,z)$, and $z(x,y,z)$ for the inverse transformation.

## Coordinate Transformation

Equations eq. (A.7a) through eq. (A.7c) and eq. (A.8a) through eq. (A.8c) contain the elements of a matrix known as the Jacobian matrix. This matrix, also referred to as the functional matrix, describes how the infinitesimal volume element of a coordinate system scales under the following mapping:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \longmapsto \boldsymbol{f}(x_1, \dots x_n) = \begin{pmatrix} f_1(x_1, \dots x_n) \\ \vdots \\ f_m(x_1, \dots x_n) \end{pmatrix}$$

We can illustrate why this is the case using a simple example. To do so, we consider the transformation:

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} \longmapsto \boldsymbol{f}(u, v, w) = \begin{pmatrix} x(u, v, w) \\ y(u, v, w) \\ z(u, v, w) \end{pmatrix}$$

Fig. A.2 illustrates how a small cuboid in the $(u, v, w)$ coordinate system is mapped to the $(x, y, z)$ coordinate system by such a transformation. In general, such a transformation is non-linear, and the result is a curvilinear volume element of arbitrary shape, as sketched in fig. A.2.

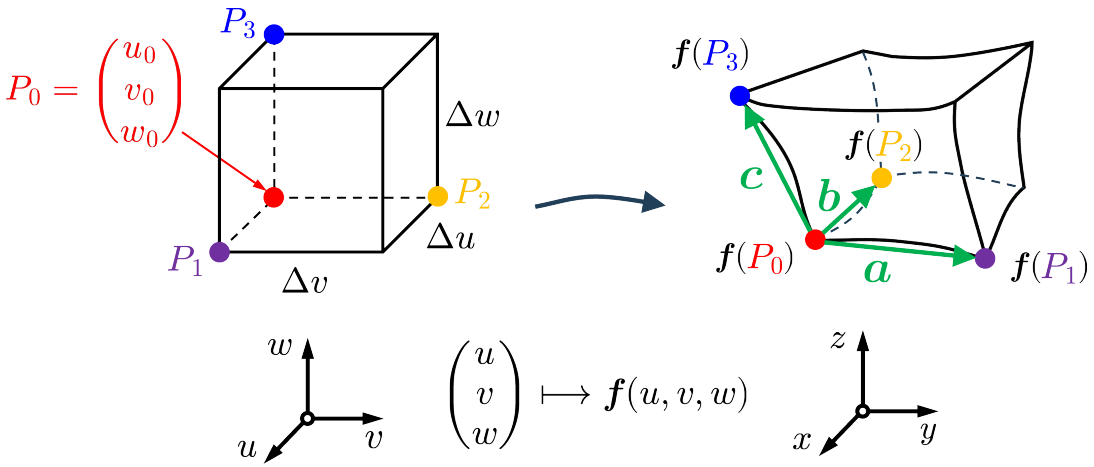



**Fig. A.2:** Transformation of a small cuboid with side lengths $\Delta u$, $\Delta v$, and $\Delta w$ under the mapping from a coordinate system $(u, v, w)$ to a coordinate system $(x, y, z)$.

Fig. A.2 shows how four vertices $P_i$ of the cuboid with side lengths $\Delta u$, $\Delta v$, and $\Delta w$ in the $(u, v, w)$ coordinate system are mapped to the corresponding vertices $\boldsymbol{f}(P_i)$ of the curvilinear volume element in $(x, y, z)$:

$$
\begin{aligned}
P_0 = (u_0, v_0, w_0) &\quad\mapsto\quad \boldsymbol{f}(u_0, v_0, w_0) \\
P_1 = (u_0 + \Delta u, v_0, w_0) &\quad\mapsto\quad \boldsymbol{f}(u_0 + \Delta u, v_0, w_0) \\
P_2 = (u_0, v_0 + \Delta v, w_0) &\quad\mapsto\quad \boldsymbol{f}(u_0, v_0 + \Delta v, w_0) \\
P_2 = (u_0, v_0, w_0 + \Delta w) &\quad\mapsto\quad \boldsymbol{f}(u_0, v_0, w_0 + \Delta w)
\end{aligned}
$$

Since we are considering a very small volume element, we can linearize and estimate the size of the curvilinear volume element using the vectors $\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$ shown in fig. A.2. We obtain the direction and length of these vectors by subtracting the starting point from the respective endpoint of the vector, that is:

$$
\begin{aligned}
\boldsymbol{a} &= \boldsymbol{f}(u_0 + \Delta u, v_0, w_0) - \boldsymbol{f}(u_0, v_0, w_0) \approx \frac{\partial \boldsymbol{f}}{\partial u} \cdot \Delta u = \left( \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right) \cdot \Delta u \\
\boldsymbol{b} &= \boldsymbol{f}(u_0, v_0 + \Delta v, w_0) - \boldsymbol{f}(u_0, v_0, w_0) \approx \frac{\partial \boldsymbol{f}}{\partial v} \cdot \Delta v = \left( \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right) \cdot \Delta v \\
\boldsymbol{c} &= \boldsymbol{f}(u_0, v_0, w_0 + \Delta w) - \boldsymbol{f}(u_0, v_0, w_0) \approx \frac{\partial \boldsymbol{f}}{\partial w} \cdot \Delta w = \left( \frac{\partial x}{\partial w}, \frac{\partial y}{\partial w}, \frac{\partial z}{\partial w} \right) \cdot \Delta w
\end{aligned}
$$

From vector calculus, we know that the volume spanned by three vectors $\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$ is the magnitude of their so-called scalar triple product. We can therefore approximate the curvilinear volume element by:

$$
\Delta \widetilde{V} = |(\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c}| = \left| \det \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \right|
$$

The vectors $\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$ appear in the columns of this determinant. The side lengths $\Delta u$, $\Delta v$, and $\Delta w$ of the cuboid in $(u, v, w)$ are all positive. We can therefore factor them out of the absolute value when substituting the three vectors $\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$ from above, and thereby obtain:

$$
\Delta \widetilde{V} = \left| \left( \frac{\partial \boldsymbol{f}}{\partial u} \times \frac{\partial \boldsymbol{f}}{\partial v} \right) \cdot \frac{\partial \boldsymbol{f}}{\partial w} \right| \cdot \Delta u \Delta v \Delta w = \left| \det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{bmatrix} \right| \cdot \Delta u \Delta v \Delta w
$$

The determinant in this equation is the so-called volume distortion factor, also known as the functional determinant or Jacobian determinant. If we consider the infinitesimal volume element, then $\Delta u \Delta v \Delta w$ becomes $\mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w$, and we obtain:

$$
\mathrm{d}\widetilde{V} = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| \cdot \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w = \left| \det \left( \underline{\underline{J}}_f(u, v, w) \right) \right| \cdot \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w
$$

Here, we have used two common abbreviated notations for the volume distortion factor, one of them being $\underline{\underline{J}}_f(u,v,w)$, i.e., the Jacobian matrix already mentioned above. For example, when switching from Cartesian coordinates to spherical coordinates for the volume integration of a function $g(x,y,z)$ over a volume $\Omega_{xyz}$ to an integration over the corresponding volume $\Omega_{r\vartheta\varphi}$, one must transform not only the limits of integration accordingly but also, crucially, the volume distortion factor of the differential volume element, that is:

$$\int_{\Omega_{xyz}} g(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \;=\; \int_{\Omega_{r\vartheta\varphi}} g\big(x(r,\vartheta,\varphi),y(r,\vartheta,\varphi),z(r,\vartheta,\varphi)\big)\left|\frac{\partial(x,y,z)}{\partial(r,\vartheta,\varphi)}\right|\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$$

In general, the Jacobian matrix $\underline{\underline{J}}_f(x_1,\dots x_n)$ of a vector-valued function $\boldsymbol{f}(x_1,\dots,x_n)$, which transforms a point with coordinates $(x_1,\dots,x_n)$ into a point with coordinates $(f_1,\dots,f_n)$ is given by:

$$\boldsymbol{f}(x_1,\dots x_n) \;=\; \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix} \quad\Rightarrow\quad \underline{\underline{J}}_f(x_1,\dots x_n) \;=\; \begin{pmatrix} \dfrac{\partial f_1}{\partial x_1} & \cdots & \dfrac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial f_m}{\partial x_1} & \cdots & \dfrac{\partial f_m}{\partial x_n} \end{pmatrix} \tag{A.9}$$

In this $n \times m$ Jacobian matrix, the first row contains the gradient of $f_1$ in row form, the second row contains the gradient of $f_2$ in row form, and so on, up to the last row, which contains the gradient of $f_m$ in row form. A comparison of eq. (A.9) with eq. (2.25) shows that the Jacobian matrix is the directional gradient of $\boldsymbol{f}(x_1,\dots,x_n)$. An infinitesimal change in the coordinates $x_1,\dots,x_n$ has the components $\mathrm{d}x_1,\dots,\mathrm{d}x_n$. However, since the new coordinates $f_i$ are functions of $x_1,\dots,x_n$, mixed terms must be taken into account for the corresponding infinitesimal change in the coordinates $(f_1,\dots,f_n)$. The change in direction in the infinitesimal neighborhood of the point $(f_1,\dots,f_n)$ remains linear, but each component $f_i$ undergoes changes not only in the direction of $x_i$ but also in directions other than $x_i$. As an example, let us calculate the Jacobian matrix for the transformation from Cartesian coordinates to spherical coordinates. Here, using eq. (A.7a) through eq. (A.7c), we obtain:

$$\underline{\underline{J}}_f(r,\vartheta,\varphi) \;=\; \frac{\partial(x,y,z)}{\partial(r,\vartheta,\varphi)} \;=\; \begin{bmatrix} \sin\vartheta\cos\varphi & r\cos\vartheta\cos\varphi & -r\sin\vartheta\sin\varphi \\ \sin\vartheta\sin\varphi & r\cos\vartheta\sin\varphi & r\sin\vartheta\cos\varphi \\ \cos\vartheta & -r\sin\vartheta & 0 \end{bmatrix}$$

The magnitude of the determinant of this Jacobian matrix is:

$$\left|\det\left(\underline{\underline{J}}_f(r,\vartheta,\varphi)\right)\right| \;=\; r^2\sin\vartheta$$

We have thus derived the volume distortion factor that must be taken into account when transitioning from Cartesian coordinates to spherical coordinates. During volume integration, the angle $\vartheta$ ranges from 0 to $\pi$, so $\sin\vartheta \geq 0$ always holds. Furthermore, since $r$ is always positive, we can omit the absolute value bars for the transformation, from which follows:

$$\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \longmapsto r^2 sin\vartheta\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$$

Similarly, one obtains the distortion factor for the transformation from Cartesian to cylindrical coordinates, or one can calculate the distortion factor for any transformation between orthogonal coordinate systems.

## Gradient, Divergence, and Curl

Next, we examine how the three differential operators, i.e., gradient, divergence, and curl, behave under coordinate transformations. The behavior of the Laplace operator under coordinate transformation then follows automatically once we understand that of the divergence and gradient. Let us start with the gradient. We already have everything needed to express the gradient operator in spherical coordinates. To do this, we write out the gradient of the scalar function $U = U(r, \vartheta, \varphi)$ explicitly in terms of Cartesian unit vectors and obtain:

$$\begin{aligned}
\mathrm{grad}\,U &= \frac{\partial U}{\partial x}\cdot\hat{\boldsymbol{i}} + \frac{\partial U}{\partial y}\cdot\hat{\boldsymbol{j}} + \frac{\partial U}{\partial z}\cdot\hat{\boldsymbol{k}} \\
&= \left(\frac{\partial U}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial U}{\partial \vartheta}\frac{\partial \vartheta}{\partial x} + \frac{\partial U}{\partial \varphi}\frac{\partial \varphi}{\partial x}\right)\cdot\hat{\boldsymbol{i}} + \left(\frac{\partial U}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial U}{\partial \vartheta}\frac{\partial \vartheta}{\partial y} + \frac{\partial U}{\partial \varphi}\frac{\partial \varphi}{\partial y}\right)\cdot\hat{\boldsymbol{j}} + \\
&\quad \left(\frac{\partial U}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial U}{\partial \vartheta}\frac{\partial \vartheta}{\partial z} + \frac{\partial U}{\partial \varphi}\frac{\partial \varphi}{\partial z}\right)\cdot\hat{\boldsymbol{k}}
\end{aligned}$$

Now we simply need to substitute the partial derivatives of $r$, $\vartheta$, and $\varphi$ with respect to Cartesian coordinates, calculated above from eq. (A.7a), eq. (A.7b), and eq. (A.7c), and perform some calculations. In doing so, we group together the partial derivatives of $U$ with respect to $r$, $\vartheta$, and $\varphi$ and obtain:

$$\begin{aligned}
\mathrm{grad}\,U &= \frac{\partial U}{\partial r}\left(\sin\vartheta\cos\varphi\cdot\hat{\boldsymbol{i}} + \sin\vartheta\sin\varphi\cdot\hat{\boldsymbol{j}} + \cos\vartheta\cdot\hat{\boldsymbol{k}}\right) + \\
&\quad \frac{1}{r}\frac{\partial U}{\partial \vartheta}\left(\cos\vartheta\cos\varphi\cdot\hat{\boldsymbol{i}} + \cos\vartheta\sin\varphi\cdot\hat{\boldsymbol{j}} + \sin\vartheta\cdot\hat{\boldsymbol{k}}\right) + \\
&\quad \frac{1}{r\sin\vartheta}\frac{\partial U}{\partial \varphi}\left(-\sin\varphi\cdot\hat{\boldsymbol{i}} + \cos\varphi\cdot\hat{\boldsymbol{j}} + 0\cdot\hat{\boldsymbol{k}}\right)
\end{aligned}$$

A comparison with eq. (A.4) shows that the terms in parentheses are simply the spherical coordinate unit vectors, hence:

$$\operatorname{grad} U = \frac{\partial U}{\partial r} \cdot \hat{\boldsymbol{e}}_r + \frac{1}{r} \frac{\partial U}{\partial \vartheta} \cdot \hat{\boldsymbol{e}}_\vartheta + \frac{1}{r \sin \vartheta} \frac{\partial U}{\partial \varphi} \cdot \hat{\boldsymbol{e}}_\varphi \tag{A.10}$$

We have thus derived the expression for the gradient operator in spherical coordinates. Naturally, we can proceed in the same way to derive the divergence operator in spherical coordinates. To do this, we write the divergence of a vector quantity $\boldsymbol{A}$ as

$$\begin{aligned} \operatorname{div} \boldsymbol{A} &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \\ &= \left( \frac{\partial A_x}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial A_x}{\partial \vartheta} \frac{\partial \vartheta}{\partial x} + \frac{\partial A_x}{\partial \varphi} \frac{\partial \varphi}{\partial x} \right) + \left( \frac{\partial A_y}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial A_y}{\partial \vartheta} \frac{\partial \vartheta}{\partial y} + \frac{\partial A_y}{\partial \varphi} \frac{\partial \varphi}{\partial y} \right) + \\ &\quad \left( \frac{\partial A_z}{\partial r} \frac{\partial r}{\partial z} + \frac{\partial A_z}{\partial \vartheta} \frac{\partial \vartheta}{\partial z} + \frac{\partial A_z}{\partial \varphi} \frac{\partial \varphi}{\partial z} \right) \end{aligned}$$

To evaluate this equation, we need to know the components $A_x$, $A_y$, and $A_z$ as functions of the corresponding quantities in spherical coordinates, namely, $A_r$, $A_\vartheta$, and $A_\varphi$. For the vector quantity $\boldsymbol{A}$, the following applies:

$$\boldsymbol{A} = A_x \cdot \hat{\boldsymbol{i}} + A_y \cdot \hat{\boldsymbol{j}} + A_z \cdot \hat{\boldsymbol{k}} = A_r \cdot \hat{\boldsymbol{e}}_r + A_\vartheta \cdot \hat{\boldsymbol{e}}_\vartheta + A_\varphi \cdot \hat{\boldsymbol{e}}_\varphi$$

To obtain, for example, $A_x$ as a function of $A_r$, $A_\vartheta$, and $A_\varphi$, we simply need to multiply this equation by the unit vector $\hat{\boldsymbol{i}}$. Similarly, we obtain $A_y$ and $A_z$ as functions of $A_r$, $A_\vartheta$, and $A_\varphi$. The result is:

$$A_x = A_r \sin \vartheta \cos \varphi + A_\vartheta \cos \vartheta \cos \varphi - A_\varphi \sin \varphi \tag{A.11a}$$

$$A_y = A_r \sin \vartheta \sin \varphi + A_\vartheta \cos \vartheta \sin \varphi + A_\varphi \cos \varphi \tag{A.11b}$$

$$A_z = A_r \cos \vartheta - A_\vartheta \sin \vartheta \tag{A.11c}$$

This allows us to determine the partial derivatives of $A_x$, $A_y$, and $A_z$ with respect to $r$, $\vartheta$, and $\varphi$:

$$\frac{\partial A_x}{\partial r} = \frac{\partial A_r}{\partial r} \sin \vartheta \cos \varphi + \frac{\partial A_\vartheta}{\partial r} \cos \vartheta \cos \varphi - \frac{\partial A_\varphi}{\partial r} \sin \varphi \tag{A.12a}$$

$$\frac{\partial A_y}{\partial r} = \frac{\partial A_r}{\partial r} \sin \vartheta \sin \varphi + \frac{\partial A_\vartheta}{\partial r} \cos \vartheta \sin \varphi + \frac{\partial A_\varphi}{\partial r} \cos \varphi \tag{A.12b}$$

$$\frac{\partial A_z}{\partial r} = \frac{\partial A_r}{\partial r} \cos \vartheta - \frac{\partial A_\vartheta}{\partial r} \sin \vartheta \tag{A.12c}$$

$$\frac{\partial A_x}{\partial \vartheta} = \frac{\partial A_r}{\partial \vartheta}\sin\vartheta\cos\varphi + A_r\cos\vartheta\cos\varphi + \frac{\partial A_\vartheta}{\partial \vartheta}\cos\vartheta\cos\varphi - A_\vartheta\sin\vartheta\cos\varphi - \frac{\partial A_\varphi}{\partial \vartheta}\sin\varphi \tag{A.12d}$$

$$\frac{\partial A_y}{\partial \vartheta} = \frac{\partial A_r}{\partial \vartheta}\sin\vartheta\sin\varphi + A_r\cos\vartheta\sin\varphi + \frac{\partial A_\vartheta}{\partial \vartheta}\cos\vartheta\sin\varphi - A_\vartheta\sin\vartheta\sin\varphi + \frac{\partial A_\varphi}{\partial \vartheta}\cos\varphi \tag{A.12e}$$

$$\frac{\partial A_z}{\partial \vartheta} = \frac{\partial A_r}{\partial \vartheta}\cos\vartheta - A_r\sin\vartheta - \frac{\partial A_\vartheta}{\partial \vartheta}\sin\vartheta - A_\vartheta\cos\vartheta \tag{A.12f}$$

$$\frac{\partial A_x}{\partial \varphi} = \frac{\partial A_r}{\partial \varphi}\sin\vartheta\cos\varphi - A_r\sin\vartheta\sin\varphi + \frac{\partial A_\vartheta}{\partial \varphi}\cos\vartheta\cos\varphi - A_\vartheta\cos\vartheta\sin\varphi - \frac{\partial A_\varphi}{\partial \varphi}\sin\varphi - A_\varphi\cos\varphi \tag{A.12g}$$

$$\frac{\partial A_y}{\partial \varphi} = \frac{\partial A_r}{\partial \varphi}\sin\vartheta\sin\varphi + A_r\sin\vartheta\cos\varphi + \frac{\partial A_\vartheta}{\partial \varphi}\cos\vartheta\sin\varphi + A_\vartheta\cos\vartheta\cos\varphi + \frac{\partial A_\varphi}{\partial \varphi}\cos\varphi - A_\varphi\sin\varphi \tag{A.12h}$$

$$\frac{\partial A_z}{\partial \varphi} = \frac{\partial A_r}{\partial \varphi}\cos\vartheta - \frac{\partial A_\vartheta}{\partial \varphi}\sin\vartheta \tag{A.12i}$$

By substituting eq. (A.12a) through eq. (A.12i) and eq. (A.7a) through eq. (A.7c) into the expression above for $\operatorname{div}\boldsymbol{A}$, we then obtain:

$$\operatorname{div}\boldsymbol{A} = \frac{2}{r}A_r + \frac{\partial A_r}{\partial r} + \frac{\cos\vartheta}{r\sin\vartheta}A_\vartheta + \frac{1}{r\sin\vartheta}\frac{\partial A_\vartheta}{\partial\vartheta} + \frac{1}{r\sin\vartheta}\frac{\partial A_\varphi}{\partial\varphi}$$

We can simplify this expression further to:

$$\operatorname{div}\boldsymbol{A} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 A_r\right) + \frac{1}{r\sin\vartheta}\frac{\partial}{\partial\vartheta}(\sin\vartheta A_\vartheta) + \frac{1}{r\sin\vartheta}\frac{\partial A_\varphi}{\partial\varphi} \tag{A.13}$$

With eq. (A.13) we have derived the expression for the divergence operator in spherical coordinates. Combining eq. (A.10) and eq. (A.13) also yields the Laplacian operator in spherical coordinates. One simply substitutes the vector components from eq. (A.10) into eq. (A.13), thereby forming $\Delta U = \nabla(\nabla U)$:

$$\Delta U = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right) + \frac{1}{r\sin\vartheta}\frac{\partial}{\partial\vartheta}\left(\sin\vartheta\frac{1}{r}\frac{\partial U}{\partial\vartheta}\right) + \frac{1}{r\sin\vartheta}\frac{\partial}{\partial\varphi}\left(\frac{1}{r\sin\vartheta}\frac{\partial U}{\partial\varphi}\right)$$

Factoring out the quantities that are not differentiated then yields the familiar form of the Laplace operator in spherical coordinates:

$$\Delta U = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right) + \frac{1}{r^2\sin\vartheta}\frac{\partial}{\partial\vartheta}\left(\sin\vartheta\frac{\partial U}{\partial\vartheta}\right) + \frac{1}{r^2\sin^2\vartheta}\frac{\partial^2 U}{\partial\varphi^2} \tag{A.14}$$

**Tab. A.1:** Differential operators in cylindrical, spherical, and Cartesian coordinates.

**Cartesian coordinates** $(x, y, z)$

$$\begin{aligned}
\operatorname{grad} U &= \frac{\partial U}{\partial x}\cdot\hat{\boldsymbol{i}} + \frac{\partial U}{\partial y}\cdot\hat{\boldsymbol{j}} + \frac{\partial U}{\partial z}\cdot\hat{\boldsymbol{k}} \\
\operatorname{div}\boldsymbol{A} &= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \\
\Delta U &= \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} + \frac{\partial^2 U}{\partial z^2} \\
\operatorname{rot}\boldsymbol{A} &= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\cdot\hat{\boldsymbol{i}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\cdot\hat{\boldsymbol{j}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\cdot\hat{\boldsymbol{k}}
\end{aligned}$$

**Spherical coordinates** $(r, \vartheta, \varphi)$

$$\begin{aligned}
\operatorname{grad} U &= \frac{\partial U}{\partial r}\cdot\hat{\boldsymbol{e}}_r + \frac{1}{r}\frac{\partial U}{\partial \vartheta}\cdot\hat{\boldsymbol{e}}_\vartheta + \frac{1}{r\sin\vartheta}\frac{\partial U}{\partial \varphi}\cdot\hat{\boldsymbol{e}}_\varphi \\
\operatorname{div}\boldsymbol{A} &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 A_r\right) + \frac{1}{r\sin\vartheta}\frac{\partial}{\partial \vartheta}(\sin\vartheta A_\vartheta) + \frac{1}{r\sin\vartheta}\frac{\partial A_\varphi}{\partial \varphi} \\
\Delta U &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial U}{\partial r}\right) + \frac{1}{r^2\sin\vartheta}\frac{\partial}{\partial \vartheta}\left(\sin\vartheta\frac{\partial U}{\partial \vartheta}\right) + \frac{1}{r^2\sin^2\vartheta}\frac{\partial^2 U}{\partial \varphi^2} \\
\operatorname{rot}\boldsymbol{A} &= \frac{1}{r\sin\vartheta}\left[\frac{\partial}{\partial \vartheta}(\sin\vartheta A_\varphi) - \frac{\partial A_\vartheta}{\partial \varphi}\right]\cdot\hat{\boldsymbol{e}}_r + \frac{1}{r}\left[\frac{1}{\sin\vartheta}\frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r}(rA_\varphi)\right]\cdot\hat{\boldsymbol{e}}_\vartheta \\
&\quad + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\vartheta) - \frac{\partial A_r}{\partial \vartheta}\right]\cdot\hat{\boldsymbol{e}}_\varphi
\end{aligned}$$

**Cylindrical coordinates** $(R, \varphi, z)$

$$\begin{aligned}
\operatorname{grad} U &= \frac{\partial U}{\partial R}\cdot\hat{\boldsymbol{e}}_R + \frac{1}{R}\frac{\partial U}{\partial \varphi}\cdot\hat{\boldsymbol{e}}_\varphi + \frac{\partial U}{\partial z}\cdot\hat{\boldsymbol{e}}_z \\
\operatorname{div}\boldsymbol{A} &= \frac{1}{R}\frac{\partial}{\partial R}(RA_R) + \frac{1}{R}\frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z} \\
\Delta U &= \frac{1}{R}\frac{\partial}{\partial R}\left(R\frac{\partial U}{\partial R}\right) + \frac{1}{R^2}\frac{\partial^2 U}{\partial \varphi^2} + \frac{\partial^2 U}{\partial z^2} \\
\operatorname{rot}\boldsymbol{A} &= \left[\frac{1}{R}\frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z}\right]\cdot\hat{\boldsymbol{e}}_R + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial R}\right]\cdot\hat{\boldsymbol{e}}_\varphi \\
&\quad + \frac{1}{R}\left[\frac{\partial}{\partial R}(RA_\varphi) - \frac{\partial A_R}{\partial \varphi}\right]\cdot\hat{\boldsymbol{e}}_z
\end{aligned}$$

Now, all that remains is the expression for $\operatorname{rot}\mathbf{A}$ in spherical coordinates. To derive this, we proceed in the same way as for $\operatorname{div}\mathbf{A}$, meaning that we write out the vector components of $\operatorname{rot}\mathbf{A}$ explicitly:

$$\begin{aligned}
\operatorname{rot}\boldsymbol{A} &= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\cdot\hat{\boldsymbol{i}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\cdot\hat{\boldsymbol{j}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\cdot\hat{\boldsymbol{k}} \\
&= \left(\frac{\partial A_z}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial A_z}{\partial \vartheta}\frac{\partial \vartheta}{\partial y} + \frac{\partial A_z}{\partial \varphi}\frac{\partial \varphi}{\partial y} - \frac{\partial A_y}{\partial r}\frac{\partial r}{\partial z} - \frac{\partial A_y}{\partial \vartheta}\frac{\partial \vartheta}{\partial z} - \frac{\partial A_y}{\partial \varphi}\frac{\partial \varphi}{\partial z}\right)\cdot\hat{\boldsymbol{i}} + \\
&\quad \left(\frac{\partial A_x}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial A_x}{\partial \vartheta}\frac{\partial \vartheta}{\partial z} + \frac{\partial A_x}{\partial \varphi}\frac{\partial \varphi}{\partial z} - \frac{\partial A_z}{\partial r}\frac{\partial r}{\partial x} - \frac{\partial A_z}{\partial \vartheta}\frac{\partial \vartheta}{\partial x} - \frac{\partial A_z}{\partial \varphi}\frac{\partial \varphi}{\partial x}\right)\cdot\hat{\boldsymbol{j}} + \\
&\quad \left(\frac{\partial A_y}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial A_y}{\partial \vartheta}\frac{\partial \vartheta}{\partial x} + \frac{\partial A_y}{\partial \varphi}\frac{\partial \varphi}{\partial x} - \frac{\partial A_x}{\partial r}\frac{\partial r}{\partial y} - \frac{\partial A_x}{\partial \vartheta}\frac{\partial \vartheta}{\partial y} - \frac{\partial A_x}{\partial \varphi}\frac{\partial \varphi}{\partial y}\right)\cdot\hat{\boldsymbol{k}}
\end{aligned}$$

Substituting the derivatives from eq. (A.12a) through eq. (A.12i) and eq. (A.7a) through eq. (A.7c) back into the right-hand side of this equation, then, after some calculation and rearrangement, one obtains in spherical coordinates:

$$\begin{aligned}
\operatorname{rot}\boldsymbol{A} &= \frac{1}{r\sin\vartheta}\left[\frac{\partial}{\partial\vartheta}(\sin\vartheta A_\varphi) - \frac{\partial A_\vartheta}{\partial\varphi}\right]\cdot\hat{\boldsymbol{e}}_r + \frac{1}{r}\left[\frac{1}{\sin\vartheta}\frac{\partial A_r}{\partial\varphi} - \frac{\partial}{\partial r}(rA_\varphi)\right]\cdot\hat{\boldsymbol{e}}_\vartheta \\
&\quad + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\vartheta) - \frac{\partial A_r}{\partial\vartheta}\right]\cdot\hat{\boldsymbol{e}}_\varphi
\end{aligned} \tag{A.15}$$

Just as we derived the expressions for the differential operators grad, div, rot, and the Laplace operator $\Delta$ here in spherical coordinates, the same can be done to obtain the respective expressions for these differential operators in cylindrical coordinates. Table A.1 provides a convenient reference summarizing these operators in Cartesian, spherical, and cylindrical coordinates.

## Cross Product of a Vector with the Curl of a Vector

We consider the vector quantity:

$$\boldsymbol{A}\times(\nabla\times\boldsymbol{B})$$

This expression can be evaluated using the Grassmann identity for $\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})$ (see tab. 2.1). However, one must take into account that $\boldsymbol{b}$ is an operator here acting on the vector $\boldsymbol{c}$. We must therefore write $\boldsymbol{c}$ to the right of the vector $\boldsymbol{b}$ in the Grassmann identity. The Grassmann identity thus reads:

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\boldsymbol{c}) - (\boldsymbol{a}\boldsymbol{b})\boldsymbol{c} \tag{A.16}$$

Furthermore, we must ensure that the $\nabla$-operator acts only on the vector $\boldsymbol{B}$. This we can do by writing this operator as $\nabla_{\boldsymbol{B}}$ when it appears to the left of $\boldsymbol{A}$, signifying that it only acts on $\boldsymbol{B}$. In this way, we obtain the following expression for the cross product of a vector $\boldsymbol{A}$ with the curl of another vector $\boldsymbol{B}$:

$$\boldsymbol{A} \times (\nabla \times \boldsymbol{B}) \;=\; \nabla_{\boldsymbol{B}}(\boldsymbol{A}\boldsymbol{B}) - (\boldsymbol{A}\nabla)\boldsymbol{B} \tag{A.17}$$

In the case of $\boldsymbol{B} = \boldsymbol{A}$, this expression simplifies to:

$$\boldsymbol{A} \times (\nabla \times \boldsymbol{A}) \;=\; \frac{1}{2}\nabla(\boldsymbol{A}^2) - (\boldsymbol{A}\nabla)\boldsymbol{A} \tag{A.18}$$

## The $\delta$-Function in Spherical and Cylindrical Coordinates

Next, we will look into how the $\delta$-function behaves under a coordinate transformation. To understand what happens in such a transformation, we use the integration property of the three-dimensional $\delta$-function, namely:

$$\delta^{(3)}_{xyz}(\boldsymbol{r} - \boldsymbol{r}_0) \;=\; \delta(x - x_0)\delta(y - y_0)\delta(z - z_0)$$

To this end, we integrate the product of a function $f(\boldsymbol{r})$ and $\delta^{(3)}_{xyz}(\boldsymbol{r} - \boldsymbol{r}_0)$ over a volume $\Omega_{xyz}$ containing $\boldsymbol{r}_0$, yielding the result:

$$\int_{\Omega_{xyz}} f(\boldsymbol{r})\delta^{(3)}_{xyx}(\boldsymbol{r} - \boldsymbol{r}_0)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \;=\; f(\boldsymbol{r}_0)$$

If we set here $f(\boldsymbol{r}) = 1$, then integrating $1 \cdot \delta^{(3)}_{xyz}(\boldsymbol{r} - \boldsymbol{r}_0)$ yields the result 1. When switching this integration from Cartesian coordinates to spherical coordinates, the corresponding transformed integral must, of course, also yield the value 1. The only question is: what does the three-dimensional $\delta$-function look like in the transformed integral? The answer is simple: it looks just as we would like it to, i.e., a $\delta$-function that is non-zero only at the spherical coordinates $(r_0, \vartheta_0, \varphi_0)$ and takes the value 1 there. We therefore make the following ansatz:

$$\int_{\Omega_{xyz}} \delta^{(3)}_{xyz}(\boldsymbol{r} - \boldsymbol{r}_0)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \;=$$

$$\int_{\Omega_{r\vartheta\varphi}} g(r, \vartheta, \varphi)\delta(r - r_0)\delta(\vartheta - \vartheta_0)\delta(\varphi - \varphi_0)\left|\frac{\partial(x, y, z)}{\partial(r, \vartheta, \varphi)}\right| \mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \;=\; 1$$

The distortion factor of the differential volume element for the transformation from Cartesian to spherical coordinates appears once again in the integral over $\Omega_{r\vartheta\varphi}$. If the function $g(r, \vartheta, \varphi)$ were not present in the integral over $\Omega_{r\vartheta\varphi}$, the integral would have the value

$r_0^2 \sin\vartheta_0$ rather than the value 1. We introduced $g(r,\vartheta,\varphi)$ to ensure that the integration over the $\delta$-function indeed yields the value 1. The function $g(r,\vartheta,\varphi)$ must therefore be exactly 1 divided by the distortion factor. This means, however, that the $\delta$-function in spherical coordinates is given by:

$$\delta^{(3)}_{r\vartheta\varphi}(\boldsymbol{r}-\boldsymbol{r}_0) \;=\; \frac{1}{r^2\sin\vartheta}\delta(r-r_0)\delta(\vartheta-\vartheta_0)\delta(\varphi-\varphi_0) \tag{A.19}$$

Provided that $\Omega_{r\vartheta\varphi}$ contains the point $\boldsymbol{r}_0$, this always ensures that:

$$\int_{\Omega_{r\vartheta\varphi}} f(\boldsymbol{r})\delta^{(3)}_{r\vartheta\varphi}(\boldsymbol{r}-\boldsymbol{r}_0) r^2\sin\vartheta\,\mathrm{d}r\,\mathrm{d}\vartheta\,\mathrm{d}\varphi \;=\; f(\boldsymbol{r}_0)$$

An analogous consideration for the delta function in cylindrical coordinates $\boldsymbol{r}(R,\varphi,z)$ yields:

$$\delta^{(3)}_{R\varphi z}(\boldsymbol{r}-\boldsymbol{r}_0) \;=\; \frac{1}{R}\delta(R-R_0)\delta(\varphi-\varphi_0)\delta(z-z_0) \tag{A.20}$$

As long as $\Omega_{R\varphi z}$ contains the point $\boldsymbol{r}_0$, the corresponding result holds:

$$\int_{\Omega_{R\varphi z}} f(\boldsymbol{r})\delta^{(3)}_{R\varphi z}(\boldsymbol{r}-\boldsymbol{r}_0) R\,\mathrm{d}R\,\mathrm{d}\varphi\,\mathrm{d}z \;=\; f(\boldsymbol{r}_0)$$

## Symmetry of the Green's Function

The defining equation for the Green's function $G(\boldsymbol{r},\boldsymbol{r}')$ is:

$$\Delta_{\boldsymbol{r}} G(\boldsymbol{r},\boldsymbol{r}') \;=\; -\frac{1}{\epsilon_0}\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}') \tag{A.21}$$

If the coordinates are interchanged such that $G(\boldsymbol{r},\boldsymbol{r}') = G(\boldsymbol{r}',\boldsymbol{r})$, then this Green's function is symmetric. The Dirichlet boundary condition for a Green's function is:

$$G(\boldsymbol{r},\boldsymbol{r}') \;=\; 0 \qquad \text{if} \qquad \boldsymbol{r}\in\partial V \tag{A.22}$$

In this case, Green's function is always symmetric. To prove this, one uses Green's second identity in the form

$$\int_\Omega (\varphi\Delta\psi - \psi\Delta\varphi)\,\mathrm{d}^3r \;=\; \int_{\partial V}\left(\varphi\frac{\partial\psi}{\partial n} - \psi\frac{\partial\varphi}{\partial n}\right)\mathrm{d}\sigma \tag{A.23}$$

and substitutes the two Green's functions for $\varphi$ and $\psi$, that is

$$\varphi(\boldsymbol{r}) \;=\; G(\boldsymbol{r},\boldsymbol{r}_1) \qquad \text{and} \qquad \psi(\boldsymbol{r}) \;=\; G(\boldsymbol{r},\boldsymbol{r}_2)$$

For the left-hand side (LHS) of eq. (A.23), one thus obtains:

$$\text{LHS} \;=\; \int_\Omega \left(G(\boldsymbol{r},\boldsymbol{r}_1)\Delta G(\boldsymbol{r},\boldsymbol{r}_2) - G(\boldsymbol{r},\boldsymbol{r}_2)\Delta G(\boldsymbol{r},\boldsymbol{r}_1)\right) \mathrm{d}^3 r$$

Using eq. (A.21), this becomes:

$$\begin{aligned}\text{LHS} \;&=\; -\frac{1}{\epsilon_0}\int_\Omega \left(G(\boldsymbol{r},\boldsymbol{r}_1)\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_2) - G(\boldsymbol{r},\boldsymbol{r}_2)\delta^{(3)}(\boldsymbol{r}-\boldsymbol{r}_1)\right) \mathrm{d}^3 r \\ &=\; -\frac{1}{\epsilon_0}\left(G(\boldsymbol{r}_2,\boldsymbol{r}_1) - G(\boldsymbol{r}_1,\boldsymbol{r}_2)\right)\end{aligned}$$

However, for the right-hand side (RHS) of eq. (A.23), one obtains after substitution

$$\text{RHS} \;=\; \int_{\partial V}\left(G(\boldsymbol{r},\boldsymbol{r}_1)\frac{\partial G(\boldsymbol{r},\boldsymbol{r}_2)}{\partial n} - G(\boldsymbol{r},\boldsymbol{r}_2)\frac{\partial G(\boldsymbol{r},\boldsymbol{r}_1)}{\partial n}\right)\mathrm{d}\sigma \;=\; 0$$

because both $G(\boldsymbol{r},\boldsymbol{r}_1)$ and $G(\boldsymbol{r},\boldsymbol{r}_2)$, as solutions to eq. (A.21), must vanish on $\partial V$ in accordance with eq. (A.22). Consequently, it follows from LHS = RHS that

$$G(\boldsymbol{r_2},\boldsymbol{r}_1) - G(\boldsymbol{r}_1,\boldsymbol{r}_2) \;=\; 0$$

and therefore $G(\boldsymbol{r}_2,\boldsymbol{r}_1)$ and $G(\boldsymbol{r}_1,\boldsymbol{r}_2)$ must be identical.

## Taylor Expansion

For a simple scalar function of a single variable $f(x)$, the first terms of the Taylor expansion about a point $x = a$ are:

$$f(x) \;=\; f(a) \;+\; (x-a)\cdot\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x=a} \;+\; (x-a)^2\cdot\left.\frac{\mathrm{d}^2 f}{\mathrm{d}x^2}\right|_{x=a} \;+\; \ldots \tag{A.24}$$

This Taylor expansion in the neighborhood of a point $x = a$ is usually part of the mathematics curriculum in secondary schools and therefore likely familiar to many readers. For most applications, the first two terms are generally sufficient for a good approximation. Here, we are interested in the case of a scalar function of three variables, $f(x, y, z)$. For the sake of clarity, we write the partial derivatives as:

$$\frac{\partial f}{\partial x_i} \;=\; f_{x_i} \qquad \text{and} \qquad \frac{\partial^2 f}{\partial x_i \partial x_j} \;=\; f_{x_i x_j}$$

Furthermore, instead of writing $\boldsymbol{r} = \boldsymbol{a}$ for the point at which the derivatives are to be evaluated, we simply write $\boldsymbol{a}$, with the understanding that the derivatives are to be evaluated at this point. Thus, the first terms of the Taylor expansion for the scalar function $f(x, y, z)$ about a point $\boldsymbol{a} = (a_1, a_2, a_3)$ are:

$$f(x,y,z) \;=\; f(\boldsymbol{a}) \;+\; (x-a_1)\cdot f_x\big|_{\boldsymbol{a}} \;+\; (y-a_2)\cdot f_y\big|_{\boldsymbol{a}} \;+\; (z-a_3)\cdot f_z\big|_{\boldsymbol{a}} \;+\; \frac{1}{2}\cdot\Big[$$
$$(x-a_1)^2\cdot f_{xx}\big|_{\boldsymbol{a}} + (x-a_1)(y-a_2)\cdot f_{xy}\big|_{\boldsymbol{a}} + (x-a_1)(z-a_3)\cdot f_{xz}\big|_{\boldsymbol{a}} +$$
$$(x-a_1)(y-a_2)\cdot f_{yx}\big|_{\boldsymbol{a}} + (y-a_2)^2\cdot f_{yy}\big|_{\boldsymbol{a}} + (y-a_2)(z-a_3)\cdot f_{yz}\big|_{\boldsymbol{a}} +$$
$$(x-a_1)(z-a_3)\cdot f_{zx}\big|_{\boldsymbol{a}} + (y-a_2)(z-a_3)\cdot f_{zy}\big|_{\boldsymbol{a}} + (z-a_3)^2\cdot f_{zz}\big|_{\boldsymbol{a}}\Big] \;+\; \ldots$$

We could simplify this expression further, since Schwarz's theorem implies $f_{x_ix_j} = f_{x_jx_i}$, but instead we rewrite this equation as:

$$f(x,y,z) \;=\; f(\boldsymbol{a}) + (\boldsymbol{r}-\boldsymbol{a})\nabla f\big|_{\boldsymbol{r}=\boldsymbol{a}} + (\boldsymbol{r}-\boldsymbol{a})\cdot\underline{\underline{\boldsymbol{H}}}\cdot(\boldsymbol{r}-\boldsymbol{a}) \;+\; \ldots \tag{A.25}$$

Here, $\underline{\underline{\boldsymbol{H}}}$ is the so-called Hessian matrix of partial derivatives, for which holds:

$$\underline{\underline{\boldsymbol{H}}} \;=\; \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} \tag{A.26}$$

In our case, this matrix is to be evaluated at the point $\boldsymbol{r} = \boldsymbol{a}$. Eq. (A.25) has a form very similar to eq. (A.24), except that eq. (A.25) involves the gradient of $f$ and mixed derivatives rather than simple derivatives with respect to a single variable; in addition, the order of evaluation for the quadratic term in eq. (A.25) is significant. We can use eq. (A.25) to determine the Taylor expansion of a vector function $\boldsymbol{B}(\boldsymbol{r}) = (B_1, B_2, B_3)$, where $B_i = B_i(x,y,z)$ for all $i$. Thus, we can apply our result for $f(x,y,z)$ to each component $B_i$, while restricting ourselves to the first two terms on the right-hand side of eq. (A.25). This yields the following Taylor expansion for $\boldsymbol{B}(\boldsymbol{r})$ about the point $\boldsymbol{a} = (a_1, a_2, a_3)$:

$$\boldsymbol{B}(\boldsymbol{r}) \;=\; \boldsymbol{B}(\boldsymbol{a}) + (\boldsymbol{r}-\boldsymbol{a})\cdot\begin{pmatrix} \nabla B_1\big|_{\boldsymbol{r}=\boldsymbol{a}} \\ \nabla B_2\big|_{\boldsymbol{r}=\boldsymbol{a}} \\ \nabla B_3\big|_{\boldsymbol{r}=\boldsymbol{a}} \end{pmatrix} \;+\; \ldots \tag{A.27}$$

We can pull out the gradient and thus obtain:

$$\boldsymbol{B}(\boldsymbol{r}) \;=\; \boldsymbol{B}(\boldsymbol{a}) + \big[(\boldsymbol{r}-\boldsymbol{a})\cdot\nabla\big]\boldsymbol{B}(\boldsymbol{r})\big|_{\boldsymbol{r}=\boldsymbol{a}} \;+\; \ldots \tag{A.28}$$

In the special case of $\boldsymbol{a} = 0$, i.e., for the Taylor expansion about the origin, we obtain:

$$\boldsymbol{B}(\boldsymbol{r}) \;=\; \boldsymbol{B}(\boldsymbol{0}) + \big(\boldsymbol{r}\cdot\nabla\big)\boldsymbol{B}(\boldsymbol{r})\big|_{\boldsymbol{r}=0} \;+\; \ldots \tag{A.29}$$

In contrast to the Taylor expansion of the scalar function $f(x, y, z)$ in eq. (A.25), the second term of the Taylor expansion here involves the directional derivative of the vector function $\boldsymbol{B}(\boldsymbol{r})$ rather than the gradient of the function.

# Appendix B

## International System of Units (SI)[1]

### Base Units

| Base quantity | Unit | Symbol |
|---|---|---|
| Length | meter | m |
| Mass | kilogram | kg |
| Time | second | s |
| Electric current | Ampere | A |
| Temperature (thermodynamic) | Kelvin | K |
| Substance amount | mole | mol |
| Luminous intensity | candela | cd |

### Derived Units (selection)

| Derived quantity | Unit | Symbol | Dimension |
|---|---|---|---|
| Frequency | Hertz | Hz | $s^{-1}$ |
| Force | Newton | N | $kg \cdot m \cdot s^{-2}$ |
| Pressure | Pascal | Pa | $N \cdot m^{-2}$ |
| Energy, work, heat | Joule | J | $N \cdot m$ |
| Power | Watt | W | $J \cdot s^{-1}$ |
| Electric charge | Coulomb | C | $A \cdot s$ |
| Electric voltage | Volt | V | $W \cdot A^{-1}$ |
| Electric resistance | Ohm | Ω | $V \cdot A^{-1}$ |
| Electric capacitance | Farad | F | $C \cdot V^{-1}$ |
| Electric conductance | Siemens | S | $A \cdot V^{-1}$ |
| Magnetic flux | Weber | Wb | $J \cdot A^{-1}$ |
| Magnetic flux density | Tesla | T | $Wb \cdot m^{-2}$ |
| Inductance | Henry | H | $V \cdot s \cdot A^{-1}$ |
| Angle | Radian | rad | – |
| Solid angle | Steradian | sr | – |
| Temperature relative to 273.15 K | Celsius | °C | K |
| Luminous flux | Lumen | lm | $cd \cdot sr$ |
| Illuminance | Lux | lx | $lm \cdot m^{-2}$ |
| Radioactivity | Becquerel | Bq | $s^{-1}$ |

[1] SI is the abbreviation for Système international d'unités

| Derived quantity | Name | Symbol | Dimension |
|---|---|---|---|
| Absorbed dose | Gray | Gy | $J \cdot kg^{-1}$ |
| Equivalent dose | Sievert | Sv | $A \cdot kg^{-1}$ |
| Catalytic activity | Katal | kat | $mol \cdot s^{-1}$ |

# SI Prefixes for Powers of Ten

| Power | Prefix | Symbol | Power | Prefix | Symbol |
|---|---|---|---|---|---|
| $10^{12}$ | tera | T | $10^{-2}$ | centi | c |
| $10^{9}$ | giga | G | $10^{-3}$ | milli | m |
| $10^{6}$ | mega | M | $10^{-6}$ | micro | $\mu$ |
| $10^{3}$ | kilo | k | $10^{-9}$ | nano | n |
| $10^{2}$ | hecto | h | $10^{-12}$ | pico | p |
| $10^{1}$ | deca | da | $10^{-15}$ | femto | f |
| $10^{-1}$ | deci | d | $10^{-18}$ | atto | a |

# Unit Conversions

### Length and area conversions:

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Parallax second (Parsec) | pc | $1\,\text{pc} = 3.084 \cdot 10^{16}\,\text{m}$ |
| Light-year | ly | $1\,\text{ly} = 9.46 \cdot 10^{15}\,\text{m}$ |
| Astronomic unit | AU | $1\,\text{AE} = 1.496 \cdot 10^{11}\,\text{m}$ |
| English or US mile | mi | $1\,\text{mi} = 1.6093\,\text{km}$ |
| Yard | yd | $1\,\text{yd} = 0.9144\,\text{m}$ |
| Inch | in | $1\,\text{in} = 2.540\,\text{cm}$ |
| Angstrom | Å | $1\,\text{Å} = 10^{-10}\,\text{m}$ |
| Fermi | F | $1\,\text{F} = 10^{-15}\,\text{m}$ |
| Barn | b | $1\,\text{b} = 10^{-28}\,\text{m}^2$ |

### Angle conversions:

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Radian | rad | $1\,\text{rad} = 360°/2\pi = 57.2958°$ |
| Arc minute | $1'$ | $1' = 1°/60 = 2.91 \cdot 10^{-4}\text{rad}$ |
| Arc second | $1''$ | $1'' = 1°/3600 = 4.85 \cdot 10^{-6}\text{rad}$ |

### Time conversions:

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Minute | min | $1\,\text{min} = 60\,\text{s}$ |
| Hour | h | $1\,\text{h} = 3\,600\,\text{s}$ |
| Day | d | $1\,\text{d} = 86\,400\,\text{s}$ |
| Year | a | $1\,\text{a} = 365.26\,\text{d} = 3.156{\cdot}10^{7}\,\text{s}$ |

### Mass conversions:

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| English or US pound | lb | $1\,\text{lb} = 0.4536\,\text{kg}$ |
| Atomic mass unit | amu | $1\,\text{amu} = 1.6605{\cdot}10^{-27}\text{kg}$ |

### Force conversions:

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Dyne | dyn | $1\,\text{dyn} = 10^{-5}\,\text{N}$ |
| Kilopond | kp | $1\,\text{kp} = 1\,\text{kg}{\cdot}g = 9.81\,\text{N}$ |

### Energy conversions:

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Calorie | cal | $1\,\text{cal} = 4.187\,\text{J}$ |
| Ergon | erg | $1\,\text{erg} = 10^{-7}\,\text{J}$ |
| Electronvolt | eV | $1\,\text{eV} = 1.602{\cdot}10^{-19}\,\text{J}$ |

### Mass-Energy equivalents:

| Reference measure | Relation | Equivalence |
|---|---|---|
| Electron mass | $E = mc^2$ | $m_e c^2 = 0.511\,\text{MeV}$ |
| Proton mass | $E = mc^2$ | $m_p c^2 = 1.6726\,\text{MeV}$ |
| Neutron mass | $E = mc^2$ | $m_n c^2 = 1.6749\,\text{MeV}$ |
| Atomic mass | $E = mc^2$ | $1\,\text{amu} \mathrel{\widehat{=}} 931.5\,\text{MeV}$ |
| Temperature | $E = k_B T$ | $1\,\text{K} \mathrel{\widehat{=}} 0.8618{\cdot}10^{-4}\,\text{eV}$ |
| Frequency | $E = hf$ | $1\,\text{Hz} \mathrel{\widehat{=}} 4.136{\cdot}10^{-15}\,\text{eV}$ |

### Power conversions:

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Horsepower | hp | $1\,\text{hp} = 735.5\,\text{W}$ |

**Temperature conversions:**

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Celsius | °C | $°C = K - 273.15$ |
| Fahrenheit | °F | $°F = 32 + (K - 273.15) \cdot 9/5$ |

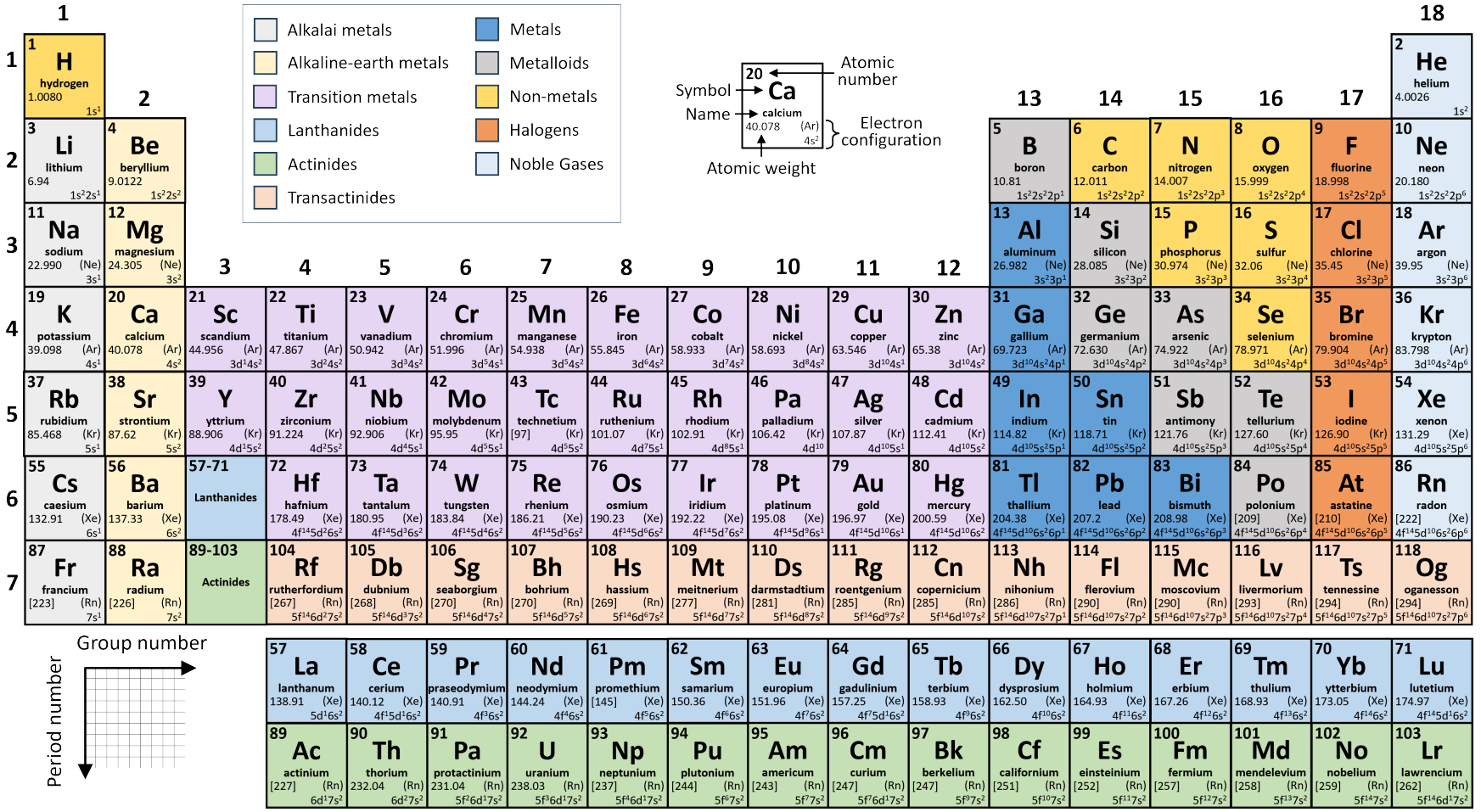
**Pressure conversions:**

| Unit of measurement | Symbol | Conversion |
|---|---|---|
| Bar | bar | $1\,\text{bar} = 10^5\,\text{Pa}$ |
| Atmosphere (standard) | atm | $1\,\text{atm} = 1.013\,\text{bar} = 1.013 \cdot 10^5\,\text{Pa}$ |
| Atmosphere (technical) | at | $1\,\text{at} = 0.981\,\text{bar} = 9.81 \cdot 10^4\,\text{Pa}$ |
| Torr | Torr | $1\,\text{Torr} = 1.333\,\text{mbar} = 1.333\,\text{hPa}$ |
| Millimeter of mercury | mm Hg | $1\,\text{mm Hg} \mathrel{\widehat{=}} 1\,\text{Torr}$ |

# Fundamental Constants

| Constant | Symbol | Value | Dimension |
|---|---|---|---|
| Elementary charge | $e$ | $1.6022 \cdot 10^{-19}$ | C |
| Vacuum speed of light | $c$ | $2.9979 \cdot 10^{8}$ | $\text{m} \cdot \text{s}^{-1}$ |
| Planck constant | $h$ | $6.6262 \cdot 10^{-34}$ | J·s |
| Reduced Planck constant | $\hbar = h/2\pi$ | $1.0546 \cdot 10^{-34}$ | J·s |
| Vacuum magnetic permeability | $\mu_0$ | $1.2566 \cdot 10^{-6}$ | $\text{N} \cdot \text{A}^{-2}$ |
| Vacuum electric permittivity | $\epsilon_0 = 1/\mu_0 c^2$ | $8.8542 \cdot 10^{-12}$ | $\text{C}^2 \cdot \text{m}^{-2} \cdot \text{N}^{-1}$ |
| Coulomb constant | $k_e = 1/4\pi\epsilon_0$ | $8.9876 \cdot 10^{9}$ | $\text{N} \cdot \text{m}^2 \cdot \text{C}^{-2}$ |
| Electron rest mass | $m_e$ | $9.1094 \cdot 10^{-31}$ | kg |
| Proton rest mass | $m_p$ | $1.6726 \cdot 10^{-27}$ | kg |
| Neutron rest mass | $m_n$ | $1.6749 \cdot 10^{-27}$ | kg |
| Bohr radius | $a_0$ | $5.2918 \cdot 10^{-11}$ | m |
| Classical electron radius | $r_e$ | $2.8179 \cdot 10^{-15}$ | m |
| Avogadro number | $N_A$ | $6.0221 \cdot 10^{23}$ | $\text{mol}^{-1}$ |
| Boltzmann constant | $k_B$ | $1.3806 \cdot 10^{-23}$ | $\text{J} \cdot \text{K}^{-1}$ |
| Gas constant | $R = N_A k_B$ | 8.3145 | $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$ |
| Gravitational constant | $G$ | $6.6743 \cdot 10^{-11}$ | $\text{m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$ |
| Gravity acceleration (standard) | $g_0$ | 9.80665 | $\text{m} \cdot \text{s}^{-2}$ |
| Gravity acceleration (50°) | $g$ | 9.810 | $\text{m} \cdot \text{s}^{-2}$ |
| Gravity acceleration (poles) | $g$ | 9.832 | $\text{m} \cdot \text{s}^{-2}$ |
| Gravity acceleration (equator) | $g$ | 9.780 | $\text{m} \cdot \text{s}^{-2}$ |

**Tab. A.1:** The periodic table of elements. The atomic weight is given in atomic mass units. If instead of the atomic weight a number in square brackets is given this number refers to the most stable isotope of an element.



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